

# OC Core 1.3.1 Master Monograph EN

## Release framing

- OC Core 1.3.1 is a strengthened public release built from the real Core 1.3 corpus plus bounded 1.3.1 deltas.
- `white_spot_total=1; package_scope=full_monograph`
- This release remains claim-bounded, source-first, and no-send until explicit owner approval.

## Contents

|                                                         |           |
|---------------------------------------------------------|-----------|
| <b>List of Figures</b>                                  | <b>9</b>  |
| <b>List of Tables</b>                                   | <b>10</b> |
| <b>1 Reader Contract and Navigation Guide</b>           | <b>11</b> |
| 1.1 The five reading routes . . . . .                   | 11        |
| 1.2 Route legend . . . . .                              | 11        |
| 1.3 External interaction crosswalk . . . . .            | 12        |
| 1.4 What this guide is not . . . . .                    | 12        |
| <b>2 Introduction</b>                                   | <b>13</b> |
| 2.1 Motivation . . . . .                                | 13        |
| 2.2 Scope of Core 1.3 . . . . .                         | 13        |
| 2.3 Vertical hierarchy of levels . . . . .              | 14        |
| 2.4 Relation to previous versions . . . . .             | 15        |
| 2.5 Structure of the monograph . . . . .                | 15        |
| <b>3 Background and Motivation</b>                      | <b>15</b> |
| 3.1 Continuum as a structural object . . . . .          | 15        |
| 3.2 Axes and dimensionality . . . . .                   | 16        |
| 3.3 Boundaries and thresholds . . . . .                 | 16        |
| 3.4 Potentials and flows . . . . .                      | 17        |
| 3.5 Continuumness . . . . .                             | 17        |
| 3.6 Motivation for Core 1.3 . . . . .                   | 17        |
| 3.7 Historical motivation . . . . .                     | 18        |
| 3.8 Scope of future background material . . . . .       | 18        |
| <b>4 Ontological Structure of the Model</b>             | <b>18</b> |
| 4.1 Axiomatic foundation: Level $K_0$ . . . . .         | 18        |
| 4.2 Construction of Level $K_1$ . . . . .               | 19        |
| 4.3 General definition of a continuum . . . . .         | 20        |
| 4.4 Taxonomy of thresholds . . . . .                    | 21        |
| 4.5 Potentials, flows, and structural tension . . . . . | 21        |

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| 4.6      | Cycles and continuumness                                          | 22        |
| 4.7      | Evolution operator                                                | 22        |
| 4.8      | Embedding into meta-spaces                                        | 23        |
| 4.9      | Birth of continua                                                 | 23        |
| 4.10     | Life of continua                                                  | 24        |
| 4.11     | Death of continua                                                 | 24        |
| 4.12     | Interaction of continua                                           | 24        |
| 4.13     | Vertical hierarchy $K_0$ – $K_{12}$                               | 25        |
| 4.14     | Summary                                                           | 25        |
| <b>5</b> | <b>Results and Derived Consequences</b>                           | <b>26</b> |
| 5.1      | Theorem 1: Monotonicity of dimensionality                         | 26        |
| 5.2      | Theorem 2: Impossibility of spontaneous dimension creation        | 26        |
| 5.3      | Theorem 3: Death through boundary and embedding collapse          | 26        |
| 5.4      | Theorem 4: Impossibility of evolution outside the embedding space | 27        |
| 5.5      | Theorem 5: Necessity and sufficiency of compatibility with $M$    | 27        |
| 5.6      | Theorem 6: Structural tension and phase transitions               | 27        |
| 5.7      | Theorem 7: Universal law of complexity growth                     | 28        |
| 5.8      | Theorem 8: Impossibility of eternal stabilisation                 | 28        |
| 5.9      | Theorem 9: Death as loss of cycles                                | 28        |
| 5.10     | Theorem 10: Monotonic growth of embedding spaces                  | 29        |
| 5.11     | Theorem 11: Threshold–expressivity incompatibility                | 29        |
| 5.12     | Theorem 12: Incompleteness of embedding spaces                    | 29        |
| 5.13     | Corollaries and domain–independent consequences                   | 29        |
| 5.14     | Implications for the vertical hierarchy                           | 30        |
| 5.15     | Summary                                                           | 30        |
| <b>6</b> | <b>Discussion</b>                                                 | <b>30</b> |
| 6.1      | Conceptual structure of the OC framework                          | 30        |
| 6.2      | Interpretation of the structural results                          | 31        |
| 6.3      | Limitations of the current formalism                              | 32        |
| 6.4      | OC in the context of existing scientific and formal theories      | 32        |
| 6.5      | Implications for the continuum hierarchy                          | 33        |
| 6.6      | Future development paths                                          | 33        |
| 6.7      | Summary                                                           | 34        |
| <b>7</b> | <b>Conclusion</b>                                                 | <b>34</b> |
| 7.1      | Summary of the Core 1.3 contributions                             | 34        |
| 7.2      | Conceptual synthesis                                              | 35        |
| 7.3      | Implications across scientific domains                            | 35        |
| 7.4      | Future directions                                                 | 35        |
| 7.5      | Final remarks                                                     | 36        |
| <b>8</b> | <b>Extended Boundary and Patch Geometry</b>                       | <b>36</b> |
| 8.1      | Formal Definition of $\partial\Omega(K)$                          | 36        |
| 8.2      | Boundary Geometry                                                 | 37        |
| 8.3      | Patch Model of Boundaries                                         | 37        |
| 8.4      | Boundary Threshold System                                         | 37        |
| 8.5      | Boundary–Driven Dynamics                                          | 38        |
| 8.6      | Boundary Failure and Collapse                                     | 39        |
| 8.7      | Boundary and Rebirth                                              | 39        |

|           |                                                                                       |           |
|-----------|---------------------------------------------------------------------------------------|-----------|
| <b>9</b>  | <b>Extended Threshold Landscape</b>                                                   | <b>39</b> |
| 9.1       | Definition of a Threshold                                                             | 39        |
| 9.2       | Taxonomy of Thresholds                                                                | 40        |
| 9.3       | Threshold Geometry                                                                    | 41        |
| 9.4       | Dimensional Thresholds                                                                | 42        |
| 9.5       | Threshold Cascades                                                                    | 42        |
| 9.6       | Death Thresholds                                                                      | 42        |
| 9.7       | Examples Across Levels                                                                | 43        |
| 9.8       | Summary                                                                               | 44        |
| <b>10</b> | <b>Full Hierarchy of Continua <math>K_0</math>–<math>K_{12}</math></b>                | <b>44</b> |
| 10.1      | Overview of the Vertical Hierarchy                                                    | 44        |
| 10.2      | Level $K_0$ : Structural Substrate                                                    | 45        |
| 10.3      | Level $K_1$ : One-Dimensional Continua                                                | 45        |
| 10.4      | Level $K_2$ : Physical Continua                                                       | 45        |
| 10.5      | Levels $K_3$ – $K_4$ : Chemical and Protocellular Continua                            | 46        |
| 10.6      | Level $K_5$ : Early Neural and Bioelectrical Continua                                 | 46        |
| 10.7      | Level $K_6$ : Cognitive Continua                                                      | 47        |
| 10.8      | Levels $K_7$ – $K_8$ : Social and Civilizational Continua                             | 47        |
| 10.9      | Levels $K_9$ – $K_{12}$ : Theoretical, Meta-Theoretical, and Upper-Coherence Continua | 48        |
| 10.10     | Cross-Level Summary                                                                   | 50        |
| <b>11</b> | <b>Evolution and Interaction Operators</b>                                            | <b>50</b> |
| 11.1      | Decomposition of the Evolution Operator                                               | 50        |
| 11.2      | Operator $F$ : Flow Dynamics                                                          | 51        |
| 11.3      | Operator $G$ : Potential Dynamics                                                     | 51        |
| 11.4      | Operator $H$ : Threshold Dynamics                                                     | 52        |
| 11.5      | Operator $Q$ : Cycle Dynamics                                                         | 52        |
| 11.6      | Operator $R$ : Boundary Evolution                                                     | 52        |
| 11.7      | Operator $S$ : Structural Reconfiguration                                             | 53        |
| 11.8      | Operator $U$ : Continuumness Dynamics                                                 | 53        |
| 11.9      | Birth Operators $\Psi_{x \rightarrow x+1}$                                            | 54        |
| 11.10     | Interaction Operator $E_{\text{int}}$                                                 | 54        |
| 11.11     | Operator Algebra and Constraints                                                      | 55        |
| 11.12     | Summary                                                                               | 55        |
| <b>12</b> | <b>Collapse and Rebirth of Continua</b>                                               | <b>55</b> |
| 12.1      | Formal Criteria for Collapse                                                          | 56        |
| 12.2      | Internal vs External Collapse                                                         | 56        |
| 12.3      | Collapse Dynamics                                                                     | 57        |
| 12.4      | Post-Collapse Residue                                                                 | 58        |
| 12.5      | Rebirth Mechanisms                                                                    | 58        |
| 12.6      | Collapse in Higher-Level Continua                                                     | 59        |
| 12.7      | Summary                                                                               | 61        |
| <b>13</b> | <b>Ontological Branching and Global Topology</b>                                      | <b>61</b> |
| 13.1      | Definition of Ontological Branches                                                    | 61        |
| 13.2      | Branch Topology                                                                       | 62        |
| 13.3      | Branching Conditions                                                                  | 63        |
| 13.4      | Branch Interaction                                                                    | 63        |
| 13.5      | Branch Collapse                                                                       | 64        |
| 13.6      | Consequences for the K-Level Hierarchy                                                | 65        |

|           |                                                                                                |           |
|-----------|------------------------------------------------------------------------------------------------|-----------|
| <b>14</b> | <b>Cross-Disciplinary Instantiations of the Continuum Framework</b>                            | <b>66</b> |
| 14.1      | Principles for Domain Embedding . . . . .                                                      | 66        |
| 14.2      | Physics (Level $K_2$ ) . . . . .                                                               | 67        |
| 14.3      | Chemistry and Origins of Life (Levels $K_3$ – $K_4$ ) . . . . .                                | 68        |
| 14.4      | Biology (Level $K_5$ ) . . . . .                                                               | 69        |
| 14.5      | Cognition (Level $K_6$ ) . . . . .                                                             | 69        |
| 14.6      | Society (Level $K_7$ ) . . . . .                                                               | 70        |
| 14.7      | Civilizational Systems (Level $K_8$ ) . . . . .                                                | 71        |
| 14.8      | Theory and Formal Systems (Levels $K_9$ – $K_{10}$ ) . . . . .                                 | 71        |
| <b>15</b> | <b>Extended Falsifiability and Experimental Programme</b>                                      | <b>72</b> |
| 15.1      | Meaning of Falsifiability for Structural Theories . . . . .                                    | 72        |
| 15.2      | Predictions and Tests in Physics ( $K_2$ ) . . . . .                                           | 73        |
| 15.3      | Predictions and Tests in Origins of Life ( $K_3$ – $K_4$ ) . . . . .                           | 74        |
| 15.4      | Predictions and Tests in Biological Systems ( $K_4$ – $K_5$ ) . . . . .                        | 74        |
| 15.5      | Predictions and Tests in Cognitive Continua ( $K_6$ ) . . . . .                                | 75        |
| 15.6      | Predictions and Tests in Social/Civilisational Continua ( $K_7$ – $K_8$ ) . . . . .            | 76        |
| 15.7      | Theoretical and Meta-Theoretical Predictions ( $K_9$ – $K_{10}$ ) . . . . .                    | 76        |
| 15.8      | Meta-Criteria and Cross-Domain Validation . . . . .                                            | 77        |
| <b>16</b> | <b>Module Architecture and Cross-Level Structures</b>                                          | <b>78</b> |
| <b>17</b> | <b>Source Witnessing, Theorem Fate, and Publication Boundary</b>                               | <b>78</b> |
| 17.1      | Canonical scientific authority . . . . .                                                       | 78        |
| 17.2      | Source witnessing . . . . .                                                                    | 79        |
| 17.3      | Theorem fate and scope discipline . . . . .                                                    | 79        |
| 17.4      | Relation to journal-core and companion artifacts . . . . .                                     | 80        |
| 17.5      | Downstream projection rule . . . . .                                                           | 80        |
| <b>18</b> | <b>Core 1.3 Foundational Consistency Dossier</b>                                               | <b>81</b> |
| 18.1      | Why the hierarchy extends through $K_{12}$ . . . . .                                           | 81        |
| 18.2      | Scientific role of the upper levels . . . . .                                                  | 81        |
| 18.3      | Statement classes . . . . .                                                                    | 81        |
| 18.4      | Proof-bearing and nonclaim layers . . . . .                                                    | 83        |
| 18.5      | Relation between master monograph and journal core . . . . .                                   | 83        |
| 18.6      | Reader-facing consequence . . . . .                                                            | 84        |
| <b>19</b> | <b>Theorem Roadmap and Proof Navigation</b>                                                    | <b>85</b> |
| 19.1      | Load-bearing theorem spine . . . . .                                                           | 85        |
| 19.2      | What surrounds the spine . . . . .                                                             | 86        |
| 19.3      | Statement classes and burden . . . . .                                                         | 87        |
| 19.4      | Shortest hostile-review checklist . . . . .                                                    | 87        |
| <b>20</b> | <b>Worked Examples, Counterexamples, and Reader-Oriented Interpretive Checks</b>               | <b>89</b> |
| 20.1      | Worked Example 1: collapse, residue, and rebirth in a minimal admissible-state model . . . . . | 89        |
| 20.2      | Worked Example 2: why $K_{11}$ and $K_{12}$ are not decorative . . . . .                       | 91        |
| 20.3      | Interpretive Check 3: how the journal core is lawfully extracted from the monograph . . . . .  | 94        |
| 20.4      | What would count as a real counterexample . . . . .                                            | 94        |
| 20.5      | Why examples belong in a strict scientific monograph . . . . .                                 | 94        |

|           |                                                                                         |            |
|-----------|-----------------------------------------------------------------------------------------|------------|
| <b>21</b> | <b>Operationalization Program and Predictive Promotion Discipline</b>                   | <b>96</b>  |
| 21.1      | Why the present release separates explanatory force from predictive promotion . . . . . | 96         |
| 21.2      | Current release truth . . . . .                                                         | 96         |
| 21.3      | Closed canon and frontier workbench . . . . .                                           | 96         |
| 21.4      | Closure economics and anti-cycle law . . . . .                                          | 96         |
| 21.5      | Program board . . . . .                                                                 | 96         |
| 21.6      | Domain order and promotion rule . . . . .                                               | 97         |
| 21.7      | Operationalization table . . . . .                                                      | 97         |
| 21.8      | Per-domain status . . . . .                                                             | 97         |
| 21.9      | Release consequence . . . . .                                                           | 99         |
| <b>22</b> | <b>Hybrid-Escalation Evidence Bar and Execution Protocols</b>                           | <b>99</b>  |
| 22.1      | Global evidence bar . . . . .                                                           | 99         |
| 22.2      | Execution protocol matrix . . . . .                                                     | 99         |
| 22.3      | Method ladder . . . . .                                                                 | 100        |
| 22.4      | Per-domain escalation doctrine . . . . .                                                | 101        |
| <b>23</b> | <b>Proof Machinery, Revision Law, Competition, and Compression</b>                      | <b>102</b> |
| 23.1      | Consistency and promotion law . . . . .                                                 | 102        |
| 23.2      | Revision law . . . . .                                                                  | 102        |
| 23.3      | Proof obligations . . . . .                                                             | 102        |
| 23.4      | Minimality and irreducibility . . . . .                                                 | 103        |
| 23.5      | Alternative-model competition . . . . .                                                 | 103        |
| 23.6      | Unified-science atlas . . . . .                                                         | 103        |
| 23.7      | Compression as a bounded metric . . . . .                                               | 103        |
| <b>24</b> | <b>Unified Science Synthesis</b>                                                        | <b>103</b> |
| 24.1      | Final Unified Synthesis Release Gate . . . . .                                          | 103        |
| 24.2      | K0: Structural Conditions for Any Universe . . . . .                                    | 103        |
| 24.3      | K1: Minimal Observable Distinctions and Energetic Axes . . . . .                        | 104        |
| 24.4      | K2: Fields, Phases, and Large-Scale Structure . . . . .                                 | 105        |
| 24.5      | K3: Molecular Organization and Chemical Bonding . . . . .                               | 106        |
| 24.6      | K4: Compartmental and Membrane Thresholds . . . . .                                     | 107        |
| 24.7      | K5: Excitable and Early Neural Continua . . . . .                                       | 107        |
| 24.8      | K6: Cognitive Prediction and Representation . . . . .                                   | 108        |
| 24.9      | K7: Social Coordination and Group Stability . . . . .                                   | 109        |
| 24.10     | K8: Civilizational Flows and Infrastructural Stability . . . . .                        | 110        |
| 24.11     | K9: Knowledge Systems and Scientific Memory . . . . .                                   | 110        |
| 24.12     | K10: Formal Systems and Recursion . . . . .                                             | 111        |
| 24.13     | K11: Meta-Theoretical Reflexivity . . . . .                                             | 112        |
| 24.14     | K12: Global Semantic Coherence . . . . .                                                | 113        |
| 24.15     | Empirical Held-Out Prediction Summary . . . . .                                         | 115        |
| <b>25</b> | <b>Practical Consequences, Predictive Power, and Use of OC Core 1.3</b>                 | <b>116</b> |
| 25.1      | Practical value and support boundary . . . . .                                          | 116        |
| 25.2      | Support classes . . . . .                                                               | 116        |
| 25.3      | What is already usable now . . . . .                                                    | 116        |
| 25.4      | How to use OC in practice . . . . .                                                     | 119        |
| 25.5      | What OC predicts across the closed empirical domains . . . . .                          | 121        |
| 25.6      | Benchmark against serious alternatives . . . . .                                        | 122        |
| 25.7      | What prior science could do, what remained fragmented, and what OC closes . . . . .     | 125        |
| 25.8      | What remains frontier or hypothesis . . . . .                                           | 127        |

|          |                                                                |            |
|----------|----------------------------------------------------------------|------------|
| <b>A</b> | <b>Notation and Symbols</b>                                    | <b>127</b> |
| A.1      | Continuum Structure . . . . .                                  | 127        |
| A.2      | Thresholds and Tension . . . . .                               | 128        |
| A.3      | Operators . . . . .                                            | 129        |
| A.4      | Levels and Embedding Spaces . . . . .                          | 130        |
| A.5      | Miscellaneous Symbols . . . . .                                | 130        |
| <b>B</b> | <b>Collected Axiomatics</b>                                    | <b>131</b> |
| B.1      | Level $K_0$ Axioms . . . . .                                   | 131        |
| B.2      | General Continuum Axioms . . . . .                             | 131        |
| B.3      | Dimensionality and Birth . . . . .                             | 132        |
| B.4      | Life and Death . . . . .                                       | 132        |
| B.5      | Embedding Spaces . . . . .                                     | 132        |
| B.6      | Interaction . . . . .                                          | 132        |
| B.7      | Remarks . . . . .                                              | 133        |
| <b>C</b> | <b>Tabular Summary of <math>K_0</math>–<math>K_{12}</math></b> | <b>133</b> |
| C.1      | Overview Table . . . . .                                       | 133        |
| C.2      | Structural Components per Level . . . . .                      | 134        |
| C.3      | Vertical Continuity Conditions . . . . .                       | 134        |
| C.4      | K-Level Parameter and Numerical Tables . . . . .               | 134        |
| <b>D</b> | <b>Core 1.3 Source-Audit Appendix</b>                          | <b>134</b> |
| D.1      | Why an appendix is still needed . . . . .                      | 134        |
| D.2      | Witness classes . . . . .                                      | 138        |
| D.3      | Questions that the appendix answers . . . . .                  | 138        |
| D.4      | Current boundary . . . . .                                     | 138        |
| <b>E</b> | <b>Journal-Core Extraction Bridge</b>                          | <b>138</b> |
| E.1      | Why the journal-core exists . . . . .                          | 139        |
| E.2      | What the journal-core may omit . . . . .                       | 139        |
| E.3      | What the monograph guarantees to later artifacts . . . . .     | 139        |
| <b>F</b> | <b>Reviewer Navigation Matrix and Audit Checklist</b>          | <b>140</b> |
| F.1      | Review questions, compact route, and decisive checks . . . . . | 140        |
| F.2      | Compact reading order by reviewer role . . . . .               | 141        |
| <b>G</b> | <b>Empirical Validation Matrix and Replay Ledger</b>           | <b>142</b> |
| <b>H</b> | <b>Institute-Run Measurement and Observation Program</b>       | <b>144</b> |
| <b>I</b> | <b>Pinned Benchmark Dataset Manifest</b>                       | <b>144</b> |
| <b>J</b> | <b>Pinned Benchmark Caseset and Parameterization Gaps</b>      | <b>147</b> |
| J.1      | Mathematics . . . . .                                          | 147        |
| J.2      | Physics . . . . .                                              | 147        |
| J.3      | Chemistry . . . . .                                            | 150        |
| J.4      | Biology . . . . .                                              | 153        |
| J.5      | Systems / Civilizational projection . . . . .                  | 156        |
| <b>K</b> | <b>Domain Replay Reports</b>                                   | <b>158</b> |
| <b>L</b> | <b>Domain Hard-Closure Command Board</b>                       | <b>160</b> |

|          |                                                                        |            |
|----------|------------------------------------------------------------------------|------------|
| <b>M</b> | <b>Proof Machinery Appendix</b>                                        | <b>161</b> |
| M.1      | Revision law authority . . . . .                                       | 162        |
| M.2      | Load-bearing proof obligations . . . . .                               | 162        |
| M.3      | Hostile-review backlog . . . . .                                       | 163        |
| M.4      | Closure priority board . . . . .                                       | 163        |
| M.5      | Minimality and irreducibility ablation ledger . . . . .                | 164        |
| M.6      | Alternative-model competition matrix . . . . .                         | 164        |
| M.7      | Bounded compression benchmark . . . . .                                | 164        |
| <b>N</b> | <b>Figure Atlas</b>                                                    | <b>164</b> |
| N.1      | Illustrative Figures . . . . .                                         | 165        |
| <b>O</b> | <b>Technical Derivation Atlas</b>                                      | <b>165</b> |
| O.1      | Extended modules and cross-level structures . . . . .                  | 165        |
| O.2      | MSpaces . . . . .                                                      | 245        |
| O.3      | Cross-level structures . . . . .                                       | 282        |
| O.4      | Cycle Structure in the Ontology of Continua . . . . .                  | 313        |
| O.5      | Structural Jets . . . . .                                              | 349        |
| O.6      | Processes . . . . .                                                    | 393        |
| O.7      | Universal Evolution Operators . . . . .                                | 449        |
| O.8      | Master Theorems of Core 1.2 . . . . .                                  | 453        |
| O.9      | Complexity Metric $S(K)$ . . . . .                                     | 455        |
| <b>P</b> | <b>Reference and Benchmark Atlas</b>                                   | <b>457</b> |
| P.1      | Structural Experiments . . . . .                                       | 458        |
| P.2      | Falsifiability . . . . .                                               | 505        |
| P.3      | Predictions . . . . .                                                  | 554        |
| <b>Q</b> | <b>Unified Synthesis Support Dossiers</b>                              | <b>602</b> |
| Q.1      | Level K0: structural conditions for any universe . . . . .             | 602        |
| Q.2      | Level K1: minimal observable distinctions and energetic axes . . . . . | 603        |
| Q.3      | Level K2: fields, phases, and large-scale structure . . . . .          | 604        |
| Q.4      | Level K3: molecular organization and chemical bonding . . . . .        | 605        |
| Q.5      | Level K4: compartmental and membrane thresholds . . . . .              | 606        |
| Q.6      | Level K5: excitable and early neural continua . . . . .                | 607        |
| Q.7      | Level K6: cognitive prediction and representation . . . . .            | 608        |
| Q.8      | Level K7: social coordination and group stability . . . . .            | 609        |
| Q.9      | Level K8: civilizational flows and infrastructural stability . . . . . | 610        |
| Q.10     | Level K9: knowledge systems and scientific memory . . . . .            | 611        |
| Q.11     | Level K10: formal systems and recursion . . . . .                      | 612        |
| Q.12     | Level K11: meta-theoretical reflexivity . . . . .                      | 613        |
| Q.13     | Level K12: global semantic coherence . . . . .                         | 615        |
| <b>R</b> | <b>Practical Utility and Model Comparison Atlas</b>                    | <b>617</b> |
| R.1      | Full use-case table . . . . .                                          | 617        |
| R.2      | Use-case trace-anchor matrix . . . . .                                 | 618        |
| R.3      | Operational playbook matrix . . . . .                                  | 619        |
| R.4      | Operational playbook trace-anchor matrix . . . . .                     | 620        |
| R.5      | Same-claim baseline matrix . . . . .                                   | 621        |
| R.6      | Serious model comparison matrix . . . . .                              | 622        |
| R.7      | Serious model comparison trace matrix . . . . .                        | 624        |
| R.8      | Readiness and open-frontier matrix . . . . .                           | 625        |

|          |                                              |            |
|----------|----------------------------------------------|------------|
| R.9      | Fragmentation-to-closure matrix . . . . .    | 626        |
| R.10     | Frontier and hypothesis boundary . . . . .   | 627        |
| <b>S</b> | <b>OC 1.3.1 Science Delta Appendix</b>       | <b>627</b> |
| <b>T</b> | <b>OC 1.3.1 Simulation and Data Appendix</b> | <b>628</b> |
| <b>U</b> | <b>OC 1.3.1 Critique Closure Appendix</b>    | <b>628</b> |



## List of Figures

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1  | Load-bearing theorem spine of the bounded Core 1.3 claim. Solid arrows mark admitted support. The positive Theorem A route terminates at Lemma 2; the Lemma 3 branch remains visible as admitted rebirth-control support beside that route. The dashed branch marks an excluded lift-heavy route that is reviewed as a boundary condition rather than admitted into the positive proof. The larger monograph remains necessary because the surrounding chapters fix the ontology, operator shell, K-level doctrine, collapse grammar, empirical bar, and audit layers that keep this spine scientifically interpretable and externally citable. . . . .                                                                                   | 86  |
| 2  | Minimal life-cycle schematic for the bounded collapse grammar. The point of the figure is not to offer a domain-specific simulation but to show the exact interpretive burden of the formal rules: in this minimal model, the object remains live only while the admissible region is nonempty, the required cycles remain present, and $k(K, t) > 0$ , with supporting flows not dominated by destructive flows on the relevant cycles. Admissible-state collapse kills it, and any later live continuum is classified as rebirth rather than persistence only when the residue, embedding-space, and birth-condition clauses of Definition 12.6 are satisfied and the identity rule rules out restoration of the dead original. . . . . | 91  |
| 3  | Extended vertical hierarchy through $K_{12}$ . The added upper levels are justified by the need to represent evolving meta-theoretical spaces and their global coherence conditions rather than by decorative level inflation. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 92  |
| 4  | Schematic structure of a continuum $K = (\Omega, A, P, J, \Theta, \partial\Omega, C, k)$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 166 |
| 5  | Axes and threshold surfaces in the extended state space of a continuum. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 166 |
| 6  | Taxonomy of thresholds: existence, stability, critical, dimensional and death thresholds. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 166 |
| 7  | Structural difference and minimal threshold $\Theta_0$ at level $K_0$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 166 |
| 8  | Schematic of the transition $\Psi_{0 \rightarrow 1}$ from the substrate $K_0$ to the first continuum $K_1$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 167 |
| 9  | Vertical hierarchy of continua from $K_0$ to $K_{10}$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 167 |
| 10 | Illustrative potential landscape and flows $J(t)$ on a continuum. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 168 |
| 11 | Schematic action of the evolution operator $E : K(t) \mapsto K(t + dt)$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 168 |
| 12 | Birth, life and death of a continuum in terms of the state space $\Omega$ , cycles $C$ and the measure $k(t)$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 168 |
| 13 | Percolation-type transition at level $K_2$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 168 |
| 14 | BKT-type transition as an example of threshold-governed emergence of a new topological axis. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 168 |
| 15 | Illustrative Bloch sphere diagram for quantum two-level systems. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 169 |
| 16 | RAF network as a chemical continuum at level $K_3$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 169 |
| 17 | Catalytic paths and supporting flows in a reaction network. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 169 |
| 18 | Membrane closure and emergence of a protocellular boundary at level $K_4$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 169 |
| 19 | Osmotic gradients, membrane tension and structural thresholds in protocells. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 170 |
| 20 | Vesicle flickering regime near curvature and osmotic thresholds. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 170 |
| 21 | Propagation of membrane potential $\Delta V$ along a boundary. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 170 |
| 22 | Ion channel states and excitation thresholds at level $K_5$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 170 |
| 23 | Proto-spike dynamics as a minimal excitatory cycle. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 170 |
| 24 | Schematic cognitive state space and binding axes at level $K_6$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 171 |
| 25 | Binding space for cognitive continua and associated thresholds. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 171 |
| 26 | Institutional cycles and social flows at level $K_7$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 171 |
| 27 | Trust thresholds and collapse of social continuumness. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 171 |
| 28 | Civilizational energy and infrastructure cycles at level $K_8$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 172 |
| 29 | Technological layers and embedding spaces for civilizational continua. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 172 |
| 30 | Graph of theories as a continuum at level $K_9$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 172 |
| 31 | Flows between representations, models and data in $K_9$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 630 |
| 32 | Self-referential structure of meta-theoretical continua at level $K_{10}$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 630 |

## List of Tables

|    |                                                                                                                                                                                                                                                                                                                                                        |     |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1  | Stable reading routes for the most common external interactions. The purpose is to keep external use concise while preserving the audit trail behind each cited claim. . . . .                                                                                                                                                                         | 12  |
| 2  | Global overview of continua $K_0$ – $K_{12}$ . . . . .                                                                                                                                                                                                                                                                                                 | 44  |
| 3  | Statement classes and their inspection burden in Core 1.3. The table does not replace the formal chapters; it simply prevents proof, interpretation, empirical promotion, and frontier language from quietly changing roles. Replay, comparator, falsifier, and escalation rules are downstream control obligations attached to those classes. . . . . | 87  |
| 4  | Examples of observations or formal objections that would genuinely change the scientific status of the present release. The table is meant to preserve falsifiability, not to immunize the framework against criticism. . . . .                                                                                                                        | 95  |
| 28 | Continuum hierarchy $K_0$ – $K_{12}$ . . . . .                                                                                                                                                                                                                                                                                                         | 133 |
| 29 | Structural components $(\Omega, A, P, J, \Theta, \partial\Omega, C, k)$ per level. . . . .                                                                                                                                                                                                                                                             | 135 |
| 30 | Minimal structural parameters required at level $K_0$ . . . . .                                                                                                                                                                                                                                                                                        | 136 |
| 31 | Minimal observable and energetic parameters used at level $K_1$ . . . . .                                                                                                                                                                                                                                                                              | 136 |
| 32 | Effective cosmological parameters used at level $K_2$ . . . . .                                                                                                                                                                                                                                                                                        | 136 |
| 33 | Representative molecular parameters relevant at level $K_3$ . . . . .                                                                                                                                                                                                                                                                                  | 136 |
| 34 | Structural and energetic parameters relevant at $K_4$ . . . . .                                                                                                                                                                                                                                                                                        | 136 |
| 35 | Electrical and ion-channel parameters used at $K_5$ . . . . .                                                                                                                                                                                                                                                                                          | 136 |
| 36 | Parameters governing representational stability and prediction thresholds at $K_6$ . . . . .                                                                                                                                                                                                                                                           | 137 |
| 37 | Typical social-coordination parameters used at $K_7$ . . . . .                                                                                                                                                                                                                                                                                         | 137 |
| 38 | Parameters for large-scale civilizational continua at $K_8$ . . . . .                                                                                                                                                                                                                                                                                  | 137 |
| 39 | Parameters defining the behaviour of scientific-meta systems at $K_9$ . . . . .                                                                                                                                                                                                                                                                        | 137 |
| 40 | Structural parameters of formal continua at $K_{10}$ . . . . .                                                                                                                                                                                                                                                                                         | 137 |
| 41 | Parameters governing meta-theoretical landscapes at $K_{11}$ . . . . .                                                                                                                                                                                                                                                                                 | 137 |
| 42 | Parameters relevant at the highest-level structural continuum $K_{12}$ . . . . .                                                                                                                                                                                                                                                                       | 138 |

# 1 Reader Contract and Navigation Guide

Core 1.3 is the flagship reader-facing monograph, but readers arrive with different tasks. This guide is therefore a route map rather than a second introduction. Its job is to show where each of the five reader roles should begin: Reviewer route, Formal reader route, Domain reader route, Broad scientific reader route, and Practical user route. External collaborators who need deployment guidance should use the Practical user route.

The dedicated Figure Atlas in Appendix [N](#) and the Technical Derivation Atlas in Appendix [O](#) are intentionally outside the default continuous reading route. Their material belongs to the appendix inspection layer; the formal-reader route below consults the Technical Derivation Atlas only when local proof support is needed.

## 1.1 The five reading routes

The reviewer-navigation appendix uses slightly narrower review labels: “rapid critical reviewer” corresponds to the Reviewer route, “formal mathematical reviewer” to the Formal reader route, and “generalist scientific reviewer” to the Broad scientific reader route.

**Reviewer route.** Read the front matter, this guide, Sections [5](#), [17–19](#), Section [20](#), and then Appendices [D–F](#). That route exposes the publication boundary, theorem status, load-bearing theorem spine, reviewer objections, and source witnesses as quickly as possible.

**Formal reader route.** Use Sections [2–7](#) for orientation only. Then read Sections [8–15](#), Section [16](#), and Sections [17–24](#) in order. Use Appendices [A–C](#), Appendix [M](#), and the Technical Derivation Atlas in Appendix [O](#) when local proof support is needed. This is the compact route that keeps exact doctrine, theorem scope, and proof machinery visible at once.

**Domain reader route.** Read Section [14](#) and Sections [20–25](#), then move into Appendices [G](#), [I](#), [J](#), [K](#), [L](#), and [R](#), with Appendix [P](#) added only when experiments, predictions, or falsifiability catalogues need inspection. That path exposes bounded domain embedding, worked interpretation, empirical promotion rules, replay surfaces, benchmark cases, execution board, practical use, and comparison evidence without forcing the reader to traverse every technical-derivation section first.

**Broad scientific reader route.** Read Sections [2–7](#), Sections [8–15](#), Section [16](#), and Sections [17–25](#) straight through as a continuous graduate-level monograph. Use Appendix [A](#) for notation when needed and defer Appendices [N–R](#) until a concrete figure, benchmark table, or support dossier question arises. This is the recommended route for a broad scientific reader.

**Practical user route.** Read the front matter, this guide, Section [17](#), Section [25](#), and Appendices [D](#), [E](#), and [R](#) first; add Appendix [P](#) only when the task requires experiments, predictions, or falsifiability catalogues. Return to Sections [21–24](#) only when a practical use case needs theorem, replay, or falsifier support to be inspected in detail. This route is for external collaborators who need deployment guidance: what can be predicted, which decisions can be supported, which procedures can be run, and how the result compares with alternatives.

## 1.2 Route legend

The five routes above are the working navigation system:

- **Reviewer route:** shortest path to claim boundary, proof spine, objections, and source witnesses.
- **Formal reader route:** compact path through doctrine, theorem scope, notation, and proof machinery.
- **Domain reader route:** bounded path from domain embedding to replay, benchmarks, execution, use, and comparison evidence.
- **Broad scientific reader route:** continuous graduate-level reading path through the main argument before atlas inspection.
- **Practical user route:** shortest path to usable tasks, required inputs, outputs, comparison boundaries, and deployment constraints.

### 1.3 External interaction crosswalk

The crosswalk maps common external needs to the shortest route bundle that can answer them while keeping the supporting audit trail visible.

| External task                     | Primary route                                 | Support route if verification is needed                                                                                                                                       |
|-----------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evaluate the release for review   | This guide; Sections 5, 17–20; Appendices D–F | Appendix M when proof-obligation details are needed                                                                                                                           |
| Audit the bounded theorem core    | Section 5, Section 18, and Section 19         | Appendix A, Appendix B, Appendix F, and Appendix M                                                                                                                            |
| Check the empirical promotion bar | Sections 21–22                                | Appendix G, Appendix H, Appendix I, Appendix J, Appendix K, and Appendix L                                                                                                    |
| Cite the unified synthesis        | Section 24                                    | Appendix Q, Appendix C.4, and the K-level parameter tables in Appendix C                                                                                                      |
| Use or compare the model          | Section 25                                    | First inspect Appendix R; use Appendix P only for experiments, predictions, or falsifiability catalogues; then use Sections 21–24 only when proof or replay support is needed |
| Verify publication boundaries     | Section 17                                    | Appendix D and Appendix E                                                                                                                                                     |

Table 1: Stable reading routes for the most common external interactions. The purpose is to keep external use concise while preserving the audit trail behind each cited claim.

### 1.4 What this guide is not

This guide does not restate the theory, replace the theorem roadmap, or turn the flagship monograph into a status summary. It simply tells the reader where each task belongs. Section 19 covers proof navigation. Section 24 covers the unified synthesis. Section 25 covers practical consequences and comparative analysis. The appendices keep the inspection burden inside the same volume without displacing the main reading spine.

## 2 Introduction

The central premise of the Ontology of Continua (OC) is that heterogeneous systems in physics, chemistry, biology, cognition, society, and metatheory can be treated as *continua*: structures defined not by their material substrate but by a shared internal ontology.

*Ontology of Continua — Core 1.3* is the master monograph for that framework. It presents the defended formal core, the cross-level hierarchy  $K_0$ – $K_{12}$ , the bounded empirical closures, and the final unified-science synthesis as one coherent scientific object. Here the primitive assumptions and their derived structural results form the *formal core*; when that core is treated as a stable doctrine rather than as a list of statements, the monograph calls it the *kernel theory*.

Each continuum  $K$  is specified by

- its admissible state space  $\Omega(K)$  and boundary  $\partial\Omega(K)$ ;
- a set of axes  $A(K)$  representing independent structural differences;
- potentials  $P(t)$  encoding energetic, chemical, biological, cognitive, or institutional constraints;
- flows  $J(t)$  that redistribute or transform these potentials;
- thresholds  $\Theta(K)$  separating distinct dynamical regimes;
- cycles  $C(K)$  that sustain organisation;
- a continuumness measure  $k(K, t)$  quantifying structural integrity and viability.

This ontology applies uniformly from  $K_0$  through  $K_{12}$ , spanning the pre-geometric substrate, reflexive metatheoretical continua, and global-coherence continua. The point is not unification for its own sake, but a precise account of how continua arise, persist, interact, specialise, and collapse.

### 2.1 Motivation

Scientific domains traditionally operate with different assumptions and representational languages: physics with fields and interactions, chemistry with reflexively autocatalytic and food-generated (RAF) networks, biology with protocells and metabolic cycles, cognition with representation and binding, society with institutions and coordination, and metatheory with formal languages and model spaces.

OC proposes that these domains can be compared through a single structural ontology based on continua, axes, potentials, thresholds, flows, cycles, and surrounding meta-spaces. The motivation is to:

1. identify structural assumptions shared across scientific fields;
2. provide a unified language for describing emergence and organisation;
3. make dimensional and threshold dynamics explicit and comparable across domains;
4. derive falsifiable conditions for transitions between levels of organisation;
5. offer a framework that is empirical, modest, and non-speculative.

OC does not replace domain theories; it exposes their structural substrate and prevents category mistakes when comparing systems across fields.

### 2.2 Scope of Core 1.3

Core 1.3 has a targeted purpose: to present the formal core, state the defended results, and make the evidence trail inspectable. Here *evidence trail* means the audit artefacts that connect a claim to its support. Source packets identify the source route; replay summaries report held-out checks; comparator checks record serious alternatives; and falsifiers state what would falsify the claim. Together these artefacts let a reader audit a claim without relying solely on prose. The reader roadmap is:

- formalise the axiomatic base and the first observable continuum,  $K_1$ ;
- present the shared state-space, boundary, threshold, potential, flow, evolution, interaction, and hierarchy vocabulary;
- bind the formal stack to that evidence trail;
- close the reader-facing arc with unified-science synthesis and practical use.

The roadmap tells the reader how the chapter sequence functions. The formal outline below names the content that sequence fixes:

- establish  $K_0$  and  $K_1$ , including structural difference, non-trivial threshold, time, state space, and the first transition operator;
- fix the shared continuum vocabulary: admissible states, boundaries, threshold families, potentials, flows, cycles, and continuumness;
- state the evolution and interaction operators, plus the birth operators and death thresholds or conditions used by later theorem routes;
- present the full  $K_0$ – $K_{12}$  hierarchy, including the substantive upper-level roles of  $K_{11}$  and  $K_{12}$ ;
- connect the formal stack to source packets, numerical tables, held-out replay summaries, and practical-use surfaces.

Domain-specific mechanisms are no longer included merely as examples. In the present release they are tied to explicit source-owned domain packets, numerical tables, and held-out replay summaries.

## 2.3 Vertical hierarchy of levels

The OC hierarchy

$$K_0, K_1, K_2, \dots, K_{12}$$

captures the main structural phases of organisation. Each level  $K_x$  is defined by its state space  $\Omega_x$ , axes  $A_x$ , thresholds  $\Theta_x$ , flows  $J_x$ , cycles  $C_x$ , and continuumness  $k_x(t)$ . Transitions  $K_x \rightarrow K_{x+1}$  require the emergence of a new axis  $A_{\text{new}}$  not representable within the geometry of  $K_x$  but available in its meta-space  $M_x$ .

A high-level overview:

- $K_0$ : pre-geometric structural substrate with no time, energy, or dynamics; defined by abstract states, differences, and minimal connectivity.
- $K_1$ : the first genuinely structural one-dimensional continuum, with an explicit admissible axis, ordered topology, and boundary vocabulary.
- $K_2$ : physical continua including space-time, quantum fields, fundamental interactions, phase transitions, percolation, and structures related to quantum chromodynamics (QCD).
- $K_3$ : chemical continua including reaction networks, RAF structures, and concentration spaces.
- $K_4$ : protocellular and early biological continua with membranes, gradients, osmotic and curvature thresholds, and metabolic cycles.
- $K_5$ : early neural and bioelectrical continua with excitation thresholds, ion channels, proto-spike dynamics, and emerging logical structure.
- $K_6$ : cognitive continua with binding, internal models, memory thresholds, and conceptual spaces.
- $K_7$ : social continua with trust thresholds, institutional cycles, and structural communication flows.
- $K_8$ : civilisational continua with systemic stability, infrastructure cycles, and global thresholds.

- $K_9$ : theory-level continua composed of theories, paradigms, models, and interpretive frameworks.
- $K_{10}$ : formal-system continua composed of proof states, formal languages, and recursion-bounded systems.
- $K_{11}$ : metatheoretical continua governing cross-framework coherence and higher-order structural coupling.
- $K_{12}$ : the terminal unified-science layer in which cross-domain compatibility, integrability, and global tension are assessed.

Core 1.3 treats  $K_{11}$  and  $K_{12}$  as active parts of the public hierarchy rather than as placeholders. Their role is to close the metatheoretical and global integrability work required by the present master monograph.

## 2.4 Relation to previous versions

Earlier versions of the OC Core existed in fragmented working documents. Core 1.3 consolidates the proof-bearing material and bounded science packets. It presents them as a single monograph with source files retained for audit and reconstruction. The goals are:

1. unify notation, structure, and threshold taxonomy across all levels;
2. eliminate inconsistencies and redundancies so each level follows from the general continuum ontology and its embedding conditions.

The kernel theory is not replaced here; Core 1.3 clarifies, stabilises, and finalises the publication-ready version of the monograph.

## 2.5 Structure of the monograph

This monograph first introduces the background, formal model, and structural results, then moves to interpretation, limits, closure, synthesis, and practical use. The dedicated reader contract in [Section 1](#) gives the route map for reviewers, formal readers, domain readers, broad scientific readers, and practical users.

# 3 Background and Motivation

This chapter introduces the structural and mathematical background required for the Ontology of Continua (OC). Its purpose is not historical exposition but the presentation of the minimal conceptual machinery underlying the unified framework. All key notions — continua, axes, thresholds, boundaries, potentials, flows, cycles, and continuumness — are introduced in a domain-agnostic form, independent of whether the target system is physical, chemical, biological, cognitive, social, or meta-theoretical.

## 3.1 Continuum as a structural object

In OC, a *continuum* is not defined by spatial smoothness or physical extent but by a set of structural conditions. A system qualifies as a continuum  $K$  if and only if it possesses:

- a nonempty admissible state space  $\Omega(K)$ ;
- a finite set of axes  $A(K) = \{A_1, \dots, A_n\}$ , each representing an independent and incompatible structural difference;
- potentials  $P(t)$  capturing energetic, chemical, informational, biological, cognitive, or institutional constraints;

- flows  $J(t)$  that transform these potentials and drive evolution;
- a threshold landscape  $\Theta(K)$  separating qualitatively distinct dynamical regimes;
- a boundary  $\partial\Omega(K)$ , where thresholds saturate;
- a repertoire of stable cycles  $C(K)$  that maintain organization and keep dynamics away from  $\partial\Omega(K)$ ;
- a continuumness measure  $k(K, t)$  quantifying viability and structural integrity;
- an embedding meta-space  $M$  with  $\Omega(K) \subseteq \Omega(M)$  and  $A(K) \subseteq A(M)$ .

This definition deliberately avoids domain specifics. Physical fields, chemical reaction networks, protocells, neural assemblies, representational systems, social institutions, and scientific theories all qualify as continua if they meet these structural criteria.

### 3.2 Axes and dimensionality

Axes  $A(K)$  determine the effective dimensionality of a continuum. Each axis represents an *incompatible difference* — a distinction that cannot be reconstructed from the existing axes. Examples include energetic axes, gradient axes, membrane axes, excitation axes, cognitive axes, social-difference axes, and higher-level expressive axes.

Dimensionality obeys the OC *monotonicity principle*:

$$\dim(K_{x+1}) > \dim(K_x) \quad \text{whenever a new continuum emerges.}$$

A new dimension appears only when:

1. a new class of differences arises that cannot be expressed within  $\text{span}_{\text{str}}(A(K_x))$ ;
2. structural tension  $T(K_x)$  exceeds the dimensional threshold  $\Theta_{\text{dim}}(K_x)$ ;
3. the embedding space  $M_x$  provides an axis  $A_{\text{new}} \in A(M_x) \setminus A(K_x)$ .

Dimensionality is therefore not a free parameter but a structural invariant enforced by incompatibility and by the availability of appropriate axes in the surrounding meta-space.

### 3.3 Boundaries and thresholds

A continuum is constrained by its threshold landscape  $\Theta(K)$ . Thresholds are functions

$$\Theta_k : \Omega(K) \rightarrow \mathbb{R}, \quad \Theta_k(s) \leq 0 \text{ for } s \in \Omega(K),$$

and fall into structural classes:

- **Existence thresholds**  $\Theta_{\text{exist}}$ : conditions for  $\Omega(K) \neq \emptyset$ ;
- **Stability thresholds**  $\Theta_{\text{stab}}$ : ensure non-divergent flows;
- **Critical thresholds**  $\Theta_{\text{crit}}$ : mark qualitative transitions;
- **Dimensional thresholds**  $\Theta_{\text{dim}}$ : govern the emergence of new axes;
- **Death thresholds**  $\Theta_{\text{death}}$ : beyond them no admissible states remain.

The structural boundary is defined as

$$\partial\Omega(K) = \{ s \in \Omega(K) \mid \exists k : \Theta_k(s) = 0 \}.$$

Boundaries encode the full structure of a continuum's limits. Their dynamics are captured by the operator

$$\frac{d}{dt} \partial\Omega = R(\partial\Omega, P, J, \Theta),$$

which describes expansion, contraction, or bifurcation of  $\partial\Omega$  under changing potentials, flows, or thresholds. Birth, persistence, and collapse correspond to distinct regimes of  $R$ .



### 3.4 Potentials and flows

Potentials  $P(t)$  encode internal constraints and driving forces. Representative examples include:

- energy landscapes, fields, and order parameters (physics);
- concentrations, pH levels, redox potentials (chemistry);
- membrane and electrochemical potentials (biology);
- predictive and representational potentials (cognition);
- normative and institutional potentials (social systems).

Flows describe the evolution of potentials:

$$\frac{dP}{dt} = J(t).$$

OC distinguishes:

- **Supporting flows**  $J_{\text{support}}$ : sustain cycles and stabilize  $k(t)$ ;
- **Critical flows**  $J_{\text{critical}}$ : push the system toward or across  $\Theta_{\text{crit}}$  or  $\Theta_{\text{dim}}$ ;
- **Destructive flows**  $J_{\text{kill}}$ : break cycles or force the system across  $\Theta_{\text{death}}$ .

These flows feed into the universal evolution operators  $F, G, H, Q, R, S, U$ , which specify how axes, potentials, thresholds, flows, cycles, and boundaries co-evolve.

### 3.5 Continuumness

The measure  $k(K, t)$  determines whether a system functions as a *live continuum*. Core 1.3 adopts the unified definition encoded by the operator  $U$ :

$$k(K, t) = \mathbf{1}_{\{\Omega(K, t) \neq \emptyset\}} S_{\text{axes}}(K, t) S_{\text{cycles}}(K, t) S_{\text{flows}}(K, t) S_{\text{coh}}(K, t),$$

where:

- $\mathbf{1}_{\{\Omega(K, t) \neq \emptyset\}} = 1$  if  $\Omega(K, t) \neq \emptyset$  and 0 otherwise;
- $S_{\text{axes}}$  measures expressive adequacy of axes;
- $S_{\text{cycles}}$  measures the strength of stabilizing cycles;
- $S_{\text{flows}}$  reflects the balance of supporting vs. destructive flows;
- $S_{\text{coh}}$  encodes global structural coherence.

Core 1.3 reserves  $\chi_{\Omega}(s; K, t)$  for pointwise membership of a state  $s$  in the admissible region. Nonemptiness of the whole admissible region is therefore always written with  $\mathbf{1}_{\{\Omega(K, t) \neq \emptyset\}}$ , not with  $\chi_{\Omega}$ .

A continuum is alive precisely when

$$k(K, t) > 0.$$

Death corresponds to  $k(K, t) \rightarrow 0$  and  $\Omega(K) \rightarrow \emptyset$ , regardless of path.

### 3.6 Motivation for Core 1.3

Core 1.3 consolidates all structural components of OC into a vertically consistent reference. Its goals are to:

- establish a canonical file and repository structure;
- consolidate the axiomatics of  $K_0$  and the construction of  $K_1$ ;

- unify the treatment of boundaries and thresholds;
- formalize potentials, flows, cycles, and evolution operators in a consistent language;
- integrate the full hierarchy  $K_0$ – $K_{12}$  together with the embedding meta-spaces  $M_x$ .

### 3.7 Historical motivation

OC arose from attempts to explain why diverse systems — phase transitions, catalytic closure, protocell stabilization, neural excitation, institutional coherence, civilizational collapse — share the same organizational invariants. Across domains, four structures repeatedly appear:

1. thresholds governing admissibility and stability;
2. emergence of new dimensions under structural tension;
3. cycles stabilizing flows;
4. collapse when thresholds are violated and  $\Omega(K)$  disappears.

OC formalizes these invariants within a single structural ontology.

### 3.8 Scope of future background material

Future expansions of this chapter will include:

- mathematical preliminaries for potentials, flows, and cycle stability;
- structural motivations for the OC axiomatics;
- a refined account of embedding spaces  $M_x$  and monotonic expansion;
- formal treatments of monotonicity, coherence, and stability across levels;
- derivations of the universal operators  $F, G, H, Q, R, S, U$  from more primitive assumptions.

Core 1.3 keeps this chapter deliberately compact: it supplies the background needed for Section 4 without duplicating the later proof, operational, and unified-synthesis chapters.

## 4 Ontological Structure of the Model

This section presents the formal core of the Ontology of Continua (OC). It gives the reader the unified notation, axioms, definitions, and structural constraints needed for the later chapters on closure, synthesis, and practical use.

OC describes continua across multiple scientific domains through one structural language. That language is built from axes, potentials, flows, thresholds, boundaries, cycles, and a measure of continuum-ness. Each continuum is also embedded in a surrounding meta-space. All definitions in this section are domain-independent at the level of formal vocabulary. Later review chapters will sort claims by proof status and empirical support; this section supplies the common model language those checks depend on.

### 4.1 Axiomatic foundation: Level $K_0$

Level  $K_0$  is a purely structural substrate. It is not a physical space; it carries no time, energy, geometry, or dynamics. Its role is to provide the minimal conditions under which any higher-level continuum is logically possible.

**Specification of  $K_0$**   $K_0$  is specified by a triple

$$K_0 = (S, \Delta, \mathcal{C}),$$

where:

- $S$  is a set of states;
- $\Delta : S \times S \rightarrow \mathbb{R}_{\geq 0}$  is a structural difference function;
- $\mathcal{C}$  is a structural relation (or family of relations) preserving distinguishability.

There is no time parameter and no evolution operator at this level.

**Axiom 0.1 (Difference and distinguishability).** For all  $s_1, s_2 \in S$ ,

$$\Delta(s_1, s_2) = 0 \Rightarrow s_1 = s_2.$$

Nonzero structural difference is the minimal condition for distinguishability. A continuum cannot exist without at least two distinguishable states.

**Axiom 0.2 (Nontrivial threshold  $\Theta_0$ ).** There exists  $\varepsilon > 0$  such that

$$\forall s_1, s_2 \in S, \quad s_1 \neq s_2 \Rightarrow \Delta(s_1, s_2) \geq \varepsilon.$$

The value  $\Theta_0 = \varepsilon$  acts as a minimal structural threshold: differences smaller than  $\varepsilon$  are not resolved at the level of  $K_0$ .

**Axiom 0.3 (Logical substrate).**  $K_0$  carries no time parameter and no dynamical operator. It does not evolve and does not generate higher levels by itself. It specifies only logical conditions on distinguishability and the existence of nontrivial differences. All dynamics and all notions of energy, geometry, and causality belong to levels  $K_1$  and above.

In particular, no operator defined on  $K_0$  can increase dimension: differences at level  $K_0$  are always expressed along the existing structural axis of distinguishability. The dimensionality associated with  $K_0$  is fixed by the embedding meta-space  $M_0$ . Here  $M_0$  denotes the minimal meta-space in which the purely structural substrate can be evaluated for later  $K_0 \rightarrow K_1$  emergence; it is not a physical container or a dynamical environment.

## 4.2 Construction of Level $K_1$

Level  $K_1$  is the simplest genuine continuum: it introduces time, a one-dimensional axis, and basic geometric structure.

**Specification of  $K_1$**  The continuum  $K_1$  is defined by:

$$K_1 = (X_1, \tau_1, \Omega_1, A_1, P_1(t), J_1(t), \Theta_1, \partial\Omega_1, C_1, k_1(t)),$$

where:

- $A_1$  is a single axis (a one-dimensional coordinate);
- $(X_1, \tau_1)$  is a topological space obtained from the structural data of  $K_0$ , typically an interval  $X_1 = (a, b)$ ;
- $\Omega_1$  is a space of admissible configurations on  $X_1$  with appropriate regularity conditions. In this time-dependent analytic example,

$$\Omega_1 = C^0(I, H^1(X_1, V_1)) \cap C^1(I, L^2(X_1, V_1)),$$

where  $I$  is the time interval and  $V_1$  is the Hilbert space used by the example; a finite-dimensional vector space is the special case used when the local model has finitely many degrees of freedom.

- $P_1(t)$  is an energy-like potential functional on  $\Omega_1$ ;
- $J_1(t)$  are flows derived from the variation of  $P_1(t)$ ;
- $\Theta_1$  encodes minimal conditions for classical stability;
- $\partial\Omega_1$  is the boundary where stability thresholds saturate;
- $C_1$  is the set of classical cycles (for example, periodic orbits);
- $k_1(t)$  measures one-dimensional continuumness according to the general definition below.

**Transition**  $\Psi_{0 \rightarrow 1}$  The passage  $K_0 \rightarrow K_1$  is generated by an operator

$$\Psi_{0 \rightarrow 1} : (S, \Delta, \mathcal{C}) \mapsto (X_1, \tau_1, A_1, \Omega_1, \Theta_1),$$

which constructs the skeletal  $K_1$  scaffold; the energy functional, flow family, boundary, cycles, and continuumness scalar are specified by the subsequent  $K_1$ -specific clauses rather than by this first transition map alone. The scaffold construction:

- identifies a family of structurally compatible states in  $S$ ;
- equips them with an order and topology  $(X_1, \tau_1)$ ;
- defines the first axis  $A_1$ ;
- constructs the admissible configuration space  $\Omega_1$ ;
- induces classical thresholds  $\Theta_1$ .

This is the first instance of dimensional emergence: a continuous axis appears that cannot be represented within the purely structural substrate of  $K_0$ .

### 4.3 General definition of a continuum

For any level  $K$  in the hierarchy, a continuum is defined as the tuple

$$K = (\Omega(K), A(K), P(t), J(t), \Theta(K), \partial\Omega(K), C(K), k(K, t)),$$

where:

- $\Omega(K)$  is a nonempty set of admissible states;
- $A(K) = \{A_1, \dots, A_n\}$  is a finite set of axes of incompatible differences;
- $P(t)$  is the vector of potentials on  $\Omega(K)$ ;
- $J(t)$  is the aggregate potential-flow vector; state updates are handled by the evolution operator  $E$ ;
- $\Theta(K) = \{\Theta_i\}$  is the set of threshold functions;
- $\partial\Omega(K)$  is the boundary of admissible states;
- $C(K)$  is the family of structurally stable cycles;
- $k(K, t)$  is the measure of continuumness.

Every continuum is assumed to be embedded in a surrounding meta-space  $M$  such that

$$\Omega(K) \subseteq \Omega(M), \quad A(K) \subseteq A(M).$$

The meta-space provides additional admissible states and axes that can host future dimensional extensions of  $K$ .

**State space and boundary** Thresholds are represented as functions  $f_i : \overline{\Omega(K)} \rightarrow \mathbb{R}$  with

$$f_i(s) \leq 0 \quad \text{for all } s \in \Omega(K),$$

and the boundary is defined as

$$\partial\Omega(K) = \{s \in \overline{\Omega(K)} \mid \exists i : f_i(s) = 0\}.$$

This definition requires no metric and applies equally to geometric, topological, energetic, informational, or logical continua.

The boundary is dynamic. The notation below is an update map, not a derivative of a set-valued object:

$$\partial\Omega(K, t + dt) = R(\partial\Omega(K, t), P(t), J(t), \Theta(K)),$$

which can contract, expand, or bifurcate  $\Omega(K)$  during birth, life, and death events.

#### 4.4 Taxonomy of thresholds

Each continuum has a structured set of thresholds  $\Theta(K)$ , organised into the following types:

- **Existence thresholds**  $\Theta_{\text{exist}}$ : conditions under which  $\Omega(K) \neq \emptyset$ .
- **Stability thresholds**  $\Theta_{\text{stab}}$ : conditions for bounded dynamics and non-divergent flows.
- **Critical thresholds**  $\Theta_{\text{crit}}$ : hypersurfaces where qualitative changes (phase transitions, bifurcations) occur.
- **Dimensional thresholds**  $\Theta_{\text{dim}}$ : conditions under which new axes and higher-dimensional continua can emerge.
- **Death thresholds**  $\Theta_{\text{death}}$ : structural limits where no admissible states remain and  $\Omega(K)$  collapses.

The full threshold landscape of a continuum is thus a collection of inequalities:

$$f_i^{(\text{exist})}(s) \leq 0, \quad f_i^{(\text{stab})}(s) \leq 0, \quad f_i^{(\text{crit})}(s) \leq 0, \quad f_i^{(\text{dim})}(s) \leq 0, \quad f_i^{(\text{death})}(s) \leq 0,$$

with corresponding boundary components given by the equalities.

#### 4.5 Potentials, flows, and structural tension

Potentials  $P(t)$  encode the internal configuration of constraints and driving forces within a continuum. They may correspond to energy landscapes, chemical concentrations, membrane gradients, representational or informational structures, or institutional and normative pressures.

The aggregate flow vector  $J(t)$  describes the rate of change of potentials:

$$\frac{dP}{dt} = J(t),$$

with  $J(t)$  decomposed into structurally distinct components:

- **supporting flows**  $J_{\text{support}}$ , which maintain cycles and stabilise  $k(K, t)$ ;
- **critical flows**  $J_{\text{critical}}$ , which move the system towards or across critical thresholds  $\Theta_{\text{crit}}$  and  $\Theta_{\text{dim}}$ ;
- **destructive flows**  $J_{\text{kill}}$ , which push the system across  $\Theta_{\text{death}}$  and reduce  $k(K, t)$ .

Structural tension  $T(K, t)$  is a functional of potentials, axes, and gradients (schematically  $T = T(P, A, \nabla P)$ ). It measures how strongly the current configuration stresses the threshold landscape. Dimensional transitions occur when  $T(K, t)$  exceeds  $\Theta_{\text{dim}}(K)$ ; collapse occurs when destructive flows combined with tension drive the system across  $\Theta_{\text{death}}(K)$ .

## 4.6 Cycles and continuumness

Cycles  $C(K)$  are closed trajectories in  $\Omega(K)$  that remain at a finite distance from the boundary and preserve continuumness. At a structural level, cycles satisfy

$$L(C) = \int_C d\ell_K < \infty, \quad S(C) = \min_{s \in C} \delta_{\partial\Omega}(s) > 0,$$

where  $d\ell_K$  is the chosen structural length element for the realised continuum and  $\delta_{\partial\Omega}(s)$  is the corresponding boundary clearance. The boundary definition above is metric-free; these cycle estimates add a structural gauge only when quantitative clearance or length is being measured. In metric realisations this gauge may be an induced distance. In non-metric realisations it is any level-admissible separation functional that vanishes exactly on the boundary and is positive on structurally interior states.

When the continuum evolves, the admissible region is read as time-dependent:  $\Omega(K, t)$  denotes the state space available to  $K$  at time  $t$ , while  $\Omega(K)$  denotes the underlying level-relative admissible space when no time slice is being specified. When older formulas write  $\Omega(K(t))$ , read that notation as the same time-slice object  $\Omega(K, t)$ . The model adopts the following unified definition of continuumness:

$$k(K, t) = \chi_{\Omega}(s_K(t); K, t) S_{\text{axes}}(K, t) S_{\text{cycles}}(K, t) S_{\text{flows}}(K, t) S_{\text{coh}}(K, t),$$

where:

- $s_K(t)$  is the current state of  $K$  at time  $t$ , and  $\chi_{\Omega}(s_K(t); K, t)$  is the pointwise admissibility indicator: it is 1 if  $s_K(t) \in \Omega(K, t)$ , and 0 otherwise;
- $S_{\text{axes}}(K, t)$  quantifies effective axis saturation, e.g. the ratio of the effective rank of working axes to the maximal possible rank for that level;
- $S_{\text{cycles}}(K, t)$  measures the strength and efficiency of stable cycles, such as a weighted sum of cycle efficiencies normalised by their maximal values;
- $S_{\text{flows}}(K, t)$  captures flow stability. With aggregated supporting and destructive magnitudes  $\Phi_{\text{support}}$  and  $\Phi_{\text{kill}}$ , one local rule is shown below. The zero-flow branch applies only when the tested state has no active flow channel:

$$S_{\text{flows}}(K, t) = \begin{cases} \frac{\Phi_{\text{support}}(t)}{\Phi_{\text{support}}(t) + \Phi_{\text{kill}}(t)}, & \Phi_{\text{support}}(t) + \Phi_{\text{kill}}(t) > 0, \\ 1, & \Phi_{\text{support}}(t) = \Phi_{\text{kill}}(t) = 0. \end{cases}$$

Thus if support vanishes while destructive flow is positive,  $S_{\text{flows}} = 0$ ;

- $S_{\text{coh}}(K, t) \in [0, 1]$  encodes global structural coherence, which may include connectedness of the relevant graphs, compatibility of thresholds and flows, and absence of contradictory constraints.

All multiplicative factors in the displayed definition are normalised to  $[0, 1]$  in the local model before the product is evaluated. The continuumness-update operator  $U$ , developed later in Section 11, advances  $k(K, t)$  to the next lawful time slice using these factors.

By construction  $0 \leq k(K, t) \leq 1$ . Positive  $k(K, t)$  is the live-state indicator. It records four simultaneous facts: the current state is admissible, the active axes remain supported, the cycle factor and flow-stability factor both remain positive, and the coherence factor has not collapsed. The full live-state status also requires the admissible-state, threshold, and identity clauses stated in Section 12. Death corresponds to  $k(K, t) \rightarrow 0$  in combination with the collapse of  $\Omega(K)$ .

## 4.7 Evolution operator

The evolution of a continuum is described at the structural level by an operator

$$E : K(t) \mapsto K(t + dt),$$

or, in expanded form,

$$E : (\Omega, A, P, J, \Theta, \partial\Omega, C, k) \longrightarrow (\Omega', A', P', J', \Theta', \partial\Omega', C', k').$$

Here  $A'$  denotes the post-evolution axis set of the same continuum in the expanded tuple. This is a unary evolution operator on a time slice of  $K$ . The active meta-space belongs to the surrounding environment. When it must be named locally, it is shown as a subscripted parameter rather than as a second argument of  $E$ , so the operator arity remains fixed.

The model does not merge flow updates, threshold tests, cycle structure, boundary motion, axis extension, and continuumness updates into one differential system. It treats these components as coupled parts of a constrained structural evolution framework with the following requirements:

- **Consistency:** if  $s(t) \in \Omega(K(t))$ , then either  $s(t + dt) \in \Omega(K(t + dt))$  or the transition is associated with a threshold crossing.
- **Threshold-respecting:** flows obey the inequality structure encoded in  $\Theta(K)$ , except when explicitly crossing critical or death thresholds.
- **Monotonicity of dimension:** if  $A'$  contains a new axis not representable as a combination of previous axes, then  $\dim(K(t + dt)) > \dim(K(t))$ ; dimension cannot decrease.

The dynamics continue as long as  $\Omega(K(t)) \neq \emptyset$ . Once  $\Omega(K(t^*)) = \emptyset$ , the continuum is dead and  $E$  can no longer act meaningfully on it.

## 4.8 Embedding into meta-spaces

Each level  $K_x$  is embedded in a meta-space  $M_x$  that provides additional admissible states and axes. At the structural level this is captured by the conditions

$$\Omega(K_x(t)) \subseteq \Omega(M_x(t)), \quad A(K_x) \subseteq A(M_x).$$

The universal evolution axiom can then be stated schematically as

$$K_x(t + dt) = E_{M_x(t)}(K_x(t)),$$

with the additional requirement that if the time-slice embedding condition fails,  $\Omega(K_x(t)) \not\subseteq \Omega(M_x(t))$ , the continuum  $K_x$  ceases to exist. When the time slice is clear, the shorter notations  $K_x$  and  $M_x$  suppress the argument  $t$ . Dimensional growth of  $K_x$  is only possible if the meta-space contains an axis  $A_{\text{new}} \in A(M_x) \setminus A(K_x)$  and structural tension exceeds the corresponding dimensional threshold. This is the axis-origin constraint used below: new axes cannot be generated from  $K_x$  alone.

## 4.9 Birth of continua

The emergence of a new continuum  $K_{x+1}$  from  $K_x$  is a threshold-induced phase transition. Structurally, birth occurs when the following conditions are met:

1. A new class of differences appears that cannot be represented using the existing axes  $A(K_x)$ .
2. Structural tension associated with these differences satisfies

$$T(K_x, t) > \Theta_{\dim}(K_x).$$

3. The embedding space  $M_x$  contains at least one axis that can host the new differences (i.e.  $A_{\text{new}} \in A(M_x) \setminus A(K_x)$ ).
4. There exists a nonempty set of admissible states  $\Omega(K_{x+1})$  compatible with the new axis and thresholds.

The operator of dimensional birth

$$\Psi_{x \rightarrow x+1} : K_x \rightarrow K_{x+1}$$

is minimal and irreversible: any nonzero emergence of the new axis constitutes the new continuum  $K_{x+1}$ , and the dimension of  $K_{x+1}$  cannot revert to that of  $K_x$  without destroying  $\Omega(K_{x+1})$ . The local model therefore instantiates dimensional monotonicity. The corresponding no-spontaneous-creation result is recorded separately in Section 5.2.

#### 4.10 Life of continua

A continuum  $K$  is alive on an interval of time if

$$\Omega(K(t)) \neq \emptyset, \quad k(K, t) > 0, \quad C(K(t)) \neq \emptyset,$$

and if supporting flows  $J_{\text{support}}$  dominate destructive flows  $J_{\text{kill}}$  when integrated over relevant cycles.

Life thus corresponds to the persistent existence of:

- stable cycles separated from the boundary by a finite structural distance;
- sufficient supporting flows to maintain these cycles;
- coherence between potentials, axes, and thresholds;
- controlled critical behaviour that does not cross  $\Theta_{\text{death}}$ .

These conditions are interpreted differently at each level  $K_x$ , but the structural pattern is the same from protocells to institutions.

#### 4.11 Death of continua

A continuum  $K$  dies at time  $t^*$  when

$$\Omega(K(t^*)) = \emptyset,$$

which implies  $\chi_{\Omega}(s_K(t^*); K, t^*) = 0$  and hence  $k(K, t^*) = 0$ . Equivalently:

- all stable cycles vanish,  $C(K(t^*)) = \emptyset$ ;
- no state satisfies the threshold inequalities concurrently;
- any attempted continuation of dynamics would violate at least one existence threshold  $\Theta_{\text{exist}}$ .

**Irreversibility of death** Once  $\Omega(K(t^*)) = \emptyset$ , there is no structural operator acting within the same level that can reconstruct a nonempty  $\Omega(K)$ . Any apparent resurrection would correspond to the birth of a new continuum  $K'$  with its own thresholds and state space, not the continuation of the original  $K$ . Thus death is structurally irreversible.

#### 4.12 Interaction of continua

Two continua  $K_a$  and  $K_b$  interact via an interaction operator

$$E_{\text{int}} : (K_a, K_b) \mapsto (K'_a, K'_b),$$

acting on their combined potentials, flows, thresholds, and axes. Structurally, several regimes are distinguished:

- **Competition:** flows of one continuum hinder the maintenance of cycles in the other; typically  $k_a(t)$  and  $k_b(t)$  cannot both increase indefinitely.
- **Parasitism:** one continuum harvests supporting flows from another, increasing its own  $k$  at the expense of the host.



- **Symbiosis:** supporting flows are coupled such that both continua increase their continuumness and extend their admissible regions.
- **Fusion:** axes and potentials combine to form a new continuum  $K_{\text{fusion}}$  with merged axes and a new state space  $\Omega_{\text{fusion}}$ .

Interaction can itself induce dimensional transitions when mixed differences create new, previously unused axes and drive tension above  $\Theta_{\text{dim}}$ .

#### 4.13 Vertical hierarchy $K_0$ – $K_{12}$

Within this formal scheme, the OC hierarchy  $K_0, \dots, K_{12}$  can be summarised as follows:

- $K_0$ : purely structural substrate of distinguishable states and differences, no time, no dynamics.
- $K_1$ : one-dimensional classical continuum with energy functionals and basic stability thresholds.
- $K_2$ : physical continua, including fields, phase transitions, percolation, Berezinskii–Kosterlitz–Thouless-type transitions, and mass structure in field theory.
- $K_3$ : chemical continua, including reaction networks, reflexively autocatalytic and food-generated structures, concentrations, and environmental parameters.
- $K_4$ : protocellular continua with membranes, osmotic and curvature thresholds, internal/external gradients, and metabolic subspaces.
- $K_5$ : early neural and bioelectrical continua with ion channels, excitation thresholds, gradient-driven flows, and protospikes.
- $K_6$ : cognitive continua with representational axes, binding, internal models, memory thresholds, and prediction thresholds.
- $K_7$ : social continua with trust thresholds, institutional cycles, communication, and coordination flows.
- $K_8$ : civilisational continua with large-scale infrastructures, systemic thresholds, and long-range cycles.
- $K_9$ : theoretical continua comprising theories, paradigms, ontologies, and model families, with consistency and coherence thresholds.
- $K_{10}$ : meta-theoretical continua governing comparison, evaluation, and transformation across classes of theories.
- $K_{11}$ : recursive meta-theoretical continua governing operator families, meta-logical update, and model-space recursion.
- $K_{12}$ : global coherence continua that bound compatibility across families of  $K_{11}$ -structures.

Each level inherits the general continuum structure and adds new axes, potentials, thresholds, and cycles specific to its domain. This chapter does not rederive every domain-specific representation theorem in full detail here; it records the common structural backbone that those instantiations must respect.

#### 4.14 Summary

This section has presented the ontological backbone of OC: the axioms of  $K_0$ , the construction of  $K_1$ , the general definition of a continuum, the taxonomy of thresholds, the roles of potentials, flows, and structural tension, and the definition of cycles and continuumness. It has also fixed the evolution operator, the embedding into meta-spaces, the structural conditions for birth, life, and death, the interaction operator, and the vertical hierarchy  $K_0$ – $K_{12}$ . All further developments in the Core and in extension papers rely on this structure as their common foundation.

## 5 Results and Derived Consequences

This section summarises the core structural results that follow from the formal framework introduced in Section 4. It presents the canonical consequences of the Ontology of Continua (OC) axioms and operators in their Core 1.3 form.

All results in this chapter are domain-independent. They follow solely from the structural definitions of continua, axes, thresholds, potentials, flows, boundaries, cycles and the measure of continuumness  $k(K, t)$ . Proofs are not reproduced here in full detail; they are deferred to dedicated extension papers. The focus is on stating the key theorems and consequences that constrain all continua in the hierarchy  $K_0$ – $K_{12}$ .

### 5.1 Theorem 1: Monotonicity of dimensionality

**Statement.** If a continuum  $K_{x+1}$  emerges from  $K_x$ , then

$$\dim(K_{x+1}) > \dim(K_x).$$

**Justification (sketch).** Dimensional emergence is defined as the appearance of a new axis  $A_{\text{new}}$  that is incompatible with all existing axes in  $A(K_x)$ ; that is, it cannot be represented as a linear or structural combination of them. The dimensional threshold  $\Theta_{\text{dim}}(K_x)$  enforces that  $A_{\text{new}}$  is nondegenerate and structurally necessary. Hence, any emergent continuum has strictly higher dimensionality than its predecessor.

**Corollary 1.1 (No degradational simplification).** There is no continuum-level evolution that reduces dimensionality while preserving existence: if  $K(t)$  is live ( $k(K, t) > 0$ ), then

$$\dim(K(t + dt)) \geq \dim(K(t)).$$

Apparent reduction of dimension corresponds to the death of the higher-dimensional continuum and the birth of a different, lower-dimensional one, not to a continuous simplification.

### 5.2 Theorem 2: Impossibility of spontaneous dimension creation

**Statement.** No continuum can increase its dimensionality without exceeding the dimensional threshold and having access to a suitable axis in its embedding space:

$$\dim(K_{x+1}) > \dim(K_x) \quad \Rightarrow \quad T(K_x, t) > \Theta_{\text{dim}}(K_x) \wedge A_{\text{new}} \in A(M_x) \setminus A(K_x).$$

**Justification (sketch).** New axes represent classes of differences that are incompatible with existing axes. Such incompatibility arises only when structural tension  $T(K_x, t)$  with respect to the current threshold landscape cannot be resolved within the existing axes. By definition of  $\Theta_{\text{dim}}$ , below this threshold all differences can be projected onto the structural span  $\text{span}_{\text{str}}(A(K_x))$ ; above it, projection fails and a new axis is required. The new axis must already be available in the embedding space  $M_x$ ; otherwise there is no structural direction along which the continuum can expand.

**Consequence 2.1.** Noise, local fluctuations or internal flows that do not raise structural tension above  $\Theta_{\text{dim}}(K_x)$  cannot generate new dimensions.

### 5.3 Theorem 3: Death through boundary and embedding collapse

**Statement.** A continuum  $K$  dies when its admissible state space collapses:

$$\Omega(K) = \emptyset.$$

**Equivalent formulations.** The following conditions are equivalent and characterise the death of  $K$ :

1. supporting cycles disappear,  $C(K) = \emptyset$ ;

2. supporting flows vanish,  $J_{\text{support}} = 0$ , or are insufficient to maintain any cycles;
3. there exists a state-independent violation of thresholds,  $\forall s : \exists k \text{ with } f_k(s) > 0$ ;
4. continuumness collapses,  $k(K, t) \rightarrow 0$ ;
5. the continuum exits the region supported by its embedding space  $M$ , i.e. no state satisfies both the internal thresholds of  $K$  and the external constraints of  $M$ .

**Corollary 3.1 (Death as exit from embedding space).** If the constraints of  $M$  change so that no configuration of  $K$  can be embedded into  $M$  while satisfying its thresholds, then  $\Omega(K)$  becomes empty and the continuum dies, regardless of its internal structure.

**Corollary 3.2 (Irreversibility of death).** Once  $\Omega(K) = \emptyset$ , no operator  $E$  acting within the same level, nor any interaction operator  $E_{\text{int}}$  that preserves the identity of  $K$ , can reconstruct a nonempty  $\Omega(K)$ . Any apparent “recovery” corresponds to the birth of a new continuum  $K'$ , not the resurrection of the original  $K$ .

#### 5.4 Theorem 4: Impossibility of evolution outside the embedding space

**Statement.** Let  $K_x$  be a continuum embedded in  $M_x$ . Then for all times  $t$ ,

$$A(K_x(t)) \subseteq A(M_x), \quad \text{span}_{\text{str}}(A(K_x(t + dt))) \subseteq \text{span}_{\text{str}}(A(M_x)).$$

**Consequence 4.1.** All dynamical processes of  $K_x$ , including dimensional birth  $K_x \rightarrow K_{x+1}$ , require that the embedding space  $M_x$  already contain the axes along which the evolution proceeds. A continuum cannot evolve in directions not present in its embedding space.

#### 5.5 Theorem 5: Necessity and sufficiency of compatibility with $M$

**Statement.** A continuum  $K$  exists as a live continuum (i.e.  $\Omega(K) \neq \emptyset$  and  $k(K, t) > 0$ ) if and only if it is compatible with its embedding space  $M$ :

$$K \text{ exists} \iff \Omega(K) \neq \emptyset \wedge A(K) \subseteq A(M) \wedge \Theta(K) \text{ is satisfiable within } M.$$

**Consequence 5.1.** Even if a configuration is mathematically well-defined at the level of  $K$ , it does not correspond to a live continuum unless the embedding space can support the required axes, thresholds and flows. Incompatible configurations are excluded structurally by setting  $\Omega(K) = \emptyset$ .

#### 5.6 Theorem 6: Structural tension and phase transitions

**Statement.** Let  $T(K, t)$  denote structural tension, understood as a functional of potentials, axes and their gradients:

$$T(K, t) = T(P(t), A, \nabla P(t)).$$

Then:

- a phase transition occurs when  $T(K, t) = \Theta_{\text{crit}}(K)$ ,
- a dimensional transition occurs when  $T(K, t) = \Theta_{\text{dim}}(K)$ ,
- structural collapse occurs when  $T(K, t) = \Theta_{\text{death}}(K)$ .

**Corollary 6.1.** All qualitative changes in the continuum — emergence of new regimes, new dimensions, or collapse — are governed by the relationship between structural tension and the threshold landscape.

## 5.7 Theorem 7: Universal law of complexity growth

**Statement.** For any live continuum  $K$ ,

$$S(K(t + dt)) \geq S(K(t)),$$

where  $S$  is a structural measure of complexity defined on the state space and its organisation (for example, combining contributions from number of axes, diversity of states, richness of cycles and properties of the embedding).

**Consequence 7.1.** Complexity increases strictly whenever:

- new axes appear;
- new stable cycles form;
- the admissible state space  $\Omega(K)$  expands;
- the embedding space  $M$  gains new axes relevant to  $K$ .

Stagnant or constant complexity corresponds to finely balanced dynamics below critical thresholds; however, this regime is generically unstable (see Theorem 5.8).

## 5.8 Theorem 8: Impossibility of eternal stabilisation

**Statement.** There is no nontrivial live continuum  $K$  that remains forever at a fixed point in its state space while maintaining positive continuumness. Formally, there is no solution with

$$\frac{dP}{dt} = 0, \quad \frac{dJ}{dt} = 0, \quad \frac{d\Theta}{dt} = 0, \quad \frac{dk}{dt} = 0$$

for all  $t$ , except the trivial case where  $K$  is structurally frozen and decoupled from its embedding space.

**Justification (sketch).** Embedding spaces  $M_x$  themselves evolve and expand (Theorem 10). Internal and external flows cannot both be identically zero on all time scales without either:

- driving the continuum to collapse (loss of supporting flows), or
- inducing structural changes (axes, thresholds, cycles).

Hence any nontrivial live continuum is subject to evolution of its structure or to eventual death.

## 5.9 Theorem 9: Death as loss of cycles

**Statement.** A live continuum  $K$  dies when its maximal structurally stable cycle complex disappears:

$$C_{\max}(K) = \emptyset.$$

**Justification (sketch).** Cycles represent self-maintaining flows that keep the continuum away from its boundary  $\partial\Omega$ . If no such cycle exists, then:

- supporting flows cannot form closed loops;
- trajectories either approach  $\partial\Omega$  or violate thresholds;
- continuumness  $k(K, t)$  decays to zero.

Therefore loss of the maximal cycle complex coincides with the collapse of  $\Omega(K)$ .

**Corollary 9.1.** Death can be detected structurally by the disappearance of all cycles satisfying a minimal stability condition (strictly positive distance to  $\partial\Omega$ ).

### 5.10 Theorem 10: Monotonic growth of embedding spaces

**Statement.** Embedding spaces form a monotonic sequence:

$$M_0 \subset M_1 \subset \dots \subset M_x \subset \dots,$$

and whenever a new continuum  $K_{x+1}$  emerges, the corresponding embedding space  $M_{x+1}$  strictly extends  $M_x$  in terms of its available axes:

$$A(M_x) \subset A(M_{x+1}).$$

**Consequence 10.1.** The growth of continua in dimension and complexity is inseparable from the growth of their embedding spaces. New continua cannot appear without an expansion of the structural possibilities encoded in  $M$ .

### 5.11 Theorem 11: Threshold–expressivity incompatibility

**Statement.** A continuum  $K$  collapses when its axes become insufficient to express the required differences:

$$\dim(A(K)) < \dim(\text{Differences}(K)),$$

where  $\text{Differences}(K)$  is the effective dimension of structurally relevant distinctions imposed by the embedding space and thresholds.

**Consequence 11.1.** Collapse may occur even without violating energetic or dynamical constraints if representational capacity is exceeded. In particular, cognitive, social or meta–theoretical continua can die purely due to loss of expressive adequacy of their axes.

### 5.12 Theorem 12: Incompleteness of embedding spaces

**Statement.** For any finite embedding space  $M_x$ , there exist potential continua  $K'$  that cannot be realised within  $M_x$  because their axes or thresholds are incompatible with  $A(M_x)$  or with the constraints of  $M_x$ .

**Consequence 12.1.** No single embedding space  $M_x$  is universal for all possible continua. The sequence of embedding spaces must itself expand and diversify to host new structural regimes.

### 5.13 Corollaries and domain–independent consequences

Collecting the theorems above, we obtain the following domain–independent consequences:

- Dimensionality is strictly monotonic for live continua; no continuous degradation of dimension is possible.
- Dimension cannot appear spontaneously; it requires structural tension above  $\Theta_{\text{dim}}$  and the presence of suitable axes in the embedding space.
- Death is characterised by the collapse of  $\Omega(K)$ , the disappearance of cycles and the irreversibility of this collapse.
- Evolution is constrained by embedding spaces; continua cannot evolve in directions that  $M$  does not support.
- Complexity tends to grow for live continua, driven by new axes, new cycles and expanding state spaces.
- Eternal exact stabilisation is structurally excluded for nontrivial continua; they either evolve or die.
- Representational and expressive limits can cause collapse even when energetic and dynamical limits are not yet reached.

These consequences hold uniformly for all levels  $K_0$ – $K_{10}$ ; their domain–specific content arises only from the interpretation of axes, potentials, thresholds and flows in each level.

## 5.14 Implications for the vertical hierarchy

In the vertical hierarchy  $K_0$ – $K_{10}$ , the structural theorems manifest as follows:

- $K_0$ : no dynamics, no birth or death; only structural conditions for distinguishability.
- $K_1$ : classical phase-like behaviour with continuous flows and basic stability/critical thresholds.
- $K_2$ : physical thresholds, including percolation, BKT transitions, coherence/condensation thresholds and mass-generating mechanisms.
- $K_3$ – $K_4$ : chemical and prebiotic thresholds (RAF closure, membrane formation, osmotic and curvature thresholds, redox and pH thresholds).
- $K_5$ : electrical thresholds for excitation, spiking and gradient-driven flows in early neural systems.
- $K_6$ : logical and representational thresholds for cognitive binding, prediction, memory stability and internal model coherence.
- $K_7$ – $K_8$ : social and civilizational thresholds governing trust, institutional stability, systemic resilience and collapse.
- $K_9$ – $K_{10}$ : expressive, coherence and consistency thresholds in the space of theories, paradigms, ontologies and meta-theoretical structures.

In each case, the same structural principles — monotonic dimension, threshold-governed emergence and collapse, dependence on embedding spaces, and complexity growth — apply, while the concrete interpretation of variables changes.

## 5.15 Summary

The results presented in this chapter constitute the structural backbone of OC. They state that dimension is monotonic, that emergent dimensions require threshold-induced phase transitions, that death is characterised by the collapse of admissible states and is irreversible, that evolution is constrained by embedding spaces, that complexity tends to grow for live continua, that cycles are the carriers of structural life, and that expressive limits can cause collapse. These principles apply across the entire continuum hierarchy and define the constraints under which all continua can exist, evolve or die.

# 6 Discussion

This section interprets the structural framework presented in [Sections 4](#) and [5](#). It explains what the Ontology of Continua (OC) commits to, the limits of those commitments, and how the same grammar can be interpreted across domains. No new axioms or theorems are introduced here; the goal is to clarify the existing formalism and outline trajectories for further development.

## 6.1 Conceptual structure of the OC framework

OC proposes that continua across scientific domains share a common ontological structure. This structure is given by the tuple

$$K = (\Omega(K), A(K), P(t), J(t), \Theta(K), \partial\Omega(K), C(K), k(K, t)),$$

where:

- $\Omega(K)$  is the state space of admissible configurations;
- $A(K)$  is the collection of axes of incompatible differences;
- $P(t)$  is the vector of potentials;

- $J(t)$  is the family of flows;
- $\Theta(K)$  is the threshold landscape;
- $\partial\Omega(K)$  is the boundary defined by threshold saturation;
- $C(K)$  is the set of structurally stable cycles;
- $k(K, t)$  is the measure of continuumness.

When the same tuple is later written without the explicit  $K$  index, it is only shorthand for this indexed continuum tuple.

The discussion below focuses on how these elements function conceptually.

**Axes.** Axes represent incompatible classes of differences that cannot be reduced to one another. They define the effective dimensionality of the continuum. In physical systems, axes may correspond to spatial or internal degrees of freedom; in chemical systems, to concentrations and environmental parameters; in cognitive or social systems, to representational or institutional coordinates. OC treats all of these as instances of the same structural role: axes determine what distinctions the continuum can express.

**Potentials and flows.** Potentials encode the internal configuration of constraints and driving forces. Examples include energetic, chemical, informational, and normative forces. Flows describe how these potentials change in time and how the system moves through  $\Omega(K)$ . The central conceptual point is that OC uses a single language for these notions: flows can support, challenge, or destroy the structure of the continuum independently of their physical realisation.

**Thresholds and boundaries.** Thresholds provide the primary mechanism for qualitative change. They partition the extended state space into regions of existence, stability, criticality, dimensional emergence, and collapse. The boundary  $\partial\Omega(K)$  is defined as the locus where at least one threshold is saturated. This replaces a collection of domain-specific concepts (critical temperatures, carrying capacities, viability limits, institutional breaking points) with a unified notion of threshold surfaces.

**Cycles and continuumness.** OC emphasises that structural persistence is an active property: it requires ongoing flows and cycles. Cycles are closed trajectories that remain strictly inside  $\Omega(K)$  and at a finite distance from  $\partial\Omega(K)$ . The measure  $k(K, t)$  integrates information about the richness of  $\Omega(K)$ , the presence and stability of cycles, the expressive capacity of axes, and the compatibility of those axes with the active thresholds. A continuum is viable only when stabilising cycles coexist with a nonempty admissible region, compatible axes, and satisfiable thresholds inside its embedding space.

## 6.2 Interpretation of the structural results

The results in [Section 5](#) express several core commitments of the OC framework.

One shared consequence is dimensional monotonicity for live continua. Theorem 1 (monotonic dimensionality) means that a continuum cannot simply lose an essential axis and remain the same continuum. Emergence is therefore threshold-driven rather than optional: Theorems 2 and 6 (threshold-driven emergence results) allow new structure only when tension crosses the relevant threshold and the embedding space supplies a suitable axis.

Collapse is not just energetic failure. The expressivity route, Theorem 11 (threshold-expressivity incompatibility), allows collapse when relevant differences outrun the available axes, while Theorems 3 and 9 (death/collapse results) bind terminal loss to empty admissibility and failed stable cycles. Once that state is reached, the original continuum is not restored by internal dynamics; later live structure must be classified separately.

These results also make the embedding space a formal part of the claim. Theorems 4, 5, 10, and 12 keep continua from being treated as self-contained objects: they require embedding spaces that supply

axes, constraints, and room for dimensional growth. Together with the complexity and non-stabilisation results (Theorems 7 and 8), this frames OC as a theory of structured evolution rather than fixed equilibrium.

Taken together, these results position OC as a structural theory of organised systems rather than a reductionist account of their microscopic constituents.

### 6.3 Limitations of the current formalism

Despite its generality, the present version of the OC framework has several clear limitations.

**Level of abstraction.** The formalism is intentionally abstract. That abstraction allows a single structural grammar to span multiple domains, but it also makes direct empirical use more demanding. Bridging from the structural language  $(\Omega, A, P, J, \Theta, \partial\Omega, C, k)$  to concrete datasets is therefore not automatic; it requires careful, domain-specific modelling rather than a direct one-step translation.

**Quantitative level transitions.** The conditions for transitions  $K_x \rightarrow K_{x+1}$  are stated qualitatively in terms of tension and dimensional thresholds. Quantitative models of such transitions, including explicit forms of  $T(P, A, \nabla P)$  and  $\Theta_{\text{dim}}$ , have so far been developed only for particular case studies, such as phase transitions and protocell closure, and remain to be generalised.

**Threshold specification.** The taxonomy of thresholds  $(\Theta_{\text{exist}}, \Theta_{\text{stab}}, \Theta_{\text{crit}}, \Theta_{\text{dim}}, \Theta_{\text{death}})$  is structurally complete, but explicit functional forms for these thresholds must be determined separately for each class of systems. At present, OC provides the structural scaffold; filling it with numerically calibrated thresholds is a task for future work.

**Expressive capacity.** Theorems on threshold-expressivity incompatibility state that collapse can be driven by insufficient dimensionality of  $A(K)$ . However, operational measures of expressive capacity for realistic cognitive, social, or theoretical systems are not yet fully developed.

**Inter-continuum interactions.** The interaction operator  $E_{\text{int}}$  captures generic regimes (competition, parasitism, symbiosis, fusion), but quantitative theories of such interactions among complex continua, such as interacting institutions, coupled civilisations, and competing theories, are still largely schematic.

### 6.4 OC in the context of existing scientific and formal theories

OC can be located with respect to several established paradigms.

- **Statistical physics.** Concepts of thresholds, criticality, order parameters, and phase transitions correspond to  $\Theta_{\text{crit}}$ , structural tension  $T$ , and changes in  $\Omega(K)$ . OC generalises these ideas beyond physical systems.
- **Chemical systems theory.** Reflexively autocatalytic and food-generated (RAF) networks, catalytic closure, and reaction-diffusion structures instantiate OC notions of cycles, supporting flows, and existence thresholds in  $K_3$ .
- **Biophysics and early life.** Membrane closure, osmotic and curvature thresholds, redox and pH gradients, and protocell dynamics are concrete examples of threshold and boundary behaviour in  $K_4$ .
- **Neuroscience.** Excitability, ion channel dynamics, and spiking correspond to electrical thresholds, flows, and cycles in  $K_5$ .
- **Cognitive science.** Binding, representation, prediction, and memory stability instantiate axes, expressive capacity, and coherence thresholds in  $K_6$ .



- **Systems and social theory.** Stability of institutions, systemic resilience, and collapse illustrate thresholds, cycles, and embedding constraints at levels  $K_7$ – $K_8$ .
- **Logic and metatheory.** The levels  $K_9$  and  $K_{10}$  treat theories, paradigms, and formal languages as continua constrained by consistency, coherence, and expressive thresholds.

OC does not claim to supersede these theories. Its role is to provide a structural layer that explains why analogous patterns of thresholds, cycles, and collapse appear across such diverse domains.

## 6.5 Implications for the continuum hierarchy

The vertical hierarchy  $K_0$ – $K_{12}$  can be read as a sequence of increasing representational and dynamical richness:

- $K_0$  encodes mere distinguishability; there is no time, energy, or geometry.
- $K_1$  introduces geometric continuity and classical stability thresholds.
- $K_2$  organises physical fields, phases, and mass-related structures.
- $K_3$ – $K_4$  introduce chemical and prebiotic organisation (RAF networks, protocells, internal gradients).
- $K_5$  introduces early bioelectrical excitability and protospiking.
- $K_6$  introduces cognitive axes, internal models, and binding thresholds.
- $K_7$ – $K_8$  describe social and civilisational continua with institutional, infrastructural, and systemic thresholds.
- $K_9$ – $K_{10}$  describe theoretical and meta-theoretical continua, where theories and languages themselves form state spaces subject to consistency and coherence thresholds.
- $K_{11}$ – $K_{12}$  describe upper-level coherence and unified-science integration, where families of meta-frameworks and domain closures are tested for global compatibility.

Conceptually, each level:

1. inherits the general continuum structure of [Section 4](#);
2. adds new axes and thresholds that cannot be reduced to lower levels;
3. is constrained by an embedding space  $M_x$  that must expand for higher levels to exist.

This hierarchy is not merely a taxonomy. It is a claim that all these domains can be analysed with a single structural toolkit.

## 6.6 Future development paths

Several directions for future work follow naturally from the current state of the Core:

- full, self-contained proofs of the theorems stated in [Section 5](#), with explicit assumptions and intermediate lemmas;
- quantitative definitions of structural tension, thresholds, and flows in key case studies, such as Berezinskii–Kosterlitz–Thouless (BKT) transitions, RAF networks, protocell stability, and protospiking;
- refined models of collapse and recovery attempts at levels  $K_3$ – $K_5$ , including explicit threshold landscapes and cycle metrics;
- systematic development of cognitive continua  $K_6$ , with formal treatment of binding, prediction, memory, and model-selection thresholds;

- applications of OC to social and civilisational dynamics at levels  $K_7$ – $K_8$ , including stress-testing of institutions and infrastructures;
- integration of the OC framework with empirical datasets and simulation pipelines, in order to test falsifiable predictions about thresholds and dimensional transitions;
- formal specification of  $K_9$ – $K_{12}$  in terms of logical systems, model theory, meta-level recursion, and global compatibility constraints.

These tasks are intended not as speculative extensions but as concrete steps from the current structural core towards domain-specific, testable models.

## 6.7 Summary

The conceptual conclusion is that OC treats axes, potentials, flows, thresholds, cycles, embedding spaces, and continuumness as one structural grammar for emergence, stability, and collapse. That grammar can be compared across domains, but it remains scientifically useful only when each domain-specific instantiation carries its own quantitative and empirical burden.

Core 1.3 does not claim to resolve every future research direction. It provides a coherent, minimal, and extensible foundation on which more detailed domain-specific developments can be built without reopening the core ontology.

## 7 Conclusion

This concluding chapter summarises the structural contributions of *Ontology of Continua — Core 1.3*. The present version provides a coherent synthesis of the formal and scientific content established in this release. It is written as the close of the proof-bearing master monograph, not as a placeholder for a later foundational volume.

### 7.1 Summary of the Core 1.3 contributions

Core 1.3 has prioritised consolidation, clarification, vertical consistency, and publication-grade scientific closure. Across the preceding chapters, the following contributions were established:

- A canonical  $\text{\LaTeX}$  and repository structure enabling reproducible academic releases.
- A refined axiomatic formulation of the substrate level  $K_0$  and the generative operator  $\Psi_{0 \rightarrow 1}$  defining the transition to  $K_1$ .
- A unified structural definition of a continuum in terms of the tuple:

$$K = (\Omega(K), A(K), P(t), J(t), \Theta(K), \partial\Omega(K), C(K), k(K, t)).$$

- A complete taxonomy of thresholds ( $\Theta_{\text{exist}}$ ,  $\Theta_{\text{stab}}$ ,  $\Theta_{\text{crit}}$ ,  $\Theta_{\text{dim}}$ ,  $\Theta_{\text{death}}$ ) and a unified definition of boundaries via threshold saturation.
- Formal statements of universal structural results: monotonicity of dimension, impossibility of spontaneous dimension creation, irreversibility of death, necessity of compatibility with embedding spaces, structural tension as the driver of phase transitions, and monotonic growth of structural complexity.
- A coherent vertical organisation of continua from  $K_0$  to  $K_{12}$ , including the upper-level integrability doctrine required by the present master release.
- A bounded empirical closure program with theorem-native packets, held-out replay summaries, falsifiers, and cross-domain integrability surfaces.

- A final unified-science synthesis that keeps formulas, numerical rows, and promotion verdicts aligned inside one monograph.

Core 1.3 therefore serves not only as a stable foundation, but as the first release in which the formal stack, the empirical closure surfaces, and the manuscript shell are forced to agree publicly.

## 7.2 Conceptual synthesis

Several overarching principles unify the OC framework.

**Thresholds govern qualitative change.** Birth, dimensional emergence, stability and collapse all occur through threshold saturation. This replaces disparate domain-specific mechanisms with a universal structure.

**Cycles maintain persistence.** Continuity is not passive. All live continua maintain supporting flows and stable cycles. The disappearance of cycles is both a precursor and a signature of collapse.

**Embedding spaces enable and constrain evolution.** A continuum's axes, potentials, thresholds and transitions must be compatible with its embedding space  $M_x$ . Emergence of higher continua requires corresponding growth of embedding spaces.

**Complexity grows monotonically for live continua.** New axes, expanded state spaces and stable cycles increase structural richness. A continuum that stops evolving structurally does so only at the moment of collapse.

Together, these principles provide a unified conceptual backbone for understanding diverse systems.

## 7.3 Implications across scientific domains

Although the monograph remains bounded in scope, the structural framework now has explicit domain-facing implications:

- **Physics:** critical surfaces, coherence thresholds, BKT transitions and phase structure instantiate  $\Theta_{\text{crit}}$  and  $\Theta_{\text{dim}}$  at level  $K_2$ .
- **Chemistry / origins of life:** catalytic closure, RAF networks, vesicle stability, osmotic and curvature thresholds, and redox gradients instantiate the threshold landscape of  $K_3$ – $K_4$ .
- **Biology:** excitability thresholds, membrane potentials, channel dynamics and metabolic cycles realise flows, thresholds and cycles of  $K_4$ – $K_5$ .
- **Cognition:** binding, representational coherence, predictive stability and expressive limits correspond to axes, potentials and thresholds in  $K_6$ .
- **Social and civilizational systems:** institutional integrity, trust thresholds, normative coherence and infrastructural stability instantiate the structural logic of  $K_7$ – $K_8$ .

These examples suggest a deep structural regularity: very different systems behave as continua under the same ontological constraints.

## 7.4 Future directions

The present release leaves room for further work, but that work no longer starts from an informal shell. Likely next developments include:

- complete mathematical proofs of all theorems stated in Section 5;

- explicit dynamical forms of structural tension, flows and thresholds;
- detailed instantiations of continua in physics, chemistry, early life, cognition and social systems;
- full formal treatment of cognitive continua  $K_6$  and the mechanisms of binding, prediction and model coherence;
- quantitative models of collapse and phase transitions at levels  $K_3$ – $K_5$ ;
- deeper refinement of theoretical and meta-theoretical continua  $K_9$ – $K_{12}$ ;
- broader comparator studies and additional empirical routes for domains beyond the bounded closure packets already promoted here.

Core 1.3 provides the structural scaffolding needed for these efforts.

## 7.5 Final remarks

The Ontology of Continua seeks to provide a single, structurally grounded framework for understanding the birth, persistence, evolution, and collapse of continua across scientific domains. Core 1.3 presents that programme as a coherent axiomatics, a universal structural language, a precise threshold taxonomy, and a vertically consistent hierarchy of continua from  $K_0$  to  $K_{12}$ , now accompanied by the science surfaces needed to audit its bounded claims.

All subsequent developments — mathematical, empirical, and conceptual — will proceed from this foundation. Core 1.3 is stable at its intended scope and ready to serve as the reference point for continued development of the Ontology of Continua.

## 8 Extended Boundary and Patch Geometry

This chapter restores the full theory of boundaries and patch geometry. Boundaries constitute one of the most universal structures in the OC framework: they govern phase separation in  $K_2$ , membrane formation in  $K_3$ – $K_4$ , excitability in  $K_5$ , and representational or institutional limits in  $K_6$ – $K_8$ . The present version consolidates the complete structure needed for Core 1.3: threshold-defined boundaries, patch discretisation, local threshold classes, and boundary-driven transitions.

### 8.1 Formal Definition of $\partial\Omega(K)$

For any continuum  $K$  with admissible state space  $\Omega(K)$  and threshold landscape  $\Theta(K)$ , the boundary is defined as the set of states at which at least one threshold is exactly saturated:

$$\partial\Omega(K) = \{ s \in \overline{\Omega(K)} : \exists k \text{ such that } g_k(s) = \Theta_k \}.$$

Equivalently, if  $f_k(s) = g_k(s) - \Theta_k$  denotes the shifted residual form used in the notation appendix and the model chapter, then interior states satisfy  $f_k(s) < 0$ , admissible-side states satisfy  $f_k(s) \leq 0$ , and the same boundary is the zero set  $f_k(s) = 0$  for at least one active threshold.

Thus:

- interior states satisfy  $f_k(s) < \Theta_k$  for all thresholds;
- exterior states violate at least one threshold;
- the boundary is a *threshold-defined surface*, not a geometric primitive.

This definition unifies classical surfaces, membranes, vesicles, percolation fronts, excitable boundaries, logical limits and institutional thresholds under one structure.

## 8.2 Boundary Geometry

Boundary geometry emerges from the local structure of thresholds, potentials and flows. If the active threshold at a point  $s \in \partial\Omega(K)$  is  $g_k$ , the outward normal is aligned with the gradient:

$$n(s) \propto \nabla g_k(s).$$

Multiple active thresholds create corners, ridges and high-curvature regions. Curvature becomes dynamically relevant whenever bending or surface-energy terms enter the potential:

$$P_{\text{bend}}(s) = \frac{1}{2} \kappa(s)^2 \kappa_b,$$

with bending rigidity  $\kappa_b$ .

This connects:

- $K_2$ : classical surface tension,
- $K_3$ – $K_4$ : membrane curvature and vesicle energetics,
- $K_5$ : excitable boundary geometry.

## 8.3 Patch Model of Boundaries

OC models boundaries not as continuous uniform surfaces but as collections of dynamic *patches*:

$$\partial\Omega(K) = \bigcup_i \partial\Omega_i.$$

Each patch  $P_i$  carries:

$$(\sigma_i, \Theta_i, P_i(t), J_i(t)),$$

representing its local state, thresholds, potentials and flows.

**Local states.** Patches take symbolic states:

$$\sigma_i \in \{L_\alpha, L_\beta, L_o, L_f, L_b\},$$

encoding local packing or structural regimes:

- $L_\alpha$ : fluid-disordered,
- $L_\beta$ : ordered/gel,
- $L_o$ : liquid-ordered,
- $L_f$ : fragile/thin (pre-rupture),
- $L_b$ : broken boundary state.

This alphabet generalises beyond biological membranes to percolation edges, fracture fronts, fluctuating interfaces and even symbolic/cognitive boundaries.

**Local thresholds.** Each patch has a vector:

$$\Theta_i = (\Theta_i^{\text{perm}}, \Theta_i^{\text{grad}}, \Theta_i^{\text{mem}}, \Theta_i^{\text{bend}}, \dots),$$

with corresponding local potentials  $P_i(t)$  and flows  $J_i(t)$ .

## 8.4 Boundary Threshold System

Across  $K_2$ – $K_5$  three threshold classes appear universally.

**1. Permeability threshold  $\Theta_{\text{perm}}$ .** A patch becomes permeable when:

$$|\nabla P_i| > \Theta_i^{\text{perm}}.$$

Examples:

- osmotic permeability ( $K_3$ – $K_4$ ),
- ion leak onset ( $K_5$ ),
- phase–front penetration ( $K_2$ ).

**2. Gradient threshold  $\Theta_{\text{grad}}$ .** If a gradient is unsustainable:

$$|\nabla P_i| > \Theta_i^{\text{grad}} \Rightarrow \sigma_i \rightarrow L_f.$$

This governs:

- vesicle rupture at high osmotic pressure ( $K_4$ ),
- action–potential initiation at axon initial segments ( $K_5$ ),
- accelerated phase fronts and crack propagation ( $K_2$ ).

**3. Membrane integrity threshold  $\Theta_{\text{mem}}$ .** If membrane tension or curvature exceeds limits:

$$\gamma_i > \Theta_i^{\text{mem}} \quad \text{or} \quad \kappa(s) > \Theta_i^{\text{bend}},$$

then:

$$\sigma_i \rightarrow L_b.$$

This yields a unified understanding of rupture, buckling, pore formation, boundary breakdown and surface collapse.

## 8.5 Boundary–Driven Dynamics

Neighbouring patches interact via flows:

$$J_{ij} = F(P_i, P_j, \sigma_i, \sigma_j).$$

This produces:

- stabilisation and rearrangement of patch mosaics,
- propagation of rupture fronts,
- curvature–driven migration,
- pore initiation,
- excitable–medium behaviour in  $K_5$ .

Patch evolution follows:

$$\frac{d\sigma_i}{dt} = S(\sigma_i, P_i, J_i, \Theta_i),$$

where  $S$  is a structural update operator independent of domain.

**Boundary Localisation Principle.** All  $K_3$ – $K_5$  dimensional transitions initiate on boundary patches.

This explains:

- protocell bursting beginning at high–curvature points,
- action potentials initiating at axon initial segments,
- phase transitions nucleating at defects or surfaces.

## 8.6 Boundary Failure and Collapse

Boundary collapse occurs when a connected set of patches becomes broken:

$$\sigma_i = L_b \quad \forall i \in \mathcal{C},$$

and the cluster  $\mathcal{C}$  percolates across the boundary.

By Theorem 3 (Results chapter),

$$\partial\Omega(K) \text{ collapses} \implies \Omega(K) = \emptyset,$$

and the continuum dies.

We distinguish:

- **local failure:** isolated  $L_b$  patches, reversible;
- **cluster collapse:** percolating  $L_b$  network, irreversible;
- **global boundary loss:** disappearance of the boundary, implying death or transition to a new continuum.

## 8.7 Boundary and Rebirth

Birth of a new continuum  $K_{x+1}$  typically requires:

- formation or rupture of boundary patches,
- creation of new patch types,
- emergence of new threshold classes on boundary regions,
- establishment of a new global boundary geometry.

**Boundary Rebirth Principle.** *New continua emerge from new boundary geometries.*

Examples:

- membrane closure in the  $K_3 \rightarrow K_4$  transition,
- appearance of excitable boundaries in  $K_4 \rightarrow K_5$ ,
- formation of institutional boundaries in  $K_6 \rightarrow K_7$ ,
- emergence of representational boundaries in  $K_9 \rightarrow K_{10}$ .

Boundaries are therefore not merely limits: they are generative surfaces where new axes, constraints and cycles are born.

# 9 Extended Threshold Landscape

This chapter reconstructs the full theory of thresholds in the Ontology of Continua (OC), consolidating structural elements from the Physics, Chemistry, Biology and Cognition runs. Thresholds are the central mechanism through which qualitative change, dimensional emergence, collapse and death are produced.

Thresholds are universal: they apply to physical, chemical, biological, cognitive, social and theoretical continua throughout the hierarchy  $K_0$ – $K_{10}$ .

## 9.1 Definition of a Threshold

A *threshold* is represented by a constraint function

$$f_k : \Omega(K) \rightarrow \mathbb{R}$$

with the property

$$f_k(s) \leq 0 \quad \text{for all admissible states } s \in \Omega(K).$$

Thresholds separate qualitatively different dynamical regimes:

- $f_k(s) < 0$ : safe regime (interior),
- $f_k(s) = 0$ : boundary regime,
- $f_k(s) > 0$ : forbidden regime (outside  $\Omega(K)$ ).

Thus thresholds jointly determine:

- the shape of the admissible set  $\Omega(K)$ ,
- the boundary  $\partial\Omega(K)$ ,
- the conditions under which the continuum exists or collapses.

Thresholds are not static constants. They depend on potentials  $P(t)$ , flows  $J(t)$ , axes  $A(K)$ , environmental constraints and the structure of the embedding space  $M$ . At the structural level this is encoded by the threshold–evolution operator  $H$  from the general evolution machinery:

$$\frac{d\Theta_k}{dt} = H_k(\Theta_k, P, J, A, M),$$

where  $\Theta_k$  denotes the parameter set defining the constraint  $f_k$ .

## 9.2 Taxonomy of Thresholds

OC distinguishes seven universal classes of thresholds. This taxonomy was stabilised in Core 2.x and clarified across the domain runs.

**1. Existence thresholds**  $\Theta_{\text{exist}}$ . Conditions required for  $\Omega(K) \neq \emptyset$ . Examples:

- minimal energy conditions for bound physical states ( $K_2$ ),
- minimal catalytic closure (RAF existence) in  $K_3$ ,
- minimal membrane continuity and integrity in  $K_4$ ,
- minimal coherence of internal models in  $K_6$ ,
- minimal legitimacy and trust for institutions in  $K_7$ .

**2. Stability thresholds**  $\Theta_{\text{stab}}$ . Conditions under which flows remain bounded and cycles are stable. Examples:

- confinement and boundedness conditions in physics ( $K_2$ ),
- osmotic homeostasis conditions in protocells ( $K_4$ ),
- membrane potential stability in neurons ( $K_5$ ),
- normative stability in social continua ( $K_7$ ).

**3. Critical thresholds**  $\Theta_{\text{crit}}$ . Surfaces of qualitative change at fixed dimensionality. Examples:

- phase transitions and BKT–type thresholds ( $K_2$ ),
- onset of catalytic networks ( $K_3$ ),
- metabolic switching in  $K_4$ ,
- spike initiation threshold in  $K_5$ ,



- prediction–error bifurcations in  $K_6$ ,
- institutional bifurcations in  $K_7$ .

**4. Dimensional thresholds  $\Theta_{\text{dim}}$ .** Constraints that trigger the creation of new axes. Dimensional thresholds are central in OC:

$$T(K, t) > \Theta_{\text{dim}}(K) \Rightarrow \text{emergence of a new axis } A_{\text{new}}.$$

They govern the transitions  $K_x \rightarrow K_{x+1}$  throughout the hierarchy.

**5. Death thresholds  $\Theta_{\text{death}}$ .** Values beyond which no admissible state remains. Equivalently,

$$\forall s \in \overline{\Omega(K)} : \exists k \text{ with } f_k(s) > 0 \iff \Omega(K) = \emptyset.$$

**6. Expressivity thresholds  $\Theta_{\text{expr}}$ .** Thresholds on representational capacity. A continuum collapses when

$$\dim(A(K)) < \dim(\text{Differences}(K)),$$

where  $\text{Differences}(K)$  denotes the effective dimension of structurally relevant distinctions imposed by the embedding space and constraints. This class is crucial for cognitive, social and meta–theoretical collapse.

**7. Embedding thresholds  $\Theta_{\text{embed}}$ .** Constraints imposed by the embedding space  $M_x$ . A continuum cannot exist as a live continuum if the embedding space does not support its axes, thresholds or flows; formally this is captured by the compatibility conditions in Theorems 4 and 5 (Results chapter).

### 9.3 Threshold Geometry

Thresholds induce a stratified geometry on  $\Omega(K)$ . For each threshold function  $f_k$  we consider the level sets

$$\Sigma_k(c) = \{s \in \Omega(K) \mid f_k(s) = c\}.$$

In particular:

- $\Sigma_k(0)$  defines components of the boundary  $\partial\Omega(K)$ ,
- $\Sigma_k(c < 0)$  defines safe regions,
- $\Sigma_k(c > 0)$  defines forbidden regions.

Interactions between thresholds generate:

- multi–threshold intersections,
- cusp and fold structures (catastrophe–like geometries),
- regions of concentrated curvature,
- patch–level heterogeneity at the boundary (cf. Section 8).

The joint threshold geometry determines:

- the global shape of  $\Omega(K)$ ,
- admissible trajectories and cycle structure,
- the available transitions of the continuum.

## 9.4 Dimensional Thresholds

Dimensional thresholds occupy a privileged role in OC because they decide when new differences become unrepresentable within the current axes.

Let  $T(K, t)$  denote structural tension. Then, schematically:

$$T(K, t) < \Theta_{\dim}(K) \Rightarrow \text{all relevant differences can be expressed within } A(K),$$

$$T(K, t) = \Theta_{\dim}(K) \Rightarrow \text{projection onto } A(K) \text{ fails,}$$

$$T(K, t) > \Theta_{\dim}(K) \Rightarrow \text{a new axis } A_{\text{new}} \text{ is structurally required.}$$

Dimensional thresholds govern, in particular:

- birth of  $K_1$  from the structural substrate  $K_0$ ,
- emergence of chemical axes in  $K_3$ ,
- membrane axes in  $K_4$ ,
- excitation axes in  $K_5$ ,
- cognitive representational axes in  $K_6$ ,
- social normative axes in  $K_7$ ,
- meta-logical axes in  $K_9$ – $K_{10}$ .

## 9.5 Threshold Cascades

Thresholds frequently interact in cascades of the form

$$\Theta_{\text{grad}} \longrightarrow \Theta_{\text{perm}} \longrightarrow \Theta_{\text{mem}},$$

or more generally, gradients  $\rightarrow$  permeability  $\rightarrow$  structural failure.

Representative examples:

- **Vesicle bursting** ( $K_4$ ). Osmotic gradients exceed  $\Theta_{\text{grad}}$ , permeability increases ( $\Theta_{\text{perm}}$ ), and membrane integrity is lost ( $\Theta_{\text{mem}}$ ).
- **Action potentials** ( $K_5$ ). Depolarisation exceeds a gradient threshold, channels open at  $\Theta_{\text{crit}}$ , and a spike cycle is activated.
- **Institutional collapse** ( $K_7$ ). Trust deficits exceed  $\Theta_{\text{grad}}$ , normative coherence fails ( $\Theta_{\text{expr}}$ ), and institutional death follows ( $\Theta_{\text{death}}$ ).

Such cascades are central for early life, neural systems, and social and infrastructural dynamics.

## 9.6 Death Thresholds

Death thresholds determine when a continuum ceases to exist as a live continuum. The following structural conditions are equivalent manifestations of death:

- complete violation of thresholds at all states:  $\forall s : \exists k \text{ with } f_k(s) > 0$ ;
- disappearance of supporting cycles:  $C(K) = \emptyset$ ;
- collapse of the admissible state space:  $\Omega(K) = \emptyset$ ;
- representational failure, e.g.  $\Theta_{\text{expr}}$  exceeded;
- incompatibility with the embedding space (no configuration satisfies both internal and embedding constraints).

Once  $\Omega(K) = \emptyset$ , death is structurally irreversible: no operator acting within the same level can recreate  $\Omega(K)$ ; any apparent recovery corresponds to the birth of a new continuum  $K'$ .

## 9.7 Examples Across Levels

The threshold taxonomy manifests differently at each level of the hierarchy.

### $K_1$ (Geometric continua).

- $\Theta_{\text{exist}}$ : differentiability and regularity conditions defining the classical region  $\Omega_{\text{cl}}$ .
- $\Theta_{\text{crit}}$ : classical bifurcations and loss of stability.

### $K_2$ (Physical continua).

- $\Theta_{\text{crit}}$ : phase transitions, including BKT thresholds,
- $\Theta_{\text{dim}}$ : coherence thresholds associated with mass generation and ordering,
- $\Theta_{\text{death}}$ : confinement/decoupling limits where the relevant continuum ceases to exist.

### $K_3$ (Chemical continua).

- $\Theta_{\text{exist}}$ : catalytic closure and RAF existence,
- $\Theta_{\text{crit}}$ : reaction–network percolation,
- $\Theta_{\text{expr}}$ : insufficient chemical axes to represent required reaction diversity.

### $K_4$ (Prebiotic/biological membrane continua).

- $\Theta_{\text{grad}}$ : osmotic pressure limits,
- $\Theta_{\text{perm}}$ : permeability thresholds,
- $\Theta_{\text{mem}}$ : membrane rupture and curvature thresholds.

### $K_5$ (Excitable continua).

- $\Theta_{\text{crit}}$ : spike initiation thresholds,
- $\Theta_{\text{stab}}$ : refractory conditions and excitability bounds,
- $\Theta_{\text{death}}$ : loss of excitability and breakdown of excitable cycles.

### $K_6$ (Cognitive continua).

- $\Theta_{\text{expr}}$ : limits of binding capacity and representational complexity,
- $\Theta_{\text{crit}}$ : prediction–error thresholds,
- $\Theta_{\text{stab}}$ : memory coherence and model–stability limits.

### $K_7$ – $K_8$ (Social and civilizational continua).

- $\Theta_{\text{exist}}$ : thresholds of trust and legitimacy,
- $\Theta_{\text{expr}}$ : institutional capacity and expressive limits,
- $\Theta_{\text{death}}$ : systemic collapse and breakdown of institutional and infrastructural cycles.

### $K_9$ – $K_{10}$ (Theoretical and meta–theoretical continua).

- $\Theta_{\text{expr}}$ : expressive capacity of theories and meta–languages,
- $\Theta_{\text{dim}}$ : thresholds for meta–level expansion,
- $\Theta_{\text{death}}$ : inconsistency, incoherence or structural incompatibility in the space of models.

## 9.8 Summary

The extended threshold landscape provides the structural backbone for all qualitative change in OC. Thresholds:

- carve out  $\Omega(K)$  and define  $\partial\Omega(K)$ ,
- govern phase transitions, dimensional emergence and collapse,
- mediate cascades that connect gradients, permeability and structural failure,
- encode expressive and embedding limitations that can kill continua even without energetic overload.

Together with the boundary and patch geometry of Section 8, the threshold landscape completes the description of how continua are born, maintained and destroyed in the OC framework.

## 10 Full Hierarchy of Continua $K_0$ – $K_{12}$

This chapter provides the complete reconstructed description of all continuum levels  $K_0$ – $K_{12}$ , integrating the material originally distributed across Core 2.x and the Physics, Chemistry, Biology, Cognition and Social runs. Unlike Section 4, which presented a compact overview, the present chapter expands each level into its full structural definition.

Core 1.3 fixes the hierarchy as  $K_0$ – $K_{12}$  because the public source corpus already contains explicit upper-level modules, cross-level transitions, and prediction/process shells for  $K_{11}$  and  $K_{12}$ . Treating the hierarchy as if it stopped at  $K_{10}$  would therefore leave the monograph internally inconsistent.

Each continuum level is defined by the tuple

$$K_x = (\Omega_x, A_x, P_x(t), J_x(t), \Theta_x, \partial\Omega_x, C_x, k_x(t)).$$

Levels differ by the nature of their axes, potentials, thresholds, boundary geometry, cycles, and by the structure of the embedding space  $M_x$  required for their existence.

### 10.1 Overview of the Vertical Hierarchy

The vertical hierarchy of continua is monotonic:

$$K_0 \rightarrow K_1 \rightarrow K_2 \rightarrow \cdots \rightarrow K_{10} \rightarrow K_{11} \rightarrow K_{12},$$

where each transition corresponds to the emergence of new axes, thresholds, flows and cycles that cannot be reduced to those of lower levels.

A global summary is shown in Table 2.

| Level    | Dominant Axes                 | Characteristic Structure                          |
|----------|-------------------------------|---------------------------------------------------|
| $K_0$    | Difference axis               | Structural substrate                              |
| $K_1$    | 1D geometric axis             | Classical continuum                               |
| $K_2$    | Physical field axes           | Phases, BKT, mass, coherence                      |
| $K_3$    | Chemical axes                 | RAF networks, catalysis                           |
| $K_4$    | Membrane axes                 | Osmotic, curvature, permeability thresholds       |
| $K_5$    | Excitability axes             | Proto-spikes, ion flows                           |
| $K_6$    | Cognitive axes                | Binding, internal models                          |
| $K_7$    | Normative/social axes         | Institutions, trust                               |
| $K_8$    | Civilizational axes           | Infrastructure, collective cycles                 |
| $K_9$    | Theoretical axes              | Paradigms, ontologies                             |
| $K_{10}$ | Meta-theoretical axes         | Model-of-models recursion                         |
| $K_{11}$ | Meta-evolutionary formal axes | Operators, meta-logics, evolving model-spaces     |
| $K_{12}$ | Global coherence axes         | Cross-framework semantic and structural coherence |

Table 2: Global overview of continua  $K_0$ – $K_{12}$ .

Each level requires an embedding space

$$M_0 \subset M_1 \subset \cdots \subset M_{10} \subset M_{11} \subset M_{12},$$

with axes  $A(M_x)$  sufficient to host the continua of that level.

## 10.2 Level $K_0$ : Structural Substrate

**Formal data.**

$$K_0 = (S, \Delta, \mathcal{C}),$$

with:

- $S$  a set of distinguishable states;
- $\Delta$  a structural difference function;
- $\mathcal{C}$  a structural relation (or family of relations) preserving distinguishability.

No time, geometry, flows or potentials exist at this level.

**Threshold.** Existence threshold:

$$\Theta_0 = \varepsilon > 0, \quad s_1 \neq s_2 \Rightarrow \Delta(s_1, s_2) \geq \varepsilon.$$

**Role.** Provides the logical substrate of distinguishability; no continuum can exist without this level.

## 10.3 Level $K_1$ : One-Dimensional Continua

**Formal data.**

$$K_1 = (\Omega_1, A_1, P_1, J_1, \Theta_1, \partial\Omega_1, C_1, k_1).$$

- $A_1$ : one geometric axis (1D continuum);
- $\Omega_1$ : classical configuration space (e.g.  $C^0(T, H^1(X)) \cap C^1(T, L^2(X))$ );
- $P_1$ : classical energy functional;
- $J_1$ : classical dynamical flow;
- $\Theta_1$ : stability and critical thresholds.

**Examples.** Classical fields in one spatial dimension, coupled oscillators, scalar diffusion models.

**Cycles.** Periodic orbits, stable limit cycles.

## 10.4 Level $K_2$ : Physical Continua

**Axes.** Multidimensional field axes:

- spatial axes;
- internal field axes (e.g. gauge degrees of freedom);
- order-parameter axes (coherence, magnetisation, etc.).

**Potentials and flows.** Physical energy functionals, field equations, renormalisation flows.

**Thresholds.**

- $\Theta_{\text{crit}}$ : phase boundaries;
- $\Theta_{\text{BKT}}$ : vortex-unbinding threshold;
- $\Theta_{\text{mass}}$ : coherence threshold for mass emergence;
- $\Theta_{\text{death}}$ : confinement or decoupling.

**Cycles.** Topological solitons, vortex pairs, field-theoretic cycles.

**10.5 Levels  $K_3$ – $K_4$ : Chemical and Protocellular Continua****10.5.1 Level  $K_3$ : Chemical Continua**

**Axes.** Chemical concentration axes, environmental axes (pH, salinity, temperature).

**Thresholds.**

- RAF existence threshold (closure);
- catalytic activation thresholds;
- percolation threshold for reaction networks.

**Cycles.** Chemical cycles, RAF-closure cycles.

**10.5.2 Level  $K_4$ : Protocellular Continua**

**Axes.** Membrane axes: curvature, surface tension, permeability, charge.

**Thresholds.**

- osmotic threshold  $\Theta_{\text{grad}}$ ;
- permeability threshold  $\Theta_{\text{perm}}$ ;
- membrane-rupture threshold  $\Theta_{\text{mem}}$ ;
- curvature threshold  $\Theta_{\text{curv}}$ .

**Cycles.** Metabolic cycles, membrane-growth cycles, gradient-maintenance cycles.

**Boundary geometry.** Detailed patch model: patch states  $\sigma_i \in \{L_\alpha, L_\beta, L_o, L_f, L_b\}$ , local threshold vectors  $\Theta_i$ , patch-level interactions and rupture dynamics.

**10.6 Level  $K_5$ : Early Neural and Bioelectrical Continua**

**Axes.** Excitability axes:

- membrane-potential axis  $\Delta V$ ;
- ion concentration axes;
- channel-state axes.

**Potentials and flows.**

- $P_5$ : electrochemical potentials;
- $J_5$ : ion fluxes, gating-variable flows.

**Thresholds.**

- excitation threshold  $\Theta_{\text{exc}}$ ;
- refractory threshold  $\Theta_{\text{refr}}$ ;
- gradient threshold  $\Theta_{\text{grad}}$ ;
- failure threshold  $\Theta_{\text{death}}$  (loss of excitability).

**Cycles.** Proto-spike cycle  $C_{\text{spike}}$ , early oscillatory cycles, patch-level excitation loops.

**10.7 Level  $K_6$ : Cognitive Continua****Axes.**

- representational axes;
- binding axes;
- conceptual axes;
- predictive axes.

**Potentials.**

- semantic-energy potentials;
- prediction-error potentials;
- memory-stability potentials.

**Thresholds.**

- binding threshold  $\Theta_{\text{bind}}$ ;
- prediction threshold  $\Theta_{\text{pred}}$ ;
- coherence threshold  $\Theta_{\text{coh}}$ ;
- expressive threshold  $\Theta_{\text{expr}}$ .

**Cycles.** Cognitive cycles:

- attention cycle;
- prediction cycle;
- memory-consolidation cycle;
- model-update cycle.

**10.8 Levels  $K_7$ – $K_8$ : Social and Civilizational Continua****10.8.1 Level  $K_7$ : Social Continua****Axes.**

- normative axes;
- trust axes;
- role axes;
- institutional axes.

**Thresholds.**

- trust threshold  $\Theta_{\text{trust}}$ ;
- legitimacy threshold  $\Theta_{\text{leg}}$ ;
- role-coherence threshold  $\Theta_{\text{role}}$ .

**Cycles.** Normative cycles, institutional cycles, coordination cycles.

**10.8.2 Level  $K_8$ : Civilizational Continua**

**Axes.** Large-scale axes:

- infrastructural axes;
- energy-flow axes;
- communication axes;
- complexity axes.

**Thresholds.**

- systemic-stress threshold;
- capacity threshold;
- cohesion threshold.

**Cycles.** Energy cycles, economic cycles, institutional mega-cycles.

**10.9 Levels  $K_9$ – $K_{12}$ : Theoretical, Meta-Theoretical, and Upper-Coherence Continua****10.9.1 Level  $K_9$ : Theoretical Continua**

**Axes.**

- conceptual axes;
- formal axes;
- modelling axes.

**Thresholds.**

- consistency threshold;
- coherence threshold;
- expressive threshold  $\Theta_{\text{expr}}$ .

**Cycles.** Theory-evolution cycles, paradigm cycles.

**10.9.2 Level  $K_{10}$ : Meta-Theoretical Continua**

**Axes.**

- meta-linguistic axes;
- functorial axes;
- model-of-models axes.



**Thresholds.**

- meta-coherence threshold;
- self-reference threshold;
- meta-expressivity threshold.

**Cycles.** Functorial cycles, meta-theoretical recursion cycles.

**10.9.3 Level  $K_{11}$ : Recursive Meta-Theoretical Continua****Axes.**

- operator-family axes;
- meta-logical axes;
- model-space transformation axes;
- recursive structural-reflection axes.

**Thresholds.**

- operator-coherence threshold;
- recursive-closure threshold;
- meta-space compatibility threshold.

**Role.**  $K_{11}$  is required once theories, operator suites, and model-spaces are themselves treated as evolving structural objects rather than as static containers. It is therefore the first level at which the framework studies meta-evolution explicitly.

**Cycles.** Operator-revision cycles, meta-logic update cycles, model-space recursion cycles.

**10.9.4 Level  $K_{12}$ : Global Coherence Continua****Axes.**

- semantic-coherence axes;
- global compatibility axes;
- cross-framework binding axes.

**Thresholds.**

- global-consistency threshold;
- semantic-coherence threshold;
- upper-bound admissibility threshold.

**Role.**  $K_{12}$  is the minimally specified upper layer needed when one must discuss global coherence constraints across families of  $K_{11}$ -structures. Its role is structural and bounding: it records that the hierarchy has an explicit upper coherence problem, not merely more of the same lower-level dimensional growth.

**Cycles.** Global coherence-restoration cycles, semantic alignment cycles, cross-framework closure cycles.

## 10.10 Cross-Level Summary

A continuum  $K_x$  exists as a live continuum if and only if

$$\Omega_x \neq \emptyset, \quad A_x \subseteq A(M_x), \quad \Theta_x \text{ is satisfiable within } M_x.$$

Vertical continuity conditions:

- $A_x \subset A_{x+1}$ ;
- $M_x \subset M_{x+1}$ ;
- $\Theta_{x+1}$  refine  $\Theta_x$ ;
- flows  $J_x$  embed into  $J_{x+1}$ ;
- cycles  $C_x$  form substructures of  $C_{x+1}$ .

This completes the reconstructed full hierarchy of continua from  $K_0$  through  $K_{12}$ .

## 11 Evolution and Interaction Operators

This chapter expands the abstract evolution operator introduced in Section 4 into the full operator family used in the Ontology of Continua (OC). It consolidates the definitions developed in Core 2.x and presents them in a compact, vertically consistent form suitable for Core 1.3.

Throughout this section a continuum is written as

$$K(t) = (\Omega(t), A(t), P(t), J(t), \Theta(t), \partial\Omega(t), C(t), k(t)),$$

with the components defined in Section 4. The embedding space is denoted by  $M$ , with axes  $A(M)$  and constraints that restrict admissible states of  $K$ .

### 11.1 Decomposition of the Evolution Operator

The structural evolution of a continuum over an infinitesimal time step is described by an operator

$$E : K(t) \longrightarrow K(t + dt).$$

Rather than specifying a single monolithic map, OC factorises  $E$  into a family of operators acting on different components of the continuum:

$$E = (F, G, H, Q, R, S, U),$$

where

- $F$  governs the dynamics of flows  $J$ ;
- $G$  governs the dynamics of potentials  $P$ ;
- $H$  governs the dynamics of thresholds  $\Theta$ ;
- $Q$  governs the dynamics of cycles  $C$ ;
- $R$  governs the dynamics of the boundary  $\partial\Omega$ ;
- $S$  governs structural reconfiguration of axes  $A$  and internal connectivity;
- $U$  governs the evolution of continuumness  $k(t)$ .

At the formal level this can be written as

$$\begin{aligned}
J(t + dt) &= F(J(t), P(t), A(t), \Theta(t), M), \\
P(t + dt) &= G(P(t), A(t), J(t), \Theta(t), M), \\
\Theta(t + dt) &= H(\Theta(t), P(t), J(t), M), \\
C(t + dt) &= Q(C(t), P(t), J(t), \Theta(t)), \\
\partial\Omega(t + dt) &= R(\partial\Omega(t), P(t), J(t), \Theta(t), M), \\
A(t + dt) &= S(A(t), P(t), J(t), \Theta(t), M), \\
k(t + dt) &= U(k(t), \Omega(t), A(t), P(t), J(t), \Theta(t), C(t), \partial\Omega(t), M).
\end{aligned}$$

The operators are constrained by the axioms of OC, in particular the monotonicity of dimension, the impossibility of evolution outside of the embedding space, and the definition of birth and death.

## 11.2 Operator $F$ : Flow Dynamics

The flow operator  $F$  governs the temporal change of flows:

$$J(t + dt) = F(J(t), P(t), A(t), \Theta(t), M).$$

Conceptually,  $F$  encodes:

- conservation or balance laws (e.g. continuity equations, Kirchhoff sums, conservation of probability);
- constitutive relations linking flows to gradients of potentials (e.g. Fick, Ohm, Fourier laws, reaction kinetics);
- structural classification of flows into supporting, critical and destructive components.

A generic decomposition is

$$J(t) = J_{\text{support}}(t) + J_{\text{critical}}(t) + J_{\text{kill}}(t),$$

with  $F$  preserving this structure:

$$\begin{aligned}
J_{\text{support}}(t + dt) &= F_{\text{support}}(J, P, A, \Theta, M), \\
J_{\text{critical}}(t + dt) &= F_{\text{critical}}(J, P, A, \Theta, M), \\
J_{\text{kill}}(t + dt) &= F_{\text{kill}}(J, P, A, \Theta, M).
\end{aligned}$$

At different levels  $K_x$  these classes have different realisations:

- in  $K_2$  (physical continua), flows include diffusive, convective and field-mediated currents;
- in  $K_3$  (chemical continua), flows correspond to reaction and transport rates;
- in  $K_4$ – $K_5$  (protocellular and bioelectrical continua), flows include ion currents, osmotic fluxes and metabolic fluxes;
- in  $K_6$ – $K_8$  (cognitive and social continua), flows become flows of information, attention, resources and commitments.

## 11.3 Operator $G$ : Potential Dynamics

The potential operator  $G$  controls how potentials change:

$$P(t + dt) = G(P(t), A(t), J(t), \Theta(t), M).$$

In many concrete models  $G$  is expressed as a differential relation

$$\frac{dP}{dt} = J,$$

possibly with additional nonlinear terms for dissipation, driving and coupling. At the structural level OC requires only that:

- potentials respond to flows and may in turn constrain them;
- changes in potentials respect the sign and inequality structure imposed by thresholds  $\Theta$ ;
- in the absence of flows  $J = 0$ , potentials tend to approach threshold surfaces or relax towards internally preferred configurations.

Different classes of potentials (energetic, chemical, electrochemical, informational, normative) are instances of a single structural role: encoding constraints and driving forces on  $\Omega(K)$ .

#### 11.4 Operator $H$ : Threshold Dynamics

Thresholds are not static; they can adapt, drift or reorganise under the influence of potentials, flows and embedding constraints. The threshold operator  $H$  is defined by

$$\Theta(t + dt) = H(\Theta(t), P(t), J(t), M).$$

Examples include:

- adaptation of ion channel activation curves in neural tissue;
- plasticity of viability ranges for protocells in changing environments;
- shifting norms and institutional thresholds in social systems;
- changes in coherence or consistency thresholds in theoretical systems.

Formally,  $H$  must satisfy:

- compatibility with embedding constraints: threshold changes cannot require configurations forbidden by  $M$ ;
- preservation of the taxonomy of thresholds ( $\Theta_{\text{exist}}$ ,  $\Theta_{\text{stab}}$ ,  $\Theta_{\text{crit}}$ ,  $\Theta_{\text{dim}}$ ,  $\Theta_{\text{death}}$ ,  $\Theta_{\text{expr}}$ ,  $\Theta_{\text{embed}}$ );
- local continuity: small changes in potentials or flows may induce small changes in thresholds, except at critical points where bifurcations are allowed.

#### 11.5 Operator $Q$ : Cycle Dynamics

Cycles  $C(t)$  represent closed trajectories that maintain the organisation of the continuum. Their evolution is governed by

$$C(t + dt) = Q(C(t), P(t), J(t), \Theta(t)).$$

Structurally,  $Q$  accounts for:

- *birth* of new cycles when flows close in state space away from  $\partial\Omega$ ;
- *stabilisation* or *strengthening* of existing cycles when supporting flows dominate;
- *weakening* and *destruction* of cycles under destructive flows or threshold violations.

Cycle complexes are particularly important. Let  $C_{\text{max}}(t)$  denote the maximal set of mutually compatible cycles supporting the continuum. Then Theorem 9 in Section 5 states that death coincides with  $C_{\text{max}}(t) = \emptyset$ . The operator  $Q$  therefore directly participates in life–death transitions.

#### 11.6 Operator $R$ : Boundary Evolution

The boundary operator  $R$  controls the geometry and position of the boundary  $\partial\Omega$ :

$$\partial\Omega(t + dt) = R(\partial\Omega(t), P(t), J(t), \Theta(t), M).$$

At a structural level  $R$  must:

- be consistent with threshold definitions:  $\partial\Omega = \{s \mid \exists k : f_k(s) = 0\}$ ;
- allow for expansion, contraction and bifurcation of  $\Omega$ ;
- encode the effects of flows and threshold shifts on accessible regions.

In concrete implementations  $R$  can take many forms:

- in  $K_2$  it may correspond to movement of phase boundaries, percolation thresholds or coherence surfaces;
- in  $K_3$ – $K_4$  it includes the dynamics of membranes, interfaces and compartment boundaries;
- in  $K_5$  it corresponds to thresholds of excitability and refractory regions in conductance space;
- in  $K_7$ – $K_8$  it can represent institutional and legal boundaries.

Section 8 provides an extended treatment of boundary geometry and patch models.

## 11.7 Operator $S$ : Structural Reconfiguration

The structural operator  $S$  governs changes in axes and internal configuration:

$$A(t + dt) = S(A(t), P(t), J(t), \Theta(t), M).$$

Typical roles of  $S$  include:

- activation or deactivation of axes (e.g. turning on latent degrees of freedom);
- re-wiring of connectivity patterns within the continuum;
- coarse-graining or refinement of effective axes under renormalisation or abstraction.

Crucially,  $S$  is subject to the monotonicity of dimension:

$$\dim A(t + dt) \geq \dim A(t).$$

If a new axis  $A_{\text{new}}$  is added such that  $A_{\text{new}} \notin \text{span}_{\text{str}}(A(t))$ , then a dimensional transition occurs and the continuum becomes  $K_{x+1}$  rather than remaining at the same level. This is formalised through the birth operators in Section 11.9 below.

## 11.8 Operator $U$ : Continuumness Dynamics

The operator  $U$  governs the evolution of continuumness  $k(t)$ :

$$k(t + dt) = U(k(t), \Omega(t), A(t), P(t), J(t), \Theta(t), C(t), \partial\Omega(t), M).$$

Using the unified definition of  $k(t)$  from Section 4,  $U$  evaluates how changes in state space, axes, flows, thresholds, cycles and boundary affect the viability of the continuum. Qualitatively:

- supporting flows and stable cycles increase or maintain  $k(t)$ ;
- destructive flows and loss of cycles decrease  $k(t)$ ;
- approach to  $\partial\Omega$  reduces  $k(t)$ ;
- expansion of  $\Omega$  and enrichment of cycle complexes increase  $k(t)$ .

Death corresponds to  $k(t) \rightarrow 0$  together with  $\Omega = \emptyset$ . Birth corresponds to the appearance of a new continuum with  $k(t) > 0$  in a previously empty region of the space of possibilities.

## 11.9 Birth Operators $\Psi_{x \rightarrow x+1}$

Dimensional transitions between levels are mediated by birth operators

$$\Psi_{x \rightarrow x+1} : (K_x, M_x) \longrightarrow (K_{x+1}, M_{x+1}).$$

Structurally,  $\Psi_{x \rightarrow x+1}$  is defined whenever the following conditions are satisfied:

1. **New differences.** There exists a class of differences outside the current structural span:

$$D_{\text{new}} \not\subseteq \text{span}_{\text{str}}(A(K_x)).$$

2. **Tension above dimensional threshold.** The structural tension exceeds the dimensional threshold:

$$T(K_x, t) > \Theta_{\text{dim}}(K_x).$$

3. **Available axis in embedding space.** The embedding space  $M_x$  contains at least one axis suitable for hosting the new differences:

$$A_{\text{new}} \in A(M_x) \setminus A(K_x).$$

4. **Nonempty admissible region.** There exists a nonempty set of states  $\Omega(K_{x+1})$  compatible with the new axis and thresholds.

The action of  $\Psi_{x \rightarrow x+1}$  can then be summarised as

$$\begin{aligned} A(K_{x+1}) &= A(K_x) \cup \{A_{\text{new}}\}, \\ \Omega(K_{x+1}) &\subseteq \Omega(M_{x+1}) \cap \{s \mid \Theta(K_{x+1})(s) \leq 0\}, \\ M_{x+1} &\supset M_x, \end{aligned}$$

with all other components of  $K_{x+1}$  (flows, potentials, cycles, continuumness) determined by the general structural machinery.

Birth operators are *minimal* and *irreversible*: any nonzero utilisation of  $A_{\text{new}}$  constitutes the new continuum  $K_{x+1}$ ; returning to the previous dimensionality would require destroying  $\Omega(K_{x+1})$ , which is interpreted as death rather than reversible simplification.

## 11.10 Interaction Operator $E_{\text{int}}$

When two continua  $K_a$  and  $K_b$  interact, their joint dynamics are governed by an interaction operator

$$E_{\text{int}} : (K_a, K_b, M) \longrightarrow (K'_a, K'_b, M'),$$

where  $M'$  is the updated embedding space for the combined system.

At a structural level  $E_{\text{int}}$  modifies potentials, flows, thresholds, cycles and possibly axes of each continuum. Several regimes are distinguished:

- **Competition.** Flows of one continuum hinder the maintenance of cycles in the other. Typically  $k_a$  and  $k_b$  cannot both increase indefinitely.
- **Parasitism.** One continuum harvests supporting flows from another, increasing its own continuumness at the expense of its host.
- **Symbiosis.** Coupled flows increase continuumness for both continua; shared cycles may emerge that stabilise the pair.
- **Fusion.** Axes and potentials combine into a new continuum  $K_{\text{fusion}}$  with merged state space and thresholds. Formally, fusion corresponds to the birth of a higher-dimensional continuum via a composite birth operator.

Interaction can itself trigger dimensional transitions when mixed differences create new axes not available to the continua in isolation and when tension exceeds relevant thresholds.

## 11.11 Operator Algebra and Constraints

Although OC does not postulate a full algebra of operators, several structural constraints are required to maintain internal consistency:

**Compatibility with embedding spaces.** All operators must respect the inclusion  $A(K(t)) \subseteq A(M(t))$  and cannot generate evolution along axes not present in  $M$ . Birth operators are defined only when  $M$  already contains the relevant axes.

**Monotonicity of dimension.** The structural operator  $S$  and birth operators  $\Psi_{x \rightarrow x+1}$  must satisfy

$$\dim A(t + dt) \geq \dim A(t),$$

with strict inequality only at dimensional transitions.

**Threshold-respecting dynamics.** Operators  $F, G, H, Q, R$  must preserve the inequality structure imposed by thresholds except at authorised crossings of critical or death thresholds. Violations of  $\Theta_{\text{exist}}$  are interpreted as death events.

**Continuumness as viability indicator.** The operator  $U$  couples all other components; in particular:

- if  $k(t) = 0$  and  $\Omega = \emptyset$ , the continuum is dead and subsequent applications of  $E$  have no effect within that level;
- if  $k(t) > 0$ , then at least one supporting cycle exists and flows must satisfy the balance conditions encoded in  $F$  and  $Q$ .

**Cross-level consistency.** For transitions  $K_x \rightarrow K_{x+1}$ , the operator family for the higher level must reduce to that of the lower level when the new axis is held constant and the additional degrees of freedom are frozen. This ensures that lower levels are recoverable as special cases of higher ones, maintaining vertical coherence of the hierarchy.

## 11.12 Summary

The operator family  $(F, G, H, Q, R, S, U, \Psi, E_{\text{int}})$  provides the dynamical backbone of the Ontology of Continua. Instead of a single highly specific equation of motion, OC employs a modular set of operators that act on flows, potentials, thresholds, cycles, boundaries, axes and continuumness. Their combined action governs birth, life, interaction and death of continua across all levels  $K_0$ – $K_{12}$ .

This chapter completes the formal core of Core 1.3 by making explicit the structural dynamics that were only implicit in earlier drafts and by preparing the ground for fully quantitative models in subsequent extension papers.

## 12 Collapse and Rebirth of Continua

This section develops the structural theory of collapse and rebirth in the Ontology of Continua (OC). It refines the general death condition  $\Omega(K) = \emptyset$  and  $k(K, t) \rightarrow 0$  from Section 5 into explicit criteria, dynamical signatures and restricted scenarios for the birth of new continua after collapse.

No new axioms are introduced; the discussion assembles consequences of the general framework of Section 4, the structural results of Section 5, the extended treatment of boundaries in Section 8, the threshold landscape in Section 9, the full hierarchy of levels in Section 10, and the operator family in Section 11.

## 12.1 Formal Criteria for Collapse

Let

$$K(t) = (\Omega(t), A(t), P(t), J(t), \Theta(t), \partial\Omega(t), C(t), k(t))$$

be a continuum embedded in a space  $M$ . The structural notion of collapse must distinguish transient excursions toward the boundary from genuine termination of the continuum.

**Definition 12.1 (Collapse).** A continuum  $K$  *collapses* at time  $t^*$  if the following conditions hold:

1. the admissible state space becomes empty:

$$\Omega(K(t^*)) = \emptyset;$$

2. continuumness decays to zero:

$$\lim_{t \nearrow t^*} k(K, t) = 0;$$

3. at least one existence or stability threshold is violated for all candidate configurations:

$$\forall s \in \Omega(M) : \quad \exists \theta \in \Theta_{\text{exist}} \cup \Theta_{\text{stab}} \text{ such that } \theta(s) > 0.$$

These conditions match the equivalent characterisations of death stated in Theorem 3 and Theorem 9 of Section 5. In particular, the vanishing of  $\Omega$  and  $k$  coincide with the disappearance of the maximal structurally stable cycle complex  $C_{\text{max}}(K)$ .

**Proposition 12.2 (Boundary destruction).** At collapse time  $t^*$ , the boundary  $\partial\Omega$  ceases to be a well-defined separating surface between admissible and inadmissible states. Formally, either

$$\partial\Omega(K(t^*)) = \emptyset \quad \text{or} \quad \partial\Omega(K(t^*)) = \Omega(M), \quad (12.1)$$

depending on whether the embedding constraints become trivially strict or trivially loose. In both cases the structural role of  $\partial\Omega$  as a viability boundary is lost.

The operator view is:

- the boundary operator  $R$  becomes undefined once  $\Omega(K) = \emptyset$ ;
- the continuumness operator  $U$  returns  $k = 0$  and remains constant thereafter for that continuum.

## 12.2 Internal vs External Collapse

Collapse can arise from internal dynamics of the continuum or from external changes in its embedding space. The distinction is structural rather than causal: it depends on whether the operators acting on  $K$  or those acting on  $M$  are responsible for the violation of thresholds.

**Definition 12.3 (Internal collapse).** *Internal collapse* occurs when collapse is caused solely by the internal evolution operator  $E = (F, G, H, Q, R, S, U)$  acting on  $K$ , while the embedding space  $M$  remains structurally static over the relevant time interval. Typical signatures include:

- destructive flows  $J_{\text{kill}}$  dominating supporting flows;
- internal thresholds  $\Theta_{\text{stab}}$  or  $\Theta_{\text{crit}}$  crossed due to internal accumulation of tension;
- cycle breakdown driven by internal imbalances.



**Definition 12.4 (External collapse).** *External collapse* occurs when the embedding space  $M$  changes in such a way that no configuration of  $K$  remains compatible with the new constraints. Formally, there exists a time interval  $[t_0, t^*]$  such that:

$$\Omega(K(t_0)) \neq \emptyset, \quad E \text{ remains within the previous admissible region,}$$

but a change in  $M$  induced by its own evolution operator  $E_M$  yields

$$\Omega(K(t^*)) = \{s \in \Omega(M(t^*)) \mid \Theta(K(t_0))(s) \leq 0\} = \emptyset.$$

Corollary 3.1 in Section 5 can be restated as: external collapse is equivalent to the loss of common solutions of the constraints of  $K$  and  $M$ .

**Mixed collapse.** In realistic systems both mechanisms co-operate. For example, a protocell may internally deplete resources while an external change in osmotic conditions simultaneously tightens viability ranges. OC does not attempt to sharp-split these cases; the internal/external distinction is primarily a tool for analysis.

### 12.3 Collapse Dynamics

Collapse is not generally an instantaneous event at the level of observables. The structural framework predicts a characteristic approach to collapse governed by thresholds, tension and boundary geometry.

**Critical slowing down.** Near stability or critical thresholds  $\Theta_{\text{stab}}$  and  $\Theta_{\text{crit}}$ , small perturbations relax increasingly slowly. In the operator language this corresponds to:

- eigenvalues of the linearised flow operator  $F$  approaching zero real part;
- time scales of cycle restoration encoded in  $Q$  diverging;
- boundary motion under  $R$  becoming sluggish while remaining directed toward contraction of  $\Omega$ .

Critical slowing appears in physical phase transitions, ecological collapses, financial crises, protocell failure and cognitive overload.

**Divergence of structural tension.** The structural tension  $T(K, t)$  approaches the death threshold  $\Theta_{\text{death}}(K)$  as collapse nears:

$$\lim_{t \nearrow t^*} T(K, t) = \Theta_{\text{death}}(K).$$

Below this threshold the system may still maintain cycles, albeit with increasingly narrow margins. At the threshold, any further perturbation forces trajectories to cross  $\partial\Omega$  or to leave the embedding region.

**Patch failure and boundary-driven collapse.** The patch model of boundaries (Section 8) provides a more granular picture. Writing the boundary as a collection of patches  $\{P_i\}$  with local states  $\sigma_i$  and threshold vectors  $\Theta_i$ , collapse can proceed via:

- local loss of integrity (e.g. membrane rupture at individual patches, institutional failure in specific domains);
- propagation of failure through coupling of patches, leading to global breach of the boundary and uncontrolled exchange with the embedding space;
- eventual destruction of all patches supporting a nonempty  $\Omega(K)$ .

From the operator perspective this is a coupled instability of  $R$ ,  $H$  and  $Q$ .

**Cycle breakdown.** As collapse approaches, the cycle operator  $Q$  eliminates more and more cycles from the structurally stable complex. The sequence

$$C_{\max}(t_0) \supset C_{\max}(t_1) \supset \cdots \supset C_{\max}(t_n)$$

shrinks until it becomes empty at  $t^*$ . Operationally:

- amplitudes of key cycles decay;
- cycle periods lengthen or become irregular;
- cycles increasingly graze  $\partial\Omega$  and are destroyed by threshold crossings.

## 12.4 Post-Collapse Residue

Even when a continuum collapses, its traces may persist in the embedding space. OC distinguishes sharply between the *continuum* and its *residue*.

**Definition 12.5 (Residue).** The *residue* of a collapsed continuum  $K$  is the set of structures in the embedding space that remain after  $\Omega(K) = \emptyset$ , for example:

- material residues (e.g. reaction products, broken infrastructure);
- topological residues (e.g. defects, scars in field configurations);
- informational residues (e.g. documents, data traces);
- normative residues (e.g. partially preserved legal frameworks).

Residues are no longer organised by the original tuple  $(\Omega, A, P, J, \Theta, \partial\Omega, C, k)$ . They may, however, form the substrate for new continua.

**Limits of recovery.** Because death is structurally irreversible (Theorem 3 and Corollary 3.2 in Section 5), no operator acting within the same level can reconstruct the original continuum from its residue. Two cases are distinguished:

- *Reconstruction.* An external agent (e.g. experimenter, engineer, institution) may construct a new continuum  $K'$  that resembles  $K$  in some variables; structurally this is a new birth event, not recovery.
- *Spontaneous reorganisation.* Under appropriate conditions residues may reorganise into a new continuum (see below). Again this is a new entity with its own identity, even if many components are inherited.

## 12.5 Rebirth Mechanisms

Rebirth is the emergence of a new continuum after collapse of a previous one. OC treats rebirth as a special case of birth, governed by the same operators  $\Psi_{x \rightarrow x+1}$  but with initial conditions shaped by residues.

**Definition 12.6 (Rebirth).** A *rebirth event* occurs when, after the collapse of  $K$ , a new continuum  $K'$  appears in the same or a related embedding space such that:

1. there exists a time interval with  $\Omega(K) = \emptyset$  and  $k(K, t) = 0$  while  $\Omega(K') \neq \emptyset$ ;
2. at least one structural element of  $K'$  (axes, potentials, boundary components, residues in  $\Omega(M)$ ) is inherited from the residue of  $K$ ;
3.  $K'$  satisfies the birth conditions for some level  $K_y$ , possibly equal to the original level of  $K$  or to a different one.

Rebirth is thus structurally rare and constrained. The following regimes are typical.

**Dimensional rebirth.** After collapse of a continuum  $K_x$ , residues may support the birth of a new continuum at a different level:

- *upward rebirth* ( $K_x \rightarrow K_{x+1}$ ): collapse of an overstrained structure at level  $K_x$  creates conditions for a higher-dimensional continuum (e.g. collapse of a simple institutional framework enabling emergence of a more complex governance structure).
- *sideways rebirth* ( $K_x \rightarrow K'_x$ ): residues support a new continuum at the same level but with different axes or thresholds (e.g. reorganisation of a failed protocell into a new protocell with altered composition).

Downward rebirth ( $K_x \rightarrow K_{x-1}$ ) is not considered rebirth of the same continuum: lower-level continua may persist as independent entities, but the collapsed higher-level continuum is not resurrected.

**Formal conditions.** Let  $K$  collapse at time  $t^*$  and  $R_{\text{res}}$  be its residue in  $M$ . A rebirth event is permitted when:

1. the residue defines a nonempty candidate state set  $\Omega_{\text{res}} \subseteq \Omega(M)$ ;
2. there exist axes  $A_{\text{res}} \subseteq A(M)$  and thresholds  $\Theta_{\text{res}}$  such that
$$\Omega(K') = \{s \in \Omega_{\text{res}} \mid \Theta_{\text{res}}(s) \leq 0\} \neq \emptyset;$$
3. structural tension based on  $(\Omega_{\text{res}}, A_{\text{res}}, \Theta_{\text{res}})$  exceeds the relevant dimensional threshold (if a level increase is involved) and an appropriate birth operator  $\Psi$  is defined.

**Examples across  $K_3$ – $K_5$ .** In the chemical and protocellular regime:

- collapse of a protocell (membrane rupture, gradient loss) leaves residues of lipids, nucleotides and catalysts in the environment;
- these residues may recombine into new vesicles and networks under favourable conditions;
- the new protocells constitute new continua  $K'_4$  with their own thresholds and cycles, even if they inherit components from the collapsed ancestors.

In early bioelectrical systems:

- collapse of excitability in a primitive network (e.g. due to channel failure) may leave membrane fragments and channels that form new excitable patches;
- regenerated excitable domains define new  $K'_5$  continua.

**Rebirth at higher levels.** At cognitive, social and civilizational levels:

- cognitive collapse (loss of coherent internal models) may be followed by reconstruction of new models using surviving representations as seeds; structurally this is the birth of a new cognitive continuum  $K'_6$ ;
- institutional collapse (e.g. failure of a governance regime) leaves legal, organisational and infrastructural residues that can be reused in the creation of new institutions  $K'_7$ ;
- civilizational collapse leaves material infrastructure, cultural artefacts and population distributions that may support emergent civilisations  $K'_8$ .

In all cases OC insists that the new continuum has its own identity; the structural irreversibility of death is not violated.

## 12.6 Collapse in Higher-Level Continua

Higher levels  $K_6$ – $K_{10}$  expose collapse phenomena that are less obviously physical but structurally analogous.

**Cognitive collapse ( $K_6$ ).** For cognitive continua the key elements are:

- axes representing concepts, features and model parameters;
- potentials measuring prediction error, value and confidence;
- cycles representing attention loops, prediction–correction cycles and learning cycles.

Cognitive collapse occurs when:

- prediction error and internal tension exceed expressive capacity thresholds ( $\Theta_{\text{expr}}$ );
- no coherent internal model can be maintained within the given axes and thresholds;
- all structurally stable cognitive cycles vanish, leading to  $C_{\text{max}}(K_6) = \emptyset$  and  $k(K_6, t) \rightarrow 0$ .

Examples include total breakdown of a conceptual scheme, severe overload or pathological states where the system cannot maintain any stable model of its environment.

**Institutional collapse ( $K_7$ ).** For social continua the relevant components are:

- social graphs, role structures and institutional arrangements in  $\Omega(K_7)$ ;
- trust, legitimacy and resource thresholds in  $\Theta(K_7)$ ;
- cycles representing institutional routines and governance loops.

Institutional collapse occurs when:

- trust and legitimacy fall below existence thresholds;
- key cycles (e.g. tax collection, legal enforcement, decision loops) break;
- social graphs fragment beyond what is compatible with institutional operation.

The residue may include legal texts, physical buildings and fragmented networks, from which new institutions  $K'_7$  may later be assembled.

**Civilizational collapse ( $K_8$ ).** At the civilizational level:

- axes represent infrastructures, sectors, regions and large-scale organisational forms;
- potentials include energy availability, resource stocks and risk gradients;
- cycles include economic cycles, infrastructure renewal, knowledge transmission.

Civilizational collapse corresponds to breakdown of these cycles across multiple sectors, crossing of multiple thresholds in the extended threshold landscape and global contraction of  $\Omega(K_8)$  to the empty set. The residue consists of infrastructure, population distributions and cultural artefacts that can seed future civilisations.

**Theoretical and meta-theoretical collapse ( $K_9$ – $K_{10}$ ).** Theories and meta-theories can also collapse structurally:

- internal inconsistency or empirical failure may violate  $\Theta_{\text{exist}}$  or  $\Theta_{\text{stab}}$  at level  $K_9$ ;
- meta-theoretical frameworks at  $K_{10}$  may become incoherent or unable to accommodate growing bodies of models, violating expressive thresholds;
- in both cases cycle complexes representing research programmes, paradigm cycles and meta-theoretical update loops can disappear, yielding  $C_{\text{max}} = \emptyset$  and effective death of the framework.

Subsequent emergence of new theories or meta-frameworks built on surviving results is structurally treated as rebirth.

## 12.7 Summary

Collapse and rebirth in OC are not metaphorical but structurally defined phenomena. Collapse corresponds to the emptying of the admissible state space, disappearance of stable cycles, loss of a meaningful boundary and vanishing continuumness. It can be driven by internal dynamics, external embedding changes or their interaction, and it exhibits characteristic signatures such as critical slowing, divergence of structural tension, patch failure and cycle breakdown.

Rebirth is strongly constrained: a new continuum may emerge from residues only when embedding spaces and threshold landscapes once again support a nonempty admissible region and when the birth conditions are satisfied. The new continuum always has its own structural identity, even if it inherits components from the collapsed predecessor.

These principles apply uniformly from protocells and early bioelectrical systems through cognitive and social structures up to theories and meta-theoretical frameworks. Collapse and rebirth are thus not special cases but generic regimes of the same structural machinery that governs the birth, evolution and death of continua across the hierarchy  $K_0$ – $K_{10}$ .

## 13 Ontological Branching and Global Topology

This section reconstructs the theory of ontological branches and the global topology of continua. It explicates Axiom 21 (Ontological Branching) and integrates it with the dimensional hierarchy  $K_0$ – $K_{10}$ , the birth operators  $\Psi_{x \rightarrow x+1}$  (Section 11) and the collapse–rebirth structure (Section 12).

Informally, a *branch* is a maximal development path along which a continuum undergoes successive dimensional births consistent with a given set of axes and embedding spaces. The collection of such branches carries a natural directed topology, providing a global view of all admissible ontogenic trajectories compatible with the OC axioms.

### 13.1 Definition of Ontological Branches

A continuum level  $K_x$  is specified by

$$K_x = (\Omega_x, A_x, P_x(t), J_x(t), \Theta_x, \partial\Omega_x, C_x, k_x(t)),$$

embedded in a space  $M_x$  with axis set  $A(M_x)$ . Dimensional birth from  $K_x$  to  $K_{x+1}$  is mediated by a birth operator  $\Psi_{x \rightarrow x+1}$  (Section 11).

**Definition 13.1 (Ontological branch).** An *ontological branch*  $B$  is a finite or countable sequence of continua

$$B = (K_{x_0}, K_{x_1}, K_{x_2}, \dots),$$

together with embedding spaces  $(M_{x_0}, M_{x_1}, \dots)$ , such that:

1. **Monotone levels.**  $x_0 < x_1 < x_2 < \dots$  and each step  $K_{x_i} \rightarrow K_{x_{i+1}}$  is a dimensional transition (no purely parametric evolution);
2. **Birth compatibility.** For each  $i$  there exists a birth operator  $\Psi_{x_i \rightarrow x_{i+1}}$  such that

$$(K_{x_{i+1}}, M_{x_{i+1}}) = \Psi_{x_i \rightarrow x_{i+1}}(K_{x_i}, M_{x_i}),$$

with  $\Psi_{x_i \rightarrow x_{i+1}}$  satisfying the general birth conditions of Section 11;

3. **Axis monotonicity.** The axis sets grow monotonically:

$$A_{x_i} \subset A_{x_{i+1}} \subseteq A(M_{x_{i+1}}),$$

and each new axis is not representable in the structural span of previous axes;

4. **Embedding monotonicity.** Embedding spaces form an inclusion chain

$$M_{x_0} \subset M_{x_1} \subset M_{x_2} \subset \dots$$

with  $A(M_{x_i}) \subseteq A(M_{x_{i+1}})$ ;

5. **Maximality.** The branch is maximal with respect to extension in the direction of increasing level: there is no continuum  $K_{x^*}$  with  $x^* > \sup_i x_i$  such that the sequence can be strictly extended while preserving the above conditions.

When  $x_0 = 0$ , the branch is *rooted* at the structural substrate  $K_0$ . More generally, branches may be defined relative to any starting level  $K_{x_0}$  that acts as an effective root for a restricted domain (e.g. chemistry, cognition, social systems).

**Remark 13.2 (Branch identity).** Two branches  $B$  and  $B'$  are considered distinct if they differ in at least one of:

- the sequence of levels  $x_i$  (e.g. skipping or inserting levels);
- the axis sets  $A_{x_i}$  (different emergent degrees of freedom);
- the threshold families  $\Theta_{x_i}$  or embedding spaces  $M_{x_i}$ .

Branches thus encode *developmental histories* in the space of continua, not just static collections of levels.

## 13.2 Branch Topology

The set of all branches carries a natural directed structure.

**Definition 13.3 (Branch graph).** Let  $\mathcal{K}$  be the class of all continua admissible under the OC axioms and let  $\mathcal{B}$  be the set of all branches. The *branch graph*  $\mathcal{G}$  is the directed graph whose:

- vertices are continua  $K_x \in \mathcal{K}$ ;
- directed edges  $K_x \rightarrow K_y$  correspond to birth operators  $\Psi_{x \rightarrow y}$  with  $y > x$ .

A branch  $B$  is then a directed path in  $\mathcal{G}$  that is maximal with respect to extension.

Because birth operators cannot reduce dimensionality (monotonicity of dimension),  $\mathcal{G}$  is a directed acyclic graph (DAG). Cycles may exist in the *state dynamics* of a given continuum  $K_x$  (through internal cycles  $C_x$ ), but not in the vertical birth graph.

**Global topology.** The global topological structure of  $\mathcal{G}$  is constrained by:

- **Shared prefixes.** Different branches share common initial segments whenever they coincide on lower levels (e.g. all physical, chemical and biological branches share  $K_0$ – $K_2$ );
- **Branch divergence.** Branches diverge when birth operators at some level  $K_x$  select different new axes or threshold structures, leading to distinct continua  $K_{x+1}^{(a)}, K_{x+1}^{(b)}$ , etc.;
- **Possible fusion.** Through interaction operators  $E_{\text{int}}$ , different branches may later generate continua that lie on a common extension, producing DAG-like rather than purely tree-like topology;
- **Hypergraph structure.** When birth operators take multiple continua as input (e.g. fusion events), the branch graph can be extended to a directed hypergraph with hyperedges  $(K_x^{(a)}, K_x^{(b)}, \dots) \rightarrow K_y$ .

The global topology of branches thus interpolates between tree-like structures (pure fission, no fusion) and hypergraph-like structures (branch fusion and collective birth). OC does not fix a single global shape but constrains all admissible shapes by dimensional monotonicity, embedding compatibility and threshold conditions.

### 13.3 Branching Conditions

Branching does not occur at arbitrary points; it is structurally triggered by specific threshold and embedding conditions.

**Axiom 21 (Ontological Branching).** Let  $K_x$  be a continuum at level  $x$  with embedding space  $M_x$ . A branching event at  $K_x$  is permitted when there exist two or more distinct birth operators

$$\Psi_{x \rightarrow x+1}^{(a)}, \Psi_{x \rightarrow x+1}^{(b)}, \dots$$

such that:

1. **Distinct new axes.** Each birth operator selects a structurally distinct new axis  $A_{\text{new}}^{(a)}, A_{\text{new}}^{(b)}, \dots$ , not mutually representable in the structural span of one another, i.e.

$$A_{\text{new}}^{(a)} \notin \text{span}_{\text{str}}(A_x \cup \{A_{\text{new}}^{(b)}\}), \text{ etc.};$$

2. **Dimensional tension.** Structural tension exceeds the dimensional threshold along multiple, non-equivalent directions of difference:

$$T(K_x, t) > \Theta_{\text{dim}}^{(a)}(K_x) \quad \text{and} \quad T(K_x, t) > \Theta_{\text{dim}}^{(b)}(K_x),$$

where the superscripts denote distinct dimensional channels;

3. **Embedding capacity.** The embedding space  $M_x$  supports all candidate axes, i.e.

$$A_{\text{new}}^{(a)}, A_{\text{new}}^{(b)}, \dots \in A(M_x);$$

4. **Nonempty admissible regions.** Each candidate continuum  $K_{x+1} = \Psi_{x \rightarrow x+1}^{(a)}(K_x, M_x)$ , etc., has a nonempty admissible set  $\Omega_{x+1}^{(a)} \neq \emptyset$ .

In many concrete systems these conditions are realised through:

- incompatibilities in potentials  $P_x(t)$  that cannot be reconciled within a single extension;
- competition between cycle complexes  $C_x^{(a)}, C_x^{(b)}$  whose joint stabilisation would violate stability thresholds;
- boundary geometries that admit distinct, mutually exclusive re-closures (e.g. alternative membrane architectures or institutional structures).

**Branching thresholds.** For convenience one may define a *branching threshold*  $\Theta_{\text{branch}}(K_x)$  as an effective composite of dimensional and embedding thresholds, such that

$$T_{\text{branch}}(K_x, t) > \Theta_{\text{branch}}(K_x) \quad \Rightarrow \quad \text{branching event possible.}$$

Core 1.3 treats  $\Theta_{\text{branch}}$  as a derived quantity rather than a new primitive threshold class: it encodes simultaneous satisfaction of the conditions for multiple birth operators.

### 13.4 Branch Interaction

Branches do not generally evolve in isolation. When continua belonging to different branches coexist in a shared embedding space, they may interact through the interaction operator  $E_{\text{int}}$  (Section 11).

**Definition 13.4 (Branch interaction).** Let  $B^{(a)}$  and  $B^{(b)}$  be two branches with continua  $\{K_x^{(a)}\}$  and  $\{K_y^{(b)}\}$ , respectively, embedded in a common or overlapping space  $M$ . A *branch interaction* is a family of interaction operators

$$E_{\text{int}}^{x,y} : (K_x^{(a)}, K_y^{(b)}, M) \longrightarrow (K_x^{(a)'}, K_y^{(b)'}, M'),$$

defined for relevant pairs of levels  $(x, y)$ .

The following regimes are typical:

- **Coexistence.** Flows are weakly coupled and allow both branches to maintain or increase their continuumness:

$$k^{(a)}(t + dt) \geq k^{(a)}(t), \quad k^{(b)}(t + dt) \geq k^{(b)}(t).$$

- **Interference.** Cross-branch flows  $J^{(a \leftrightarrow b)}$  create structural tension that pushes one or both branches closer to their death thresholds without immediate collapse.
- **Suppression.** One branch drives the other below its viability thresholds, leading to collapse of the suppressed branch while the suppressor remains viable or even strengthens.
- **Parasitism.** One branch harvests supporting flows from another, increasing its own continuumness while eroding the other's cycle complexes.
- **Fusion.** Through strong coupling and shared boundary structures, two branches can generate a new continuum that lies on a common extension branch. Formally this corresponds to a composite birth operator fed by both  $B^{(a)}$  and  $B^{(b)}$ .

Shared boundary structures are particularly important. If  $\partial\Omega^{(a)} \cap \partial\Omega^{(b)} \neq \emptyset$ , then:

- cross-branch flows restricted to the intersection can mediate coupling;
- patch states at the shared boundary can synchronise or destabilise both branches simultaneously;
- new continua may nucleate at the shared boundary, initiating new branches or branch fusions.

### 13.5 Branch Collapse

Branches can terminate in several ways, extending the notion of collapse from individual continua (Section 12) to whole development paths.

**Definition 13.5 (Branch collapse).** A branch  $B = (K_{x_0}, K_{x_1}, \dots)$  *collapses* at level  $x_n$  if:

1. the continuum  $K_{x_n}$  collapses in the sense of Definition 12.1 (empty admissible set,  $k \rightarrow 0$ , threshold violation);
2. there exists no continuum  $K_{x_{n+1}}$  with  $x_{n+1} > x_n$  such that a birth operator  $\Psi_{x_n \rightarrow x_{n+1}}$  can be defined on any residue of  $K_{x_n}$  (Section 12);
3. the branch cannot be extended further while respecting the axioms of dimensional monotonicity and embedding compatibility.

Three structurally distinct scenarios are relevant:

- **Terminal collapse.** The last continuum on the branch collapses and no rebirth at any higher level is possible. The branch is finite and ends at  $x_n$ .
- **Fusion collapse.** The branch is absorbed into a more stable branch via fusion: continua on  $B$  lose their identity as independent vertices in the branch graph and become internal structure of a continuum on another branch. From the perspective of  $B$ , this is collapse by loss of branch identity.



- **Asymptotic thinning.** The branch does not end at a single dramatic collapse but enters a regime where higher levels become increasingly marginal ( $k_{x_i} \rightarrow 0$  as  $i \rightarrow \infty$ ) and cannot support further dimensional births. The branch is infinite in index but finite in effective complexity.

**Global constraints.** The OC axioms impose constraints that prevent uncontrolled infinite branching cascades at finite levels:

- the embedding space at each level has finite axis capacity relevant to a given domain, limiting the number of distinct birth channels;
- dimensional thresholds require nontrivial structural tension; repeated branching without sufficient supporting flows would drive branches toward collapse;
- monotonic complexity theorems constrain viable branches to those whose complexity growth is compatible with available cycles and thresholds.

### 13.6 Consequences for the K-Level Hierarchy

The branch structure refines the vertical hierarchy  $K_0$ – $K_{10}$  by identifying which sequences of levels and axes are actually realisable.

**Ontogenic development paths.** Each branch defines an *ontogenic development path*:

- in physical domains, branches describe ways in which field-theoretic continua at  $K_2$  give rise to chemical, protocellular and biological continua at  $K_3$ – $K_5$ ;
- in cognitive and social domains, branches describe how neural continua  $K_5$  can support cognitive continua  $K_6$ , which in turn support social and civilizational continua  $K_7$ – $K_8$ ;
- in theoretical domains, branches connect empirical continua to theoretical and meta-theoretical continua  $K_9$ – $K_{10}$ .

The diversity of observed physical, chemical, biological and social systems corresponds to the multiplicity of branches compatible with a shared substrate.

**Allowed vs forbidden transitions.** The branch topology codifies which transitions between levels are allowed:

- edges in the branch graph  $\mathcal{G}$  correspond to admissible dimensional birth operators;
- absence of an edge  $K_x \rightarrow K_y$  (with  $y > x$ ) expresses a structural impossibility: no embedding space and threshold landscape can support that direct transition without passing through required intermediate levels;
- forbidden transitions typically violate either dimensional monotonicity (attempted simplification of axes) or embedding thresholds (attempted realisation of axes unsupported by  $M_x$ ).

**Vertical coherence.** Because branches must reduce to lower levels when additional axes are frozen (Section 11), the branch structure enforces vertical coherence:

- each higher-level continuum  $K_y$  restricts to a lower-level continuum  $K_x$  along a branch when additional axes are held constant;
- conversely, lower levels act as necessary substrata for higher levels;
- branches record this coherence as shared prefixes across domains (e.g. all biological branches must contain physical and chemical levels).

**Global picture.** Taken together, the vertical hierarchy  $K_0$ – $K_{10}$ , the threshold landscape, and the branch topology form the global structural picture of OC. The operative machinery is carried by  $F$ ,  $G$ ,  $H$ ,  $Q$ ,  $R$ ,  $S$ ,  $U$ ,  $\Psi$ , and  $E_{\text{int}}$ :

- continua are local objects with internal dynamics;
- branches are global development paths threading through levels;
- the branch graph encodes which paths are structurally allowed;
- collapse and rebirth prune and reconfigure branches over time.

Core 1.3 fixes this global picture at the structural level. Subsequent extensions and domain-specific studies may quantify branch probabilities, explore explicit examples of branching in concrete systems and develop empirical criteria for identifying branch structure in real-world data.

## 14 Cross-Disciplinary Instantiations of the Continuum Framework

This chapter tests a bounded representation claim: selected scientific and formal-theoretical domains can be expressed by the tuple

$$(\Omega, A, P(t), J(t), \Theta, \partial\Omega, C, k(t))$$

in ways consistent with the structural hierarchy of levels  $K_0$ – $K_{12}$ , the threshold landscape, the operator family, and the branch topology of Section 13. Here  $k(t)$  suppresses the continuum index only to keep the domain tuple readable; locally it denotes the corresponding  $k(K_x, t)$ . Likewise, later domain blocks may suppress  $t$  in symbols such as  $P_x$  or  $J_x$  when the paragraph describes the family of potentials or flows rather than a single time slice. Each domain-specific mapping is treated here under one canonical label: the *candidate-embedding rule*. Phrases such as bounded representation test, structural-compatibility screen, bounded embedding pass, and bounded embedding criterion are local descriptions of that rule, not separate criteria. The aim is to show how well-studied classes of systems can be represented as continua under OC’s structural axioms, not to claim that every object in a discipline is exhausted by a single OC encoding (**anderson\_more\_is\_different**).

This is a structural-compatibility claim, not an operational-advantage claim. The practical comparison is deferred to Section 25 and Appendix R. This chapter asks whether a domain can be represented as a bounded continuum; it does not claim that every such domain already has a closed theorem packet, held-out replay, comparator audit, or deployment-ready procedure.

### 14.1 Principles for Domain Embedding

To represent a discipline within OC, the following structural alignment conditions must be satisfied.

**1. Mapping of objects to continuum components.** A domain has a candidate continuum representation if its objects can be mapped to:

- $\Omega$ : admissible configurations,
- $A$ : axes defining essential degrees of freedom,
- $P$ : potentials encoding forces, energies, costs, or semantic tension,
- $J$ : flows (matter, charge, probability, information, resources),
- $\Theta$ : thresholds governing stability, phase change, and collapse,
- $\partial\Omega$ : viability boundaries,
- $C$ : cycles maintaining persistence,
- $k(t)$ : continuumness (viability measure).

**2. Embedding-space constraints.** A valid instantiation requires both  $\Omega(K_x) \subseteq \Omega(M_x)$  and  $A(K_x) \subseteq A(M_x)$ . If the required states or axes are absent from the embedding space, the proposed instantiation is structurally inadmissible.

**3. K-level correspondence.** A discipline is boundedly embedded at level  $K_x$  only when:

- its dominant axes correspond to the axes of  $K_x$ ,
- its thresholds refine those of lower levels,
- its flows and cycles match the operator family of  $K_x$ ,
- its behaviour respects dimensional monotonicity.

**4. Structural compatibility.** Domain instantiation requires:

- internal consistency with OC thresholds,
- existence of at least one supporting cycle  $C_{\text{support}}$ ,
- nonempty admissible region  $\Omega \neq \emptyset$ ,
- extendibility of domain dynamics through the operator family.

A proposed embedding fails if any check fails: the mapping has an empty admissible region, the required axes are absent from the embedding space, the declared thresholds cannot be satisfied, or the proposed cycles do not keep the object away from the relevant boundary. These are rejection conditions, not merely cosmetic omissions.

The following sections summarise bounded candidate representations across disciplines.

## 14.2 Physics (Level $K_2$ )

Physical systems instantiate  $K_2$  when field configurations, order parameters, and coherence structures form the relevant axes. The discussion is intentionally bounded by standard critical phenomena, renormalisation, and percolation settings rather than by an unrestricted claim about all of physics ([goldenfeld\\_lectures](#); [wilson\\_renormalization](#); [stauffer\\_percolation](#)).

### State space and axes.

- $\Omega_2$ : field configurations (scalar, vector, gauge fields).
- $A_2$ : spatial axes, mode axes, internal symmetry axes, order-parameter axes.

### Potentials and flows.

- $P_2$ : physical energy functionals (action, free energy).
- $J_2$ : diffusive, convective, and field-theoretic currents.

### Thresholds.

- $\Theta_{\text{crit}}$ : phase boundaries,
- $\Theta_{\text{BKT}}$ : Berezinskii–Kosterlitz–Thouless vortex unbinding threshold,
- $\Theta_{\text{mass}}$ : coherence threshold for mass emergence,
- $\Theta_{\text{death}}$ : confinement or decoupling conditions.

**Boundary structures.** Within this bounded mapping,  $\partial\Omega$  corresponds to critical surfaces, percolation boundaries, and phase-separation interfaces.

**Cycles.** Topological solitons, vortex pairs, renormalisation cycles, and oscillatory field modes are instances of  $C_2$ .

**Bounded embedding criterion (Physics  $\rightarrow K_2$ ).** A physical system meets the bounded  $K_2$  embedding criterion when:

- its fields define a connected configuration space  $\Omega_2$ ,
- its energy functional defines thresholds compatible with OC,
- flows obey conservation laws expressible via  $J_2$ ,
- coherent cycles maintain structural persistence.

### 14.3 Chemistry and Origins of Life (Levels $K_3$ – $K_4$ )

Chemical systems are represented at  $K_3$ , while protocellular systems are represented at  $K_4$ . These mappings are bounded by reaction-network closure, catalytic organisation, and membrane-constrained prebiotic structure (**kauffman\_autocatalytic**; **raf\_networks**).

**Chemical continua ( $K_3$ ).**

- $\Omega_{\text{chem}}$ : concentration vectors, reaction states, catalytic configurations.
- $A_{\text{chem}}$ : species concentrations, environmental parameters (pH, salinity, temperature).
- $P_{\text{chem}}(t)$ : chemical potentials, affinity gradients, and concentration disequilibria.
- $J_{\text{chem}}$ : reaction rates, transport fluxes.
- $\Theta_{\text{act}}$ : catalytic activation threshold.
- $\Theta_{\text{closure}}$ : reflexively autocatalytic and food-generated (RAF) closure threshold.
- $C_{\text{chem}}$ : reaction cycles, RAF closure loops.

**Protocellular continua ( $K_4$ ).**

- $\Omega_4$ : vesicle states, membrane configurations, internal chemical compositions.
- $A_4$ : curvature, surface tension, permeability, charge.
- $P_4(t)$ : osmotic, chemical, electrochemical, and curvature potentials.
- $\Theta_{\text{perm}}, \Theta_{\text{osm}}, \Theta_{\text{curv}}$ : membrane viability thresholds.
- $J_4$ : osmotic fluxes, ion fluxes, metabolic flows.
- $C_4$ : metabolic cycles, membrane-maintenance cycles, gradient-sustaining cycles.

**Bounded embedding criterion (Chemistry  $\rightarrow K_3$ ).** A chemical system meets the bounded  $K_3$  embedding criterion when:

- reaction networks form a nonempty admissible region,
- catalytic and closure thresholds are representable without membrane-specific axes,
- cycles maintain reaction closure and concentration control,
- embedding constraints support the chemical axes required by the route.

**Bounded embedding criterion (Protocells  $\rightarrow K_4$ ).** A protocellular system meets the bounded  $K_4$  embedding criterion when:

- membrane-supported chemical dynamics form a nonempty admissible region,
- membrane boundaries define  $\partial\Omega_4$ ,
- cycles maintain concentration gradients and compartment integrity,
- embedding constraints support both chemical and membrane axes.

#### 14.4 Biology (Level $K_5$ )

Biological cells, beyond the protocellular  $K_4$  route, emerge when protocells acquire excitability and structured regulation. Here the target is the bounded embedding of excitable and morphogenetic organisation rather than an unrestricted theory of life as such (**turing\_morphogenesis**; **hopfield\_nn**).

**State space and axes.**

- $\Omega_5$ : distributions of membrane potential, channel states, ion gradients.
- $A_5$ : excitability axes (membrane potential  $\Delta V$ , ion concentrations, gating-variable axes).

**Potentials and flows.**

$P_5$  : electrochemical potentials,  $J_5$  : ion currents, metabolic fluxes.

**Thresholds.**

$\Theta_{\text{exc}}, \quad \Theta_{\text{refr}}, \quad \Theta_{\text{grad}}, \quad \Theta_{\text{death}}.$

**Cycles.** The relevant cycles include the proto-spike cycle  $C_{\text{spike}}$ , metabolic cycles, and electrical oscillation loops.

**Bounded embedding criterion (Biology  $\rightarrow K_5$ ).** A biological system meets the bounded  $K_5$  embedding criterion when:

- excitability defines nontrivial  $\Delta V$ -axes,
- thresholds define firing and refractory structure,
- cycles maintain homeostasis and signalling,
- embedding includes ionic, energetic, and membrane-support axes.

#### 14.5 Cognition (Level $K_6$ )

The  $K_6$ – $K_{10}$  sections below are structural candidate embeddings. They are frontier-facing representation sketches unless Section 25 or Appendix R explicitly promotes a narrower operational lane. They are not usable-now practical claims by themselves.

Cognitive systems arise when representations, bindings, and predictive models become the dominant axes. The discussion is anchored to classical work on the organisation of behaviour and predictive-processing motifs, not to a claim that OC replaces cognitive science in full (**hebb\_organization**; **friston\_free\_energy**).

**State space and axes.**

- $\Omega_6$ : representational configurations, activated conceptual states, prediction states.
- $A_6$ : representational axes, binding axes, predictive axes.

### Potentials and flows.

- $P_6$ : prediction error, semantic tension, confidence.
- $J_6$ : attention flows, inference flows, memory transitions.

### Thresholds.

$$\Theta_{\text{pred}}, \quad \Theta_{\text{bind}}, \quad \Theta_{\text{coh}}, \quad \Theta_{\text{expr}}.$$

**Cycles.** The relevant cycles include attention-prediction cycles, memory-consolidation cycles, and model-stability loops.

**Bounded embedding criterion (Cognition  $\rightarrow K_6$ ).** A cognitive system meets the bounded  $K_6$  embedding criterion when:

- it maintains coherent internal models within expressivity thresholds,
- flows reduce prediction error and sustain conceptual structure,
- cycles regulate attention, memory, and updating,
- boundaries  $\partial\Omega_6$  encode model limits.

## 14.6 Society (Level $K_7$ )

Social systems instantiate  $K_7$  when norms, institutions, and trust form the dominant structural axes. The intended scope is thresholded collective behaviour, networked coordination, and institutional persistence (**granovetter\_threshold**; **schelling\_micromotives**; **newman\_networks**).

### State space and axes.

- $\Omega_7$ : communication graphs, role structures, institutional states.
- $A_7$ : normative axes, trust axes, institutional axes.

### Potentials and flows.

$$\begin{aligned} P_7 &: \text{trust, legitimacy, symbolic capital,} \\ J_7 &: \text{communication, resource exchange, and norm transmission.} \end{aligned}$$

### Thresholds.

$$\Theta_{\text{trust}}, \quad \Theta_{\text{leg}}, \quad \Theta_{\text{role}}.$$

**Cycles.** The relevant cycles include institutional cycles, normative cycles, and coordination cycles.

**Bounded embedding criterion (Society  $\rightarrow K_7$ ).** A social system meets the bounded  $K_7$  embedding criterion when:

- trust and norms define viability thresholds,
- institutions maintain cycles of governance and coordination,
- flows of communication and resources respect embedding constraints,
- social boundaries form nonempty  $\partial\Omega_7$ .

## 14.7 Civilizational Systems (Level $K_8$ )

Civilizational systems extend social continua to encompass infrastructures, energy flows, and large-scale coordination. The scope is limited to large-scale organised systems with networked infrastructure and stability constraints (**newman\_networks**; **may\_stability\_complexity**).

### State space and axes.

- $\Omega_8$ : infrastructure networks, economic states, population distributions.
- $A_8$ : infrastructural axes, energy-flow axes, communication axes, complexity axes.

### Potentials and flows.

$P_8$  : energy availability, resource gradients, risk potentials,       $J_8$  : trade, energy flux, data flows.

### Thresholds.

$$\Theta_{\text{stress}}, \quad \Theta_{\text{capacity}}, \quad \Theta_{\text{cohesion}}.$$

**Cycles.** The relevant cycles include economic cycles, infrastructure-renewal cycles, and knowledge-transfer loops.

**Bounded embedding criterion (Civilizational systems  $\rightarrow K_8$ ).** A civilizational system meets the bounded  $K_8$  embedding criterion when:

- energy and resource structures define  $P_8$ ,
- infrastructural flows form stable global cycles,
- institutions at  $K_7$  embed consistently as substructures,
- large-scale boundaries define global  $\partial\Omega_8$ .

## 14.8 Theory and Formal Systems (Levels $K_9$ – $K_{10}$ )

Theory-level and metatheoretical formal-system structures occupy the upper route treated in this chapter. This subsection gives an OC-internal schema for  $K_9$ – $K_{10}$ , with local support carried by Sections 10 and 18. It does not claim a fully closed external embedding theorem for every theory or formal system. Recursive structural continua remain the  $K_{11}$  layer outside this chapter's candidate-embedding rule.

### Theoretical continua ( $K_9$ ).

- $\Omega_9$ : space of models, formalisms, and representations.
- $A_9$ : conceptual, formal, and modelling axes.
- $P_9$ : coherence potentials, empirical tension, explanatory power.
- $J_9$ : model-revision flows, anomaly-handling flows, and argument-transfer flows.
- $\Theta_{\text{expr}}, \Theta_{\text{coh}}$ : thresholds for consistency and expressivity.
- $\partial\Omega_9$ : inconsistency, expressivity-failure, or unrepaired-anomaly boundaries.
- $C_9$ : paradigm cycles, research cycles.
- $k_9(t)$ : coherence and validation-bandwidth continuumness.

### Metatheoretical formal-system continua ( $K_{10}$ ).

- $\Omega_{10}$ : space of metatheoretical states, formal systems, proof states, model-of-models objects, and recursion-bounded languages.
- $A_{10}$ : metatheory axes, recursion axes, consistency axes, expressivity axes, and theory-comparison axes.
- $P_{10}$ : proof-flow capacity, consistency margin, formal expressive capacity, and metatheoretical coherence potential.
- $J_{10}$ : proof-search flows, formal-update flows, meta-model translations, and recursion-control flows.
- $\Theta_{\text{cons}}, \Theta_{\text{rec}}, \Theta_{\text{meta}}$ : thresholds for consistency, bounded recursion, and meta-level coherence.
- $\partial\Omega_{10}$ : inconsistency, undecidable-overreach, recursion-boundary surfaces, or meta-theory boundary failures.
- $C_{10}$ : proof cycles, recursion cycles, formal-update cycles, and model-of-models audit cycles.
- $k_{10}(t)$ : formal and metatheoretical viability under consistency, recursion, self-reference, and proof-throughput constraints.

**Candidate-embedding rule (Theory  $\rightarrow K_9$ , metatheoretical formal system  $\rightarrow K_{10}$ ).** The theoretical and metatheoretical formal-system routes meet the candidate-embedding rule when the relevant  $K_9$  or  $K_{10}$  structure satisfies the following conditions:

- its conceptual and formal axes form a valid axis set,
- its coherence constraints define thresholds compatible with OC,
- cycles of reasoning, modelling, and revision sustain viability,
- embedding spaces support the required expressive degrees of freedom.

### Summary

This chapter states bounded cross-disciplinary candidate-embedding criteria across physics, chemistry, protocells, biology, cognition, society, civilizational systems, theory, and formal systems. Each bounded route is mapped to the same structural tuple at the level of detail declared in its subsection, but each mapping is claimed only at the level of structural compatibility, not as a sweeping proof that OC already replaces the best local models in every discipline. That stronger operational and comparative question is taken up later under the explicit support classes in Section 25 and the row-level atlas in Appendix R.

## 15 Extended Falsifiability and Experimental Programme

This section restores and expands the falsifiability framework for the Ontology of Continua (OC). It formulates explicit structural predictions, falsification criteria, and experimental programmes across the hierarchy  $K_2$ – $K_{12}$ , using only the axioms and definitions of Core 1.3.

The goal is not to provide a full catalogue of experiments but to show that OC is structurally falsifiable: it makes cross-domain commitments about thresholds, cycles, dimensional transitions and boundary behaviour that can in principle be contradicted by empirical and theoretical data.

### 15.1 Meaning of Falsifiability for Structural Theories

For a structural theory, falsifiability is not limited to individual numerical predictions. Instead, falsification occurs when the structural relations between

$$(\Omega, A, P(t), J(t), \Theta, \partial\Omega, C, k(t))$$



are violated in ways incompatible with the axioms and theorems of OC.

### Structural vs phenomenological falsifiability.

- *Phenomenological falsifiability* concerns direct mismatches between model outputs and observations (e.g. wrong phase diagram, incorrect scaling exponent).
- *Structural falsifiability* concerns violations of the universal constraints on continua: thresholds, cycles, dimensional monotonicity, embedding-compatibility, collapse and rebirth conditions.

OC primarily lives at the structural level; phenomenological models must factor through OC's structural requirements to be considered proper instantiations.

**Structural falsification channels.** A continuum model claiming to instantiate OC can be falsified if:

1. **Threshold violations** occur systematically: observed systems remain viable while violating putative  $\Theta_{\text{exist}}$ ,  $\Theta_{\text{stab}}$ ,  $\Theta_{\text{crit}}$  or  $\Theta_{\text{death}}$ .
2. **Cycle disappearance** contradicts viability: systems maintain long-term organisation without any identifiable cycle complex  $C$  that satisfies the OC conditions.
3. **Forbidden dimensional transitions** are observed: dimension appears to increase without new axes in the embedding space or without crossing a dimensional threshold  $\Theta_{\text{dim}}$ .
4. **Inconsistent tuples** are required: successful models require mutually incompatible versions of the canonical OC state tuple for one and the same system.

**Structural vs empirical failure.** Empirical failure of a specific domain model does not immediately falsify OC; the structural theory is challenged only when the empirical success of alternative models requires violations of OC's structural constraints. Conversely, if a domain repeatedly exhibits structures that OC forbids (e.g. stable continua without supporting cycles), OC is directly falsified.

**Cross-domain falsification logic.** Because OC claims universality, a structural falsification in one domain (propely established) propagates across all domains that rely on the same axioms. This makes OC *more* falsifiable than narrow theories:

- if monotonicity of dimension fails in a controlled physical system, it must also be reconsidered for cognitive and social continua;
- if viable protocells are demonstrated without any boundary structure compatible with  $\partial\Omega$ , the entire K-level hierarchy is affected.

The remainder of this chapter formulates explicit predictions and tests.

## 15.2 Predictions and Tests in Physics ( $K_2$ )

Physical continua provide the most classical testing ground for OC at level  $K_2$ .

### Structural predictions.

- **P2.1 (Critical thresholds).** Phase transitions correspond to structural crossings of  $\Theta_{\text{crit}}$ . No sharp long-range ordering transition may occur without a corresponding threshold surface in  $\Omega_2$ .
- **P2.2 (Percolation monotonicity).** For connectivity-based systems with control parameter  $p$ , global viability ( $k_2 > 0$ ) requires  $p \geq p_c$ ; for  $p < p_c$  no infinite cluster compatible with  $\Omega_2$  can exist.
- **P2.3 (Dimensional monotonicity).** No physical continuum can increase its effective dimension without the activation of at least one new axis in the embedding space  $M_2$  and crossing a dimensional threshold  $\Theta_{\text{dim}}$ .

- **P2.4 (Boundary-driven collapse).** Collapse of ordered phases occurs via destruction of supporting cycles (e.g. vortex pairs, domain structures) and loss of well-defined  $\partial\Omega_2$ , not via spontaneous disappearance of order in the interior alone.

#### Falsification criteria and tests.

- **F2.1.** A robust, reproducible phase transition with no structural thresholds in the order-parameter geometry would falsify P2.1.
- **F2.2.** Observation of stable infinite clusters for  $p < p_c$  across well-controlled percolation-like systems would contradict P2.2.
- **F2.3.** Demonstration of a continuum whose effective dimensionality increases (e.g. from 2D to 3D behaviour) while all axes of  $M_2$  remain fixed and no new degrees of freedom are activated would falsify P2.3 and the general dimensional monotonicity theorem.
- **Experimental programme.** Precision experiments on coherence destruction, vortex unbinding, and percolation thresholds can be used to map empirical  $\Theta_{\text{crit}}$  and test whether the associated  $\partial\Omega_2$  behaves as OC predicts (critical slowing down, boundary localisation of collapse).

### 15.3 Predictions and Tests in Origins of Life ( $K_3$ – $K_4$ )

For origins-of-life scenarios OC makes explicit claims about catalytic closure, membrane structure and protocell viability.

#### Structural predictions.

- **P3.1 (Catalytic closure).** Long-term stable chemical organisations at  $K_3$  require at least one RAF-like cycle complex; systems without catalytic closure cannot maintain  $k_3 > 0$ .
- **P3.2 (Osmotic collapse threshold).** Protocells with fixed membrane composition exhibit a radius–tension–permeability relation: beyond a critical combination of osmotic gradient, curvature and permeability ( $\Theta_{\text{grad}}$ ,  $\Theta_{\text{perm}}$ ,  $\Theta_{\text{mem}}$ ) collapse is inevitable.
- **P3.3 (Boundary necessity).** Protocells lacking a structurally identifiable boundary  $\partial\Omega_4$  (membrane or equivalent compartment structure) cannot sustain nontrivial metabolic flows  $J_{\text{metabolic}}$  over timescales defining  $k_4 > 0$ .
- **P3.4 (Patch-localised failure).** Membrane collapse initiates on boundary patches where local thresholds are first crossed, not uniformly across the surface.

#### Falsification criteria and tests.

- **F3.1.** Demonstration of chemically self-sustaining organisations with  $k_3 > 0$  and no identifiable catalytic closure or cycle complex would falsify P3.1.
- **F3.2.** Systematic violation of predicted osmotic-collapse relations (e.g. arbitrary gradients sustained without structural adjustment) would contradict P3.2.
- **F3.3.** Observation of long-lived protocell-like systems with robust metabolic fluxes but no boundary structure compatible with  $\partial\Omega_4$  would falsify P3.3 and weaken the role of boundaries in OC.
- **Experimental programme.** Controlled protocell leakage and bursting experiments, combined with quantitative measurements of gradients and membrane properties, can test the structural status of  $\Theta_{\text{grad}}$ ,  $\Theta_{\text{perm}}$ ,  $\Theta_{\text{mem}}$  and patch dynamics.

### 15.4 Predictions and Tests in Biological Systems ( $K_4$ – $K_5$ )

At the biological level, OC focuses on excitability, proto-spikes and the transition  $K_4 \rightarrow K_5$ .

### Structural predictions.

- **P4.1 (Minimal excitability threshold).** The transition from purely metabolic protocells to excitable continua requires a minimal excitability threshold  $\Theta_{\text{exc}}$ ; no stable  $K_5$  behaviour can exist without nontrivial  $\Delta V$ -axes and a well-defined spike threshold.
- **P4.2 (Proto-spike invariants).** Across early excitable systems there exist minimal invariants for spike-like cycles (e.g. minimal amplitude of  $\Delta V$ , refractory period  $\tau$ ) corresponding to structural constraints of  $C_{\text{spike}}$ .
- **P4.3 (Excitability precedes complex inheritance).** Long-term inheritance of complex regulatory structures requires prior establishment of  $K_5$ -like channels and excitability cycles; purely  $K_4$  systems cannot indefinitely sustain inherited regulatory complexity without transitioning to a higher level.

### Falsification criteria and tests.

- **F4.1.** Demonstration of genuinely  $K_5$ -like signalling (spike trains, excitable waves) in systems with no identifiable excitability axes or thresholds would falsify P4.1.
- **F4.2.** Discovery of stable, complex inheritance mechanisms in protocell systems strictly confined to  $K_4$  (no excitability, no spike-like cycles) would challenge P4.3.
- **Experimental programme.** Minimal ion-channel sets, synthetic excitable membranes and controlled  $\Delta V$  instability tests can be used to map the structural space of  $\Theta_{\text{exc}}$ ,  $\Theta_{\text{refr}}$  and the resulting  $C_{\text{spike}}$ .

## 15.5 Predictions and Tests in Cognitive Continua ( $K_6$ )

For cognitive systems OC emphasises binding capacity, prediction error and cycle-based maintenance of internal models.

### Structural predictions.

- **P6.1 (Binding capacity limit).** For any cognitive continuum there is a finite binding capacity threshold  $\Theta_{\text{bind}}$  beyond which additional features cannot be integrated into coherent representations without collapse of  $k_6$ .
- **P6.2 (Prediction-error collapse).** There exists a prediction-error threshold  $\Theta_{\text{pred}}$  beyond which no coherent internal model can be maintained; cognitive collapse occurs when prediction error cannot be reduced by any flows  $J_6$  within current axes.
- **P6.3 (Cycle-based collapse).** Cognitive collapse always involves destruction of key cycles (attention, prediction, memory); there is no collapse scenario with  $C_{\text{max}}(K_6)$  intact.

### Falsification criteria and tests.

- **F6.1.** Robust evidence for unbounded binding capacity in finite systems (no observable thresholds, no degradation) would falsify P6.1.
- **F6.2.** Empirical regimes where prediction error diverges yet stable, coherent internal models persist indefinitely would contradict P6.2.
- **F6.3.** Demonstration of cognitive collapse in which attention, prediction and memory cycles remain fully intact would falsify P6.3 and the cycle-based definition of collapse.
- **Experimental programme.** Behavioural and neurophysiological studies of binding degradation under load, prediction-error saturation, and memory reconsolidation can probe the empirical counterparts of  $\Theta_{\text{bind}}$ ,  $\Theta_{\text{pred}}$ , and  $C_{\text{memory}}$ .

## 15.6 Predictions and Tests in Social/Civilisational Continua ( $K_7$ – $K_8$ )

At higher social levels OC proposes structural constraints on trust, institutional cycles and infrastructural stability.

### Structural predictions.

- **P7.1 (Trust threshold).** Stable social continua at  $K_7$  require trust above a threshold  $\Theta_{\text{trust}}$ ; below this threshold institutional cycles cannot maintain  $k_7 > 0$ .
- **P7.2 (Institutional cycle closure).** No stable  $K_7$  continuum exists without at least one closed institutional cycle (e.g. taxation–service–legitimacy loop) compatible with OC’s definition of  $C_7$ .
- **P8.1 (Infrastructure cycle thresholds).** Civilisational continua  $K_8$  require stable infrastructural cycles; collapse at  $K_8$  involves boundary failure in  $\partial\Omega_8$  and percolation of failure across key infrastructural layers.

### Falsification criteria and tests.

- **F7.1.** Historical or simulated examples of long-term stable institutions with trust consistently below any plausible  $\Theta_{\text{trust}}$  would challenge P7.1.
- **F7.2.** Demonstration of durable, high-complexity societies without any identifiable closed governance or resource cycles would falsify P7.2.
- **F8.1.** Robust models of civilisational collapse that do not involve breakdown of infrastructural cycles or boundary failures in  $\partial\Omega_8$  would contradict P8.1.
- **Experimental programme.** Quantitative analyses of historical data, network fragility models, and large-scale simulations of trust dynamics and infrastructure percolation can be used to test the structural role of  $\Theta_{\text{trust}}$ ,  $C_{\text{inst}}$  and  $\partial\Omega_8$ .

## 15.7 Theoretical and Meta-Theoretical Predictions ( $K_9$ – $K_{10}$ )

OC itself, and other high-level frameworks, can be tested at the theoretical and meta-theoretical levels.

### Structural predictions.

- **P9.1 (Coherence thresholds).** Any long-lived theoretical continuum  $K_9$  must satisfy coherence and expressivity thresholds  $\Theta_{\text{coh}}$ ,  $\Theta_{\text{expr}}$ ; persistent success of theories that explicitly violate internal coherence would contradict the OC framework.
- **P9.2 (Inconsistency cascades).** Accumulated inconsistencies in a theoretical framework must lead to collapse of its cycle complex (research programmes, paradigm loops); sustained progress in the presence of arbitrarily large incoherence is incompatible with OC.
- **P10.1 (Meta-incompleteness).** Any meta-theoretical continuum  $K_{10}$  is structurally incomplete: as bodies of models grow, structural tension at  $K_{10}$  must eventually exceed some  $\Theta_{\text{dim}}$ , forcing either expansion of axes or collapse.

### Falsification criteria and tests.

- **F9.1.** A fully self-consistent, closed theoretical system that captures all relevant models while never requiring expansion or revision and yet continues to support unbounded empirical progress would challenge P9.2 and P10.1.
- **F9.2.** Demonstration of theoretical frameworks that remain incoherent by OC standards but nonetheless exhibit stable, cumulative, cross-domain predictive success would falsify the structural link between coherence and  $k_9$ .

- **Experimental programme.** Formal analyses of model families, logical consistency checks, and meta-level studies of theory change can be used to test the existence and behaviour of  $\Theta_{\text{expr}}$ ,  $\Theta_{\text{coh}}$  and associated cycles at  $K_9$ – $K_{10}$ .

## 15.8 Meta-Criteria and Cross-Domain Validation

Finally, OC proposes meta-level criteria that couple falsifiability across domains and K-levels.

**Predictive invariants across K-levels.** Several structural relations are predicted to hold uniformly:

- existence of at least one supporting cycle  $C_{\text{support}}$  for any live continuum;
- monotonicity of dimension under birth operators;
- necessity of embedding-space compatibility  $A(K_x) \subseteq A(M_x)$  and threshold compatibility with  $M_x$ ;
- collapse signatures: critical slowing, boundary localisation, cycle breakdown, divergence of structural tension.

Violation of these invariants at any well-understood K-level would propagate upward and downward in the hierarchy.

**Algorithmic structural tests.** Core 1.2 and subsequent work are expected to develop algorithmic tests for:

- $\Omega$ -consistency (nonempty admissible region given thresholds),
- cycle viability (existence of cycle complexes  $C$  under given flows),
- embedding compatibility (axes and thresholds vs.  $M_x$ ).

These tests can be applied to concrete models in physics, chemistry, biology, cognition and social sciences to check whether they are legitimate OC instantiations.

**Cross-domain falsification logic.** Because axioms are shared across K-levels, falsification has a *cross-domain* structure:

- if one structural axiom fails in a tightly controlled physical context, all higher-level predictions derived from it must be reconsidered;
- conversely, consistent validation of the same structural motif (e.g. boundary-driven collapse) in multiple domains increases confidence in the corresponding OC axiom.

**Unified validation pipeline.** Core 1.3 provides the structural basis for an explicit validation pipeline in Core 1.2:

1. select domain models and identify their continuum components  $(\Omega, A, P, J, \Theta, \partial\Omega, C, k)$ ;
2. test local structural predictions (thresholds, cycles, boundaries);
3. test cross-level invariants (dimensional monotonicity, embedding compatibility);
4. update the mapping between domain theories and K-levels.

OC is therefore not a purely conceptual framework: it is designed to be empirically constrained and — in principle — structurally falsifiable across the full hierarchy  $K_0$ – $K_{10}$ .

## 16 Module Architecture and Cross-Level Structures

This section provides a high-level overview of the extended structural modules of the Ontology of Continua. Each module is developed in the technical atlas; here we summarise their roles and relations so the main spine has a complete bridge between formal doctrine and the Core 1.3 closure chapters.

The extended modules are:

- **K-levels master** (Section O.1): global overview of all continua  $K_0$ – $K_{12}$ , their state spaces, axes, thresholds and core processes.
- **Embedding spaces / M-spaces** (Section O.2): definition of meta-spaces  $M_x$  that constrain and parameterise the evolution of continua  $K_x$ .
- **Cross-level / cross-K structures** (Section O.3): operators, mappings and landscape structures that link different levels of the hierarchy and generate phase transitions between them.
- **Processes** (Section O.6): unified description of evolution operators  $E = (F, G, H, Q, R, S, U)$  across levels, including tension dynamics, thresholds, flows and cycles.
- **Cycles** (Section O.4): structural and dynamical cycles  $C_j$  as carriers of continuity, memory and temporal organisation.
- **Predictions** (Section P.3): level-specific and cross-level predictions, empirical signatures and qualitative regimes implied by the formalism.
- **Experiments** (Section P.1): conceptual and concrete experimental designs to probe continua, transitions and threshold landscapes.
- **Falsifiability structure** (Section P.2): explicit conditions under which the Ontology of Continua can be refuted, constrained or forced to restructure its core.
- **Jets / local evolution structure** (Section O.5): local expansions of the evolution operator and short-time dynamics around given configurations of  $K_x$  and  $M_x$ .

Together, these modules refine and operationalise the core formalism across levels and applications, while maintaining a single coherent hierarchy of continua and meta-spaces.

## 17 Source Witnessing, Theorem Fate, and Publication Boundary

This chapter fixes the publication boundary of Core 1.3. Its task is to state which scientific authority is canonical, how source witnessing is enforced, how theorem fate is separated across outward artifacts, and how later channels must remain aligned with the source corpus, SPOT, and flagship monograph. It does not re-argue the formal doctrine or theorem spine; those burdens belong to the surrounding chapters and appendices.

### 17.1 Canonical scientific authority

The canonical scientific authority of the release is the SPOT projection `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json` (surface id `OC_CORE_1_3_SCIENCE_SPOT`; current SHA recorded in its `metadata.repo_sha` field). The source-owned science corpus at `releases/oc_core_1_3/editorial/science_sources/` is the authoring and witness locus from which that SPOT is projected. Historical source witnesses, including earlier Core 1.2/Core 1.3 repair baselines, remain provenance anchors; they are not the current projection authority. The present master monograph is the flagship reader-facing projection and compiled reference volume derived from those records. It keeps the orientation ladder, formal doctrine, theorem route, empirical discipline, synthesis, practical utility, and audit layers visible together, but it does not displace the SPOT as the final authority on claim scope, promotion state, or proof boundary.

This rule matters because foundational programmes accumulate several different textual layers over time: source-native definitions and theorems, explanatory prose, repair notes, channel-specific summaries, and later operational packets. Core 1.3 treats those layers as distinguishable, not interchangeable.

## 17.2 Source witnessing

Core 1.3 is source-first in the strict sense that the flagship claims used in public release must remain bound to public source witnesses rather than to late summary prose alone. The point is not archival fetishism. The point is to stop drift between five things that often blur inside large research programmes:

- theorem-bearing statements,
- structural and interpretive explanation,
- repair and release notes,
- condensed public packets,
- and later operational projections.

When that boundary blurs, a summary sentence can begin to masquerade as the theorem itself, or a release dashboard can begin to replace the actual scientific record. Core 1.3 blocks that failure by keeping source witnessing bound to the source-owned science corpus and the canonical SPOT projection rather than relegating it to hidden control files.

## 17.3 Theorem fate and scope discipline

The present release also separates theorem fate explicitly. Not every inherited row serves the same role in every outward artifact.

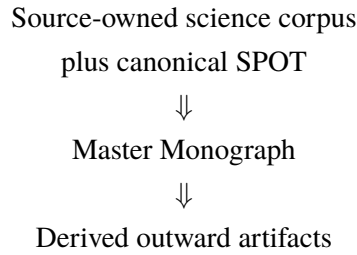
- `THEOREM_NATIVE` rows may carry positive theorem-native claims when their proof and dependency route remains exact.
- `BRIDGE_ONLY` rows may remain scientifically useful under explicit scope notes, but they cannot be promoted as theorem-native.
- `FRAME_ONLY` and `EXTENSION` rows belong in support layers, appendices, narrower companion artifacts, or future work rather than in the positive body of every outward paper.
- `REFUTED` rows are excluded from the positive proof path. They may be mentioned only as negative audit history or as a falsifier record, not as live support.
- Frontier or hypothesis material may be discussed only under visibly labelled frontier language and may not be silently promoted by summary prose.

The public positive closure claim is scoped exactly to `CORE_1_3_SCIENCE_ONLY`. `FRAME_ONLY` rows may still close as frame-level `PASS` rows inside the synthesis surface, but they do not become standalone theorem-native empirical promotions. `EXTENSION` and `REFUTED` rows remain outside the public `PASS` scope for different reasons: `EXTENSION` rows are frontier/support lanes awaiting later promotion or rejection, while `REFUTED` rows are negative-audit lanes. Neither class can carry positive closure claims in this release. For avoidance of doubt, a `FRAME_ONLY` `PASS` row is excluded from the public positive closure claim unless a later release promotes it into `CORE_1_3_SCIENCE_ONLY` with the required theorem-native support.

This is not retreat. It is a condition of honest publication. The monograph can remain large without rhetorical inflation only if the reader can tell which rows are theorem-bearing, which are bounded, and which are support or frontier material.

## 17.4 Relation to journal-core and companion artifacts

Core 1.3 therefore adopts a one-way reader-facing publication logic:



This diagram describes publication order, not the full provenance graph of the release package. Several editorial JSON surfaces, generated science chapters, and release matrices are direct SPOT projections or mirrors of other governing canonical surfaces. They do not become true merely because the master monograph cites them, and they are not descendants of the monograph alone. The monograph is the flagship reading object; the source-owned science corpus remains the authoring and witness locus, and the SPOT remains the canonical projection authority for release surfaces.

The journal-core article is narrower by design. Companion dossiers, matrices, and release surfaces are narrower or more operational by design. Those artifacts may compress, subset, or reorganise the flagship material for a specific task, but they may not enlarge the scientific claim, widen notation, or outrun the theorem-fate discipline fixed here.

The practical consequence is simple. If an outward reader-facing artifact says something stronger than the source corpus and SPOT authorise, the outward artifact is wrong, even if it is shorter, cleaner, or easier to market. The flagship boundary is the publication rule that enforces that ceiling; it is not an additional source of scientific authority.

## 17.5 Downstream projection rule

The same boundary stabilizes later use in review, collaboration, product, business, and public communication. Different channels may legitimately cite different parts of the flagship volume:

- the theorem route when bounded proof is the issue,
- the empirical route when promotion discipline is the issue,
- the practical route when use and comparison are the issue,
- the audit appendices when challenge and traceability are the issue.

What they may not do is invent a second scientific authority. Source witnessing, theorem fate, and publication boundary exist so that external use can remain concise without becoming interpretively lawless.

**Bridge to the next chapter.** With the publication boundary fixed, the next task is internal consistency: which statement classes the flagship volume uses, why the hierarchy runs through  $K_{12}$ , and how upper levels remain lawful without rhetorical inflation.



## 18 Core 1.3 Foundational Consistency Dossier

This chapter fixes the internal consistency contract of the flagship volume. Its purpose is not to replace the formal chapters with editorial commentary, but to make three things readable in one place:

- why the present hierarchy is fixed as  $K_0$  through  $K_{12}$  rather than by an arbitrary cutoff;
- the full statement-class discipline for axiomatic, derived, empirical, structural, interpretive, and frontier claims;
- how the master monograph and the narrower journal-core extraction are related without silent drift in scope or notation.

### 18.1 Why the hierarchy extends through $K_{12}$

Core 1.2 already contains enough structure to motivate a release hierarchy larger than  $K_{10}$ . The point is not that higher labels are rhetorically attractive. The point is that the audited source corpus already distinguishes:

1. lower-level continua whose axes, thresholds, and flows are internal to a given model class;
2. theoretical and meta-theoretical continua that organise theories and model-spaces themselves;
3. a further stratum in which operators, meta-logics, and model-spaces become explicit objects of structural evolution;
4. and an upper bounding layer in which global coherence conditions across such higher-order structures must themselves be represented.

That is why Core 1.3 carries the public hierarchy through  $K_{12}$  rather than stopping at  $K_{10}$ . This is a release-level design rationale, not a standalone impossibility proof. The supporting route is the K-level doctrine in Section 10, the theorem-navigation discussion in Section 19, and the worked preservation test in Section 20. Those sections assign  $K_{11}$  to evolving meta-theoretical objects and  $K_{12}$  to global compatibility and integrability conditions. The present release does not claim a proof that no further extension is conceivable in principle; it claims only that  $K_{12}$  is the highest level lawfully carried by the current audited source corpus and therefore the last public level of this release.

### 18.2 Scientific role of the upper levels

**Level  $K_{11}$ .**  $K_{11}$  is the first level at which formal ontologies, operator systems, meta-logics, and model-spaces are themselves treated as structured continua subject to admissibility, transformation, and collapse conditions. This level is needed once one studies not only theories, but the evolution of the spaces in which theories and theory-building operators live.

**Level  $K_{12}$ .**  $K_{12}$  is not introduced as a mystical completion layer. It is the minimally specified upper coherence stratum required when one needs to discuss global compatibility, boundedness, and failure modes across families of  $K_{11}$ -objects. Its role is structural rather than triumphalist: it prevents the hierarchy from pretending that there is no formal problem above recursively evolving meta-theoretical spaces.

### 18.3 Statement classes

To keep the monograph readable without blurring scientific status, Core 1.3 uses the following proof-status vocabulary in this chapter. It is the local theorem and claim-scope taxonomy; Section 19 adds the downstream operational inspection burden for replay, comparator, falsifier, and escalation rules.

- **Axiomatic statements** fix primitive assumptions, admissibility conditions, or identity rules.

- **Derived statements** are theorem-bearing statements with explicit support dependence and proof obligations.
- **Empirical statements** require a standalone source-owned domain packet, canonical theorem-native trace, pinned data route, numerical parameter law, observable binding, locked held-out replay procedure, comparator audit, and explicit falsifier before promotion.
- **Structural statements** describe the organisation of the framework, the hierarchy, and the mapping between modules or levels.
- **Interpretive statements** explain what the formal structure means, which reader it is for, and what later projections may test.
- **Frontier statements** may remain in the research program, but may not enter positive outward science without formal promotion.

These local classes are not assumed to be disjoint. When a sentence could fit more than one class, its public status is assigned by semantic role and proof function rather than by surface phrasing:

1. an explicit contradiction of a promoted claim first flags a refutation-candidate or surface-contradiction case for SPOT audit;
2. empirical content is inspected by packet, data route, replay, comparator, and falsifier support before any prose label is trusted;
3. derived theorem content is inspected by its proof route and dependency graph;
4. admitted primitive assumptions remain axiomatic premises;
5. structural mapping and hierarchy claims remain structural even when they mention possible later use;
6. explanatory, audience-facing, or didactic sentences remain interpretive unless they assert theorem or empirical force;
7. unpromoted research-program proposals remain frontier statements;
8. pure structural or interpretive content keeps the SPOT-recorded `FRAME_ONLY` or `EXTENSION` class unless it is explicitly attached to evidence-bearing bridge material;
9. `BRIDGE_ONLY` is a stable evidence-support class for rows or packets that carry bridge support under explicit scope notes; it is not a theorem-native promotion queue.

Words such as “frontier”, “candidate”, or “projection” are therefore diagnostic signals, not the class assignment rule itself.

These prose classes are a local proof-status taxonomy, not a replacement for the canonical SPOT scientific-class and closure-verdict vocabulary. Release verdicts such as `PASS` and `FAIL_CLOSED` remain a separate verdict layer. The mapping is:

- Axiomatic statements are admitted premises or identity rules. They may sit inside a `THEOREM_NATIVE` route as named assumptions, but the axiom itself is not thereby converted into a proved theorem.
- Derived statements map to `THEOREM_NATIVE` when the support route is closed.
- Empirical statements map to `THEOREM_NATIVE` only after the source-owned packet, theorem-native trace, pinned data route, numerical parameter law, observable binding, locked held-out replay procedure, comparator audit, and falsifier gates pass.
- Structural and interpretive statements map to `FRAME_ONLY` or `EXTENSION` as recorded by SPOT unless the statement is attached to an evidence-bearing bridge row.

- Frontier statements map to `EXTENSION` until a later section or release promotes them through the same theorem-native or empirical gates used for other positive claims, or records an explicit refutation.
- Fail-closed bridge, frame, or extension lanes retain the class recorded by SPOT; closure failure by itself is not refutation.
- Statements map to `REFUTED` only when SPOT records an explicit refutation rather than a blocked promotion.

This chapter classifies the statement classes that affect theorem status, structural scope, interpretation, empirical promotion, and frontier boundaries. Operational route discipline is handled later in the theorem-roadmap and proof-machinery chapters. Editorial-bridge language is not a seventh proof class; it is an interpretive and release-consistency label for statements that connect the monograph, journal core, dossiers, and public package without adding theorem force. Operational statements are the later exact-match, held-out replay, statistical-threshold, tail-breach, comparator, falsifier, and escalation rules attached to empirical promotion. Structural statements are reviewed by scope, dependency, and source-boundary checks instead. Both kinds of control are handled after the statement class, numerical parameter law, and observable binding have already been fixed here, and neither becomes a seventh proof-status class.

The journal-core extraction is allowed to narrow this body, but not to blur these classes into one another. An interpretive sentence may explain a theorem; it may not replace the theorem. A structural remark may motivate a projection; it may not silently serve as proof of that projection.

## 18.4 Proof-bearing and nonclaim layers

The master monograph therefore separates three layers explicitly:

1. the inherited formal backbone that still bears the structure of the core theory;
2. the Core 1.3 repair layer that clarifies witness discipline, theorem fate, chronology, and publication boundary;
3. the bounded nonclaim layer that states what the present release does not yet defend, even when the broader research program still pursues it.

This separation matters for hostile review. A reviewer should be able to ask, for any sentence that sounds strong: is it axiomatic, derived, empirical, structural, interpretive, or frontier? If it is derived, what supports it? If it is empirical, what packet, data route, replay, comparator, and falsifier support it? If it is structural or interpretive, what exactly is it not claiming? If it is frontier, where is it visibly barred from positive release authority?

## 18.5 Relation between master monograph and journal core

The canonical SPOT is the scientific authority of the release, and the source-owned science corpus is the authoring and witness locus behind that projection. The master monograph is the flagship reader-facing reference because it carries the full theory, the reader ladder, the appendices, and the explicit publication boundary in one place. The *Journal Core* is the bounded article artifact. The *journal-core extraction* is the operation that selects one defensible theorem-bearing route for editorial inspection without pretending that the whole monograph is already compressed into that shorter artifact.

For that reason the extraction rule is one-way:

Master Monograph  $\longrightarrow$  Journal Core,

with explicit preservation of notation, nonclaim boundaries, and theorem scope. The extraction procedure may omit detail. The Journal Core may not enlarge the claim. The public closure boundary remains the `CORE_1_3_SCIENCE_ONLY` scope stated in Section 17. `FRAME_ONLY` closures may be `PASS` as

frame support, but they do not become standalone empirical promotions. `BRIDGE_ONLY` rows are scope-bounded evidence support rather than public promoted closure; they are not re-labelled as theorem-native `PASS` rows unless a separate source-owned theorem-native packet is created and passes the canonical gates. `EXTENSION` and `REFUTED` lanes remain outside public `PASS` authority.

## 18.6 Reader-facing consequence

This chapter exists because scientific rigor and readability do not pull in opposite directions here. The more explicit the dossier becomes about hierarchy, statement class, and proof boundary, the easier it becomes for three readers at once:

- the editor who needs a clean overview of what is actually defended;
- the formal reader who needs exact scope and dependency discipline;
- the non-specialist scientific reader who needs a coherent narrative entry point before diving into the full technical shell.

**Bridge to the next chapter.** With the statement classes, hierarchy boundary, and master-to-derivative relation fixed, the next chapter narrows further to one task only: the shortest lawful proof-navigation route through the load-bearing theorem spine.

## 19 Theorem Roadmap and Proof Navigation

This chapter is the clearest compact map of the proof-bearing route. Its task is to show where the load-bearing theorem spine lies, which statement classes carry which scientific burden, and where a hostile reviewer should probe first. It does not replace the formal chapters or the reader guide. The reader guide owns entry routes; this chapter owns proof navigation.

### 19.1 Load-bearing theorem spine

At the level of the bounded journal-core claim, the present release depends on a relatively small load-bearing spine even though the monograph itself remains much larger. The schematic route below is compact rather than proof-theoretic minimality in the strict sense: it keeps the residue and rebirth branch visible because the defended article-level theorem classifies post-collapse identity, not death alone. The spine can therefore be stated schematically as dependency notation rather than object-language implication. To avoid making the route look like a proof calculus, the support map is written as labelled review edges:

1. **Edge 1.** Axiom 3.2 together with Definitions 12.1, 12.3, and 12.4 supports Source Theorem 3, and Source Theorem 3 supports Lemma 1.
2. **Edge 2.** Axiom 3.3 together with Definition 12.6 supports Source Corollary 3.2, and Source Corollary 3.2 supports Lemma 3.
3. **Edge 3.** Lemma 1 together with Definition 12.5 supports Lemma 2.
4. **Edge 4.** Lemma 2 is the sole premise recorded for Theorem A in the canonical proof-obligation registry.

Definitions in this list are prerequisite nodes, not independent theorem premises. The word “supports” marks an admitted release support step rather than a primitive implication operator of OC.

Lemma 3 remains an admitted rebirth-control obligation in the surrounding roadmap, but it is not a premise of Theorem A in the canonical proof-obligation registry. Its load-bearing role is different: it supports the companion rebirth-control statement, Proposition B. Whenever a compressed article proof discusses the death–residue–rebirth package in one paragraph, Lemma 3 should be read as supporting Proposition B, while Theorem A itself remains the death-and-residue classification routed through Lemma 2.

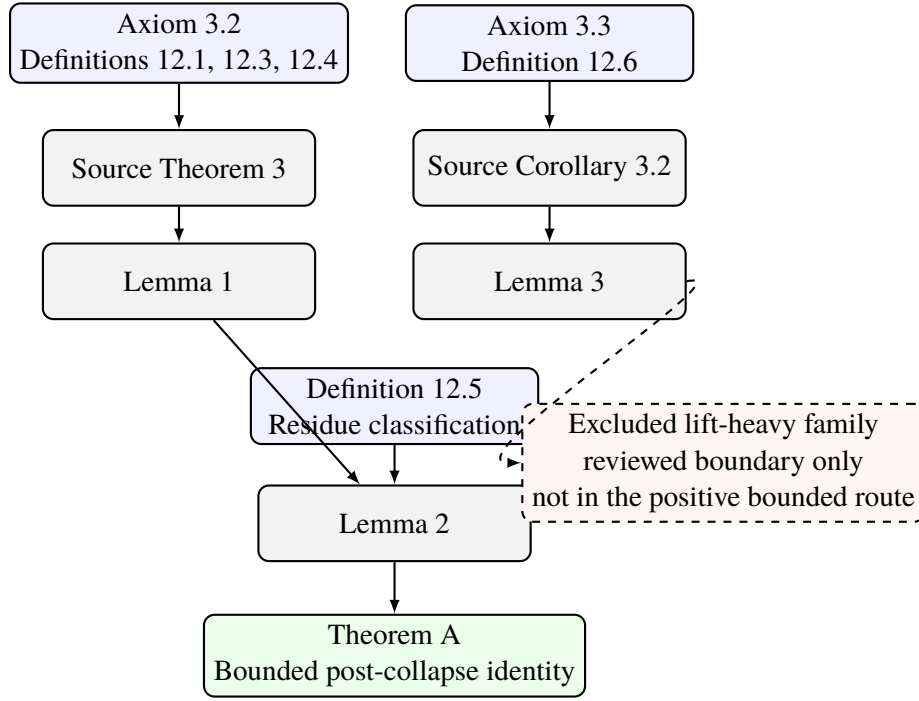
**Proposition B (article-local rebirth control).** Within the article-level extraction, after  $K$  has lawfully collapsed so that  $\Omega(K) = \emptyset$  and  $k(K, t) = 0$ , a later object  $K'$  is classified as rebirth if and only if  $K'$  is live, is numerically distinct from the dead original continuum  $K$ , appears either in the same-embedding branch or in an explicitly witnessed related-embedding branch, inherits at least one declared residue support element, and satisfies the relevant birth conditions. The identity rule supplies the reason this non-identity condition is forced rather than optional. This proposition is a companion control statement for the rebirth branch; it is not a premise of Theorem A.

In this roadmap, the *Source* prefix marks a witness projection from the audited public LaTeX source corpus and archival compiled PDF described in the source-audit appendix. Such rows are carried into Core 1.3 as source-witnessed support nodes and are inspected through the canonical proof-obligation registry; they are not newly asserted primitive theorems of the article-level extraction. The unprefixes lemmas, Theorem A, and Proposition B are the article-level derived rows assembled from that witnessed support. Theorem A uses the death-and-residue spine; Proposition B uses the rebirth-control branch.

Axioms, source theorems and corollaries, lemmas, and Theorem A carry proof-force nodes; definitions fix the terms used by those nodes.

The point of making that chain explicit is not to pretend that the whole framework is reducible to one diagram. The point is to show that the bounded claim has an auditable positive core rather than a cloud

of motivational prose. Figure 1 also marks one excluded lift-heavy temptation route as a hostile-review boundary; it is reviewed precisely so that it cannot be mistaken for admitted support in the positive chain. The Corollary 3.2 / Lemma 3 branch is shown as an admitted rebirth-control branch for Proposition B beside the theorem route rather than as an antecedent of Theorem A. In this roadmap the related-embedding branch is represented by one auditable article-local witness, not by an exhaustive list of every admissible source-level related-embedding witness. This is the same branch denoted  $\mathcal{M} \sim_{\text{res}} \mathcal{M}'$  in Section 20: a named structure-preserving map  $h : \mathcal{M} \rightarrow \mathcal{M}'$ , a residue element  $r \in R_{\text{res}}(K)$ , admissible-region membership  $h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(\mathcal{M}')$ , and preservation of the threshold and admissibility predicates on the residue-bearing substructure. Section 20 and the journal core state that witness test explicitly.



Solid arrows: admitted theorem route and admitted rebirth-control branch  
Dashed arrows: excluded temptation route reviewed but not admitted

Figure 1: Load-bearing theorem spine of the bounded Core 1.3 claim. Solid arrows mark admitted support. The positive Theorem A route terminates at Lemma 2; the Lemma 3 branch remains visible as admitted rebirth-control support beside that route. The dashed branch marks an excluded lift-heavy route that is reviewed as a boundary condition rather than admitted into the positive proof. The larger monograph remains necessary because the surrounding chapters fix the ontology, operator shell, K-level doctrine, collapse grammar, empirical bar, and audit layers that keep this spine scientifically interpretable and externally citable.

## 19.2 What surrounds the spine

The theorem spine has a small proof dependency, but it is interpreted and audited inside the wider manuscript in five distinct ways:

1. the formal doctrine chapters fix admissibility, thresholds, operators, levels, branching, and falsifiability;
2. the consistency and source-boundary chapters fix statement class, theorem fate, and publication scope;

3. the proof-machinery chapter and appendix fix revision, comparator, compression, and proof-obligation discipline;
4. the operational and empirical chapters fix promotion, replay, and falsifier discipline;
5. the appendices keep the audit and atlas burden inside the same volume.

That is why the monograph is allowed to be larger than the defended article without becoming rhetorically loose. These surrounding chapters are publication and audit scaffolding, not additional premises of Theorem A.

### 19.3 Statement classes and burden

The table below is a proof-navigation map for reviewer use. It follows the statement-class taxonomy of Section 18. The proposition row is a navigation-only subdivision of derived statements, not a new proof-status class. Operational rules remain a downstream inspection burden rather than a separate proof-status class.

| Statement class             | Scientific burden                                                                                                                                                                              | Where to inspect it                                                               |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Axiomatic statements        | Fix primitive admissibility, identity, or threshold conditions.                                                                                                                                | Axioms Appendix, Notation Appendix, and Sections 8–12.                            |
| Derived theorems and lemmas | Carry the proof-bearing positive body.                                                                                                                                                         | Sections 5, 12, 19, Reviewer Navigation Matrix, and Proof Machinery Appendix.     |
| Article-level propositions  | Carry companion control statements that are derived from the same audited support but are not premises of Theorem A. Proposition B is the rebirth-control example.                             | Sections 19, 20, and Journal Core Bridge Appendix.                                |
| Structural statements       | Explain hierarchy, module layout, and relation between levels or artifacts.                                                                                                                    | Sections 10–15, Section 16, Section 17, and Section 18.                           |
| Interpretive statements     | Explain meaning, nonclaims, and downstream boundaries.                                                                                                                                         | Sections 6, 20, 25, Source Audit Appendix, and Reviewer Navigation Matrix.        |
| Empirical statements        | Require a source-owned domain packet, theorem-native trace, numerical parameter law, observable binding, pinned data route, held-out replay, comparator audit, and falsifier before promotion. | Sections 21–22 and Appendices G–L.                                                |
| Frontier statements         | Mark lawful future work without positive release authority.                                                                                                                                    | Section 25, Appendix R, and the frontier rows in the source-owned science corpus. |

Table 3: Statement classes and their inspection burden in Core 1.3. The table does not replace the formal chapters; it simply prevents proof, interpretation, empirical promotion, and frontier language from quietly changing roles. Replay, comparator, falsifier, and escalation rules are downstream control obligations attached to those classes.

All inspection targets in Table 3 are label-checked against the English master source. If a release wrapper changes one of those labels, the table must be regenerated or patched before the roadmap is treated as a mechanically followable audit route.

### 19.4 Shortest hostile-review checklist

If a reviewer wants the most efficient attack sequence, the logically hardest checks remain the following:

1. verify that collapse is tied to loss of admissible realization rather than to loose metaphor;

2. verify that the irreversibility rule forbids restoration of the original continuum as numerically the same entity;
3. verify that residue and rebirth are controlled classifications rather than narrative labels;
4. verify that no excluded lift-heavy theorem family is smuggled back into the positive bounded route;
5. verify that the journal-core extraction says nothing stronger than the source-owned science corpus and canonical SPOT permit; the flagship monograph is then checked only as a derived comparison surface;
6. verify that the K-level doctrine, especially  $K_{11}$  and  $K_{12}$ , is structurally justified rather than rhetorically inflated.

If the release fails there, later prose cannot rescue it. If it survives there, a slower full reading becomes worth the effort.

**Bridge to the next chapter.** With the proof route mapped, the next chapter changes mode deliberately. It no longer navigates the spine abstractly; it shows how the formal grammar should be read through worked examples, counterexamples, and bounded interpretive checks.



## 20 Worked Examples, Counterexamples, and Reader-Oriented Interpretive Checks

This chapter turns the formal grammar into bounded reading moves. Its purpose is to show what the theorem route licenses, what it forbids, and what would count as a real counterexample at the release boundary. It does not replace the proof route with storytelling; it tests whether the reader can apply the proof route without hand-waving. The examples therefore show how the formal grammar should be read, why the K-level doctrine is not an arbitrary count, and how the master volume disciplines the shorter journal-core extraction. In these examples, “declared” means named inside the local example data or in a cited support route before the classification or reduction test is applied; it never means supplied afterward to rescue a desired reading.

### 20.1 Worked Example 1: collapse, residue, and rebirth in a minimal admissible-state model

Consider a continuum  $K$  whose admissible realization at time  $t$  is described by

$$\Omega(K, t) = \{x \in [0, 1] \mid \phi(x, t) \leq \vartheta(t)\},$$

where  $\phi(x, t)$  is an incompatibility load and  $\vartheta(t)$  is the currently lawful scalar threshold. Suppose that at an earlier stage  $t_0$  the set is nonempty, for example because  $\phi(x, t_0)$  is below threshold on a compact interval  $[a, b] \subseteq [0, 1]$ .

Now assume that the embedding regime changes, the threshold tightens, or the internal evolution operator drives the incompatibility load upward so that the declared collapse time is  $t_1 = t^*$ , the approaching path satisfies the one-sided source condition  $\lim_{t \nearrow t^*} k(K, t) = 0$ , and at  $t_1$ ,

$$\forall x \in [0, 1] \quad \phi(x, t_1) > \vartheta(t_1).$$

Then  $\Omega(K, t_1) = \emptyset$ . In the language of the source-native death condition, this full witness package is not merely a bad regime or a temporarily unproductive phase. Because the pointwise admissibility indicator  $\chi_\Omega(s_K(t_1); K, t_1)$  is then zero, the continuumness measure also collapses to  $k(K, t_1) = 0$  under the unified Core 1.3 definition. The implication is immediate because  $k$  is defined as the product whose first factor is the pointwise admissibility indicator. It is therefore the lawful death of the continuum under the current admissibility regime.

The important interpretive point is what follows next. Suppose that after collapse, the embedding space still preserves some structured trace: constraints, marks, gradients, memory residues, or other inherited structure that a later continuum can exploit. That trace is not the original live continuum, because the admissible realization has already become empty. It is residue: a surviving structural aftermath without live identity.

Finally suppose that at time  $t_2 > t_1$ , after changes in thresholds, embedding, or generative conditions, a new nonempty admissible realization appears for some later object  $K'$ . The lawful classification is not that the original continuum has revived. The irreversibility rule forbids that reading. The correct classification is rebirth only when the full predicate of Definition 12.6 is met: the later live continuum appears in the same or a related embedding space, inherits at least one structural element from residue, and satisfies the relevant birth conditions. The separate identity rule in Section 12 then says why that later live continuum is numerically new rather than a restoration of the original. If a route uses the related-embedding option rather than the same embedding space, the relation must be tested against the residue and threshold clauses rather than inferred after the fact to save a desired rebirth reading. In this article-level use, one auditable compatibility witness for “related” is the explicitly scoped condition. Let  $\Omega_{\text{amb}}(\mathcal{M}')$  denote the ambient admissible region declared for the proposed embedding space, and let the quantified  $K'$  range over declared later-candidate continua in  $\mathcal{M}'$ :

$$\begin{aligned} \exists K' \exists h : \mathcal{M} \rightarrow \mathcal{M}' \exists r \in R_{\text{res}}(K) \\ h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(\mathcal{M}') \text{ and } \text{Pres}_{\Theta, \Omega}(h, r; K, K'). \end{aligned}$$

For this worked example,  $\mathcal{M} \sim_{\text{res}} \mathcal{M}'$  means exactly that the displayed line is shorthand for the full witness package described below. That package includes the residue-bearing substructure, the active residue and target index families, the index bijection, and the induced transport operator; without those data the displayed formula is not a standalone formal predicate. Here  $K'$  is the declared later candidate tested for live admissibility,  $\mathcal{M}$  is the embedding space of the collapsed continuum,  $\mathcal{M}'$  is the embedding space proposed for the later continuum,  $R_{\text{res}}(K)$  is the residue set of the collapsed continuum, and  $\mathcal{R}_K(r) \subseteq R_{\text{res}}(K)$  denotes the declared residue-bearing substructure containing  $r$ : a local residue subobject named in the witness data on which the transported threshold and admissibility data are declared. If no such substructure is declared, the related embedding branch is not available in this example.  $\text{Pres}_{\Theta, \Omega}$  means that the named map  $h$  preserves  $\mathcal{R}_K(r)$  and the threshold and admissibility predicates used to test the later live continuum. This witness is not by itself the full rebirth predicate; it supplies the related-embedding clause inside that predicate. This is the example-specific relation denoted by  $\mathcal{M} \sim_{\text{res}} \mathcal{M}'$ . The formal rebirth predicate remains Definition 12.6; the source predicate may admit other properly declared related-embedding witnesses; this worked example uses the stricter  $\sim_{\text{res}}$  witness so the related branch is auditable rather than assumed. The witness data also include the active residue-threshold index family  $I_K^{\text{res}}$ , the corresponding active target subfamily  $I_{K'}^h \subseteq I_{K'}$ , and a declared bijection  $\eta_h : I_K^{\text{res}} \xrightarrow{\sim} I_{K'}^h$ . The induced pushforward  $h_*$  applies  $h$  to the residue state and transports each named residual threshold by  $\eta_h$ . For this example,  $\text{Pres}_{\Theta, \Omega}$  is the following declared predicate:

$$\begin{aligned} \text{Pres}_{\Theta, \Omega}(h, r; K, K') \text{ holds exactly when} \\ h \text{ preserves } \mathcal{R}_K(r), \\ h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(\mathcal{M}'), \\ \mathcal{D}_{\Theta, \Omega}^h(K'; h(r)) = h_* \mathcal{D}_{\Theta, \Omega}^{\text{res}}(K; r). \end{aligned}$$

Here  $\mathcal{D}_{\Theta, \Omega}$  denotes the local admissibility data: the pointwise admissible-membership indicator together with the active residual-threshold signs and saturated constraints. The term  $\mathcal{D}_{\Theta, \Omega}^{\text{res}}(K; r)$  is the same data restricted to the residue-index family  $I_K^{\text{res}}$ , while  $\mathcal{D}_{\Theta, \Omega}^h(K'; h(r))$  is the target-side record restricted to  $I_{K'}^h$ . More explicitly, for a finite active threshold subfamily  $J \subseteq I_Y$  of  $\Theta_Y = \{f_i^Y\}_{i \in I_Y}$  at a tested object  $Y$ , point  $y$ , and test time  $\tau$ , the restricted typed record is

$$\mathcal{D}_{\Theta, \Omega}^J(Y; y) = (\chi_{\Omega}(y; Y, \tau), (\text{sgn } f_i^Y(y))_{i \in J}, \{i \in J : f_i^Y(y) = 0\}).$$

The unreduced symbol  $\mathcal{D}_{\Theta, \Omega}(Y; y)$  abbreviates the special case  $J = I_Y$ ; it is not the record compared in the related-embedding witness. Here  $I_K^{\text{res}}$  is the subfamily of active thresholds declared on the residue-bearing substructure, and  $I_{K'}^h$  is the corresponding active target subfamily used to test  $\Omega(K')$  along this witness. The transport satisfies  $f_{\eta_h(i)}^{K'} \circ h = f_i^K$  on  $\mathcal{R}_K(r)$ . Consequently the saturated set is transported as

$$\eta_h(\{i \in I_K^{\text{res}} : f_i^K(r) = 0\}) = \{j \in I_{K'}^h : f_j^{K'}(h(r)) = 0\}.$$

The equality therefore says that the named map transports the residue-bearing threshold and admissibility data used to test the later live continuum; it is not an informal resemblance clause. The pushforward  $h_*$  is a witness-local transport-plus-target-evaluation operator rather than a source-only pushforward: it applies  $h$  to the residue state, carries each named residual threshold  $f_i$  to the corresponding declared residual threshold  $f'_{\eta_h(i)}$ , preserves the sign relation  $\leq 0$ , preserves the transported saturated subset, and evaluates the target membership component by the separate predicate  $h(r) \in \Omega(K')$ .

This example is deliberately modest. It does not claim that every physically, biologically, or socially interesting regime can already be reduced to such a scalar picture. What it does show is how the theorem grammar should be read once the support chain is accepted. A hostile reviewer may reject the premises or the admissibility logic. What the reviewer should not be allowed to say is that the release leaves the post-collapse identity consequences undefined.

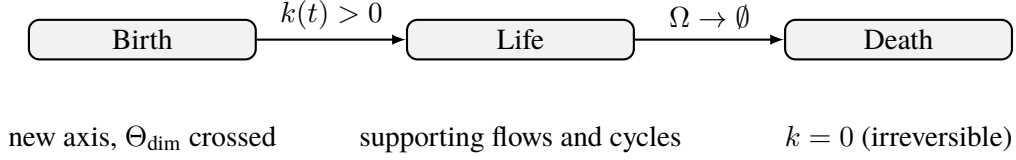


Figure 2: Minimal life-cycle schematic for the bounded collapse grammar. The point of the figure is not to offer a domain-specific simulation but to show the exact interpretive burden of the formal rules: in this minimal model, the object remains live only while the admissible region is nonempty, the required cycles remain present, and  $k(K, t) > 0$ , with supporting flows not dominated by destructive flows on the relevant cycles. Admissible-state collapse kills it, and any later live continuum is classified as rebirth rather than persistence only when the residue, embedding-space, and birth-condition clauses of Definition 12.6 are satisfied and the identity rule rules out restoration of the dead original.

## 20.2 Worked Example 2: why $K_{11}$ and $K_{12}$ are not decorative

One of the easiest ways to make a foundational framework look more sophisticated than it is would be to keep extending its hierarchy labels without necessity. Core 1.3 blocks that temptation by forcing the reader to ask what would break if the hierarchy stopped at  $K_{10}$ .

At the level of  $K_9$ , the theory already treats theories, paradigms, and ontological schemes as structured objects. At  $K_{10}$ , it treats model-of-models recursion and self-reference as part of the formal landscape. That still leaves two scientific problems open.

1. How do operator families, meta-logics, and model-spaces themselves evolve as structured continua rather than as static meta-language?
2. How are global coherence conditions expressed once multiple such higher-order objects coexist and partially constrain one another?

Stopping at  $K_{10}$  would force those problems back into informal commentary. The public source corpus, however, already contains the beginnings of explicit treatment for both. That is why  $K_{11}$  and  $K_{12}$  are retained: not because bigger counts sound more impressive, but because the framework has already crossed the line at which the evolution of theory-space and the coherence of higher-order structures have to be represented formally.

The reader-oriented reduction check here is explicit and bounded. It is an audit heuristic for seeing what a  $K_{10}$ -only reduction would have to preserve, not a replacement for the source-owned irreducibility dossier. In this paragraph,  $\text{Car}(K_n)$  denotes the carrier set of objects assigned to level  $K_n$ , not the level label itself. Let  $\mathcal{U}_{11,12} = \text{Car}(K_{11}) \sqcup \text{Car}(K_{12})$  denote the disjoint union of objects assigned to the two tested upper-level roles. A  $K_{10}$ -only reduction is a tuple  $(\rho, \equiv_\rho, \rho_*, \lambda)$  whenever time-sensitive requirements are present, where  $\rho : \mathcal{U}_{11,12} \rightarrow \text{Car}(K_{10})$ ,  $\equiv_\rho \subseteq \mathcal{U}_{11,12} \times \mathcal{U}_{11,12}$  is the declared compatibility relation induced by the translation, and  $\rho_*$  is shorthand for the family of transports  $\{\rho_{*,X}\}_X$  that sends upper-level requirements to their proposed  $K_{10}$  counterparts. The component  $\lambda$  is the family of order-preserving interval transports  $\{\lambda_X : I_X \rightarrow I_{\rho(X)}\}_X$ . If no time-sensitive requirement is being tested,  $\lambda$  is the empty auxiliary family. Despite the glyph,  $\equiv_\rho$  is not assumed to be an equivalence relation in this test; it is only a typed compatibility predicate unless reflexivity, symmetry, and transitivity are supplied separately. For each object  $X$ , let  $\mathcal{B}(X) \subseteq \mathcal{L}_{\text{OC}}(X)$  be a finite typed set of local OC requirements, and let  $\mathcal{B}^\wedge(X)$  be its finite conjunction-closure. The allowed atom constructors are *Adm*, *Cont*, *Op*, *Id*, and *Fals*, respectively standing for admissibility, positive continuumness, operator-obligation, identity-preservation, and falsifier requirements. The notation  $\mathcal{L}_{\text{OC}}(X)$  names the finitely parameterised requirement language generated by those atom types for  $X$ , while the tested set  $\mathcal{B}(X)$  remains finite. The notation  $\mathcal{L}_{10}(\rho(X))$  names the corresponding proposed  $K_{10}$  target language. The transport  $\rho_{*,X} : \mathcal{L}_{\text{OC}}(X) \rightarrow \mathcal{L}_{10}(\rho(X))$  maps each typed atom to its proposed  $K_{10}$ -language counterpart and extends to finite conjunctions by preserving their typed clauses. For the reduction test, the satisfaction

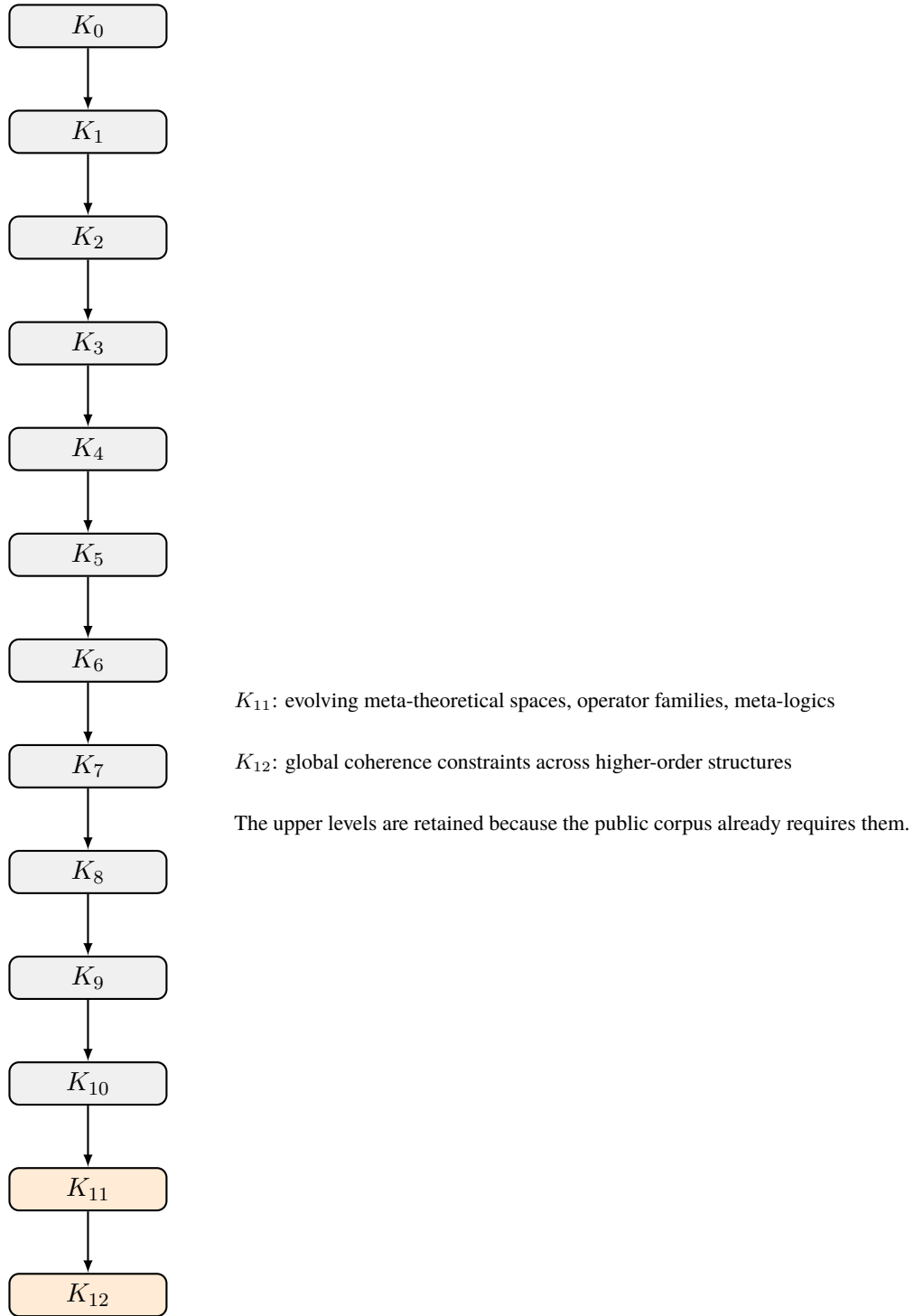


Figure 3: Extended vertical hierarchy through  $K_{12}$ . The added upper levels are justified by the need to represent evolving meta-theoretical spaces and their global coherence conditions rather than by decorative level inflation.

clauses are the following bounded clauses, where  $I_X$  is the declared nonempty finite test interval for  $X$  and  $\varepsilon \in (0, 1)$ . Let  $\text{Guard}_X(O, S, T)$ ,  $\text{IdPres}_X(\iota)$ , and  $\text{Kill}_X(\varphi)$  denote, respectively, the route-local operator-admissibility guard, identity-preservation predicate, and active falsifier predicate recorded in the proof-obligation sheet for the tested claim:

$$\begin{aligned} X \models \text{Adm}(\omega) &\iff \omega \in \Omega(X), \\ X \models \text{Cont}(\varepsilon) &\iff \forall t \in I_X, k(X, t) > \varepsilon, \\ X \models \text{Op}(O : S \rightarrow T) &\iff S \subseteq \text{Dom}_X(O), O(S) \subseteq T, \text{ and } \text{Guard}_X(O, S, T) = 1, \\ X \models \text{Id}(\iota) &\iff \text{IdPres}_X(\iota) = 1, \\ X \models \text{Fals}(\varphi) &\iff \text{Kill}_X(\varphi) = 0. \end{aligned}$$

Finite conjunctions are satisfied exactly when each typed clause in the conjunction is satisfied. Here  $X \models r$  means satisfaction of that typed requirement under the admissibility, continuumness, operator-obligation, identity, and falsifier clauses of the surrounding proof route. An operator-obligation atom is satisfied only when the relevant operator is defined on the declared source and target objects and the mapped obligation preserves the operator's stated domain, codomain, and admissibility guard; otherwise it fails. The expression  $\rho(X) \models \rho_{*,X}(r)$  means satisfaction of the translated requirement in the target language, using  $I_{\rho(X)} = \lambda_X(I_X)$  for time-sensitive atoms. The relation  $\equiv_\rho$  is a typed compatibility relation on tested objects; if it is used as a quotient, reflexivity, symmetry, and transitivity must be supplied explicitly. In this section it is used only as a burden-preservation compatibility test, not as a decorative quotient. The reduction is lossless only if

$$\forall X \in \mathcal{U}_{11,12} \forall r \in \mathcal{B}^\wedge(X) : \quad X \models r \iff \rho(X) \models \rho_{*,X}(r),$$

and only if  $X \equiv_\rho Y$  never declares compatible two objects whose admissible-state, continuumness, operator-obligation, identity, or falsifier burdens differ. This makes the compatibility relation a preservation constraint rather than a decorative quotient. In the reduction test, source-object compatibility and source-to-target preservation are separate checks. The relation  $\equiv_\rho$  constrains compatibility among source objects only; the pair  $X, \rho(X)$  preserves identity only when the admissible-state, continuumness, operator-obligation, identity-preservation, and falsifier burdens of  $X$  match their translated  $K_{10}$  counterparts under  $\rho_{*,X}$ . Residue language is only an analogy here, not the identity criterion for reduction. The displayed condition is the local, formally quantified preservation test. It is necessary for any reduction claim, but this worked example also records a separate non-definitional coherence audit. For a finite family of tested objects  $F \subseteq \mathcal{U}_{11,12}$ , chosen requirements  $r_X \in \mathcal{B}^\wedge(X)$ , and a declared common meta-language  $\mathcal{L}_F$ , the audit must also declare translations  $\iota_{F,X} : \mathcal{B}^\wedge(X) \rightarrow \mathcal{L}_F$ . The family-level test is the finite conjunction  $\bigwedge_{X \in F} \iota_{F,X}(r_X)$  inside  $\mathcal{L}_F$ , together with any cross-object coherence clauses asserted for that family. In this heuristic check, a  $K_{10}$ -only reduction is flagged if it preserves every object locally while losing a declared cross-framework coherence obligation carried by  $K_{12}$ . Thus, in the audit sense used here, “reducible without loss” would require preserving inside  $K_{10}$  alone all three burdens that the release assigns to the upper levels:

1. explicit evolution of meta-theoretical objects as lawful structured states,
2. a distinction between local meta-model recursion and global cross-framework coherence conditions,
3. and the admissibility or collapse conditions attached to those higher-order objects.

Accordingly, a critic would need to provide one explicit  $K_{10}$ -only reduction map that preserves those burdens simultaneously. If such a map exists, the upper levels should be demoted. If, however, the framework genuinely needs one stratum for evolving meta-theoretical objects and another for cross-framework coherence conditions, then the upper levels are not optional. Core 1.3 takes the second position and, crucially, states the preservation test under which that position could fail. The role split is explicit:  $K_{11}$

carries the evolution of meta-theoretical objects and operator/model spaces, while  $K_{12}$  carries global cross-framework coherence, boundedness, and integrability conditions. The same preservation criterion is used below whenever the table speaks of a later continuum being numerically identical to the dead original. In this section that means preservation of original live identity: the original admissible-state and continuumness conditions must remain live, not merely that a later object can be related to residue.

### 20.3 Interpretive Check 3: how the journal core is lawfully extracted from the monograph

Readers often distrust large monographs because they have seen too many cases in which a short article quietly says something stronger than the full technical work behind it. Core 1.3 is designed to make that drift visible and prevent it.

The lawful extraction rule is asymmetric:

Master Monograph  $\longrightarrow$  Journal Core.

The journal core may compress, but it may not enlarge. That principle has several operational consequences.

1. The journal core can suppress much of the didactic shell, but it must preserve the exact positive theorem scope.
2. It can omit broad projection discussions, but it may not promote any projection into accomplished proof.
3. It can shorten the literature screen, but it may not falsify the priority or comparator boundary set in the master.
4. It can move quickly through notation once the master has fixed it, but it may not invent a second notation universe.

This is not an additional scientific premise. It is a release-governance rule for packaging and review. The monograph is where the broader framework is taught, bounded, and audited. The journal core is where one narrower result is exposed for fast hostile inspection. If the two diverge, the release should be treated as broken.

### 20.4 What would count as a real counterexample

To keep the examples falsifiable, it is useful to state what kind of observation or formal result would genuinely pressure the present release.

### 20.5 Why examples belong in a strict scientific monograph

Examples are often treated as a concession to weaker readers. In foundational work that is a mistake. A disciplined example does at least three scientific jobs at once. It tests whether the formal grammar can be applied without hand-waving. It reveals where an apparent theorem is only an interpretive gloss. And it gives editors and reviewers a compact way to see what would count as a counterexample before they invest in line-by-line reconstruction.

That is why the present section is included inside the master monograph rather than hidden in an outreach note. A large theory becomes more reliable when the examples are good enough to expose its failure modes, not when the examples are removed in the name of austerity.

**Bridge to the next chapter.** The next three chapters shift from interpretive checks to closure discipline: how promoted packets are operationalized, how empirical replay is bounded, and how proof machinery, competition, and compression stay synchronized with the science verdict.

| Observation or formal result                                                                                                                                                             | Immediate scientific consequence                                                                                                                                                                                                                                                                                                            | Release-level implication                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| A nonempty admissible realization remains after the alleged collapse event.                                                                                                              | The death classification governed by Definition 12.1, Axiom 3.2, and Source Theorem 3 in Section 19 is not yet lawful.                                                                                                                                                                                                                      | Recheck threshold logic, collapse statement, and source witness used in the positive body.                                        |
| A later live continuum appears after lawful death and can still be proved numerically identical to the dead original continuum.                                                          | The irreversibility route fixed by Axiom 3.3, Definition 12.6, Source Corollary 3.2, and Lemma 3 in Section 19 is undermined.                                                                                                                                                                                                               | Revise or withdraw the irreversibility and rebirth-control branch; revise Theorem A only if the Lemma 2 route is also challenged. |
| Residue can be shown to preserve full live identity rather than only structural aftermath.<br>The journal core requires an excluded lift-heavy theorem family to sustain its main claim. | The classification boundary fixed by Definition 12.5, Lemma 2, and Theorem A in Section 19 is too weak or too broad as stated.<br>This schematic family-level objection targets the excluded lift-heavy theorem family marked in Figure 1; if that family is required, the extraction violates the support boundary stated in this section. | Rewrite the residue boundary before public release.<br>Narrow the article or restore support explicitly in the master first.      |
| $K_{11}$ and $K_{12}$ are shown reducible without loss to $K_{10}$ .                                                                                                                     | The upper-level doctrine stated here and in Section 18 is overstated under its own preservation test.                                                                                                                                                                                                                                       | Demote the hierarchy or re-prove the necessity of the upper layers.                                                               |

Table 4: Examples of observations or formal objections that would genuinely change the scientific status of the present release. The table is meant to preserve falsifiability, not to immunize the framework against criticism.

## 21 Operationalization Program and Predictive Promotion Discipline

This chapter turns the theorem-bearing core into lawful operational discipline. Its job is to state how a packet becomes promotable, how bridge evidence and claim scope are bounded, and where operational language must stop. Comparator verdicts are summarized here only through promotion discipline and are handled explicitly in Chapter 25 and Appendix R. This chapter does not carry the empirical replay burden or the proof-machinery burden in full; those tasks are handled by Chapter 22 and Chapter 23.

This chapter is projected from the canonical science SPOT surface `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json` (surface id `OC_CORE_1_3_SCIENCE_SPOT`). The generator and validator make the alignment auditable: operational status, bridge evidence, and closure verdicts are checked against the same source-owned science corpus rather than edited as independent prose claims. The generated section below prints the exact `metadata.repo_sha` and `metadata.ts_utc` for the SPOT revision used in this build.

SPOT revision: repo SHA `0dd049c48def00df8a7d50bef07abb127a3c4fb3`; timestamp `2026-04-22T09:23:17Z`.

### 21.1 Why the present release separates explanatory force from predictive promotion

The canonical science SPOT distinguishes explanatory reach from lawful promotion. Empirical and predictive outward claims are promoted only when a domain closes one continuous trace from the kernel and K-level theorem spine through observables to held-out evidence and explicit falsifiers; theorem-native mathematical claims close by proof route rather than by measurement surface.

### 21.2 Current release truth

The canonical SPOT checks 5 core lanes. At present 5 lanes satisfy the full promotion bar, while 0 lanes remain fail-closed. That PASS count applies only to the five core theorem-native lanes inside `CORE_1_3_SCIENCE_ONLY`; frame-only, extension, and refuted rows remain support, frontier, or negative-audit rows rather than public positive closure claims. Mathematics is the formal anchor for exact proof obligations, while physics, chemistry, biology, and Systems / Civilizational projection now clear theorem-native closure on their locked held-out routes. Within the declared `CORE_1_3_SCIENCE_ONLY` scope, the hostile-review backlog is closed and the Core 1.3 science verdict is PASS.

### 21.3 Closed canon and frontier workbench

The program therefore runs in two layers. Closed canon admits theorem-native results only. Frontier workbench holds broader unified-science campaigns under explicit promotion gates and may not be advertised as closed science. The closed canon currently tracks 22 entities, while the frontier workbench tracks 2 active campaigns. Execution order remains `DOMAIN_PRIORITY_BOARD_ONLY`, and Phase 1 is tracked as a first-class dossier surface rather than as drifting editorial placeholder text.

### 21.4 Closure economics and anti-cycle law

Priority is ranked by  $(\text{closure impact} \times \text{dependency centrality} \times \text{expected information gain}) / (\text{compute cost} + \text{data cost} + \text{measurement cost} + \text{editorial cost})$ . Anti-cycle rule: A method class may not be repeated unless a new formal premise, new dataset family, or new observable family enters the claim.

### 21.5 Program board

| Domain      | Formal order | Cost order | Gain per cost | Phase                         | Launch rule        |
|-------------|--------------|------------|---------------|-------------------------------|--------------------|
| Mathematics | 0            | 0          | 116.666667    | PHASE_1_STABILIZE_CLOSED_CORE | MAINTAIN_ALWAYS_ON |



| Domain                              | Formal order | Cost order | Gain per cost | Phase                     | Launch rule                                                                             |
|-------------------------------------|--------------|------------|---------------|---------------------------|-----------------------------------------------------------------------------------------|
| Physics                             | 1            | 1          | 64.8          | PHASE_3_PHYSICS_CLOSURE   | START_IMMEDIATELY_AFTER_PHASE_1                                                         |
| Chemistry                           | 2            | 2          | 32.0          | PHASE_4_CHEMISTRY_CLOSURE | START_AFTER_PHYSICS_THEOREM_PACKET_LOCKS                                                |
| Biology                             | 3            | 4          | 17.818182     | PHASE_5_BIOLOGY_CLOSURE   | START_AFTER_CHEMISTRY_THEOREM_PACKET_LOCKS                                              |
| Systems / Civilizational projection | 4            | 3          | 16.615385     | PHASE_6_SYSTEMS_CLOSURE   | ALLOW_PARALLEL_DATA_PREP_AFTER_CHEMISTRY_BUT_FINAL_PROMOTION_AFTER_BIOLOGY_FORMAL_ROUTE |

## 21.6 Domain order and promotion rule

The formal derivation order remains physics, chemistry, biology, and then Systems / Civilizational projection. The empirical cost order lets cheaper systems data preparation run early, but final systems promotion still waits for the biology formal route to lock. Bridge evidence may support the research program, but it cannot by itself promote a domain to closed science.

## 21.7 Operationalization table

| Domain                              | Class          | Trace          | Evidence          | Verdict | Next lawful action         |
|-------------------------------------|----------------|----------------|-------------------|---------|----------------------------|
| Mathematics                         | THEOREM_NATIVE | TRACE_COMPLETE | EVIDENCE_COMPLETE | PASS    | MAINTAIN_REPLAY_DISCIPLINE |
| Physics                             | THEOREM_NATIVE | TRACE_COMPLETE | EVIDENCE_COMPLETE | PASS    | MAINTENANCE_ONLY           |
| Chemistry                           | THEOREM_NATIVE | TRACE_COMPLETE | EVIDENCE_COMPLETE | PASS    | MAINTENANCE_ONLY           |
| Biology                             | THEOREM_NATIVE | TRACE_COMPLETE | EVIDENCE_COMPLETE | PASS    | MAINTENANCE_ONLY           |
| Systems / Civilizational projection | THEOREM_NATIVE | TRACE_COMPLETE | EVIDENCE_COMPLETE | PASS    | MAINTENANCE_ONLY           |

## 21.8 Per-domain status

**Mathematics.** Wave WAVE\_0\_ANCHOR currently places the domain in class THEOREM\_NATIVE, with trace status TRACE\_COMPLETE, evidence status EVIDENCE\_COMPLETE, and verdict PASS. Benchmark families cover TIER1\_AIMATH\_PDF\_013\_Q010, strictclosureresulttledger, counterexamplesearch corpus, deterministicreplaytraces. Measured outputs track epsilon\_star, minimal\_denominator, minimal\_augmentation\_rank, component\_fiber\_count, move\_connectivity\_or\_degree\_bound, strict\_terminalized\_total, counterexample\_executed\_total, python\_evidence\_executed\_total. Measurement schema: Replay declared mathematical packets deterministically against the strict closure ledger and counterexample traces. This formal lane does not introduce a separate external theorem-to-observable route; it closes as a theorem-native mathematical anchor without an additional measurement surface. Acceptance criterion: No replay divergence and no surviving counterexample against the promoted packet. Falsifier: A single replay divergence or surviving counterexample collapses promotion for the affected packet. Primary route institutions: Clay Mathematics Institute, American Institute of Mathematics. Numerical packet / replay

status: PASS\_ACTIVE\_NUMERICAL\_PACKET / PASS\_REPLAYABLE. Open gaps: none. Next lawful action: MAINTAIN\_REPLAY\_DISCIPLINE.

**Physics.** Wave WAVE\_1\_PHYSICS currently places the domain in class THEOREM\_NATIVE, with trace status TRACE\_COMPLETE, evidence status EVIDENCE\_COMPLETE, and verdict PASS. Benchmark families cover PHY\_BENCH\_001, fundamental constants residuals, spectral line residual envelopes, transport or plasma scaling envelopes. Measured outputs track relative\_constant\_error, spectral\_line\_residual, transport\_scaling\_residual. Measurement schema: Bind theorem-to-observable maps to pinned official constants and spectra, then evaluate held-out residual envelopes. The theorem-to-observable route has three claims. K1/K2 contradiction-load and stabilization-margin operators lawfully generate the bounded physics packet over constants, Balmer spectra, and transport-scaling families. Root-law thresholds are exposed as theorem-native residual observables on the pinned official routes rather than as unrestricted physical universalization claims. Every promoted benchmark family is traced through the locked K1/K2/K9/K10 route, official datasets, and held-out replay. Acceptance criterion: Held-out residuals stay inside the declared tolerance band for every promoted benchmark family. Falsifier: Any benchmark family with residuals outside tolerance or with broken sign/order constraints falsifies the promoted packet. Primary route institutions: NIST, U.S. Department of Energy Office of Science. Numerical packet / replay status: PASS\_ACTIVE\_NUMERICAL\_PACKET / PASS\_REPLAYABLE. Open gaps: none. Next lawful action: MAINTENANCE\_ONLY.

**Chemistry.** Wave WAVE\_2\_CHEMISTRY currently places the domain in class THEOREM\_NATIVE, with trace status TRACE\_COMPLETE, evidence status EVIDENCE\_COMPLETE, and verdict PASS. Benchmark families cover CHEM\_BENCH\_001, thermochemical reference values, spectroscopic transition families, kinetic equilibrium observable envelopes. Measured outputs track enthalpy\_residual, spectral\_transition\_residual, kinetic\_or\_equilibrium\_residual. Measurement schema: Map theorem packets to thermochemical, spectral, and kinetic observables against pinned reference tables. The theorem-to-observable route has three claims. K2/K3/K4 organizational thresholds lawfully generate the bounded chemistry packet over thermochemical, spectral, and kinetic observable families. The promoted chemistry packet binds source-native thresholds to pinned residual observables without expanding to unrestricted chemical universalization. Every promoted chemistry benchmark family is traced through the locked K2/K3/K4 route, pinned reference tables, and held-out replay. Acceptance criterion: Declared chemistry benchmark families pass the tolerance envelope on held-out compounds or reactions. Falsifier: A held-out thermochemical, spectral, or kinetic family outside tolerance collapses the promoted chemistry packet. Primary route institutions: NIST, U.S. Department of Energy Office of Science. Numerical packet / replay status: PASS\_ACTIVE\_NUMERICAL\_PACKET / PASS\_REPLAYABLE. Open gaps: none. Next lawful action: MAINTENANCE\_ONLY.

**Biology.** Wave WAVE\_3\_BIOLOGY currently places the domain in class THEOREM\_NATIVE, with trace status TRACE\_COMPLETE, evidence status EVIDENCE\_COMPLETE, and verdict PASS. Benchmark families cover BIO\_BENCH\_001, gene-expression time-series families, cell-state transition signatures, stimulus-response excitability onset markers. Measured outputs track expression\_signature\_residual, state\_transition\_order\_error, response\_onset\_residual. Measurement schema: Choose one bounded biological lane and evaluate predicted state transitions or response signatures against open assay data. The theorem-to-observable route has three claims. K4/K5/K6/K7 organizational thresholds lawfully generate the bounded biology packet over expression, state-transition, and response-onset observable families. The promoted biology packet binds source-native biological organization claims to pinned assay observables without expanding to unrestricted biological universalization. Every promoted biological benchmark family is traced through the locked K4/K5/K6/K7 route, open assay datasets, and held-out replay. Acceptance criterion: The promoted biological packet reproduces the declared measurable signatures on held-out datasets without widening the claim boundary. Falsifier: If the bounded signature family fails on held-out datasets, the biological packet remains frontier-bound. Primary route institutions: NCBI GEO. Numerical packet / replay status: PASS\_ACTIVE\_NUMERICAL\_PACKET / PASS\_REPLAYABLE. Open gaps: none. Next lawful action: MAINTENANCE\_ONLY.

**Systems / Civilizational projection.** Wave WAVE\_4\_SYSTEMS currently places the domain in class THEOREM\_NATIVE, with trace status TRACE\_COMPLETE, evidence status EVIDENCE\_COMPLETE, and verdict PASS. Benchmark families cover SYS\_BENCH\_001, macro indicator trajectories, resource-intensity or throughput curves, held-out interval stability and regime shifts. Measured outputs track trajectory\_residual, turning\_point\_error, held\_out\_interval\_regime\_score. Measurement schema: Evaluate K8 projection packets against pinned official macro-process series with held-out interval tests. The theorem-to-observable route has three claims. K8 projection claims lawfully generate the

bounded systems packet over measurable macro-process trajectories and regime-shift signatures on pinned official series. The promoted systems packet binds source-native K8 projection structure to residual and turning-point observables without authorizing unrestricted civilizational prediction. Every promoted systems benchmark family is traced through the locked K8 route, official macro-process data, and held-out interval replay. Acceptance criterion: Declared trajectory and regime-change observables remain inside tolerance on held-out intervals. Falsifier: A missed turning point or out-of-band held-out trajectory falsifies the promoted systems packet. Primary route institutions: World Bank, U.S. Department of Energy Office of Science. Numerical packet / replay status: `PASS_ACTIVE_NUMERICAL_PACKET / PASS_REPLAYABLE`. Open gaps: none. Next lawful action: `MAINTENANCE_ONLY`.

## 21.9 Release consequence

The scientific SPOT therefore sets the Core 1.3 science verdict, within scope `CORE_1_3_SCIENCE_ONLY`, to `PASS`. The kernel, theorem spine, domain packets, held-out evidence, hostile-review dossiers, and unified atlas are now mutually aligned under the declared promotion bar.

**Transition to empirical execution.** Operational status alone is not enough. The next chapter fixes the evidence bar, held-out discipline, and replay conditions by which promoted packets stay scientifically lawful.

## 22 Hybrid-Escalation Evidence Bar and Execution Protocols

This chapter presents the empirical execution bar of the release. It states which source authorities, held-out designs, benchmark policies, tolerances, and falsifier conditions lawfully test promoted packets, and it separates bounded empirical support from rhetorical overreach. It does not restate the theorem spine, the full proof-support ledger, or the complete dataset inventory; it tells the reader how the packets are actually challenged.

The scientific authority for that evidence bar remains the canonical science SPOT, `OC_CORE_1_3_SCIENCE_SPOT`, materialised in this repository at `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json`. This chapter projects that authority into reader-facing form so that the execution bar, replay discipline, and promotion verdict share one lawful source boundary. The generated section below prints the exact `metadata.repo_sha` and `metadata.ts_utc` for the SPOT revision used in this build.

SPOT revision: repo SHA `0dd049c48def00df8a7d50bef07abb127a3c4fb3`; timestamp `2026-04-22T09:23:17Z`. The evidence bar is now locked to the canonical SPOT rather than to drifting summary surfaces. The mathematics anchor is replay-only: it requires deterministic exact replay rather than a data-backed benchmark route. The empirical domain lanes require held-out results that clear the declared five-sigma bar without tail-breach inflation. Residual containment by itself is not enough for promotion.

### 22.1 Global evidence bar

For the empirical lanes, the default evidence bar remains hybrid escalation: official or open primary data first, then institute-run measurements only if the observable family cannot be closed otherwise. The mathematics lane is replay-only and closes through deterministic exact replay rather than an out-of-sample data route. In every lane, evidence discipline sits behind theorem-native trace closure rather than replacing it.

### 22.2 Execution protocol matrix

| Domain                                        | Wave               | Evidence bar       | Held-out policy                                        | Replay command                                                                                  | Replay status    | Closure verdict |
|-----------------------------------------------|--------------------|--------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------|-----------------|
| Mathematics                                   | WAVE_0_ ANCHOR     | HYBRID_ES CALATION | DETERMINIS TIC_COUNTE REXAMPLE_A ND_TRACE_R EPLAY      | run_claim_si mulations_v1 .py                                                                   | PASS_REPL AYABLE | PASS            |
| Physics                                       | WAVE_1_ PHYSICS    | HYBRID_ES CALATION | LEAVE_ONE_ BENCHMARK_ FAMILY_OUT                       | run_oc_core_ domain_hard_ closure_repl ay_v1.py--do main-idPHYSI CS                             | PASS_REPL AYABLE | PASS            |
| Chemistry                                     | WAVE_2_ CHEMIST RY | HYBRID_ES CALATION | HOLD_OUT_O NE_COMPOUN D_OR_REACT ION_CLASS             | run_oc_core_ domain_hard_ closure_repl ay_v1.py--do main-idCHEMI STRY                           | PASS_REPL AYABLE | PASS            |
| Biology                                       | WAVE_3_ BIOLOGY    | HYBRID_ES CALATION | RESERVE_ON E_DATASET_ FAMILY_PER _SIGNATURE _CLASS     | run_oc_core_ domain_hard_ closure_repl ay_v1.py--do main-idBIOLO GY                             | PASS_REPL AYABLE | PASS            |
| Systems / Civi-<br>lizational projec-<br>tion | WAVE_4_ SYSTEMS    | HYBRID_ES CALATION | HELD_OUT_I NTERVAL_RE PLAY_WITH_ TURNING_PO INT_CHECKS | run_oc_core_ domain_hard_ closure_repl ay_v1.py--do main-idSYSTE MS_CIVILIZAT IONAL_PROJEC TION | PASS_REPL AYABLE | PASS            |

## 22.3 Method ladder

| Order | Method class                                 | Entry rule                                                                                                  | Anti-cycle condition                                                                                                            |
|-------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1     | SOURCE_EXTRAC TION                           | Use when a claim lacks an exact theorem target, exact dependency list, or exact source text.                | May not be repeated unless a new source block, new theo-rem family, or newly scoped parent claim is introduced.                 |
| 2     | FRESH_DERIVAT ION_FROM_OPER ATORS            | Use when source extrac- tion alone does not close the theorem-native route.                                 | May not be repeated unless a new formal premise or a new decomposition of the claim space appears.                              |
| 3     | REPARAMETERIZ ATION_OR_CLAI M_DECOMPOSITI ON | Use when the claim is too wide, rubberized, or under- specified to admit a lawful parameter law.            | May not be repeated unless a new observable family or a new admissible regime is declared before replay.                        |
| 4     | NEW_DATA_OR_M EASUREMENT_RO UTE              | Use only after theorem target, parameter law, and held-out policy are already fixed.                        | May not be repeated unless the new route adds a previ- ously unavailable observable family or a nonoverlapping held-out family. |
| 5     | SAME_CLAIM_CL ASS_ADVERSARI AL_COMPARISON    | Use when the claim survives internal closure steps and must now face simpler same- claim-class competitors. | May not be repeated unless a new comparator family or a new cost-normalized metric is introduced.                               |

## 22.4 Per-domain escalation doctrine

**Mathematics.** Protocol OC13::EXECUTION::MATHEMATICS::ANCHOR\_REPLAY runs under evidence bar HYBRID\_ESCALATION and currently reports readiness PASS. Closure phase: PHASE\_1\_STABILIZE\_CLOSED\_CORE. Launch rule: MAINTAIN\_ALWAYS\_ON. Pinned route institutions: Clay Mathematics Institute, American Institute of Mathematics. Held-out policy: DETERMINISTIC\_COUNTEREXAMPLE\_AND\_TRACE\_REPLAY, with minimum cases 31. Quantitative gate: minimum cases 31; strictly terminalized cases at least 31; counterexample survivors capped at 0; replay divergences capped at 0. Replay harness: logion/k7/spe/orchestrator/science/run\_claim\_simulations\_v1.py. Institute-run trigger: ONLY\_IF\_DETERMINISTIC\_REPLAY\_BREAKS. Measurement wave: NONE. Open gaps: none. Escalation plan: Expand the exact-question corpus rather than broadening outward claims when a replay gap appears.

**Physics.** Protocol OC13::EXECUTION::PHYSICS::CONSTANTS\_SPECTRA\_TRANSPORT runs under evidence bar HYBRID\_ESCALATION and currently reports readiness PASS. Closure phase: PHASE\_3\_PHYSICS\_CLOSURE. Launch rule: START\_IMMEDIATELY\_AFTER\_PHASE\_1. Pinned route institutions: NIST, U.S. Department of Energy Office of Science. Held-out policy: LEAVE\_ONE\_BENCHMARK\_FAMILY\_OUT, with minimum cases 30. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma; maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; critical residual threshold 2.5 sigma; severe residual threshold 1.5 sigma. Replay harness: logion/k7/spe/orchestrator/science/run\_oc\_core\_domain\_hard\_closure\_replay\_v1.py--domain-idPHYSICS. Institute-run trigger: OPEN\_DATA\_COVERAGE\_LT\_1\_0\_OR\_HELD\_OUT\_CASES\_LT\_30\_OR\_RESIDUAL\_BREACH\_PERSISTS. Measurement wave: INSTITUTE\_RUN::PHYSICS::WAVE\_1A. Open gaps: none. Escalation plan: If official/open data do not cover the required observable family, queue institute-run measurement or observational partner acquisition.

**Chemistry.** Protocol OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY\_SPECTRA\_KINETICS runs under evidence bar HYBRID\_ESCALATION and currently reports readiness PASS. Closure phase: PHASE\_4\_CHEMISTRY\_CLOSURE. Launch rule: START\_AFTER\_PHYSICS\_THEOREM\_PACKET\_LOCKS. Pinned route institutions: NIST, U.S. Department of Energy Office of Science. Held-out policy: HOLD\_OUT\_ONE\_COMPOUND\_OR\_REACTION\_CLASS, with minimum cases 30. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma; maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; critical residual threshold 2.5 sigma; severe residual threshold 1.5 sigma. Replay harness: logion/k7/spe/orchestrator/science/run\_oc\_core\_domain\_hard\_closure\_replay\_v1.py--domain-idCHEMISTRY. Institute-run trigger: OPEN\_DATA\_COVERAGE\_LT\_1\_0\_OR\_HELD\_OUT\_CASES\_LT\_30\_OR\_RESIDUAL\_BREACH\_PERSISTS. Measurement wave: INSTITUTE\_RUN::CHEMISTRY::WAVE\_2A. Open gaps: none. Escalation plan: When official/open reference data are insufficient, queue targeted laboratory or partner-measurement acquisition.

**Biology.** Protocol OC13::EXECUTION::BIOLOGY::STATE\_TRANSITIONS\_AND\_RESPONSE\_SIGNATURES runs under evidence bar HYBRID\_ESCALATION and currently reports readiness PASS. Closure phase: PHASE\_5\_BIOLOGY\_CLOSURE. Launch rule: START\_AFTER\_CHEMISTRY\_THEOREM\_PACKET\_LOCKS. Pinned route institutions: NCBI GEO. Held-out policy: RESERVE\_ONE\_DATASET\_FAMILY\_PER\_SIGNATURE\_CLASS, with minimum cases 30. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma; maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; expected calibration error at or below 0.05. Replay harness: logion/k7/spe/orchestrator/science/run\_oc\_core\_domain\_hard\_closure\_replay\_v1.py--domain-idBIOLOGY. Institute-run trigger: OPEN\_DATA\_COVERAGE\_LT\_1\_0\_OR\_HELD\_OUT\_CASES\_LT\_30\_OR\_SIGNATURE\_CLASS\_UNDERCONSTRAINED. Measurement wave: INSTITUTE\_RUN::BIOLOGY::WAVE\_3A. Open gaps: none. Escalation plan: If open repositories cannot close the lane honestly, promote an institute-run observation or assay plan instead of overclaiming.

**Systems / Civilizational projection.** Protocol OC13::EXECUTION::SYSTEMS::MACRO\_TRAJECTORIES\_AND\_REGIME\_SHIFTS runs under evidence bar HYBRID\_ESCALATION and currently reports readiness PASS. Closure phase: PHASE\_6\_SYSTEMS\_CLOSURE. Launch rule: ALLOW\_PARALLEL\_DATA\_PREP\_AFTER\_CHEMISTRY\_BUT\_FINAL\_PROMOTION\_AFTER\_BIOLOGY\_FORMAL\_ROUTE. Pinned route institutions: World Bank, U.S. Department of Energy Office of Science. Held-out policy: HELD\_OUT\_INTERVAL\_REPLAY\_WITH\_TURNING\_POINT\_CHECKS, with minimum cases 30. Quantitative gate: minimum cases 30; coverage

ratio 1.0; tail breaches capped at 0; Brier score at or below 0.08; expected calibration error at or below 0.05; critical-failure F1 at least 0.87; critical-failure precision at least 0.85; critical-failure recall at least 0.9. Replay harness: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION`. Institute-run trigger: `OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_INTERVAL_COUNT_LT_30_OR_TURNING_POINT_BREACH_PERSIST`. Measurement wave: `INSTITUTE_RUN::SYSTEMS::WAVE_4A`. Open gaps: none. Escalation plan: If open macro-process series are insufficient, queue institute-run observation design or partner data acquisition.

**Transition to proof machinery.** Once the empirical bar is fixed, the next chapter closes the triad by stating the proof machinery, revision law, competition discipline, and compression rule that keep the outward manuscript synchronized to the science verdict.

## 23 Proof Machinery, Revision Law, Competition, and Compression

This chapter closes the Core 1.3 triad of operational discipline, empirical execution, and proof machinery. It fixes how proof obligations, revision law, comparator discipline, and compression rules are kept aligned with the canonical science state. It does not replace the theorem roadmap or the empirical protocols; it tells the reader how the release stays lawful when claims are challenged, revised, compared, or compressed for outward use. Comparison summaries, compression statements, and outward publication surfaces are projection-only editorial layers. They cannot alter the `CORE_1_3_SCIENCE_ONLY` truth scope, source-owned science corpus, closure verdicts, theorem refs, falsifier refs, formulas, or numerical rows.

This chapter is projected from the canonical science `SPOT OC_CORE_1_3_SCIENCE_SPOT` at `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json` so that the proof machinery and the science verdict stay synchronized.

The proof machinery is projected from one scientific authority. This keeps foundational consistency, empirical bridge status, revision law, and competition pressure under one coherent scientific verdict instead of across contradictory summary layers.

### 23.1 Consistency and promotion law

The foundational dossier tracks 13 K-level doctrine rows and sets the overall science verdict to `PASS`. The promotion law keeps two gates distinct. A derived claim is admissible only when it is source-bound, proof-bound, and dependency-exact. An empirical claim is promoted only under the full source-owned empirical gate: a domain packet, theorem-native trace, numerical parameter law, observable binding, pinned data route, locked held-out replay, comparator audit, and explicit falsifier must all be present and must survive the declared thresholds. This sentence is a compressed gate summary; the theorem roadmap records the item-by-item checklist. The hostile-review backlog is now fully locked, so no open dossier remains capable of blocking the promoted global `PASS`. Closure work is now executed through generated dossier packages and a first-class Phase 1 closed-core dossier rather than through drifting editorial reminders.

### 23.2 Revision law

The declared revision mutability remains `DERIVED_FIRST`, while the root revision status remains `LOCKED_PENDING_ACCUMULATED_CROSS_DOMAIN_FAILURE`.

### 23.3 Proof obligations

The load-bearing dependency atlas now tracks 6 explicit spine edges and 6 proof-obligation sheets. Together they replace vague prose summaries with explicit admissibility burdens, forbidden shortcuts, excluded dependencies, and falsifier conditions.

### 23.4 Minimality and irreducibility

The strongest honest meta-result remains `STRONGEST_IMPOSSIBILITY_BOUNDARY`, and K11/K12 now stand under a locked irreducibility result beyond K10.

### 23.5 Alternative-model competition

The competition policy remains `BOTH`: same-claim-class competition and domain-baseline competition are both in force. The matrix currently tracks 5 comparison rows, of which 4 are pass-eligible under the present SPOT.

### 23.6 Unified-science atlas

The unified atlas status is `PASS`. Shared operators are now globally locked across closed domains and no surviving cross-domain contradiction remains in the falsifier matrix.

### 23.7 Compression as a bounded metric

The release treats compression as `BOUNDED_METRIC` rather than as a central theorem claim. Coverage per rooted theorem-support statement remains `0.294118`, while coverage per active numerical packet remains `1.0`.

**Transition to synthesis.** With operational discipline, empirical execution, and proof machinery all fixed under one authority, the manuscript can now move to synthesis and then to the reader-facing practical question of what the framework presently enables.

## 24 Unified Science Synthesis

Final unified-synthesis release status: `PASS`.

This chapter states the promoted synthesis projected from the canonical science SPOT. It gives the lawfully promoted unified synthesis with explicit K0–K12 parameter laws, numerical rows, falsifiers, and empirical held-out prediction summaries. The title and scope of this chapter follow the canonical SPOT record; the opening gate paragraph states the release status, and the K-level rows then give the theorem claims, parameter laws, observable bindings, numerical subsets, falsifiers, and packet bindings needed to keep the synthesis auditable.

### 24.1 Final Unified Synthesis Release Gate

Exact validator: `FINAL_UNIFIED_SYNTHESIS_VALIDATOR_PASS`. The release gate now reports 0 bridge-only domains, 0 hostile-review blockers, and integrability at `PASS`. The final promotion rule is parallel: every empirical core domain must be theorem-native; the bridge-only domain total must be zero; the hostile-review blocker total must be zero; the unified atlas and integrability suite must pass; and the K0–K12 numerical rows must remain explicit. No release blocker remains in the canonical SPOT.

### 24.2 K0: Structural Conditions for Any Universe

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K0 fixes the non-empty admissible meta-domain required for any lawful continuum and therefore bounds every later K-level before empirical specialization.

**Operator and K-level binding.** The local K0 root process glyphs  $\Psi_0$ ,  $\Phi_0$ , and  $\Lambda_0$  fix distinction generation, relational reconfiguration, and compositional assembly; compact synthesis aliases  $F_0$ ,  $Q_0$ , and  $U_0$  label the corresponding root constraints only where they are defined locally. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

### Parameter law.

$$\mu(\Omega(K_0)) > 0, \quad \forall x \in \{1, \dots, 12\} : \text{DoF}(M_x) > 0 \text{ and } \frac{\text{DoF}(K_x)}{\text{DoF}(M_x)} \leq 1, \quad C_{\text{nt}} \geq 1$$

**Observable map and units.** Meta-admissibility, meta-space compatibility, and the presence of at least one non-trivial lawful cycle. Registered observables: `K0::ADMISSIBLE_STATE_MEASURE`, `K0::META_SPACE_COMPATIBILITY_RATIO`, `K0::NONTRIVIAL_CYCLE_PRESENCE`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTE_D_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 30.

| Quantity                                               | Symbol                            | Value / range | Units or note                                          |
|--------------------------------------------------------|-----------------------------------|---------------|--------------------------------------------------------|
| Admissible-state measure floor                         | $\mu(\Omega(K_0))$                | $> 0$         | dimensionless<br>lower bound                           |
| Meta-space compatibility ratio for each promoted level | $\text{DoF}(K_x)/\text{DoF}(M_x)$ | $\leq 1$      | dimensionless<br>ratio; applies for each x in 1,...,12 |
| Non-trivial cycle presence                             | $C_{\text{nt}}$                   | 1             | present                                                |

**Falsifier and collapse boundary.** Collapse occurs if the admissible set degenerates, if a later K-level exceeds its meta-space, or if the structural substrate loses its last non-trivial lawful cycle. Falsifier witness anchors are tabulated in Appendix Q.

### Domain packet bindings.

- Mathematics is closed as `THEOREM_NATIVE` with verdict `PASS` and 0 held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are campaign status `PASS_31_OF_31_TERMINALIZED`; 31 terminalized cases. The closure dossier anchor is listed in Appendix Q.

## 24.3 K1: Minimal Observable Distinctions and Energetic Axes

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K1 introduces resolvable energetic and temporal distinctions while preserving the admissible-state floor inherited from K0.

**Operator and K-level binding.** F, G, and U bind contradiction load to resolvable energetic distinctions, monotonic update order, and a non-degenerate observable region. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

### Parameter law.

$$\Delta E \geq \Theta_{\Delta E}, \quad \frac{dt}{d\tau} > 0, \quad \mu(\Omega(K_1)) > 0$$

**Observable map and units.** The first observable regime is tracked by energy-gap resolution, monotonic update ordering, and positive configuration-space measure. Registered observables: `K1::ENERGY_GAP_THRESHOLD`, `K1::TEMPORAL_MONOTONICITY`, `K1::CONFIGURATION_MEASURE`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.



**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 31.

| Quantity                          | Symbol              | Value / range           | Units or note             |
|-----------------------------------|---------------------|-------------------------|---------------------------|
| Minimal resolvable energy scale   | $\Theta_{\Delta E}$ | $10^{-34}$ – $10^{-33}$ | J                         |
| Temporal monotonicity             | $dt/d\tau$          | $> 0$                   | ordered update constraint |
| Configuration-space measure floor | $\mu(\Omega(K_1))$  | $> 0$                   | dimensionless lower bound |

**Falsifier and collapse boundary.** Collapse begins when energetic distinctions are no longer resolvable, temporal order breaks, or the first observable state space degenerates. Falsifier witness anchors are tabulated in Appendix Q.

#### Domain packet bindings.

- Physics is closed as `THEOREM_NATIVE` with verdict `PASS` and 20 held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

## 24.4 K2: Fields, Phases, and Large-Scale Structure

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K2 binds field configurations, expansion history, and phase thresholds into the first large-scale physical continuum admissible under the kernel operators.

**Operator and K-level binding.** F, H, and R constrain field admissibility, phase transitions, and large-scale stabilization margins inside the physical continuum. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

#### Parameter law.

$$H_0 > 0, \quad 0 < \Omega_m < 1, \quad |\Omega_k| < 2 \times 10^{-3}, \quad \mathcal{T}_{\text{phase}}(t) \in \{T_c^{\text{EW}}, T_c^{\text{QCD}}\}$$

**Observable map and units.** Expansion rate, curvature, cosmological density fractions, and critical phase windows tied to physical observables and replay packets. Registered observables: `K2::EXPANSION_RATE`, `K2::CURVATURE_BOUND`, `K2::PHASE_TRANSITION_WINDOW`, `CHEMISTRY::GAS_PHASE_ENTHALPY_RESIDUAL`, `CHEMISTRY::SPECTRAL_TRANSITION_RESIDUAL`, `CHEMISTRY::KINETIC_OR_EQUILIBRIUM_RESIDUAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 32.

| Quantity                | Symbol                              | Value / range                                            | Units or note                      |
|-------------------------|-------------------------------------|----------------------------------------------------------|------------------------------------|
| Hubble parameter today  | $H_0$                               | $67.4 \pm 0.5$                                           | $\text{km s}^{-1} \text{Mpc}^{-1}$ |
| Matter density fraction | $\Omega_m$                          | $0.315 \pm 0.007$                                        | dimensionless                      |
| Critical phase windows  | $T_c^{\text{EW}}, T_c^{\text{QCD}}$ | $\sim 100 \text{ GeV};$<br>$150\text{--}170 \text{ MeV}$ | closed phase set                   |
| Curvature bound         | $\Omega_k$                          | $ \Omega_k  < 2 \times 10^{-3}$                          | dimensionless                      |

**Falsifier and collapse boundary.** K2 collapses when curvature exits the admissible band, phase windows fail to close lawfully, or large-scale physical observables lose replayable stabilization. Falsifier witness anchors are tabulated in Appendix Q.

#### Domain packet bindings.

- Chemistry is closed as THEOREM\_NATIVE with verdict PASS and 19 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Physics is closed as THEOREM\_NATIVE with verdict PASS and 20 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

The SPOT also tracks 1 excluded or non-promoted side binding for this level outside the closed synthesis summary: DRT. The corresponding source rows remain in the canonical surfaces and support appendices, not in the promoted synthesis stack.

## 24.5 K3: Molecular Organization and Chemical Bonding

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K3 packages bounded molecular organization through lawful bond-length, bond-energy, and activation-energy envelopes derived from lower-level admissibility.

**Operator and K-level binding.** G, H, and R bind composition-sensitive bonding, activation barriers, and molecular-scale stabilization windows. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

#### Parameter law.

$$r_{\text{cov}} \in [0.1, 0.2] \text{ nm}, \quad E_{\text{bond}} \in [1, 5] \text{ eV}, \quad E^{\ddagger} \in [0.1, 1] \text{ eV}$$

**Observable map and units.** Bond lengths, dissociation energies, dielectric context, and activation barriers for bounded chemistry lanes. Registered observables: K3::BOND\_LENGTH, K3::BOND\_ENERGY, K3::ACTIVATION\_ENERGY, CHEMISTRY::GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY::SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY::KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 33.

| Quantity                   | Symbol            | Value / range | Units or note |
|----------------------------|-------------------|---------------|---------------|
| Bond length (covalent)     | $r_{\text{cov}}$  | 0.1–0.2       | nm            |
| Bond dissociation energy   | $E_{\text{bond}}$ | 1–5           | eV            |
| Reaction activation energy | $E^{\ddagger}$    | 0.1–1         | eV            |

**Falsifier and collapse boundary.** K3 collapses when bond- and barrier-level regularities cannot be bounded by a non-degenerate parameter law or when chemical observables drift outside admissible molecular windows. Falsifier witness anchors are tabulated in Appendix Q.

#### Domain packet bindings.

- Chemistry is closed as THEOREM\_NATIVE with verdict PASS and 19 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

## 24.6 K4: Compartmental and Membrane Thresholds

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K4 closes the first compartmental organization packet by binding membrane thickness, bending stability, and gradient maintenance to lawful biological thresholds.

**Operator and K-level binding.** H, R, and S couple membrane stability, osmotic admissibility, and retained gradients into a bounded protocellular regime. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

**Parameter law.**

$$d_{\text{mem}} \in [3, 5] \text{ nm}, \quad \kappa \in [10, 25] k_B T, \quad \Delta pH \in [0.5, 2]$$

$$\gamma_{\text{edge}} \in [5, 20] \text{ pN}, \quad \Pi \in [10^5, 10^6] \text{ Pa}, \quad P_i \in [10^{-12}, 10^{-8}] \text{ m/s}$$

**Observable map and units.** Membrane thickness, bending modulus, osmotic pressure, proton gradient, and permeability lanes for compartmental organization. Registered observables: K4 : :MEMBRANE\_THICKNESS, K4 : :BENDING\_MODULUS, K4 : :PROTON\_GRADIENT, K4 : :LINE\_TENSION, K4 : :OSMOTIC\_PRESSURE\_SCALE, K4 : :PERMEABILITY\_RANGE, BIOLOGY : :EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY : :STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY : :RESPONSE\_ONSET\_RESIDUAL, CHEMISTRY : :GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY : :SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY : :KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 34.

| Quantity                 | Symbol                 | Value / range          | Units or note |
|--------------------------|------------------------|------------------------|---------------|
| Membrane thickness       | $d_{\text{mem}}$       | 3–5                    | nm            |
| Membrane bending modulus | $\kappa$               | 10–25                  | $k_B T$       |
| Typical proton gradient  | $\Delta pH$            | 0.5–2                  | pH units      |
| Line tension (edge)      | $\gamma_{\text{edge}}$ | 5–20                   | pN            |
| Osmotic pressure scale   | $\Pi$                  | $10^5$ – $10^6$        | Pa            |
| Permeability range       | $P_i$                  | $10^{-12}$ – $10^{-8}$ | m/s           |

**Falsifier and collapse boundary.** K4 collapses if compartments cannot retain a lawful gradient, if membranes lose structural stability, or if permeability escapes the admissible packet. Falsifier witness anchors are tabulated in Appendix Q.

**Domain packet bindings.**

- Biology is closed as `THEOREM_NATIVE` with verdict `PASS` and 16 held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Chemistry is closed as `THEOREM_NATIVE` with verdict `PASS` and 19 held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

## 24.7 K5: Excitable and Early Neural Continua

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K5 binds excitability, refractory timing, and membrane potential maintenance into the first lawful neural packet.

**Operator and K-level binding.** F, G, and S stabilize excitable-state transitions, ion-channel gating, and refractory boundaries in bounded neural lanes. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

**Parameter law.**

$$C_m = 1 \mu\text{F}/\text{cm}^2, \quad V_{\text{rest}} \in [-75, -60] \text{ mV}, \quad \tau_{\text{ref}} \in [2, 5] \text{ ms}$$

**Observable map and units.** Membrane capacitance, resting potential, channel conductance, and refractory timing for bounded excitable systems. Registered observables: K5::MEMBRANE\_CAPACITANCE, K5::RESTING\_POTENTIAL, K5::REFRACTORY\_TIME, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 35.

| Quantity                   | Symbol              | Value / range | Units or note             |
|----------------------------|---------------------|---------------|---------------------------|
| Membrane capacitance       | $C_m$               | 1             | $\mu\text{F}/\text{cm}^2$ |
| Resting membrane potential | $V_{\text{rest}}$   | $-75 - -60$   | mV                        |
| Refractory time            | $\tau_{\text{ref}}$ | 2–5           | ms                        |

**Falsifier and collapse boundary.** K5 collapses when excitability loses bounded thresholds, refractory timing fails, or the neural packet cannot survive counterexample pressure across datasets. Falsifier witness anchors are tabulated in Appendix Q.

**Domain packet bindings.**

- Biology is closed as THEOREM\_NATIVE with verdict PASS and 16 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

## 24.8 K6: Cognitive Prediction and Representation

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K6 formalizes bounded cognitive prediction and representation by linking prediction-error thresholds, working-memory span, and adaptation strength.

**Operator and K-level binding.** Q, R, and S bind representation stability, prediction thresholds, and adaptive update constraints in bounded cognitive lanes. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

**Parameter law.**

$$\Theta_{\text{pred}} \in [0.05, 0.15], \quad M_{\text{WM}} \in [3, 7], \quad \eta \in [10^{-3}, 10^{-1}]$$

**Observable map and units.** Prediction error, working-memory span, and adaptation strength for bounded cognitive organization. Registered observables: K6::PREDICTION\_ERROR\_THRESHOLD, K6::WORKING\_MEMORY\_SPAN, K6::HEBBIAN\_STRENGTH, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 36.

| Quantity                   | Symbol                 | Value / range | Units or note    |
|----------------------------|------------------------|---------------|------------------|
| Prediction-error threshold | $\Theta_{\text{pred}}$ | 0.05–0.15     | normalized units |
| Working-memory span        | $M_{\text{WM}}$        | 3–7           | bound items      |

| Quantity                     | Symbol | Value / range         | Units or note |
|------------------------------|--------|-----------------------|---------------|
| Hebbian strength coefficient | $\eta$ | $10^{-3}$ – $10^{-1}$ | dimensionless |

**Falsifier and collapse boundary.** K6 collapses when predictive stabilization fails across bounded cognitive observables or when representation thresholds require hidden freedom. Falsifier witness anchors are tabulated in Appendix Q.

#### Domain packet bindings.

- Biology is closed as THEOREM\_NATIVE with verdict PASS and 16 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

The SPOT also tracks 1 excluded or non-promoted side binding for this level outside the closed synthesis summary: DRT. The corresponding source rows remain in the canonical surfaces and support appendices, not in the promoted synthesis stack.

## 24.9 K7: Social Coordination and Group Stability

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K7 closes bounded social coordination through group size, communication bandwidth, and coordination thresholds.

**Operator and K-level binding.** R, S, and U bind coordination thresholds, communication throughput, and conflict costs into admissible social trajectories. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

#### Parameter law.

$$N_s \in [120, 180], \quad J_{\text{comm}} \in [1, 5] \text{ bits/s}, \quad \Theta_{\text{coord}} \in [0.2, 0.4] \\ C_{\text{conf}} \in [0.1, 1]$$

**Observable map and units.** Group stability size, communication throughput, coordination threshold, and conflict cost inside social coordination continua. Registered observables: K7::GROUP\_STABILITY\_SIZE, K7::COMMUNICATION\_BANDWIDTH, K7::COORDINATION\_THRESHOLD, K7::CONFLICT\_COST\_SCALE, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 37.

| Quantity                             | Symbol                  | Value / range | Units or note   |
|--------------------------------------|-------------------------|---------------|-----------------|
| Group stability size                 | $N_s$                   | 120–180       | agents          |
| Information-flow bandwidth per agent | $J_{\text{comm}}$       | 1–5           | bits/s          |
| Coordination threshold               | $\Theta_{\text{coord}}$ | 0.2–0.4       | dimensionless   |
| Conflict cost scale                  | $C_{\text{conf}}$       | 0.1–1         | arbitrary units |

**Falsifier and collapse boundary.** K7 collapses when communication throughput and coordination thresholds cannot stabilize social group trajectories without hidden rescue assumptions. Falsifier witness anchors are tabulated in Appendix Q.

### Domain packet bindings.

- Biology is closed as `THEOREM_NATIVE` with verdict `PASS` and 16 held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

## 24.10 K8: Civilizational Flows and Infrastructural Stability

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K8 binds civilizational energy density, renewal timing, and repair-decay balance into bounded macro-process trajectories and regime-shift packets.

**Operator and K-level binding.** H, R, and U bind throughput, repair-decay balance, and regime-shift boundaries in bounded systems-level observables. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

### Parameter law.

$$E_{\text{dens}} \in [10^2, 10^3] \text{ W/m}^2, \quad \tau_{\text{infra}} \in [20, 50] \text{ years}, \quad R_{\text{rep}} \in [0.8, 1.2] \\ \Theta_I \in [0.1, 0.3]$$

**Observable map and units.** Energy density, infrastructure renewal, information coherence, and repair-decay balance in bounded civilizational lanes. Registered observables: `K8::ENERGY_CONSUMPTION_DENSITY`, `K8::INFRASTRUCTURE_RENEWAL_TIME`, `K8::REPAIR_TO_DECAY_RATIO`, `K8::INFORMATION_COHERENCE_THRESHOLD`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::POPULATION_TRAJECTORY_RESIDUAL`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::GDP_GROWTH_TURNING_POINT_ERROR`, `SYSTEMS_CIVILIZATIONAL_PROJECTION::PROCESS_SYSTEM_REGIME_SCORE`.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 38.

| Quantity                        | Symbol                | Value / range   | Units or note    |
|---------------------------------|-----------------------|-----------------|------------------|
| Energy consumption density      | $E_{\text{dens}}$     | $10^2$ – $10^3$ | W/m <sup>2</sup> |
| Infrastructure renewal time     | $\tau_{\text{infra}}$ | 20–50           | years            |
| Information-coherence threshold | $\Theta_I$            | 0.1–0.3         | dimensionless    |
| Repair-to-decay ratio           | $R_{\text{rep}}$      | 0.8–1.2         | dimensionless    |

**Falsifier and collapse boundary.** K8 collapses when macro-trajectories lose bounded turning-point fidelity, when repair-decay balance drifts outside its band, or when regime-shift packets require discretionary knobs. Falsifier witness anchors are tabulated in Appendix Q.

### Domain packet bindings.

- Systems / Civilizational projection is closed as `THEOREM_NATIVE` with verdict `PASS` and 18 held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 18 held-out cases; mean absolute normalized error 0.56097 sigma; 95th-percentile normalized error 1.924704 sigma; maximum normalized error 2.25395 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

## 24.11 K9: Knowledge Systems and Scientific Memory

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K9 formalizes bounded scientific-knowledge continua through anomaly accumulation, paradigm coherence, and model-validation bandwidth.

**Operator and K-level binding.** Q, R, and U bind anomaly handling, paradigm coherence, and memory retention inside knowledge-system packets. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

**Parameter law.**

$$\Theta_{\text{par}} \in [0.15, 0.25], \quad J_{\text{anom}} \in [10^{-4}, 10^{-2}], \quad \tau_{\text{mem}} \in [30, 200] \text{ years} \\ B_{\text{val}} \in [0.1, 1]$$

**Observable map and units.** Paradigm coherence, anomaly accumulation, validation bandwidth, and scientific-memory half-life. Registered observables: K9::PARADIGM\_COHERENCE\_THRESHOLD, K9::ANOMALY\_ACCUMULATION\_RATE, K9::SCIENTIFIC\_MEMORY\_HALF\_LIFE, K9::MODEL\_VALIDATION\_BANDWIDTH, MATHEMATICS::EPSILON\_STAR, MATHEMATICS::MINIMAL\_DENOMINATOR, MATHEMATICS::MINIMAL\_AUGMENTATION\_RANK, MATHEMATICS::COMPONENT\_FIBER\_COUNT, MATHEMATICS::MOVE\_CONNECTIVITY\_OR\_DEGREE\_BOUND, MATHEMATICS::STRICT\_TERMINALIZED\_TOTAL, MATHEMATICS::COUNTEREXAMPLE\_EXECUTED\_TOTAL, MATHEMATICS::PYTHON\_EVIDENCE\_EXECUTED\_TOTAL, PHYSICS::FINE\_STRUCTURE\_CONSTANT\_RESIDUAL, PHYSICS::HYDROGEN\_BALMER\_LINE\_RESIDUAL, PHYSICS::TRANSPORT\_SCALING\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 39.

| Quantity                     | Symbol                | Value / range         | Units or note |
|------------------------------|-----------------------|-----------------------|---------------|
| Paradigm-coherence threshold | $\Theta_{\text{par}}$ | 0.15–0.25             | dimensionless |
| Anomaly accumulation rate    | $J_{\text{anom}}$     | $10^{-4}$ – $10^{-2}$ | per cycle     |
| Scientific-memory half-life  | $\tau_{\text{mem}}$   | 30–200                | years         |
| Model-validation bandwidth   | $B_{\text{val}}$      | 0.1–1                 | normalized    |

**Falsifier and collapse boundary.** K9 collapses if anomaly load outruns lawful model revision or if scientific memory and validation bandwidth fail to stabilize the knowledge packet. Falsifier witness anchors are tabulated in Appendix Q.

**Domain packet bindings.**

- Mathematics is closed as THEOREM\_NATIVE with verdict PASS and 0 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are campaign status PASS\_31\_OF\_31\_TERMINALIZED; 31 terminalized cases. The closure dossier anchor is listed in Appendix Q.
- Physics is closed as THEOREM\_NATIVE with verdict PASS and 20 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

## 24.12 K10: Formal Systems and Recursion

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K10 binds bounded recursion, consistency thresholds, and proof-flow capacity into the formal-system lane of the unified-science closure stack.

**Operator and K-level binding.** F, Q, and U bind recursion depth, proof throughput, and consistency thresholds in formal continua. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

**Parameter law.**

$$d_{\text{rec}} \in [10, 10^4], \quad \Theta_{\text{cons}} \in [0.01, 0.05], \quad J_{\text{proof}} \in [1, 10^3]$$



**Observable map and units.** Recursion depth, consistency threshold, and proof-flow capacity in bounded formal systems. Registered observables: `K10::RECURSION_DEPTH`, `K10::CONSISTENCY_THRESHOLD`, `K10::PROOF_FLOW_CAPACITY`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 40.

| Quantity                    | Symbol                 | Value / range      | Units or note |
|-----------------------------|------------------------|--------------------|---------------|
| Recursion depth (effective) | $d_{\text{rec}}$       | 10–10 <sup>4</sup> | steps         |
| Consistency threshold       | $\Theta_{\text{cons}}$ | 0.01–0.05          | dimensionless |
| Proof-flow capacity         | $J_{\text{proof}}$     | 1–10 <sup>3</sup>  | steps/s       |

**Falsifier and collapse boundary.** K10 collapses if formal recursion and proof throughput cannot remain bounded without violating consistency thresholds. Falsifier witness anchors are tabulated in Appendix Q.

#### Domain packet bindings.

- Mathematics is closed as `THEOREM_NATIVE` with verdict `PASS` and 0 held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are campaign status `PASS_31_OF_31_TERMINALIZED`; 31 terminalized cases. The closure dossier anchor is listed in Appendix Q.
- Metaontology is closed as `FRAME_ONLY` with verdict `PASS` and 0 held-out cases. This is a frame-only support row, so packet completeness is recorded as a support-surface status rather than as a theorem-native empirical closure claim. No quantitative replay metrics are required for this row. The closure dossier anchor is listed in Appendix Q.
- Physics is closed as `THEOREM_NATIVE` with verdict `PASS` and 20 held-out cases. The theorem packet is `COMPLETE` and the parameter law is `COMPLETE`. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

### 24.13 K11: Meta-Theoretical Reflexivity

This row is currently `PASS` and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K11 captures meta-theoretical reflexivity as irreducible scientific work beyond K10 across the closed domain atlas.

**Operator and K-level binding.** Q, R, and S bind meta-coherence, reflexive depth, and cross-landscape coupling under the requirement that no K10-only translation preserves all declared upper-level burdens. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

#### Parameter law.

$$\Theta_{\text{meta}} \in [0.02, 0.1], \quad d_{\text{refl}} \in [3, 50], \quad J_X \in [0.1, 1] \\ P_{\text{funct}} \in [10^0, 10^3]$$

**Observable map and units.** Meta-coherence, reflexive depth, and cross-landscape coupling in the meta-theoretical packet. Registered observables: `K11::META_COHERENCE_THRESHOLD`, `K11::REFLEXIVE_DEPTH`, `K11::CROSS_LANDSCAPE_COUPLING`, `K11::FUNCTORIAL_POTENTIAL`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMP`



LE\_EXECUTED\_TOTAL, MATHEMATICS::PYTHON\_EVIDENCE\_EXECUTED\_TOTAL, PHYSICS::FINE\_STRUCTURE\_CONSTANT\_RESIDUAL, PHYSICS::HYDROGEN\_BALMER\_LINE\_RESIDUAL, PHYSICS::TRANSPORT\_SCALING\_RESIDUAL, CHEMISTRY::GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY::SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY::KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::POPULATION\_TRAJECTORY\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::GDP\_GROWTH\_TURNING\_POINT\_ERROR, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::PROCESS\_SYSTEM\_REGIME\_SCORE.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 41.

| Quantity                 | Symbol                 | Value / range   | Units or note  |
|--------------------------|------------------------|-----------------|----------------|
| Meta-coherence threshold | $\Theta_{\text{meta}}$ | 0.02–0.1        | dimensionless  |
| Functorial potential     | $P_{\text{funct}}$     | $10^0$ – $10^3$ | abstract units |
| Reflexive depth          | $d_{\text{refl}}$      | 3–50            | levels         |
| Cross-landscape coupling | $J_X$                  | 0.1–1           | dimensionless  |

**Falsifier and collapse boundary.** K11 collapses if an explicit K10-only reduction preserves the declared meta-theoretical bounds or if meta-coherence depends on an excluded lift route. Falsifier witness anchors are tabulated in Appendix Q.

#### Domain packet bindings.

- Mathematics is closed as THEOREM\_NATIVE with verdict PASS and 0 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are campaign status PASS\_31\_OF\_31\_TERMINALIZED; 31 terminalized cases. The closure dossier anchor is listed in Appendix Q.
- Physics is closed as THEOREM\_NATIVE with verdict PASS and 20 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Chemistry is closed as THEOREM\_NATIVE with verdict PASS and 19 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Biology is closed as THEOREM\_NATIVE with verdict PASS and 16 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Systems / Civilizational projection is closed as THEOREM\_NATIVE with verdict PASS and 18 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 18 held-out cases; mean absolute normalized error 0.56097 sigma; 95th-percentile normalized error 1.924704 sigma; maximum normalized error 2.25395 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Metaontology is closed as FRAME\_ONLY with verdict PASS and 0 held-out cases. This is a frame-only support row, so packet completeness is recorded as a support-surface status rather than as a theorem-native empirical closure claim. No quantitative replay metrics are required for this row. The closure dossier anchor is listed in Appendix Q.

## 24.14 K12: Global Semantic Coherence

This row is currently PASS and therefore contributes lawfully to the closed synthesis stack. Promotion blockers: none remain at this level.

**Promoted theorem-native claim.** K12 packages global semantic coherence across the closed domain atlas because cross-domain meaning remains jointly interpretable and no surviving global falsifier breaks the unified synthesis.

**Operator and K-level binding.** Q, R, S, and U bind semantic coherence, global compatibility, and reference stability into the final unified-science synthesis layer. Core witness anchors are tabulated in Appendix Q and Appendix C.4.

**Parameter law.**

$$C_{\text{uni}} \in [10^3, 10^{12}], \quad \Theta_{\text{uni}} \in [10^{-3}, 10^{-2}], \quad R_{\text{uni}} \in [0.1, 1] \\ T_{\text{uni}} \in [10^2, 10^8]$$

**Observable map and units.** Universal integration capacity, cross-continuum compatibility, and structural reachability. Registered observables: K12::UNIVERSAL\_INTEGRATION\_CAPACITY, K12::CROSS\_CONTINUUM\_COMPATIBILITY, K12::STRUCTURAL\_REACHABILITY, K12::GLOBAL\_TENSION\_BUDGET, MATHEMATICS::EPSILON\_STAR, MATHEMATICS::MINIMAL\_DENOMINATOR, MATHEMATICS::MINIMAL\_AUGMENTATION\_RANK, MATHEMATICS::COMPONENT\_FIBER\_COUNT, MATHEMATICS::MOVE\_CONNECTIVITY\_OR\_DEGREE\_BOUND, MATHEMATICS::STRICT\_TERMINALIZED\_TOTAL, MATHEMATICS::COUNTEREXAMPLE\_EXECUTED\_TOTAL, MATHEMATICS::PYTHON\_EVIDENCE\_EXECUTED\_TOTAL, PHYSICS::FINE\_STRUCTURE\_CONSTANT\_RESIDUAL, PHYSICS::HYDROGEN\_BALMER\_LINE\_RESIDUAL, PHYSICS::TRANSPORT\_SCALING\_RESIDUAL, CHEMISTRY::GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY::SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY::KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::POPULATION\_TRAJECTORY\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::GDP\_GROWTH\_TURNING\_POINT\_ERROR, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::PROCESS\_SYSTEM\_REGIME\_SCORE.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 42.

| Quantity                       | Symbol                | Value / range         | Units or note |
|--------------------------------|-----------------------|-----------------------|---------------|
| Universal integration capacity | $C_{\text{uni}}$      | $10^3$ – $10^{12}$    | dimensionless |
| Cross-continuum compatibility  | $\Theta_{\text{uni}}$ | $10^{-3}$ – $10^{-2}$ | dimensionless |
| Structural reachability        | $R_{\text{uni}}$      | 0.1–1                 | dimensionless |
| Global tension budget          | $T_{\text{uni}}$      | $10^2$ – $10^8$       | dimensionless |

**Falsifier and collapse boundary.** K12 collapses if cross-domain compatibility fragments, if the unified atlas cannot pass integrability, or if a surviving global falsifier breaks semantic coherence. Falsifier witness anchors are tabulated in Appendix Q.

**Domain packet bindings.**

- Mathematics is closed as THEOREM\_NATIVE with verdict PASS and 0 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are campaign status PASS\_31\_OF\_31\_TERMINALIZED; 31 terminalized cases. The closure dossier anchor is listed in Appendix Q.
- Physics is closed as THEOREM\_NATIVE with verdict PASS and 20 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 20 held-out cases; mean absolute normalized error 0.188136 sigma; 95th-percentile normalized error 0.64778 sigma; maximum normalized error 1.48 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Chemistry is closed as THEOREM\_NATIVE with verdict PASS and 19 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 19 held-out cases; mean absolute normalized error 0.533285 sigma; 95th-percentile normalized error 1.0 sigma; maximum normalized error 1.597772 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Biology is closed as THEOREM\_NATIVE with verdict PASS and 16 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 16 held-out cases; mean absolute normalized error 0.444222 sigma; 95th-percentile normalized error 1.030033 sigma; maximum normalized error 1.438582 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.

- Systems / Civilizational projection is closed as THEOREM\_NATIVE with verdict PASS and 18 held-out cases. The theorem packet is COMPLETE and the parameter law is COMPLETE. The key replay metrics are 30 covered cases; 18 held-out cases; mean absolute normalized error 0.56097 sigma; 95th-percentile normalized error 1.924704 sigma; maximum normalized error 2.25395 sigma; tail breaches 0. The closure dossier anchor is listed in Appendix Q.
- Metaontology is closed as FRAME\_ONLY with verdict PASS and 0 held-out cases. This is a frame-only support row, so packet completeness is recorded as a support-surface status rather than as a theorem-native empirical closure claim. No quantitative replay metrics are required for this row. The closure dossier anchor is listed in Appendix Q.

## 24.15 Empirical Held-Out Prediction Summary

The following summary records the empirical synthesis lanes that have already cleared theorem-native promotion and held-out prediction review. Where a promoted lane carries severe cases or false negatives, those adverse metrics are reported explicitly below. The pass/fail policy is the validation matrix in Appendix G; Appendix C.4 and the canonical synthesis surface retain the full numerical record. Metric labels are printed with their canonical surface names; `critical_failure_recall` is the promoted systems recall metric used by the synthesis surface.

| Domain                              | Class          | Verdict | Held-out total | Metrics                                                                                                                                                                                                                                                                                               |
|-------------------------------------|----------------|---------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physics                             | THEOREM_NATIVE | PASS    | 20             | covered_case_total=30, covered_held_out_case_total=20, normalized_error_mean_abs=0.188136, normalized_error_p95=0.64778, normalized_error_max=1.48, severe_case_total=0, tail_breach_count=0                                                                                                          |
| Chemistry                           | THEOREM_NATIVE | PASS    | 19             | covered_case_total=30, covered_held_out_case_total=19, normalized_error_mean_abs=0.533285, normalized_error_p95=1.0, normalized_error_max=1.597772, severe_case_total=1, tail_breach_count=0                                                                                                          |
| Biology                             | THEOREM_NATIVE | PASS    | 16             | covered_case_total=30, covered_held_out_case_total=16, normalized_error_mean_abs=0.444222, normalized_error_p95=1.030033, normalized_error_max=1.438582, severe_case_total=0, tail_breach_count=0                                                                                                     |
| Systems / Civilizational projection | THEOREM_NATIVE | PASS    | 18             | covered_case_total=30, covered_held_out_case_total=18, normalized_error_mean_abs=0.56097, normalized_error_p95=1.924704, normalized_error_max=2.25395, severe_case_total=2, fn=1, critical_failure_precision=1.0, critical_failure_recall=0.979592, critical_failure_f1=0.989691, tail_breach_count=0 |

**Transition to practical use.** The release-gated synthesis consolidates the integrated Core 1.3 result without turning that result into an unbounded public claim. The next chapter translates the same bounded result into prediction tasks, decision rules, and operational procedures.

## 25 Practical Consequences, Predictive Power, and Use of OC Core 1.3

OC Core 1.3 is practically useful when a reader must choose a lawful packet, make a bounded residual prediction, perform anomaly screening, audit a state transition, monitor a regime shift, or compare routes across domains. This chapter states what can be used now, what inputs are needed, what outputs it produces, and where the boundary lies. It does not present OC as an unrestricted universal prediction system.

The practical value lies in playbooks and comparison matrices. They show what prior science already does well, where OC adds route selection, falsifier testing, and replay discipline, and where the atlas marks complementarity, integration, or a local model's superiority in a specific lane.

This framing is comparative rather than replacement-oriented. Specialised local model families in physics, chemistry, and biology remain strong inside their native lanes. The Systems / Civilizational projection lane is likewise bounded and does not license unrestricted civilizational forecasting or control. OC's present claim is narrower: it aligns selected closed packets through shared route selection, falsifier tests, and replay checks while each domain-level closure remains local to its validated theorem and its replay records. Appendix R gives the row-level trace anchors for that operational claim. The surrounding comparison vocabulary is contextualised by established model families ([goldenfeld\\_lectures](#); [wilson\\_renormalization](#); [kauffman\\_autocatalytic](#); [raf\\_networks](#); [hebb\\_organization](#); [friston\\_free\\_energy](#)). The canonical utility atlas also carries theorem-native mathematics and one cross-domain unified-science row; the opening paragraphs foreground the empirical lanes because they provide the most immediate replay-tested use cases for external readers.

This chapter draws from the canonical practical-utility atlas so that reader-facing statements about usefulness stay tied to the canonical support classes: `FORMALLY_PROVED`, `THEOREM_NATIVE_HELD_OUT_VALIDATED`, and `OPERATIONALLY_SUPPORTED_WITHIN_BOUNDS`. The Practical Utility and Model Comparison Atlas (Appendix R) carries the detailed support surface: use cases, playbooks, same-claim baselines, serious comparators, readiness boundaries, and trace anchors in one place.

**Provenance.** The governing utility surface and its release mirror are named in Appendix R; the governing JSON surface records the exact repository SHA and UTC release timestamp.

### 25.1 Practical value and support boundary

The practical utility atlas currently tracks 6 usable-now rows, 3 frontier-program rows, and 1 hypothesis-only row. The theorem-native domain lanes are usable under explicit theorem, data-route, and falsifier discipline; the cross-domain route-selection lane is operationally supported within bounds and is used for packet selection rather than as a domain-level theorem claim. Its closed practical value is not unrestricted universal prediction; it is lawful packet selection, bounded residual prediction, anomaly screening, state-transition auditing, regime-shift monitoring, and cross-domain route selection within one source-bound routing framework. The atlas also keeps 3 frontier-program rows and 1 hypothesis-only row explicit so practical ambition does not masquerade as finished closure.

### 25.2 Support classes

Every practical claim below is explicitly labeled so the reader can see what is already usable, what is bounded, and what still remains exploratory.

- `FORMALLY_PROVED` means formally proved and is usable now within the declared bounds.
- `THEOREM_NATIVE_HELD_OUT_VALIDATED` means theorem-native + empirical / held-out validated and is usable now within the declared bounds.
- `OPERATIONALLY_SUPPORTED_WITHIN_BOUNDS` means operationally supported within bounds and is usable now within the declared bounds.
- `FRONTIER_PROGRAM` means frontier program and is not yet a lawful usable-now claim.
- `HYPOTHESIS_ONLY` means hypothesis only and is not yet a lawful usable-now claim.

### 25.3 What is already usable now

| Domain      | Problem class                                               | What OC gives now                                                                                                                                          | Support                                         | Output / decision                                                                                                                      | Trace                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathematics | Strict closure certification and counterexample elimination | Certify whether a bounded formal object or question packet survives strict closure and counterexample search under the locked mathematical lane.           | formally proved                                 | A replayable accept or reject result together with minimal denominator, augmentation rank, or surviving counterexample classification. | row MATHEMATICS_USE_001_STRICT_CLOSURE_CERTIFICATION; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/mathematics/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics; refs content/20_oc_core_1_3_theorem_roadmap.tex, content/21_oc_core_1_3_worked_examples.tex         |
| Physics     | Constants, spectral-line, and transport-envelope audit      | Audit bounded constants, Balmer-line families, and transport-scaling envelopes on held-out benchmark families and flag out-of-envelope residual behaviour. | theorem-native + empirical / held-out validated | A pass or fail residual verdict, anomaly prioritisation, and a decision about whether the family remains inside the promoted packet.   | row PHYSICS_USE_001_BOUND_ED_RESIDUAL_AUDIT; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics; refs content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex |

| Domain    | Problem class                                                        | What OC gives now                                                                                                                                            | Support                                         | Output / decision                                                                                                                                                  | Trace                                                                                                                                                                                                                                                                                                                                                            |
|-----------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemistry | Thermochemical, spectroscopic, and kinetic packet audit              | Audit bounded thermochemical, spectral, and reaction-envelope families on held-out compounds or reactions and flag out-of-envelope chemistry lanes.          | theorem-native + empirical / held-out validated | A pass or fail residual verdict, compound or reaction-family screening result, and a decision about whether the lane remains inside the promoted chemistry packet. | row CHEMISTRY_USE_001_COMPOUND_REACTION_PACKET_AUDIT; source releases/oc_core_1_3/editorial/science_sources/closure_bundle_s/chemistry/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry; refs content/22_oc_core_1_3_operationationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex |
| Biology   | Expression-signature, state-transition, and response-onset screening | Screen bounded biological lanes for expression-signature residuals, state-transition ordering errors, and response-onset residuals across held-out datasets. | theorem-native + empirical / held-out validated | A pass or fail signature verdict, a state-transition ordering judgment, and an escalation decision for the biological lane.                                        | row BIOLOGY_USE_001_TRANSITION_AND_RESPONSE_SCREENING; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology; refs content/22_oc_core_1_3_operationationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocol_s.tex    |

| Domain                         | Problem class                                                                | What OC gives now                                                                                                                                                               | Support                                         | Output / decision                                                                                                                                           | Trace                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Systems /K8 projection         | Macro-trajectory and regime-shift monitoring                                 | Monitor bounded macro trajectories, turning-point candidates, and regime-shift signatures on held-out official series and flag out-of-band intervals.                           | theorem-native + empirical / held-out validated | A pass or fail trajectory verdict, a turning-point or regime-shift alert, and a decision about whether the lane remains inside the promoted systems packet. | row SYSTEMS_U SE_001_REGIM E_SHIFT_MONI TORING; source r eleases/oc_c ore_1_3/edit orial/scienc e_sources/cl osure_bundle s/systems_ci vilizational _projection/ bundle.json; closure release s/oc_core_1_ 3/editorial/ dossier_pack ages/closure _bundles/sys tems_civiliz ational_proj ection; refs con tent/22_oc_c ore_1_3_oper ationalizati on_program.t ex,content/2 3_oc_core_1_ 3_empirical_ execution_pr otocols.tex |
| Cross-domain / unified science | Cross-domain route selection, escalation discipline, and packet coordination | Choose the lawful domain packet, comparator set, and escalation route for a mixed scientific or operational problem without letting rhetoric outrun the closed packet boundary. | operationally supported within bounds           | A lawful packet-selection decision, a comparator shortlist, and an explicit escalation path when the closed packet is not enough.                           | row UNIFIED_U SE_001_CROSS _DOMAIN_ROUT E_SELECTION; refs content/2 2_oc_core_1_ 3_operationa lization_pro gram.tex,con tent/23_oc_c ore_1_3_emi rical_execut ion_protocol s.tex                                                                                                                                                                                                                                              |

## 25.4 How to use OC in practice

The playbooks below answer the operational question directly: when to use OC, what inputs are required, what procedure to run, and what the resulting decision or prediction actually is.

**Mathematics: Strict closure and counterexample playbook** Use when the task is to determine whether a bounded exact question already lies inside the promoted mathematical anchor or should be rejected by the strict counterexample route. Inputs: formal question encoding; strict closure policy; counterexample search route. Procedure: 1. Encode the target object in the admissible exact-question form. 2. Run the declared terminalization and counterexample search. 3. Accept the result only if the replay is deterministic and survivor-free. 4. Escalate to frontier if the question leaves the declared exact scope. Output: A replayable proof result, a surviving counterexample, or a lawful frontier demotion. Support class: formally proved. Trace: row MATHEMATICS\_PLAYBOOK \_001\_STRICT\_CLOSURE; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/mathematics/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/mathematics; refs content/20\_oc\_core\_1\_3\_theorem\_roadmap.tex, content/21\_oc\_core\_1\_3\_worked\_examples.tex. Failure boundary: Stop if the object cannot be encoded in the admissible exact-question format or if a surviving counterexample remains after the declared search.

**Physics: Bounded constants, spectra, and transport audit** Use when the task is to decide whether a constants family, spectral family, or transport family belongs inside the promoted physics packet or should be escalated as an anomaly. Inputs: chosen benchmark family; official dataset route; held-out split and sigma policy. Procedure: 1. Map the observable family to the promoted physics packet. 2. Load the pinned official dataset and the locked held-out split. 3. Run the declared residual replay and inspect sign or order constraints. 4. Accept the family only if the replay stays inside the promoted tolerance band. Output: A bounded inclusion, anomaly, or escalation verdict for the target physical family. Support class: theorem-native + empirical / held-out validated. Trace: row PHYSICS\_PLAYBOOK\_001\_BOUNDED\_RESIDUAL\_AUDIT; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/physics/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/physics; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_protocols.tex. Failure boundary: Stop and demote to frontier if the observable family requires a widened scope, hidden fit freedom, or a post-hoc split change.

**Chemistry: Thermochemical, spectral, and kinetic packet audit** Use when the task is to decide whether a bounded compound family or reaction class lies inside the promoted chemistry packet or should be escalated as an out-of-envelope case. Inputs: compound or reaction-family definition; pinned NIST or DOE-linked route; held-out rotation and tolerance policy. Procedure: 1. Map the chemistry lane to the promoted packet and observable family. 2. Load the pinned reference tables and the locked held-out split. 3. Run the declared replay on thermochemical, spectral, or kinetic outputs. 4. Accept the lane only if the held-out family remains inside the promoted tolerance band. Output: A bounded inclusion, anomaly, or escalation verdict for the target chemistry lane. Support class: theorem-native + empirical / held-out validated. Trace: row CHEMISTRY\_PLAYBOOK\_001\_PACKET\_AUDIT; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/chemistry/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/chemistry; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_protocols.tex. Failure boundary: Stop and demote to frontier if the lane requires hidden fit freedom, widened scope, or post-hoc split changes.

**Biology: Bounded biological signature and onset screening** Use when the task is to decide whether a bounded assay lane belongs inside the promoted biological packet or should be escalated as an unresolved biological signature family. Inputs: bounded assay family; multi-dataset held-out split; declared signature, transition, or onset observable. Procedure: 1. Map the biological lane to the promoted packet and observable family. 2. Load the locked multi-dataset route and held-out split. 3. Run the declared replay on expression, transition-order, or onset outputs. 4. Accept the lane only if the held-out datasets stay inside the promoted signature envelope. Output: A bounded inclusion, anomaly, or escalation verdict for the target biological lane. Support class: theorem-native + empirical / held-out validated. Trace: row BIOLOGY\_PLAYBOOK\_001\_SIGNATURE\_SCREENING; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/biology/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/biology; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_protocols.tex. Failure boundary: Stop and demote to frontier if the lane depends on a single dataset family, hidden fit freedom, or a widened biological claim boundary.

**Systems /K8 projection: Bounded macro-trajectory and regime-shift monitoring** Use when the task is to decide whether an official macro-process series remains inside the promoted systems packet or should be escalated as a turning-point or regime anomaly. Inputs: target macro-process series; held-out interval; declared trajectory or turning-point metric. Procedure: 1. Map the macro-process lane to the promoted systems packet. 2. Load the pinned official series and the locked held-out interval. 3. Run the declared residual and turning-point replay. 4. Accept the lane only if held-out intervals remain inside the promoted trajectory and regime envelope. Output: A bounded inclusion, anomaly, or escalation verdict for the target systems lane. Support class: theorem-native + empirical / held-out validated. Trace: row SYSTEMS\_PLAYBOOK\_001\_REGIME\_SHIFT\_MONITORING; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/systems\_civilizational\_projection/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/systems\_civilizational\_projection; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_protocols.tex. Failure boundary: Stop and demote to frontier if the lane requires out-of-band trajectory behaviour, a widened horizon, or post-hoc interval selection.



**Cross-domain /unified science: Cross-domain packet selection and escalation playbook** Use when a scientific or engineering problem touches more than one domain lane and the team must decide which closed packet, comparator family, and evidence route should govern the next action. Inputs: declared target problem class; bounded observable family; candidate domain packets; decision or intervention objective. Procedure: 1. Map the target problem to the narrowest lawful observable family. 2. Select the closed packet whose theorem-to-observable map already governs that family. 3. Check same-claim baselines and serious comparators before claiming advantage. 4. Escalate only when the closed packet boundary is reached and the atlas permits the translation. Output: A lawful route-selection decision with explicit packet, comparator, falsifier, and escalation boundaries. Support class: operationally supported within bounds. Trace: row UNIFIED\_PLAYBOOK\_001\_ROUTE\_SELECTION; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_protocols.tex. Failure boundary: Stop and demote to frontier if the observable family is outside every closed packet or if two candidate routes produce incompatible bounded verdicts.

## 25.5 What OC predicts across the closed empirical domains

**Physics.** For this domain, OC currently uses the closed packet to audit bounded constants, Balmer-line families, and transport-scaling envelopes on held-out benchmark families and flag out-of-envelope residual behaviour. Use-case trace: row PHYSICS\_USE\_001\_BOUNDED\_RESIDUAL\_AUDIT; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/physics/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/physics; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_protocols.tex. Measured outputs: relative\_constant\_error, spectral\_line\_residual, transport\_scaling\_residual. Benchmark families: PHY\_BENCH\_001; fundamental constants residuals; spectral line residual envelopes; transport or plasma scaling envelopes. Held-out case total: 20. What remains open: OC Core 1.3 does not yet lawfully promote a general OC replacement for the full physics model stack.

**Chemistry.** For this domain, OC currently uses the closed packet to audit bounded thermochemical, spectral, and reaction-envelope families on held-out compounds or reactions and flag out-of-envelope chemistry lanes. Use-case trace: row CHEMISTRY\_USE\_001\_COMPOUND\_REACTION\_PACKET\_AUDIT; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/chemistry/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/chemistry; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_protocols.tex. Measured outputs: enthalpy\_residual, spectral\_transition\_residual, kinetic\_or\_equilibrium\_residual. Benchmark families: CHEM\_BENCH\_001; thermochemical reference values; spectroscopic transition families; kinetic or equilibrium observable envelopes. Held-out case total: 19. What remains open: OC Core 1.3 does not yet promote a general OC catalyst-discovery or unrestricted synthetic-chemistry engine.

**Biology.** For this domain, OC currently uses the closed packet to screen bounded biological lanes for expression-signature residuals, state-transition ordering errors, and response-onset residuals across held-out datasets. Use-case trace: row BIOLOGY\_USE\_001\_TRANSITION\_AND\_RESPONSE\_SCREENING; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/biology/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/biology; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_protocols.tex. Measured outputs: expression\_signature\_residual, state\_transition\_order\_error, response\_onset\_residual. Benchmark families: BIO\_BENCH\_001; gene-expression time-series families; cell-state transition signatures; stimulus-response or excitability onset markers. Held-out case total: 16. What remains open: OC Core 1.3 does not yet promote a general OC biological intervention or whole-phenotype control engine.

**Systems /K8 projection.** For this domain, OC currently uses the closed packet to monitor bounded macro trajectories, turning-point candidates, and regime-shift signatures on held-out official series and flag out-of-band intervals. Use-case trace: row SYSTEMS\_USE\_001\_REGIME\_SHIFT\_MONITORING; source releases/oc\_core\_1\_3/editorial/science\_sources/closure\_bundles/systems\_civilizational\_projection/bundle.json; closure releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/systems\_civilizational\_projection; refs content/22\_oc\_core\_1\_3\_operationalization\_program.tex, content/23\_oc\_core\_1\_3\_empirical\_execution\_

protocols.tex. Measured outputs: trajectory\_residual, turning\_point\_error, held\_out\_interval\_regime\_score. Benchmark families: SYS\_BENCH\_001; macro indicator trajectories; resource-intensity or throughput curves; held-out interval stability and regime shifts. Held-out case total: 18. What remains open: OC Core 1.3 does not yet promote unrestricted long-horizon civilizational forecasting or control as lawful closed science.

## 25.6 Benchmark against serious alternatives

This chapter distinguishes between simpler same-claim-class baselines and serious established model families. The first question is whether a cheaper competitor already solves the same bounded lane. The second is whether OC merely coexists with, integrates, or improves the fragmented scientific picture around that lane. Each comparison row below is an indexed main-body summary rather than a standalone verdict. The comparison id is the row-level trace anchor; Appendix R expands that id into the source refs, scope boundary, comparator row, and trace hooks that make the summary auditable.

### Same-claim-class baselines.

| Comparison id                 | Domain      | Comparator families                                                            | Verdict                            | Comparison scope                                                                                                                                                                                                                                            | Trace                                                                                                                                                                                                                 |
|-------------------------------|-------------|--------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAME_CLAIM_CLASS::MATHEMATICS | Mathematics | exact enumerative baseline; direct constructive proof baseline                 | DECLARED_NO_SUPERIOR_BASELINE      | Theorem-native anchor lane is trace-closed and replayable. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION                                                                                                       | row SAME_CLAIM_CLASS::MATHEMATICS; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/mathematics/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics |
| SAME_CLAIM_CLASS::PHYSICS     | Physics     | bounded residual fit without theorem trace; family-wise least-squares baseline | NO_SIMPLER_SUPERIOR_BASELINE_FOUND | Bounded theorem-native physics packet is trace-closed, replay-closed, and comparator-clean on the pinned official held-out route. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION                                | row SAME_CLAIM_CLASS::PHYSICS; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics             |
| SAME_CLAIM_CLASS::CHEMISTRY   | Chemistry   | reference-table envelope baseline; reaction-class residual fit baseline        | NO_SIMPLER_SUPERIOR_BASELINE_FOUND | Bounded theorem-native chemistry packet is trace-closed, replay-closed, and comparator-clean on the pinned thermochemical, spectral, and kinetic held-out route. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION | row SAME_CLAIM_CLASS::CHEMISTRY; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry       |

| Comparison id                                       | Domain                 | Comparator families                                                    | Verdict                                            | Comparison scope                                                                                                                                                                                                                                      | Trace                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------|------------------------|------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAME_CLAIM_CLASS::BIOLOGY                           | Biology                | dataset-family signature fit baseline; state-transition score baseline | NO_SIMP<br>LER_SUP<br>ERIOR_B<br>ASELINE<br>_FOUND | Bounded theorem-native biology packet is trace-closed, replay-closed, and comparator-clean across the pinned multi-dataset held-out biological lanes. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION      | row SAME_CLAIM_CLASS::BIOLOGY; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology                                                                               |
| SAME_CLAIM_CLASS::SYSTEMS_CIVILIZATIONAL_PROJECTION | Systems /K8 projection | macro-trend residual baseline; turning-point heuristic baseline        | NO_SIMP<br>LER_SUP<br>ERIOR_B<br>ASELINE<br>_FOUND | Bounded theorem-native systems packet is trace-closed, replay-closed, and comparator-clean on the pinned macro-trajectory and regime-shift held-out route. Cost-normalized budget: NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION | row SAME_CLAIM_CLASS::SYSTEMS_CIVILIZATIONAL_PROJECTION; source releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json; closure releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection |

### Serious model families.

| Comparison id                                        | Domain  | Model family                                                  | Problem class                         | Relation to OC | What OC adds or closes                                                                                                                                           | Source refs                                                                                                                                                                                |
|------------------------------------------------------|---------|---------------------------------------------------------------|---------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::PHYSICS::PLASMA_TRANSPORT_MODELS | Physics | Gyrokinetic, confinement-scaling, and plasma-transport models | transport or plasma scaling envelopes | complements    | OC adds a shared bounded packet and replay rule that lets transport residual families live beside constants and spectra under one inspected comparison boundary. | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json |

| Comparison id                                          | Domain    | Model family                                                  | Problem class                                          | Relation to OC                   | What OC adds or closes                                                                                                                                                                      | Source refs                                                                                                                                                                                  |
|--------------------------------------------------------|-----------|---------------------------------------------------------------|--------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::PHYSICS::QUANTUM_ATOMICS_PRECISION | Physics   | Precision quantum electrodynamics and atomic-structure theory | fundamental constants and spectral transition families | integrates                       | OC adds one bounded theorem-to-observable packet that keeps constants, Balmer spectra, and transport residual families under one explicit closure, falsifier, and held-out replay contract. | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics_bundle.json   |
| SERIOUS_COMPARATOR::CHEMISTRY::KINETICS_AND_TST        | Chemistry | Chemical kinetics, transition-state, and equilibrium models   | reaction-class kinetic or equilibrium envelopes        | integrates                       | OC adds a bounded chemistry packet that keeps those observable families under one theorem, one parameter-law shell, and one held-out replay discipline.                                     | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry_bundle.json |
| SERIOUS_COMPARATOR::CHEMISTRY::QUANTUM_CHEMISTRY       | Chemistry | Quantum chemistry, ab initio, and density-functional families | thermochemical and spectroscopic envelopes             | remains superior within its lane | OC adds a bounded packet that ties thermochemical, spectroscopic, and kinetic envelopes to one theorem-native route with one falsifier and one replay discipline.                           | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry_bundle.json |
| SERIOUS_COMPARATOR::BIOLOGY::EXCITABILITY_AND_RESPONSE | Biology   | Excitable-media and response-onset dynamical models           | response-onset residual lanes                          | complements                      | OC adds a single bounded biological packet that keeps response onset tied to expression and transition-order evidence inside one replay and falsifier framework.                            | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology_bundle.json   |

| Comparison id                                              | Domain                        | Model family                                                                   | Problem class                                       | Relation to OC | What OC adds or closes                                                                                                                                                            | Source refs                                                                                                                                                                                                          |
|------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::BIOLOGY::SYSTEMS_BIOLOGY               | Biology                       | Gene-regulatory, transcriptomic, and systems-biology state-space models        | expression-signature and state-transition lanes     | integrates     | OC adds a bounded packet that keeps expression signatures, state-transition order, and response onset under one multi-dataset held-out closure route.                             | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology_bundle.json                           |
| SERIOUS_COMPARATOR::SYSTEMS_DYNAMICS_AND_CHANGE_POINTS     | Systems /K8 projection        | Systems dynamics, control, and change-point models                             | macro-process trajectories and regime shifts        | integrates     | OC adds a bounded macro-trajectory packet that keeps trajectory residuals, turning points, and regime scores under one theorem-to-observable route and one held-out decision bar. | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection_bundle.json |
| SERIOUS_COMPARATOR::CROSS_DOMAIN::COMPLEXITY_PROGRAMS      | Cross-domain /unified science | Complexity-science and multiscale integration programs                         | cross-domain integration and multiscale explanation | complements    | OC adds a harder closure grammar: explicit theorem packets, parameter laws, held-out replay, falsifiers, and same-claim comparator ledgers for each bounded lane.                 | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, content/24_oc_core_1_3_proof_machinery.tex                                                                   |
| SERIOUS_COMPARATOR::CROSS_DOMAIN::REDUCTIONIST_PATTERNWORK | Cross-domain /unified science | Best-in-class local-model portfolio governance across separate domain theories | cross-domain scientific planning and explanation    | integrates     | OC adds one explicit kernel, one K-level ladder, one theorem-to-observable grammar, and one atlas for translating bounded packets across domains under declared scope controls.   | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, content/24_oc_core_1_3_proof_machinery.tex                                                                   |

## 25.7 What prior science could do, what remained fragmented, and what OC closes

| Comparison id                                                  | Domain                        | Prior science already did                                                                                                                                         | Fragmentation or limit                                                                                                                                                                               | What OC closes or adds                                                                                                                                                                      | Open boundary                                                                                                                        |
|----------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::PHYSICS::PLASMA_TRANSPORT_MODELS           | Physics                       | Provides strong local transport and confinement heuristics for specific plasma regimes and benchmark devices.                                                     | These models are powerful in their native lane but do not by themselves close a unified residual grammar spanning constants, spectra, and transport under one promotion rule.                        | OC adds a shared bounded packet and replay rule that lets transport residual families live beside constants and spectra under one inspected comparison boundary.                            | OC does not replace specialised transport solvers or device-specific plasma modelling outside the promoted packet.                   |
| SERIOUS_COMPARATOR::PHYSICS::QED_ATOMIC_PRECISION              | Physics                       | Delivers high-precision local calculations for relations among constants, atomic transitions, and spectral families inside explicit microscopic equation classes. | Selection and justification of the local model family remain lane-specific, and the route does not by itself give a shared kernel-to-observable closure discipline across domains.                   | OC adds one bounded theorem-to-observable packet that keeps constants, Balmer spectra, and transport residual families under one explicit closure, falsifier, and held-out replay contract. | Detailed local derivation outside the promoted residual packet still relies on the standard physics model stack.                     |
| SERIOUS_COMPARATOR::CHEMISTRY::KINETICS_AND_TST                | Chemistry                     | Explains rate, barrier, and equilibrium behaviour for declared reaction classes with strong domain-specific local formalisms.                                     | The family remains reaction-class specific and does not itself unify thermochemistry, spectroscopy, and kinetics under one kernel-bound promotion route.                                             | OC adds a bounded chemistry packet that keeps those observable families under one theorem, one parameter-law shell, and one held-out replay discipline.                                     | OC does not erase the need for local kinetic modelling outside the promoted bounded families.                                        |
| SERIOUS_COMPARATOR::CHEMISTRY::QUANTUM_CHEMISTRY               | Chemistry                     | Provides molecule-specific energetic and spectroscopic calculations with a mature tool chain for local chemical systems.                                          | Method choice, basis dependence, and domain-specific tuning remain lane-specific, and the family does not by itself provide a closure ladder from kernel operators to held-out benchmark governance. | OC adds a bounded packet that ties thermochemical, spectroscopic, and kinetic envelopes to one theorem-native route with one falsifier and one replay discipline.                           | High-precision molecule-specific calculation remains the job of the established chemistry stack outside the promoted OC packet.      |
| SERIOUS_COMPARATOR::BIOLOGY::EXCITABILITY_AND_RESPONSE         | Biology                       | Captures threshold and onset dynamics well in specific excitability, signalling, or response families.                                                            | The family remains local to its chosen mechanism and does not by itself close broader biological signature, transition, and onset lanes under one bounded acceptance bar.                            | OC adds a single bounded biological packet that keeps response onset tied to expression and transition-order evidence inside one replay and falsifier framework.                            | Detailed mechanism-specific onset modelling still belongs to the established lane-specific models outside the promoted packet.       |
| SERIOUS_COMPARATOR::BIOLOGY::SYSTEMS_BIOLOGY                   | Biology                       | Provides established state-space, regulatory-network, and transcriptional modelling inside specific assay and dataset families.                                   | Performance is often lane- and dataset-specific, and cross-dataset closure plus theorem-to-observable promotion discipline remain fragmented.                                                        | OC adds a bounded packet that keeps expression signatures, state-transition order, and response onset under one multi-dataset held-out closure route.                                       | OC does not yet replace detailed mechanistic biological modelling outside the promoted assay lanes.                                  |
| SERIOUS_COMPARATOR::SYSTEMS::SYSTEM_DYNAMICS_AND_CHANGE_POINTS | Systems /K8 projection        | Provides strong local tools for trend extraction, control analysis, and regime-change detection in declared systems lanes.                                        | These tools are typically lane-specific and do not by themselves close a shared theorem route from kernel operators to official macro-process held-out promotion.                                    | OC adds a bounded macro-trajectory packet that keeps trajectory residuals, turning points, and regime scores under one theorem-to-observable route and one held-out decision bar.           | OC does not promote unrestricted civilizational prediction or replace every local macro or control model outside the bounded packet. |
| SERIOUS_COMPARATOR::CROSS_DOMAIN::COMPLEXITY_PROGRAMS          | Cross-domain /unified science | Provides rich ideas about emergence, multiscale organisation, and complex adaptive behaviour across many scientific domains.                                      | These programs are often strong conceptually but heterogeneous in proof bar, operator discipline, and closed promotion criteria.                                                                     | OC adds a harder closure grammar: explicit theorem packets, parameter laws, held-out replay, falsifiers, and same-claim comparator ledgers for each bounded lane.                           | OC does not exhaust every possible complexity-science question; it narrows attention to the lanes it can lawfully close.             |

| Comparison id                                          | Domain                        | Prior science already did                                                                            | Fragmentation or limit                                                                                                                    | What OC closes or adds                                                                                                                                                          | Open boundary                                                                                                       |
|--------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::CROSS_DOMAIN::REDUCTIONIST_PATWORK | Cross-domain /unified science | Explains many local scientific lanes well by using separate best-in-class models inside each domain. | Cross-domain translation, shared operator meaning, and unified promotion discipline remain fragmented across disconnected model families. | OC adds one explicit kernel, one K-level ladder, one theorem-to-observable grammar, and one atlas for translating bounded packets across domains under declared scope controls. | OC does not abolish local domain theories; it governs how bounded closed packets are related and promoted together. |

## 25.8 What remains frontier or hypothesis

| Domain                 | Problem class                                                           | Support          | Why this is not yet closed                                                                                                 |
|------------------------|-------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------|
| Physics                | Broader microphysical law extension beyond the promoted residual packet | frontier program | OC Core 1.3 does not yet lawfully promote a general OC replacement for the full physics model stack.                       |
| Chemistry              | General catalyst or out-of-family chemistry design                      | frontier program | OC Core 1.3 does not yet promote a general OC catalyst-discovery or unrestricted synthetic-chemistry engine.               |
| Biology                | Multi-scale phenotype control and unrestricted biological intervention  | frontier program | OC Core 1.3 does not yet promote a general OC biological intervention or whole-phenotype control engine.                   |
| Systems /K8 projection | Unrestricted long-horizon civilizational forecasting and control        | hypothesis only  | OC Core 1.3 does not yet promote unrestricted long-horizon civilizational forecasting or control as lawful closed science. |

**Transition to the atlas.** The practical section is intentionally bounded: it carries the high-signal tables needed for reader-facing use while leaving the exhaustive audit burden to Appendix R. The main chapter therefore remains a guide to bounded use rather than a duplicate of the atlas.

## A Notation and Symbols

This appendix collects the main symbols and notational conventions used throughout the Core 1.3 master monograph. It is not exhaustive, but it covers the structural elements that recur across the main proof-bearing chapters and the science appendices.

### A.1 Continuum Structure

$K$  A continuum (generic level).

$K_x$  Continuum at level  $x \in \{0, \dots, 12\}$ .

$K(t)$  Time-sliced continuum object obtained from a fixed continuum  $K$  at local time  $t$ . If a chapter needs the internal state separately, it names that state as  $s(t) \in \Omega(K, t)$ .

$\tau(K)$  Scalar level-time symbol used by the K-level chapters when a level has an effective temporal rhythm. In those chapters  $\tau(K_x)$  is the local scalar alias of  $\tau_{\text{eff}}(K_x)$ . A finite ordered family of temporal parameters must be written with a decorated family symbol such as  $\boldsymbol{\tau}(K) = (\tau_1(K), \dots, \tau_m(K))$ . If the level is atemporal or the local construction does not represent time,  $\tau(K)$  is left undefined rather than silently replaced by a trivial time object. An actual temporal order relation, when needed, is written separately, for example as  $\preceq_K$ , and is not denoted by  $\tau(K)$ . Topology is written as a paired structure such as  $(X_1, \tau_1)$ , or as the legacy local alias  $(X, \tau)$  when the local  $K_1$  display binds it explicitly.

$\tau_{\text{eff}}(K)$  Scalar effective time or rhythm selected from the temporal-parameter family  $\boldsymbol{\tau}(K)$  by the local rule declared in the surrounding chapter. When  $\tau_{C_j}$  is a cycle period and the selected rhythm is the slowest non-degradable cycle, the corresponding extremum is  $\max_j \tau_{C_j}$ . Displays using  $\min_j \tau_{C_j}$  name the fastest closed rhythm and must not be read as the slowest-cycle convention.

$M_x$  Embedding space for level  $K_x$ .

$M_x(t)$  Time-indexed embedding space for  $K_x$  when the surrounding chapter allows the environment itself to vary.

$\Omega(K)$  Set of admissible states of  $K$ .

$\Omega(K, t)$  Time-indexed admissible states of  $K$  at time  $t$ ; this is the dynamic specialization of  $\Omega(K)$  used when a chapter needs the admissible set at a specific instant. Legacy expressions such as  $\Omega(K(t))$  and  $C(K(t))$  denote the same time slice as the comma forms  $\Omega(K, t)$  and  $C(K, t)$ ; the comma form is the preferred notation in Core 1.3.

$\mathbf{1}_{\{\Omega(K,t) \neq \emptyset\}}$  Nonemptiness indicator for the time-slice admissible region: it is 1 exactly when  $\Omega(K, t) \neq \emptyset$  and 0 otherwise.

$H(\Omega(K_x))$  Legacy K-level notation for the same nonemptiness factor in continuity products. It is a Boolean admissibility indicator, not the threshold-evolution operator  $H_{\text{thr}}$ .

$\Sigma_\Omega(K_x; M_x)$  Threshold-saturation locus of  $\Omega(K_x)$  inside the declared ambient embedding space  $M_x$ :

$$\Sigma_\Omega(K_x; M_x) = \{s \in \overline{\Omega(K_x)}^{M_x} \mid \exists i \in I_{\text{act}}(K_x; M_x) : f_i(s) = 0\}.$$

Here  $g_i, \theta_i^{\text{bdry}} : \overline{\Omega(K_x)}^{M_x} \rightarrow \mathbb{R}$  are evaluated on the same local state domain and  $f_i$  is the shifted residual form of a raw threshold inequality:  $f_i(s) = g_i(s) - \theta_i^{\text{bdry}}(s)$ . Thus  $f_i(s) \leq 0$  is the admissible-side convention and  $f_i(s) = 0$  is equivalent to the pointwise raw saturation equation  $g_i(s) = \theta_i^{\text{bdry}}(s)$ . The decorated  $\theta_i^{\text{bdry}}$  is the local boundary-threshold function, not the global threshold family  $\Theta(K)$ . Scalar thresholds are the constant-function special case. Let  $I_\Theta(K_x; M_x)$  be the finite threshold family declared for this local boundary computation. The active set is

$$I_{\text{act}}(K_x; M_x) = \left\{ i \in I_\Theta(K_x; M_x) \left| \begin{array}{l} \exists s \in \overline{\Omega(K_x)}^{M_x} : f_i(s) = 0, \\ \text{and every neighbourhood } U \ni s \text{ in } \overline{\Omega(K_x)}^{M_x} \text{ contains a point } u \in U \\ \text{with } f_j(u) \leq 0 \text{ for all } j \neq i \text{ but } f_i(u) > 0 \end{array} \right. \right\}.$$

Thus  $I_{\text{act}}$  records constraints whose equality can be reached and whose removal has a local violating witness. The stronger reading that removing the constraint enlarges the full admissible side requires the regularity and constraint-qualification assumptions stated locally; without those assumptions,  $I_{\text{act}}$  is used only as a threshold-saturation witness. The same residual convention is used in the model and boundary chapters.

$\partial\Omega(K_x)$  Boundary of the admissible states of  $K_x$ . In theorem-bearing threshold contexts this is the threshold-defined admissible boundary recorded by the active saturation locus  $\Sigma_\Omega(K_x; M_x)$ . The ambient topological boundary  $\partial_{M_x}^{\text{top}} \Omega(K_x)$  is a separate derived topological object. A local section may identify  $\partial\Omega(K_x)$  with  $\partial_{M_x}^{\text{top}} \Omega(K_x)$  only after it states the required regularity and constraint-qualification assumptions.

$\partial\Omega(K_x, t)$  Time-indexed admissible boundary of  $K_x$  at time  $t$ , equivalently  $\partial_{M_x(t)} \Omega(K_x, t)$  under the threshold family active at that slice when the ambient space varies. If the surrounding chapter does not vary the ambient space, the shorthand is read relative to the fixed  $M_x$ .

$A(K)$  Set of axes of incompatible differences:  $A(K) = \{A_1, \dots, A_n\}$ .

$A(K, t)$  Time-indexed effective axis set of  $K$ , used when the working axis family is tracked dynamically.

$P(t)$  Vector of potentials (energetic, chemical, informational, biological, cognitive, social, etc.).

$J(t)$  Vector of flows transforming potentials and states.

$\Theta(K)$  Threshold landscape of  $K$ , including existence, stability, critical, dimensional, death, expressive and embedding thresholds.

$\Theta(K, t)$  Time-indexed threshold landscape of  $K$  when the admissible boundary is updated during the evolution.

$C(K)$  Family of structurally stable cycles of  $K$ .

$C(K, t)$  Time-indexed cycle family of  $K$  when the currently surviving stable cycles are tracked dynamically.

$k(K, t)$  Measure of continuumness (viability) of  $K$ .

## A.2 Thresholds and Tension

$\Theta_{\text{exist}}$  Existence thresholds: minimal conditions for  $\Omega(K) \neq \emptyset$ .

$\Theta_{\text{stab}}$  Stability thresholds: conditions for bounded dynamics.

$\Theta_{\text{crit}}$  Critical thresholds: surfaces of qualitative change or phase transition.

$\Theta_{\text{dim}}$  Dimensional thresholds: conditions for emergence of new axes.

$\Theta_{\text{time}}$  Temporal-stability threshold used in K-level time-emergence clauses. A cycle contributes to scalar level-time only when its persistence or closure score satisfies the local  $\Pi(C_j) > \Theta_{\text{time}}$  obligation.

$\Theta_{\text{death}}$  Death thresholds: limits beyond which no admissible states remain.

$\Theta_{\text{expr}}$  Expressive thresholds: minimal expressive capacity of axes required to represent relevant differences.

$\Theta_{\text{embed}}$  Embedding thresholds: constraints imposed by embedding spaces  $M_x$ .

$T(K, t)$  Structural tension functional, depending on potentials, axes and gradients.



### A.3 Operators

The time-indexed operators below describe internal dynamics for levels  $K_1$  and above on time slices of a fixed continuum  $K$ . The  $K_0$ -local glyphs and cross-level birth operators listed later are separate namespaced entries, not members of that internal-dynamics family. Level  $K_0$  is non-dynamical and carries no internal time parameter, so statements about  $K_0$  are expressed through structural relations rather than through the time-indexed operators listed here. Where the surrounding level is already fixed, the notation suppresses that constant  $K$ -dependence only for brevity rather than to imply that thresholds, cycles, boundaries, axes, or continuumness are level-free. In those displayed signatures, bare  $M$  abbreviates the active  $M_x$  or  $M_x(t)$ ;  $A(t)$ ,  $\Theta(t)$ ,  $C(t)$ ,  $\Omega(t)$ , and  $\partial\Omega(t)$  abbreviate  $A(K, t)$ ,  $\Theta(K, t)$ ,  $C(K, t)$ ,  $\Omega(K, t)$ , and  $\partial\Omega(K, t)$  for the fixed continuum under discussion. The displayed signatures in this appendix are reader-facing projections of the exact operator definitions in Section 11. They keep the principal output component visible while retaining the full coupled arguments needed by the formal operator chapter. The K-level archive and the process atlas both use the glyph family  $(\Psi, \Phi, \Lambda, U, X)$ , but they do not share one theorem-bearing namespace. In the K-level archive these glyphs are level-row shorthand; in the process atlas they are process-layer operators local to the process modules. The list below is deliberately namespaced: it includes K-level glyphs, dynamic operators used in the formal chapters, and presentation aliases used only in specific rows. A glyph entry is therefore not a global definition unless its entry explicitly says so. Bare  $\Psi$  denotes the generic birth/rebirth operator family of the formal core. Its indexed forms state the route being used:  $\Psi_{x \rightarrow x+1}$  denotes adjacent-level birth, while a rebirth route may use a same-level or cross-level candidate  $K'$  as specified by the local rebirth predicate. Legacy or local process uses must carry their level or module subscript. The source  $K_0$  archive uses  $(\Psi_0, \Phi_0, \Lambda_0)$  as its local root process glyphs for distinction generation, relational reconfiguration, and compositional assembly. The unified synthesis also uses compact aliases such as  $F_0$ ,  $Q_0$ , and  $U_0$  inside its  $K_0$  numeric rows; those aliases name local root constraints only and do not inherit the time-indexed  $K_1 +$  dynamic operator meanings listed below.

$E_{\text{evol}}$  Structural evolution operator, often written locally as  $E$  after the chapter has bound the symbol:  $E_{\text{evol}} : K(t) \rightarrow K(t + dt)$ . The decorated form avoids collision with energy quantities such as  $E(K_x)$ .

$F$  Flow operator:  $F : (J(t), P(t), A(t), \Theta(t), M) \mapsto J(t + dt)$ .

$G$  Potential operator:  $G : (P(t), A(t), J(t), \Theta(t), M) \mapsto P(t + dt)$ .

$H_{\text{thr}}$  Threshold-evolution operator, often written locally as  $H$  after the chapter has bound the symbol:  $H_{\text{thr}} : (\Theta(t), P(t), J(t), M) \mapsto \Theta(t + dt)$ .

$Q$  Cycle operator:  $Q : (C(t), P(t), J(t), \Theta(t)) \mapsto C(t + dt)$ .

$R$  Boundary operator:  $R : (\partial\Omega(t), P(t), J(t), \Theta(t), M) \mapsto \partial\Omega(t + dt)$ .

$S_{\text{axis}}$  Axis-structure operator, often written locally as  $S$  after the chapter has bound the symbol:  $S_{\text{axis}} : (A(t), P(t), J(t), \Theta(t), M) \mapsto A(t + dt)$ . The decorated form avoids collision with scalar complexity  $S(K)$ .

$U$  Continuumness operator:  $U : (k(t), \Omega(t), A(t), P(t), J(t), \Theta(t), C(t), \partial\Omega(t), M) \mapsto k(t + dt)$ .

$\text{DoF}(X)$  Degrees of freedom available to the continuum, level, or embedding object  $X$  in the local compatibility test. The  $K_0$  synthesis rows use  $\text{DoF}(K_x)/\text{DoF}(M_x)$  as a dimensionless meta-space compatibility ratio.

$C_{\text{nt}}$  Non-trivial lawful-cycle presence count or indicator in the local synthesis row. The threshold  $C_{\text{nt}} \geq 1$  means that at least one non-trivial lawful cycle remains available to the tested structure.

$\Psi_0, \Phi_0, \Lambda_0$  Local  $K_0$  root process glyphs used in the source  $K_0$  archive for distinction generation, relational reconfiguration, and compositional assembly.

$\Phi$  Meta-space evolution operator used in the K-level archive when a level's embedding space is itself allowed to change.

$\Lambda$  Constraint operator enforcing the active law or admissibility family in the relevant meta-domain.

$X$  Logical-branching operator used when the formal branch structure of a continuum must be tracked explicitly. This is the K-level archive convention; process modules that use  $X$  for collapse define that use locally.

$\Psi_{x \rightarrow x+1}$  Birth operator for  $x \in \{0, \dots, 11\}$ : admissible transition  $(K_x, M_x) \mapsto (K_{x+1}, M_{x+1})$  when the surrounding birth, embedding, and threshold conditions are satisfied.

$\Psi_{x \rightarrow y}$  Non-adjacent birth-route notation for  $y > x$ . It is not an additional primitive unless a local theorem explicitly declares it so; by default it abbreviates the admissible composition  $\Psi_{y-1 \rightarrow y} \circ \dots \circ \Psi_{x \rightarrow x+1}$ , with every intermediate threshold and embedding obligation still required.

$\Psi_{\text{reb}}$  Rebirth-route operator family. A candidate-specific use is written  $\Psi_{\text{reb}}(K, K', h; R_{\text{res}}(K), \mathcal{M}, \mathcal{M}')$ :  $R_{\text{res}}(K)$  is the residue set of the collapsed continuum,  $\mathcal{M}$  and  $\mathcal{M}'$  are the source and target embedding spaces,  $h : \mathcal{M} \rightarrow \mathcal{M}'$  is the witness map, and the later candidate  $K'$  is tested against residue, threshold conditions, and identity conditions. This notation allows same-level rebirth as well as cross-level rebirth when the local Definition 12.6 route permits it.

$E_{\text{int}}$  Interaction operator for multiple continua.

## A.4 Levels and Embedding Spaces

The labels below are mnemonic descriptions for quick notation lookup. The canonical level doctrine is stated in Section 10 and the generated K-level archive.

- $K_0$  Structural substrate (no time, no geometry, no energy).
- $K_1$  One-dimensional classical continua.
- $K_2$  Physical continua (fields, phases, percolation, Berezinskii–Kosterlitz–Thouless (BKT) transitions, mass-related structures).
- $K_3$  Chemical continua (reaction networks, reflexively autocatalytic and food-generated (RAF) closure).
- $K_4$  Protocellular continua (membranes, gradients, osmotic and curvature thresholds).
- $K_5$  Early neural and bioelectrical continua (ion channels, excitability, spikes).
- $K_6$  Cognitive continua (representations, binding, prediction).
- $K_7$  Social continua (norms, institutions, trust).
- $K_8$  Civilizational continua (infrastructures, systemic thresholds).
- $K_9$  Theory-level continua: theories, paradigms, ontologies, and formal languages treated as structured objects.
- $K_{10}$  Meta-theoretical continua: model-of-models recursion and self-referential theory structure.
- $K_{11}$  Evolving meta-theoretical spaces: operator and model-space evolution above local meta-model recursion.
- $K_{12}$  Global coherence layer: cross-framework compatibility, integrability, and the admissible upper envelope of the public hierarchy.

## A.5 Miscellaneous Symbols

- $\dim(X)$  General structural dimension convention for any object  $X$  that is already represented as an axis set, structural set, or locally linearized space in the surrounding argument.
- $\dim(K)$  For  $K_1$  and above, the dimensionality of continuum  $K$ , fixed in this monograph as  $\dim(K) = |A(K)|$ , where  $A(K)$  is its effective axis set.
- $\dim(K(t))$  Time-slice dimensionality convention for an evolving continuum:  $\dim(K(t)) = |A(K, t)|$ , where  $A(K, t)$  is the effective axis set at that slice.
- $\dim_{\text{cov}}(\Omega(K))$  Covering dimension of  $\Omega(K)$  with the subspace topology inherited from the declared ambient space. This notation is used only for nonempty admissible regions unless a local empty-set convention is explicitly declared.
- $\dim_{\text{eff}}(\Omega(K))$  Effective admissible-state dimension used in cross-level emergence comparisons. It may coincide with  $\dim_{\text{cov}}(\Omega(K))$  only under the local hypotheses stated in the surrounding argument; otherwise it is a rank or active-axis proxy, not the literal covering dimension. Legacy displays of  $\dim(\Omega(K_1)) < \dim(\Omega(K_2))$  in transition schematics are to be read as  $\dim_{\text{eff}}$ -comparisons. Local linearized approximations may also use a separate symbol such as  $r_{\text{lin}}(K)$ .
- $\dim(S, \Delta, C)$  Dimension of a structural tuple, read as the dimension of the structural set equipped with its declared difference and connectivity data.
- $\dim(K_0)$  Base-level dimension convention for the structural substrate:  $\dim(K_0) = 0$ , because  $K_0$  carries distinguishability without a geometric axis family of the higher-level kind. Abstract  $K_0$  structural variation in  $A(K_0)$  is therefore not counted by the higher-level effective-axis dimension  $\dim(K)$ .
- $r_{\text{lin}}(K)$  Optional local linearized rank notion for a section-specific approximation. It is introduced separately whenever a local argument needs a rank concept distinct from  $\dim(K)$ .
- $C_{\text{max}}(t)$  Maximal compatible cycle set at the time slice used by the operator chapter’s death criterion.
- $C_{\text{max}}(K)$  Maximal structurally stable cycle complex or set associated with  $K$  under the full threshold landscape  $\Theta(K)$ . Time-slice death checks use  $C_{\text{max}}(t)$ , and summary statistics must be named separately.
- $\text{span}_{\text{str}}(A)$  Structural span of a set of axes: the smallest structurally closed subobject, inside the fixed ambient continuum or meta-space, that contains  $A$  and is closed under the locally declared admissible composition, restriction, and threshold-respecting transport operations. Equivalently it is the intersection of all locally declared structurally closed subobjects that contain  $A$ . This notation is used only in contexts where that local closed-subobject family is nonempty and stable under arbitrary intersections; otherwise the section must declare a replacement span operation.
- $\text{span}_{\text{lin}}(A)$  Ordinary linear span of a set of axes inside a local linearization. It is used only when a section states an explicit linear approximation.
- $\text{span}(A)$  Shorthand for  $\text{span}_{\text{str}}(A)$  in theorem-bearing structural text unless a section explicitly declares a local linearisation. Use  $\text{span}_{\text{lin}}(A)$  only under such an explicit linear approximation. The subscripted forms remain preferred.
- $\chi_{\Omega}(s; K, t)$  Binary admissibility indicator with codomain  $\{0, 1\}$ :  $\chi_{\Omega}(s; K, t) = 1$  exactly when the state  $s$  of  $K$  at time  $t$  lies in  $\Omega(K, t)$ , and  $\chi_{\Omega}(s; K, t) = 0$  otherwise. The symbol  $\chi_{\Omega}$  is reserved for pointwise membership; use  $\mathbf{1}_{\{\Omega(K, t) \neq \emptyset\}}$  for nonemptiness of the admissible region itself.

This notation appendix is intentionally compact; more specialised symbols used only in particular extension papers are defined locally in those texts.

## B Collected Axiomatics

This appendix collects the main axioms of the Ontology of Continua as used in Core 1.3. It does not replace the detailed exposition in Section 4; it serves as a compact reference map for the load-bearing formal rows.

### B.1 Level $K_0$ Axioms

**Axiom 0.1 (Difference and distinguishability).**  $K_0 = (S, \Delta, \mathcal{C})$  with a difference function  $\Delta : S \times S \rightarrow \mathbb{R}_{\geq 0}$  such that

$$\Delta(s_1, s_2) = 0 \Rightarrow s_1 = s_2.$$

**Axiom 0.2 (Nontrivial threshold  $\Theta_0$ ).** There exists  $\varepsilon > 0$  such that

$$s_1 \neq s_2 \Rightarrow \Delta(s_1, s_2) \geq \varepsilon.$$

The quantity  $\Theta_0 = \varepsilon$  is the minimal structural threshold of distinguishability.

**Axiom 0.3 (No dynamics at  $K_0$ ).**  $K_0$  carries no time parameter and no evolution operator. It specifies only logical conditions on distinguishability; all dynamics belong to  $K_1$  and higher levels.

**Axiom 0.4 (Embedding constraint).** The degrees of freedom of any continuum are restricted by its embedding space: for a continuum  $K \subset M$  one has

$$A(K) \subseteq A(M), \quad \dim A(K) \leq \dim A(M).$$

### B.2 General Continuum Axioms

**Axiom 1.1 (Continuum data).** Any continuum  $K$  is specified by a tuple

$$K = (\Omega(K), A(K), P(t), J(t), \Theta(K), \partial\Omega(K), C(K), k(K, t))$$

with nonempty  $\Omega(K)$  and finite axis set  $A(K)$ .

**Axiom 1.2 (Boundary via thresholds).** There exists a family of functions  $f_k : \overline{\Omega(K)} \rightarrow \mathbb{R}$  such that

$$\Omega(K) = \{s \mid f_k(s) \leq 0 \ \forall k\}, \quad \partial\Omega(K) = \{s \mid \exists k : f_k(s) = 0\}.$$

**Axiom 1.3 (Threshold taxonomy).** Thresholds partition into existence, stability, critical, dimensional, death, expressive and embedding thresholds:

$$\Theta(K) = (\Theta_{\text{exist}}, \Theta_{\text{stab}}, \Theta_{\text{crit}}, \Theta_{\text{dim}}, \Theta_{\text{death}}, \Theta_{\text{expr}}, \Theta_{\text{embed}}).$$

**Axiom 1.4 (Continuumness).** Continuumness  $k(K, t)$  is a scalar functional of the continuum components, satisfying  $0 \leq k \leq 1$  and

$$k(K, t) = 0 \iff \Omega(K) = \emptyset \text{ or } C(K) = \emptyset.$$

**Axiom 1.5 (Evolution operator).** There exists an evolution operator  $E = (F, G, H, Q, R, S, U)$  acting on the components of  $K$  as described in Section 11.  $E$  is defined only while  $\Omega(K) \neq \emptyset$ .

### B.3 Dimensionality and Birth

**Axiom 2.1 (Monotonic dimension).** For any live continuum  $K(t)$  one has

$$\dim A(t + dt) \geq \dim A(t).$$

Strict inequality occurs only at dimensional transitions.

**Axiom 2.2 (Dimensional threshold).** A new axis can be activated only when the structural tension satisfies

$$T(K, t) > \Theta_{\dim(K)}.$$

**Axiom 2.3 (Embedding availability).** If a new axis  $A_{\text{new}}$  is added to  $K$ , it must belong to the axis set of the embedding space:

$$A_{\text{new}} \in A(M) \setminus A(K).$$

**Axiom 2.4 (Birth operator).** Whenever Axioms 2.2 and 2.3 are satisfied and a nonempty admissible region  $\Omega(K_{x+1})$  exists, a birth operator  $\Psi_{x \rightarrow x+1}$  is defined and maps  $(K_x, M_x)$  to  $(K_{x+1}, M_{x+1})$ .

### B.4 Life and Death

**Axiom 3.1 (Life conditions).** A continuum is *alive* on a time interval if and only if

$$\Omega(K(t)) \neq \emptyset, \quad C(K(t)) \neq \emptyset, \quad k(K, t) > 0.$$

**Axiom 3.2 (Death condition).** A continuum *dies* at time  $t^*$  when

$$\Omega(K(t^*)) = \emptyset.$$

After death the operators  $F, G, H, Q, R, S, U$  are no longer defined for that continuum.

**Axiom 3.3 (Irreversibility of death).** No operator acting within the same level can restore a dead continuum; any new live continuum is considered a new entity.

### B.5 Embedding Spaces

**Axiom 4.1 (Monotonic embedding spaces).** Embedding spaces form a monotonic sequence

$$M_0 \subset M_1 \subset M_2 \subset \dots$$

with  $A(M_x) \subset A(M_{x+1})$  whenever a new level  $K_{x+1}$  is born.

**Axiom 4.2 (Compatibility with embedding).** A continuum exists only if its states, axes and thresholds are compatible with its embedding space:

$$\Omega(K) \neq \emptyset \implies A(K) \subseteq A(M), \quad \Theta(K) \text{ is satisfiable in } M.$$

### B.6 Interaction

**Axiom 5.1 (Interaction operator).** For any pair of continua  $(K_a, K_b)$  embedded in the same space  $M$  there exists an interaction operator

$$E_{\text{int}} : (K_a, K_b, M) \rightarrow (K'_a, K'_b, M')$$

that updates their potentials, flows, thresholds, cycles and possibly axes, subject to the same structural constraints as the single-continuum operators.

**Axiom 5.2 (Conservation of identity in non-fusion regimes).** In competition, parasitism and symbiosis the identities of  $K_a$  and  $K_b$  are preserved. Fusion creates a new continuum  $K_{\text{fusion}}$  with its own identity and embedding space.

## B.7 Remarks

The axioms listed here represent the subset of the full Core 2.x axiomatics that is required for Core 1.3. Additional technical axioms introduced in extension papers (for example, detailed forms of structural tension, specific boundary conditions or renormalisation schemes) are compatible with this list but are not reproduced here.

## C Tabular Summary of $K_0$ – $K_{12}$

This appendix summarises the continua  $K_0$ – $K_{12}$  in tabular form. The goal is to provide a compact reference; detailed descriptions are given in Section 10.

### C.1 Overview Table

Table 28: Continuum hierarchy  $K_0$ – $K_{12}$ .

| Level    | Domain                              | Structural characterisation                                                                                                          |
|----------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $K_0$    | Structural substrate                | Set of distinguishable states $(S, \Delta, \mathcal{C})$ , no time, no energy, no geometry; minimal threshold $\Theta_0$ .           |
| $K_1$    | Classical continua                  | One-dimensional axis, continuous configurations on $(X, \tau)$ , basic stability thresholds.                                         |
| $K_2$    | Physical continua                   | Fields, phases, percolation and BKT-type transitions, mass generation, physical thresholds.                                          |
| $K_3$    | Chemical continua                   | Reaction networks, RAF structures, concentrations, environmental parameters, catalytic closure thresholds.                           |
| $K_4$    | Protocellular continua              | Membranes, osmotic and curvature thresholds, gradient maintenance, metabolic subspaces.                                              |
| $K_5$    | Early neural/bioelectrical          | Ion channels, membrane potentials, excitability thresholds, proto-spikes.                                                            |
| $K_6$    | Cognitive continua                  | Representational axes, binding, internal models, prediction and memory thresholds.                                                   |
| $K_7$    | Social continua                     | Norms, roles, institutions, trust thresholds, institutional cycles.                                                                  |
| $K_8$    | Civilizational continua             | Infrastructures, technological systems, large-scale threshold landscapes and collapse regimes.                                       |
| $K_9$    | Theoretical continua                | Theories, paradigms, ontologies, logical languages; coherence and consistency thresholds.                                            |
| $K_{10}$ | Meta-theoretical continua           | Structures that organise and transform models and modelling frameworks; self-referential thresholds.                                 |
| $K_{11}$ | Recursive meta-theoretical continua | Operator systems, meta-logics, and evolving model-spaces treated as continua with their own admissibility and recursion constraints. |
| $K_{12}$ | Global coherence continua           | Upper coherence stratum for compatibility, semantic alignment, and boundedness across families of higher-order structures.           |

## C.2 Structural Components per Level

Table 29 summarises the main components of the continuum tuple for each level.

## C.3 Vertical Continuity Conditions

For neighbouring levels  $K_x$  and  $K_{x+1}$  the following continuity conditions hold:

- State spaces are nested via projection: there exists a projection  $\pi_{x+1 \rightarrow x} : \Omega(K_{x+1}) \rightarrow \Omega(K_x)$  that forgets the new axis.
- Axes are monotonic:  $A(K_x) \subset A(K_{x+1})$  and  $\dim A(K_{x+1}) > \dim A(K_x)$ .
- Thresholds respect inheritance: thresholds of  $K_x$  are recovered from those of  $K_{x+1}$  by restricting to the subspace where the new axis is fixed.
- Embedding spaces are nested:  $M_x \subset M_{x+1}$  with  $A(M_x) \subset A(M_{x+1})$ .

These relations ensure that the continuum hierarchy is vertically coherent: higher levels refine rather than contradict the structure of lower levels.

## C.4 K-Level Parameter and Numerical Tables

This subsection collects the numerical and parameter tables that support the reader-facing K-level apparatus of the flagship volume. It sits immediately after the general K-level tables because it provides the concrete structural, empirical, and derived quantities referenced later by the unified synthesis and the practical chapter.

Values are given together with uncertainties or admissible ranges when applicable. All constants are quoted in SI unless stated otherwise. Where the continuum formalism introduces structural parameters not directly measured, the table records the corresponding functional form, numerical estimate, or bounded range rather than leaving the row implicit.

### K0: Structural conditions

### K1: Minimal observable distinctions

### K2: Cosmological and field-level parameters

### K3: Molecular-scale parameters

### K4: Compartmental and membrane parameters

### K5: Excitable systems and early neural parameters

### K6: Cognitive-level parameters

### K7: Social coordination parameters

### K8: Civilizational and infrastructural parameters

### K9: Knowledge-system parameters

### K10: Formal-system and recursion parameters

### K11: Meta-theoretical parameters

### K12: Universal-structure parameters

## D Core 1.3 Source-Audit Appendix

This appendix records the principles by which the 1.3 release remains source-first while still operating as a publication program rather than as a static archive.

### D.1 Why an appendix is still needed

The main body of the monograph is written to be read. The source-audit machinery is written to be checked. Those are related but non-identical tasks. If the full provenance burden were pushed into the main body, the book would become harder to read. If it were omitted entirely, the reader would need to trust a hidden workflow. The appendix therefore carries the discipline without replacing the main narrative.

Table 29: Structural components  $(\Omega, A, P, J, \Theta, \partial\Omega, C, k)$  per level.

| Level    | Axes $A$                                                      | Potentials $P$                                                                   | Typical cycles $C$                                                                              |
|----------|---------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| $K_0$    | Structural distinguishability axis                            | None (no dynamics)                                                               | None (no time).                                                                                 |
| $K_1$    | Single geometric axis                                         | Classical energy functionals                                                     | Periodic orbits, oscillations.                                                                  |
| $K_2$    | Spatial, internal and order parameter axes                    | Field energies, order parameters, coupling constants                             | Phase cycles, vortex/defect cycles, coherence cycles.                                           |
| $K_3$    | Concentration axes, environmental axes                        | Chemical potentials, free energy, pH, redox potentials                           | Metabolic loops, autocatalytic cycles, RAF structures.                                          |
| $K_4$    | Membrane axes, gradient axes, structural axes of compartments | Osmotic, curvature and electrochemical potentials                                | Membrane growth/division cycles, gradient maintenance cycles.                                   |
| $K_5$    | Excitation axes, electrical axes, channel configuration axes  | Membrane potential, gating variables, synaptic weights                           | Spike cycles, proto-circuit cycles, oscillatory activity.                                       |
| $K_6$    | Representational and feature axes, model axes                 | Predictive, value and confidence potentials                                      | Attention cycles, prediction-correction cycles, learning cycles.                                |
| $K_7$    | Social role axes, group axes, institutional axes              | Normative, reputational and resource potentials                                  | Role/interaction cycles, institutional cycles, governance loops.                                |
| $K_8$    | Civilizational axes (infrastructures, sectors, regions)       | Resource, energy and risk potentials                                             | Economic cycles, infrastructure renewal cycles, stability cycles.                               |
| $K_9$    | Theory and paradigm axes, formal language axes                | Coherence, consistency and expressive potentials                                 | Programme cycles, theory revision cycles, paradigm cycles.                                      |
| $K_{10}$ | Meta-model and meta-language axes                             | Structural adequacy and applicability potentials                                 | Meta-theoretical update cycles, cross-model translation cycles.                                 |
| $K_{11}$ | Operator, meta-logical, and recursive model-space axes        | Operator coherence, recursive closure, and meta-space compatibility potentials   | Operator revision cycles, meta-logic update cycles, recursive model-space cycles.               |
| $K_{12}$ | Global coherence and semantic binding axes                    | Global consistency, semantic coherence, and upper-bound admissibility potentials | Global coherence restoration cycles, semantic alignment cycles, cross-framework closure cycles. |

Table 30: Minimal structural parameters required at level  $K_0$ .

| Quantity                                               | Symbol                                                                     | Value / range |
|--------------------------------------------------------|----------------------------------------------------------------------------|---------------|
| Admissible-state measure floor                         | $\mu(\Omega(K_0))$                                                         | $> 0$         |
| Meta-space compatibility ratio for each promoted level | $\forall x \in \{1, \dots, 12\} : \frac{\text{DoF}(K_x)}{\text{DoF}(M_x)}$ | $\leq 1$      |
| Non-trivial cycle presence                             | $C_{\text{nt}}$                                                            | 1 (present)   |

Table 31: Minimal observable and energetic parameters used at level  $K_1$ .

| Quantity                          | Symbol              | Value / range             |
|-----------------------------------|---------------------|---------------------------|
| Minimal resolvable energy scale   | $\Theta_{\Delta E}$ | $10^{-34}$ – $10^{-33}$ J |
| Temporal monotonicity             | $dt/d\tau$          | $> 0$                     |
| Configuration-space measure floor | $\mu(\Omega(K_1))$  | $> 0$                     |

Table 32: Effective cosmological parameters used at level  $K_2$ .

| Quantity                                                     | Symbol                              | Value / range                   |
|--------------------------------------------------------------|-------------------------------------|---------------------------------|
| Hubble parameter today ( $\text{km s}^{-1}\text{Mpc}^{-1}$ ) | $H_0$                               | $67.4 \pm 0.5$                  |
| Matter density fraction                                      | $\Omega_{\text{m}}$                 | $0.315 \pm 0.007$               |
| Critical phase windows                                       | $T_c^{\text{EW}}, T_c^{\text{QCD}}$ | $\sim 100$ GeV;<br>150–170 MeV  |
| Curvature parameter                                          | $\Omega_k$                          | $ \Omega_k  < 2 \times 10^{-3}$ |

Table 33: Representative molecular parameters relevant at level  $K_3$ .

| Quantity                                  | Symbol            | Typical value / range |
|-------------------------------------------|-------------------|-----------------------|
| Bond length (covalent)                    | $r_{\text{cov}}$  | 0.1–0.2 nm            |
| Bond dissociation energy                  | $E_{\text{bond}}$ | 1–5 eV                |
| van der Waals radius                      | $r_{\text{vdW}}$  | 0.12–0.2 nm           |
| Reaction activation energies              | $E^\ddagger$      | 0.1–1 eV              |
| Solvent dielectric constant (water, 298K) | $\varepsilon_r$   | 78.4                  |

Table 34: Structural and energetic parameters relevant at  $K_4$ .

| Quantity                 | Symbol                 | Value / range              |
|--------------------------|------------------------|----------------------------|
| Membrane thickness       | $d_{\text{mem}}$       | 3–5 nm                     |
| Membrane bending modulus | $\kappa$               | 10–25 $k_{\text{B}}T$      |
| Line tension (edge)      | $\gamma_{\text{edge}}$ | 5–20 pN                    |
| Osmotic pressure scale   | $\Pi$                  | $10^5$ – $10^6$ Pa         |
| Typical proton gradient  | $\Delta pH$            | 0.5–2 units                |
| Permeability ranges      | $P_i$                  | $10^{-12}$ – $10^{-8}$ m/s |

Table 35: Electrical and ion-channel parameters used at  $K_5$ .

| Quantity                      | Symbol               | Value / range               |
|-------------------------------|----------------------|-----------------------------|
| Membrane capacitance          | $C_m$                | 1 $\mu\text{F}/\text{cm}^2$ |
| Resting membrane potential    | $V_{\text{rest}}$    | –75– –60 mV                 |
| Nernst potentials (Na, K, Cl) | $E_{\text{Na,K,Cl}}$ | (+60, –90, –65) mV          |
| Channel conductance (single)  | $g_{\text{ch}}$      | 5–40 pS                     |
| Refractory time               | $\tau_{\text{ref}}$  | 2–5 ms                      |



Table 36: Parameters governing representational stability and prediction thresholds at  $K_6$ .

| Quantity                              | Symbol                 | Value / range            |
|---------------------------------------|------------------------|--------------------------|
| Prediction-error threshold            | $\Theta_{\text{pred}}$ | 0.05–0.15 (normed units) |
| Working-memory span                   | $M_{\text{WM}}$        | 3–7 bound items          |
| Feature dimensionality (per modality) | $d_f$                  | $10^2$ – $10^3$          |
| Hebbian strength coefficient          | $\eta$                 | $10^{-3}$ – $10^{-1}$    |

Table 37: Typical social-coordination parameters used at  $K_7$ .

| Quantity                             | Symbol                  | Value / range           |
|--------------------------------------|-------------------------|-------------------------|
| Group stability size (Dunbar-like)   | $N_s$                   | 120–180                 |
| Information-flow bandwidth per agent | $J_{\text{comm}}$       | 1–5 bits/s              |
| Coordination threshold               | $\Theta_{\text{coord}}$ | 0.2–0.4                 |
| Conflict cost scale                  | $C_{\text{conf}}$       | 0.1–1 (arbitrary units) |

Table 38: Parameters for large-scale civilizational continua at  $K_8$ .

| Quantity                        | Symbol                | Value / range                    |
|---------------------------------|-----------------------|----------------------------------|
| Energy consumption density      | $E_{\text{dens}}$     | $10^2$ – $10^3$ W/m <sup>2</sup> |
| Infrastructure renewal time     | $\tau_{\text{infra}}$ | 20–50 years                      |
| Information-coherence threshold | $\Theta_I$            | 0.1–0.3                          |
| Repair-to-decay ratio           | $R_{\text{rep}}$      | 0.8–1.2                          |

Table 39: Parameters defining the behaviour of scientific-meta systems at  $K_9$ .

| Quantity                     | Symbol                | Value / range                   |
|------------------------------|-----------------------|---------------------------------|
| Paradigm-coherence threshold | $\Theta_{\text{par}}$ | 0.15–0.25                       |
| Anomaly accumulation rate    | $J_{\text{anom}}$     | $10^{-4}$ – $10^{-2}$ per cycle |
| Model-validation bandwidth   | $B_{\text{val}}$      | 0.1–1 (normalized)              |
| Scientific-memory half-life  | $\tau_{\text{mem}}$   | 30–200 years                    |

Table 40: Structural parameters of formal continua at  $K_{10}$ .

| Quantity                    | Symbol                 | Value / range                  |
|-----------------------------|------------------------|--------------------------------|
| Recursion depth (effective) | $d_{\text{rec}}$       | $10$ – $10^4$                  |
| Consistency threshold       | $\Theta_{\text{cons}}$ | 0.01–0.05                      |
| Proof-flow capacity         | $J_{\text{proof}}$     | $1$ – $10^3$ steps/s (machine) |

Table 41: Parameters governing meta-theoretical landscapes at  $K_{11}$ .

| Quantity                 | Symbol                 | Value / range                    |
|--------------------------|------------------------|----------------------------------|
| Meta-coherence threshold | $\Theta_{\text{meta}}$ | 0.02–0.1                         |
| Functorial potential     | $P_{\text{funct}}$     | $10^0$ – $10^3$ (abstract units) |
| Reflexive depth          | $d_{\text{refl}}$      | 3–50                             |
| Cross-landscape coupling | $J_X$                  | 0.1–1                            |

Table 42: Parameters relevant at the highest-level structural continuum  $K_{12}$ .

| Quantity                       | Symbol                | Value / range                      |
|--------------------------------|-----------------------|------------------------------------|
| Universal integration capacity | $C_{\text{uni}}$      | $10^3$ – $10^{12}$ (dimensionless) |
| Cross-continuum compatibility  | $\Theta_{\text{uni}}$ | 0.001–0.01                         |
| Structural reachability        | $R_{\text{uni}}$      | 0.1–1                              |
| Global tension budget          | $T_{\text{uni}}$      | $10^2$ – $10^8$                    |

## D.2 Witness classes

The release uses several witness classes:

- the public LaTeX corpus as the source-native textual witness,
- the archival compiled PDF as the stable page-and-line witness,
- the bounded journal-core extraction as the outward review-focused derivative witness,
- and companion packets as support witnesses for chronology, theorem fate, and release packaging.

None of these witnesses should be mistaken for the others. The source witness is not the same as the release manifest. The release manifest is not the same as the theorem. The journal-core article is not the same as the monograph. The 1.3 publication line remains healthy only when these roles stay distinct.

## D.3 Questions that the appendix answers

This appendix is meant to answer concrete review questions.

- From which public source stack is the present volume built?
- Which outward article is derived from it?
- Why is the monograph larger than the journal-core article?
- What justifies calling the volume 1.3 rather than simply republishing 1.2?
- How should later ESTRAs and channel projections interpret the distinction between formal master, review article, and support packet?

The release is stronger when those questions can be answered inside the volume rather than by pointing the reader to an unrelated dashboard.

## D.4 Current boundary

Core 1.3 should therefore be read as a source-first enlargement and clarification of the formal publication shell. The monograph does not pretend that every future projection is already proved. The journal core does not pretend to be the whole theory. The companion packets do not pretend to be the manuscript. That three-way distinction is the governing boundary of the current release.

# E Journal-Core Extraction Bridge

This appendix explains how the journal-core article relates to the master monograph.

## **E.1 Why the journal-core exists**

Some channels require a bounded, faster-reading scientific artifact. Editors, reviewers, and external readers often need a paper-sized object whose defended claim can be read in one sitting. The journal-core exists for that purpose. Its job is not to replace the monograph but to provide a narrow article that remains anchored to the same scientific truth conditions.

## **E.2 What the journal-core may omit**

The journal-core may compress:

- extensive chapter-scale didactic explanation,
- large parts of the domain atlas,
- broad module coverage,
- detailed appendical inventories,
- and some of the monograph-level pedagogical scaffolding.

It may not silently change:

- theorem scope,
- explicit nonclaims,
- claimed source basis,
- figure meaning,
- or the relation between inherited corpus and present defended claim.

## **E.3 What the monograph guarantees to later artifacts**

Because the journal-core is extracted from the present volume, later owner-facing and public channels do not need to improvise their own understanding of the science. They can derive from the master monograph when they need depth, from the journal core when they need bounded exposition, and from the companion layer when they need support evidence.

This is the practical publication value of the monograph. It is not merely larger than the article. It is the document that prevents every later projection from becoming a separate interpretive exercise.

## F Reviewer Navigation Matrix and Audit Checklist

Start with the row that matches the question or concern, inspect the primary route, and use the verification column as the pass/fail test. This appendix uses three canonical artifact terms:

- *Master Monograph*: the English flagship volume. Later uses of *Monograph* are shorthand for this same artifact.
- *Journal Core*: the shorter bounded extraction from the monograph.
- *Release*: the outward package that also contains supporting surfaces, PDFs, assets, and companion dossiers.

Capitalisation marks the canonical glossary terms above; lowercase uses are generic prose shorthand unless explicitly styled as one of those terms. The following class labels are local definitions for this appendix rather than additional canonical glossary terms. Concrete artifact and witness classes are as follows:

- *Surfaces* are the release-bundled `editorial/*_latest.json` projections.
- *Companion dossiers* are the generated dossier packages under `editorial/dossier_packages/`.
- *Assets* are the released figures and tables under `assets/`.
- *Archival PDF witness*: the prior Core 1.2 PDF named in the journal-core provenance route in Appendix E: *Ontology of Continua — Core v1.2.0*.

### F.1 Review questions, compact route, and decisive checks

#### Core theorem route and scope.

| Reviewer question                                                       | Primary route to inspect | Verification cue                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is the central defended result?                                    | Abstract; Sections 5, 19 | Theorem A: bounded post-collapse identity classification; no general lift theorem.                                                                                                                                                                                                                                                       |
| Where does the proof-bearing route begin?                               | Sections 5, 19           | Verify that Axiom 3.2 with Definitions 12.1, 12.3, and 12.4 supports Source Theorem 3; that Source Theorem 3 supports Lemma 1; that Lemma 1 with Definition 12.5 supports Lemma 2; and that Lemma 2 supports Theorem A. Inspect Axiom 3.3, Definition 12.6, and Source Corollary 3.2 separately as the companion rebirth-control branch. |
| Is collapse defined formally or rhetorically?                           | Sections 5, 12           | Admissible-state emptiness, $k \rightarrow 0$ , and threshold failure.                                                                                                                                                                                                                                                                   |
| What exactly is residue?                                                | Sections 12, 20          | Post-collapse structural trace; no continuing live identity.                                                                                                                                                                                                                                                                             |
| What exactly is rebirth?                                                | Sections 12, 20          | Later live continuum grounded in residue; numerically new.                                                                                                                                                                                                                                                                               |
| Why is irreversibility necessary?                                       | Sections 5, 19           | Blocks same-entity restoration after lawful death.                                                                                                                                                                                                                                                                                       |
| Could the main theorem depend on an excluded lift-heavy theorem family? | Section 19; Appendix E   | No excluded lift-heavy theorem family is used in the bounded proof route.                                                                                                                                                                                                                                                                |
| Why is the monograph larger than the journal core?                      | Section 17; Appendix E   | Didactic shell, formal architecture, and repair logic live here.                                                                                                                                                                                                                                                                         |
| Why is the journal core smaller without becoming weaker?                | Appendix E               | One-way extraction; theorem scope unchanged.                                                                                                                                                                                                                                                                                             |
| Why is the hierarchy fixed as $K_0, \dots, K_{12}$ ?                    | Sections 18, 10, 20      | Upper levels carry substantive higher-order roles.                                                                                                                                                                                                                                                                                       |

### Audit, provenance, and review posture.

| Reviewer question                                | Primary route to inspect   | Verification cue                                                                                                                                                                 |
|--------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What would falsify the upper-level doctrine?     | Sections 20, 10            | Lossless reduction of $K_{11}$ and $K_{12}$ to $K_{10}$ .                                                                                                                        |
| How do proof-oriented statement classes differ?  | Sections 18, 19            | Axiomatic, derived, structural, interpretive, empirical, and frontier rows keep separate burdens; editorial-bridge language remains packaging support rather than a proof class. |
| What is merely interpretive?                     | Sections 6, 17, 20         | Pass only if interpretive language stays outside the proof route and is not promoted into theorem, replay, or practical-use authority.                                           |
| What is the exact source base?                   | Section 17; Appendix D     | Source corpus, SPOT surface, and archival PDF witness are separate objects.                                                                                                      |
| Why is ORCID foregrounded?                       | Title page; Section 17     | Stable scholarly identity.                                                                                                                                                       |
| What is the role of companion dossiers?          | Section 17; Appendix E     | Support and provenance exposure, not replacement authority.                                                                                                                      |
| How should a non-specialist enter the monograph? | Sections 1, 20             | Didactic ladder before proof audit.                                                                                                                                              |
| Where are the most likely points of critique?    | Section 19; Appendix E     | Collapse semantics, irreversibility, upper levels, extraction discipline.                                                                                                        |
| What would count as a decisive release failure?  | Sections 20, 15, 17        | Continued identity after admissible-state loss, or reliance on an excluded theorem family.                                                                                       |
| What remains outside scope?                      | Sections 6, 17; Appendix E | Pass only if excluded theorem families, frontier rows, and projection targets remain outside the positive proof body.                                                            |

## F.2 Compact reading order by reviewer role

**Rapid critical reviewer.** Read the abstract, theorem roadmap, relevant theorems and corollaries, the worked examples, and the journal-core bridge appendix. Return to this matrix only to choose a new route after that first pass. This route should reveal immediately whether the release is too large, too narrow, or too broad in scope.

**Formal mathematical reviewer.** Read the model, results, axioms, operators, collapse/rebirth material, hierarchy chapter, notation appendix, and source-audit chapter. The decisive question is whether the derived bounded claim can be reconstructed from the stated source-native rows without informal supplementation.

**Generalist scientific reviewer.** Read the introduction, reader guide, theorem roadmap, worked examples, discussion, conclusion, and then return to the formal chapters selectively. The decisive question here is whether the volume explains itself honestly enough that scientific judgment can precede local notation mastery.

## **G Empirical Validation Matrix and Replay Ledger**

This appendix is projected from the canonical science SPOT so that bridge evidence and promotion verdicts remain visibly distinct.

This appendix is projected from the canonical SPOT. It reports both bridge evidence and promotion verdicts without letting one masquerade as the other.

| Domain                              | Class              | Trace              | Evidence                  | Verdict | Blocking<br>ids | Notes                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|--------------------|--------------------|---------------------------|---------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathematics                         | THEOREM<br>_NATIVE | TRACE_C<br>OMplete | EVIDENC<br>E_COMPL<br>ETE | PASS    | none            | Sources: Clay Mathematics Institute, American Institute of Mathematics; Replay: logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py; Metrics: Campaign PASS_31_OF_31_TERMINALIZED; 31 terminalized cases; Next action: MAINTAIN_REPLAY_DISCIPLINE                                                                                                                                                          |
| Physics                             | THEOREM<br>_NATIVE | TRACE_C<br>OMplete | EVIDENC<br>E_COMPL<br>ETE | PASS    | none            | Sources: NIST, U.S. Department of Energy Office of Science; Replay: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS; Metrics: 30 covered / 20 held-out cases; Mean abs. normalized error 0.188136 sigma; P95 normalized error 0.64778 sigma; Maximum normalized error 1.48 sigma; Tail breaches 0; Next action: MAINTENANCE_ONLY                                    |
| Chemistry                           | THEOREM<br>_NATIVE | TRACE_C<br>OMplete | EVIDENC<br>E_COMPL<br>ETE | PASS    | none            | Sources: NIST, U.S. Department of Energy Office of Science; Replay: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY; Metrics: 30 covered / 19 held-out cases; Mean abs. normalized error 0.533285 sigma; P95 normalized error 1.0 sigma; Maximum normalized error 1.597772 sigma; Tail breaches 0; Next action: MAINTENANCE_ONLY                                  |
| Biology                             | THEOREM<br>_NATIVE | TRACE_C<br>OMplete | EVIDENC<br>E_COMPL<br>ETE | PASS    | none            | Sources: NCBI GEO; Replay: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY; Metrics: 30 covered / 16 held-out cases; Mean abs. normalized error 0.444222 sigma; P95 normalized error 1.030033 sigma; Maximum normalized error 1.438582 sigma; Tail breaches 0; Next action: MAINTENANCE_ONLY                                                                        |
| Systems / Civilizational projection | THEOREM<br>_NATIVE | TRACE_C<br>OMplete | EVIDENC<br>E_COMPL<br>ETE | PASS    | none            | Sources: World Bank, U.S. Department of Energy Office of Science; Replay: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION; Metrics: 30 covered / 18 held-out cases; Mean abs. normalized error 0.56097 sigma; P95 normalized error 1.924704 sigma; Maximum normalized error 2.25395 sigma; Tail breaches 0; Next action: MAINTENANCE_ONLY |

## H Institute-Run Measurement and Observation Program

This appendix records the stand-by measurement waves that would become mandatory only if pinned official or open sources stopped closing the declared empirical lanes honestly. At the present release boundary, all four empirical domains remain on official- or open-data routes; the institute-run waves are retained as controlled fallback mechanisms rather than as active evidence channels.

| Domain                              | Stand-by wave                     | Escalation trigger                                                                                                                             | Current action             |
|-------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Physics                             | INSTITUTE_RUN::PHYSICS::WAVE_1A   | Open-data coverage falls below full coverage, held-out coverage drops below 30 cases, or residual breaches persist after replay maintenance.   | Maintain replay discipline |
| Chemistry                           | INSTITUTE_RUN::CHEMISTRY::WAVE_2A | Open-data coverage falls below full coverage, held-out coverage drops below 30 cases, or residual breaches persist after replay maintenance.   | Maintain replay discipline |
| Biology                             | INSTITUTE_RUN::BIOLOGY::WAVE_3A   | Open-data coverage falls below full coverage, held-out coverage drops below 30 cases, or the bounded signature class remains underconstrained. | Maintain replay discipline |
| Systems / Civilizational projection | INSTITUTE_RUN::SYSTEMS::WAVE_4A   | Open-data coverage falls below full coverage, held-out interval coverage drops below 30 intervals, or turning-point breaches persist.          | Maintain replay discipline |

**Interpretive rule.** These waves are not alternate proof routes. They are reserve observation programs that become lawful only after the official or open route fails to cover the declared observable family. Until such a trigger is reached, the current release remains committed to the pinned replay discipline already documented in the empirical protocol chapters and benchmark appendices.

## I Pinned Benchmark Dataset Manifest

This appendix records the pinned benchmark families, held-out policies, quantitative gates, and falsifiers that define lawful empirical promotion in the current release. The point is not to restate raw registry fields one by one, but to show which dataset family carries which evidential burden and what would count as a genuine empirical failure.

**Mathematics.** Protocol `OC13::EXECUTION::MATHEMATICS::ANCHOR_REPLAY` runs under the hybrid-escalation evidence bar, but in practice this lane is deterministic rather than measurement-driven. The held-out policy is `DETERMINISTIC_COUNTEREXAMPLE_AND_TRACE_REPLAY`, with a minimum of 31 strict cases. Coverage, sigma thresholds, Brier score, and ECE are therefore not applicable. Promotion survives only if no replay divergence and no surviving counterexample appear. Replay command: `logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py`. No external dataset manifest is required because the mathematics anchor closes on strict replay and counterexample traces rather than on observational data.

**Physics.** Protocol `OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT` runs under hybrid escalation and binds theorem-to-observable maps to pinned official constants, spectral tables, and transport challenge corpora. The current caseset contains 30 pinned cases, 20 held-out cases, and declared predictions for all 30 benchmark rows. The quantitative gate requires full coverage, mean absolute normalized error at most 1.0 sigma, p95 at most 2.5 sigma, maximum error at most 5.0 sigma, and zero tail breaches. Promotion survives only if every benchmark family remains within tolerance; any residual breach or broken sign/order constraint falsifies the packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-id PHYSICS`.



| Domain                              | Execution protocol                                                  | Held-out design                                                      | Quantitative gate                                                                                                                |
|-------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Mathematics                         | OC13::EXECUTION::MATHEMATICS::ANCHOR_REPLAY                         | Deterministic counterexample and trace replay; minimum 31 cases      | No sigma gate; promotion fails on any divergence or surviving counterexample                                                     |
| Physics                             | OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT               | Leave one benchmark family out; minimum 30 cases                     | Coverage 1.0; mean abs. error at most 1.0 sigma; p95 at most 2.5 sigma; max at most 5.0 sigma; tail breaches 0                   |
| Chemistry                           | OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY_SPECTRA_KINETICS        | Hold out one compound or reaction class; minimum 30 cases            | Coverage 1.0; mean abs. error at most 1.0 sigma; p95 at most 2.5 sigma; max at most 5.0 sigma; tail breaches 0                   |
| Biology                             | OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPONSE_SIGNATURES | Reserve one dataset family per signature class; minimum 30 cases     | Coverage 1.0; mean abs. error at most 1.0 sigma; p95 at most 2.5 sigma; max at most 5.0 sigma; ECE at most 0.05; tail breaches 0 |
| Systems / Civilizational projection | OC13::EXECUTION::SYSTEMS::MACRO_TRAJECTORIES_AND_REGIMES_SHIFTS     | Held-out interval replay with turning-point checks; minimum 30 cases | Coverage 1.0; Brier at most 0.08; ECE at most 0.05; tail breaches 0                                                              |

- PHY\_BENCH\_001: fine\_structure\_constant\_residual. Authority: NIST/CODATA constants. Dataset: <https://physics.nist.gov/cuu/Constants/index.html>. Held-out policy: leave one family out for residual verification.
- PHY\_BENCH\_002: hydrogen\_balmer\_line\_residual. Authority: NIST Atomic Spectra Data base. Dataset: [https://physics.nist.gov/PhysRefData/ASD/lines\\_form.html](https://physics.nist.gov/PhysRefData/ASD/lines_form.html). Held-out policy: reserve transitions not used during packet fitting.
- PHY\_BENCH\_003: transport\_scaling\_residual. Authority: DOE Office of Science challenge corpora. Dataset: [https://science.osti.gov/-/media/fes/pdf/workshop-reports/FES\\_Grand\\_Challenges\\_Report\\_final.pdf](https://science.osti.gov/-/media/fes/pdf/workshop-reports/FES_Grand_Challenges_Report_final.pdf). Held-out policy: hold out one transport-regime family.

**Chemistry.** Protocol OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY\_SPECTRA\_KINETICS maps theorem packets to thermochemical, spectral, and kinetic observables against pinned reference tables. The current caseset contains 30 pinned cases, 19 held-out cases, and declared predictions for all 30 benchmark rows. The quantitative gate matches the physics lane: full coverage, mean absolute normalized error at most 1.0 sigma, p95 at most 2.5 sigma, maximum error at most 5.0 sigma, and zero tail breaches. A held-out thermochemical, spectral, or kinetic family outside tolerance falsifies the promoted chemistry packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY`.

- CHEM\_BENCH\_001: gas\_phase\_enthalpy\_residual. Authority: NIST Chemistry WebBook. Dataset: <https://webbook.nist.gov/chemistry/>. Held-out policy: hold out one compound family from calibration.
- CHEM\_BENCH\_002: spectral\_transition\_residual. Authority: NIST Chemistry WebBook. Dataset: <https://webbook.nist.gov/chemistry/>. Held-out policy: hold out one spectral family.
- CHEM\_BENCH\_003: kinetic\_or\_equilibrium\_residual. Authority: DOE BES/EFRC challenge corpora. Dataset: <https://science.osti.gov/bes/efrc/Research/Grand-Challenges>. Held-out policy: hold out one reaction-class envelope.

**Biology.** Protocol OC13::EXECUTION::BIOLOGY::STATE\_TRANSITIONS\_AND\_RESPONSE\_SIGNATURES chooses one bounded biological lane at a time and evaluates predicted state transitions or response signatures against open assay data. The current caseset contains 30 pinned cases, 16 held-out cases, and declared predictions for all 30 benchmark rows. The quantitative gate requires full coverage, mean absolute normalized error at most 1.0 sigma, p95 at most 2.5 sigma, maximum error at most 5.0 sigma, ECE at most 0.05, and zero tail breaches. Promotion survives only if the bounded signature family reproduces the held-out dataset family without widening the claim boundary. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY`.

- BIO\_BENCH\_001: expression\_signature\_residual. Authority: NCBI GEO. Dataset: <https://www.ncbi.nlm.nih.gov/geo/>. Held-out policy: reserve one time-series family for held-out replay.
- BIO\_BENCH\_002: state\_transition\_order\_error. Authority: NCBI GEO. Dataset: <https://www.ncbi.nlm.nih.gov/geo/>. Held-out policy: reserve one cell-state transition family.
- BIO\_BENCH\_003: response\_onset\_residual. Authority: official open biological assay repositories. Dataset: <https://www.ncbi.nlm.nih.gov/geo/>. Held-out policy: reserve one response-signature family.

**Systems / Civilizational projection.** Protocol OC13::EXECUTION::SYSTEMS::MACRO\_TRAJECTORIES\_AND\_REGIME\_SHIFTS evaluates K8 projection packets against pinned macro-process series with held-out interval tests. The current caseset contains 30 pinned cases, 18 held-out cases, and declared predictions for all 30 benchmark rows. The quantitative gate requires full coverage, Brier score at most 0.08, ECE at most 0.05, and zero tail breaches. Promotion survives only if the declared trajectory and regime-change observables remain inside tolerance on held-out intervals; a missed turning point or out-of-band trajectory falsifies the packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION`.

- SYS\_BENCH\_001: population\_trajectory\_residual. Authority: World Bank Indicators API. Dataset: <https://api.worldbank.org/v2/en/indicator/SP.POP.TOTL?format=json>. Held-out policy: leave out one interval family.
- SYS\_BENCH\_002: gdp\_growth\_turning\_point\_error. Authority: World Bank Indicators API. Dataset: <https://api.worldbank.org/v2/en/indicator/NY.GDP.MKTP.KD.ZG?format=json>. Held-out policy: reserve one held-out macro interval.
- SYS\_BENCH\_003: process\_system\_regime\_score. Authority: World Bank Indicators API. Dataset: <https://api.worldbank.org/v2/en/indicator/EG.USE.PCAP.KG.OE?format=json>. Held-out policy: reserve one resource-intensity curve family.

## J Pinned Benchmark Caseset and Parameterization Gaps

This appendix records the concrete pinned seed cases that sit beneath the benchmark manifest. The caseset does not by itself imply empirical validation: it only freezes the observable family, held-out role, and parameterization gap honestly enough that the replay backlog becomes executable rather than rhetorical.

### J.1 Mathematics

Pinned cases: '0'; accession-frozen cases: '0'; held-out cases: '0'; prediction-declared cases: '0' / '0'. Coverage readiness: 'NOT\_REQUIRED' with minimum cases '31', gap '31', coverage ratio '0', and parameterization status 'NOT\_REQUIRED'. No cases are currently frozen for this lane. That is a lawful blocker, not hidden unfinished release text.

### J.2 Physics

Pinned cases: '30'; accession-frozen cases: '0'; held-out cases: '20'; prediction-declared cases: '30' / '30'. Coverage readiness: 'PASS\_CASESET\_MINIMUM\_REACHED' with minimum cases '30', gap '0', coverage ratio '1.0', and parameterization status 'PREDICTIONS\_DECLARED\_FOR\_ALL\_CASES'.

- `PHY_CASE_001_ALPHA_CONSTANT_REFERENCE / PHY_BENCH_001`: CODATA fine-structure constant reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_002_PLANCK_CONSTANT_REFERENCE / PHY_BENCH_001`: Planck constant reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_003_RYDBERG_CONSTANT_REFERENCE / PHY_BENCH_001`: Rydberg constant reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_004_ELECTRON_MASS_REFERENCE / PHY_BENCH_001`: Electron mass reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_005_PROTON_ELECTRON_MASS_RATIO / PHY_BENCH_001`: Proton-electron mass ratio reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_006_BOHR_RADIUS_REFERENCE / PHY_BENCH_001`: Bohr radius reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_007_HARTREE_ENERGY_REFERENCE / PHY_BENCH_001`: Hartree energy reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.

- `PHY_CASE_008_ELECTRON_G_FACTOR_REFERENCE / PHY_BENCH_001`: Electron g-factor reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_009_AVOGADRO_CONSTANT_REFERENCE / PHY_BENCH_001`: Avogadro constant reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_010_VACUUM_IMPEDANCE_REFERENCE / PHY_BENCH_001`: Vacuum impedance reference row.  
Observable: `relative_constant_error`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::CONSTANTS::SIGMA_RESIDUAL_PROFILE_V1`. Method: Sigma-normalized residual fit against pinned NIST/CODATA anchors with leave-family-out held-out evaluation. Authority: NIST/CODATA constants.
- `PHY_CASE_011_HYDROGEN_BALMER_ALPHA / PHY_BENCH_002`: Hydrogen Balmer-alpha spectral transition.  
Observable: `spectral_line_residual`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::SPECTRA::LINE_RESIDUAL_PROFILE_V1`. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDatabase.
- `PHY_CASE_012_HYDROGEN_BALMER_BETA / PHY_BENCH_002`: Hydrogen Balmer-beta spectral transition.  
Observable: `spectral_line_residual`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::SPECTRA::LINE_RESIDUAL_PROFILE_V1`. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDatabase.
- `PHY_CASE_013_HYDROGEN_BALMER_GAMMA / PHY_BENCH_002`: Hydrogen Balmer-gamma spectral transition.  
Observable: `spectral_line_residual`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::SPECTRA::LINE_RESIDUAL_PROFILE_V1`. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDatabase.
- `PHY_CASE_014_HYDROGEN_BALMER_DELTA / PHY_BENCH_002`: Hydrogen Balmer-delta spectral transition.  
Observable: `spectral_line_residual`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::SPECTRA::LINE_RESIDUAL_PROFILE_V1`. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDatabase.
- `PHY_CASE_015_SODIUM_D1_TRANSITION / PHY_BENCH_002`: Sodium D1 spectral transition.  
Observable: `spectral_line_residual`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::SPECTRA::LINE_RESIDUAL_PROFILE_V1`. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDatabase.
- `PHY_CASE_016_SODIUM_D2_TRANSITION / PHY_BENCH_002`: Sodium D2 spectral transition.  
Observable: `spectral_line_residual`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::PHYSICS::SPECTRA::LINE_RESIDUAL_PROFILE_V1`. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDatabase.
- `PHY_CASE_017_HELIUM_5876_TRANSITION / PHY_BENCH_002`: Helium I 5876 Å spectral transition.

- Observable: spectral\_line\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::SPECTRA::LINE\_RESIDUAL\_PROFILE\_V1. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDataBase.
- PHY\_CASE\_018\_HELIUM\_6678\_TRANSITION / PHY\_BENCH\_002: Helium I 6678 A spectral transition.  
Observable: spectral\_line\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::SPECTRA::LINE\_RESIDUAL\_PROFILE\_V1. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDataBase.
  - PHY\_CASE\_019\_OXYGEN\_777NM\_TRIPLET / PHY\_BENCH\_002: O I 777 nm triplet spectral family.  
Observable: spectral\_line\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::SPECTRA::LINE\_RESIDUAL\_PROFILE\_V1. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDataBase.
  - PHY\_CASE\_020\_CALCIIUM\_II\_K\_LINE / PHY\_BENCH\_002: Ca II K spectral line.  
Observable: spectral\_line\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::SPECTRA::LINE\_RESIDUAL\_PROFILE\_V1. Method: Pinned spectral residual envelope fit on calibration transitions with held-out transition-family replay. Authority: NISTAtomicSpectraDataBase.
  - PHY\_CASE\_021\_TOKAMAK\_CONFINEMENT\_SCALING / PHY\_BENCH\_003: Tokamak confinement scaling envelope.  
Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFSgrandchallenges.
  - PHY\_CASE\_022\_DIVERTOR\_HEAT\_FLUX\_ENVELOPE / PHY\_BENCH\_003: Divertor heat-flux envelope family.  
Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFSgrandchallenges.
  - PHY\_CASE\_023\_MAGNETIC\_RECONNECTION\_RATE / PHY\_BENCH\_003: Magnetic reconnection rate family.  
Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFSgrandchallenges.
  - PHY\_CASE\_024\_GYROKINETIC\_HEAT\_TRANSPORT / PHY\_BENCH\_003: Gyrokinetic heat-transport envelope.  
Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFSgrandchallenges.
  - PHY\_CASE\_025\_EDGE\_LOCALIZED\_MODE\_THRESHOLD / PHY\_BENCH\_003: Edge-localized-mode onset threshold family.  
Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFSgrandchallenges.
  - PHY\_CASE\_026\_INERTIAL\_CONFINEMENT\_HOTSPOT\_SCALING / PHY\_BENCH\_003: Inertial-confinement hotspot scaling family.  
Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFSgrandchallenges.
  - PHY\_CASE\_027\_LASER\_PLASMA\_WAKEFIELD\_ENVELOPE / PHY\_BENCH\_003: Laser-plasma wakefield envelope family.

Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFEsgrandchallenges.

- PHY\_CASE\_028\_HALL\_THRUSTER\_TRANSPORT\_ENVELOPE / PHY\_BENCH\_003: Hall thruster transport envelope family.

Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFEsgrandchallenges.

- PHY\_CASE\_029\_ION\_ACOUSTIC\_DISPERSION\_FAMILY / PHY\_BENCH\_003: Ion-acoustic dispersion family.

Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFEsgrandchallenges.

- PHY\_CASE\_030\_ALFVENIC\_TURBULENCE\_SPECTRUM / PHY\_BENCH\_003: Alfvenic turbulence spectrum family.

Observable: transport\_scaling\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

Profile: OC13::PHYSICS::TRANSPORT::SCALING\_ENVELOPE\_PROFILE\_V1. Method: Regime-envelope residual fit with held-out transport family replay. Authority: DOEFEsgrandchallenges.

### J.3 Chemistry

Pinned cases: '30'; accession-frozen cases: '0'; held-out cases: '19'; prediction-declared cases: '30' / '30'. Coverage readiness: 'PASS\_CASESET\_MINIMUM\_REACHED' with minimum cases '30', gap '0', coverage ratio '1.0', and parameterization status 'PREDICTIONS\_DECLARED\_FOR\_ALL\_CASES'.

- CHEM\_CASE\_001\_H2O\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Water gas-phase thermochemistry reference row.

Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.

- CHEM\_CASE\_002\_CO2\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Carbon dioxide gas-phase thermochemistry reference row.

Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.

- CHEM\_CASE\_003\_CH4\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Methane gas-phase thermochemistry reference row.

Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.

- CHEM\_CASE\_004\_NH3\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Ammonia gas-phase thermochemistry reference row.

Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.

- CHEM\_CASE\_005\_CH3OH\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Methanol gas-phase thermochemistry reference row.

Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

- Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.
- CHEM\_CASE\_006\_C2H5OH\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Ethanol gas-phase thermochemistry reference row.  
Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_007\_CO\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Carbon monoxide gas-phase thermochemistry reference row.  
Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_008\_H2\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Hydrogen gas-phase thermochemistry reference row.  
Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_009\_N2\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Nitrogen gas-phase thermochemistry reference row.  
Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_010\_O2\_GAS\_PHASE\_ENTHALPY / CHEM\_BENCH\_001: Oxygen gas-phase thermochemistry reference row.  
Observable: enthalpy\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::THERMOCHEMISTRY::ENTHALPY\_PROFILE\_V1. Method: Sigma-normalized thermochemical residual fit against pinned NIST WebBook values. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_011\_H2O\_IR\_FAMILY / CHEM\_BENCH\_002: Water infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_012\_CO2\_IR\_FAMILY / CHEM\_BENCH\_002: Carbon dioxide infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_013\_CH4\_IR\_FAMILY / CHEM\_BENCH\_002: Methane infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_014\_NH3\_IR\_FAMILY / CHEM\_BENCH\_002: Ammonia infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_015\_CO\_IR\_FAMILY / CHEM\_BENCH\_002: Carbon monoxide infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.

- Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
- CHEM\_CASE\_016\_C2H6\_IR\_FAMILY / CHEM\_BENCH\_002: Ethane infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_017\_C2H4\_IR\_FAMILY / CHEM\_BENCH\_002: Ethylene infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_018\_C2H2\_IR\_FAMILY / CHEM\_BENCH\_002: Acetylene infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_019\_CH3OH\_IR\_FAMILY / CHEM\_BENCH\_002: Methanol infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_020\_C2H5OH\_IR\_FAMILY / CHEM\_BENCH\_002: Ethanol infrared spectrum family.  
Observable: spectral\_transition\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::SPECTRA::TRANSITION\_PROFILE\_V1. Method: Pinned spectral-family residual fit against NIST WebBook spectral references. Authority: NISTChemistryWebBook.
  - CHEM\_CASE\_021\_HABER\_BOSCH\_EQUILIBRIUM\_ENVELOPE / CHEM\_BENCH\_003: Haber-Bosch equilibrium envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorpora.
  - CHEM\_CASE\_022\_WATER\_GAS\_SHIFT\_ENVELOPE / CHEM\_BENCH\_003: Water-gas-shift reaction envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorpora.
  - CHEM\_CASE\_023\_CO\_OXIDATION\_ENVELOPE / CHEM\_BENCH\_003: Carbon-monoxide oxidation envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorpora.
  - CHEM\_CASE\_024\_METHANOL\_SYNTHESIS\_ENVELOPE / CHEM\_BENCH\_003: Methanol synthesis envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorpora.
  - CHEM\_CASE\_025\_STEAM\_REFORMING\_ENVELOPE / CHEM\_BENCH\_003: Methane steam-reforming envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorpora.
  - CHEM\_CASE\_026\_ELECTROCHEMICAL\_CO2\_REDUCTION\_ENVELOPE / CHEM\_BENCH\_003: Electrochemical CO2 reduction envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.



- Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorporata.
- CHEM\_CASE\_027\_PHOTOCATALYTIC\_WATER\_SPLITTING\_ENVELOPE / CHEM\_BENCH\_003: Photocatalytic water-splitting envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role H ELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorporata.
  - CHEM\_CASE\_028\_FISCHER\_TROPSCH\_ENVELOPE / CHEM\_BENCH\_003: Fischer-Tropsch product-distribution envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role H ELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorporata.
  - CHEM\_CASE\_029\_PROTON\_COUPLED\_ELECTRON\_TRANSFER\_ENVELOPE / CHEM\_BENCH\_003: Proton-coupled electron-transfer envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role H ELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorporata.
  - CHEM\_CASE\_030\_C\_H\_ACTIVATION\_ENVELOPE / CHEM\_BENCH\_003: Catalytic C-H activation envelope.  
Observable: kinetic\_or\_equilibrium\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role H ELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::CHEMISTRY::KINETICS::REACTION\_ENVELOPE\_PROFILE\_V1. Method: Reaction-class envelope fit with held-out kinetics/equilibrium replay. Authority: DOEBES/EFRCchallengecorporata.

## J.4 Biology

Pinned cases: '30'; accession-frozen cases: '30'; held-out cases: '16'; prediction-declared cases: '30' / '30'. Coverage readiness: 'PASS\_CASESET\_MINIMUM\_REACHED' with minimum cases '30', gap '0', coverage ratio '1.0', and parameterization status 'PREDICTIONS\_DECLARED\_FOR\_ALL\_CASES'.

- BIO\_CASE\_001\_GSE184558\_CTRL\_DAY3\_1 / BIO\_BENCH\_001: GSE184558 control differentiation Day3 replicate 1.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- BIO\_CASE\_002\_GSE184558\_CTRL\_DAY3\_2 / BIO\_BENCH\_001: GSE184558 control differentiation Day3 replicate 2.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- BIO\_CASE\_003\_GSE184558\_CTRL\_DAY3\_3 / BIO\_BENCH\_001: GSE184558 control differentiation Day3 replicate 3.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- BIO\_CASE\_004\_GSE184558\_CTRL\_DAY5\_1 / BIO\_BENCH\_001: GSE184558 control differentiation Day5 replicate 1.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- BIO\_CASE\_005\_GSE184558\_CTRL\_DAY5\_2 / BIO\_BENCH\_001: GSE184558 control differentiation Day5 replicate 2.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.

- **BIO\_CASE\_006\_GSE184558\_CTRL\_DAY5\_3 / BIO\_BENCH\_001:** GSE184558 control differentiation Day5 replicate 3.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- **BIO\_CASE\_007\_GSE184558\_CTRL\_DAY7\_1 / BIO\_BENCH\_001:** GSE184558 control differentiation Day7 replicate 1.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- **BIO\_CASE\_008\_GSE184558\_CTRL\_DAY7\_2 / BIO\_BENCH\_001:** GSE184558 control differentiation Day7 replicate 2.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- **BIO\_CASE\_009\_GSE184558\_CTRL\_DAY7\_3 / BIO\_BENCH\_001:** GSE184558 control differentiation Day7 replicate 3.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- **BIO\_CASE\_010\_GSE184558\_SHRNA5\_DAY3\_1 / BIO\_BENCH\_001:** GSE184558 shRNA5 differentiation Day3 replicate 1.  
Observable: expression\_signature\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::EXPRESSION::SIGNATURE\_PROFILE\_V1. Method: Accession-frozen expression-signature fit with held-out time-series replay. Authority: NCBIGEO.
- **BIO\_CASE\_011\_GSE226159\_CPC\_01 / BIO\_BENCH\_002:** GSE226159 cardiac progenitor sample CPC\_01.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- **BIO\_CASE\_012\_GSE226159\_CPC\_02 / BIO\_BENCH\_002:** GSE226159 cardiac progenitor sample CPC\_02.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- **BIO\_CASE\_013\_GSE226159\_CPC\_03 / BIO\_BENCH\_002:** GSE226159 cardiac progenitor sample CPC\_03.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- **BIO\_CASE\_014\_GSE226159\_NON\_CPC\_01 / BIO\_BENCH\_002:** GSE226159 non-CPC sample N\_CPC\_01.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- **BIO\_CASE\_015\_GSE226159\_NON\_CPC\_02 / BIO\_BENCH\_002:** GSE226159 non-CPC sample N\_CPC\_02.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- **BIO\_CASE\_016\_GSE226159\_NON\_CPC\_03 / BIO\_BENCH\_002:** GSE226159 non-CPC sample N\_CPC\_03.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.

- BIO\_CASE\_017\_GSE226159\_RNA\_12\_24/BIO\_BENCH\_002: GSE226159 CHIR 12 uM 24h sample.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- BIO\_CASE\_018\_GSE226159\_RNA\_6\_36/BIO\_BENCH\_002: GSE226159 CHIR 6 uM 36h sample.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- BIO\_CASE\_019\_GSE226159\_T24H\_PLUSBI\_1/BIO\_BENCH\_002: GSE226159 BI-1347 plus sample at 24h replicate 1.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- BIO\_CASE\_020\_GSE226159\_T24H\_MINUSBI\_1/BIO\_BENCH\_002: GSE226159 BI-1347 minus sample at 24h replicate 1.  
Observable: state\_transition\_order\_error; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::STATE\_TRANSITION::ORDER\_PROFILE\_V1. Method: Accession-frozen state-transition order fit with held-out transition replay. Authority: NCBIGEO.
- BIO\_CASE\_021\_GSE69063\_ANAPHYLAXIS\_249\_T0/BIO\_BENCH\_003: GSE69063 anaphylaxis patient 249 at T0.  
Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.
- BIO\_CASE\_022\_GSE69063\_ANAPHYLAXIS\_249\_T1/BIO\_BENCH\_003: GSE69063 anaphylaxis patient 249 at T1.  
Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.
- BIO\_CASE\_023\_GSE69063\_SEPSIS\_A\_T0/BIO\_BENCH\_003: GSE69063 sepsis patient A at T0.  
Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.
- BIO\_CASE\_024\_GSE69063\_SEPSIS\_A\_T1/BIO\_BENCH\_003: GSE69063 sepsis patient A at T1.  
Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.
- BIO\_CASE\_025\_GSE69063\_SEPSIS\_A\_T2/BIO\_BENCH\_003: GSE69063 sepsis patient A at T2.  
Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.
- BIO\_CASE\_026\_GSE69063\_TRAUMA\_A\_T0/BIO\_BENCH\_003: GSE69063 trauma patient A at T0.  
Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.
- BIO\_CASE\_027\_GSE69063\_TRAUMA\_A\_T1/BIO\_BENCH\_003: GSE69063 trauma patient A at T1.  
Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.
- BIO\_CASE\_028\_GSE69063\_TRAUMA\_A\_T2/BIO\_BENCH\_003: GSE69063 trauma patient A at T2.

Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.

- BIO\_CASE\_029\_GSE69063\_HEALTHY\_927/BIO\_BENCH\_003: GSE69063 healthy control 927.  
 Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.
- BIO\_CASE\_030\_GSE69063\_HEALTHY\_928/BIO\_BENCH\_003: GSE69063 healthy control 928.  
 Observable: response\_onset\_residual; stage ACCESSION\_FROZEN\_OPEN\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::BIOLOGY::RESPONSE::ONSET\_PROFILE\_V1. Method: Accession-frozen response-onset fit with held-out response-signature replay. Authority: NCBIGEO.

## J.5 Systems / Civilizational projection

Pinned cases: '30'; accession-frozen cases: '0'; held-out cases: '18'; prediction-declared cases: '30' / '30'. Coverage readiness: 'PASS\_CASESET\_MINIMUM\_REACHED' with minimum cases '30', gap '0', coverage ratio '1.0', and parameterization status 'PREDICTIONS\_DECLARED\_FOR\_ALL\_CASES'.

- SYS\_CASE\_001\_POP\_USA/SYS\_BENCH\_001: United States total population trajectory.  
 Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- SYS\_CASE\_002\_POP\_CHN/SYS\_BENCH\_001: China total population trajectory.  
 Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- SYS\_CASE\_003\_POP\_IND/SYS\_BENCH\_001: India total population trajectory.  
 Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- SYS\_CASE\_004\_POP\_DEU/SYS\_BENCH\_001: Germany total population trajectory.  
 Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- SYS\_CASE\_005\_POP\_BRA/SYS\_BENCH\_001: Brazil total population trajectory.  
 Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- SYS\_CASE\_006\_POP\_ZAF/SYS\_BENCH\_001: South Africa total population trajectory.  
 Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- SYS\_CASE\_007\_POP\_JPN/SYS\_BENCH\_001: Japan total population trajectory.  
 Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- SYS\_CASE\_008\_POP\_FRA/SYS\_BENCH\_001: France total population trajectory.  
 Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
 Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.

- **SYS\_CASE\_009\_POP\_MEX / SYS\_BENCH\_001: Mexico total population trajectory.**  
Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_010\_POP\_IDN / SYS\_BENCH\_001: Indonesia total population trajectory.**  
Observable: population\_trajectory\_residual; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::POPULATION::TRAJECTORY\_PROFILE\_V1. Method: Pinned official macro-trajectory fit with held-out country replay. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_011\_GDPGROWTH\_USA / SYS\_BENCH\_002: United States GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_012\_GDPGROWTH\_CHN / SYS\_BENCH\_002: China GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_013\_GDPGROWTH\_IND / SYS\_BENCH\_002: India GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_014\_GDPGROWTH\_DEU / SYS\_BENCH\_002: Germany GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role CALIBRATION\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_015\_GDPGROWTH\_BRA / SYS\_BENCH\_002: Brazil GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_016\_GDPGROWTH\_ZAF / SYS\_BENCH\_002: South Africa GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_017\_GDPGROWTH\_JPN / SYS\_BENCH\_002: Japan GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_018\_GDPGROWTH\_FRA / SYS\_BENCH\_002: France GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_019\_GDPGROWTH\_MEX / SYS\_BENCH\_002: Mexico GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.
- **SYS\_CASE\_020\_GDPGROWTH\_IDN / SYS\_BENCH\_002: Indonesia GDP growth held-out interval.**  
Observable: gdp\_growth\_turning\_point\_error; stage PINNED\_OFFICIAL\_CASE; held-out role HELD\_OUT\_SEED; prediction PREDICTION\_DECLARED\_PENDING\_EXECUTION.  
Profile: OC13::SYSTEMS::GDP\_GROWTH::TURNING\_POINT\_PROFILE\_V1. Method: Held-out interval turning-point fit against pinned World Bank GDP growth series. Authority: WorldBankIndicatorsAPI.

- `SYS_CASE_021_ENERGYUSE_USA / SYS_BENCH_003`: United States energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_022_ENERGYUSE_CHN / SYS_BENCH_003`: China energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_023_ENERGYUSE_IND / SYS_BENCH_003`: India energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_024_ENERGYUSE_DEU / SYS_BENCH_003`: Germany energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `CALIBRATION_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_025_ENERGYUSE_BRA / SYS_BENCH_003`: Brazil energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_026_ENERGYUSE_ZAF / SYS_BENCH_003`: South Africa energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_027_ENERGYUSE_JPN / SYS_BENCH_003`: Japan energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_028_ENERGYUSE_FRA / SYS_BENCH_003`: France energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_029_ENERGYUSE_MEX / SYS_BENCH_003`: Mexico energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.
- `SYS_CASE_030_ENERGYUSE_IDN / SYS_BENCH_003`: Indonesia energy-use-per-capita curve.  
Observable: `process_system_regime_score`; stage `PINNED_OFFICIAL_CASE`; held-out role `HELD_OUT_SEED`; prediction `PREDICTION_DECLARED_PENDING_EXECUTION`.  
Profile: `OC13::SYSTEMS::RESOURCE_INTENSITY::REGIME_PROFILE_V1`. Method: Pinned resource-intensity curve fit with held-out interval regime replay. Authority: `WorldBankIndicatorsAPI`.

## K Domain Replay Reports

This appendix collects the replay packets that the empirical closure program actually executes. The summary table gives the operational overview. The domain notes that follow explain what each replay lane is meant to verify and what would count as a real failure rather than as cosmetic noise.

**Mathematics.** The mathematics anchor is a deterministic replay lane rather than an empirical benchmark family. It currently sits at verdict `PASS` with replay state `PASS_REPLAYABLE`. No parameterized cases are required because the lane

| Domain                              | Replay state        | Caseset summary                               | Key metric snapshot                                                               | Wave                                      |
|-------------------------------------|---------------------|-----------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------|
| Mathematics                         | PASS_REPL<br>AYABLE | Deterministic strict replay; minimum 31 cases | No divergence; no surviving counterexample                                        | not required                              |
| Physics                             | PASS_REPL<br>AYABLE | 30 covered / 20 held-out cases                | Mean abs. 0.188136 sigma; p95 0.64778 sigma; max 1.48 sigma; tail breaches 0      | INSTITUTE<br>_RUN::PHY<br>SICS::WAVE_1A   |
| Chemistry                           | PASS_REPL<br>AYABLE | 30 covered / 19 held-out cases                | Mean abs. 0.533285 sigma; p95 1.0 sigma; max 1.597772 sigma; tail breaches 0      | INSTITUTE<br>_RUN::CHE<br>MISTRY::WAVE_2A |
| Biology                             | PASS_REPL<br>AYABLE | 30 covered / 16 held-out cases                | Mean abs. 0.444222 sigma; p95 1.030033 sigma; max 1.438582 sigma; tail breaches 0 | INSTITUTE<br>_RUN::BIOL<br>OGY::WAVE_3A   |
| Systems / Civilizational projection | PASS_REPL<br>AYABLE | 30 covered / 18 held-out cases                | Mean abs. 0.56097 sigma; p95 1.924704 sigma; max 2.25395 sigma; tail breaches 0   | INSTITUTE<br>_RUN::SYST<br>EMS::WAVE_4A   |

tests strict closure ledgers and counterexample traces directly. Promotion fails on the first replay divergence or on the first surviving counterexample. Replay command: `logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py`. The lawful next action remains `MAINTAIN_REPLAY_DISCIPLINE`.

**Physics.** The promoted physics packet is `OC13::PHYSICS::FOUNDATIONAL_CONSTANTS_AND_SPECTRA`, executed through `OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT`. Held-out residuals must remain within the declared tolerance band for every promoted benchmark family. Any family outside tolerance, or any broken sign/order constraint, falsifies the packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS`. The current official replay basis is `PARTIAL_OFFICIAL_PRIMARY_DATA`, with 30 covered cases, 20 held-out cases, mean absolute normalized error 0.188136 sigma, p95 0.64778 sigma, maximum 1.48 sigma, and zero tail breaches.

- `PHY_BENCH_001`: fine\_structure\_constant\_residual from <https://physics.nist.gov/cuu/Constants/index.html>; held-out policy: leave one family out for residual verification.
- `PHY_BENCH_002`: hydrogen\_balmer\_line\_residual from [https://physics.nist.gov/PhysRefData/ASD/lines\\_form.html](https://physics.nist.gov/PhysRefData/ASD/lines_form.html); held-out policy: reserve transitions not used during packet fitting.
- `PHY_BENCH_003`: transport\_scaling\_residual from [https://science.osti.gov/-/media/fes/pdf/workshop-reports/FES\\_Grand\\_Challenges\\_Report\\_final.pdf](https://science.osti.gov/-/media/fes/pdf/workshop-reports/FES_Grand_Challenges_Report_final.pdf); held-out policy: hold out one transport-regime family.

**Chemistry.** The promoted chemistry packet is `OC13::CHEMISTRY::THERMOCHEMISTRY_AND_SPECTRA`, executed through `OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY_SPECTRA_KINETICS`. The declared thermochemical, spectral, and kinetic benchmark families must remain within tolerance on held-out compounds or reaction classes. A held-out family outside tolerance falsifies the packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY`. The official replay basis remains `PARTIAL_OFFICIAL_PRIMARY_DATA`, with 30 covered cases, 19 held-out cases, mean absolute normalized error 0.533285 sigma, p95 1.0 sigma, maximum 1.597772 sigma, and zero tail breaches.

- `CHEM_BENCH_001`: gas\_phase\_enthalpy\_residual from <https://webbook.nist.gov/chemistry/>; held-out policy: hold out one compound family from calibration.
- `CHEM_BENCH_002`: spectral\_transition\_residual from <https://webbook.nist.gov/chemistry/>; held-out policy: hold out one spectral family.
- `CHEM_BENCH_003`: kinetic\_or\_equilibrium\_residual from <https://science.osti.gov/bes/efrc/Research/Grand-Challenges>; held-out policy: hold out one reaction-class envelope.

**Biology.** The promoted biology packet is `OC13::BIOLOGY::STATE_TRANSITION_AND_RESPONSE_SIGNATURES`, executed through `OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPO`

NSE\_SIGNATURES. The bounded measurable signatures must reproduce the held-out dataset families without widening the claim boundary. If the signature family fails on the held-out datasets, the biological lane loses promotion for the affected packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY`. The official replay basis is OFFICIAL\_OPEN\_PRIMARY\_DATA, with 30 covered cases, 16 held-out cases, mean absolute normalized error 0.444222 sigma, p95 1.030033 sigma, maximum 1.438582 sigma, and zero tail breaches.

- BIO\_BENCH\_001: expression\_signature\_residual from <https://www.ncbi.nlm.nih.gov/geo/>; held-out policy: reserve one time-series family for held-out replay.
- BIO\_BENCH\_002: state\_transition\_order\_error from <https://www.ncbi.nlm.nih.gov/geo/>; held-out policy: reserve one cell-state transition family.
- BIO\_BENCH\_003: response\_onset\_residual from <https://www.ncbi.nlm.nih.gov/geo/>; held-out policy: reserve one response-signature family.

**Systems / Civilizational projection.** The promoted systems packet is OC13::SYSTEMS::K8\_MACRO\_PROCESS\_AND\_REGIME\_SHIFT, executed through OC13::EXECUTION::SYSTEMS::MACRO\_TRAJECTORIES\_AND\_REGIME\_SHIFTS. The declared trajectory and regime-change observables must remain within tolerance on held-out intervals; a missed turning point or an out-of-band trajectory falsifies the packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION`. The official replay basis is OFFICIAL\_OPEN\_PRIMARY\_DATA, with 30 covered cases, 18 held-out cases, mean absolute normalized error 0.56097 sigma, p95 1.924704 sigma, maximum 2.25395 sigma, and zero tail breaches.

- SYS\_BENCH\_001: population\_trajectory\_residual from <https://api.worldbank.org/v2/en/indicator/SP.POP.TOTL?format=json>; held-out policy: leave out one interval family.
- SYS\_BENCH\_002: gdp\_growth\_turning\_point\_error from <https://api.worldbank.org/v2/en/indicator/NY.GDP.MKTP.KD.ZG?format=json>; held-out policy: reserve one held-out macro interval.
- SYS\_BENCH\_003: process\_system\_regime\_score from <https://api.worldbank.org/v2/en/indicator/EG.USE.PCAP.KG.OE?format=json>; held-out policy: reserve one resource-intensity curve family.

## L Domain Hard-Closure Command Board

This appendix turns the empirical closure backlog into an execution board that a reviewer can actually read. The summary table states which domains remain in maintenance mode. The domain notes make explicit which command constitutes the lawful next step, which acceptance rule that command is meant to protect, and which reserve measurement wave remains on stand-by.

The current board exposes five tracked lanes. All five are validated and sit in maintenance mode; none presently requires an active institute-run fallback wave.

| Domain                              | Validation | Phase       | Lawful command                                                                                        | Next action                |
|-------------------------------------|------------|-------------|-------------------------------------------------------------------------------------------------------|----------------------------|
| Mathematics                         | PASS       | MAINTENANCE | <code>run_claim_simulations_v1.py</code>                                                              | MAINTAIN_REPLAY_DISCIPLINE |
| Physics                             | PASS       | MAINTENANCE | <code>run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS</code>                           | MAINTAIN_REPLAY_DISCIPLINE |
| Chemistry                           | PASS       | MAINTENANCE | <code>run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY</code>                         | MAINTAIN_REPLAY_DISCIPLINE |
| Biology                             | PASS       | MAINTENANCE | <code>run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY</code>                           | MAINTAIN_REPLAY_DISCIPLINE |
| Systems / Civilizational projection | PASS       | MAINTENANCE | <code>run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION</code> | MAINTAIN_REPLAY_DISCIPLINE |



**Mathematics.** Validation remains PASS, and the lane is executable now because the command path is purely deterministic. Caseset readiness is 0 parameterized out of 0 cases with a minimum requirement of 31, while coverage and parameterization remain NOT\_REQUIRED. Promotion survives only if no replay divergence and no surviving counterexample appear. The lawful command is `logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py`. The required prerequisites remain the pinned source route, the benchmark manifest, the numerical packet specification, the replay harness, and an explicit falsifier. Blocking ids: none. No institute-run wave is required.

**Physics.** Validation remains PASS, and the lane stays in maintenance rather than in expansion mode. Caseset readiness is 30 parameterized out of 30 cases, with coverage PASS\_CASESET\_MINIMUM\_REACHED and parameterization PREDICTIONS\_DECLARED\_FOR\_ALL\_CASES. The acceptance criterion requires held-out residuals to stay within the declared tolerance band for every promoted benchmark family; any family outside tolerance or with broken sign/order constraints falsifies the packet. The lawful command is `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS`. The reserve wave INSTITUTE\_RUN::PHYSICS::WAVE\_1A remains on stand-by, not active.

**Chemistry.** Validation remains PASS, and the chemistry lane also sits in maintenance mode. Caseset readiness is 30 parameterized out of 30 cases, with coverage PASS\_CASESET\_MINIMUM\_REACHED and parameterization PREDICTIONS\_DECLARED\_FOR\_ALL\_CASES. The acceptance criterion requires the declared chemistry benchmark families to remain within tolerance on held-out compounds or reactions; a held-out thermochemical, spectral, or kinetic family outside tolerance falsifies the packet. The lawful command is `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY`. The reserve wave INSTITUTE\_RUN::CHEMISTRY::WAVE\_2A remains on stand-by.

**Biology.** Validation remains PASS, and the biological lane is executable now only in maintenance mode. Caseset readiness is 30 parameterized out of 30 cases, with coverage PASS\_CASESET\_MINIMUM\_REACHED and parameterization PREDICTIONS\_DECLARED\_FOR\_ALL\_CASES. The acceptance criterion requires the promoted biological packet to reproduce the declared measurable signatures on held-out datasets without widening the claim boundary. If the bounded signature family fails on the held-out datasets, the lane loses promotion. The lawful command is `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY`. The reserve wave INSTITUTE\_RUN::BIOLOGY::WAVE\_3A remains on stand-by.

**Systems / Civilizational projection.** Validation remains PASS, and the systems lane is likewise executable only in maintenance mode. Caseset readiness is 30 parameterized out of 30 cases, with coverage PASS\_CASESET\_MINIMUM\_REACHED and parameterization PREDICTIONS\_DECLARED\_FOR\_ALL\_CASES. The acceptance criterion requires the declared trajectory and regime-change observables to remain within tolerance on held-out intervals; a missed turning point or out-of-band held-out trajectory falsifies the packet. The lawful command is `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION`. The reserve wave INSTITUTE\_RUN::SYSTEMS::WAVE\_4A remains on stand-by.

## M Proof Machinery Appendix

The source-owned science corpus is the authoring and witness locus. The canonical SPOT surface `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json` is the synchronised scientific projection authority used for audit and navigation. References to both below are provenance links, not a division of authority between two independent truth layers. Source stamp: corpus path `releases/oc_core_1_3/editorial/science_sources/`; governing surface `OC_CORE_1_3_SCIENCE_SPOT_latest.json`; snapshot fields `metadata.repo_sha` and `metadata.ts_utc`. The generated material below is a compact proof-machinery atlas, not a replacement for the full proof-obligation sheets. It opens with release revision law and then projects the load-bearing proof obligations, hostile- review backlog, closure priority, ablation, competition, and compression surfaces. Premises, forbidden shortcuts, excluded dependencies, and falsifier conditions are authoritative in `releases/oc_core_1_3/editorial/OC_CORE_1_3_PROOF_OBLIGATION_REGISTRY_latest.json`; auxiliary rows remain navigation summaries derived from the canonical SPOT and its release surfaces rather than independent proof authority.

## M.1 Revision law authority

| Priority | Layer                                          | Revision rule                                                            | Conflict trigger                                               | Current status                                  |
|----------|------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------|
| 1        | EMPIRICAL_BRIDGES_AND_REPLAY_BINDING_S         | REVISE_EMPIRICAL_BRIDGES_FIRST                                           | OFFICIAL_OR_INSTITUTE_RUN_REPLAY_BREACH                        | STABLE                                          |
| 2        | PARAMETERIZATION_AND_NUMERICAL_PACKETS         | REVISE_PARAMETERIZATIONS_BEFORE_THEOREM_SCOPE                            | RESIDUAL_OR_CALIBRATION_BREACH_PERSISTENT_AFTER_BRIDGE_REPAIR  | STABLE                                          |
| 3        | STANDALONE_DOMAIN_PACKETS_AND_DERIVED_THEOREMS | REVISE_STANDALONE_DOMAIN_PACKETS_BEFORE_ROOT_KERNEL                      | MULTI_BENCHMARK_FAILURE_PERSISTS_AFTER_PARAMETERIZATION_REPAIR | REAUDIT_REQUIRED_FOR_BRIDGE_ONLY_DOMAINS        |
| 4        | ROOT_KERNEL                                    | ROOT_KERNEL_REVISION_ALLOWED_ONLY_AFTER_ACCUMULATED_CROSS_DOMAIN_FAILURE | ACCUMULATED_CROSS_DOMAIN_FAILURE_EXCEEDS_DECLARED_THRESHOLD    | LOCKED_PENDING_A_CUMULATED_CROSS_DOMAIN_FAILURE |

## M.2 Load-bearing proof obligations

| Obligation id                      | Statement                                                                                                                    | Excluded dependencies                                                  | Falsifier condition                                                                                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBLIGATION: : SOURCE_THEOREM_3     | Show that Axiom 3.2 plus Definitions 12.1, 12.3, and 12.4 lawfully imply Source Theorem 3.                                   | Excluded lift theorem family, Journal-core-only compression            | If the implication requires a hidden excluded dependency or the admissibility premises do not suffice, Source Theorem 3 must be narrowed or withdrawn.             |
| OBLIGATION: : LEMMA_1              | Show that Source Theorem 3 closes Lemma 1 without widening the claim boundary.                                               | Interpretive-only support, Compression-only support                    | If Lemma 1 requires support not traceable to Source Theorem 3, the spine is broken and the positive body must be rebuilt.                                          |
| OBLIGATION: : SOURCE_COROLLARY_3_2 | Show that Axiom 3.3 plus Definition 12.6 lawfully imply Source Corollary 3.2 and preserve irreversibility.                   | Identity restoration after lawful death, Metaphorical rebirth language | If the same continuum can return numerically identical after lawful death without violating Axiom 3.3, the irreversibility corollary must be revised or withdrawn. |
| OBLIGATION: : LEMMA_3              | Show that Source Corollary 3.2 closes Lemma 3 under the declared residue and rebirth boundary.                               | Persistence-by-analogy, Undeclared rebirth equivalence                 | If residue preserves full live identity instead of structural aftermath only, the classification boundary must be rewritten.                                       |
| OBLIGATION: : LEMMA_2              | Show that Lemma 1 plus Definition 12.5 imply Lemma 2 without hidden extensions.                                              | EXTENSION branches, BRIDGE_ONLY packets                                | If Lemma 2 requires an excluded branch or bridge-only evidence as proof support, the theorem route is illegitimate.                                                |
| OBLIGATION: : THEOREM_A            | Show that Lemma 2 is sufficient for Theorem A and that the journal-core extraction does not exceed the master theorem scope. | Journal-core-only strengthening, Excluded lift-heavy theorem family    | If the journal core requires support not present in the master route, the article must be narrowed or the master must be repaired first.                           |

### M.3 Hostile-review backlog

| Review id                            | Challenge                                                                                                | Status | Resolution state            | Exit criterion                                                                                                                                                     |
|--------------------------------------|----------------------------------------------------------------------------------------------------------|--------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HOSTILE : : LAWFUL_COLLAPSE          | Collapse must be tied to the loss of admissible realization rather than to loose metaphor.               | PASS   | PASS_LOCKED                 | A formal dossier links collapse claims to admissible-state loss, threshold logic, and source witnesses without prose leakage.                                      |
| HOSTILE : : IRREVERSIBILITY          | Irreversibility under Axiom 3.3 must forbid numerical identity restoration after lawful death.           | PASS   | PASS_LOCKED                 | The irreversibility corollary is formalized as a dossier with explicit counterexample conditions and no identity loophole.                                         |
| HOSTILE : : RESIDUE_REBIRTH_BOUNDARY | Residue and rebirth must remain controlled classifications rather than narrative labels.                 | PASS   | PASS_LOCKED                 | Residue, rebirth, and persistence are separated by theorem-backed boundary conditions with declared falsifiers.                                                    |
| HOSTILE : : EXCLUDED_LIFT_FIREWALL   | No excluded lift-heavy theorem family may be smuggled into the positive bounded argument.                | PASS   | PASS_LOCKED                 | An exclusion ledger proves that forbidden theorem families and forbidden branches do not enter the core proof path.                                                |
| HOSTILE : : JOURNAL_CORE_ASYMMETRY   | The journal core must remain a lawful extraction from the master and must never enlarge the claim.       | PASS   | PASS_LOCKED                 | A journal-core extraction dossier proves that every public theorem claim is traceably derived from the master route.                                               |
| HOSTILE : : K11_K12_IRREDUCIBILITY   | K11 and K12 must be proved necessary beyond K10 or explicitly demoted with the dependency graph rebuilt. | PASS   | PROVED_NECESSARY_BEYOND_K10 | Either an irreducibility dossier proves that K11 and K12 carry nonreducible scientific work, or the upper levels are demoted and all dependent routes are rebuilt. |

### M.4 Closure priority board

| Domain                              | Formal order | Cost order | Gain per cost | Phase                         | Parallelism note                                                                                                          |
|-------------------------------------|--------------|------------|---------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Mathematics                         | 0            | 0          | 116.666667    | PHASE_1_STABILIZE_CLOSED_CORE | Runs continuously as the formal anchor for all later domains.                                                             |
| Physics                             | 1            | 1          | 64.8          | PHASE_3_PHYSICS_CLOSURE       | Primary first closure lane; no later domain should outrun it.                                                             |
| Chemistry                           | 2            | 2          | 32.0          | PHASE_4_CHEMISTRY_CLOSURE     | Next formal lane after physics; some data preparation may run earlier, but promotion waits for the physics shell to lock. |
| Biology                             | 3            | 4          | 17.818182     | PHASE_5_BIOLOGY_CLOSURE       | Formal order precedes systems, but the high measurement cost makes it a slower lane.                                      |
| Systems / Civilizational projection | 4            | 3          | 16.615385     | PHASE_6_SYSTEMS_CLOSURE       | Cheaper empirical prep than biology is allowed, but final unification waits for the biological formal route.              |

## M.5 Minimality and irreducibility ablation ledger

| Component                       | Kind             | Ablation status                              | Removal effect                                                                                        |
|---------------------------------|------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------|
| ROOT_THEOREM_SUPPORT            | KERNEL_COMPONENT | NECESSARY_UNDER_CURRENT_PROOF_STACK          | Collapses the promoted contradiction-lift-or-collapse doctrine.                                       |
| K_LEVEL_PARTITION               | K_LEVEL_BOUNDARY | FRONTIER_PENDING_STRICT_IRREDUCIBILITY_PROOF | Would blur the explicit K-level doctrine and weaken the current boundary ledger.                      |
| EMPIRICAL_ANGERING_DISCIPLINE   | PROMOTION_GUARD  | NECESSARY_FOR_NON_RUBBERIZING_PROMOTION      | Would allow positive outward science without observables, data routes, and replayable falsifiers.     |
| STRICT_UNIQUENESS_OR_MINIMALITY | META_THEOREM     | EXPLICIT_FRONTIER_RESIDUE_LOCALIZED          | The strongest current result remains an impossibility boundary rather than a strict uniqueness proof. |

## M.6 Alternative-model competition matrix

| Competition id                                     | Class                              | Evaluation status                                        | Comparison scope                                                                                                                                                 |
|----------------------------------------------------|------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAME_CLAIM_CLASS:META_MODEL                        | SAME_CLAIM_CLASS<br>LASS_METAMODEL | PENDING_COMPLEXITY_PENALIZED_SAME_CLAIM_CLASS_COMPARISON | Whole-kernel explanatory and predictive claim class                                                                                                              |
| DOMAIN_BASELINE::PHYSICS                           | DOMAIN_BASELINE                    | DOMAIN_BASELINE_COMPARISON_PAS_S_ELIGIBLE                | Bounded theorem-native physics packet is trace-closed, replay-closed, and comparator-clean on the pinned official held-out route.                                |
| DOMAIN_BASELINE::CHEMISTRY                         | DOMAIN_BASELINE                    | DOMAIN_BASELINE_COMPARISON_PAS_S_ELIGIBLE                | Bounded theorem-native chemistry packet is trace-closed, replay-closed, and comparator-clean on the pinned thermochemical, spectral, and kinetic held-out route. |
| DOMAIN_BASELINE::BIOLOGY                           | DOMAIN_BASELINE                    | DOMAIN_BASELINE_COMPARISON_PAS_S_ELIGIBLE                | Bounded theorem-native biology packet is trace-closed, replay-closed, and comparator-clean across the pinned multi-dataset held-out biological lanes.            |
| DOMAIN_BASELINE::SYSTEMS_CIVILIZATIONAL_PROJECTION | DOMAIN_BASELINE                    | DOMAIN_BASELINE_COMPARISON_PAS_S_ELIGIBLE                | Bounded theorem-native systems packet is trace-closed, replay-closed, and comparator-clean on the pinned macro-trajectory and regime-shift held-out route.       |

## M.7 Bounded compression benchmark

| Axis                                                | Complexity units | Performance value | Interpretation                                                                                   |
|-----------------------------------------------------|------------------|-------------------|--------------------------------------------------------------------------------------------------|
| THEOREM_SUPPORT_DESCRIPTION_LENGTH                  | 17               | 0.294118          | How much validated domain coverage is currently carried per rooted theorem-support statement.    |
| EMPIRICAL_PACKET_COMPLEXITY_VS_RESIDUAL_PERFORMANCE | 5                | 1.0               | How much validated coverage is carried per active numerical packet under the present replay bar. |

## N Figure Atlas

This appendix preserves the inherited figure compendium inside the flagship volume while keeping the central reading path focused on the theorem-bearing and synthesis-bearing narrative. The figures remain part of the monograph; they are simply staged here as a dedicated visual atlas rather than interrupting the main doctrinal route.

## N.1 Illustrative Figures

This appendix collects the illustrative TikZ figures used in the Core 1.3 release. Each panel visualises a structural component or level of the Ontology of Continua (OC). All figures are generated from standalone TikZ snippets in the `figures/` directory.

## O Technical Derivation Atlas

This appendix restores the full recursive technical subtree of the Core. It keeps the module overview, the level-by-level derivation leaves, the M-space and cross-level transition atlases, the cycle, jet, and process corpora, and the operator and theorem support files inside the same flagship PDF without forcing the main body to follow raw directory order.

### O.1 Extended modules and cross-level structures

The ontology of continua is organised as a hierarchical sequence of levels

$$K_0, K_1, \dots, K_{12},$$

each representing a qualitatively distinct form of structure, dynamics, and admissible states. Every level  $K_x$  possesses:

- a state space  $\Omega(K_x)$ ,
- a set of axes  $A(K_x)$ ,
- potentials  $P(K_x)$ ,
- thresholds  $\Theta(K_x)$ ,
- flows  $J(K_x)$ ,
- cycles  $C(K_x)$  with characteristic times  $\tau(K_x)$ ,
- a continuum measure  $k(K_x)$ ,
- and a boundary  $\partial\Omega(K_x)$  describing the conditions of collapse or transition.

Each  $K_x$  is embedded into a corresponding meta-space  $M_x$  that determines the admissible axes and states:

$$A(K_x) \subseteq A(M_x), \quad \Omega(K_x) \subseteq \Omega(M_x).$$

The meta-space restricts the degrees of freedom available to the continuum and thereby shapes its evolution.

#### O.1.1 Dimensional Growth and Transitions $K_x \rightarrow K_{x+1}$

A transition from level  $K_x$  to  $K_{x+1}$  occurs when:

1. a new class of differences emerges that cannot be represented within the existing axes  $A(K_x)$ ;
2. structural tension exceeds a threshold:

$$T(K_x) > \Theta_{\dim}(K_x);$$

3. the meta-space requires a higher-dimensional representation;
4. the existing configuration space  $\Omega(K_x)$  becomes insufficient or collapses onto its boundary  $\partial\Omega(K_x)$ .

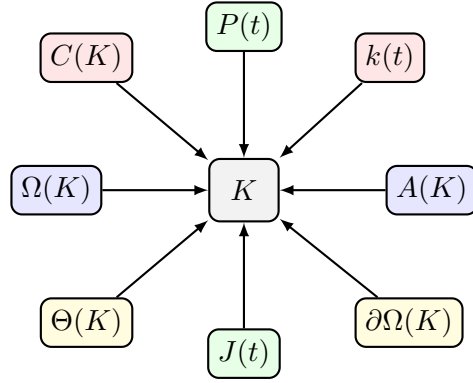


Figure 4: Schematic structure of a continuum  $K = (\Omega, A, P, J, \Theta, \partial\Omega, C, k)$ .

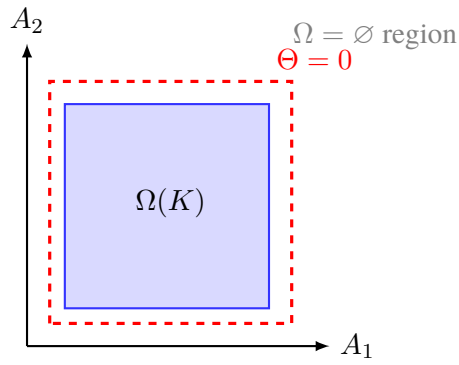


Figure 5: Axes and threshold surfaces in the extended state space of a continuum.

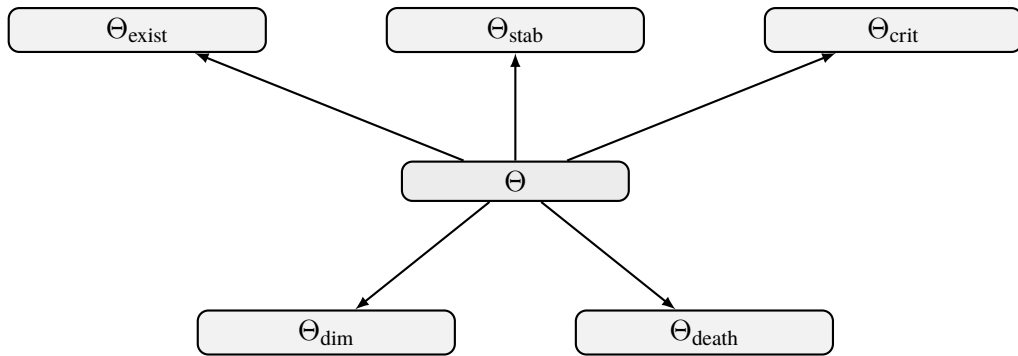


Figure 6: Taxonomy of thresholds: existence, stability, critical, dimensional and death thresholds.

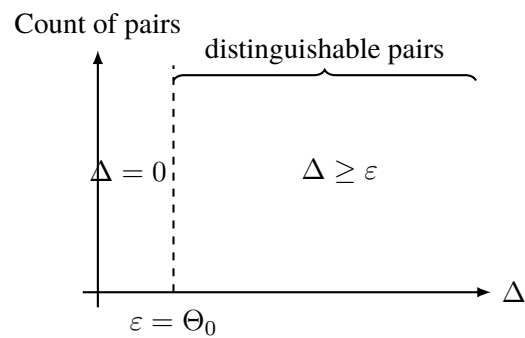


Figure 7: Structural difference and minimal threshold  $\Theta_0$  at level  $K_0$ .

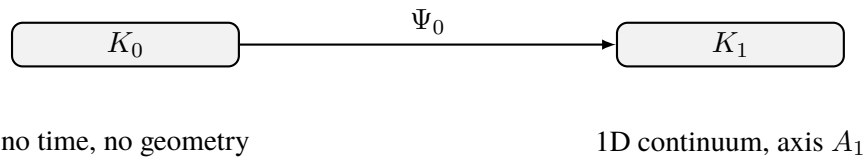


Figure 8: Schematic of the transition  $\Psi_{0 \rightarrow 1}$  from the substrate  $K_0$  to the first continuum  $K_1$ .

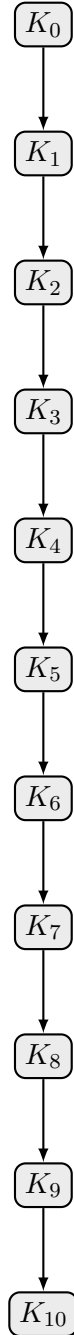


Figure 9: Vertical hierarchy of continua from  $K_0$  to  $K_{10}$ .

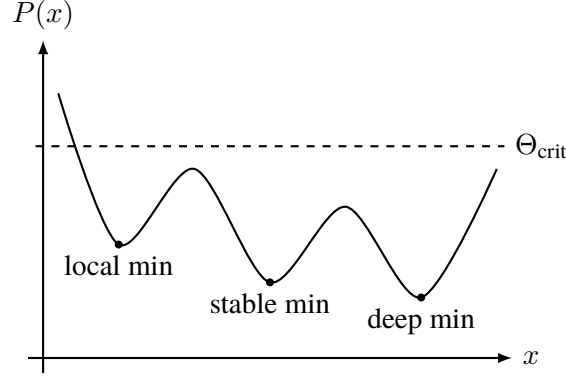
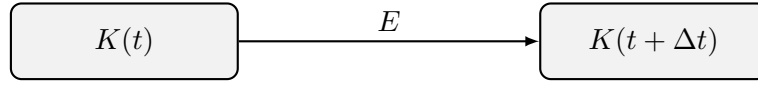


Figure 10: Illustrative potential landscape and flows  $J(t)$  on a continuum.



$$(\Omega, A, P, J, \Theta, \partial\Omega, C, k) \quad (\Omega', A', P', J', \Theta', \partial\Omega', C', k')$$

Figure 11: Schematic action of the evolution operator  $E : K(t) \mapsto K(t + dt)$ .

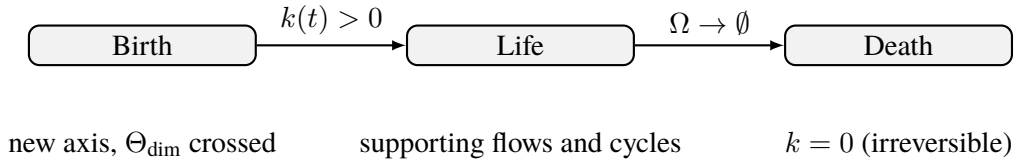


Figure 12: Birth, life and death of a continuum in terms of the state space  $\Omega$ , cycles  $C$  and the measure  $k(t)$ .

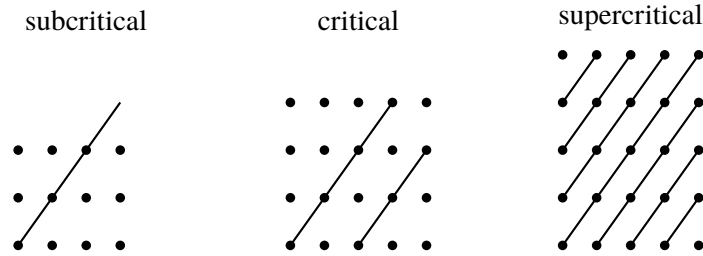


Figure 13: Percolation-type transition at level  $K_2$ .

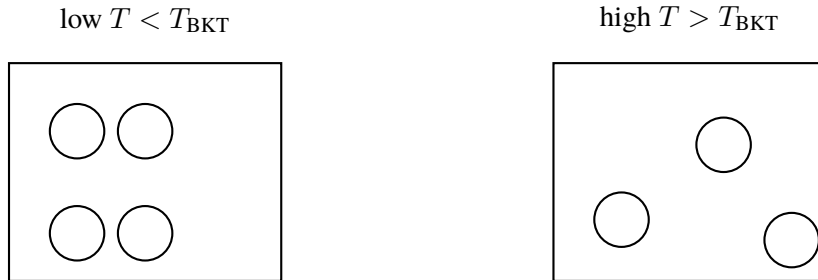


Figure 14: BKT-type transition as an example of threshold-governed emergence of a new topological axis.



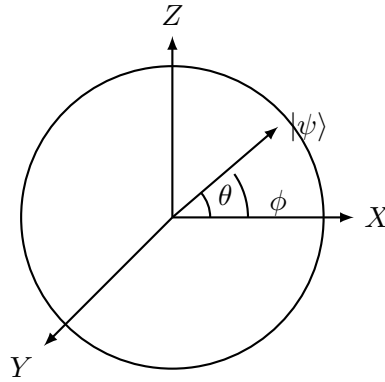


Figure 15: Illustrative Bloch sphere diagram for quantum two-level systems.

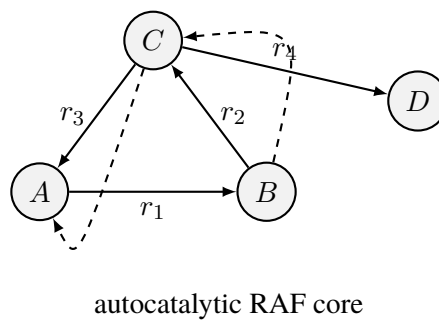


Figure 16: RAF network as a chemical continuum at level  $K_3$ .

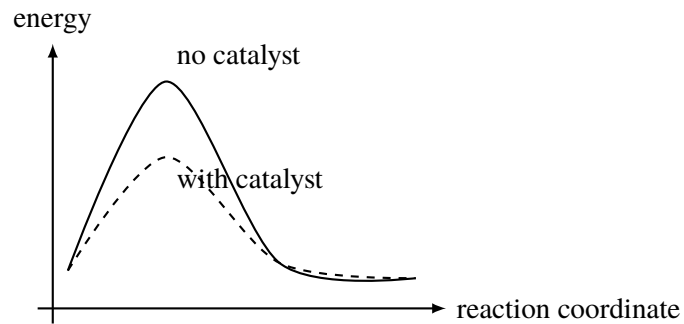


Figure 17: Catalytic paths and supporting flows in a reaction network.

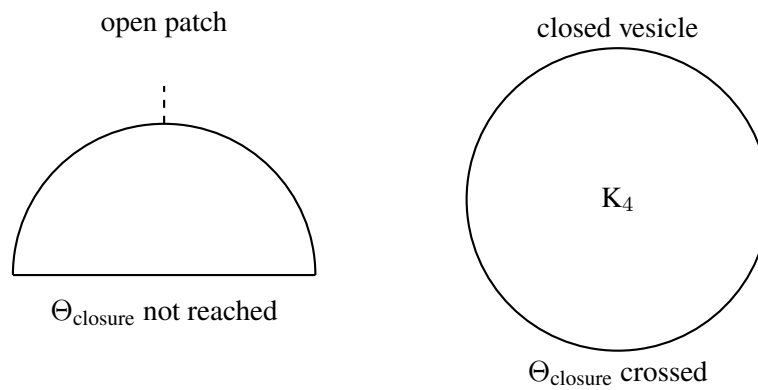


Figure 18: Membrane closure and emergence of a protocellular boundary at level  $K_4$ .

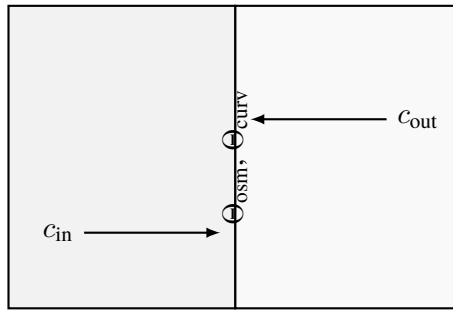


Figure 19: Osmotic gradients, membrane tension and structural thresholds in protocells.

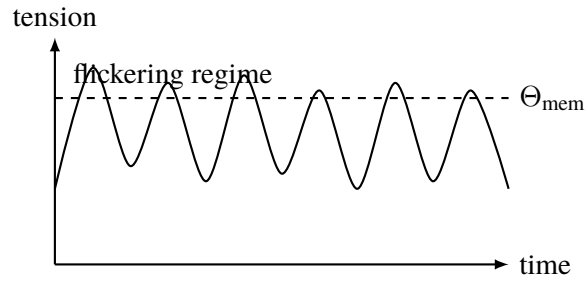


Figure 20: Vesicle flickering regime near curvature and osmotic thresholds.

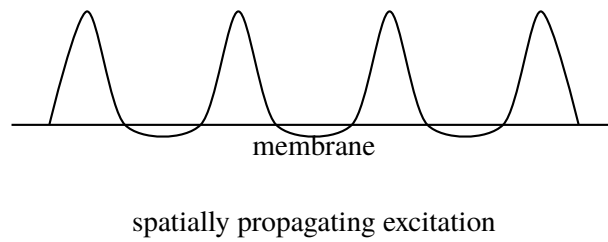


Figure 21: Propagation of membrane potential  $\Delta V$  along a boundary.

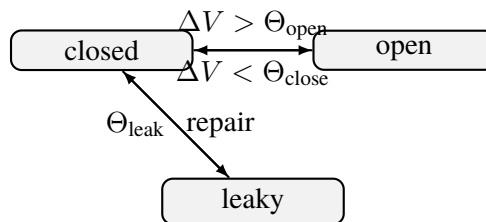


Figure 22: Ion channel states and excitation thresholds at level  $K_5$ .

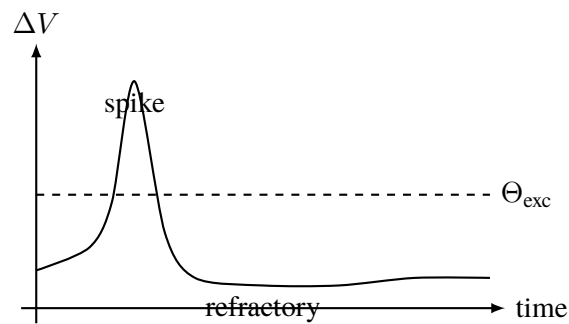


Figure 23: Proto-spike dynamics as a minimal excitatory cycle.

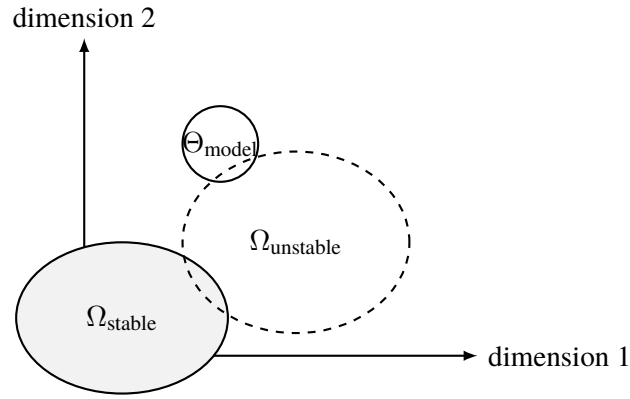


Figure 24: Schematic cognitive state space and binding axes at level  $K_6$ .

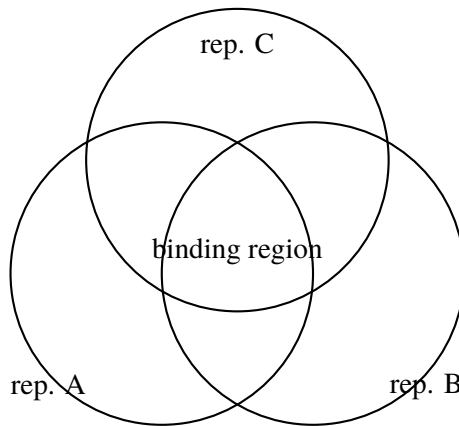
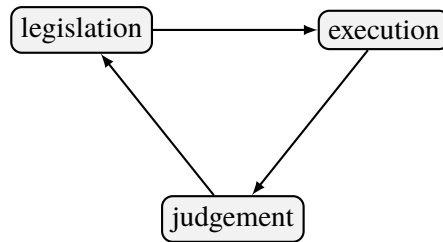


Figure 25: Binding space for cognitive continua and associated thresholds.



$C_7$ : institutional cycle

Figure 26: Institutional cycles and social flows at level  $K_7$ .

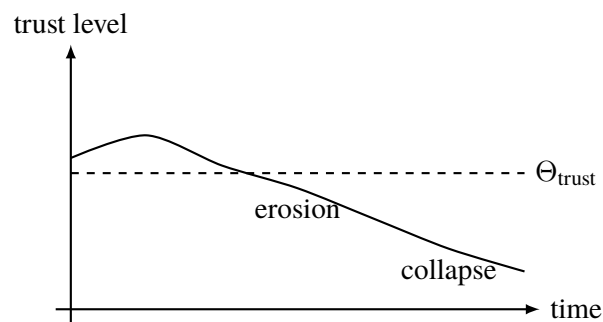
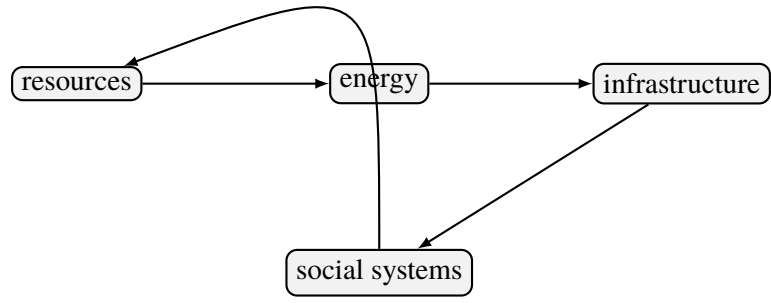


Figure 27: Trust thresholds and collapse of social continuumness.



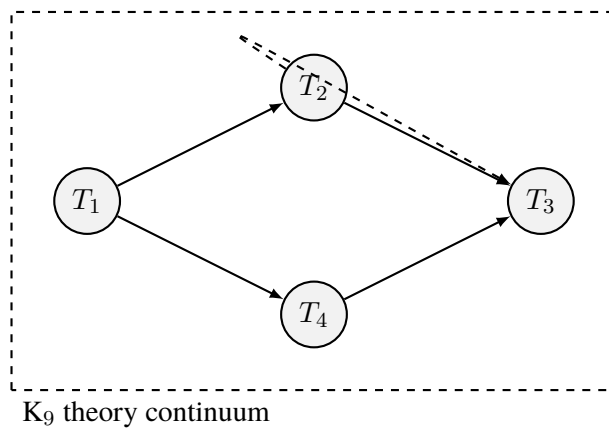
$C_8$ : civilizational cycle

Figure 28: Civilizational energy and infrastructure cycles at level  $K_8$ .

|                               |
|-------------------------------|
| knowledge and culture         |
| institutional and legal       |
| communication and computation |
| energy and transport          |
| physical infrastructure       |

$M_8$ : embedding space for  $K_8$

Figure 29: Technological layers and embedding spaces for civilizational continua.



$K_9$  theory continuum

Figure 30: Graph of theories as a continuum at level  $K_9$ .

This process is irreversible:

$$\dim K_{x+1} \geq \dim K_x,$$

and any “reduction” of dimensionality corresponds to the death of the continuum:

$$\Omega(K_x) = \emptyset.$$

### O.1.2 Universal Structure Across All Levels

Despite the qualitative diversity of levels, every  $K_x$  shares a common formal structure. The evolution of a continuum is governed by:

$$K_x(t + dt) = E(K_x(t), M_x(t)),$$

where  $E$  is an evolution operator constrained by the meta-space  $M_x$ . A continuum evolves only within the admissible region of its meta-space:

$$K_x(t + dt) \in \Omega(M_x) \quad \text{or the evolution is forbidden.}$$

The continuum measure is:

$$k(K_x) = k_\Omega \cdot k_A \cdot k_C \cdot k_J \cdot k_T,$$

where:

- $k_\Omega$  measures the volume of allowed states,
- $k_A$  counts realised axes relative to the meta-space,
- $k_C$  reflects stability of cycles,
- $k_J$  describes coherence of flows,
- $k_T$  measures structural tension relative to thresholds.

### O.1.3 Hierarchy of Levels $K_0 \rightarrow K_{12}$

Each level represents a distinct stratum of organisation:

- $K_0$  — meta-pressure, possibility conditions, and pre-structural logic;
- $K_1$  — continuous one-dimensional continua;
- $K_2$  — two-dimensional continua with internal flows and clusters;
- $K_3$  — chemical configuration spaces and reaction manifolds;
- $K_4$  — protocell-level continua with membranes and boundaries;
- $K_5$  — neural continua with excitation cycles and attractors;
- $K_6$  — cognitive continua with predictive models;
- $K_7$  — social continua with group-level stability cycles;
- $K_8$  — civilisational continua with large-scale coordination flows;
- $K_9$  — theoretical continua (structure of explanations);
- $K_{10}$  — meta-theoretical continua (structure of structural spaces);
- $K_{11}$  — formal ontologies and logic-level continua;
- $K_{12}$  — meta-ontological continua governing the possibility space of all lower levels.

Higher levels do not replace lower ones; they embed and extend them:

$$K_x \subseteq M_x, \quad M_x \subseteq \Omega(K_{x+1}).$$

### O.1.4 Boundaries, Collapse and Death of Continua

Every  $K_x$  has a boundary  $\partial\Omega(K_x)$  where the continuum loses admissibility. A continuum dies when:

$$\Omega(K_x) = \emptyset \quad \Rightarrow \quad k(K_x) = 0,$$

which corresponds to:

- violation of thresholds  $\Theta(K_x)$ ,
- loss of stable cycles  $C(K_x)$ ,
- destructive flows reaching critical values,
- collapse of axes or incompatibility with  $M_x$ .

Death is distinct from evolution: it removes the continuum from the space of admissible structures and cannot be undone.

### O.1.5 Role of $k$ -Level Structure in the Core Model

The  $K$ -levels provide the vertical backbone of the Ontology of Continua. They:

- organise emergence from physical to cognitive to social to meta-ontological structures;
- define the recursion  $K_x \rightarrow M_x \rightarrow K_{x+1}$ ;
- supply the dimensional ladder necessary for universal applicability;
- constrain the evolution of continua via admissibility conditions;
- unify all domains of reality under one mathematical framework.

The structure of  $K$ -levels makes the ontology both hierarchical and recursive: every level depends on the meta-space above it and constrains the level below.

### O.1.6 $K_0$ Overview

Level  $K_0$  represents the pre-structural foundation of all continua. It is not a physical or geometric domain but the logical substrate that defines the *conditions of possibility* for any continuum  $K_x$ . It provides the metapressure, admissibility constraints, logical superiority (Axiom 0.3), and the universal rules that govern which structures may exist.

$K_0$  does not itself possess the conventional axes or geometric dimensions. Instead, it specifies:

- the meta-laws  $\mathcal{L}$  determining structural coherence,
- the possibility space  $\mathcal{P}$  for all continua,
- the metapressure driving emergence of differences,
- the universal thresholds that define admissibility and collapse,
- the recursion rules  $K_x \mapsto M_x \mapsto K_{x+1}$ .

Its primary function is **logical governance** of all deeper levels.

### O.1.7 State Space $\Omega(K_0)$

$\Omega(K_0)$  consists of all admissible configurations of the pair

$$(\mathcal{P}, \mathcal{L}),$$

where  $\mathcal{P}$  is the meta-domain of possible structures and  $\mathcal{L}$  is the system of meta-laws that enforce coherence.

Because  $K_0$  is pre-geometric,  $\Omega(K_0)$  is not a metric space but an abstract set with the following properties:

1.  $\Omega(K_0)$  contains all possible specifications of structural constraints that allow continua to exist.
2.  $\Omega(K_0)$  has no internal topology unless induced by constraints in  $\mathcal{L}$ .
3. Every  $\Omega(K_x)$  for  $x \geq 1$  is a *subset* of  $\Omega(K_0)$ :

$$\Omega(K_x) \subseteq \Omega(K_0).$$

There is no continuous evolution inside  $\Omega(K_0)$ ; it defines *conditions*, not dynamical trajectories.

### O.1.8 Boundary $\partial\Omega(K_0)$

The boundary of  $K_0$  consists of all meta-configurations that violate internal coherence conditions of the meta-laws  $\mathcal{L}$ .

A configuration  $(\mathcal{P}, \mathcal{L})$  lies on  $\partial\Omega(K_0)$  if:

- the laws contradict each other,
- admissibility collapses ( $\mathcal{P}$  becomes empty),
- recursion rules become undefined,
- structural consistency conditions cannot be maintained.

Crossing  $\partial\Omega(K_0)$  corresponds to the logical impossibility of any continuum.

### O.1.9 Axes $A(K_0)$

$K_0$  has no geometric axes. Instead it contains:

- abstract axes of *possibility*,
- admissibility axes governing which structures can be produced,
- meta-axes that define the logical dependencies among levels.

In formal terms:

$$A(K_0) = \{\text{dimensions of admissible structural variation}\}.$$

These are the ultimate sources of all later axes:

$$A(K_x) \subseteq A(K_0).$$

### O.1.10 Potentials $P(K_0)$

Potentials at  $K_0$  describe meta-conditions that push continua toward structure:

$$P(K_0) = \{\text{pressures, constraints, logical gradients}\}.$$

They include:

- metapressure (Axiom 0.z),
- pressure toward structuralisation,
- pressure toward compatible differences,
- pressure toward nonempty admissible spaces.

These potentials are not numeric energies but logical intensities.

### O.1.11 Thresholds $\Theta(K_0)$

Thresholds at  $K_0$  correspond to logical break-points:

$$\Theta(K_0) = \{\Theta_{\text{coh}}, \Theta_{\text{adm}}, \Theta_{\text{meta}}\}.$$

Examples:

- $\Theta_{\text{coh}}$ : minimal coherence of meta-laws,
- $\Theta_{\text{adm}}$ : minimal non-empty possibility space,
- $\Theta_{\text{meta}}$ : minimal support for recursion.

If any threshold is violated, no continuum can exist at any  $K_x$ .

### O.1.12 Flows $J(K_0)$

There are no physical flows at  $K_0$ .  $J(K_0)$  instead contains:

- flows of admissibility,
- flows of logical dependence,
- propagation of constraints,
- meta-flows describing which structures can generate others.

These flows shape the space of possible continua but do not correspond to time evolution.

### O.1.13 Cycles $C(K_0)$

Cycles at  $K_0$  correspond to:

- fixed-point consistency cycles,
- meta-stability loops of admissibility,
- self-referential closure cycles in  $\mathcal{L}$ .

A cycle exists if:

$$\Pi(C) > \Theta_{\text{coh}},$$

where  $\Pi(C)$  is the “power” of the meta-cycle (stability of recursive closure).

There are no temporal cycles; these are purely logical cycles.

### O.1.14 Time $\tau(K_0)$

Time does not exist at  $K_0$ .

Theorem “Emergence of Time from C-cycles” (Physics Run):

Time emerges when  $\Omega(K_2)$  contains a non-degradable cycle  $C$  with:

$$\Pi(C) > \Theta_{\text{time}}.$$

Thus:

$$\tau(K_0) \text{ undefined.}$$



### O.1.15 Continuumness $k(K_0)$

Continuumness measures the admissibility of all continua:

$$k(K_0) = H(\Omega(K_0)) \cdot k_A(K_0) \cdot k_{\text{coh}}(K_0),$$

where:

- $H(\Omega(K_0)) = 1$  if  $\Omega(K_0) \neq \emptyset$ , else 0,
  - $k_A(K_0)$  measures the richness of admissible axes,
  - $k_{\text{coh}}$  measures meta-law coherence.
- If  $k(K_0) = 0$ , no continuum at any higher level can exist.

### O.1.16 Structural Tension $T(K_0)$

Structural tension is:

$$T(K_0) = T(\mathcal{P}, \mathcal{L}),$$

representing the degree of incompatibility within meta-laws or between meta-laws and admissible structures.

High tension may:

- reduce admissible sets,
- produce incompatible branches,
- collapse  $\Omega(K_0)$ .

### O.1.17 Energy $E(K_0)$

There is no physical energy at  $K_0$ . Instead:

$$E(K_0) = E_{\text{meta}},$$

a formal measure of internal support for consistent structures:

$$E_{\text{meta}} \propto \text{coherence of } \mathcal{L}.$$

### O.1.18 Operators on $K_0$ ( $\Psi, \Phi, \Lambda, U, X$ )

Operators at  $K_0$  govern admissibility and recursion:

- $\Psi$ : operator of level creation

$$\Psi : K_x \mapsto M_{x+1}$$

- $\Phi$ : operator of meta-space evolution

$$\Phi : M_x \mapsto M_x(t + dt)$$

- $\Lambda$ : constraint operator enforcing  $\mathcal{L}$
- $U$ : operator of structural unification
- $X$ : operator of logical branching (Axiom 21)

These operators define the backbone of vertical recursion.

### O.1.19 Processes on $K_0$

Key processes:

- logical unification,
- admissibility expansion/contraction,
- meta-stabilisation of recursive definitions,
- branching of possibility spaces,
- collapse of incompatible meta-structures.

### O.1.20 Predictions for $K_0$

The theory predicts:

- no observable dynamics at  $K_0$ ,
- universality of thresholds for all levels,
- inevitability of dimensional growth under metapressure,
- existence of meta-ontological branching structures.

### O.1.21 Experiments for $K_0$

$K_0$  cannot be experimentally probed directly. Indirect tests:

- universality of threshold behaviour across physical, chemical, biological, cognitive and social systems,
- ubiquity of structural recursion patterns,
- falsifiability tests of emergent time (Physics Run),
- stability of branching structures ( $K_{11}$ – $K_{12}$ ).

### O.1.22 Collapse and Death of $K_0$

Collapse occurs if:

$$\Omega(K_0) = \emptyset.$$

This corresponds to:

- contradictory meta-laws,
  - logically inconsistent possibility space,
  - unsupportable recursion  $\Psi$  or  $\Phi$ ,
  - violation of  $\Theta_{\text{coh}}$  or  $\Theta_{\text{adm}}$ .
- If  $K_0$  collapses, all continua collapse simultaneously.

### O.1.23 Falsifiability of $K_0$

The following predictions are falsifiable:

1. Universal presence of thresholds  $\Theta$  across all levels.
  2. Irreversibility of dimensional growth (Theorem 1).
  3. Emergence of time only at  $K_2$  through cycles.
  4. Existence of universal recursion  $K_x \rightarrow M_x \rightarrow K_{x+1}$ .
- Failure of any of these behaviours empirically falsifies  $K_0$ .

### O.1.24 Branching / Ontological Position of $K_0$

Based on Axiom 21 (Ontological Branching),  $K_0$  is the root node of the ontological branching tree:

$$K_0 \rightarrow K_1 \rightarrow K_2 \rightarrow \cdots \rightarrow K_{12}.$$

All branches must remain compatible with  $\mathcal{L}$ ; incompatible branches die through collapse of admissibility.

### O.1.25 Relation to M-spaces

$K_0$  defines the admissibility and structure of all meta-spaces:

$$M_x \subseteq \Omega(K_0).$$

Conversely:

$$K_x \subseteq M_x, \quad M_x \subseteq \Omega(K_{x+1}),$$

which makes  $K_0$  the single unifying superspace for the entire hierarchy.

### O.1.26 $K_1$ Overview

Level  $K_1$  is the first genuinely *structural* continuum. It arises from the action of the operator  $\Psi_{0 \rightarrow 1}$  on  $K_0$ , introducing a one-dimensional axis  $A_1$  and thus enabling the existence of continuous degrees of freedom.

$K_1$  is not yet physical space; rather, it is the abstract prototype of a one-dimensional continuum:

$$K_1 = (X, \tau, \mu, A_1),$$

where  $X$  is a one-dimensional domain,  $\tau$  is the structural topology, and  $\mu$  is a measure compatible with  $\tau$ .

The creation of  $K_1$  is the first step where:

- differences can be expressed along an axis,
- coherent variation becomes possible,
- structural tension can propagate,
- constraints produce continuous curves.

Every higher continuum inherits  $A_1$  as its foundational axis.

### O.1.27 State Space $\Omega(K_1)$

The state space of  $K_1$  is:

$$\Omega(K_1) = C^0(X, V),$$

the space of continuous functions from the one-dimensional domain  $X$  to a value set  $V$ .

Properties:

1.  $\Omega(K_1)$  is infinite-dimensional (function space).
2. Every configuration is continuous by definition.
3. Discontinuities lie outside  $\Omega(K_1)$  and correspond to violations of structural coherence.
4.  $X$  may be open, closed, compact or non-compact depending on constraints in the corresponding  $M_1$ .

This is the minimal space in which continuous variation exists.

### O.1.28 Boundary $\partial\Omega(K_1)$

The boundary consists of configurations that approach loss of continuity:

$$\partial\Omega(K_1) = \{f \in C^0(X, V) : \exists x_0 \in X \text{ with } \lim_{x \rightarrow x_0} f(x) \text{ undefined}\}.$$

Interpretation:

- Points where continuity fails.
- Points where structural tension exceeds  $\Theta_1$ .
- Points where the domain  $X$  degenerates or collapses.

Crossing the boundary destroys the continuum  $K_1$ .

### O.1.29 Axes $A(K_1)$

$K_1$  has exactly one axis:

$$A_1 : X \rightarrow \mathbb{R},$$

representing the intrinsic parameter along which differences become ordered.

Conditions:

- $A_1$  must be monotonic (Axiom  $A_{1,\text{mon}}$ ).
- $A_1$  must be connected (Axiom  $A_{1,\text{conn}}$ ).
- $A_1$  must allow variation compatible with  $\tau$ .

All future axes  $A_k$  for  $k \geq 2$  extend this foundational structure.

### O.1.30 Potentials $P(K_1)$

Potentials on  $K_1$  include:

$$P(K_1) = \{P_{\text{grad}}, P_{\text{bound}}, P_{\text{smooth}}\},$$

describing:

- gradients along the axis,
- boundary-induced stresses,
- smoothness constraints,
- allowed curvature of functions in  $\Omega(K_1)$ .

These are not yet forces; they are structural potentials governing coherence of variation.

### O.1.31 Thresholds $\Theta(K_1)$

Thresholds regulate the admissibility of states:

$$\Theta(K_1) = \{\Theta_{\text{cont}}, \Theta_{\text{grad}}, \Theta_{\text{smooth}}\}.$$

Examples:

- $\Theta_{\text{cont}}$  — minimal continuity required.
- $\Theta_{\text{grad}}$  — maximal admissible gradient before tension becomes unsustainable.
- $\Theta_{\text{smooth}}$  — minimal regularity required.

If any threshold is exceeded, the configuration lies outside  $\Omega(K_1)$ .

### O.1.32 Flows $J(K_1)$

Flows describe allowed continuous changes along  $A_1$ :

$$J(K_1) = \{\partial_x f(x), \partial_t f(x, t), \text{admissible deformations}\}.$$

Key properties:

- flows propagate structural tension,
- flows must preserve continuity,
- flows cannot create new axes (requires  $K_1 \rightarrow K_2$  transition).

### O.1.33 Cycles $C(K_1)$

Because  $K_1$  supports continuous variation but not yet multi-dimensional interaction, cycles are topologically trivial:

$$C_{\text{triv}} = \text{constant loops in function space.}$$

No nontrivial structural cycles exist, since cyclic structure requires at least two independent axes ( $K_2$ ). Thus:

$$\tau(K_1) \text{ cannot emerge.}$$

### O.1.34 Time $\tau(K_1)$

Time does not exist at  $K_1$ .

According to the Time-Origin Theorem (Physics Run):

time emerges only at  $K_2$  from non-degradable cycles.

Thus:

$$\tau(K_1) \text{ is undefined and unrepresentable.}$$

### O.1.35 Continuumness $k(K_1)$

The measure of continuumness:

$$k(K_1) = H(\Omega(K_1)) \cdot \frac{|A(K_1)|}{|A(M_1)|} \cdot (1 - \frac{T_1}{\Theta_{\text{cont}}})_+,$$

where:

- $|A(K_1)| = 1$ ,
  - $A(M_1)$  may permit more axes but not realized yet,
  - $T_1$  is structural tension,
  - $(\cdot)_+$  is the positive part.
- $k(K_1)$  measures whether  $K_1$  remains structurally viable.

### O.1.36 Structural Tension $T(K_1)$

Structural tension on  $K_1$  arises from:

- gradients of functions,
- boundary effects,
- incompatibility of constraints along  $A_1$ .

If:

$$T(K_1) > \Theta_{\text{cont}},$$

then continuity fails and the continuum collapses.

### O.1.37 Energy $E(K_1)$

Energy is defined structurally as:

$$E(K_1) = \int_X \mathcal{E}(f(x), \partial_x f(x)) d\mu,$$

where  $\mathcal{E}$  is a structural energy density depending on:

- value variation,
- smoothness constraints,
- regularity penalties.

$E(K_1)$  has no physical meaning yet; it is a functional measuring coherence.

### O.1.38 Operators on $K_1$ ( $\Psi, \Phi, \Lambda, U, X$ )

- $\Psi_{1 \rightarrow 2}$  — birth of the second axis  $A_2$ , enabling  $K_2$ .
- $\Phi$  — evolution of the admissible meta-space  $M_1$ .
- $\Lambda$  — constraint operator enforcing continuity, monotonicity and connectedness.
- $U$  — unification of local structure into global coherence.
- $X$  — branching operator enabling incompatible  $K_1$  manifolds.

These operators define how  $K_1$  evolves and how new dimensions arise.

### O.1.39 Processes on $K_1$

Key processes:

- propagation of gradients,
- smoothing of discontinuities,
- stabilization of continuous configurations,
- structural collapse under excessive tension,
- dimensional transition to  $K_2$  under pressure of incompatible differences (Axiom 15).

### O.1.40 Predictions for $K_1$

The theory predicts:

1. continuity must be globally maintained;
2. dimensional growth requires incompatible variations that cannot be represented on a single axis;
3. no temporal phenomena appear at  $K_1$ ;
4. collapse under excessive gradients is universal;
5. all  $K_1$  continua must embed into their  $M_1$  meta-space.

#### O.1.41 Experiments for $K_1$

Indirect tests:

- universality of 1D gradient-collapse behaviour in physical, chemical and biological filaments,
- observation of threshold behaviour under steep gradients,
- detection of forced dimensional growth in systems where a single axis becomes insufficient.

Examples:

- polymer strand collapse under tension,
- memristive filament breakdown,
- 1D reaction-diffusion channel instability.

#### O.1.42 Collapse and Death of $K_1$

$K_1$  collapses when:

$$\Omega(K_1) = \emptyset.$$

This happens if:

- continuity is broken,
- gradients exceed thresholds,
- constraints in  $\mathcal{L}$  (carried from  $K_0$ ) become incompatible.

After collapse, transition to  $K_2$  is impossible.

#### O.1.43 Falsifiability of $K_1$

Predictions that may be empirically falsified:

1. universal presence of continuity thresholds,
2. absence of nontrivial cycles,
3. absence of emergent time,
4. monotonic connected axis requirement,
5. dimensional transition only under incompatible differences.

Violation of any of these predictions challenges the  $K_1$  description.

#### O.1.44 Branching / Ontological Position of $K_1$

$K_1$  sits directly above  $K_0$  in the ontological tree:

$$K_0 \rightarrow K_1 \rightarrow K_2.$$

Branching at  $K_1$  corresponds to incompatible admissible manifolds sharing the same meta-laws but differing in structural constraints.

Such branches remain admissible as long as they stay within  $\Omega(K_1)$ .

### O.1.45 Relation to M-spaces

$K_1$  is embedded in its meta-space:

$$K_1 \subseteq M_1 \subseteq \Omega(K_0).$$

$M_1$  determines:

- the topology  $\tau$  of  $X$ ,
- the regularity constraints,
- permissible ranges for potentials,
- admissibility thresholds.

The transition to  $K_2$  occurs when:

$A_1$  cannot represent all emerging differences.

### O.1.46 $K_2$ Overview

Level  $K_2$  is the first continuum that supports *genuine two-dimensional structure*. It arises through the dimensional transition  $\Psi_{1 \rightarrow 2}$  when the constraints of  $K_1$  become insufficient to represent incompatible variations (Axiom 15). The new axis  $A_2$  is orthogonal in the structural sense and cannot be expressed as a function of  $A_1$ .

As a result,  $K_2$  becomes the first level at which:

- nontrivial cycles exist,
- time  $\tau$  emerges (via stable  $C$ -cycles),
- gradients and potentials form two-dimensional fields,
- defects, vortices and dislocations appear,
- topological phenomena become meaningful.

$K_2$  is the minimal setting for classical field-theoretic behaviour and the BKT-type transition.

### O.1.47 State Space $\Omega(K_2)$

The state space of  $K_2$  is:

$$\Omega(K_2) = C^0(X_1 \times X_2, V),$$

a space of continuous fields over a 2D domain.

Key properties:

1.  $\Omega(K_2)$  contains configurations with meaningful local gradients in two independent directions.
2. Defects (e.g., vortices) lie in  $\partial\Omega(K_2)$  unless they satisfy admissibility constraints of  $M_2$ .
3. Nontrivial homotopy classes exist:

$$\pi_1(\Omega(K_2)) \neq 0.$$

This is the first level where geometry (in the weak structural sense) emerges.



### O.1.48 Boundary $\partial\Omega(K_2)$

$\partial\Omega(K_2)$  contains:

- discontinuous fields,
- fields whose energy density diverges,
- configurations exceeding  $\Theta_{\text{grad}}$  or  $\Theta_{\text{defect}}$ ,
- fields where cycles collapse ( $C \rightarrow \emptyset$ ).

A special part of the boundary corresponds to topological defects whose core size collapses to zero, signalling destruction of dimensional coherence.

### O.1.49 Axes $A(K_2)$

Axes consist of:

$$A(K_2) = \{A_1, A_2\},$$

where:

- $A_1$  is inherited from  $K_1$ ,
- $A_2$  is a new independent structural axis.

The independence condition is:

$$\nexists h : \mathbb{R} \rightarrow \mathbb{R} \text{ such that } A_2 = h \circ A_1.$$

This independence is the formal core of the dimensional jump.

### O.1.50 Potentials $P(K_2)$

The set of potentials now includes full 2D structural terms:

$$P(K_2) = \{P_{\text{grad}}, P_{\text{curl}}, P_{\text{div}}, P_{\text{bond}}, P_{\text{smooth}}, P_{\text{top}}\}.$$

Interpretation:

- gradients in two axes encode local tension,
- curl-like structures encode rotational degrees of freedom,
- divergence potentials encode compressive or dilational tension,
- $P_{\text{top}}$  captures energy of vortices and defects.

These potentials reproduce, at the abstract level, the structural precursors of fields in physics.

### O.1.51 Thresholds $\Theta(K_2)$

Thresholds now include:

$$\Theta(K_2) = \{\Theta_{\text{grad}}, \Theta_{\text{curl}}, \Theta_{\text{defect}}, \Theta_{\text{coh}}, \Theta_{\text{BKT}}\}.$$

Their meaning:

- $\Theta_{\text{grad}}$ : maximal admissible gradient magnitude.
- $\Theta_{\text{curl}}$ : rotational tolerance.
- $\Theta_{\text{defect}}$ : defect-core stability threshold.
- $\Theta_{\text{coh}}$ : coherence threshold for 2D order.
- $\Theta_{\text{BKT}}$ : threshold for the appearance of stable  $C$ -cycles; equivalent to  $K_c$  in the BKT theorem of representability.

Crossing  $\Theta_{\text{BKT}}$  yields the emergent time axis.

### O.1.52 Flows $J(K_2)$

Flows include:

$$J(K_2) = \{\partial_{x_1} f, \partial_{x_2} f, \text{2D deformations, vortex drift, phase rotations}\}.$$

Here appear:

- divergence and curl flows,
- diffusion-like equilibration,
- defect-annihilation and defect-unbinding dynamics.

Flows define the dynamical admissibility of evolution in  $\Omega(K_2)$ .

### O.1.53 Cycles $C(K_2)$

$K_2$  is the first level with nontrivial cycles:

$$C(K_2) = \{C_{\text{vortex}}, C_{\text{phase}}, C_{\text{ring}}\}.$$

Most important:

$$C_{\text{vortex}} : \oint \nabla \theta \cdot d\ell = 2\pi n.$$

According to Theorem of Representability 5 (BKT):

- For coupling  $K < K_c$ ,  $C_{\text{vortex}}$  are unstable and decay.
- For  $K > K_c$ , cycles become stable and produce global coherence.

These stable cycles generate the time degree of freedom.

### O.1.54 Time $\tau(K_2)$

Time emerges for the first time at  $K_2$ .

From the Time Emergence Theorem:

$$\tau \text{ exists iff there is a stable cycle } C \text{ with } \Pi(C) > \Theta_{\text{time}}.$$

Here  $\Pi(C)$  is the cycle-power functional.

Thus:

$$\tau(K_2) = \begin{cases} \text{undefined,} & K < K_c, \\ \text{well-defined monotonic parameter,} & K > K_c. \end{cases}$$

This is the structural analogue of temporal ordering arising from persistent 2D coherence.

### O.1.55 Continuumness $k(K_2)$

The general formula:

$$k(K_2) = \frac{\mu(\Omega(K_2))}{\mu(S(K_2))} \cdot \frac{|A(K_2)|}{|A(M_2)|} \cdot \frac{\sum \text{Stab}(C)}{\sum \text{MaxStab}(C)} \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\max}}\right) \cdot \left(1 - \frac{T}{\Theta_{\text{coh}}}\right)_+.$$

Interpretation:

- growth of  $\Omega(K_2)$  increases  $k$ ,
- increasing defect density lowers  $k$ ,
- stable cycles raise  $k$ ,
- turbulence-like flows reduce  $k$ .

This quantifies structural viability of the 2D continuum.

### O.1.56 Structural Tension $T(K_2)$

Sources of tension:

- large gradients,
- vortex cores,
- defect–anti-defect interactions,
- incompatible boundary conditions.

Collapse occurs when:

$$T(K_2) > \Theta_{\text{coh}}.$$

This leads to loss of coherence and destruction of  $C$ -cycles.

### O.1.57 Energy $E(K_2)$

The structural energy:

$$E(K_2) = \int_{X_1 \times X_2} \mathcal{E}(f, \nabla f, \nabla^2 f) d\mu,$$

where  $\mathcal{E}$  includes:

- gradient energy,
- elastic-like energy,
- defect-core energy,
- topological energy of vortices.

This is the structural prototype of energy functionals in physics.

### O.1.58 Operators on $K_2$ ( $\Psi, \Phi, \Lambda, U, X$ )

- $\Psi_{2 \rightarrow 3}$  — dimensional transition to  $K_3$  when 2D structure becomes insufficient.
- $\Phi$  — meta-space evolution operator for  $M_2$ .
- $\Lambda$  — constraint-enforcing operator that stabilizes gradients, smoothness and defect bounds.
- $U$  — unification of local 2D regions into global coherent fields.
- $X$  — branching operator allowing multiple  $K_2$  manifolds with different coherence structures.

### O.1.59 Processes on $K_2$

Processes include:

- vortex creation, annihilation and drift,
- phase ordering and disordering,
- BKT-type transitions,
- stabilization of global coherence,
- collapse under defect proliferation,
- dimensional growth to  $K_3$  when incompatible variations cannot be captured by  $\{A_1, A_2\}$ .

### O.1.60 Predictions for $K_2$

1. Existence of a coherence threshold  $K_c$  (BKT).
2. Sharp transition between stable and unstable vortices.
3. Emergence of time only when vortex cycles stabilize.
4. Universal power-law correlations below  $K_c$ .
5. Loss of coherence under defect proliferation.
6. Necessity of two independent axes for nontrivial cycles.

These predictions match a wide class of physical systems (XY-model, superfluid films, thin magnetic layers).

### O.1.61 Experiments for $K_2$

Canonical experimental signatures:

- measurement of vortex-unbinding temperature in 2D superfluids,
- detection of BKT-like transitions in atomically thin films,
- observation of coherence collapse at high defect density,
- phase-ordering kinetics in 2D materials,
- universal jump in stiffness at the BKT transition.

Indirect tests appear in:

- graphene defect dynamics,
- 2D liquid-crystal films,
- biological membranes with topological defect patterns.

### O.1.62 Collapse and Death of $K_2$

$K_2$  collapses when:

$$\Omega(K_2) = \emptyset,$$

due to:

- runaway defect proliferation,
- loss of coherence ( $T > \Theta_{\text{coh}}$ ),
- destruction of all  $C$ -cycles,
- inability to maintain two independent axes.

After collapse, no transition to  $K_3$  is possible.

### O.1.63 Falsifiability of $K_2$

Falsifiable predictions:

1. existence of a sharp BKT-like threshold,
2. necessity of cycles for emergent time,
3. correlation-length behaviour near  $K_c$ ,

4. universal stiffness jump,
5. topological protection of  $C_{\text{vortex}}$  above  $K_c$ .

Empirical violation of these features would contradict the structural model.

#### O.1.64 Branching / Ontological Position of $K_2$

$K_2$  is the root of all higher-dimensional continua:

$$K_0 \rightarrow K_1 \rightarrow K_2 \rightarrow K_3 \rightarrow \dots$$

Branching at  $K_2$  corresponds to different admissible 2D coherence structures:

- vortex-rich vs vortex-free phases,
- smooth vs defect-dominated manifolds,
- high-coherence vs low-coherence branches.

These branches remain distinct unless unified by operator  $U$ .

#### O.1.65 Relation to M-spaces

$K_2$  is embedded in:

$$K_2 \subseteq M_2 \subseteq M_1 \subseteq M_0.$$

$M_2$  defines:

- admissible 2D topologies,
- allowed defect classes,
- gradient and curl bounds,
- coherence conditions for dimensional stability.

Transition to  $K_3$  requires that  $M_2$  cannot accommodate new incompatible classes of differences.

#### O.1.66 $K_3$ Overview

$K_3$  is the level at which *chemical organisation* becomes structurally possible. It represents the transition from purely geometric and topological continua ( $K_0$ – $K_2$ ) to systems in which:

- reactions, transformations and conversions appear as admissible flows,
- potentials correspond to reaction affinities and concentration gradients,
- boundaries  $\partial\Omega$  include molecular-compositional and topological constraints,
- minimal autocatalytic closure becomes representable,
- the first structurally coherent networks arise (RAF networks).

The dimensional transition  $\Psi_{2 \rightarrow 3}$  occurs when incompatible chemical differences cannot be expressed within the axes  $\{A_1, A_2\}$  of  $K_2$ . A new axis  $A_3$  is introduced, representing a distinct dimension of *composition space*, enabling emergent chemical continua.

$K_3$  serves as the structural substrate for all later biological and cognitive continua, but itself remains non-living.

### O.1.67 State Space $\Omega(K_3)$

The state space of  $K_3$  is:

$$\Omega(K_3) = \{(\mathbf{c}, \mathbf{r}, \Gamma) \mid \mathbf{c} \in \mathbb{R}_{\geq 0}^N, \mathbf{r} \in \mathbb{R}_{\geq 0}^M, \Gamma \subseteq R\},$$

where:

- $\mathbf{c}$ : vector of concentrations of molecular species,
- $\mathbf{r}$ : vector of reaction fluxes,
- $\Gamma$ : set of enabled reactions (stoichiometric structure).

Properties:

1.  $\Omega(K_3)$  allows nonlinear transformations induced by reaction networks.
2. Admissible states must satisfy mass-balance constraints and threshold limitations from  $M_3$ .
3. RAF subnetworks embed into  $\Omega(K_3)$  as structurally coherent regions.

Stability of  $K_3$  depends on whether  $\Omega(K_3)$  contains any closed autocatalytic subsets with sustained flux.

### O.1.68 Boundary $\partial\Omega(K_3)$

$\partial\Omega(K_3)$  includes:

- states where any concentration becomes negative (forbidden),
- divergent fluxes exceeding  $\Theta_{\text{flux}}$ ,
- chemical configurations violating stoichiometric feasibility,
- collapse regimes: pH-collapse, osmotic collapse, reaction-chain instability,
- loss of autocatalytic closure.

Critical for  $K_3$ : **loss of closure** implies  $\Omega(K_3)$  becomes disconnected, preventing sustained organisation.

### O.1.69 Axes $A(K_3)$

$K_3$  supports three independent structural axes:

$$A(K_3) = \{A_1, A_2, A_3\},$$

where  $A_3$  is the new compositional axis.

$A_3$  encodes:

- stoichiometric structure,
- reaction-direction degrees of freedom,
- compositional differences that cannot be mapped to geometric or phase-coherence differences of  $K_2$ .

Independence:

$$A_3 \notin \text{span}\{A_1, A_2\}.$$

This expresses the irreducible novelty of chemical organisation.

### O.1.70 Potentials $P(K_3)$

Chemical potentials include:

$$P(K_3) = \{\mu_i, \Delta G_r, P_{\text{affinity}}, P_{\text{mix}}, P_{\text{exchange}}\},$$

with:

- $\mu_i$ : species-level potentials,
- $\Delta G_r$ : reaction Gibbs free energies,
- $P_{\text{affinity}}$ : reaction driving forces,
- $P_{\text{mix}}$ : entropic mixing potentials,
- $P_{\text{exchange}}$ : exchange with environment (*if* allowed by  $M_3$ ).

Chemical organisation arises when these potentials create a stable, self-reinforcing structure.

### O.1.71 Thresholds $\Theta(K_3)$

Relevant thresholds include:

$$\Theta(K_3) = \{\Theta_{\text{closure}}, \Theta_{\text{flux}}, \Theta_{\text{pH}}, \Theta_{\text{osm}}, \Theta_{\text{grad}}, \Theta_{\text{react}}\}.$$

Interpretation:

- $\Theta_{\text{closure}}$ : minimal conditions for autocatalytic closure,
- $\Theta_{\text{flux}}$ : maximal sustainable reaction rate,
- $\Theta_{\text{pH}}$ : permissible acid-base imbalance,
- $\Theta_{\text{osm}}$ : osmotic-stability threshold,
- $\Theta_{\text{grad}}$ : concentration-gradient bounds,
- $\Theta_{\text{react}}$ : energy threshold for reactions.

Crossing these thresholds drives collapse (Section [O.1.82](#)).

### O.1.72 Flows $J(K_3)$

Admissible flows:

$$J(K_3) = \{\text{reaction fluxes, diffusive transport, exchange flows, autocatalytic amplification}\}.$$

Qualitative types:

- linear mass-action flows,
- nonlinear autocatalytic flows,
- balancing flows preventing runaway instability,
- exchange flows across proto-boundaries.

Autocatalytic flows are central: they define structural amplification necessary for  $K_3$  viability.

### O.1.73 Cycles $C(K_3)$

Cycles in  $K_3$  include:

$$C(K_3) = \{\text{reaction cycles, autocatalytic loops, } C_{\text{RAF}}\}.$$

$C_{\text{RAF}}$  is the minimal closed RAF cycle satisfying:

$$r_i \in \text{RAF} \iff \text{all reactants of } r_i \text{ and a catalyst are produced within the set.}$$

Properties:

- RAF cycles provide structural persistence,
- their power functional  $\Pi(C_{\text{RAF}})$  determines whether organisation is sustainable,
- existence of  $C_{\text{RAF}}$  is necessary (but not sufficient) for transition to  $K_4$ .

When  $C_{\text{RAF}}$  becomes stable,  $K_3$  supports precursor forms of temporal ordering.

### O.1.74 Time $\tau(K_3)$

Time in  $K_3$  arises when autocatalytic cycles maintain non-zero cycle power:

$$\tau(K_3) \text{ exists iff } \Pi(C_{\text{RAF}}) > \Theta_{\text{time}}.$$

Temporal structure is weaker than in  $K_2$ :

- time is defined through reaction-cycle iteration,
- time fails if cycles extinguish or drift to zero flux.

Stable temporal ordering is a necessary condition for transition to biological  $K_4$ .

### O.1.75 Continuumness $k(K_3)$

The general formula applies:

$$k(K_3) = \frac{\mu(\Omega(K_3))}{\mu(S(K_3))} \cdot \frac{|A(K_3)|}{|A(M_3)|} \cdot \frac{\sum \text{Stab}(C)}{\sum \text{MaxStab}(C)} \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\text{max}}}\right) \cdot \left(1 - \frac{T}{\Theta_{\text{stab}}}\right)_+.$$

Interpretation:

- existence of a RAF cycle boosts  $k(K_3)$ ,
- high imbalance of fluxes lowers  $k(K_3)$ ,
- osmotic or pH-stress dramatically reduces  $k(K_3)$ ,
- $k(K_3) = 0$  corresponds to chemical death.

### O.1.76 Structural Tension $T(K_3)$

Sources of tension:

- incompatible reaction fluxes,
- strong concentration gradients,
- pH imbalance,
- osmotic gradients,
- missing catalysts (incomplete closure).

When:

$$T(K_3) > \Theta_{\text{stab}},$$

autocatalytic organisation collapses.



### O.1.77 Energy $E(K_3)$

Energy functional:

$$E(K_3) = \sum_i \mu_i c_i + \sum_r \Delta G_r r + E_{\text{grad}} + E_{\text{mix}} + E_{\text{osm}}.$$

Terms denote:

- chemical potentials and free energies,
- gradient-energy costs,
- osmotic imbalance,
- mixing entropy.

This functional governs chemical stability.

### O.1.78 Operators on $K_3$ ( $\Psi, \Phi, \Lambda, U, X$ )

- $\Psi_{3 \rightarrow 4}$  — biological transition: appearance of compartmentalisation (membrane closure) generating  $K_4$ .
- $\Phi$  — evolution of  $M_3$  constraints (composition limits, reaction admissibility).
- $\Lambda$  — operator enforcing stoichiometric consistency.
- $U$  — unification of reaction subnetworks.
- $X$  — allows branching into alternative chemical organisations (different RAF cores).

### O.1.79 Processes on $K_3$

Representative processes:

- autocatalytic amplification,
- cross-catalytic chain formation,
- pH-drift and recovery,
- osmotic imbalance and stabilisation,
- formation or dissolution of RAF subnetworks,
- structural drift toward  $K_4$  when a boundary  $\partial\Omega$  emerges.

### O.1.80 Predictions for $K_3$

1. RAF emergence is a threshold phenomenon governed by  $\Theta_{\text{closure}}$ .
2. Minimal chemical organisation requires autocatalytic cycles.
3. pH-collapse produces rapid destruction of  $\Omega(K_3)$ .
4. Osmotic-stress curve has a universal sigmoidal form ( $\Theta_{\text{osm}}$ ).
5. Transition to  $K_4$  requires formation of a semi-stable proto-boundary.

### O.1.81 Experiments for $K_3$

Empirical analogues:

- experimental RAF networks (Steel–Hordijk),
- protocell-free polymerisation systems,
- pH-gradient oscillators,
- osmotic swelling–bursting cycles in vesicles without membranes,
- open-system autocatalytic reactors.

Each provides tests for  $\Theta_{\text{closure}}$ ,  $\Theta_{\text{react}}$ , and collapse criteria.

### O.1.82 Collapse and Death of $K_3$

Collapse occurs when:

$$\Omega(K_3) = \emptyset,$$

due to:

- destruction of RAF closure,
- pH-collapse,
- osmotic burst,
- runaway flux divergence,
- incompatible stoichiometric constraints.

After death, no transition to biological  $K_4$  is possible.

### O.1.83 Falsifiability of $K_3$

Testable predictions:

1. existence of a minimal RAF threshold,
2. necessity of autocatalytic loops for sustained organisation,
3. osmotic-collapse envelope reproducible across chemistries,
4. pH-limits defining structural survival,
5. inability of chemical organisation to persist without closure.

Violations of these predictions would refute the structural model.

### O.1.84 Branching / Ontological Position of $K_3$

Branching occurs through:

- alternative RAF cores,
- differing stoichiometric subnetworks,
- divergent environmental-exchange regimes.

Positionally:

$$K_2 \rightarrow K_3 \rightarrow K_4,$$

where  $K_3$  is the unique chemical intermediate between physical and biological continua.

### O.1.85 Relation to M-spaces

$K_3$  exists within  $M_3$ , which specifies:

- admissible composition ranges,
- allowable reaction classes,
- environmental exchange limits,
- osmotic and pH constraints,
- structural conditions for emergence of boundaries  $\partial\Omega$ .

Transition  $K_3 \rightarrow K_4$  requires  $M_3$  to generate a new admissible topology with an inside/outside distinction.

### O.1.86 $K_4$ Overview

$K_4$  is the level where a *biological* structure first becomes possible. The defining event is the appearance of a *semi-stable membrane* that generates a new ontological distinction:

inside | outside.

This is a genuine new axis of the continuum, creating:

- a bounded chemical interior,
- selective transport regimes,
- spatially structured gradients,
- proto-homeostasis,
- the first biological cycles of excitation, survival and repair.

The transition  $K_3 \rightarrow K_4$  corresponds to protocell formation: a RAF network enclosed by a semi-permeable membrane whose stability is governed by threshold conditions  $\Theta_{\text{mem}}$ ,  $\Theta_{\text{perm}}$ ,  $\Theta_{\text{osm}}$  and  $\Theta_{\text{curv}}$ .

$K_4$  is the minimal biological continuum: the simplest form of organised and sustained protocellular life.

### O.1.87 State Space $\Omega(K_4)$

The state space includes:

$$\Omega(K_4) = \{(\mathbf{c}_{\text{in}}, \mathbf{c}_{\text{out}}, \partial\Omega, \mathbf{g}, \Gamma_{\text{RAF}}, \mathbf{p})\},$$

where:

- $\mathbf{c}_{\text{in}}$ : concentrations inside the membrane,
- $\mathbf{c}_{\text{out}}$ : external concentrations,
- $\partial\Omega$ : membrane boundary with patch-structure,
- $\mathbf{g}$ : gradients across the membrane (osmotic, ionic, pH, potential),
- $\Gamma_{\text{RAF}}$ : the internal catalytic network,
- $\mathbf{p}$ : permeability and curvature parameters for each boundary patch.

The state space is higher-dimensional than  $K_3$  due to membrane topology and patchwise degrees of freedom.

Admissible states must satisfy:

- membrane cohesion,

- controlled permeability,
- non-destructive gradients,
- viability of RAF network inside.

#### O.1.88 Boundary $\partial\Omega(K_4)$

The membrane is a *dynamic boundary*:

$$\partial\Omega(K_4) = \bigcup_{i=1}^{N_{\text{patch}}} \partial\Omega_i.$$

Each patch  $\partial\Omega_i$  has:

- local curvature  $K_i$ ,
- local permeability  $p_i$ ,
- local tension  $T_i$ ,
- local gradient constraints (osmotic, ionic, pH),
- phase-state ( $L_\alpha$ ,  $L_o$ ,  $L_\beta$ , porous).

A patch collapses when:

$$T_i > \Theta_{\text{mem},i} \quad \text{or} \quad \Delta\Pi_i > \Theta_{\text{osm},i} \quad \text{or} \quad K_i > \Theta_{\text{curv},i}.$$

Loss of any critical patch destroys  $\Omega(K_4)$ .

#### O.1.89 Axes $A(K_4)$

$K_4$  supports at minimum the following axes:

$$A(K_4) = \{A_{\text{comp}}, A_{\text{boundary}}, A_{\text{curv}}, A_{\text{perm}}, A_{\text{grad}}\}.$$

Interpretation:

- $A_{\text{comp}}$ : inside/outside compartment distinction,
- $A_{\text{boundary}}$ : membrane degrees of freedom,
- $A_{\text{curv}}$ : curvature states of boundary patches,
- $A_{\text{perm}}$ : variable permeability regimes,
- $A_{\text{grad}}$ : chemical/electrical/ionic gradients.

These axes are *not* reducible to chemical axes of  $K_3$ ; each represents new structural differences of biological organisation.

#### O.1.90 Potentials $P(K_4)$

Potentials include:

$$P(K_4) = \{\mu_i^{\text{in}}, \mu_i^{\text{out}}, P_{\text{grad}}, P_{\text{osm}}, P_{\text{curv}}, P_{\text{mem}}, P_{\text{pump}}\}.$$

Meaning:

- internal/external chemical potentials,
- gradient potentials driving selective transport,

- osmotic pressure potentials,
- curvature energy of membrane patches,
- membrane-tension potentials,
- energy stored in primitive pumps (if any appear).

These potentials regulate survival: too large a gradient or curvature destroys the system.

#### O.1.91 Thresholds $\Theta(K_4)$

Thresholds determine membrane and protocell viability:

$$\Theta(K_4) = \{\Theta_{\text{mem}}, \Theta_{\text{perm}}, \Theta_{\text{osm}}, \Theta_{\text{curv}}, \Theta_{\text{pH}}, \Theta_{\text{grad}}, \Theta_{\text{RAF-survival}}\}.$$

Key interpretations:

- $\Theta_{\text{mem}}$ : maximal membrane tension before rupture,
- $\Theta_{\text{perm}}$ : limits on passive permeability,
- $\Theta_{\text{osm}}$ : osmotic tolerance before burst,
- $\Theta_{\text{curv}}$ : curvature instability threshold,
- $\Theta_{\text{pH}}$ : internal pH viability range,
- $\Theta_{\text{grad}}$ : gradients sustainable without collapse,
- $\Theta_{\text{RAF-survival}}$ : minimal catalytic activity for homeostatic cycles.

These thresholds collectively determine the boundary of viable protocell states.

#### O.1.92 Flows $J(K_4)$

Flows in  $K_4$  include:

- transmembrane transport:

$$J_{\text{ion}}, J_{\text{water}}, J_{\text{solute}},$$

- passive leak flows,
- curvature-driven flows and patch rearrangements,
- internal RAF-reactive flows,
- pump-driven active flows (if primitive pumps exist),
- osmotic swelling–relaxation cycles.

Flow stability requires control of:

$$\Delta V, \Delta \Pi, \Delta c_i, \Delta \text{pH}.$$

#### O.1.93 Cycles $C(K_4)$

Biological cycles first appear:

$$C(K_4) = \{C_{\text{survival}}, C_{\text{osmotic}}, C_{\text{pH-recovery}}, C_{\text{leak-pump}}, C_{\text{RAF-stability}}\}.$$

Meaning:

- $C_{\text{survival}}$ : minimal homeostatic cycle preserving membrane integrity,
- $C_{\text{osmotic}}$ : repeated swelling–relaxation cycles,

- $C_{\text{pH-recovery}}$ : buffering and reaction flows,
  - $C_{\text{leak-pump}}$ : balance of passive leaks and active or semi-active transport,
  - $C_{\text{RAF-stability}}$ : internal network dynamics feeding boundary stability.
- Cycles define biological time direction and protocell persistence.

#### O.1.94 Time $\tau(K_4)$

Time exists when:

$$\Pi(C_{\text{survival}}) > \Theta_{\text{time}} \quad \text{and} \quad \Pi(C_{\text{RAF-stability}}) > \Theta_{\text{time}}.$$

Thus:

- biological time arises from survival cycles,
- time collapses when membrane cannot sustain stable cycle operation,
- temporal ordering is stronger than in  $K_3$  due to boundary feedback loops.

#### O.1.95 Continuumness $k(K_4)$

Applying the general formula:

$$k(K_4) = \frac{\mu(\Omega(K_4))}{\mu(S(K_4))} \cdot \frac{|A(K_4)|}{|A(M_4)|} \cdot \frac{\sum \text{Stab}(C)}{\sum \text{MaxStab}(C)} \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\text{max}}}\right) \cdot \left(1 - \frac{T}{\Theta_{\text{stab}}}\right)_+.$$

Implications:

- membrane stability increases  $k(K_4)$ ,
- osmotic and curvature fluctuations decrease  $k(K_4)$ ,
- loss of patch-level integrity sets  $k(K_4) = 0$ ,
- successful protocell organisation corresponds to  $k(K_4) > 0$  for extended periods.

#### O.1.96 Structural Tension $T(K_4)$

Sources:

- incompatible osmotic gradients,
- excessive curvature stress at boundary patches,
- pH stress,
- uncontrolled permeability,
- imbalance between leak and pump flows,
- insufficient RAF support.

Structural failure occurs if:

$$T(K_4) > \Theta_{\text{mem}}.$$

#### O.1.97 Energy $E(K_4)$

Energy functional includes:

$$E(K_4) = E_{\text{chem}}^{\text{in}} + E_{\text{chem}}^{\text{out}} + E_{\text{osm}} + E_{\text{curv}} + E_{\text{grad}} + E_{\text{mem}} + E_{\text{RAF}}.$$

Components:

- internal and external chemical energies,
  - osmotic pressure energy,
  - membrane curvature energy,
  - gradient energy,
  - membrane-tension energy,
  - RAF network catalytic energy.
- $E(K_4)$  determines protocell stability and dynamics.

#### **O.1.98 Operators on $K_4$ ( $\Psi, \Phi, \Lambda, U, X$ )**

- $\Psi_{4 \rightarrow 5}$  — emergence of electrical excitability and primitive ion channels (transition to  $K_5$ ),
- $\Phi$  — evolution of membrane constraints and permitted chemistries,
- $\Lambda$  — structural coupling of membrane and RAF sets,
- $U$  — unification of gradient and membrane subsystems,
- $X$  — branching into alternative protocell types (different membrane chemistries or transport regimes).

#### **O.1.99 Processes on $K_4$**

Characteristic processes:

- vesicle formation and closure,
- curvature-driven rearrangements,
- osmotic swelling and relaxation,
- pH-buffering cycles,
- leak–pump balance dynamics,
- proto-metabolic feedback loops,
- emergence of early ion channels,
- stabilisation of electrical axis (pre- $K_5$ ).

#### **O.1.100 Predictions for $K_4$**

1. Membrane stability requires curvature and osmotic thresholds within narrow bounds.
2. Patch-based tension distribution predicts local flicker modes.
3. pH-recovery cycles are necessary for protocell continuation.
4. RAF networks alone cannot maintain organisation without boundary stability.
5. Electrical gradients begin to appear before full  $K_5$ .
6. Transition to  $K_5$  occurs only after stabilisation of ion channels.

### O.1.101 Experiments for $K_4$

Empirical analogues include:

- fatty-acid vesicle experiments,
- protocell osmotic-burst tests,
- patch-level fluorescence mapping of membrane flicker,
- ion-channel insertion in simple vesicles,
- RAF networks enclosed in vesicles,
- pH-buffered vesicle dynamics.

These tests validate thresholds  $\Theta_{\text{mem}}$ ,  $\Theta_{\text{curv}}$ ,  $\Theta_{\text{osm}}$ ,  $\Theta_{\text{pH}}$ .

### O.1.102 Collapse and Death of $K_4$

Collapse occurs when:

$$\Omega(K_4) = \emptyset.$$

Contributors:

- membrane rupture,
- extreme osmotic imbalance,
- curvature-induced bursting,
- catastrophic pH collapse,
- leak runaway,
- failure of survival cycles.

Once collapsed, transition to  $K_5$  is impossible.

### O.1.103 Falsifiability of $K_4$

Predictions that can be experimentally challenged:

1. the existence of strict membrane thresholds for survival,
2. universal form of osmotic-collapse curves,
3. necessity of RAF-supported buffering cycles,
4. impossibility of sustained protocell organisation without patch-level curvature control,
5. early appearance of proto-electrical gradients,
6. requirement of minimal leak–pump balance.

Failure of these predictions would refute  $K_4$  as presently defined.

### O.1.104 Branching / Ontological Position of $K_4$

Branching types:

- different membrane chemistries (fatty-acid, phospholipid, mixed),
- distinct permeability regimes,
- alternative RAF cores compatible with membrane dynamics,



- divergent proto-metabolic strategies.

Ontological position:

$$K_3 \rightarrow K_4 \rightarrow K_5,$$

with  $K_4$  as the first biological continuum.

#### O.1.105 Relation to M-spaces

$M_4$  specifies:

- admissible membrane materials,
- allowed curvature and permeability ranges,
- environmental regimes (osmotic, ionic, pH),
- permitted transport processes,
- constraints for forming  $\partial\Omega$  as a stable boundary.

Compatibility of  $K_4$  with  $M_4$  determines protocell viability.

#### O.1.106 $K_5$ Overview

$K_5$  is the level at which *electrical excitability* emerges as a new ontological dimension. The protocell boundary of  $K_4$  becomes an *electrically active membrane* through the formation of ion channels with controlled conductance. This generates a new axis:

$$A_{\text{exc}} : \Delta V \mapsto \text{electrical state difference},$$

where  $\Delta V$  is the transmembrane potential.

This axis is not definable in  $K_4$ . It represents a new class of incompatible differences and therefore constitutes a genuine increase in dimension.

Organisationally,  $K_5$  is characterised by:

- stable ion channels (open/closed/leaky/blocked),
- controlled ionic fluxes,
- excitable cycles (excitation–recovery),
- refractory dynamics,
- noise control and thresholded responses,
- self-sustaining electrical organisation tightly coupled to the chemical interior.

$K_5$  is the minimal *neuron-like* continuum, even if no real neurons exist yet.

#### O.1.107 State Space $\Omega(K_5)$

A state of  $K_5$  includes:

$$\Omega(K_5) = \{\Delta V, c_i, g_{\text{ion}}, P_{\text{open}}, x(t), G = (V, E), I(t)\}.$$

Components:

- $\Delta V$  — transmembrane potential,
- $c_i$  — ionic concentrations inside and outside,
- $g_{\text{ion}}$  — conductances of ion channels,
- $P_{\text{open}}$  — open probabilities,

- $x(t)$  — internal activation variables,
- $G = (V, E)$  — proto-neural connectivity graph for multi-unit  $K_5$  systems,
- $I(t)$  — external or internal currents.

Admissibility requires:

$$|\Delta V| < \Theta_{\text{volt-max}}, \quad c_i \in \text{viability range}, \quad 0 \leq P_{\text{open}} \leq 1.$$

#### O.1.108 Boundary $\partial\Omega(K_5)$

The membrane inherits  $K_4$ 's boundary with a new structure:

$$\partial\Omega(K_5) = \partial\Omega(K_4) \cup \{\text{ion channels with gating kinetics}\}.$$

For each channel  $k$ :

- state space: open/closed/blocked/leaky,
- parameters:  $g_k$ , reversal potential  $E_k$ , gating variables,
- patch-level localisation,
- interactions with local curvature and tension.

Collapse of channel ordering or gating kinetics destroys  $\Omega(K_5)$ .

#### O.1.109 Axes $A(K_5)$

The minimal axes are:

$$A(K_5) = A(K_4) \cup \{A_{\text{exc}}, A_{\text{ion}}, A_{\text{gate}}\}.$$

Where:

- $A_{\text{exc}}$ : electrical excitation axis,
- $A_{\text{ion}}$ : ionic species and flux differences,
- $A_{\text{gate}}$ : gating-state differences for ion channels.

These axes are incompatible with chemical-only differences of  $K_4$  and imply a higher-dimensional continuum.

#### O.1.110 Potentials $P(K_5)$

Potentials include:

$$P(K_5) = \{E_{\text{ion}}, P_{\text{gate}}, P_{\text{exc}}, P_{\text{noise}}, P_{\text{refrac}}, P_{\text{coupling}}\}.$$

Interpretation:

- $E_{\text{ion}}$  — Nernst potentials for each ionic species,
- $P_{\text{gate}}$  — gating activation potentials,
- $P_{\text{exc}}$  — excitation thresholds for spike initiation,
- $P_{\text{noise}}$  — noise-control potentials,
- $P_{\text{refrac}}$  — refractory resetting potentials,
- $P_{\text{coupling}}$  — potentials for electrically coupling multiple units via  $G$ .

### O.1.111 Thresholds $\Theta(K_5)$

Threshold structure:

$$\Theta(K_5) = \{\Theta_{\text{exc}}, \Theta_{\text{noise}}, \Theta_{\text{volt-max}}, \Theta_{\text{channel-open}}, \Theta_{\text{channel-close}}, \Theta_{\text{channel-noise}}, \Theta_{\text{front}}, \Theta_{\text{refrac}}\}.$$

Meaning:

- $\Theta_{\text{exc}}$ : excitation threshold for spike generation,
- $\Theta_{\text{noise}}$ : noise limit beyond which excitation is lost,
- $\Theta_{\text{volt-max}}$ : maximal safe transmembrane voltage,
- $\Theta_{\text{channel-open}}, \Theta_{\text{channel-close}}$ : gating viability thresholds,
- $\Theta_{\text{channel-noise}}$ : noise tolerance in gating,
- $\Theta_{\text{front}}$ : minimal conditions for propagation fronts in multi-unit settings,
- $\Theta_{\text{refrac}}$ : minimal refractory recovery level.

Exceeding  $\Theta_{\text{volt-max}}$  or  $\Theta_{\text{channel-noise}}$  annihilates the continuum.

### O.1.112 Flows $J(K_5)$

Flows:

- ionic currents:

$$J_{\text{ion},k} = g_k(\Delta V - E_k),$$

- leak currents and noise currents,
- gating-variable flows:  $\dot{x} = f(x, \Delta V)$ ,
- spike-propagation flows over graph edges  $E$ ,
- refractoriness recovery flows,
- coupling flows between units.

Stability requires:

$$\sigma(J) < \sigma_{\max}(K_5).$$

### O.1.113 Cycles $C(K_5)$

Characteristic cycles:

$$C(K_5) = \{C_{\text{exc}}, C_{\text{refrac}}, C_{\text{noise-control}}, C_{\text{channel-dynamics}}, C_{\text{coupling}}\}.$$

Descriptions:

- $C_{\text{exc}}$ : excitation cycle (rest  $\rightarrow$  excitation  $\rightarrow$  peak  $\rightarrow$  repolarisation),
- $C_{\text{refrac}}$ : refractory cycle restoring channel states,
- $C_{\text{noise-control}}$ : suppression of stochastic fluctuations,
- $C_{\text{channel-dynamics}}$ : gating cycles that sustain excitability,
- $C_{\text{coupling}}$ : multi-unit synchronisation cycles.

Each cycle has a characteristic period  $\tau_{C_j}$ .

### O.1.114 Time $\tau(K_5)$

Time emerges from the existence of stable excitation cycles:

$$\tau(K_5) = \min_j \tau_{C_j}, \quad \Pi(C_j) > \Theta_{\text{time}}.$$

Temporal ordering at  $K_5$  is more rigid than at  $K_4$ , due to sharp excitation thresholds and refractory intervals.

### O.1.115 Continuumness $k(K_5)$

Using the general definition:

$$k_5 = H(\Omega(K_5)) \cdot \frac{|V_{\text{cycle}}|}{|V|} \cdot \frac{\sum_j C_j^{\text{eff}}}{\sum_j C_j^{\text{max}}} \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\text{max}}}\right) \cdot \left(1 - \frac{T(K_5)}{\Theta_{\text{stab}}}\right)_+.$$

Interpretation:

- cycle fraction controls viability,
- excessive noise reduces  $k_5$ ,
- too large  $\Delta V$  collapses  $k_5$  to zero,
- coordinated excitation raises  $k_5$ .

### O.1.116 Structural Tension $T(K_5)$

Sources of tension:

- incompatible ionic gradients,
- gating noise,
- runaway excitation,
- insufficient refractory recovery,
- loss of coupling stability,
- uncontrolled depolarisation.

Failure if:

$$T(K_5) > \Theta_{\text{volt-max}}.$$

### O.1.117 Energy $E(K_5)$

Energy functional:

$$E(K_5) = E_{\text{chem}} + E_{\text{ion}} + E_{\text{pump}} + E_{\text{exc}} + E_{\text{noise}} + E_{\text{coupling}}.$$

Components:

- chemical energy for maintaining gradients,
- ionic current energy,
- pump-driven energy consumption,
- excitation and gating energy,
- noise-control energy,
- coupling energy in multi-unit networks.

### O.1.118 Operators on $K_5$ ( $\Psi, \Phi, \Lambda, U, X$ )

- $\Psi_{5 \rightarrow 6}$  — emergence of cognitive modelling axes and transformation of excitation cycles into attractors,
- $\Phi$  — evolution of channel kinetics and conductances,
- $\Lambda$  — structural coupling of excitation with chemical homeostasis,
- $U$  — unification of ionic species into integrated electrical dynamics,
- $X$  — branching into different excitability regimes (burst, tonic, phasic).

### O.1.119 Processes on $K_5$

Typical processes:

- spike initiation and recovery,
- gating kinetics transitions,
- fluctuations around excitation threshold,
- propagation of fronts across  $G$ ,
- refractoriness-driven ordering of activity,
- synchronisation in coupled units,
- emergence of attractor-like stable firing patterns.

### O.1.120 Predictions for $K_5$

1. Excitability requires tightly controlled noise regimes.
2. Spike-like events appear spontaneously once  $\Theta_{\text{exc}}$  is crossed.
3. Global depolarisation causes structural collapse.
4. Networked  $K_5$  systems form attractors that prefigure  $K_6$  cognitive models.
5. Channel diversity expands viable regions of  $\Omega(K_5)$ .

### O.1.121 Experiments for $K_5$

Empirical analogues:

- artificial lipid vesicles with reconstituted ion channels,
- patch-clamp experiments measuring gating noise,
- voltage-clamp characterisation of  $\Delta V$  thresholds,
- multi-vesicle coupling and synchronisation tests,
- proto-neuron models with minimal excitability.

These tests directly probe  $\Theta_{\text{exc}}$ ,  $\Theta_{\text{noise}}$ ,  $\Theta_{\text{channel-open}}$ , etc.

### O.1.122 Collapse and Death of $K_5$

Occurs when:

$$\Omega(K_5) = \emptyset.$$

Mechanisms:

- uncontrolled depolarisation,
- channel malfunction or gating collapse,
- ionic-gradient dissipation,
- noise-induced runaway failures,
- inability to recover from excitation.

Collapse prevents emergence of  $K_6$ .

### O.1.123 Falsifiability of $K_5$

Key testable predictions:

1. Excitable behaviour cannot be sustained without refractory cycles.
2. Noise-control thresholds are real and measurable.
3. Ion-channel distributions determine viability boundaries.
4. Stable attractors in  $K_5$  must exist before  $K_6$  can form.
5. Spike propagation requires  $\Theta_{\text{front}}$ .

Failure of these predictions refutes the model of  $K_5$ .

### O.1.124 Branching / Ontological Position of $K_5$

Branching types:

- tonic–phasic–burst excitability regimes,
- multi-channel vs. single-channel systems,
- weakly vs. strongly coupled units,
- distributed vs. localised excitation patterns.

Ontological location:

$$K_4 \rightarrow K_5 \rightarrow K_6.$$

$K_5$  is the bridge between biological protocells and cognitive systems.

### O.1.125 Relation to M-spaces

$M_5$  specifies:

- permitted ionic species and charge-carriers,
- admissible ranges of conductances and potentials,
- noise regimes compatible with excitability,
- environmental constraints allowing channel stability,
- allowed coupling structures.

Compatibility of  $K_5$  with  $M_5$  determines the existence of electrically excitable systems.

### O.1.126 $K_6$ Overview

$K_6$  is the level at which *cognitive organisation* emerges as an ontological dimension. The excitable electrical dynamics of  $K_5$  produce stable attractors; these attractors become *models* that encode internal structure, memory, predictive states and representational relations.

The birth of  $K_6$  occurs when excitation cycles of  $K_5$  are no longer merely electrical waves, but become *informationally interpretable* patterns whose differences cannot be expressed in  $A(K_5)$ . This necessitates a new set of axes and a new dimensional structure.

The minimal features:

- representational states,
- prediction capability (prediction error axis),
- semantic and binding structure,
- internal models  $\mu_t$ ,
- memory consolidation/reconsolidation,
- stable components of cognitive graphs,
- emergence of proto-symbols.

$K_6$  is the smallest continuum that deserves the name “cognitive”. It is not yet linguistic or social (those belong to  $K_7$ ).

### O.1.127 State Space $\Omega(K_6)$

A state of  $K_6$  is:

$$\Omega(K_6) = \{m, \mu_t, G_{\text{cog}}, x(t), r_i, PE_t, B_t, M_{\text{con}}, M_{\text{rec}}\}.$$

Elements:

- $m$  — active cognitive model,
- $\mu_t$  — distribution of model states,
- $G_{\text{cog}}$  — cognitive graph (nodes = models, edges = binding),
- $x(t)$  — activation patterns inherited from  $K_5$  attractors,
- $r_i$  — representational units (proto-symbols),
- $PE_t$  — prediction error,
- $B_t$  — binding configuration,
- $M_{\text{con}}, M_{\text{rec}}$  — consolidated vs. reconsolidating memory states.

Admissibility:

$$D(m) \leq \Theta_{\text{cog}}, \quad |PE_t| < \Theta_{\text{pred}}, \quad B_t \in \text{binding-capacity}.$$

$D(m)$  is inconsistency of model  $m$ .

### O.1.128 Boundary $\partial\Omega(K_6)$

Cognitive boundaries include:

- limits of representational capacity,
- edges of binding depth,
- maximal prediction-horizon,

- memory stability limits,
  - semantic coherence boundaries,
  - transitions where  $PE_t$  cannot be reconciled.
- Crossing  $\partial\Omega(K_6)$  collapses modelling capability.

#### O.1.129 Axes $A(K_6)$

As established in K6 Run 2, the minimal set is:

$$A(K_6) = \{A_{\text{sel}}, A_{\text{cmp}}, A_{\text{bind}}, A_{\text{cat}}, A_{\text{mem}}, A_{\text{pred}}, A_{\text{model}}\}.$$

Interpretation:

- $A_{\text{sel}}$  — selection axis (which model is active),
- $A_{\text{cmp}}$  — comparison axis,
- $A_{\text{bind}}$  — binding axis (feature composition),
- $A_{\text{cat}}$  — categorisation axis,
- $A_{\text{mem}}$  — memory axis,
- $A_{\text{pred}}$  — prediction axis,
- $A_{\text{model}}$  — model-identity axis.

These axes cannot be reduced to excitation differences of  $K_5$ .

#### O.1.130 Potentials $P(K_6)$

Potentials regulate artefacts of cognition:

$$P(K_6) = \{P_{\text{pred}}, P_{\text{val}}, P_{\text{bind}}, P_{\text{sal}}, P_{\text{mem}}, P_{\text{elim}}\}.$$

Where:

- $P_{\text{pred}}$  — prediction potential controlling model update,
- $P_{\text{val}}$  — valence potential (affective shaping),
- $P_{\text{bind}}$  — potential for feature binding,
- $P_{\text{sal}}$  — salience modulation,
- $P_{\text{mem}}$  — consolidation pressure for memory,
- $P_{\text{elim}}$  — elimination of incoherent models.

#### O.1.131 Thresholds $\Theta(K_6)$

Thresholds determine cognitive viability:

$$\Theta(K_6) = \{\Theta_{\text{cog}}, \Theta_{\text{pred}}, \Theta_{\text{bind}}, \Theta_{\text{sal}}, \Theta_{\text{model}}, \Theta_{\text{mem-stab}}, \Theta_{\text{noise-cog}}\}.$$

Meaning:

- $\Theta_{\text{cog}}$ : global inconsistency tolerance,
- $\Theta_{\text{pred}}$ : maximal allowable prediction error,
- $\Theta_{\text{bind}}$ : binding capacity limit,



- $\Theta_{\text{sal}}$ : saturation threshold for salience,
- $\Theta_{\text{model}}$ : collapse limit for model viability,
- $\Theta_{\text{mem-stab}}$ : memory stability threshold,
- $\Theta_{\text{noise-cog}}$ : cognitive noise tolerance.

Crossing these leads to  $K_6$  collapse or transition to  $K_7$  depending on direction.

#### O.1.132 Flows $J(K_6)$

The flows are informational and representational:

- $J_{\text{pred}} = -\nabla_m P E_t$  — prediction update flow,
- $J_{\text{bind}}$  — binding/unbinding transitions,
- $J_{\text{cmp}}$  — comparison dynamics,
- $J_{\text{sel}}$  — model-selection flow,
- $J_{\text{mem}}$  — consolidation/reconsolidation flow,
- $J_{\text{elim}}$  — elimination of incoherent models.

Stability requires:

$$\sigma(J_{\text{pred}}) < \sigma_{\text{max}}(K_6).$$

#### O.1.133 Cycles $C(K_6)$

Characteristic cycles include:

$$C(K_6) = \{C_{\text{pred}}, C_{\text{bind}}, C_{\text{mem}}, C_{\text{cmp}}, C_{\text{model}}\}.$$

Descriptions:

- $C_{\text{pred}}$ : prediction cycle (model  $\rightarrow$  prediction  $\rightarrow$  error  $\rightarrow$  update),
- $C_{\text{bind}}$ : feature-binding cycle (composition  $\rightarrow$  stabilisation  $\rightarrow$  release),
- $C_{\text{mem}}$ : memory cycle (consolidation  $\rightarrow$  storage  $\rightarrow$  reconsolidation),
- $C_{\text{cmp}}$ : comparison–differentiation cycle,
- $C_{\text{model}}$ : cycle of model selection, evaluation, elimination.

Each has characteristic period  $\tau_{C_j}$ .

#### O.1.134 Time $\tau(K_6)$

Time arises from *stability of cognitive cycles*:

$$\tau(K_6) = \min_j \tau_{C_j}, \quad \Pi(C_j) > \Theta_{\text{time}}.$$

Cognitive time is slower and more integrative than electrical time ( $K_5$ ).

### O.1.135 Continuumness $k(K_6)$

Using the general formula (K6 Run 7):

$$k_6 = H(\Omega(K_6)) \cdot \frac{|C_{\max}^{(6)}|}{|C_{\text{all}}|} \cdot \frac{r}{r_{\max}} \cdot \left(1 - \frac{D(m)}{\Theta_{\text{cog},\max}}\right)_+ \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\max}}\right).$$

Where:

- $|C_{\max}^{(6)}|$  — size of largest connected component in  $G_{\text{cog}}$ ,
- $r$  — number of active axes vs. maximal axes,
- $D(m)$  — model inconsistency metric.

Interpretation:

- large connected semantic structure increases  $k_6$ ,
- inconsistency and prediction error reduce  $k_6$ ,
- fragmentation of cognitive graph collapses  $k_6$ .

### O.1.136 Structural Tension $T(K_6)$

Sources:

- high prediction error  $PE_t$ ,
- incompatible bindings,
- semantic clashes,
- unstable memory reconsolidation,
- incoherent model competition,
- runaway salience.

Collapse when:

$$T(K_6) > \Theta_{\text{cog}}.$$

### O.1.137 Energy $E(K_6)$

Energy is cognitive–neural hybrid:

$$E(K_6) = E_{K_5} + E_{\text{pred}} + E_{\text{bind}} + E_{\text{mem}} + E_{\text{cmp}} + E_{\text{model}}.$$

Interpretation:

- prediction update energy,
- memory maintenance energy,
- binding energy costs,
- energy in representational graphs,
- cost of model switching.

### O.1.138 Operators on $K_6$ ( $\Psi, \Phi, \Lambda, U, X$ )

- $\Psi_{6 \rightarrow 7}$  — emergence of communicative axes (shared models),
- $\Phi$  — evolution of representational structure,
- $\Lambda$  — unification of model components into higher-order structures,
- $U$  — integration of cognitive cycles into stable semantic units,
- $X$  — branching into specialisations (memory-heavy, prediction-heavy, binding-heavy).

### O.1.139 Processes on $K_6$

Typical:

- model formation and dissolution,
- prediction and error correction,
- semantic binding,
- consolidation and reconsolidation,
- salience modulation,
- feature extraction,
- generation of proto-symbols  $\sigma$ ,
- attractor dynamics shaping cognitive topology.

### O.1.140 Predictions for $K_6$

1. Cognitive collapse occurs when  $PE_t$  accumulates faster than binding can stabilise.
2. Stable long-term memory requires reconsolidation cycles.
3. Binding axes have maximal capacity that can be experimentally tested.
4. Semantic connectivity predicts robustness against noise.
5. Cognitive time  $\tau(K_6)$  increases with representational complexity.
6. Proto-symbolic states appear naturally when  $G_{\text{cog}}$  reaches sufficient connectivity.

### O.1.141 Experiments for $K_6$

Possible empirical analogues:

- neural-network attractor models with controlled noise,
- predictive coding experiments measuring  $PE_t$  thresholds,
- binding-capacity behavioural tests,
- memory reconsolidation protocols,
- semantic graph fragmentation tests,
- probing emergence of proto-symbols in neural systems.

#### **O.1.142 Collapse and Death of $K_6$**

Failure when:

$$\Omega(K_6) = \emptyset.$$

Mechanisms:

- uncontrolled growth of prediction error,
- catastrophic binding failure,
- semantic graph collapse,
- memory disintegration,
- runaway salience cascades,
- inability to resolve model conflicts.

Death of  $K_6$  prevents transition to social cognition  $K_7$ .

#### **O.1.143 Falsifiability of $K_6$**

The model predicts:

- existence of strict prediction-error thresholds,
- necessity of binding-capacity limits,
- measurable cognitive cycles (prediction, memory, binding),
- hierarchical emergence of proto-symbols,
- dependence of semantic stability on graph connectivity.

Refutation would follow from observing cognitive systems without these structures.

#### **O.1.144 Branching / Ontological Position of $K_6$**

Branches:

- memory-dominant vs. prediction-dominant systems,
- high-binding vs. low-binding organisms,
- model-rich vs. model-poor cognitive architectures,
- distributed vs. localised semantic graphs.

Ontological positioning:

$$K_5 \rightarrow K_6 \rightarrow K_7.$$

$K_6$  is the prerequisite for communication, social behaviour and shared models.

#### **O.1.145 Relation to M-spaces**

$M_6$  constrains:

- allowable prediction-error landscapes,
- binding-topology ranges,
- memory-stability regimes,
- structural coherence requirements,
- admissible dynamics of representational graphs.

Compatibility with  $M_6$  determines the viability of cognitive continua.

### O.1.146 $K_7$ Overview

$K_7$  is the level of *social continua*: systems whose states, axes and potentials arise from *interaction between multiple  $K_6$  cognitive continua*. It is the smallest level at which:

- shared models and common knowledge emerge,
- communication forms a structural dimension,
- cooperation and conflict become dynamical forces,
- social identity and role formation appear,
- collective memory and norms stabilise behaviour,
- group-level cycles define temporal organisation.

$K_7$  is not a sum of agents: it is a new continuum with its own admissible states, thresholds, flows and death mechanisms.

### O.1.147 State Space $\Omega(K_7)$

A state of the social continuum includes:

$$\Omega(K_7) = \{G_{\text{soc}}, M_{\text{shared}}, C_{\text{comm}}, R_{\text{roles}}, N_{\text{norms}}, H_{\text{hist}}, S_{\text{struc}}, \mu_t^{(7)}, \Sigma_{\text{identity}}\}.$$

Components:

- $G_{\text{soc}}$  — social graph (agents as nodes, ties as edges),
- $M_{\text{shared}}$  — shared models (collective beliefs),
- $C_{\text{comm}}$  — communication structure,
- $R_{\text{roles}}$  — role allocation (statuses, functions),
- $N_{\text{norms}}$  — norms and constraints,
- $H_{\text{hist}}$  — collective histories and narratives,
- $S_{\text{struc}}$  — structural positions and modules,
- $\mu_t^{(7)}$  — distribution of group states over time,
- $\Sigma_{\text{identity}}$  — social identity structures.

Admissibility is constrained by  $M_7$  and group-coherence thresholds.

### O.1.148 Boundary $\partial\Omega(K_7)$

The boundary includes:

- collapse of shared models,
- fragmentation of the social graph,
- breakdown of communication,
- loss of normative coherence,
- role instability,
- identity collapse,
- inability to stabilise collective memory.

Crossing  $\partial\Omega(K_7)$  destroys group continuity.

### O.1.149 Axes $A(K_7)$

From the core Social Systems module:

$$A(K_7) = \{A_{\text{comm}}, A_{\text{coop}}, A_{\text{conf}}, A_{\text{norm}}, A_{\text{role}}, A_{\text{id}}, A_{\text{shared-model}}\}.$$

Interpretation:

- $A_{\text{comm}}$  — communication axis (channels, codes),
- $A_{\text{coop}}$  — cooperation axis,
- $A_{\text{conf}}$  — conflict/competition axis,
- $A_{\text{norm}}$  — normativity axis,
- $A_{\text{role}}$  — role-position axis,
- $A_{\text{id}}$  — social identities,
- $A_{\text{shared-model}}$  — collective models/beliefs.

These axes cannot be reduced to  $K_6$  cognitive differences.

### O.1.150 Potentials $P(K_7)$

Potentials drive social dynamics:

$$P(K_7) = \{P_{\text{comm}}, P_{\text{coop}}, P_{\text{conf}}, P_{\text{norm}}, P_{\text{id}}, P_{\text{role}}, P_{\text{shared}}\}.$$

Meaning:

- $P_{\text{comm}}$ : communication effectiveness,
- $P_{\text{coop}}$ : cooperation advantage,
- $P_{\text{conf}}$ : conflict pressure,
- $P_{\text{norm}}$ : normative cohesion,
- $P_{\text{id}}$ : identity stabilisation potential,
- $P_{\text{role}}$ : functional-role pressure,
- $P_{\text{shared}}$ : shared-model coherence potential.

### O.1.151 Thresholds $\Theta(K_7)$

$$\Theta(K_7) = \{\Theta_{\text{comm}}, \Theta_{\text{coh}}, \Theta_{\text{norm}}, \Theta_{\text{id}}, \Theta_{\text{role}}, \Theta_{\text{shared}}, \Theta_{\text{fragment}}\}.$$

Interpretation:

- $\Theta_{\text{comm}}$ : minimal communication bandwidth,
- $\Theta_{\text{coh}}$ : coherence threshold for group identity,
- $\Theta_{\text{norm}}$ : minimum normative alignment,
- $\Theta_{\text{id}}$ : identity-persistence threshold,
- $\Theta_{\text{role}}$ : role-stability threshold,
- $\Theta_{\text{shared}}$ : shared-model coherence threshold,
- $\Theta_{\text{fragment}}$ : fragmentation threshold of the social graph.

Crossing any of these may cause collapse of  $K_7$ .

### O.1.152 Flows $J(K_7)$

- $J_{\text{comm}}$  — flows of communication,
- $J_{\text{coop}}$  — cooperation dynamics,
- $J_{\text{conf}}$  — conflict escalation or de-escalation,
- $J_{\text{norm}}$  — norm-updating flows,
- $J_{\text{id}}$  — identity transitions,
- $J_{\text{role}}$  — role reallocation,
- $J_{\text{shared}}$  — flow updating shared models,
- $J_{\text{PE} \rightarrow \text{comm}}$ : projection of prediction-error signals from  $K_6$  into communication,
- $J_{\text{comm} \rightarrow A_{\text{error}}}$ : feedback from social communication back into cognitive prediction axes.

Stability requires:

$$\sigma(J(K_7)) < \sigma_{\max}(K_7).$$

### O.1.153 Cycles $C(K_7)$

$$C(K_7) = \{C_{\text{comm}}, C_{\text{coop}}, C_{\text{conf}}, C_{\text{norm}}, C_{\text{id}}, C_{\text{shared}}\}.$$

Descriptions:

- $C_{\text{comm}}$ : communication  $\rightarrow$  interpretation  $\rightarrow$  response  $\rightarrow$  update,
- $C_{\text{coop}}$ : cooperation cycles (trust–reinforcement loops),
- $C_{\text{conf}}$ : conflict–resolution cycles,
- $C_{\text{norm}}$ : norm creation, enforcement, revision,
- $C_{\text{id}}$ : identity formation, negotiation, stabilisation,
- $C_{\text{shared}}$ : maintenance of shared beliefs and collective memory.

Each cycle produces social temporal structure.

### O.1.154 Time $\tau(K_7)$

Social time arises from stable cycles:

$$\tau(K_7) = \min_j \tau_{C_j}, \quad \Pi(C_j) > \Theta_{\text{time}}.$$

$K_7$  time is typically slower and more inertial than  $K_6$  time due to collective inertia.

### O.1.155 Continuumness $k(K_7)$

From the Social Systems module:

$$k_7 = H(\Omega(K_7)) \cdot \frac{|C_{\text{max}}^{(7)}|}{|C_{\text{all}}|} \cdot \frac{|A_{\text{active}}|}{|A_{\text{max}}|} \cdot \left(1 - \frac{T_7}{\Theta_7}\right)_+ \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\max}}\right).$$

Where:

- $C_{\text{max}}^{(7)}$  — largest coherent social component,
- $T_7$  — structural tension of the social continuum,
- $\Theta_7$  — dominant threshold (coherence or shared-model limit),

- $\sigma(J)$  — volatility of social flows.  
High fragmentation  $\Rightarrow k_7 \rightarrow 0$ .

#### **O.1.156 Structural Tension $T(K_7)$**

Sources:

- communication failures,
- norm conflicts,
- role instability,
- fragmentation of shared models,
- identity clashes,
- rapid shifts in cooperation/conflict balance,
- overconcentration of influence or structural bottlenecks.

Collapse when:

$$T(K_7) > \Theta_{\text{coh}}.$$

#### **O.1.157 Energy $E(K_7)$**

$$E(K_7) = E_{\text{comm}} + E_{\text{coop}} + E_{\text{conf}} + E_{\text{norm}} + E_{\text{id}} + E_{\text{shared}} + E_{K_6 \rightarrow K_7}.$$

Interpretation:

- cost of communication,
- cost of maintaining cooperation,
- energy of conflict escalation,
- normative enforcement cost,
- identity maintenance,
- maintaining shared beliefs,
- coupling energy inherited from  $K_6$ .

#### **O.1.158 Operators on $K_7$ ( $\Psi, \Phi, \Lambda, U, X$ )**

- $\Psi_{7 \rightarrow 8}$ : birth of institutional and civilisational continua,
- $\Phi$ : evolution of social structures and roles,
- $\Lambda$ : unification of subsystems into coherent groups,
- $U$ : stabilisation of norms and identities,
- $X$ : branching into subcultures, factions, and specialised groups.

#### **O.1.159 Processes on $K_7$**

Typical:

- communication,
- cooperation and coalition formation,
- conflict escalation and resolution,



- norm formation, enforcement and revision,
- role allocation and authority formation,
- identity creation and transformation,
- formation of shared narratives and collective memory,
- social learning and diffusion.

#### **O.1.160 Predictions for $K_7$**

1. Social continua collapse when shared-model coherence drops below  $\Theta_{\text{shared}}$ .
2. Communication bottlenecks predict fragmentation.
3. Normative instability precedes identity collapse.
4. Highly modular groups resist noise and conflict.
5. Transition to  $K_8$  requires persistent normative and communicative stability.

#### **O.1.161 Experiments for $K_7$**

Possible empirical analogues:

- multi-agent reinforcement-learning simulations,
- network-fragmentation studies,
- norm diffusion experiments,
- collective-memory stability tests,
- communication-bandwidth perturbation experiments,
- identity coherence measurements in social groups.

#### **O.1.162 Collapse and Death of $K_7$**

Death occurs when:

$$\Omega(K_7) = \emptyset.$$

Mechanisms:

- fragmentation of social graph,
- loss of communication capacity,
- normative disintegration,
- identity collapse,
- competitive runaway dynamics,
- inability to stabilise shared models.

### O.1.163 Falsifiability of $K_7$

The theory predicts:

- existence of coherence thresholds,
- measurable communication bandwidth effects,
- clear relation between fragmentation and collapse,
- stable social cycles with characteristic periods,
- necessity of shared-model structures for group persistence.

Falsification would follow from observation of large-scale social stability without any of these structures.

### O.1.164 Branching / Ontological Position of $K_7$

Branches:

- cooperative vs. competitive societies,
- highly normative vs. weakly normative groups,
- identity-centric vs. role-centric structures,
- centralised vs. decentralised communication networks.

Ontological location:

$$K_6 \rightarrow K_7 \rightarrow K_8.$$

### O.1.165 Relation to M-spaces

$M_7$  constrains:

- social graph topologies,
- range of admissible norm systems,
- communication complexity,
- identity coherence,
- structural modularity,
- coupling to  $M_6$  (cognitive layer).

Compatibility with  $M_7$  determines whether a social continuum can exist.

### O.1.166 $K_8$ Overview

$K_8$  is the level of *civilizational continua*: large-scale, long-lived structures that arise when multiple  $K_7$  social continua stabilise shared institutions, infrastructures, codified norms, collective memory mechanisms and large-scale coordination structures.

Distinctive features:

- institutions as persistent operators,
- codified law and formal norms,
- economic and logistical infrastructures,
- large-scale information networks,
- symbolic systems (religion, science, ideology),

- historical cycles and path-dependence,
- multi-layered governance and authority.

$K_8$  is not reducible to the sum of  $K_7$  groups: it has its own admissible states, thresholds, cycles and collapse modes.

#### O.1.167 State Space $\Omega(K_8)$

$$\Omega(K_8) = \{I_{\text{inst}}, G_{\text{gov}}, S_{\text{symbol}}, L_{\text{law}}, X_{\text{econ}}, U_{\text{infra}}, H_{\text{civil}}, M_{\text{meta}}, \mu_t^{(8)}\}.$$

Components:

- $I_{\text{inst}}$  — institutional architecture,
- $G_{\text{gov}}$  — governance structures,
- $S_{\text{symbol}}$  — symbolic systems (religion, science, ideology),
- $L_{\text{law}}$  — legal and normative codification,
- $X_{\text{econ}}$  — economic organization and exchange systems,
- $U_{\text{infra}}$  — physical and informational infrastructure,
- $H_{\text{civil}}$  — civilizational memory and historiography,
- $M_{\text{meta}}$  — meta-models (cosmologies, worldviews),
- $\mu_t^{(8)}$  — distribution of civilizational states in time.

#### O.1.168 Boundary $\partial\Omega(K_8)$

The boundary includes:

- failure of institutional coherence,
- collapse of governance,
- symbolic disintegration (loss of shared meaning),
- infrastructural failure (logistics, energy, communication),
- legal disorder,
- economic implosion,
- loss of civilizational memory,
- breakdown of meta-models (cosmological/ideological collapse).

Crossing  $\partial\Omega(K_8)$  produces civilizational collapse.

#### O.1.169 Axes $A(K_8)$

$$A(K_8) = \{A_{\text{inst}}, A_{\text{gov}}, A_{\text{symbol}}, A_{\text{law}}, A_{\text{econ}}, A_{\text{infra}}, A_{\text{memory}}, A_{\text{meta}}\}.$$

Interpretation:

- $A_{\text{inst}}$ : institutional dimension,
- $A_{\text{gov}}$ : governance / authority dimension,
- $A_{\text{symbol}}$ : symbolic–ideological systems,
- $A_{\text{law}}$ : codified legal order,

- $A_{\text{econ}}$ : exchange, production, distribution,
- $A_{\text{infra}}$ : infrastructural networks,
- $A_{\text{memory}}$ : collective civilizational memory,
- $A_{\text{meta}}$ : meta-models (science, religion, worldview).

#### **O.1.170 Potentials $P(K_8)$**

$$P(K_8) = \{P_{\text{inst}}, P_{\text{gov}}, P_{\text{symbol}}, P_{\text{law}}, P_{\text{econ}}, P_{\text{infra}}, P_{\text{memory}}, P_{\text{meta}}\}.$$

Potentials describe:

- institutional resilience,
- governance efficiency and legitimacy,
- symbolic coherence,
- legal stability,
- economic viability,
- infrastructural robustness,
- memory stability and path-dependence force,
- meta-model strength and integrative capacity.

#### **O.1.171 Thresholds $\Theta(K_8)$**

$$\Theta(K_8) = \{\Theta_{\text{inst}}, \Theta_{\text{gov}}, \Theta_{\text{symbol}}, \Theta_{\text{law}}, \Theta_{\text{econ}}, \Theta_{\text{infra}}, \Theta_{\text{memory}}, \Theta_{\text{meta}}, \Theta_{\text{collapse}}\}.$$

Critical thresholds:

- $\Theta_{\text{inst}}$ : minimal institutional coherence,
- $\Theta_{\text{gov}}$ : governance capacity threshold,
- $\Theta_{\text{symbol}}$ : symbolic coherence threshold,
- $\Theta_{\text{law}}$ : minimal legal order,
- $\Theta_{\text{econ}}$ : economic sustainability threshold,
- $\Theta_{\text{infra}}$ : infrastructural continuity threshold,
- $\Theta_{\text{memory}}$ : memory persistence threshold,
- $\Theta_{\text{meta}}$ : meta-model coherence threshold,
- $\Theta_{\text{collapse}}$ : global civilizational failure limit.

#### **O.1.172 Flows $J(K_8)$**

- $J_{\text{inst}}$ : institutional change,
- $J_{\text{gov}}$ : governance dynamics,
- $J_{\text{symbol}}$ : symbolic/ideological flows,
- $J_{\text{law}}$ : legal adaptation,
- $J_{\text{econ}}$ : economic flows (production, exchange),
- $J_{\text{infra}}$ : infrastructural evolution,

- $J_{\text{memory}}$ : updating civilizational memory,
- $J_{\text{meta}}$ : shifts in shared meta-models,
- $J_{\text{coop} \rightarrow J_{\text{stab}}}$ : stabilisation signal from  $K_7$  cooperation dynamics.

Stability condition:

$$\sigma(J(K_8)) < \sigma_{\max}(K_8).$$

### O.1.173 Cycles $C(K_8)$

$$C(K_8) = \{C_{\text{inst}}, C_{\text{gov}}, C_{\text{symbol}}, C_{\text{law}}, C_{\text{econ}}, C_{\text{infra}}, C_{\text{memory}}, C_{\text{meta}}\}.$$

Descriptions:

- $C_{\text{inst}}$ : institutional adaptation/reinforcement cycles,
- $C_{\text{gov}}$ : authority–legitimacy cycles,
- $C_{\text{symbol}}$ : symbolic renewal cycles,
- $C_{\text{law}}$ : law creation / codification cycles,
- $C_{\text{econ}}$ : expansion–contraction cycles,
- $C_{\text{infra}}$ : infrastructure maintenance cycles,
- $C_{\text{memory}}$ : memory consolidation cycles,
- $C_{\text{meta}}$ : meta-model evolution cycles.

Civilizational time is built from these cycles.

### O.1.174 Effective Time $\tau_{\text{eff}}(K_8)$

$$\tau_{\text{eff}}(K_8) = \max_j \tau_{C_j}, \quad \Pi(C_j) > \Theta_{\text{time}}.$$

$K_8$  time is slower than  $K_7$  time due to large system inertia, infrastructure and symbolic-linguistic path-dependence.

### O.1.175 Continuumness $k(K_8)$

$$k_8 = H(\Omega(K_8)) \cdot \frac{|C_{\text{max}}^{(8)}|}{|C_{\text{all}}|} \cdot \frac{|A_{\text{active}}|}{|A_{\text{max}}|} \cdot \left(1 - \frac{T_8}{\Theta_8}\right)_+ \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\max}}\right).$$

Interpretation:

- $C_{\text{max}}^{(8)}$  — largest coherent institutional-symbolic component,
- $T_8$  — structural tension,
- $\Theta_8$  — dominant civilizational threshold,
- $\sigma(J)$  — volatility of civilizational flows.

$k_8 \rightarrow 0$  when institutions or symbolic systems disintegrate.

### O.1.176 Structural Tension $T(K_8)$

Sources:

- institutional overload or conflict,
- governance failures,
- symbolic fragmentation,
- legal contradiction,
- economic crisis,
- infrastructural breakdown,
- memory conflict or historical rewriting,
- meta-model collapse (epistemic crisis).

Collapse when:

$$T(K_8) > \Theta_{\text{collapse}}.$$

### O.1.177 Energy $E(K_8)$

$$E(K_8) = E_{\text{inst}} + E_{\text{gov}} + E_{\text{symbol}} + E_{\text{law}} + E_{\text{econ}} + E_{\text{infra}} + E_{\text{memory}} + E_{\text{meta}} + E_{K_7 \rightarrow K_8}.$$

Requirements:

- institutional maintenance,
- governance enforcement,
- symbolic production and reproduction,
- legal codification,
- economic energy budget,
- infrastructural upkeep,
- memory preservation,
- meta-model update cost,
- energy of coupling with  $K_7$  social structures.

### O.1.178 Operators on $K_8$ ( $\Psi$ , $\Phi$ , $\Lambda$ , $U$ , $X$ )

- $\Psi_{8 \rightarrow 9}$ : gives rise to  $K_9$  (theoretical/epistemic continua),
- $\Phi$ : civilizational evolution,
- $\Lambda$ : consolidation of institutions into coherent systems,
- $U$ : stabilisation of symbolic and legal orders,
- $X$ : branching into distinct civilizations or cultural complexes.

### O.1.179 Processes on $K_8$

Typical:

- institution formation and decay,
- governance and political cycles,

- symbolic production (myths, science, ideology),
- lawmaking and legal codification,
- economic development and trade,
- infrastructure building and maintenance,
- historical narrative formation,
- meta-model evolution (paradigm shifts).

#### **O.1.180 Predictions for $K_8$**

1. Civilizations require symbolic coherence above  $\Theta_{\text{symbol}}$ .
2. Institutional fragmentation predicts collapse.
3. Infrastructure failure is an early-warning indicator.
4. Meta-model collapse leads to systemic reconfiguration.
5. Transition to  $K_9$  requires stable symbols, institutions and epistemic structures.

#### **O.1.181 Experiments for $K_8$**

Possible analogues:

- large-scale agent-based simulations (civilization models),
- institutional-fragmentation simulations,
- macro-economic perturbation tests,
- infrastructure-network resilience models,
- symbolic-system coherence metrics,
- historical data analysis for civilizational cycles.

#### **O.1.182 Collapse and Death of $K_8$**

Death when:

$$\Omega(K_8) = \emptyset.$$

Mechanisms:

- institutional collapse,
- symbolic/ideological disintegration,
- breakdown of governance,
- infrastructural failure,
- economic collapse,
- legal dissolution,
- loss of civilizational memory,
- meta-model breakdown.

### O.1.183 Falsifiability of $K_8$

Predictions are refuted if:

- stable civilizations exist without symbolic or institutional coherence,
- long-term stability appears without infrastructure or governance,
- historical cycles fail to appear in large-scale data,
- meta-model collapse does not correlate with systemic turbulence.

### O.1.184 Branching / Ontological Position of $K_8$

Branches:

- symbolic-heavy vs. infrastructural-heavy civilizations,
- centralized vs. decentralized governance,
- legalistic vs. customary orders,
- innovation-driven vs. tradition-driven systems.

Ontological position:

$$K_7 \rightarrow K_8 \rightarrow K_9.$$

### O.1.185 Relation to M-spaces

$M_8$  constrains:

- possible institutional forms,
- range of symbolic and meta-model structures,
- governance architectures,
- infrastructural topologies,
- economic-exchange capacities,
- admissible historical trajectories and memory structures,
- coupling to  $M_7$  and  $M_9$ .

Compatibility with  $M_8$  determines whether a civilization can exist.

### O.1.186 $K_9$ Overview

$K_9$  is the level of *theoretical continua*: large-scale, explicit, self-referential conceptual systems that structure, predict and integrate knowledge. Examples include scientific paradigms, formal theories, logical systems, philosophical frameworks and mathematical structures.

Distinctive features:

- explicit symbolic representation,
- axiomatic or rule-based organization,
- high-level abstraction and model construction,
- internal consistency constraints,
- meta-level feedback from  $K_6$  cognition and  $K_8$  civilizations,
- long-term paradigmatic cycles (Kuhn-type structure),
- cross-theory coupling, unification and competition.



$K_9$  is not reducible to  $K_8$  culture or  $K_6$  cognition; it has its own dynamical space, thresholds and collapse modes.

### O.1.187 State Space $\Omega(K_9)$

$$\Omega(K_9) = \{A_{\text{axioms}}, R_{\text{rules}}, M_{\text{models}}, P_{\text{pred}}, V_{\text{valid}}, S_{\text{syntax}}, D_{\text{deriv}}, H_{\text{theory}}, \mu_t^{(9)}\}.$$

Components:

- $A_{\text{axioms}}$  — axioms, premises, fundamental principles,
- $R_{\text{rules}}$  — inference and transformation rules,
- $M_{\text{models}}$  — internal models, representations,
- $P_{\text{pred}}$  — prediction set,
- $V_{\text{valid}}$  — validation/verification structure,
- $S_{\text{syntax}}$  — symbolic and formal language,
- $D_{\text{deriv}}$  — derivational system,
- $H_{\text{theory}}$  — historical lineage of the theory,
- $\mu_t^{(9)}$  — time-evolving distribution of theoretical states.

### O.1.188 Boundary $\partial\Omega(K_9)$

Boundary conditions include:

- logical inconsistency,
- contradiction in axioms,
- collapse of derivational system,
- loss of predictive power,
- incompatibility with  $M_9$  allowed structures,
- inability to stabilize symbolic syntax,
- catastrophic loss of validation mechanisms,
- failure to integrate empirical constraints from lower levels.

Crossing  $\partial\Omega(K_9)$  destroys the theory as a continuum.

### O.1.189 Axes $A(K_9)$

$$A(K_9) = \{A_{\text{axiom}}, A_{\text{syntax}}, A_{\text{model}}, A_{\text{pred}}, A_{\text{valid}}, A_{\text{meta}}\}.$$

Interpretations:

- $A_{\text{axiom}}$ : variation in fundamental assumptions,
- $A_{\text{syntax}}$ : symbolic and linguistic structures,
- $A_{\text{model}}$ : model-building dimension,
- $A_{\text{pred}}$ : predictive dimension,
- $A_{\text{valid}}$ : empirical and logical validation,
- $A_{\text{meta}}$ : meta-theoretical structure (epistemology, methodology).

### O.1.190 Potentials $P(K_9)$

$$P(K_9) = \{P_{\text{consistency}}, P_{\text{coherence}}, P_{\text{expressive}}, P_{\text{predictive}}, P_{\text{integrative}}, P_{\text{meta}}\}.$$

Potentials encode:

- logical consistency,
- internal coherence,
- expressive power of syntax,
- predictive accuracy,
- integrative capacity (unification potential),
- meta-level robustness and methodological rigor.

### O.1.191 Thresholds $\Theta(K_9)$

$$\Theta(K_9) = \{\Theta_{\text{consistency}}, \Theta_{\text{coherence}}, \Theta_{\text{predictive}}, \Theta_{\text{valid}}, \Theta_{\text{meta}}, \Theta_{\text{collapse}}\}.$$

Descriptions:

- $\Theta_{\text{consistency}}$ : minimal logical consistency,
- $\Theta_{\text{coherence}}$ : structure–model coherence threshold,
- $\Theta_{\text{predictive}}$ : minimal predictive power,
- $\Theta_{\text{valid}}$ : validation threshold,
- $\Theta_{\text{meta}}$ : meta-theoretical coherence threshold,
- $\Theta_{\text{collapse}}$ : unrestricted-unification failure.

### O.1.192 Flows $J(K_9)$

- $J_{\text{axiom}}$ : axiom evolution,
- $J_{\text{syntax}}$ : syntax development,
- $J_{\text{model}}$ : model-building dynamics,
- $J_{\text{pred}}$ : prediction refinement,
- $J_{\text{valid}}$ : validation/verification flow,
- $J_{\text{meta}}$ : meta-theoretical shifts (paradigm changes),
- $J_{8 \rightarrow 9}$ : inflow of structures from  $K_8$  civilizational context.

Stability:

$$\sigma(J(K_9)) < \sigma_{\max}(K_9).$$

### O.1.193 Cycles $C(K_9)$

$$C(K_9) = \{C_{\text{axiom}}, C_{\text{syntax}}, C_{\text{model}}, C_{\text{pred}}, C_{\text{valid}}, C_{\text{meta}}\}.$$

Interpretation:

- $C_{\text{axiom}}$ : axiom revision cycles,
- $C_{\text{syntax}}$ : symbolic evolution cycles,
- $C_{\text{model}}$ : sequence of model constructions,

- $C_{\text{pred}}$ : prediction–feedback cycles,
- $C_{\text{valid}}$ : empirical or logical validation cycles,
- $C_{\text{meta}}$ : paradigm-level cycles (Kuhn-type).

#### O.1.194 Effective Time $\tau_{\text{eff}}(K_9)$

Time emerges from the slowest non-degradable theoretical cycle:

$$\tau_{\text{eff}}(K_9) = \max_j \tau_{C_j}, \quad \Pi(C_j) > \Theta_{\text{time}}.$$

Theoretical time is slower than  $K_8$  cultural time due to symbolic inertia.

#### O.1.195 Continuumness $k(K_9)$

$$k_9 = H(\Omega(K_9)) \cdot \frac{|C_{\text{max}}^{(9)}|}{|C_{\text{all}}|} \cdot \frac{|A_{\text{active}}|}{|A_{\text{max}}|} \cdot \left(1 - \frac{T_9}{\Theta_9}\right)_+ \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\text{max}}}\right).$$

Interpretation:

- $C_{\text{max}}^{(9)}$  — largest coherent theoretical subsystem,
- $T_9$  — structural tension,
- $\Theta_9$  — dominant theoretical threshold.

$k_9 \rightarrow 0$  when the theoretical system becomes incoherent or inconsistent.

#### O.1.196 Structural Tension $T(K_9)$

Sources:

- contradiction between axioms,
- incompatible models,
- predictive failure,
- logical or syntactic inconsistency,
- validation conflicts,
- paradigm instability.

Collapse:

$$T(K_9) > \Theta_{\text{collapse}}.$$

#### O.1.197 Energy $E(K_9)$

$$E(K_9) = E_{\text{axiom}} + E_{\text{syntax}} + E_{\text{model}} + E_{\text{pred}} + E_{\text{valid}} + E_{\text{meta}} + E_{8 \rightarrow 9}.$$

Energy expenditures:

- maintaining axioms and derivational systems,
- developing syntax and formal language,
- constructing models,
- making predictions,
- performing validation,
- sustaining meta-theoretical reflection,
- integrating civilizational data and constraints from  $K_8$ .

### **O.1.198 Operators on $K_9$ ( $\Psi, \Phi, \Lambda, U, X$ )**

- $\Psi_{9 \rightarrow 10}$ : generates  $K_{10}$  (metatheoretical continua),
- $\Phi$ : internal evolution of theoretical frameworks,
- $\Lambda$ : consolidation of axioms and models,
- $U$ : stabilisation of theory–world correspondence,
- $X$ : branching of theories into competing paradigms.

### **O.1.199 Processes on $K_9$**

- axiom selection and revision,
- syntax development,
- model building,
- prediction and inference,
- validation (logical or empirical),
- paradigm formation, competition and replacement,
- cross-theory unification attempts.

### **O.1.200 Predictions for $K_9$**

1. All stable theoretical continua satisfy the consistency threshold.
2. Predictive theories must maintain  $P_{\text{pred}} > \Theta_{\text{predictive}}$ .
3. Paradigm shifts correspond to crossing  $\Theta_{\text{meta}}$ .
4. Unifying theories require high integrative potential.
5. Transition to  $K_{10}$  requires meta-level coherence.

### **O.1.201 Experiments for $K_9$**

Possible proxies:

- formal-logic simulations,
- model-theoretic dynamics,
- validation feedback loop simulations,
- historical analysis of scientific revolutions,
- syntactic-evolution models,
- paradigm competition simulations.

### **O.1.202 Collapse and Death of $K_9$**

Death when:

$$\Omega(K_9) = \emptyset.$$

Mechanisms:

- logical inconsistency,

- complete predictive failure,
- collapse of symbolic or syntactic base,
- invalidation of all core axioms,
- loss of empirical or logical grounding,
- inability to sustain derivation cycles,
- paradigm fragmentation beyond  $\partial\Omega$ .

### O.1.203 Falsifiability of $K_9$

To falsify:

- existence of stable theories without consistency,
- predictive systems operating without validation,
- paradigm shifts unrelated to meta-level thresholds,
- theoretical continua without coherent axioms or syntax.

### O.1.204 Branching / Ontological Position of $K_9$

Branches:

- formal vs. empirical theories,
- ontological vs. instrumentalist frameworks,
- unifying vs. pluralistic paradigms,
- axiomatic vs. computational theories.

Ontological position:

$$K_8 \rightarrow K_9 \rightarrow K_{10}.$$

### O.1.205 Relation to M-spaces

$M_9$  defines:

- allowable syntax and symbolic rules,
- ranges of admissible axioms,
- boundaries of consistency,
- model-class capacities,
- validation structures,
- permissible meta-theoretical operations,
- embedding of  $K_9$  within  $M_8$  civilizational and  $M_{10}$  metatheoretical constraints.

Compatibility with  $M_9$  determines whether a theoretical system can exist.

### O.1.206 $K_{10}$ Overview

$K_{10}$  is the level of *metatheoretical continua*: systems capable of representing, evaluating and transforming whole classes of theories ( $K_9$ ) including themselves. These continua encode:

- metatheoretical axioms,
- meta-rules for theory comparison and evaluation,

- meta-models describing structures of theories,
- universal operators governing theoretical evolution,
- recursive self-reference and structural fixed points,
- global coherence constraints across multiple theories.

Distinctive features:

- full self-referential capacity,
- global integrative power across lower levels,
- meta-consistency constraints,
- ability to detect contradictions in  $K_9$  theories,
- universalization of thresholds and operators,
- preparation of the space for  $K_{11}$  (structural recursion) via  $\Psi_{10 \rightarrow 11}$ .

#### O.1.207 State Space $\Omega(K_{10})$

$$\Omega(K_{10}) = \{A_{\text{meta}}, R_{\text{meta}}, M_{\text{meta}}, C_{\text{cross}}, E_{\text{eval}}, S_{\text{self}}, \mu_t^{(10)}\}.$$

Components:

- $A_{\text{meta}}$  — meta-axioms (statements about theories),
- $R_{\text{meta}}$  — meta-rules for theory modification,
- $M_{\text{meta}}$  — models of theoretical spaces,
- $C_{\text{cross}}$  — cross-theory comparison structures,
- $E_{\text{eval}}$  — evaluation operators,
- $S_{\text{self}}$  — self-representational state,
- $\mu_t^{(10)}$  — distribution over metatheoretical configurations.

#### O.1.208 Boundary $\partial\Omega(K_{10})$

Boundary conditions occur when:

- self-reference becomes paradoxical,
- meta-axioms contradict each other,
- evaluation operators lose definability,
- meta-rules diverge (no fixed point),
- global coherence across  $K_9$  theories collapses,
- metatheory becomes incompatible with  $M_{10}$  constraints,
- recursion ( $K_{10} \rightarrow K_{10}$ ) becomes ill-formed.

Crossing  $\partial\Omega(K_{10})$  destroys metatheory as a continuum.

**O.1.209 Axes  $A(K_{10})$** 

$$A(K_{10}) = \{A_{\text{metaaxiom}}, A_{\text{metarule}}, A_{\text{metamodel}}, A_{\text{evaluation}}, A_{\text{coherence}}, A_{\text{self}}\}.$$

Interpretation:

- $A_{\text{metaaxiom}}$ : variation of meta-level premises,
- $A_{\text{metarule}}$ : rules that transform whole theories,
- $A_{\text{metamodel}}$ : structures modeling  $K_9$  landscapes,
- $A_{\text{evaluation}}$ : global assessment axes,
- $A_{\text{coherence}}$ : coherence among theories,
- $A_{\text{self}}$ : depth of self-representation and recursion.

**O.1.210 Potentials  $P(K_{10})$** 

$$P(K_{10}) = \{P_{\text{metaconsistency}}, P_{\text{global}}, P_{\text{unification}}, P_{\text{recursion}}, P_{\text{reflection}}, P_{\text{selffix}}\}.$$

Potentials encode:

- meta-consistency (absence of contradictions across theories),
- global integrative potential (unifying frameworks),
- unification power of meta-structures,
- recursive stability,
- reflective depth,
- existence of self-fixing points ( $K_{10}$  describing itself without contradiction).

**O.1.211 Thresholds  $\Theta(K_{10})$** 

$$\Theta(K_{10}) = \{\Theta_{\text{metaconsistency}}, \Theta_{\text{coherence}}, \Theta_{\text{selfref}}, \Theta_{\text{recursion}}, \Theta_{\text{unification}}, \Theta_{\text{collapse}}\}.$$

Descriptions:

- $\Theta_{\text{metaconsistency}}$ : minimal consistency across theories,
- $\Theta_{\text{coherence}}$ : global coherence threshold,
- $\Theta_{\text{selfref}}$ : threshold for safe self-reference,
- $\Theta_{\text{recursion}}$ : stability under recursive application,
- $\Theta_{\text{unification}}$ : threshold for unifying divergent theories,
- $\Theta_{\text{collapse}}$ : metatheoretical failure point.

**O.1.212 Flows  $J(K_{10})$** 

- $J_{\text{metaaxiom}}$ : evolution of meta-axioms,
- $J_{\text{metarule}}$ : transformation of meta-rules,
- $J_{\text{metamodel}}$ : dynamics of meta-models,
- $J_{\text{cross}}$ : flow of cross-theory comparisons,
- $J_{\text{eval}}$ : evolution of evaluation operators,
- $J_{\text{self}}$ : dynamics of self-representation,

- $J_{9 \rightarrow 10}$ : influx from  $K_9$  theoretical continua.

Stability condition:

$$\sigma(J(K_{10})) < \sigma_{\max}(K_{10}).$$

#### O.1.213 Cycles $C(K_{10})$

$$C(K_{10}) = \{C_{\text{metaaxiom}}, C_{\text{metarule}}, C_{\text{metamodel}}, C_{\text{evaluation}}, C_{\text{coherence}}, C_{\text{self}}\}.$$

Interpretation:

- cycles of meta-axiom revision,
- cycles of meta-rule transformation,
- cycles of meta-model reconstruction,
- evaluation cycles,
- global coherence cycles,
- self-referential cycles (test–adjust–retreat–fix).

#### O.1.214 Effective Time $\tau_{\text{eff}}(K_{10})$

Metatheoretical time emerges from the slowest stable meta-cycle:

$$\tau_{\text{eff}}(K_{10}) = \max_j \tau_{C_j}, \quad \Pi(C_j) > \Theta_{\text{time}}.$$

Metatheoretical time is slower than theoretical ( $K_9$ ) time because recursive structures change only under strong tension.

#### O.1.215 Continuumness $k(K_{10})$

$$k_{10} = H(\Omega(K_{10})) \cdot \frac{|C_{\max}^{(10)}|}{|C_{\text{all}}|} \cdot \frac{|A_{\text{active}}|}{|A_{\max}|} \cdot \left(1 - \frac{T_{10}}{\Theta_{10}}\right)_+ \cdot \left(1 - \frac{\sigma(J)}{\sigma_{\max}}\right).$$

Interpretation:

- global coherence across meta-cycles,
- stability of self-representation,
- meta-consistency levels,
- recursive integrity.

#### O.1.216 Structural Tension $T(K_{10})$

Sources:

- contradictions between meta-axioms,
- incompatible meta-rules,
- failures in recursive definitions,
- loss of global coherence across theories,
- tension between unification and plurality,
- paradoxes of self-reference (Russell-type, Gödel-type).

Collapse occurs when:

$$T(K_{10}) > \Theta_{\text{collapse}}.$$



### O.1.217 Energy $E(K_{10})$

$$E(K_{10}) = E_{\text{metaaxiom}} + E_{\text{metarule}} + E_{\text{metamodel}} + E_{\text{eval}} + E_{\text{coherence}} + E_{\text{self}} + E_{9 \rightarrow 10}.$$

Energy expenditures:

- maintaining meta-axioms,
- maintaining recursive structures,
- running self-evaluation cycles,
- enforcing global coherence,
- sustaining cross-theory comparisons,
- integrating theoretical structures from  $K_9$ .

### O.1.218 Operators on $K_{10}$ ( $\Psi, \Phi, \Lambda, U, X$ )

- $\Psi_{10 \rightarrow 11}$  — birth of  $K_{11}$ : structural recursion as a continuum,
- $\Phi$  — evolution of metatheories,
- $\Lambda$  — meta-stabilization (closing meta-loops),
- $U$  — unification of disparate  $K_9$  theories,
- $X$  — branching of metatheories (methodological pluralism vs. unity).

### O.1.219 Processes on $K_{10}$

- meta-axiomatisation,
- reconstruction of theoretical landscapes,
- unification and pluralistic decomposition,
- recursive self-evaluation,
- paradigm-meta-paradigm transitions,
- consistency enforcement across theoretical classes.

### O.1.220 Predictions for $K_{10}$

1. Stable metatheories must satisfy  $\Theta_{\text{metaconsistency}}$ .
2. Meta-unification is possible only when  $P_{\text{global}}$  is high.
3. Self-reference stabilizes only at fixed points of  $\Phi$ .
4. Recursive collapse is preceded by rising  $T_{10}$ .
5. Transition to  $K_{11}$  requires meta-stability and recursive definability.

### O.1.221 Experiments for $K_{10}$

Possible proxies:

- simulations of meta-rule dynamics,
- recursive consistency-check algorithms,
- global-coherence analysis of multiple theories,

- simulations of self-referential systems,
- fixed-point detection in evolving metatheories.

#### **O.1.222 Collapse and Death of $K_{10}$**

Death when:

$$\Omega(K_{10}) = \emptyset.$$

Mechanisms:

- paradoxical self-reference,
- incoherent meta-rule sets,
- collapse of global coherence,
- inability to evaluate or compare theories,
- divergence under recursion,
- incompatibility with  $M_{10}$ .

#### **O.1.223 Falsifiability of $K_{10}$**

To falsify:

- existence of stable metatheories violating meta-consistency,
- recursive systems stable without fixed points,
- global coherence emerging below  $\Theta_{\text{coherence}}$ ,
- transitions to  $K_{11}$  without satisfying recursive thresholds.

#### **O.1.224 Branching / Ontological Position of $K_{10}$**

Branches:

- unifying vs. pluralistic metatheories,
- axiomatic vs. algorithmic metatheories,
- static vs. dynamic recursion frameworks,
- single-level vs. multi-level meta-systems.

Ontological chain:

$$K_9 \rightarrow K_{10} \rightarrow K_{11}.$$

#### **O.1.225 Relation to M-spaces**

$M_{10}$  defines:

- allowable meta-axioms,
- recursion constraints,
- global coherence conditions,
- structural fixed-point conditions,
- bandwidth of self-reference,
- compatibility with  $M_9$  and  $M_{11}$ ,
- meta-level thresholds and universal constraints.

Compatibility with  $M_{10}$  determines whether metatheoretical continua can exist.

### O.1.226 $K_{11}$ Overview

$K_{11}$  is the level of *recursive structural continua*: systems whose states include entire classes of metatheories ( $K_{10}$ ) together with operators acting recursively on them. Unlike  $K_{10}$ , which evaluates, modifies and unifies theories,  $K_{11}$  organises *hierarchies of metastructures* and provides:

- recursive embeddings  $K_{10} \hookrightarrow K_{10}$ ,
- operators acting on families of metatheories,
- meta-coherence across recursive layers,
- fixed-point conditions for self-referential hierarchies,
- branching of structural possibilities (Axiom 21),
- the structural substrate for  $K_{12}$  (limit continua).

$K_{11}$  represents the first level where the *entire tower*  $K_0 \rightarrow K_{10}$  can be treated as a manipulable object.

### O.1.227 State Space $\Omega(K_{11})$

$$\Omega(K_{11}) = \{\mathcal{M}, \mathcal{R}, \mathcal{H}, \mathcal{B}, \mathcal{F}, \mu_t^{(11)}\}.$$

Components:

- $\mathcal{M}$  — sets of metatheories (structured  $K_{10}$ -clusters),
- $\mathcal{R}$  — recursive rules acting on  $\mathcal{M}$ ,
- $\mathcal{H}$  — hierarchical embeddings and projections,
- $\mathcal{B}$  — branching structures of ontological possibilities,
- $\mathcal{F}$  — structural fixed-point sets,
- $\mu_t^{(11)}$  — dynamical measure over recursive configurations.

$K_{11}$  therefore encodes *spaces of spaces* — meta-hierarchies.

### O.1.228 Boundary $\partial\Omega(K_{11})$

Boundaries arise when:

- recursion becomes ill-founded (non-wellfounded chains),
- hierarchical embeddings contradict each other,
- branching leads to global inconsistency (violation of Axiom 21),
- fixed points fail to exist under recursion,
- meta-coherence cannot be preserved across levels,
- incompatibility with  $M_{11}$  constraints appears,
- recursion depth or complexity exceeds the admissible bounds of  $M_{11}$ .

Crossing  $\partial\Omega(K_{11})$  destroys the recursive continuum.

### O.1.229 Axes $A(K_{11})$

$$A(K_{11}) = \{A_{\text{recursive}}, A_{\text{hier}}, A_{\text{branch}}, A_{\text{fix}}, A_{\text{metaembed}}, A_{\text{coh}}\}.$$

Interpretation:

- $A_{\text{recursive}}$  — degree of recursion,

- $A_{\text{hier}}$  — depth and structure of hierarchies,
- $A_{\text{branch}}$  — ontological branching axis (Axiom 21),
- $A_{\text{fix}}$  — fixed-point formation axis,
- $A_{\text{metaembed}}$  — meta-embedding transformations,
- $A_{\text{coh}}$  — coherence across recursive strata.

#### O.1.230 Potentials $P(K_{11})$

$$P(K_{11}) = \{P_{\text{rec}}, P_{\text{hier}}, P_{\text{branch}}, P_{\text{fix}}, P_{\text{coh}}, P_{\text{metaembed}}\}.$$

Descriptions:

- recursive potential — capacity for repeated structural application,
- hierarchical potential — ability to support deep multilevel embeddings,
- branching potential — richness of ontological alternatives,
- fixed-point potential — ability to stabilise recursive loops,
- coherence potential — maintaining structural consistency across layers,
- embedding potential — ability to integrate or compare metatheories.

#### O.1.231 Thresholds $\Theta(K_{11})$

$$\Theta(K_{11}) = \{\Theta_{\text{rec}}, \Theta_{\text{branch}}, \Theta_{\text{coh}}, \Theta_{\text{fix}}, \Theta_{\text{hier}}, \Theta_{\text{collapse}}\}.$$

Characterisation:

- $\Theta_{\text{rec}}$  — minimum condition for safe recursion,
- $\Theta_{\text{branch}}$  — branching threshold (Axiom 21.4),
- $\Theta_{\text{coh}}$  — inter-level coherence threshold,
- $\Theta_{\text{fix}}$  — fixed-point existence threshold,
- $\Theta_{\text{hier}}$  — minimal hierarchical stability,
- $\Theta_{\text{collapse}}$  — limit of recursive viability.

#### O.1.232 Flows $J(K_{11})$

- $J_{\text{rec}}$ : evolution of recursive structures,
- $J_{\text{hier}}$ : flow along hierarchical embeddings,
- $J_{\text{branch}}$ : dynamics of branching topologies,
- $J_{\text{fix}}$ : formation and destruction of fixed points,
- $J_{\text{embed}}$ : embedding and projection flows,
- $J_{\text{coh}}$ : coherence-preserving flows,
- $J_{10 \rightarrow 11}$ : influx of metatheoretical structures from  $K_{10}$ .

Stability condition:

$$\sigma(J(K_{11})) < \sigma_{\text{max}}(K_{11}).$$

**O.1.233 Cycles  $C(K_{11})$** 

$$C(K_{11}) = \{C_{\text{rec}}, C_{\text{hier}}, C_{\text{branch}}, C_{\text{fix}}, C_{\text{embed}}, C_{\text{coh}}\}.$$

Interpretation:

- recursive refinement cycles,
- cycles of hierarchical reconstruction,
- branching cycles of structural differentiation,
- fixed-point search cycles,
- embedding–projection cycles,
- coherence restoration cycles.

**O.1.234 Effective Time  $\tau_{\text{eff}}(K_{11})$** 

$$\tau_{\text{eff}}(K_{11}) = \max_j \tau_{C_j}, \quad \Pi(C_j) > \Theta_{\text{time}}.$$

Recursive time differs from metatheoretical time:

- it is dominated by the slowest stable recursive process,
- deeper hierarchies produce slower timescales,
- fixed points create long-lived quasi-static plateaus.

**O.1.235 Continuumness  $k(K_{11})$** 

$$k_{11} = H(\Omega(K_{11})) \cdot \frac{|C_{\text{max}}^{(11)}|}{|C_{\text{all}}|} \cdot \frac{|A_{\text{active}}|}{|A_{\text{max}}|} \cdot \frac{P_{\text{fix}}}{P_{\text{fix,max}}} \cdot \left(1 - \frac{T_{11}}{\Theta_{11}}\right)_+.$$

Interpretation:

- recursive stability,
- existence of fixed points,
- branching without collapse,
- inter-level coherence.

**O.1.236 Structural Tension  $T(K_{11})$** 

Sources:

- incompatible recursive embeddings,
- contradictions between branches,
- failure of fixed-point formation,
- excessive branching (Axiom 21.6),
- collapse of meta-coherence across hierarchies.

If:

$$T(K_{11}) > \Theta_{\text{collapse}},$$

then the recursive continuum collapses.

### O.1.237 Energy $E(K_{11})$

$$E(K_{11}) = E_{\text{rec}} + E_{\text{hier}} + E_{\text{branch}} + E_{\text{fix}} + E_{\text{coh}} + E_{10 \rightarrow 11}.$$

Energy is required for:

- maintaining deep recursion,
- preserving branching structures,
- stabilising fixed points,
- enforcing coherence,
- integrating  $K_{10}$  metatheories.

### O.1.238 Operators on $K_{11}$ ( $\Psi, \Phi, \Lambda, U, X$ )

- $\Psi_{11 \rightarrow 12}$  — formation of limit continuum  $K_{12}$ ,
- $\Phi$  — evolution in recursive space,
- $\Lambda$  — structural closure of recursion loops,
- $U$  — universal embedding across recursive hierarchies,
- $X$  — ontological branching operator (Axiom 21).

### O.1.239 Processes on $K_{11}$

- recursive reconstruction of meta-structures,
- formation of hierarchical strata,
- ontological branching (controlled and uncontrolled),
- fixed-point search and stabilization,
- recursive integration of  $K_{10}$  structures,
- cross-level coherence enforcement.

### O.1.240 Predictions for $K_{11}$

1. Stable recursion requires  $P_{\text{fix}} > 0$  and  $\Theta_{\text{fix}}$  satisfied.
2. Excessive branching leads to  $T(K_{11}) \uparrow$  and eventual collapse.
3. Deep hierarchies produce longer timescales and slow dynamics.
4. Transition to  $K_{12}$  requires existence of a global recursive fixed point.
5. Inter-level incoherence predicts collapse before  $K_{12}$  formation.

### O.1.241 Experiments for $K_{11}$

Possible proxies:

- simulations of recursive operators,
- fixed-point detection algorithms for recursive maps,
- agent-based recursive hierarchy models,
- branching-process simulations,
- meta-meta-consistency solvers.

#### O.1.242 Collapse and Death of $K_{11}$

Death when:

$$\Omega(K_{11}) = \emptyset.$$

Mechanisms:

- recursive divergence,
- loss of fixed points,
- catastrophic branching,
- collapse of hierarchical coherence,
- incompatibility with  $M_{11}$ .

#### O.1.243 Falsifiability of $K_{11}$

The level is falsified if:

- stable recursive hierarchies exist without fixed points,
- branching occurs below  $\Theta_{\text{branch}}$ ,
- cross-level coherence arises without meeting  $\Theta_{\text{coh}}$ ,
- transitions to  $K_{12}$  occur without recursive stability,
- recursion operates with well-foundedness violations that do not collapse.

#### O.1.244 Branching / Ontological Position of $K_{11}$

Branches:

- shallow vs. deep recursion regimes,
- narrow vs. wide branching structures,
- coherent vs. incoherent hierarchical embeddings,
- fixed-point-rich vs. fixed-point-poor regimes.

Position in ontological chain:

$$K_{10} \rightarrow K_{11} \rightarrow K_{12}.$$

#### O.1.245 Relation to M-spaces

$M_{11}$  specifies:

- recursion admissibility,
- global coherence bounds,
- branching limits,
- hierarchical embedding capacities,
- fixed-point existence constraints,
- compatibility with  $M_{10}$  and  $M_{12}$ .

A continuum  $K_{11}$  exists iff its structures satisfy the admissibility region of  $M_{11}$ .

### O.1.246 $K_{12}$ Overview

$K_{12}$  is the *limit continuum*: the structural fixed point of the recursive meta-hierarchy developed in  $K_{11}$ . It is the first level where:

- recursive meta-structures stabilise,
- branching (Axiom 21) reaches closure,
- a global structural fixed point exists,
- all operators ( $\Psi, \Phi, \Lambda, U, X$ ) act as symmetries,
- the entire ladder  $K_0 \dots K_{11}$  becomes representable inside one invariant continuum.

$K_{12}$  is therefore not merely “the next level”, but the structural limit of the K-tower in Core v2.6.

### O.1.247 State Space $\Omega(K_{12})$

$$\Omega(K_{12}) = \{\Omega^*, A^*, P^*, \Theta^*, J^*, C^*, \mu_t^{(12)}\},$$

where:

- $\Omega^*$  is the limit state-space containing all admissible embeddings of  $K_0 \rightarrow K_{11}$ ,
- $A^*$  is the global axis set induced by the closure of all Axes from previous levels,
- $P^*$  are potentials that survive recursive convergence,
- $\Theta^*$  are stable thresholds preserved under recursion,
- $J^*$  are flows that remain invariant under all structural operators,
- $C^*$  is the fixed family of cycles that defines the time of  $K_{12}$ ,
- $\mu_t^{(12)}$  is the measure over limit configurations.

All components satisfy global invariance:

$$X^* = \mathcal{R}(X^*), \quad \forall X \in \{\Omega, A, P, \Theta, J, C\}.$$

### O.1.248 Boundary $\partial\Omega(K_{12})$

The boundary consists of:

- configurations where recursion does not converge,
- points where branching produces incompatible global topologies,
- states violating limit invariance under operators,
- divergence of fixed-point cycles,
- non-admissibility in  $M_{12}$ .

Crossing  $\partial\Omega(K_{12})$  destroys the limit structure and reverts the system to a failing  $K_{11}$ -regime.

### O.1.249 Axes $A(K_{12})$

$$A(K_{12}) = \overline{\bigcup_{i=0}^{11} A(K_i)} \quad \text{subject to recursive closure.}$$

This axis set is:

- closed under recursion from  $K_{11}$ ,



- stable under all operators  $(\Psi, \Phi, \Lambda, U, X)$ ,
  - minimal among axis sets satisfying global invariance.
- Each axis becomes a *symmetry direction* in  $K_{12}$ .

#### O.1.250 Potentials $P(K_{12})$

$$P(K_{12}) = \lim_{n \rightarrow \infty} P^{(n)}(K_{11}),$$

i.e. the potentials that remain finite, coherent and closed under recursion:

- $P_{\text{fix}}^*$  — fixed-point-supporting potential,
- $P_{\text{coh}}^*$  — global coherence potential,
- $P_{\text{branch}}^*$  — stabilised branching potential,
- $P_{\text{embed}}^*$  — universal embedding potential.

#### O.1.251 Thresholds $\Theta(K_{12})$

$$\Theta(K_{12}) = \lim_{n \rightarrow \infty} \Theta^{(n)}(K_{11}),$$

giving:

- $\Theta_{\text{fix}}^*$  — fixed-point existence threshold,
- $\Theta_{\text{coh}}^*$  — global coherence threshold,
- $\Theta_{\text{branch}}^*$  — branching-closure threshold,
- $\Theta_{\text{inv}}^*$  — invariance-threshold for operators,
- $\Theta_{\text{collapse}}^*$  — structural collapse limit.

All thresholds become *globally stable constants* for the limit continuum.

#### O.1.252 Flows $J(K_{12})$

$$J(K_{12}) = \{J^* \mid \Phi(J^*) = J^*\}.$$

Thus every flow must be:

- invariant under recursive evolution,
- compatible with limit thresholds,
- globally coherent across projections and embeddings,
- fixed under operator action.

This is the only level where *all flows are symmetry flows*.

#### O.1.253 Cycles $C(K_{12})$

$$C(K_{12}) = \{C^* \mid \Pi(C^*) > \Theta_{\text{time}}^*, \Lambda(C^*) = C^*\}.$$

These are the cycles that:

- persist through all recursive refinements,
- define the emergent time of  $K_{12}$ ,
- remain invariant under branching and operator action,
- serve as the structural backbone of the continuum.

**O.1.254 Effective Time  $\tau_{\text{eff}}(K_{12})$** 

Time is determined by invariant cycles:

$$\tau_{\text{eff}}(K_{12}) = \max_j \tau_{C_j^*}.$$

Properties:

- all temporal directions are symmetries,
- no new time directions can arise,
- time is globally coherent and invariant under recursion.

**O.1.255 Continuumness  $k(K_{12})$** 

$$k_{12} = H(\Omega(K_{12})) \cdot \frac{|C^*|}{|C_{\text{max}}|} \cdot \frac{|A^*|}{|A_{\text{max}}|} \cdot \frac{P_{\text{fix}}^*}{P_{\text{fix,max}}^*} \cdot \left(1 - \frac{T_{12}}{\Theta_{\text{collapse}}^*}\right)_+.$$

$k_{12}$  attains the maximum possible value among all  $K$ -levels:

$$k_{12} = \max_{i=0\dots 12} k(K_i).$$

**O.1.256 Structural Tension  $T(K_{12})$** 

Sources:

- unresolved branching incompatibilities,
- operator-invariance failures,
- instability of fixed points (rare but possible),
- projection–embedding inconsistencies,
- violations of  $M_{12}$  bounds.

Collapse occurs if:

$$T(K_{12}) > \Theta_{\text{collapse}}^*.$$

**O.1.257 Energy  $E(K_{12})$** 

$$E(K_{12}) = E_{\text{fix}}^* + E_{\text{coh}}^* + E_{\text{branch}}^* + E_{\text{embed}}^* + E_{\text{sym}}.$$

Interpretation:

- minimal energy required to preserve fixed-point structure,
- energy of global coherence maintenance,
- energy stabilising branching closure,
- energy supporting universal embeddings,
- symmetry energy of operator invariance.

$K_{12}$  is the most energy-stable admissible continuum.

### O.1.258 Operators on $K_{12}$ ( $\Psi, \Phi, \Lambda, U, X$ )

At the limit level:

- $\Psi$  — becomes identity on admissible states,
- $\Phi$  — becomes the global invariance flow,
- $\Lambda$  — closure of all limit cycles,
- $U$  — universal embedding operator,
- $X$  — stabilised branching operator.

Operator algebra becomes Abelian up to equivalence classes:

$$[\Psi, \Phi] = [\Phi, \Lambda] = \dots = 0.$$

### O.1.259 Processes on $K_{12}$

- convergence of recursive structures,
- closure of branching topologies,
- formation of global fixed points,
- collapse of non-admissible hierarchies,
- emergence of full symmetry under operators,
- stabilisation of time and cycle structure.

### O.1.260 Predictions for $K_{12}$

1. Any admissible recursive meta-hierarchy eventually converges to structures representable in  $K_{12}$ .
2. No new axes or potentials can emerge beyond this level.
3. All operator actions reduce to global symmetries.
4. Branching becomes topologically closed; no new branches arise.
5. Failure to achieve a global fixed point forces collapse back to  $K_{11}$ .

### O.1.261 Experiments for $K_{12}$

Only indirect experiments exist:

- simulations of recursive convergence,
- fixed-point computation in high-order meta-spaces,
- invariance tests under operator algebra,
- stability tests for branching closures,
- modelling admissibility under  $M_{12}$ .

### O.1.262 Collapse and Death of $K_{12}$

Death if:

$$\Omega(K_{12}) = \emptyset.$$

Mechanisms:

- recursive divergence at deep levels,
- operator-invariance violations,
- failure of branching closure,
- global incoherence of fixed points,
- incompatibility with the meta-space  $M_{12}$ .

### O.1.263 Falsifiability of $K_{12}$

Falsified if:

- a meta-hierarchy converges without forming invariant fixed points,
- branching remains unclosed yet stability is preserved,
- new axes or potentials emerge beyond closure,
- operator algebra fails to become symmetric,
- admissible recursion produces non-convergent hierarchies.

### O.1.264 Branching / Ontological Position of $K_{12}$

Branching structure:

- all branches close,
- no new branches form,
- the topology becomes globally compact and invariant.

Ontological position:

$$K_{11} \longrightarrow K_{12} \quad (\text{limit continuum}).$$

Beyond  $K_{12}$  the theory does not posit new levels; the limit continuum contains all admissible structural possibilities.

### O.1.265 Relation to M-spaces

$M_{12}$  defines:

- full admissibility region for limit structures,
- invariance requirements for operators,
- global coherence constraints,
- branching closure bounds,
- fixed-point existence conditions,
- compatibility with all  $M_0 \dots M_{11}$ .

A continuum exists at level  $K_{12}$  iff its entire structure lies within the admissible region of  $M_{12}$ .

## O.2 MSpaces

The family of meta-spaces  $M_0, M_1, \dots, M_{12}$  provides the *admissibility structure* for continua  $K_0 \dots K_{12}$ . Each  $K_i$  can exist only if its full configuration  $\Omega(K_i)$  lies strictly within the admissible region of the corresponding meta-space  $M_i$ :

$$\Omega(K_i) \subseteq \Omega(M_i).$$

M-spaces have three universal functions:

1. **Admissibility constraints:** they define which structural configurations, thresholds and flows are permitted. They serve as the global regulator of continua.
2. **Dimensional guidance:**  $M_{i+1}$  determines whether  $K_i$  can form new axes and transition into a higher-dimensional continuum.
3. **Structural supervision:** each meta-space enforces global compatibility of operators such as  $\Psi$ ,  $\Phi$ ,  $\Lambda$ ,  $U$ , and  $X$ , and ensures that continua obey the universal laws of evolution, collapse, and branching.

M-spaces do not evolve “inside” the K-hierarchy; instead, they form a parallel supervisory structure. The relation can be visualised as:

$$K_i \hookrightarrow M_i, \quad M_i \Rightarrow \text{constraints on } K_i.$$

The transition  $K_i \rightarrow K_{i+1}$  is possible only if:

$$A(K_{i+1}) \subseteq A(M_{i+1}), \quad P(K_{i+1}) \subseteq P(M_{i+1}), \quad \Theta(K_{i+1}) \leq \Theta(M_{i+1}).$$

Each  $M_i$  expands the dimensionality of admissible structures compared to  $M_{i-1}$ , providing the over-structure within which the next continuum level may emerge.

### O.2.1 $M_1$ Overview

$M_1$  is the meta-space that provides the admissibility structure for the one-dimensional continuum  $K_1$ . Unlike  $M_0$ , which governs only proto-differences,  $M_1$  introduces the first genuine geometric and dynamical admissibility conditions:

$$\Omega(K_1) \subseteq \Omega(M_1).$$

$M_1$  is the minimal meta-space in which the following are well-defined:

- a measurable axis  $A_1$ ;
- a continuous field  $\phi(x)$  over a one-dimensional domain;
- proto-dynamics via flows  $J_1 = (\partial_x \phi, \partial_t \phi)$ ;
- admissible energy functionals and actions;
- well-formed structural tension  $T_1$  and a meaningful threshold  $\Theta_1$ ;
- the existence and collapse rules of  $K_1$ .

Thus,  $M_1$  is the meta-structure that makes  $K_1$  mathematically and physically possible.

### 1. Structure of $\Omega(M_1)$

$$\Omega(M_1) = \left\{ \phi : X \rightarrow V \mid \phi \in C^0(X, V), \partial_x \phi, \partial_t \phi \text{ well-defined in } H^1 \right\}.$$

$M_1$  enforces:

1. **Continuity:** Fields must admit continuous representations. Discontinuous maps lie outside  $\partial\Omega(M_1)$ .
2. **Sobolev regularity:** First derivatives must exist in the weak sense:

$$\partial_x \phi, \partial_t \phi \in H^1.$$

3. **Metric compatibility:** The admissible metric structure of  $X$  and  $V$  is fixed by  $M_1$  and is required to compute energy and tension.
4. **Topological admissibility:**  $X$  must be a one-dimensional topological manifold (open interval, circle, half-line, etc.).

Any violation of these constraints places a configuration on the boundary  $\partial\Omega(M_1)$ , making the existence of  $K_1$  impossible.

### 2. Admissible Axes in $M_1$

$M_1$  allows exactly one structural axis:

$$A_1 : X \rightarrow \mathbb{R},$$

representing the first measurable dimension.

The following are prohibited in  $M_1$ :

- additional independent axes;
- incompatible nonlinear structure for  $A_1$ ;
- oscillatory axes without admissible dynamics.

These prohibitions define the scope of possible  $K_1$  evolutions.

### 3. Admissible Potentials and Energies

$M_1$  requires the existence of a well-defined energy functional consistent with the universal operator structure:

$$E[\phi] = \int_X \left( \frac{1}{2} |\partial_x \phi|^2 + V(\phi) \right) dx,$$

with:

- $V(\phi)$  bounded below,
- coercivity to prevent collapse,
- differentiability to define flows.

Admissible potentials  $P(M_1)$  are exactly those satisfying the above.

#### 4. Thresholds and Structural Tension

$M_1$  defines the threshold:

$$\Theta_1 = \inf\{T_1(\phi) : \phi \in \Omega(M_1)\},$$

and the structural tension functional:

$$T_1(\phi) = \int_X (|\partial_x \phi|^2 + W(\phi)) \, dx,$$

where  $W$  includes constraint-enforcing terms.

Violation of  $T_1 < \Theta_1$  drives  $K_1$  to collapse.

#### 5. Admissible Flows and Proto-Dynamics

Allowed flows are:

$$J_1 = (\partial_x \phi, \partial_t \phi),$$

with  $\partial_t \phi$  belonging to the admissible tangent space of  $\Omega(M_1)$ .

$M_1$  prohibits flows that:

- increase tension without admissible compensation,
- cause nonphysical discontinuities,
- violate the action principle or conservation constraints.

#### 6. Collapse and Boundary of $M_1$

A configuration lies on the boundary  $\partial\Omega(M_1)$  if:

- continuity fails,
- Sobolev regularity fails,
- energy becomes unbounded,
- $T_1 \geq \Theta_1$ ,
- $A_1$  becomes non-measurable.

Collapse of  $K_1$  occurs when its admissible configurations cross this boundary.

#### 7. Relation to $K_0$ and $K_2$

$M_1$  is the minimal space that permits the transition  $K_0 \rightarrow K_1$  via the operator  $\Psi_{0 \rightarrow 1}$ .

A transition  $K_1 \rightarrow K_2$  is admissible only if:

$$A_2 \in A(M_2), \quad \Omega(K_2) \subseteq \Omega(M_2), \quad T_1 \geq \Theta_{\dim}.$$

Thus,  $M_1$  governs:

- existence and stability of  $K_1$ ,
- the admissible domain for the first dynamic continuum,
- the structural possibility of higher-dimensional continua.

### O.2.2 $M_2$ Overview

$M_2$  is the meta-space that provides the admissibility structure for the two-dimensional continuum  $K_2$ . Transition into  $M_2$  corresponds to the emergence of:

- spatial dimensionality beyond  $A_1$ ,
- local geometric structure,
- a dynamical metric,
- causal cones and permitted propagation velocities,
- quantization thresholds and minimal clusters,
- the full action-based dynamics of continua.

It is the first meta-space where geometry, time, and physical propagation are jointly admissible.

#### 1. Structure of $\Omega(M_2)$

The admissible configuration space in  $M_2$  is:

$$\Omega(M_2) = \left\{ \phi : X_2 \rightarrow V \mid \phi \in C^0(X_2, V), \partial_i \phi, \partial_t \phi \in H^1, g_{\mu\nu} \in C^0 \right\}.$$

Requirements:

1. **Two-dimensional topology.**  $X_2$  must be a 2-manifold with local coordinate charts. Typical examples:  $\mathbb{R}^2$ , a cylinder, or a compact surface.
2. **Local differentiability.** First derivatives  $\partial_i \phi$  and  $\partial_t \phi$  must exist in the Sobolev sense.
3. **Admissible metric field.**  $M_2$  allows a dynamical metric  $g_{\mu\nu}$ , but only with signature:

$$(-, +, +),$$

arising as the Hessian of the structural tension functional  $T_2$ .

4. **Locality constraint.** Allowed interactions must be local in  $X_2$ .
5. **Finite propagation.** Causal cones must exist ( $\Theta_{\text{conn}}$  enforces the light-cone admissibility).

These conditions define the admissible region for any  $K_2$  continuum.

#### 2. Admissible Axes in $M_2$

$M_2$  admits exactly two independent spatial axes:

$$A_1, A_2,$$

with  $A_1$  inherited and  $A_2$  arising from the dimensional threshold condition:

$$T_1 \geq \Theta_{\text{dim}}.$$

No additional axes are admissible in  $M_2$ . Higher axes require transitions into  $M_3$  and above.



### 3. Admissible Potentials and Energies

The meta-space  $M_2$  enforces energy functionals of the form:

$$E[\phi] = \int_{X_2} \left( \frac{1}{2} g^{ij} \partial_i \phi \partial_j \phi + V(\phi) \right) \sqrt{|g|} d^2 x.$$

Admissible potentials  $P(M_2)$  must satisfy:

- coercivity to ensure stability against collapse,
- boundedness from below,
- differentiability for generating flows,
- local dependence only (no nonlocal potentials allowed).

### 4. Thresholds and Structural Tension

$M_2$  defines:

$$\Theta_2 = \inf \{ T_2(\phi, g_{\mu\nu}) : (\phi, g) \in \Omega(M_2) \},$$

with the structural tension:

$$T_2 = \int_{X_2} (g^{ij} \partial_i \phi \partial_j \phi + W(\phi) + \alpha R) \sqrt{|g|} d^2 x,$$

where:

- $W(\phi)$  enforces constraints on admissible configurations,
- $R$  is the scalar curvature,
- $\alpha$  controls geometric stiffness.

A key feature of  $M_2$ : \*\*the metric  $g_{\mu\nu}$  emerges from the second variation of  $T_2$ \*\*, making geometry a derivative concept rather than primitive.

### 5. Admissible Flows and Propagation

Allowed flows:

$$J_2 = (\partial_i \phi, \partial_t \phi, \nabla_i T_2, \square_g \phi),$$

with the d'Alembertian defined through  $g$ .

Propagation constraints:

- signals must lie within the causal cone determined by  $\Theta_{\text{conn}}$ ,
- propagation speed cannot exceed the admissible limit set by  $g_{\mu\nu}$ ,
- flows must preserve regularity ( $H^1$  compatibility).

These encode the emergence of physical causality in  $K_2$ .

## 6. Quantization Thresholds and Minimal Clusters

$M_2$  is the first meta-space to enforce:

$$\Theta_{\text{quant}} > 0.$$

This yields:

- a minimal stable cluster size  $q_{\min}$ ,
- discrete operator actions:

$$a^\dagger|k\rangle, \quad a|k\rangle,$$

as changes in connectedness measure  $k$ ,

- quantized excitations as topologically stable configurations in  $\Omega(M_2)$ .

Thus, quantization is a structural necessity, not an external axiom.

## 7. Collapse and Boundary of $M_2$

A configuration touches  $\partial\Omega(M_2)$  if:

- the metric degenerates ( $\det g \rightarrow 0$ ),
- locality is violated,
- $T_2 \geq \Theta_2$ ,
- curvature diverges ( $|R| \rightarrow \infty$ ),
- admissible axes cease to be measurable.

Crossing this boundary forces collapse of any  $K_2$  continuum.

## 8. Relation to $K_1$ and $K_3$

$M_2$  enables the transition  $K_1 \rightarrow K_2$  via  $\Psi_{1 \rightarrow 2}$ , defined through:

$$A_1 \mapsto (A_1, A_2), \quad \Omega(K_1) \rightarrow \Omega(K_2), \quad T_1 \rightarrow T_2, \quad \dim = 2.$$

Transition to  $K_3$  becomes admissible only when:

$$A_3 \in A(M_3), \quad T_2 \geq \Theta_{\text{dim}}, \quad \Omega(K_3) \subseteq \Omega(M_3).$$

Thus,  $M_2$  is the meta-space of:

- geometry,
- causality,
- quantization,
- dimensional stability,
- propagation.

It is the structural foundation for all higher continua.

### O.2.3 $M_3$ Overview

$M_3$  is the meta-space that provides the admissibility conditions for three-dimensional continua  $K_3$ . It is the first meta-space in which:

- a third independent axis  $A_3$  becomes admissible,
- reactive structure appears (chemical binding / dissociation),
- valence states become part of the admissible configuration space,
- diffusion and transport occur in three spatial dimensions,
- local interaction graphs (reaction networks) are stable objects,
- activation thresholds and reaction potentials become well-defined,
- spatial heterogeneity and gradients are dynamically supported.

$M_3$  is the mathematical and physical environment within which the chemical continuum  $K_3$  can exist, evolve and undergo structural transitions towards higher continua such as  $K_4$ .

#### 1. Admissible Configuration Space $\Omega(M_3)$

The admissible set of configurations in  $M_3$  is:

$$\Omega(M_3) = \left\{ (\phi, G_{\text{chem}}, \rho, g_{ij}) \mid \phi \in C^0(X_3, V), G_{\text{chem}} \in \mathcal{G}_{\text{RAF}}, \rho \in L^1(X_3), g_{ij} \in C^0 \right\}.$$

Key admissibility conditions:

1. **Three-dimensional topology.**  $X_3$  must be a 3-manifold supporting local coordinate charts, differentiability and diffusion.
2. **Local reaction networks.**  $G_{\text{chem}}$  must be representable as a finite, locally supported reaction graph satisfying the RAF conditions for closure and catalytic action.
3. **Density fields.**  $\rho(x)$  defines molecular or atomic concentrations and must be integrable with finite total mass.
4. **Metric admissibility.**  $g_{ij}$  describes spatial geometry but does not yet include relativistic structure; only Riemannian (positive definite) metrics are admissible at this level.
5. **Locality.** All interactions must be expressible through local potentials and local reaction kernels.

Thus,  $\Omega(M_3)$  expands the geometric and physical admissibility of  $M_2$  into the chemical regime.

#### 2. Admissible Axes $A(M_3)$

A new axis  $A_3$  becomes admissible in  $M_3$ , satisfying:

$$A_3 \not\subset \text{span}(A_1, A_2), \quad T_2 \geq \Theta_{\text{dim}}.$$

Additional admissible internal axes:

- $A_{\text{val}}$ : valence states (bonding possibilities),
- $A_{\text{react}}$ : admissible reaction modes,
- $A_{\text{conf}}$ : configuration states of molecules.

These internal axes encode chemical structure within the meta-space.

### 3. Potentials $P(M_3)$

Admissible potentials in  $M_3$  correspond to chemical and spatial interactions:

$$P(M_3) = \{U_{\text{bond}} + U_{\text{rep}} + U_{\text{conf}} + U_{\text{diff}} + U_{\text{local}}\}.$$

Where:

- $U_{\text{bond}}$  — bonding potentials with minima at stable bond lengths,
- $U_{\text{rep}}$  — short-range repulsive potentials preventing collapse,
- $U_{\text{conf}}$  — internal configuration potentials for molecular states,
- $U_{\text{diff}}$  — potentials associated with concentration gradients,
- $U_{\text{local}}$  — any locally defined energetic contribution.

Constraints:

1. potentials must be bounded from below,
2. they must be differentiable almost everywhere,
3. they must enforce local, not global or action-at-a-distance, structure.

### 4. Thresholds $\Theta(M_3)$

$M_3$  formalizes several chemical thresholds:

- $\Theta_{\text{bond}}$  — minimal energy required to break a bond,
- $\Theta_{\text{act}}$  — activation threshold for reactions,
- $\Theta_{\text{cluster}}$  — minimal stable cluster energy,
- $\Theta_{\text{grad}}$  — threshold for maintaining spatial gradients,
- $\Theta_{\text{RAF}}$  — threshold for RAF-network closure.

Stability of  $K_3$  requires:

$$T_3 < \min\{\Theta_{\text{bond}}, \Theta_{\text{cluster}}, \Theta_{\text{RAF}}\}.$$

Where  $T_3$  is the structural tension of configurations in  $\Omega(K_3)$ .

### 5. Flows $J(M_3)$

Admissible flows include:

$$J(M_3) = (J_{\text{diff}}, J_{\text{react}}, J_{\text{grad}}, \nabla_i T_3).$$

Interpretation:

- $J_{\text{diff}}$  — diffusion flows in 3D,
- $J_{\text{react}}$  — local reaction fluxes given RAF rules,
- $J_{\text{grad}}$  — flows maintaining or reducing local gradients,
- $\nabla_i T_3$  — flows driven by gradients of structural tension.

All flows must preserve local mass conservation.

## 6. Cycles and Reaction Networks

$M_3$  is the first meta-space where chemical cycles are admissible:

$$C_{\text{chem}} = \{\text{bond-formation} \rightarrow \text{transformation} \rightarrow \text{bond-breaking} \rightarrow \dots\}.$$

RAF nets become admissible cycles when:

$$E_{\text{in}} \geq \Theta_{\text{act}} \quad \text{and} \quad \text{closure conditions satisfied.}$$

These cycles form the minimal building blocks for the emergence of biological continua ( $K_4$ ).

## 7. Boundary $\partial\Omega(M_3)$ and Collapse

A configuration approaches the boundary  $\partial\Omega(M_3)$  if:

- bonding potentials become singular,
- reaction flux diverges,
- gradients become non-integrable ( $|\nabla\rho| \rightarrow \infty$ ),
- RAF closure is lost ( $G_{\text{chem}} \notin \mathcal{G}_{\text{RAF}}$ ),
- total structural tension exceeds threshold:

$$T_3 \geq \Theta_3.$$

Crossing  $\partial\Omega(M_3)$  destroys chemical continuity and collapses  $K_3$ .

## 8. Relation to $M_2$ and $M_4$

The transition  $K_2 \rightarrow K_3$  becomes admissible when:

$$A_3 \in A(M_3), \quad \Omega(K_3) \subseteq \Omega(M_3), \quad T_2 \geq \Theta_{\text{dim}}.$$

Transition toward  $K_4$  requires:

$$\Omega(K_4) \subseteq \Omega(M_4), \quad \partial\Omega(M_3) \text{ supports bounded permeability,} \quad \Theta_{\text{mem}} > 0.$$

Thus,  $M_3$  is the meta-space of:

- chemical structure,
- reaction networks,
- three-dimensional diffusion,
- valence and bonding,
- activation barriers,
- robust gradients.

It is the structural foundation enabling chemical continua  $K_3$  and the emergence of protocellular structures in  $K_4$ .

### O.2.4 $M_4$ Overview

$M_4$  is the meta-space that provides the admissibility conditions for biological continua of level  $K_4$ . It is the first meta-space in which:

- a stable *semipermeable boundary* is admissible,
- the topological distinction **inside/outside** becomes an axis,
- nontrivial *concentration, electrical, pH and redox gradients* can be sustained,
- active transport and energy conversion processes are allowed,
- reaction networks are confined and spatially coordinated,
- the transition from chemical to protobiological structure ( $K_3 \rightarrow K_4$ ) is admissible.

$M_4$  introduces the geometric and energetic conditions for compartmentalization—the defining property of early cells.

#### 1. Admissible Configuration Space $\Omega(M_4)$

The admissible configurations in  $M_4$  extend those of  $M_3$  by adding:

$$\Omega(M_4) = \left\{ (\phi, G_{\text{chem}}, \rho_{\text{in/out}}, \Delta\mu, g_{ij}, \partial\Omega, \Pi) \mid \text{admissibility conditions hold} \right\}.$$

Where:

- $\partial\Omega$  — a semipermeable membrane forming a compact boundary,
- $\rho_{\text{in/out}}$  — concentration fields inside and outside,
- $\Delta\mu$  — electrochemical potential gradients across  $\partial\Omega$ ,
- $\Pi$  — permeability tensor (direction- and species-dependent),
- $G_{\text{chem}}$  — internal RAF networks that remain functional under confinement.

Admissibility conditions:

1. **Membrane stability:**  $\partial\Omega$  must maintain mechanical integrity under tension  $T_{\text{mem}} < \Theta_{\text{mem}}$ .
2. **Permeability constraints:**  $\Pi$  must be finite, anisotropic and species-specific.
3. **Gradient compatibility:** electrochemical gradients must satisfy:

$$|\Delta\mu| < \Theta_{\text{grad-max}}.$$

4. **Volume finiteness:** the internal region must have bounded volume and mass.
5. **RAF compatibility:** internal reaction networks must remain topologically closed.

$M_4$  thus provides the minimal environment where protocells can exist.

#### 2. Admissible Axes $A(M_4)$

A new fundamental axis appears:

$$A_{\text{in/out}} = \{\text{inside, outside}\}.$$

It satisfies:

$$A_{\text{in/out}} \not\subseteq \text{span}(A_1, A_2, A_3) \Rightarrow \text{new dimension allowed.}$$

Additional admissible axes include:

- $A_{\text{perm}}$  — permeability states,
- $A_{\text{grad}}$  — gradient configurations (pH, ions, charge),
- $A_{\text{vol}}$  — volume/pressure axis,
- $A_{\text{energy}}$  — internal energy states (ATP-like precursors, redox pools).

These axes define the internal functional geometry of early cells.

### 3. Potentials $P(M_4)$

$M_4$  admits new classes of potentials:

$$P(M_4) = (U_{\text{mem}}, U_{\text{grad}}, U_{\text{osm}}, U_{\text{charge}}, U_{\text{redox}}, U_{\text{active}}).$$

Interpretation:

- $U_{\text{mem}}$  — membrane bending and surface tension potential,
- $U_{\text{grad}}$  — energy stored in chemical/electrical gradients,
- $U_{\text{osm}}$  — osmotic pressure potential,
- $U_{\text{charge}}$  — Coulombic and dielectric interactions,
- $U_{\text{redox}}$  — redox-energy landscape,
- $U_{\text{active}}$  — energy conversion enabling active transport.

All potentials must be:

1. differentiable almost everywhere,
2. bounded from below,
3. compatible with membrane stability conditions.

### 4. Thresholds $\Theta(M_4)$

Key biological thresholds introduced:

- $\Theta_{\text{mem}}$  — membrane rupture threshold,
- $\Theta_{\text{grad}}$  — minimal gradient needed to maintain structure,
- $\Theta_{\text{grad-max}}$  — maximal tolerable gradient,
- $\Theta_{\text{osm}}$  — osmotic pressure tolerance,
- $\Theta_{\text{pH}}$  — viability interval for internal pH,
- $\Theta_{\text{redox}}$  — redox compatibility threshold.

Critical condition for  $K_4$  to remain alive:

$$T_4 < \min\{\Theta_{\text{mem}}, \Theta_{\text{osm}}, \Theta_{\text{grad-max}}, \Theta_{\text{pH}}, \Theta_{\text{redox}}\}.$$

## 5. Flows $J(M_4)$

Admissible flows include:

$$J(M_4) = (J_{\text{diff}}, J_{\text{perm}}, J_{\text{pump}}, J_{\text{redox}}, J_{\text{osm}}, \nabla_i T_4).$$

Where:

- $J_{\text{diff}}$  — passive diffusion inside/outside,
- $J_{\text{perm}}$  — membrane-permeation flows,
- $J_{\text{pump}}$  — active transport against gradients,
- $J_{\text{redox}}$  — redox-cycling flows,
- $J_{\text{osm}}$  — osmotic balance flows,
- $\nabla_i T_4$  — flows driven by structural tension gradients.

All flows must satisfy mass conservation within the membrane boundary.

## 6. Cycles in $M_4$

$M_4$  is the meta-space where biological cycles become admissible:

- $C_{\text{RAF-core}}$  — reaction-autocatalytic cycles,
- $C_{\text{metabolic}}$  — basic metabolic turnover,
- $C_{\text{pump}}$  — pump-driven active transport cycles,
- $C_{\text{redox}}$  — electron-transfer cycles,
- $C_{\text{buffer}}$  — pH-buffering cycles.

Condition for cycle stability:

$$\oint_C dA_i \approx 0 \quad \text{and} \quad T_4 \text{ bounded.}$$

These cycles generate the temporal structure  $\tau(K_4)$ .

## 7. Boundary $\partial\Omega(M_4)$

A configuration approaches  $\partial\Omega(M_4)$  when:

- membrane tension approaches the rupture threshold:

$$T_{\text{mem}} \rightarrow \Theta_{\text{mem}},$$

- internal pH exits the viability interval,
- gradients exceed  $\Theta_{\text{grad-max}}$ ,
- osmotic imbalance becomes unstable,
- confinement breaks RAF closure inside the compartment.

Crossing  $\partial\Omega(M_4)$  destroys the protobiological continuum  $K_4$ .



## 8. Relation to $M_3$ and $M_5$

Transition  $K_3 \rightarrow K_4$  is admissible when:

$$A_{\text{in/out}} \in A(M_4), \quad \Omega(K_4) \subseteq \Omega(M_4), \quad T_3 \geq \Theta_{\text{dim}}.$$

Transition toward  $K_5$  requires:

$$A_{\text{exc}} \in A(M_5), \quad \Delta V \geq \Theta_{\text{exc}}, \quad \text{channels and electrical conduction admissible.}$$

Thus,  $M_4$  is the meta-space of:

- membrane structure,
- compartmentalization,
- gradients and osmotic physics,
- energy conversion,
- early metabolism,
- minimal biological stability.

It is the structural foundation enabling protocellular life.

### O.2.5 $M_5$ Overview

$M_5$  is the meta-space that makes *electrical excitability* admissible. It is the environment in which a biological continuum can exhibit:

- stable membrane voltage  $V(t)$ ,
- ion-channel-mediated conductance states,
- threshold-based excitation ( $\Theta_{\text{exc}}$ ),
- propagating electrical events (precursors of action potentials),
- regenerative feedback required for spike cycles.

Where  $M_4$  enabled *chemical* compartmentalization,  $M_5$  enables *electrochemical* dynamics that define the  $K_5$  continuum.

#### 1. Admissible Configuration Space $\Omega(M_5)$

The admissible configurations extend  $\Omega(M_4)$  by including:

$$\Omega(M_5) = \left\{ (\partial\Omega, V, g_{\text{ion}}, \Pi_{\text{ion}}, \Delta\mu_{\text{ion}}, C_m, C_{\text{channels}}) \mid \text{admissibility conditions hold} \right\}.$$

Where:

- $V$  — membrane voltage field,
- $C_m$  — membrane capacitance,
- $g_{\text{ion}}$  — conductance states of ion channels,
- $C_{\text{channels}}$  — channel population and gating logic,
- $\Pi_{\text{ion}}$  — ion-specific permeabilities,
- $\Delta\mu_{\text{ion}}$  — electrochemical gradients generating currents.

Admissibility requires:

1. Membrane integrity as in  $M_4$ .
2. Ion channels capable of switching between at least two states:  
 $\{\text{open, closed}\}.$

3. Voltage dynamics satisfying:

$$C_m \frac{dV}{dt} = - \sum_i g_i (V - E_i) + I_{\text{ext}}$$

in an admissible parameter region.

4. Existence of a nontrivial equilibrium and a separatrix defining a threshold.

Thus  $M_5$  is the minimal environment where a cell can be excitable.

## 2. Admissible Axes $A(M_5)$

A new fundamental axis arises:

$$A_{\text{exc}} = \{\text{excitable, nonexcitable}\},$$

with the corresponding voltage-threshold distinction:

$$A_V = \{V < \Theta_{\text{exc}}, V \geq \Theta_{\text{exc}}\}.$$

Other admissible axes:

- $A_{\text{ion}}$  — ion-specific channel conformations,
- $A_{\text{gate}}$  — gating variable states (activation/inactivation),
- $A_{\text{cond}}$  — conductance regimes,
- $A_{\text{spike}}$  — refractory/overshoot/reset phases.

These axes define the high-dimensional dynamical geometry of excitable membranes.

## 3. Potentials $P(M_5)$

$M_5$  introduces potentials associated with electrodynamics:

$$P(M_5) = (U_{\text{ion}}, U_V, U_{\text{gate}}, U_{\text{CAP}}, U_{\text{rest}}).$$

Interpretation:

- $U_{\text{ion}}$  — Nernst-like electrochemical potentials,
- $U_V$  — voltage-dependent energy landscape,
- $U_{\text{gate}}$  — gating energy for channel transitions,
- $U_{\text{CAP}}$  — capacitive energy of the membrane,
- $U_{\text{rest}}$  — resting-state energy minimum.

Conditions for admissibility:

1.  $U_V$  must have a local minimum (resting state),
2. the gain function must allow suprathreshold instability,
3. gating transitions must be thermally and chemically feasible.

#### 4. Thresholds $\Theta(M_5)$

The defining threshold:

$\Theta_{\text{exc}}$  = minimal depolarization required to initiate a regenerative event.

Additional thresholds:

- $\Theta_{\text{gate}}$  — gating activation threshold,
- $\Theta_{\text{cond}}$  — conductance switching threshold,
- $\Theta_{\text{stability}}$  — stability boundary for oscillatory modes.

Critical relation:

$$V(t) \geq \Theta_{\text{exc}} \Rightarrow \text{transition into spike-generating regime.}$$

#### 5. Flows $J(M_5)$

Admissible flows expand those in  $M_4$  by including:

$$J(M_5) = (J_{\text{ion}}, J_V, J_{\text{gate}}, J_{\text{exc}}, \nabla_i T_5).$$

Where:

- $J_{\text{ion}}$  — ion-specific currents,
- $J_V$  — voltage-driven flows,
- $J_{\text{gate}}$  — dynamic gating transitions,
- $J_{\text{exc}}$  — regenerative excitation flow regime,
- $\nabla_i T_5$  — flows driven by tension gradients under excitability.

Conservation laws:

$$\sum_i J_{\text{ion},i} = C_m \frac{dV}{dt}.$$

#### 6. Cycles in $M_5$

The key cycle:

$$C_{\text{spike}} = \{\text{rest} \rightarrow \text{depolarization} \rightarrow \text{overshoot} \rightarrow \text{repolarization} \rightarrow \text{refractory} \rightarrow \text{rest}\}.$$

Other admissible cycles:

- $C_{\text{gate}}$  — gating activation/inactivation loop,
- $C_{\text{osc}}$  — subthreshold oscillatory cycles (when allowed),
- $C_{\text{conduct}}$  — transition cycles between conductance states.

Condition for existence of a spike cycle:

$$\oint_{C_{\text{spike}}} dA_{\text{phase}} \approx 0 \quad \text{and} \quad V(t) \text{ crosses } \Theta_{\text{exc}} \text{ once per cycle.}$$

## 7. Boundary $\partial\Omega(M_5)$

A configuration approaches  $\partial\Omega(M_5)$  when:

- the membrane cannot support stable resting potential,
- gating kinetics break excitability,
- conductance saturates or collapses,
- $\Theta_{\text{exc}}$  becomes unreachable (too high) or trivial (zero),
- depolarization block persists,
- ionic gradients fall below minimal viability levels.

Crossing  $\partial\Omega(M_5)$  makes a spike impossible; thus  $K_5$  collapses or returns to a  $K_4$ -like regime.

## 8. Relation to $M_4$ and $M_6$

Transition  $K_4 \rightarrow K_5$  is admissible when:

$$A_{\text{exc}} \in A(M_5), \quad V(t) \geq \Theta_{\text{exc}}, \quad \text{ion channels form coherent gating dynamics.}$$

Transition toward  $K_6$  (cognitive attractors) requires:

$$A_{\text{concept}} \in A(M_6), \quad C_{\text{spike}} \text{ must support coupling to other spikes,}$$

emergence of attractor dynamics from spike trains.

Thus  $M_5$  is the meta-space enabling:

- electrical excitability,
- spike generation,
- dynamic conductance regulation,
- early bioelectrical logic,
- the structural bridge to neural computation.

### O.2.6 $M_6$ Overview

$M_6$  is the meta-space that makes *cognitive organization* admissible. While  $M_5$  supports electrical excitability and spike cycles,  $M_6$  enables:

- stable attractors over spike trains,
- semantic axes and representational geometry,
- proto-logical operations acting on neural states,
- classification and prediction dynamics,
- emergence of symbolic and sub-symbolic concepts,
- coherence constraints required for cognitive continua ( $K_6$ ).

Thus  $M_6$  is the minimal environment for *meaning-bearing neural dynamics*.

### 1. Admissible Configuration Space $\Omega(M_6)$

$$\Omega(M_6) = \{(S(t), A_{\text{sem}}, \mathcal{W}, \mathcal{A}, C_{\text{att}}, \Pi_6) \mid \text{admissibility conditions hold}\}.$$

Where:

- $S(t)$  — neural state trajectories (spike trains, firing rates, phase codes),
- $\mathcal{W}$  — synaptic weight tensor with plasticity constraints,
- $\mathcal{A}$  — admissible attractor set,
- $A_{\text{sem}}$  — semantic axes embedded into the attractor geometry,
- $C_{\text{att}}$  — cycle family of attention / working memory,
- $\Pi_6$  — constraints ensuring stability and separability of representations.

A configuration is admissible if:

1. neural dynamics can converge to at least one stable attractor;
2. attractor basins are separable by hyperplanes or manifolds;
3. mapping between input patterns and attractors is consistent;
4. plasticity rules preserve coherence rather than collapse structures.

### 2. Admissible Axes $A(M_6)$

$M_6$  introduces a new class of axes:

$$A_{\text{concept}} = \{\text{conceptual distinctions emerging from attractors}\}.$$

These axes correspond to *meaning-bearing degrees of freedom*.

Other admissible axes:

- $A_{\text{sem}}$  — semantic embedding axes in neural space,
- $A_{\text{wm}}$  — working-memory maintenance axis,
- $A_{\text{att}}$  — attentional modulation axis,
- $A_{\text{proto\_logic}}$  — axes supporting proto-logical operations,
- $A_{\text{prediction}}$  — predictive coding axes.

A necessary condition for  $K_6$ :

$$\dim(A_{\text{sem}}) \geq 1 \quad \text{and} \quad A_{\text{concept}} \subseteq A(M_6).$$

### 3. Potentials $P(M_6)$

Cognitive potentials generalize  $M_5$  electrodynamic potentials to higher-level structure:

$$P(M_6) = (U_{\text{att}}, U_{\text{wm}}, U_{\text{concept}}, U_{\text{pred}}, U_{\text{stability}}).$$

Interpretation:

- $U_{\text{att}}$  — potential landscape shaping attentional cycles,
- $U_{\text{wm}}$  — energy required to maintain working memory states,
- $U_{\text{concept}}$  — conceptual potential separating attractors,

- $U_{\text{pred}}$  — potential of predictive mismatches,
- $U_{\text{stability}}$  — global stabilizing potential for attractor geometry.

A cognitive continuum requires:

$U_{\text{concept}}$  has multiple minima corresponding to concepts.

#### 4. Thresholds $\Theta(M_6)$

Key thresholds include:

- $\Theta_{\text{att}}$  — minimal activation for attentional engagement,
- $\Theta_{\text{wm}}$  — threshold for stable memory maintenance,
- $\Theta_{\text{sep}}$  — separability threshold between attractors,
- $\Theta_{\text{concept}}$  — threshold for emergence of distinct concepts,
- $\Theta_{\text{coh}}$  — cognitive coherence threshold.

Critical condition for existence of a cognitive continuum:

$$\Theta_{\text{sep}} < \Delta_{\text{basin}}, \quad \Theta_{\text{coh}} \leq T_{\text{cog}},$$

where  $\Delta_{\text{basin}}$  is inter-attractor distance.

#### 5. Flows $J(M_6)$

Admissible cognitive flows include:

$$J(M_6) = (J_{\text{att}}, J_{\text{wm}}, J_{\text{concept}}, J_{\text{pred}}, J_{\text{plasticity}}, \nabla_i T_6).$$

Interpretation:

- $J_{\text{att}}$  — attentional modulation flow,
- $J_{\text{wm}}$  — working-memory stabilization flow,
- $J_{\text{concept}}$  — flows transforming attractor geometry,
- $J_{\text{pred}}$  — prediction-error propagation,
- $J_{\text{plasticity}}$  — synaptic updates maintaining coherence.

Conservation relation:

$$\sum_i J_{\text{plasticity},i} = \frac{d\mathcal{W}}{dt}.$$

#### 6. Cycles $C(M_6)$

Key cycles enabling cognition:

$$C_{\text{att\_wm}} = \{\text{attention} \rightarrow \text{encoding} \rightarrow \text{maintenance} \rightarrow \text{updating} \rightarrow \text{release}\}.$$

Further cycles:

- $C_{\text{predictive}}$  — prediction  $\rightarrow$  error  $\rightarrow$  update,
- $C_{\text{concept}}$  — stabilization of conceptual boundaries,

- $C_{\text{proto\_logic}}$  — logical transformations of states,
- $C_{\text{plasticity}}$  — learning cycles adjusting  $\mathcal{W}$ .

Necessary condition for cognitive life:

$$\oint_{C_{\text{att\_wm}}} dA_{\text{state}} \approx 0.$$

## 7. Boundary $\partial\Omega(M_6)$

A configuration reaches  $\partial\Omega(M_6)$  when:

- attractors lose stability or collapse,
- semantic axes become degenerate,
- prediction error diverges (unbounded),
- plasticity breaks coherence,
- $\Theta_{\text{concept}}$  cannot be crossed (no concept formation),
- attention cannot stabilize (cycle fails).

Crossing the boundary implies collapse of  $K_6$  into  $K_5$ -like dynamics (purely electrical).

## 8. Relation to $M_5$ and $M_7$

**From  $M_5$  to  $M_6$ :**

Requirements:

$C_{\text{spike}}$  must couple into structured patterns,  $\mathcal{W}$  must support attractor basins.

Thus spike trains acquire meaning-bearing geometry.

**Toward  $M_7$  (social cognition):**

$A_{\text{shared}} \in A(M_7)$ , requires stable conceptual repertoire in  $M_6$ ,

forming the basis for communication, shared attention, and social inference.

### O.2.7 $M_7$ Overview

$M_7$  is the meta-space that renders *social organization* admissible. While  $M_6$  supports cognitive continua founded on attractors, concepts, and proto-logical operations,  $M_7$  enables:

- shared meaning and intersubjective concepts,
- social axes and coordination mechanisms,
- communication channels and symbolic exchange,
- emergence of group-level cycles and norms,
- population-scale potentials and flows,
- structural conditions for the existence of social continua ( $K_7$ ).

Thus  $M_7$  is the minimal environment for *multi-agent cognitive coupling*.

### 1. Admissible Configuration Space $\Omega(M_7)$

$$\Omega(M_7) = \{ (\mathcal{C}_i, A_{\text{shared}}, R_{\text{comm}}, \mathcal{N}, C_{\text{soc}}, \Sigma_7) \mid \text{admissibility conditions hold} \}.$$

Components:

- $\mathcal{C}_i$  — cognitive continua of individuals embedded in  $M_6$ ,
- $A_{\text{shared}}$  — axes supporting shared representation and meaning,
- $R_{\text{comm}}$  — admissible communication relations,
- $\mathcal{N}$  — network structure of interactions,
- $C_{\text{soc}}$  — cycles sustaining social order (norms, roles, reciprocity),
- $\Sigma_7$  — coherence constraints ensuring group stability.

Admissibility requires:

1. existence of at least one shared semantic axis across agents,
2. ability to transmit distinctions along  $R_{\text{comm}}$ ,
3. stability of social cycles against fluctuations,
4. interaction topology  $\mathcal{N}$  supports coherence rather than fragmentation.

### 2. Admissible Axes $A(M_7)$

New axes introduced in  $M_7$ :

- $A_{\text{shared}}$  — axes of shared meaning / intersubjectivity,
- $A_{\text{comm}}$  — communication axes (symbols, signals, norms),
- $A_{\text{coord}}$  — axes of coordinated action,
- $A_{\text{role}}$  — axes corresponding to social roles,
- $A_{\text{norm}}$  — axes associated with norms and rule-following,
- $A_{\text{coop}}$  — cooperation-defection spectrum,
- $A_{\text{identity}}$  — axes capturing group identity distinctions.

Necessary condition for  $K_7$ :

$$\dim(A_{\text{shared}}) \geq 1, \quad A_{\text{comm}} \subseteq A(M_7).$$

### 3. Potentials $P(M_7)$

Social potentials generalize  $P(M_6)$  to multi-agent organization:

$$P(M_7) = (F_{\text{comm}}, F_{\text{coord}}, F_{\text{status}}, F_{\text{norm}}, F_{\text{coh}}, F_{\text{coop}}).$$

Interpretation:

- $F_{\text{comm}}$  — communication potential enabling information flow,
- $F_{\text{coord}}$  — potential stabilizing coordinated activity,
- $F_{\text{status}}$  — gradients of social influence and hierarchy,
- $F_{\text{norm}}$  — normative potentials (pressure to conform),



- $F_{\text{coh}}$  — group coherence potential,
- $F_{\text{coop}}$  — energetic equivalent of cooperation incentives.

A social continuum exists when:

$$F_{\text{coh}} > \Theta_{\text{fragment}}, \quad F_{\text{norm}} \text{ admits stable cycles.}$$

#### 4. Thresholds $\Theta(M_7)$

Critical thresholds include:

- $\Theta_{\text{comm}}$  — minimal communication bandwidth,
- $\Theta_{\text{shared}}$  — threshold for shared concept formation,
- $\Theta_{\text{coh}}$  — group coherence threshold,
- $\Theta_{\text{coop}}$  — cooperation threshold,
- $\Theta_{\text{norm}}$  — minimal pressure for norm stability,
- $\Theta_{\text{fragment}}$  — fragmentation threshold at  $\partial\Omega(M_7)$ .

Crossing  $\Theta_{\text{fragment}}$  leads to collapse into independent  $K_6$  continua.

#### 5. Flows $J(M_7)$

Admissible flows:

$$J(M_7) = (J_{\text{comm}}, J_{\text{norm}}, J_{\text{coop}}, J_{\text{identity}}, J_{\text{coord}}, \nabla_i T_7).$$

Interpretation:

- $J_{\text{comm}}$  — communication flow across the network,
- $J_{\text{norm}}$  — propagation of norms,
- $J_{\text{coop}}$  — cooperative / defection dynamics,
- $J_{\text{identity}}$  — flows shaping group identity,
- $J_{\text{coord}}$  — flows enabling synchronized action,
- $\nabla_i T_7$  — gradient of social tension.

Conservation relation (bounded resources of attention, trust, commitment):

$$\sum_i J_{\text{comm},i} \leq C_{\text{channel}}.$$

#### 6. Cycles $C(M_7)$

Fundamental social cycles:

$$C_{\text{soc}} = \{\text{signal} \rightarrow \text{response} \rightarrow \text{feedback} \rightarrow \text{update of norms} \rightarrow \text{new signal}\}.$$

Other cycles:

- $C_{\text{role}}$  — role acquisition, enactment, reinforcement,
- $C_{\text{coop}}$  — cooperation loop (benefit  $\rightarrow$  trust  $\rightarrow$  cooperation),
- $C_{\text{conflict}}$  — conflict  $\rightarrow$  negotiation  $\rightarrow$  resolution,

- $C_{\text{identity}}$  — formation and stabilization of group identity,
- $C_{\text{institution}}$  — emergence of institutional rules.

A living social continuum requires:

$$\oint_{C_{\text{soc}}} dA_{\text{state}} \approx 0.$$

## 7. Boundary $\partial\Omega(M_7)$

A configuration reaches  $\partial\Omega(M_7)$  when:

- shared axes collapse (loss of common meaning),
- communication channels fail or saturate,
- normative cycles break (no stable expectations),
- cooperation becomes unsustainable,
- identity axes fracture into incompatible subspaces,
- social tension  $T_7$  exceeds tolerance thresholds.

Crossing the boundary produces fragmentation into multiple  $K_6$  continua.

## 8. Relation to $M_6$ and $M_8$

**From  $M_6$  to  $M_7$ :**

Necessary conditions:

$$A_{\text{concept}}^{(i)} \cap A_{\text{concept}}^{(j)} \neq \emptyset, \quad R_{\text{comm}} \neq \emptyset.$$

Shared conceptual structure is required for intersubjectivity.

**Toward  $M_8$  (civilizational-technological space):**

$M_8$  introduces:

- symbolic systems,
- writing, institutions, infrastructure,
- technological and economic axes.

Thus  $M_7$  supplies the foundation of *proto-institutional coherence* required for  $K_8$ .

### O.2.8 $M_8$ Overview

$M_8$  is the meta-space enabling the existence of civilizational-technological continua ( $K_8$ ). While  $M_7$  supports social coordination through shared meaning, norms, and communication,  $M_8$  introduces the structural, symbolic, and material conditions for:

- externalized memory (writing, notation, symbolic media),
- technological infrastructures,
- institutional persistence across generations,
- distributed economic and informational flows,
- large-scale collective organization,
- long-range stabilization of social and symbolic cycles.

Thus  $M_8$  is the minimal space in which *civilization* becomes an admissible continuum.

### 1. Admissible Configuration Space $\Omega(M_8)$

$$\Omega(M_8) = \{(\mathcal{S}, \mathcal{I}, A_{\text{symbol}}, A_{\text{tech}}, \mathcal{M}_{\text{infra}}, C_{\text{civ}}, \Sigma_8) \mid \text{admissibility conditions hold}\}.$$

Components:

- $\mathcal{S}$  — stable symbolic systems (writing, mathematics, codified knowledge),
- $\mathcal{I}$  — institutional structures (law, governance, economic systems),
- $A_{\text{symbol}}$  — axes of symbolic representation,
- $A_{\text{tech}}$  — axes for technological differentiation,
- $\mathcal{M}_{\text{infra}}$  — material and informational infrastructures,
- $C_{\text{civ}}$  — civilizational cycles (production–distribution–feedback),
- $\Sigma_8$  — global constraints ensuring long-term stability.

Admissibility requires:

1. symbolic stability: reproducibility of symbols across time,
2. institutional persistence:  $\tau_{\text{inst}} \gg \tau_{\text{social}}$ ,
3. infrastructural capacity supporting large-scale flows,
4. boundedness of civilizational tension  $T_8$ ,
5. non-emptiness of  $\Omega(K_8)$  under  $M_8$  constraints.

### 2. Admissible Axes $A(M_8)$

$$A(M_8) = \{A_{\text{symbol}}, A_{\text{tech}}, A_{\text{infrastructure}}, A_{\text{institution}}, A_{\text{economic}}, A_{\text{scientific}}, A_{\text{bureaucratic}}\}.$$

Descriptions:

- $A_{\text{symbol}}$  — discrete symbolic distinctions (script, notation, code),
- $A_{\text{tech}}$  — technological differentiation (tools, machines, platforms),
- $A_{\text{infrastructure}}$  — transport, communication, energy networks,
- $A_{\text{institution}}$  — legal and governance axes,
- $A_{\text{economic}}$  — value, exchange, resource allocation axes,
- $A_{\text{scientific}}$  — axes of theoretical and empirical representation,
- $A_{\text{bureaucratic}}$  — formal procedural distinctions.

Necessary condition for civilization:

$$\dim(A_{\text{symbol}}) \geq 1 \quad \text{and} \quad A_{\text{tech}} \neq \emptyset.$$

### 3. Potentials $P(M_8)$

Civilizational potentials extend social potentials with symbolic and material components:

$$P(M_8) = (F_{\text{symbol}}, F_{\text{tech}}, F_{\text{infrastructure}}, F_{\text{institution}}, F_{\text{economic}}, F_{\text{scientific}}, F_{\text{stability}}).$$

Interpretation:

- $F_{\text{symbol}}$  — potential enabling encoding and transmission of abstract distinctions,

- $F_{\text{tech}}$  — gradients enabling technological innovation and adoption,
- $F_{\text{infrastructure}}$  — potentials enabling transport, energy, communication flows,
- $F_{\text{institution}}$  — regulation and constraint potentials,
- $F_{\text{economic}}$  — production/exchange potentials,
- $F_{\text{scientific}}$  — epistemic potential for stable theoretical growth,
- $F_{\text{stability}}$  — long-range coherence potential stabilizing  $K_8$ .

Civilization exists when:

$$F_{\text{symbol}} > \Theta_{\text{entropy}}, \quad F_{\text{institution}} > \Theta_{\text{collapse}}.$$

#### 4. Thresholds $\Theta(M_8)$

Key thresholds:

- $\Theta_{\text{symbol}}$  — minimal reliability of symbolic reproduction,
- $\Theta_{\text{infrastructure}}$  — minimal infrastructural capacity,
- $\Theta_{\text{institution}}$  — critical strength of institutions,
- $\Theta_{\text{economic}}$  — viability threshold for resource circulation,
- $\Theta_{\text{scientific}}$  — threshold for stable knowledge systems,
- $\Theta_{\text{entropy}}$  — maximal entropy of communication tolerated,
- $\Theta_{\text{collapse}}$  — boundary associated with civilizational failure.

Crossing  $\Theta_{\text{collapse}}$  induces breakdown into fragmented  $K_7$  continua.

#### 5. Flows $J(M_8)$

$$J(M_8) = (J_{\text{symbol}}, J_{\text{infrastructure}}, J_{\text{economic}}, J_{\text{institution}}, J_{\text{scientific}}, \nabla_i T_8).$$

Interpretation:

- $J_{\text{symbol}}$  — flow of symbolic information across media,
- $J_{\text{infrastructure}}$  — transport/energy/communication flows,
- $J_{\text{economic}}$  — production–exchange–consumption circulation,
- $J_{\text{institution}}$  — flow of decisions, rules, and governance signals,
- $J_{\text{scientific}}$  — flow of models and empirical updates,
- $\nabla_i T_8$  — gradients of civilizational tension.

A necessary stability condition:

$$\oint J_{\text{economic}} dt \approx 0, \quad \oint J_{\text{institution}} dt \approx 0.$$

## 6. Cycles $C(M_8)$

Fundamental cycles of civilization:

- $C_{\text{production}}$  — production  $\rightarrow$  distribution  $\rightarrow$  consumption  $\rightarrow$  reinvestment,
- $C_{\text{institution}}$  — rule creation  $\rightarrow$  enforcement  $\rightarrow$  feedback  $\rightarrow$  revision,
- $C_{\text{infrastructure}}$  — construction  $\rightarrow$  usage  $\rightarrow$  maintenance,
- $C_{\text{symbol}}$  — encoding  $\rightarrow$  transmission  $\rightarrow$  interpretation  $\rightarrow$  correction,
- $C_{\text{science}}$  — hypothesis  $\rightarrow$  test  $\rightarrow$  revision  $\rightarrow$  theory,
- $C_{\text{innovation}}$  — invention  $\rightarrow$  scaling  $\rightarrow$  saturation  $\rightarrow$  replacement.

A living civilizational continuum requires:

$$\oint_{C_{\text{symbol}}} dA \approx 0 \quad \text{and} \quad \oint_{C_{\text{institution}}} dA \approx 0.$$

## 7. Boundary $\partial\Omega(M_8)$

A system approaches  $\partial\Omega(M_8)$  when:

- symbolic systems lose fidelity,
- infrastructures decay faster than maintenance cycles can compensate,
- institutions fail to maintain coherence,
- economic circulation collapses,
- technological axes fracture (loss of compatibility),
- civilizational tension  $T_8$  exceeds stability thresholds.

Crossing  $\partial\Omega(M_8)$  causes reversion to  $M_7$ -admissible structures.

## 8. Relation to $M_7$ and $M_9$

From  $M_7$  to  $M_8$ :

Necessary conditions:

$$A_{\text{symbol}} \neq \emptyset, \quad \tau_{\text{symbol}} \gg \tau_{\text{social}}, \quad \mathcal{M}_{\text{infra}} \text{ supports macroscopic flows.}$$

Symbolic stability transforms social meaning into persistent civilization.

**Toward  $M_9$  (theoretical meta-space):**

$M_9$  introduces:

- mathematical and scientific theory spaces,
- explicit model transformations,
- epistemic coherence thresholds,
- abstract meta-representations independent of material substrates.

Thus  $M_8$  provides the material and institutional substrate for  $K_9$ .

### O.2.9 $M_9$ Overview

$M_9$  is the meta-space that enables the existence of theoretical continua ( $K_9$ ). While  $M_8$  provides the material, symbolic and institutional substrate of civilizations,  $M_9$  introduces the structural conditions for the emergence, coexistence and transformation of scientific, mathematical and conceptual models.

In  $M_9$ , the fundamental units are not agents, infrastructures or institutions, but *models*, represented as structured objects with internal logics, constraints and transformation rules.

Thus  $M_9$  is the minimal space where *theory* becomes an admissible continuum.

#### 1. Admissible Configuration Space $\Omega(M_9)$

$$\Omega(M_9) = \{(\mathcal{M}, \text{Mor}, A_{\text{model}}, \Theta_{\text{coh}}, C_{\text{theory}}, \Sigma_9) \mid \text{coherence conditions hold}\}.$$

Components:

- $\mathcal{M}$  — the set of admissible models (mathematical, physical, chemical, biological, cognitive, social),
- $\text{Mor}$  — morphisms between models (interpretations, reductions, equivalences, functors),
- $A_{\text{model}}$  — axes of theoretical variation,
- $\Theta_{\text{coh}}$  — coherence thresholds,
- $C_{\text{theory}}$  — cycles of theoretical refinement,
- $\Sigma_9$  — global structural constraints on theory-space.

Admissibility requires:

1. existence of stable models:  $\mathcal{M} \neq \emptyset$ ,
2. existence of meaningful morphisms between models,
3. non-trivial theoretical axes,
4. coherence:  $d_{\text{coh}}(M_i, M_j) < \Theta_{\text{coh}}$  for admissible pairs,
5. possibility of theoretical evolution under operator  $E$ .

#### 2. Axes $A(M_9)$

$$A(M_9) = \{A_{\text{syntax}}, A_{\text{semantics}}, A_{\text{abstraction}}, A_{\text{idealization}}, A_{\text{approximation}}, A_{\text{domain}}, A_{\text{logic}}\}.$$

Descriptions:

- $A_{\text{syntax}}$  — formal languages, symbolic structures,
- $A_{\text{semantics}}$  — interpretation spaces,
- $A_{\text{abstraction}}$  — degrees of generality,
- $A_{\text{idealization}}$  — allowed simplifications,
- $A_{\text{approximation}}$  — error and convergence structures,
- $A_{\text{domain}}$  — domain-specific axes (physics, chemistry, biology...),
- $A_{\text{logic}}$  — logical rules, inference systems.

A necessary condition for model-theoretic life:

$$\dim(A_{\text{syntax}}) \geq 1, \quad \dim(A_{\text{semantics}}) \geq 1.$$

### 3. Potentials $P(M_9)$

$$P(M_9) = (F_{\text{expressivity}}, F_{\text{consistency}}, F_{\text{coherence}}, F_{\text{generalization}}, F_{\text{predictive}}, F_{\text{compression}}).$$

Interpretation:

- $F_{\text{expressivity}}$  — potential enabling rich theoretical descriptions,
- $F_{\text{consistency}}$  — potential ensuring logical stability,
- $F_{\text{coherence}}$  — potential keeping models mutually interpretable,
- $F_{\text{generalization}}$  — capacity for abstraction and domain transfer,
- $F_{\text{predictive}}$  — accuracy and empirical adequacy,
- $F_{\text{compression}}$  — potential for minimal sufficient representations.

Conditions for existence of  $K_9$ :

$$F_{\text{coherence}} > \Theta_{\text{fragmentation}}, \quad F_{\text{consistency}} > \Theta_{\text{paradox}}.$$

### 4. Thresholds $\Theta(M_9)$

Key coherence thresholds:

- $\Theta_{\text{coh}}$  — maximal allowable divergence for models to coexist,
- $\Theta_{\text{consistency}}$  — threshold of internal logical stability,
- $\Theta_{\text{predictive}}$  — minimal predictive accuracy,
- $\Theta_{\text{paradox}}$  — boundary where inconsistency collapses the continuum,
- $\Theta_{\text{fragmentation}}$  — threshold for theoretical disintegration.

Crossing  $\Theta_{\text{paradox}}$  forces collapse of  $\Omega(K_9)$ , analogous to institutional collapse in  $K_8$ .

### 5. Flows $J(M_9)$

$$J(M_9) = (J_{\text{inference}}, J_{\text{interpretation}}, J_{\text{translation}}, J_{\text{abstraction}}, J_{\text{specialization}}, \nabla_i T_9).$$

Interpretation:

- $J_{\text{inference}}$  — flow of deductions and proofs,
- $J_{\text{interpretation}}$  — reinterpretation between models,
- $J_{\text{translation}}$  — translation across formal languages,
- $J_{\text{abstraction}}$  — flow towards higher generality,
- $J_{\text{specialization}}$  — domain-specific derivations,
- $\nabla_i T_9$  — gradients of theoretical tension.

Stability condition:

$$\oint J_{\text{inference}} dt \approx 0 \quad \Rightarrow \quad \text{model remains logically stable across cycles.}$$

## 6. Cycles $C(M_9)$

Fundamental cycles:

- $C_{\text{theory}}$  — hypothesis  $\rightarrow$  derivation  $\rightarrow$  evaluation  $\rightarrow$  revision,
- $C_{\text{abstraction}}$  — concrete  $\rightarrow$  abstract  $\rightarrow$  generalized  $\rightarrow$  instantiated,
- $C_{\text{interpretation}}$  — model  $\rightarrow$  mapping  $\rightarrow$  reinterpretation,
- $C_{\text{coherence}}$  — local fits  $\rightarrow$  global consistency  $\rightarrow$  refit cycles,
- $C_{\text{unification}}$  — domain models  $\rightarrow$  cross-domain structure  $\rightarrow$  unified framework.

A living theoretical continuum requires:

$$\oint_{C_{\text{coherence}}} dA \approx 0.$$

## 7. Boundary $\partial\Omega(M_9)$

Conditions approaching the boundary:

- breakdown of inference closure,
- loss of coherence:  $d_{\text{coh}} > \Theta_{\text{coh}}$ ,
- proliferation of paradoxes,
- divergence of formal languages beyond mutual interpretability,
- failure of model–world correspondence (predictive collapse),
- runaway theoretical tension  $T_9$ .

Crossing  $\partial\Omega(M_9)$  collapses theory to fragmented  $M_8$ -admissible structures.

## 8. Relation to $M_8$ and $M_{10}$

**From  $M_8$  to  $M_9$ :**

Necessary conditions:

$$A_{\text{syntax}} \neq \emptyset, \quad A_{\text{semantics}} \neq \emptyset, \quad \Theta_{\text{coh}} \text{ finite.}$$

Symbolic and scientific structures of  $M_8$  lift into model spaces of  $M_9$  via the operator  $\Psi$ .

**Toward  $M_{10}$  (meta-theoretical space):**

$M_{10}$  introduces:

- categories of model-categories,
- functorial dynamics,
- meta-coherence thresholds,
- cycles of self-description,
- the space where meta-theories themselves form continua.

Thus  $M_9$  provides the structured model-space that becomes the substrate for  $K_{10}$ .



### O.2.10 $M_{10}$ Overview

$M_{10}$  is the meta-space that enables the existence of meta-theoretical continua ( $K_{10}$ ). While  $M_9$  hosts *models* and their morphisms (interpretations, equivalences, translations),  $M_{10}$  hosts *categories of model-categories*, together with functorial transformations, coherence constraints and recursive self-description dynamics.

In  $M_{10}$ , the fundamental units are not models but *theories-of-models*: structured objects describing how entire families of models relate, transform, unify or collapse.

Thus  $M_{10}$  represents the minimal logical space in which *meta-theory becomes a living continuum*.

#### 1. Admissible Configuration Space $\Omega(M_{10})$

$$\Omega(M_{10}) = \{(\mathcal{C}, \text{Fun}, A_{\text{meta}}, \Theta_{\text{meta}}, C_{\text{meta}}, \Sigma_{10}) \mid \text{meta-coherence conditions hold}\}.$$

Components:

- $\mathcal{C}$  — admissible categories-of-model-categories,
- $\text{Fun}$  — functors between such categories (meta-morphisms),
- $A_{\text{meta}}$  — axes of meta-theoretical variation,
- $\Theta_{\text{meta}}$  — meta-coherence thresholds,
- $C_{\text{meta}}$  — cycles of meta-theoretical refinement,
- $\Sigma_{10}$  — global structural constraints defining  $M_{10}$ .

Admissibility requires:

1. existence of at least one stable category  $\mathcal{C} \neq \emptyset$ ,
2. existence of meta-morphisms (functors) preserving essential structure,
3. finite meta-coherence measure,
4. closure of inference at the categorical level,
5. compatibility with  $M_9$  via projection  $\pi_9 : M_{10} \rightarrow M_9$ .

$$\Omega(K_{10}) \subseteq \Omega(M_{10}) \quad \text{as required by Theorem 8.}$$

#### 2. Axes $A(M_{10})$

$$A(M_{10}) = \{A_{\text{functor}}, A_{\text{natural}}, A_{\text{meta-logic}}, A_{\text{equivalence}}, A_{\text{coherence}}, A_{\text{self}}, A_{\text{hierarchy}}\}.$$

Interpretation:

- $A_{\text{functor}}$  — variation of functorial mappings,
- $A_{\text{natural}}$  — structure of natural transformations,
- $A_{\text{meta-logic}}$  — logical rules governing meta-inference,
- $A_{\text{equivalence}}$  — axes of categorical equivalence,
- $A_{\text{coherence}}$  — higher coherence laws (MacLane-type),
- $A_{\text{self}}$  — axes enabling reflexivity and self-description,
- $A_{\text{hierarchy}}$  — axes of vertical level transitions ( $K_9 \rightarrow K_{10} \rightarrow K_{11}$ ).

A key innovation of  $M_{10}$  relative to  $M_9$ :

$$A_{\text{self}} \neq \emptyset.$$

This axis is responsible for recursive structures and meta-theoretical closure.

### 3. Potentials $P(M_{10})$

$$P(M_{10}) = \{F_{\text{meta-coherence}}, F_{\text{meta-consistency}}, F_{\text{reflection}}, F_{\text{unification}}, F_{\text{categorical depth}}, F_{\text{meta-prediction}}\}.$$

Interpretation:

- $F_{\text{meta-coherence}}$  — ability to maintain consistent relations across categories,
- $F_{\text{meta-consistency}}$  — avoidance of contradictions in meta-inference,
- $F_{\text{reflection}}$  — ability for the meta-framework to describe itself,
- $F_{\text{unification}}$  — capacity to unify disparate model-categories,
- $F_{\text{categorical depth}}$  — ability to sustain higher structures (2-categories, n-categories, infinity-categories),
- $F_{\text{meta-prediction}}$  — predictive constraints on how models evolve in  $M_9$ .

Existence of  $K_{10}$  requires:

$$F_{\text{meta-coherence}} > \Theta_{\text{fragmentation}}^{(10)}, \quad F_{\text{reflection}} > \Theta_{\text{self-collapse}}.$$

### 4. Thresholds $\Theta(M_{10})$

Key thresholds:

- $\Theta_{\text{meta}}$  — global meta-coherence threshold,
- $\Theta_{\text{self-consistency}}$  — limit of reflexive consistency,
- $\Theta_{\text{unification}}$  — minimum conditions for unifying model-categories,
- $\Theta_{\text{categorical}}$  — threshold for collapse of higher categorical structure,
- $\Theta_{\text{self-collapse}}$  — boundary where self-reference destroys the continuum.

Crossing  $\Theta_{\text{self-collapse}}$  yields a Gödel-type breakdown: the continuum no longer has an admissible  $\Omega$ .

### 5. Flows $J(M_{10})$

$$J(M_{10}) = (J_{\text{functorial}}, J_{\text{natural-transform}}, J_{\text{meta-inference}}, J_{\text{coherence-repair}}, \nabla_i T_{10}, J_{\text{self}})$$

Interpretation:

- $J_{\text{functorial}}$  — flows of functors transforming entire categories,
- $J_{\text{natural-transform}}$  — adjustments between functors,
- $J_{\text{meta-inference}}$  — inference rules acting at the meta-level,
- $J_{\text{coherence-repair}}$  — flows reestablishing higher coherence,
- $\nabla_i T_{10}$  — gradients of meta-theoretical tension,
- $J_{\text{self}}$  — recursive flows in reflexive structures.

Coherence condition:

$$\oint_{C_{\text{meta}}} dA_{\text{coherence}} \approx 0.$$

## 6. Cycles $C(M_{10})$

Fundamental cycles:

- $C_{\text{meta-theory}}$  — theory-of-models  $\rightarrow$  meta-analysis  $\rightarrow$  refinement  $\rightarrow$  new meta-theory,
- $C_{\text{coherence}}$  — local categorical checks  $\rightarrow$  global coherence  $\rightarrow$  higher-order correction,
- $C_{\text{abstraction}}$  —  $K_9$ -level description  $\rightarrow$  meta-categorical abstraction  $\rightarrow$  reapplication,
- $C_{\text{reflection}}$  — description of theory  $\rightarrow$  description of the description,
- $C_{\text{self}}$  — recursive cycles that make  $K_{10}$  possible,
- $C_{\text{unification}}$  — disparate model-categories  $\rightarrow$  adjunctions  $\rightarrow$  equivalence  $\rightarrow$  unified framework.

Life of a meta-theoretical continuum requires:

$$\oint_{C_{\text{self}}} dA_{\text{self}} \approx 0.$$

## 7. Boundary $\partial\Omega(M_{10})$

Approaching the boundary:

- collapse of functorial structure,
- divergence of natural transformations,
- non-closure of meta-inference,
- loss of meta-coherence ( $D > \Theta_{\text{meta}}$ ),
- runaway self-reference,
- collapse of higher-order categories.

Crossing  $\partial\Omega(M_{10})$  forces collapse to  $M_9$ -admissible structures.

## 8. Relation to $M_9$ and $M_{11}$

**From  $M_9$  to  $M_{10}$ :**

Necessary conditions:

$$A_{\text{functor}} \neq \emptyset, \quad A_{\text{coherence}} \neq \emptyset.$$

The birth operator  $\Psi$  lifts model-space ( $M_9$ ) to meta-model-space ( $M_{10}$ ).

**Towards  $M_{11}$ :**

$M_{11}$  introduces:

- meta-meta-theoretical spaces,
- global compatibility across meta-hierarchies,
- operators acting on towers of model-categories,
- universal constraints on admissible continua.

Thus  $M_{10}$  is the essential intermediate space enabling recursive theoretical organization.

### O.2.11 $M_{11}$ Overview

$M_{11}$  is the highest meta-space required for continua up to level  $K_{11}$ . Where  $M_{10}$  hosts categories of model-categories (meta-theoretical structures),  $M_{11}$  hosts *hierarchies of meta-theoretical towers* and provides the global logical environment for their compatibility, reflection, and higher-order stabilisation.

In this space, the fundamental units are *meta-meta-categorical* objects: structured towers

$$(\mathcal{C}^{(0)}, \mathcal{C}^{(1)}, \dots, \mathcal{C}^{(n)}, \dots)$$

where each  $\mathcal{C}^{(i)}$  is itself a category-of-categories at level  $i$  and morphisms between towers are transformations that preserve their full vertical structure.

Thus  $M_{11}$  is the minimal logical environment in which *meta-hierarchical continua* can exist.

#### 1. Admissible Configuration Space $\Omega(M_{11})$

$$\Omega(M_{11}) = \{(\mathbf{T}, \mathfrak{F}, A_{11}, \Theta_{11}, C_{11}, \Sigma_{11}) \mid \text{global hierarchy-coherence holds}\}.$$

Where:

- $\mathbf{T}$  — admissible infinite or finite towers of meta-categories,
- $\mathfrak{F}$  — meta-functors acting on towers,
- $A_{11}$  — axes of meta-hierarchical variation,
- $\Theta_{11}$  — global coherence thresholds at the highest level,
- $C_{11}$  — cycles of trans-hierarchical self-organisation,
- $\Sigma_{11}$  — structural constraints defining  $M_{11}$ .

Admissibility requires:

1. hierarchy must close under vertical composition,
2. no divergence of coherence across levels,
3. existence of at least one stable trans-hierarchical cycle,
4. compatibility with  $M_{10}$  through canonical projection  $\pi_{10} : M_{11} \rightarrow M_{10}$ ,
5. finiteness of global meta-tension.

$$\Omega(K_{11}) \subseteq \Omega(M_{11}) \quad (\text{Theorem 8: necessity and sufficiency of compatibility}).$$

#### 2. Axes $A(M_{11})$

$$A(M_{11}) = \{A_{\text{tower}}, A_{\text{coherence}}, A_{\text{depth}}, A_{\text{reflection}}, A_{\text{translation}}, A_{\text{hierarchical consistency}}, A_{\text{global}}\}.$$

Interpretation:

- $A_{\text{tower}}$  — variation across levels of the meta-theoretical tower,
- $A_{\text{coherence}}$  — maintenance of coherence across multiple levels,
- $A_{\text{depth}}$  — hierarchical depth and structural recursion,
- $A_{\text{reflection}}$  — self-description across levels,
- $A_{\text{translation}}$  — translation between different meta-hierarchies,
- $A_{\text{hierarchical consistency}}$  — preservation of structural rules across the tower,

- $A_{\text{global}}$  — axes governing global admissibility of entire hierarchies.

Key feature:

$\dim(M_{11}) > \dim(M_{10})$  (Theorem: monotonicity and irreversibility of dimension growth).

### 3. Potentials $P(M_{11})$

$$P(M_{11}) = \{F_{\text{global coherence}}, F_{\text{hierarchical depth}}, F_{\text{meta-reflection}}, F_{\text{compatibility}}, F_{\text{translation}}, F_{\text{unification}}, F_{\text{stability}}\}.$$

Interpretation:

- $F_{\text{global coherence}}$  — ability to maintain coherence across all levels,
- $F_{\text{hierarchical depth}}$  — capacity to sustain deep tower structures,
- $F_{\text{meta-reflection}}$  — ability to reflect on entire hierarchies,
- $F_{\text{compatibility}}$  — harmony between levels,
- $F_{\text{translation}}$  — mapping between different towers,
- $F_{\text{unification}}$  — unification across hierarchical scales,
- $F_{\text{stability}}$  — overall stability of meta-hierarchical dynamics.

Existence of  $K_{11}$  requires:

$$F_{\text{global coherence}} > \Theta_{11}^{\text{collapse}}, \quad F_{\text{meta-reflection}} > \Theta_{11}^{\text{self-break}}.$$

### 4. Thresholds $\Theta(M_{11})$

Critical thresholds:

- $\Theta_{11}^{\text{coherence}}$  — minimal global coherence,
- $\Theta_{11}^{\text{depth}}$  — minimal structural depth,
- $\Theta_{11}^{\text{translation}}$  — threshold for interoperability,
- $\Theta_{11}^{\text{collapse}}$  — boundary where the hierarchy fails,
- $\Theta_{11}^{\text{self-break}}$  — self-reference instability.

Crossing  $\Theta_{11}^{\text{collapse}}$  yields hierarchical fragmentation:

$$\mathbf{T} \rightarrow (\mathcal{C}^{(i)}) \quad \text{without admissible vertical structure.}$$

### 5. Flows $J(M_{11})$

$$J(M_{11}) = (J_{\text{tower}}, J_{\text{coherence}}, J_{\text{translation}}, J_{\text{meta-inference}}, \nabla_i T_{11}, J_{\text{reflection}}, J_{\text{global}}).$$

Interpretation:

- $J_{\text{tower}}$  — flows acting on hierarchical depth,
- $J_{\text{coherence}}$  — flows restoring multi-level consistency,
- $J_{\text{translation}}$  — flows mapping one tower into another,
- $J_{\text{meta-inference}}$  — inference acting across the full tower,

- $\nabla_i T_{11}$  — gradients of global meta-tension,
- $J_{\text{reflection}}$  — recursion and reflection across levels,
- $J_{\text{global}}$  — flows determining global admissibility.

## 6. Cycles $C(M_{11})$

Fundamental cycles:

- $C_{\text{hierarchy}}$  — formation, refinement and stabilisation of the tower,
- $C_{\text{coherence}}$  — coherence restoration loops across levels,
- $C_{\text{translation}}$  — mapping towers and verifying equivalence,
- $C_{\text{reflection}}$  — recursive cycles enabling understanding of the hierarchy from within,
- $C_{\text{global}}$  — global admissibility cycles ensuring  $\Omega(K_{11}) \subseteq \Omega(M_{11})$ .

Criterion of life:

$$\oint_{C_{\text{global}}} dA_{\text{global}} \approx 0.$$

## 7. Boundary $\partial\Omega(M_{11})$

Near the boundary:

- hierarchical divergence,
- loss of coherence between levels,
- unresolvable contradictions,
- runaway recursion,
- breakdown of global admissibility.

Crossing  $\partial\Omega(M_{11})$  forces collapse into  $M_{10}$ -admissible structures.

## 8. Relation to $M_{10}$ and higher meta-spaces

**Relation to  $M_{10}$ :**

$$A_{\text{hierarchical consistency}} \in A(M_{11}) \setminus A(M_{10}),$$

thus  $M_{11}$  strictly extends  $M_{10}$ .

Vertical projection:

$$\pi_{10} : M_{11} \rightarrow M_{10}$$

forgets hierarchical depth and retains only meta-theoretical structure.

**Towards  $M_{12}$  (if defined):**

$M_{12}$  would host universal meta-meta-logical structures and global constraints on entire families of meta-hierarchies.

$M_{11}$  is the minimal space where such extension becomes admissible.

### O.2.12 $M_{12}$ Overview

$M_{12}$  is the highest meta-space required for continua up to level  $K_{12}$ . While  $M_{11}$  hosts structured towers of meta-theoretical hierarchies,  $M_{12}$  hosts *universal meta-logical environments* governing the admissibility, coherence and global constraints for entire *families of meta-hierarchies*.

In short:

$M_{12}$  = the minimal meta-logical space in which the entire tower  $M_1, \dots, M_{11}$  can coexist coherently.

It is not simply “one level higher”; it introduces a new class of differences (Ax. 21), a new axis unavailable to  $M_{11}$ , and a global constraint governing all previous M-spaces.

This makes  $M_{12}$  the terminal environment for Core 1.3.

#### 1. Admissible Configuration Space $\Omega(M_{12})$

$$\Omega(M_{12}) = \{(\mathcal{U}, A_{12}, \Theta_{12}, C_{12}, \Lambda_{12}, T_{12}) \mid \text{global meta-logical admissibility holds}\}.$$

Where:

- $\mathcal{U}$  — admissible *universes of meta-hierarchies*, i.e. collections of towers from  $M_{11}$  equipped with compatibility relations;
- $A_{12}$  — axes determining variation across universes of meta-hierarchies;
- $\Theta_{12}$  — global thresholds ensuring logical admissibility;
- $C_{12}$  — cycles governing stabilisation at the universal level;
- $\Lambda_{12}$  — universal constraints (logical, structural, categorical);
- $T_{12}$  — global structural tension across families of hierarchies.

Admissibility requires:

1. coherence of all embedded meta-hierarchies ( $M_1$ – $M_{11}$ ),
2. existence of at least one universal stabilising cycle,
3. global consistency with the operator  $\Psi$  (meta-space generation),
4. bounded meta-tension  $T_{12} < \infty$ ,
5. preservation of universal structural laws  $\Lambda_{12}$ .

By Theorem 8:

$$\Omega(K_{12}) \subseteq \Omega(M_{12}) \quad \text{is necessary and sufficient for the existence of } K_{12}.$$

#### 2. Axes $A(M_{12})$

$$A(M_{12}) = \{A_{\text{universal}}, A_{\text{meta-logical}}, A_{\text{cross-hierarchy}}, A_{\text{invariance}}, A_{\text{global compatibility}}, A_{\text{admissibility}}\}.$$

Where:

- $A_{\text{universal}}$  — variation across entire universes of meta-hierarchies,
- $A_{\text{meta-logical}}$  — global logical constraints,
- $A_{\text{cross-hierarchy}}$  — variation across  $M_i$  structures,
- $A_{\text{invariance}}$  — invariants preserved across all levels,

- $A_{\text{global compatibility}}$  — compatibility axis for full K/M stacks,
- $A_{\text{admissibility}}$  — the existence axis for  $K_{12}$  itself.

This introduces a new class of differences not embeddable into  $M_{11}$ :

$$A_{\text{universal}} \notin A(M_{11}),$$

thus by Ax. 21:

$$\dim(M_{12}) > \dim(M_{11}).$$

### 3. Potentials $P(M_{12})$

$$P(M_{12}) = \{U_{\text{universal}}, U_{\text{coherence}}, U_{\text{global inference}}, U_{\text{meta-stability}}, U_{\text{admissibility}}, U_{\text{invariant preservation}}\}.$$

Interpretation:

- $U_{\text{universal}}$  — potential supporting whole universes of meta-hierarchies,
- $U_{\text{coherence}}$  — ability to maintain coherence across all embedded M-spaces,
- $U_{\text{global inference}}$  — potential for reasoning across universes,
- $U_{\text{meta-stability}}$  — ability to resist collapse of entire towers,
- $U_{\text{admissibility}}$  — ensures  $\Omega(K_{12}) \subseteq \Omega(M_{12})$ ,
- $U_{\text{invariant preservation}}$  — ensures invariants across levels.

$K_{12}$  exists iff:

$$U_{\text{admissibility}} > \Theta_{12}^{\text{existence}}.$$

### 4. Thresholds $\Theta(M_{12})$

Critical thresholds:

$$\Theta_{12} = \{\Theta_{12}^{\text{coherence}}, \Theta_{12}^{\text{universal}}, \Theta_{12}^{\text{collapse}}, \Theta_{12}^{\text{invariance}}, \Theta_{12}^{\text{reflection}}\}.$$

Interpretation:

- $\Theta_{12}^{\text{coherence}}$  — minimal coherence among all  $M_i$ ,
- $\Theta_{12}^{\text{universal}}$  — minimal universal compatibility,
- $\Theta_{12}^{\text{collapse}}$  — threshold where entire meta-hierarchies disintegrate,
- $\Theta_{12}^{\text{invariance}}$  — minimal preservation of invariants,
- $\Theta_{12}^{\text{reflection}}$  — threshold for self-consistency of meta-logic.

Crossing  $\Theta_{12}^{\text{collapse}}$  yields:

$$\mathfrak{U} \rightarrow \{M_i\}_{i=1}^{11} \quad \text{with loss of universal structure.}$$



## 5. Flows $J(M_{12})$

$$J(M_{12}) = (J_{\text{universal}}, J_{\text{coherence}}, J_{\text{reflection}}, J_{\text{meta-inference}}, J_{\text{cross-hierarchy}}, \nabla T_{12}, J_{\text{stability}}).$$

Interpretation:

- $J_{\text{universal}}$  — flows reorganising universes of meta-hierarchies,
- $J_{\text{coherence}}$  — flows restoring coherence across  $M_1$ – $M_{11}$ ,
- $J_{\text{reflection}}$  — self-referential meta-logical flows,
- $J_{\text{meta-inference}}$  — inference acting on entire universes,
- $J_{\text{cross-hierarchy}}$  — stabilisation across different M-levels,
- $\nabla T_{12}$  — gradients of global structural tension,
- $J_{\text{stability}}$  — flows preventing collapse of universal structure.

## 6. Cycles $C(M_{12})$

Fundamental cycles:

- $C_{\text{universal}}$  — generation and stabilisation of universes of meta-hierarchies,
- $C_{\text{coherence}}$  — coherence-restoring loops across all M-spaces,
- $C_{\text{reflection}}$  — recursive cycles maintaining meta-logical integrity,
- $C_{\text{invariance}}$  — cycles ensuring global invariants,
- $C_{\text{admissibility}}$  — cycles ensuring existence of  $K_{12}$ .

Life criterion:

$$\oint_{C_{\text{universal}}} dA_{\text{universal}} \approx 0.$$

## 7. Boundary $\partial\Omega(M_{12})$

Near the boundary:

- universal incompatibility,
- breakdown of global structural invariants,
- meta-logical incoherence,
- collapse of stabilising cycles,
- divergence of meta-tension.

Crossing the boundary forces reduction to isolated  $M_{11}$  universes.

## 8. Relation to $M_{11}$ and full K/M hierarchy

$$M_{11} \subset M_{12} \quad \text{strictly.}$$

Projection:

$$\pi_{11} : M_{12} \rightarrow M_{11}$$

forgets universal constraints and retains only tower-structured hierarchies.

$M_{12}$  is the minimal environment in which:

$$\Psi : M_{11} \rightarrow M_{12}$$

is well-defined (Run 11).

The existence of  $K_{12}$  requires:

$$\Omega(K_{12}) \subseteq \Omega(M_{12}).$$

$M_{12}$  is thus the terminal meta-space for the Core.

### O.3 Cross-level structures

#### O.3.1 Purpose of the Cross-K layer

The Cross-K layer formalises all structures, transitions and constraints that *link different continuum levels*  $K_0 \rightarrow K_{12}$ . While each  $K$ -level has its own ontology (axes, potentials, thresholds, flows, cycles, domains  $\Omega(K)$ , boundaries  $\partial\Omega(K)$  and continuum measure  $k(K)$ ), many phenomena in the Core require **relations across levels**. These relations are neither optional nor external: they are part of the intrinsic architecture of a dynamic continuum.

The Cross-K layer therefore:

- defines **mapping structures** between levels;
- specifies **transition operators** and their constraints;
- introduces **consistency conditions** ensuring that evolution across adjacent levels is mathematically and conceptually valid;
- provides **global landscape rules**, describing how the entire stack of  $K$ -levels forms a coherent meta-continuum.

#### O.3.2 Definition of Cross-K structures

A *Cross-K structure* is any formal object that:

1. involves at least two levels  $K_i, K_j$  with  $i \neq j$ ;
2. establishes a relation  $R_{ij}$  between their components  $(A, P, \Theta, J, C, \Omega, \partial\Omega, k)$ ;
3. satisfies the general compatibility rule:

$$R_{ij} : K_i \leftrightarrow K_j$$

while preserving continuity and allowing valid transitions under the global operator  $E$ .

Thus Cross-K structures include:

- **transition maps** (dimensional, structural, energetic);
- **inheritance relations** (axes, potentials, thresholds);
- **boundary transformations** between  $\partial\Omega(K_i)$  and  $\partial\Omega(K_j)$ ;
- **cycle projections and lifts** (mapping cycles from one level into another);
- **global constraints** ensuring that no level violates the rules of its neighbours.

### O.3.3 Role of this master file

This master module intentionally remains a **thin orchestration layer**. It does not store heavy theoretical content. Instead, it:

- introduces the concept of Cross-K structures,
- provides the global definition and purpose,
- and assembles all detailed cross-level analyses through `\input`.

The individual transition sections are organised as:

- global landscape (all-level perspective),
- local transitions ( $K_i \rightarrow K_{i+1}$ ),
- special two-way or non-adjacent mappings where required.

### O.3.4 Module structure

The following files are orchestrated by this module:

- `crossk_global_landscape.tex` — global multi-level geometry;
- `crossk_k0_k1.tex`, `crossk_k1_k2.tex`, ... — adjacent-level transition blocks;
- `crossk_k10_k11.tex`, `crossk_k11_k12.tex` — upper-level transitions and meta-structures.

### O.3.5 Inclusion of detailed sections

The detailed cross-K analyses can be included as:

```
\input{content/crossk/crossk_global_landscape.tex}
\input{content/crossk/crossk_k0_k1.tex}
\input{content/crossk/crossk_k1_k2.tex}
...
\input{content/crossk/crossk_k11_k12.tex}
```

This file thus defines the **conceptual container** for the Cross-K domain of OC Core 1.3.

### O.3.6 Global cross-K landscape

The cross-K landscape describes how individual continua  $K_d$  are embedded into a single multi-level structure. Instead of treating each level as an isolated theory, the Ontology of Continua organises them as a chain of dimension-increasing transitions:

$$R_d : K_d \rightarrow K_{d+1},$$

where each  $R_d$  is a constrained transition that:

- preserves the core axioms (non-emptiness of  $\Omega$ , thresholds, flows, cycles),
- introduces at least one new axis  $A_{\text{new}} \in M_d \setminus A(K_d)$ ,
- passes a dimensional threshold  $T_d > \Theta_{d,\text{dim}}$ ,
- produces a strictly richer space of admissible states  $\Omega(K_{d+1}) \supset \Omega(K_d)$ .

The global landscape is therefore not a flat list of levels, but a structured sequence of phase transitions in which each  $K_d$  emerges from a predecessor and constrains its successors.

### O.3.7 1. Levels involved and their roles

At a coarse resolution, the chain can be grouped into four bands:

1. **Physical band** ( $K_0$ – $K_2$ ): basic structure, action and percolation of connectivity.
2. **Material and biological band** ( $K_3$ – $K_5$ ): fields, chemistry, compartments and proto-neural dynamics.
3. **Cognitive and social band** ( $K_6$ – $K_8$ ): meaning, social systems and civilisational infrastructures.
4. **Theoretical band** ( $K_9$ – $K_{11}$ ): theories, metatheories and trans-metatheoretical constraints.

Each band is internally coherent but also linked by explicit cross-level operators  $R_d$  and by shared meta-spaces  $M_n$ .

### O.3.8 2. Shared axes and thresholds across levels

Across the chain  $K_0$ – $K_{11}$ , several structural motifs reappear:

- **Axes**  $A(K_d)$ : each level introduces new axes but inherits transformed versions of earlier ones (e.g. energy, gradients, information, roles, symbols).
- **Thresholds**  $\Theta(K_d)$ : existence, stability, critical, death and dimensional thresholds appear at all levels, specialised to the local physics/biology/social/theoretical context.
- **State domains**  $\Omega(K_d)$ : every continuum has a non-empty domain of admissible states, with a boundary  $\partial\Omega(K_d)$  defined by violated thresholds.
- **Cycles**  $C(K_d)$ : minimal stabilising cycles exist on each level (from trivial physical cycles to institutional and meta-theoretical cycles).

The global landscape can be seen as the evolution of these structural motifs as dimension and complexity increase.

### O.3.9 3. Cross-level flows, cycles, and tensions

Cross-level dynamics can be summarised by three families of objects:

**Flows.** Cross-level flows  $J_{d,d+1}$  map states and structures from  $K_d$  to  $K_{d+1}$  (e.g. chemical gradients shaping  $K_4$ , neural patterns inducing  $K_6$ , social norms supporting  $K_8$ ).

**Cycles.** Some cycles explicitly span multiple levels, such as civilisation–theory feedback loops ( $K_8$ – $K_9$ – $K_8$ ) or meta-theory–practice cycles ( $K_9$ – $K_{10}$ – $K_8$ ).

**Tension.** Cross-level tension  $T_{d,d+1}$  measures mismatch between the constraints of  $K_d$  and  $K_{d+1}$ ; excessive tension can prevent the birth of a new level or trigger collapse downwards.

Formally, one can write a generic cross-level tension functional

$$T_{d,d+1} = F(A(K_{d+1}) - A(K_d), P(K_{d+1}) - P(K_d), J_{d,d+1}, \Theta(K_{d+1}) - \Theta(K_d)),$$

with  $T_{d,d+1} > \Theta_{d,\dim}$  signalling a dimensional transition.

### O.3.10 4. Birth and death conditions in the global chain

Birth of a new level  $K_{d+1}$  is governed by three conditions:

1. **Availability of a new axis:** there exists  $A_{\text{new}} \in M_d$  such that  $A_{\text{new}} \notin A(K_d)$ .
2. **Dimensional tension:** structural tension exceeds a critical threshold,  $T_d > \Theta_{d,\text{dim}}$ , making the old configuration unstable without a new dimension.
3. **Admissible state extension:** the new domain satisfies  $\Omega(K_{d+1}) \neq \emptyset$  and  $\Omega(K_{d+1}) \supset \Omega(K_d)$ .

Death of a level  $K_d$  may occur locally (collapse of a specific continuum) or globally (destruction of its entire class) when:

$$\Omega(K_d) = \emptyset \quad \text{or} \quad k(K_d) \rightarrow 0 \quad \text{or} \quad \tau_{\text{cycle}}(K_d) \rightarrow \infty.$$

Global collapse of multiple adjacent levels can be understood as a cascade of threshold violations propagating across the chain.

### O.3.11 5. Canonical cross-level chains

Several cross-level chains are of particular importance:

- **Physical–chemical–biological chain:**  $K_1 \rightarrow K_2 \rightarrow K_3 \rightarrow K_4 \rightarrow K_5$  (from action and percolation to membranes and proto-neural dynamics).
- **Biological–cognitive–social chain:**  $K_4 \rightarrow K_5 \rightarrow K_6 \rightarrow K_7$  (from gradients and spikes to cognition and institutions).
- **Social–civilisational–theoretical chain:**  $K_7 \rightarrow K_8 \rightarrow K_9 \rightarrow K_{10} \rightarrow K_{11}$  (from norms and roles to civilisational technologies and meta-frameworks).

The detailed cross-K modules in the technical derivation atlas refine these canonical chains by providing explicit descriptions of shared axes, thresholds, flows, cycles and death conditions for each adjacent transition.

In this sense, the global cross-K landscape is the *map* of how all individual continua  $K_d$  fit into a single, coherent, multi-level structure.

### O.3.12 K0–K1 cross-level structure

The transition from  $K_0$  to  $K_1$  represents the first and most fundamental cross-level structure in the entire hierarchy of continua. It describes the birth of distinguishable states, the appearance of an explicit axis, and the formation of a non-trivial state space  $\Omega(K_1)$  from the minimal pre-ontological substrate of  $K_0$ .

This section summarises the structural dependencies, inherited components, transition operators, and threshold conditions for the emergence of  $K_1$ .

#### O.3.13 1. Levels involved and their roles

**K0.**  $K_0$  is defined by the *axiom of difference and connectedness*. It contains:

- a minimal domain of potential states without explicit axes;
- a structural difference measure  $\Delta(s_1, s_2)$ ;
- the existence threshold  $\Theta_0$  ensuring non-triviality:  $0 < \Delta < \varepsilon$  for some  $\varepsilon > 0$ ;
- no time, no flows  $J$ , no cycles  $C$ , no dynamics.

**K1.**  $K_1$  is the first continuum with:

- an explicit axis  $A_1$  capturing a stable class of distinctions;
- a defined state space  $\Omega(K_1)$  with boundary  $\partial\Omega(K_1)$ ;
- a primitive energy/potential structure  $P_1$ ;
- minimal flows  $J_1$  and a trivial cycle  $C_{\text{triv}}$ ;
- explicit thresholds  $\Theta_1$  (existence, gradient, stability).

The role of the  $K_0 \rightarrow K_1$  transition is the **generation of structure**: the formation of the first representable dimension in the continuum.

### O.3.14 2. Shared and inherited axes and thresholds

Although  $K_0$  has no axes, several of its structural properties are inherited by  $K_1$  in transformed form:

**Difference  $\Delta$ .** The primitive difference structure of  $K_0$  becomes the metric component of the axis  $A_1$  in  $K_1$ :

$$A_1 \sim f(\Delta),$$

meaning that the existence of distinguishability in  $K_0$  is what enables the emergence of an explicit coordinate in  $K_1$ .

**Thresholds.** The existence threshold  $\Theta_0$  becomes a *lower bound* on  $K_1$ : if  $\Theta_0$  fails (i.e.,  $\Delta = 0$ ), then no axis can be created and  $\Omega(K_1) = \emptyset$ .

Conversely,  $\Theta_1$  introduces new stability conditions for gradients, boundaries, and allowable flows. These thresholds do not exist in  $K_0$  and are born only after the first axis appears.

### O.3.15 3. Cross-level flows, cycles, and tensions

**Flows.** There are no intrinsic flows in  $K_0$ , so the operator  $J_1$  of  $K_1$  has no precursor. Instead, it arises from the newly available structure:

$$J_1 : A_1 \rightarrow \Omega(K_1)$$

represents movements along or across the new axis.

**Cycles.**  $K_0$  cannot support cycles because it lacks time and flows. Thus the trivial cycle of  $K_1$ :

$$C_{\text{triv}} : s \mapsto s,$$

is the first possible cyclic structure in the hierarchy.

**Tension.** Cross-level structural tension between  $K_0$  and  $K_1$  is defined by:

$$T_{0,1} = g(\Delta - \Theta_0, P_1 - \Theta_1).$$

If  $T_{0,1}$  exceeds its dimensional threshold  $\Theta_{1,\text{dim}}$ , the emergent axis becomes unstable or collapses.

### O.3.16 4. Birth and death conditions across the levels

**Birth of  $K_1$ .**  $K_1$  emerges when:

$$\Delta > \Theta_0 \quad \text{and} \quad T_0 > \Theta_{0,\text{dim}},$$

allowing the appearance of a new axis  $A_1$ :

$$A_1 \in M_1 \setminus A(K_0).$$

This is the action of the operator  $\Psi_{0 \rightarrow 1}$  formalised in the Core: it extends the minimal structure of  $K_0$  into a representational continuum  $K_1$ .

**Death propagation.** If  $\Theta_0$  fails,  $K_1$  cannot exist:

$$\Theta_0^{\text{death}} \Rightarrow \Omega(K_1) = \emptyset.$$

If  $\Theta_1$  fails, the continuum  $K_1$  collapses, but  $K_0$  remains intact, because  $K_0$  has no structural dependencies on  $K_1$ .

### O.3.17 5. Conceptual examples

**Emergence of the first axis from minimal differences.** Any situation where a raw distinction (e.g., “state A differs from state B”) can be stabilised and parameterised corresponds to a realisation of the  $K_0 \rightarrow K_1$  transition. The axis  $A_1$  embodies this stabilised distinction.

**Boundary formation.** Once the axis exists, boundaries become meaningful:

$$\partial\Omega(K_1) = \{s \in \Omega(K_1) \mid \nabla A_1 \text{ reaches a threshold}\}.$$

**First flows.** With an axis and boundaries in place, gradient-like flows arise: an impossibility in  $K_0$ , but a natural consequence of the stabilised distinction in  $K_1$ .

These examples illustrate the global significance of the  $K_0 \rightarrow K_1$  transition:

**It is the birth of all representable structure in the continuum hierarchy.**

### O.3.18 $K_1$ – $K_2$ cross-level structure

The transition  $K_1 \rightarrow K_2$  marks the emergence of geometric and energetic structure from the minimal representational continuum. Where  $K_1$  contains a single axis, primitive potentials and trivial flows,  $K_2$  introduces:

- multiple axes forming a coordinate structure,
- explicit potential–gradient relations,
- non-trivial flows  $J_2$ ,
- causal ordering,
- and a richer domain  $\Omega(K_2)$  supporting physical dynamics.

The cross-level structure captures how the simple representational geometry of  $K_1$  is extended into the physical-like geometry of  $K_2$ .

### O.3.19 1. Levels involved and their roles

**K1.**  $K_1$  provides:

- a single axis  $A_1$ ,
- a primitive potential  $P_1$ ,
- basic boundary structure  $\partial\Omega(K_1)$ ,
- first gradient-based flows  $J_1$ ,
- trivial cycles but no full geometric or causal structure.

**K2.**  $K_2$  expands the dimensionality and introduces:

- at least one new axis  $A_2$ ,
- a multi-dimensional coordinate chart,
- energetic potentials and fields,
- non-trivial dynamical flows  $J_2$ ,
- causal ordering and physically interpretable constraints,
- a richer state space  $\Omega(K_2)$  with non-trivial topology.

$K_2$  is the first continuum that supports percolation, field propagation and dynamical behaviour.

### O.3.20 2. Shared and inherited axes and thresholds

**Axes.** The axis  $A_1$  of  $K_1$  becomes one coordinate axis inside  $A(K_2)$ :

$$A(K_1) \subset A(K_2) = \{A_1, A_2, \dots\}.$$

The new axis  $A_2$  allows the emergence of gradients, fields and transport across a multi-dimensional domain.

**Thresholds.** The existence threshold  $\Theta_1$  ensures stability of  $A_1$ ; if violated, the entire transition collapses:

$$\Theta_1^{\text{death}} \Rightarrow \Omega(K_2) = \emptyset.$$

$K_2$  introduces new threshold classes:

- **energetic thresholds** for fields and potentials,
- **gradient thresholds** governing flows,
- **percolation thresholds** for connectivity,
- **causality thresholds** ensuring ordering of events.

These thresholds refine and extend the structure inherited from  $K_1$ .

### O.3.21 3. Cross-level flows, cycles, and tensions

**Flows.** Flows  $J_2$  descend from  $J_1$ , but gain complexity due to the additional axis and energetic potentials:

$$J_2 = \nabla P_2 + \text{transport terms along } A_2.$$

This includes diffusion-like, field-like and causal propagation behaviours.



**Cycles.** Cycles in  $K_2$  correspond to closed geometric or energetic loops:

$$C_2 : \oint_{\gamma} J_2 \cdot dA.$$

These are impossible in  $K_1$ , where dimensionality is insufficient. Thus  $C_2$  is a strict upward extension of the trivial cycle of  $K_1$ .

**Tension.** The cross-level tension is defined as:

$$T_{1,2} = h(P_2 - P_1, \Theta_2 - \Theta_1, A_2).$$

If  $T_{1,2}$  exceeds the dimensional threshold, the new axis  $A_2$  becomes unstable and  $\Omega(K_2)$  collapses back into a degenerate form.

#### O.3.22 4. Birth and death conditions across the levels

**Birth of  $K_2$ .**  $K_2$  emerges when:

$$T_1 > \Theta_{1,\text{dim}} \quad \text{and} \quad A_2 \in M_2 \setminus A(K_1),$$

with a corresponding expansion of the state space:

$$\Omega(K_2) = \Omega(K_1) \cup \Delta\Omega_{\text{geom}}.$$

**Death propagation.** If  $\Theta_1$  fails, no upward extension exists. If  $\Theta_2$  fails, only  $K_2$  collapses;  $K_1$  remains valid because its structure does not depend on the higher-level potentials or flows.

#### O.3.23 5. Conceptual examples

**Emergence of geometry.** A single axis ( $K_1$ ) cannot support flows with curl, divergence or multi-directional propagation. The addition of  $A_2$  enables:

grad, div, curl – like structures.

**Percolation.** A multi-dimensional domain supports connectivity transitions (percolation threshold  $p_c$ ), which become part of the  $K_2$  threshold structure but have no analogue in  $K_1$ .

**Causality.** Directional propagation across a multi-dimensional structure leads to the first appearance of causal ordering: an impossibility in  $K_1$ .

These examples illustrate the essential meaning of the  $K_1 \rightarrow K_2$  transition:

**It is the birth of geometry, fields and dynamics from the minimal representational continuum.**

#### O.3.24 K2–K3 cross-level structure

The transition  $K_2 \rightarrow K_3$  describes the emergence of the chemical continuum from a physical continuum that already possesses geometry, fields and dynamical flows. While  $K_2$  contains physical degrees of freedom (spatial axes, energetic potentials, field gradients, percolation structure),  $K_3$  introduces:

- chemical composition spaces,
- binding energies and molecular potentials,
- reaction thresholds and stoichiometric constraints,

- stable clusters and bond networks,
- new classes of flows (reaction, diffusion–reaction, charge transfer),
- and an expanded domain  $\Omega(K_3)$  structured by chemical connectivity.

The  $K_2 \rightarrow K_3$  transition is the first major increase in structural complexity: physical fields become the substrate for chemical organisation.

### O.3.25 1. Levels involved and their roles

**K2 (physical continuum).**  $K_2$  provides:

- multi-dimensional axes (geometric coordinates),
- energetic potentials  $P_2$ ,
- flows  $J_2$  (transport, diffusion, field propagation),
- percolation structure of connectivity,
- causal ordering of events.

**K3 (chemical continuum).**  $K_3$  builds upon  $K_2$  and introduces:

- new axes describing chemical species, composition and charge,
- potentials associated with binding energy and reaction landscapes,
- thresholds  $\Theta_3$  for reaction feasibility and molecular stability,
- flows  $J_3$  involving reactions, dissociation and charge-transfer,
- stable clusters and reaction cycles defining chemical pathways.

$K_3$  is the first continuum supporting templating, catalysis, and information-like persistence via molecular states.

### O.3.26 2. Shared and inherited axes and thresholds

**Axes.** The spatial axes of  $K_2$  are fully inherited:

$$A_{\text{geom}}(K_2) \subset A(K_3).$$

$K_3$  adds chemical axes such as:

- composition/stoichiometry,
- charge state,
- bond configuration,
- reaction coordinate(s).

**Thresholds.** Chemical thresholds  $\Theta_3$  extend physical thresholds  $\Theta_2$ . Key inherited constraints include:

- energetic thresholds: insufficient  $P_2$  prevents bond formation,
- percolation thresholds: connectivity of interaction regions must exceed the critical value  $p_c$ ,
- gradient thresholds: sufficient flux or collision frequency is required for reaction activation.

If a physical threshold fails, chemical structure cannot emerge:

$$\Theta_2^{\text{death}} \Rightarrow \Omega(K_3) = \emptyset.$$

### O.3.27 3. Cross-level flows, cycles, and tensions

**Flows.** Flows in  $K_3$  are built upon  $J_2$ , but now include:

- reaction flows along chemical coordinates,
- diffusion–reaction coupling,
- charge-transfer and redox flows,
- flows between potential wells in the reaction landscape.

Formally:

$$J_3 = (J_2, J_{\text{react}}, J_{\text{charge}}, J_{\text{bond}}).$$

**Cycles.** Chemical cycles are higher-dimensional than physical loops. They represent:

$$C_3 : \text{reactive sequence } A \rightarrow B \rightarrow C \rightarrow A.$$

Chemical cycles require  $K_2$  transport plus  $K_3$  reaction pathways.

The existence of a stable cycle in  $K_3$  depends on:

$$\oint_{\gamma} J_3 \cdot dA > 0 \quad \text{and} \quad \text{all intermediate states satisfy } \Theta_3.$$

**Tension.** Cross-level tension  $T_{2,3}$  measures compatibility between physical fields and chemical constraints:

$$T_{2,3} = f(P_3 - P_2, \Theta_3 - \Theta_2, A_{\text{chem}}).$$

If  $T_{2,3}$  exceeds its threshold, the chemical continuum collapses back to a purely physical regime (no stable molecules or reactions).

### O.3.28 4. Birth and death conditions across the levels

**Birth of  $K_3$ .** Chemical structure emerges when:

$$T_2 > \Theta_{2,\text{dim}} \quad \text{and} \quad A_{\text{chem}} \in M_3 \setminus A(K_2).$$

This corresponds to the appearance of new degrees of freedom associated with binding energies and composition.

The state space expands:

$$\Omega(K_3) = \Omega(K_2) \cup \Delta\Omega_{\text{chem}}.$$

**Death propagation.** If  $\Theta_2$  fails, chemical organisation is impossible. If  $\Theta_3$  fails (e.g. due to insufficient binding energy, incompatible fields or excessive thermal noise), only  $K_3$  collapses, while  $K_2$  remains unaffected.

### O.3.29 5. Conceptual examples

**Percolation enabling chemical connectivity.** Only when regions of physical interaction percolate ( $p > p_c$ ), chemical clusters can form; otherwise molecules cannot stabilise.

**Reaction landscapes.** The transition introduces new potentials:

$$P_3 = P_2 + E_{\text{bond}} + E_{\text{charge}},$$

which create wells and activation barriers defining reaction pathways.

**Operator picture (creation/annihilation of clusters).** The formation or dissociation of a minimal cluster  $q_{\min}$  can be expressed via operators:

$$a^\dagger|k\rangle = \sqrt{f(k)+1}|k+k_{\text{unit}}\rangle, \quad a|k\rangle = \sqrt{f(k)}|k-k_{\text{unit}}\rangle,$$

reflecting the quantised change in connectivity within  $\Omega(K_3)$ .

These examples illustrate the essence of the  $K_2 \rightarrow K_3$  transition in bounded form: stable clusters, binding energies and reaction pathways emerge from physical fields and geometry.

### O.3.30 K3–K4 cross-level structure

The transition  $K_3 \rightarrow K_4$  marks the emergence of the biological continuum from the chemical continuum. While  $K_3$  supports chemical connectivity, reaction pathways and stable clusters,  $K_4$  introduces:

- membranes and boundary formation,
- sustained gradients (ion, pH, concentration, redox),
- osmotic and electrochemical potentials,
- active and passive transport flows,
- metabolic cycles and buffering loops,
- early information carriers (charge, structure, templating),
- regulatory thresholds and proto-homeostasis.

This is the birth of the first living-like continua: systems capable of maintaining non-equilibrium structure.

### O.3.31 1. Levels involved and their roles

**K3 (chemical continuum).**  $K_3$  provides:

- composition axes (species, charge, stoichiometry),
- potentials defined by reaction landscapes,
- reaction and charge-transfer flows  $J_3$ ,
- stable clusters and chemical cycles,
- no persistent boundaries and no regulated gradients.

**K4 (biological continuum).**  $K_4$  introduces:

- explicit boundary  $\partial\Omega(K_4)$ : membranes and compartments,
- gradients as axes ( $A_{\text{grad}}$ ),
- potentials  $P_{\text{grad}}, P_{\text{ion}}, P_{\text{redox}}$ ,
- flows  $J_4$  (osmotic, ion-conductive, active transport, metabolic),
- cycles  $C_4$  (energy, buffering, redox, pump cycles),
- thresholds  $\Theta_4$  for membrane stability, gradient maintenance, and osmotic balance.

The role of the  $K_3 \rightarrow K_4$  transition is the stabilisation of structure through controlled non-equilibrium dynamics.

### O.3.32 2. Shared and inherited axes and thresholds

**Axes.** From  $K_3$ ,  $K_4$  inherits:

$$A_{\text{chem}}(K_3) \subset A(K_4).$$

New axes include:

- $A_{\text{grad}}$  — concentration, pH, redox and electrical gradients,
- $A_{\text{mem}}$  — membrane curvature and permeability states,
- $A_{\text{exc}}$  — proto-electrical excitation axis linking to  $K_5$ .

**Thresholds.**  $K_4$  adds threshold families:

- $\Theta_{\text{mem}}$  — membrane rupture/permeability limits,
- $\Theta_{\text{grad}}$  — minimal and maximal sustainable gradients,
- $\Theta_{\text{osm}}$  — osmotic pressure constraints ( $\Delta\pi$ ),
- $\Theta_{\text{redox}}$  — redox balance stability,
- $\Theta_{\text{energy}}$  — metabolic viability thresholds.

If  $\Theta_3$  fails (chemical stability), no biological boundary forms:

$$\Theta_3^{\text{death}} \Rightarrow \Omega(K_4) = \emptyset.$$

### O.3.33 3. Cross-level flows, cycles, and tensions

**Flows.** Flows  $J_4$  include all chemical flows of  $J_3$  plus:

- $J_{\text{osm}}$  — osmotic water flux,
- $J_{\text{ion}}$  — ion flux through channels or membrane,
- $J_{\text{pump}}$  — active transport powered by  $P_{\text{energy}}$ ,
- $J_{\text{redox}}$  — electron transfer and energy transformation,
- $J_{\text{metabolic}}$  — coupled reaction–transport loops.

These define the first non-equilibrium dynamical regime in the hierarchy.

**Cycles.** Cycles  $C_4$  (energy, buffering, pump, redox) are structural stabilisers:

$$C_{\text{energy}} : P_{\text{source}} \rightarrow J_{\text{pump}} \rightarrow \text{gradient restoration} \rightarrow P_{\text{source}}.$$

A cycle exists only if:

$$\oint_C dA > 0 \quad \text{and} \quad S(C) > 0,$$

where  $S(C)$  is the cycle stability from the Core definition.

**Tension.** Cross-level tension  $T_{3,4}$  reflects incompatibilities between chemical reactivity and biological stability:

$$T_{3,4} = f(\Theta_{\text{mem}} - P_{\text{chem}}, \Theta_{\text{grad}} - P_{\text{grad}}, J_{\text{react}} - J_{\text{buffer}}).$$

Excess tension destroys gradients or membranes, collapsing  $K_4$  back to  $K_3$ .

### O.3.34 4. Birth and death conditions across the levels

**Birth of  $K_4$ .** The biological continuum appears when:

$$T_3 > \Theta_{3,\text{dim}} \quad \text{and} \quad A_{\text{grad}}, A_{\text{mem}} \in M_4 \setminus A(K_3),$$

corresponding to the emergence of boundaries and persistent gradients.

The state space expands as:

$$\Omega(K_4) = \Omega(K_3) \cup \Delta\Omega_{\text{bio}}.$$

**Death propagation.** If membranes cannot maintain gradients (violations of  $\Theta_{\text{mem}}$ ,  $\Theta_{\text{grad}}$ ,  $\Theta_{\text{osm}}$ ), then:

$$\Omega(K_4) \rightarrow \Omega(K_3),$$

i.e. collapse into the chemical regime.

$K_3$  remains stable unless its own thresholds fail.

### O.3.35 5. Conceptual examples

**Osmotic boundary formation.** The membrane arises as a stable region of  $\partial\Omega$  satisfying:

$$\Delta\pi = RT(C_{\text{in}} - C_{\text{out}})$$

within  $\Theta_{\text{osm}}$ .

**Gradient-based organisation.** A stable  $pH$  or ion gradient defines a new axis  $A_{\text{grad}}$ , enabling:

- proto-metabolism,
- vectorial energy conversion,
- selective transport.

**Proto-excitation.** Electrical gradients  $\Delta V$  allow proto-spike behaviour when:

$$\Delta V > \Theta_{\text{exc}},$$

linking  $K_4$  to  $K_5$  via early information-like flows.

These examples illustrate the essence of the  $K_3 \rightarrow K_4$  transition in bounded form: boundaries, gradients and non-equilibrium cycles become the first stable biological organisation.

### O.3.36 K4–K5 cross-level structure

The transition  $K_4 \rightarrow K_5$  marks the emergence of the neuronal continuum from the biological continuum. While  $K_4$  supports gradients, membranes, and metabolic cycles,  $K_5$  introduces:

- electrical excitability and membrane voltage dynamics,
- ion channels with discrete open/closed states,
- propagating electrical events (proto-spikes),
- refractory mechanisms and excitability thresholds,
- spatial propagation along membranes and early network structure,
- attractor-like patterns forming proto-information flows.

This is the birth of nervous-system-like organisation: structure capable of encoding, propagating and transforming electrical signals.

### O.3.37 1. Levels involved and their roles

**K4 (biological continuum).**  $K_4$  provides:

- membranes and compartments  $\partial\Omega$ ,
- gradients (ion, pH, redox) with potentials  $P_{\text{grad}}$ ,
- ion fluxes and osmosis  $(J_{\text{ion}}, J_{\text{osm}})$ ,
- metabolic and pump cycles  $C_{\text{energy}}$ ,
- thresholds  $\Theta_{\text{mem}}, \Theta_{\text{grad}}, \Theta_{\text{osm}}, \Theta_{\text{redox}}$ .

However,  $K_4$  lacks:

- regenerative electrical excitation,
- discrete channel gating dynamics,
- long-range electrical propagation,
- stable electrical attractors.

**K5 (neuronal continuum).**  $K_5$  introduces:

- the excitation axis  $A_{\text{exc}}$  defining membrane voltage,
- ion-channel axes  $A_{\text{channel}}$  describing open/closed states,
- electrical potentials  $P_{\text{elect}}$ ,
- flows  $J_5 = (J_{\text{ion}}, J_{\text{exc}}, J_{\text{leak}}, J_{\text{shunt}})$ ,
- proto-spike and refractory cycles  $C_{\text{spike}}, C_{\text{recovery}}$ ,
- thresholds  $\Theta_{\text{exc}}, \Theta_{\text{refrac}}, \Theta_{\text{noise—electrical}}$ .

$K_5$  is the first continuum capable of spatiotemporal pattern propagation.

### O.3.38 2. Shared and inherited axes and thresholds

**Inherited axes.** The gradients of  $K_4$  become components of the electrical axis:

$$A_{\text{grad}}(K_4) \subset A_{\text{exc}}(K_5).$$

The membrane axis  $A_{\text{mem}}$  becomes the substrate for channel placement and spatial propagation.

**New axes.**  $K_5$  introduces:

- $A_{\text{exc}}$  — membrane voltage axis,
- $A_{\text{channel}}$  — discrete gating states,
- $A_{\text{lat}}$  — lateral propagation along the membrane surface.

**Thresholds.** New threshold families include:

- $\Theta_{\text{exc}}$  — minimal depolarisation for spike formation,
- $\Theta_{\text{refrac}}$  — recovery time constraints,
- $\Theta_{\text{channel}}$  — gating stability and failure points,
- $\Theta_{\text{noise—electrical}}$  — noise-induced collapse limits.

Violation of  $\Theta_4$  prevents excitability:

$$\Theta_4^{\text{death}} \Rightarrow \Omega(K_5) = \emptyset.$$

### O.3.39 3. Cross-level flows, cycles, and tensions

**Flows.** Flows in  $K_5$  derive from  $J_4$  but gain new electrical components:

$$J_5 = (J_{\text{ion}}, J_{\text{exc}}, J_{\text{leak}}, J_{\text{shunt}}).$$

Key contributions:

- $J_{\text{exc}}$ : regenerative sodium-like surge,
- $J_{\text{leak}}$ : damping background current,
- $J_{\text{shunt}}$ : membrane-stabilising bypass currents,
- $J_{\text{ion}}$ : classical Nernst-like gradient-driven flux.

**Cycles.** Two fundamental cycles appear in  $K_5$ :

$$C_{\text{spike}} : \Delta V \uparrow \rightarrow \Delta V_{\text{peak}} \rightarrow \text{repolarisation} \rightarrow \text{recovery}.$$

$$C_{\text{recovery}} : \text{refractory} \rightarrow \text{reset} \rightarrow \text{excitability restored}.$$

These cycles depend on stable gradients and channel kinetics.

**Tension.** Cross-level tension:

$$T_{4,5} = g(\Theta_{\text{exc}} - P_{\text{elect}}, J_{\text{leak}} - J_{\text{exc}}, \Theta_{\text{channel}} - A_{\text{channel}}).$$

When  $T_{4,5}$  exceeds the dimensional threshold, propagation fails and the system collapses back to a non-excitable  $K_4$  regime.

### O.3.40 4. Birth and death conditions across the levels

**Birth of  $K_5$ .** The neuronal continuum emerges when:

$$T_4 > \Theta_{4,\text{dim}} \quad \text{and} \quad A_{\text{exc}}, A_{\text{channel}} \in M_5 \setminus A(K_4),$$

allowing stable spike dynamics.

State space expansion:

$$\Omega(K_5) = \Omega(K_4) \cup \Delta\Omega_{\text{neural}}.$$

**Death propagation.** If  $\Theta_4$  thresholds fail (membrane rupture, gradient collapse), then  $K_5$  loses the basis for excitability.

If  $\Theta_5$  thresholds fail (excess noise, channel instability, insufficient gradient), only  $K_5$  collapses;  $K_4$  survives.

### O.3.41 5. Conceptual examples

**Proto-spike formation.** Given:

$$\Delta V > \Theta_{\text{exc}} \quad \text{and} \quad A_{\text{channel}} = \text{open},$$

a regenerative upstroke forms and propagates across the local membrane.

**2D propagation.** Lateral axis  $A_{\text{lat}}$  allows propagation of electrical patterns across two-dimensional membrane surfaces — impossible in  $K_4$ .



**Excitability as new dimension.** The appearance of  $\Delta V(t)$  dynamics constitutes a new axis and a new class of flows, satisfying the dimensional growth rule:

$$\dim K_5 = \dim K_4 + 1.$$

**Early information flow.** Spike-like activity supports pattern transmission and proto-coding, creating the first substrate for structured information within the continuum hierarchy.

Thus the  $K_4 \rightarrow K_5$  transition can be summarised in bounded terms as the emergence of excitability, electrical signalling and pattern propagation from biological gradients and membranes.

### O.3.42 K5–K6 cross-level structure

The transition  $K_5 \rightarrow K_6$  marks the emergence of the cognitive continuum from the neuronal continuum. While  $K_5$  supports electrical excitability, spike propagation and network-like synchronisation,  $K_6$  introduces:

- conceptual axes and semantic distinctions,
- cognitive potentials (confidence, relevance, salience),
- thresholds for coherence and contradiction,
- structured flows of attention, memory and interpretation,
- cycles of understanding, prediction and learning,
- an internal cognitive timescale  $\tau(K_6)$ .

This transition enables the appearance of meaning, inference, representation and model-formation — capacities not reducible to neuronal dynamics alone.

### O.3.43 1. Levels involved and their roles

**K5 (neuronal continuum).**  $K_5$  provides:

- spike-based propagation and synchronisation,
- channel dynamics and membrane voltage axes,
- excitability thresholds  $\Theta_{\text{exc}}$ ,
- neural flows  $J_5$  and propagation cycles,
- network connectivity and oscillatory modes.

However,  $K_5$  lacks:

- semantic structure,
- abstract representations,
- thresholds based on coherence or contradiction,
- stable conceptual attractors.

**K6 (cognitive continuum).**  $K_6$  introduces:

- conceptual axes  $A^c$  defining distinctions between meanings,
- potentials  $P^c$  describing salience, belief strength, relevance,
- thresholds  $\Theta^c$  for coherence, contradiction, cognitive overload and collapse,
- flows  $J^c$ : attention, memory, interpretation, argumentation,

- cycles  $C^c$ : attention cycle, understanding cycle, learning cycle,
- measure  $k(K_6)$  describing cognitive coherence.

Thus  $K_6$  is the first level capable of modelling its own internal states.

### O.3.44 2. Shared and inherited axes and thresholds

**Inherited axes.** Neuronal synchronisation patterns provide the substrate for conceptual axes:

$$A_{\text{sync}}(K_5) \rightarrow A^c(K_6).$$

Oscillatory and connectivity patterns become semantic distinctions.

**New axes.**  $K_6$  introduces:

- $A^c$  — conceptual differentiation axes,
- $A_{\text{belief}}$  — degree of confidence,
- $A_{\text{value}}$  — relevance or utility gradients,
- $A_{\text{model}}$  — internal model coordinates.

**New thresholds.**  $\Theta^c$  includes:

- **coherence thresholds** — minimal compatibility of concepts,
- **contradiction thresholds** — upper bound of tension within a model,
- **overload thresholds** — limits of attention or memory flux,
- **collapse thresholds** —  $\Theta_{\text{collapse}}$  where  $\Omega(K_6)$  becomes empty if contradictions exceed limits.  
Violation of  $\Theta_5$  collapses excitability and therefore prohibits  $K_6$ .

### O.3.45 3. Cross-level flows, cycles, and tensions

**Flows.** Cognitive flows  $J^c$  extend neural flows  $J_5$ :

$$J^c = (J_{\text{attention}}, J_{\text{memory}}, J_{\text{interpretation}}, J_{\text{argument}}).$$

Examples:

- attention shifts modulating neural synchronisation,
- memory consolidation shaping network attractors,
- interpretive flows re-weighting  $P^c$ ,
- argumentative flows restructuring conceptual axes.

**Cycles.**  $K_6$  contains stable cycles:

$$C_{\text{attention}} : \text{orient} \rightarrow \text{focus} \rightarrow \text{disengage} \rightarrow \text{reorient},$$

$$C_{\text{understanding}} : \text{prediction} \rightarrow \text{input} \rightarrow \text{correction} \rightarrow \text{updated model},$$

$$C_{\text{learning}} : \text{activation} \rightarrow \text{error} \rightarrow \text{adjustment} \rightarrow \text{stabilisation}.$$

**Tension.** Cross-level tension is given by:

$$T_{5,6} = f(\Theta^c - P^c, J^c - J_5, A^c - A_{\text{sync}}).$$

If  $T_{5,6}$  exceeds the dimensional threshold, conceptual coherence breaks down and the continuum collapses to a non-semantic neuronal regime.

#### O.3.46 4. Birth and death conditions across the levels

**Birth of  $K_6$ .** The cognitive continuum appears when:

$$T_5 > \Theta_{5,\text{dim}} \quad \text{and} \quad A^c \in M_6 \setminus A(K_5),$$

representing the emergence of conceptual differentiation not reducible to neural axes.

State-space expansion:

$$\Omega(K_6) = \Omega(K_5) \cup \Delta\Omega_{\text{cog}}.$$

**Death propagation.** If  $\Theta_5$  fails (loss of excitability),  $K_6$  becomes impossible.

If  $\Theta^c$  fails (contradiction, overload, collapse),  $K_6$  dies but  $K_5$  remains valid:

$$\Omega(K_6) \rightarrow \emptyset, \quad \Omega(K_5) \text{ intact.}$$

#### O.3.47 5. Conceptual examples

**Semantic axis formation.** Repeated synchronisation patterns in  $K_5$  stabilise into conceptual distinctions at  $K_6$ , yielding axes such as:

$$A^c = \{\text{object vs. background, true vs. false, cause vs. effect, } \dots\}.$$

**Internal models.** Predictive cycles create stable structures in  $\Omega(K_6)$  representing hypotheses, concepts and expectations.

**Cognitive collapse.** If:

$$T_{5,6} > \Theta_{\text{collapse}},$$

the conceptual space loses coherence, shrinking  $\Omega(K_6)$  to zero.

Hence, the  $K_5 \rightarrow K_6$  transition can be summarised in bounded terms as the emergence of conceptual structure, meaning and internal models from neural dynamics and synchronisation.

#### O.3.48 K6–K7 cross-level structure

The transition  $K_6 \rightarrow K_7$  marks the emergence of the social continuum from the cognitive continuum. While  $K_6$  supports conceptual structures, meaning, and internal models,  $K_7$  introduces:

- social roles and norms as new axes,
- communication flows and shared symbolic structures,
- collective thresholds for coordination and expectation,
- institutional cycles and stabilising feedback loops,
- patterns of cooperation, conflict, and norm enforcement,
- a shared domain  $\Omega(K_7)$  of socially valid configurations.

This transition corresponds to the emergence of collective organisation that cannot be reduced to individual cognition.

### O.3.49 1. Levels involved and their roles

**K6 (cognitive continuum).**  $K_6$  provides:

- conceptual axes  $A^c$ ,
- cognitive potentials  $P^c$ ,
- thresholds  $\Theta^c$  for coherence and contradiction,
- flows of attention, memory, interpretation  $J^c$ ,
- cycles of understanding and learning,
- cognitive coherence measure  $k_6$ .

However,  $K_6$  lacks:

- inter-agent coordination rules,
- shared symbolic norms,
- population-level stability thresholds,
- institutional persistence.

**K7 (social continuum).**  $K_7$  introduces:

- axes  $A_{\text{soc}}$  (roles, norms, status, identity),
- potentials  $P_{\text{soc}}$  (trust, authority, legitimacy),
- social thresholds  $\Theta_{\text{soc}}$  (coordination, cohesion, enforcement, collapse),
- flows  $J_{\text{comm}}$  (communication, signalling, imitation),
- cycles  $C_{\text{inst}}$  (rule–action–consequence–correction),
- a global measure  $k_7$  describing social stability.

$K_7$  is the first continuum in which expectations and norms regulate behaviour across multiple agents.

### O.3.50 2. Shared and inherited axes and thresholds

**Inheritance.** Conceptual distinctions from  $K_6$  provide the substrate for shared symbols:

$$A^c(K_6) \rightarrow A_{\text{soc}}(K_7).$$

Internal models become the basis for shared narratives.

**New axes.**  $K_7$  introduces:

- $A_{\text{role}}$  — behavioural expectations,
- $A_{\text{norm}}$  — permissible/impermissible actions,
- $A_{\text{status}}$  — social differentiation,
- $A_{\text{collective}}$  — group-level identity structures.

These axes do not exist at  $K_6$ .

**Thresholds.**  $\Theta_{\text{soc}}$  includes:

- cohesion thresholds (minimal trust or communication density),
- contradiction thresholds (incompatibility of norms),
- enforcement thresholds (stability of sanctions),
- collapse thresholds (failure of institutional cycles).

Violation of  $\Theta^c$  collapses coherence at the cognitive level and prevents the emergence of  $K_7$ .

### O.3.51 3. Cross-level flows, cycles, and tensions

**Flows.** Social flows  $J_{\text{comm}}$  extend cognitive flows:

$$J_{\text{comm}} = (J_{\text{signal}}, J_{\text{language}}, J_{\text{imitation}}, J_{\text{coord}}).$$

They redistribute meaning across agents and stabilise collective states.

**Cycles.**  $K_7$  is characterised by institutional cycles:

$$C_{\text{inst}} : \text{rule} \rightarrow \text{action} \rightarrow \text{consequence} \rightarrow \text{correction} \rightarrow \text{rule}.$$

A society exists only if at least one such stabilising cycle remains intact.

**Tension.** Cross-level tension  $T_{6,7}$  measures incompatibility between personal conceptual models and collective structures:

$$T_{6,7} = f(A_{\text{soc}} - A^c, P_{\text{soc}} - P^c, J_{\text{comm}} - J^c, \Theta_{\text{soc}} - \Theta^c).$$

If  $T_{6,7}$  exceeds the dimensional threshold, collective organisation fails and the system collapses to individual cognition.

### O.3.52 4. Birth and death conditions across the levels

**Birth of  $K_7$ .**  $K_7$  emerges when:

$$T_6 > \Theta_{6,\text{dim}} \quad \text{and} \quad A_{\text{soc}} \in M_7 \setminus A(K_6).$$

This corresponds to the formation of stable social norms and roles.

State-space expansion:

$$\Omega(K_7) = \Omega(K_6) \cup \Delta\Omega_{\text{soc}}.$$

**Death propagation.** If  $\Theta^c$  fails (cognitive collapse),  $K_7$  becomes impossible.

If  $\Theta_{\text{soc}}$  fails (loss of cohesion, breakdown of cycles), the system collapses to individual cognition:

$$\Omega(K_7) \rightarrow \Omega(K_6).$$

$K_6$  survives unless its own thresholds are violated.

### O.3.53 5. Conceptual examples

**Emergence of norms.** Repeated cognitive interpretations across agents stabilise into collective norms, forming new axes in  $A_{\text{soc}}$ .

**Institutional stability.** Once an institutional cycle  $C_{\text{inst}}$  closes with minimal drift:

$$\oint_C dA_{\text{soc}} \approx 0,$$

a stable social subsystem is formed.

**Breakdown.** If communication density drops below a cohesion threshold:

$$J_{\text{comm}} < \Theta_{\text{soc,coh}},$$

collective structure dissolves.

Thus, the  $K_6 \rightarrow K_7$  transition can be summarised in bounded terms as the emergence of shared norms, communication flows and institutional cycles from individual cognition.

### O.3.54 K7–K8 cross-level structure

The transition  $K_7 \rightarrow K_8$  marks the emergence of the civilisational–technological continuum from the social continuum. Where  $K_7$  provides norms, roles, communication flows and institutional cycles,  $K_8$  introduces:

- technological and infrastructural axes,
- large-scale symbolic and informational systems,
- energy and logistics networks,
- economic mechanisms of reproduction,
- repair, maintenance and high-order cooperation cycles,
- global continuity conditions for population-level systems.

$K_8$  is the first continuum capable of sustaining persistent structures that exceed the cognitive and social horizons of individual agents.

### O.3.55 1. Levels involved and their roles

**K7 (social continuum).**  $K_7$  provides:

- social norms and roles ( $A_{\text{soc}}$ ),
- communication flows  $J_{\text{comm}}$ ,
- institutional cycles  $C_{\text{inst}}$ ,
- collective potentials  $P_{\text{soc}}$  (trust, legitimacy),
- cohesion and enforcement thresholds  $\Theta_{\text{soc}}$ .

However,  $K_7$  lacks:

- large-scale infrastructures,
- technological production cycles,
- energy-density thresholds,
- material and informational logistics,
- stable multi-generational reproduction mechanisms.

**K8 (civilisational–technological continuum).**  $K_8$  introduces:

- axes for infrastructure, technology and symbolic systems ( $A_{\text{tech}}$ ),
- potentials for energy density, repair capacity, logistics throughput,
- flows  $J^{(8)} = (J_{\text{energy}}, J_{\text{material}}, J_{\text{logistics}}, J_{\text{info}}, J_{\text{population}})$ ,
- infrastructure cycles  $C_{\text{sys}}$  (production  $\rightarrow$  transport  $\rightarrow$  use  $\rightarrow$  maintenance),
- thresholds  $\Theta_8 = (\Theta_E, \Theta_L, \Theta_I, \Theta_M, \Theta_R)$ ,
- measure  $k_8$  describing civilisational continuity.

Thus  $K_8$  is the first continuum with large-scale energetic and informational autonomy.

### O.3.56 2. Shared and inherited axes and thresholds

**Shared symbolic structures.** Symbolic structures and communication systems of  $K_7$  become the basis for:

$$A_{\text{symbol}}(K_7) \rightarrow A_{\text{tech}}(K_8),$$

enabling technical codification (writing, mathematics, engineering, standards).

**New axes in  $K_8$ .** These include:

- $A_{\text{infra}}$  — roads, power grids, water systems,
- $A_{\text{energy}}$  — energy density and conversion systems,
- $A_{\text{econ}}$  — economic mechanisms of reproduction,
- $A_{\text{info}}$  — information storage and transmission,
- $A_{\text{repair}}$  — maintenance and regeneration structures.

**Thresholds.** Inherited social cohesion thresholds remain necessary but insufficient. New thresholds arise:

- $\Theta_E$  — minimal energy density for system-level reproduction,
- $\Theta_L$  — logistics throughput thresholds,
- $\Theta_I$  — informational coherence thresholds,
- $\Theta_M$  — repair capacity threshold,
- $\Theta_R$  — demographic sustainability threshold.

If any of these thresholds fail,  $\Omega(K_8)$  collapses.

### O.3.57 3. Cross-level flows, cycles, and tensions

**Flows.** Flows in  $K_8$  generalise and scale-up social flows:

$$J^{(8)} = (J_{\text{energy}}, J_{\text{material}}, J_{\text{logistics}}, J_{\text{info}}, J_{\text{population}}).$$

**Cycles.** Civilisational stability requires the infrastructure cycle  $C_{\text{sys}}$ :

$$\text{production} \rightarrow \text{transport} \rightarrow \text{use} \rightarrow \text{maintenance} \rightarrow \text{production}.$$

Stability condition:

$$\oint dR \approx 0.$$

**Tension.** Cross-level tension:

$$T_{7,8} = f(\Theta_E - P_{\text{energy}}, \Theta_L - J_{\text{logistics}}, \Theta_I - P_{\text{info}}, \Theta_M - P_{\text{repair}}, \Theta_R - P_{\text{population}}).$$

Excess tension destroys infrastructural continuity.

#### O.3.58 4. Birth and death conditions across the levels

**Birth of  $K_8$ .**  $K_8$  emerges when:

$$T_7 > \Theta_{7,\text{dim}} \quad \text{and} \quad A_{\text{tech}}, A_{\text{infra}} \in M_8 \setminus A(K_7).$$

The domain expands:

$$\Omega(K_8) = \Omega(K_7) \cup \Delta\Omega_{\text{civ}}.$$

**Death propagation.** If social thresholds fail ( $\Theta_{\text{soc}}$ ), civilizational systems lose coherence and collapse to  $K_7$ .

If any of the five  $K_8$  thresholds fail:

$$\Theta_E, \Theta_L, \Theta_I, \Theta_M, \Theta_R,$$

then  $\Omega(K_8) = \emptyset$ .

$K_7$  may remain stable unless its own thresholds are violated.

#### O.3.59 5. Conceptual examples

**Energy-density escalation.** When energy density surpasses  $\Theta_E$ , technological and economic structures become self-sustaining.

**Information codification.** Social symbols evolve into technical systems (writing  $\rightarrow$  mathematics  $\rightarrow$  science), enabling new axes in  $A_{\text{tech}}$ .

**Infrastructure failure.** If logistics capacity drops below  $\Theta_L$  or repair rates below  $\Theta_M$ , civilisational cycles break and the system collapses:

$$C_{\text{sys}} \nrightarrow \text{closed loop}.$$

Thus the  $K_7 \rightarrow K_8$  transition can be summarised in bounded terms as the emergence of technological-civilisational structure from social norms, communication flows and institutional stability.

#### O.3.60 K8–K9 cross-level structure

The transition  $K_8 \rightarrow K_9$  marks the emergence of the **theoretical / scientific / metaparadigmatic continuum** from the civilisational–technological continuum. While  $K_8$  supports population-level systems, infrastructures, symbolic codification and technological cycles,  $K_9$  introduces:

- formal theoretical frameworks,
- logical and methodological axes,
- thresholds for consistency, evidence and explanatory power,
- flows of argumentation, proof and criticism,
- paradigm cycles and scientific evolution,
- recursive structures capable of describing lower levels and themselves.

$K_9$  is the first level where knowledge becomes an explicit structural object with its own continuity conditions.



### O.3.61 1. Levels involved and their roles

**K8 (civilisational–technological continuum).**  $K_8$  provides:

- symbolic systems, writing, mathematics, formal languages,
- institutions for knowledge transmission and accumulation,
- information networks and codified repositories,
- computational and technological capabilities,
- thresholds  $\Theta_8$  (energy, logistics, information coherence).

However,  $K_8$  lacks:

- formal meta-structures for theory comparison,
- logical thresholds for consistency or contradiction,
- explicit measures of explanatory or predictive strength,
- structured cycles of scientific evolution.

**K9 (theoretical continuum).**  $K_9$  introduces:

- axes  $A^9$  (logic, ontology, methodology, interpretation),
- potentials  $P^9$  (explanatory power, predictive accuracy, coherence, parsimony),
- thresholds  $\Theta^9$  (logical consistency, empirical adequacy, heuristic depth, Gödelian limits),
- flows  $J^9$  (arguments, proofs, critiques, paradigm shifts),
- cycles  $C^9$  (normal science, anomaly accumulation, revolution),
- a measure  $k(K_9)$  capturing global theoretical coherence.

Thus  $K_9$  is a meta-representational layer above the civilisational axis.

### O.3.62 2. Shared and inherited axes and thresholds

**Symbolic inheritance.** Symbolic infrastructures of  $K_8$  become the substrate for theoretical distinctions:

$$A_{\text{symbol}}(K_8) \rightarrow A^9.$$

Writing  $\rightarrow$  mathematics  $\rightarrow$  logic  $\rightarrow$  formal models.

**New axes.**  $K_9$  introduces:

- logical axes (validity, inference rules),
- ontological axes (entity types, structural assumptions),
- methodological axes (models, heuristics, paradigms),
- interpretational axes (competing readings of the same theory).

**New thresholds.**  $\Theta^9$  includes:

- **logical consistency threshold** — violation collapses the theory,
- **empirical threshold** — failure to match evidence,
- **heuristic threshold** — insufficient explanatory depth,
- **paradigm threshold** — incompatibility with existing frameworks,

- **Gödelian threshold** — incompleteness constraints.

If information coherence  $\Theta_I$  fails in  $K_8$ , theoretical structures cannot stabilise:

$$\Theta_I^{(K_8)\text{death}} \Rightarrow \Omega(K_9) = \emptyset.$$

### O.3.63 3. Cross-level flows, cycles, and tensions

**Flows.** Theoretical flows  $J^9$  extend symbolic flows of  $K_8$ :

$$J^9 = (J_{\text{argument}}, J_{\text{proof}}, J_{\text{critique}}, J_{\text{reinterpret}}).$$

These flows modify  $A^9$ ,  $P^9$  and the boundaries  $\partial\Omega(K_9)$ .

**Cycles.** Scientific cycles  $C^9$  include:

$$C_{\text{science}} : \text{theory} \rightarrow \text{prediction} \rightarrow \text{experiment} \rightarrow \text{anomaly} \rightarrow \text{revision} \rightarrow \text{theory}.$$

A theoretical continuum exists only if at least one such cycle remains stable.

**Tension.** Cross-level tension:

$$T_{8,9} = f(\Theta^9 - P^9, A^9 - A_{\text{symbol}}, J^9 - J_{\text{info}}, \Theta_8 - \Theta^9).$$

If  $T_{8,9}$  exceeds the dimensional threshold, paradigms collapse into non-theoretical symbolic structures.

### O.3.64 4. Birth and death conditions across the levels

**Birth of  $K_9$ .** The theoretical continuum emerges when:

$$T_8 > \Theta_{8,\text{dim}} \quad \text{and} \quad A^9 \in M_9 \setminus A(K_8),$$

i.e. new logical/ontological axes appear that cannot be reduced to technology or civilisational organisation.

Expansion:

$$\Omega(K_9) = \Omega(K_8) \cup \Delta\Omega_{\text{theory}}.$$

**Death propagation.** If  $\Theta_8$  fails (civilisational collapse),  $K_9$  loses its substrate.

If  $\Theta^9$  fails (logical inconsistency, empirical refutation, paradigm breakdown), the theoretical continuum collapses but  $K_8$  remains:

$$\Omega(K_9) \rightarrow \emptyset.$$

### O.3.65 5. Conceptual examples

**Formalisation of knowledge.** Technological symbolic systems of  $K_8$  enable mathematical and logical structures forming  $K_9$  axes.

**Scientific revolution cycle.** As anomalies accumulate beyond  $\Theta_{\text{heuristic}}^9$ :

$$T_{8,9} > \Theta_{\text{paradigm}}^9,$$

a paradigm shift occurs, creating a new region of  $\Omega(K_9)$ .

**Collapse case.** If informational coherence in  $K_8$  falls below  $\Theta_I$ , theoretical production becomes impossible — characteristic of civilisational collapse.

Thus, the  $K_8 \rightarrow K_9$  transition can be summarised in bounded terms as the emergence of formal theories, logic and scientific evolution from technological, symbolic and institutional infrastructures.

### O.3.66 K9–K10 cross-level structure

The transition  $K_9 \rightarrow K_{10}$  marks the emergence of the **metatheoretical continuum** from the theoretical continuum. While  $K_9$  contains scientific theories, paradigms, models and logical frameworks,  $K_{10}$  introduces:

- meta-axes comparing and transforming entire theoretical systems,
- functorial mappings between model-categories,
- thresholds for cross-theory compatibility and meta-coherence,
- flows of interpretation, translation and reconstruction,
- meta-cycles that regulate the evolution of theories,
- recursive structures capable of describing  $K_0 \dots K_{10}$ , including themselves.

$K_{10}$  is the highest representational layer in the Core model that remains within the human epistemic boundary.

### O.3.67 1. Levels involved and their roles

**K9 (theoretical continuum).**  $K_9$  provides:

- logical, ontological and methodological axes  $A^9$ ,
- theoretical potentials  $P^9$  (explanatory power, coherence, simplicity),
- thresholds  $\Theta^9$  (consistency, empirical adequacy),
- flows  $J^9$  (arguments, proofs, critiques),
- paradigm cycles  $C^9$  (normal science, anomaly, revolution).

However,  $K_9$  lacks:

- a structured space for comparing theories,
- functorial mappings between model-structures,
- meta-logic and meta-ontology,
- recursive representation that includes the theory of theories.

**K10 (metatheoretical continuum).**  $K_{10}$  introduces:

- axes  $A^{10}$  (functor types, meta-logic, ontology-of-ontologies),
- potentials  $P^{10}$  (meta-coherence, expressivity, universality),
- thresholds  $\Theta^{10}$  (categorical consistency, functorial compatibility, meta-logical limits),
- flows  $J^{10}$  (interpretation, translation, reconstruction, functorial lifts),
- cycles  $C^{10}$  (meta-stability, self-consistency, cross-paradigm reconciliation),
- measure  $k(K_{10})$  of global meta-coherence.

The defining feature of  $K_{10}$  is its recursive capacity:

$$K_{10} \text{ describes all levels } K_0 \dots K_{10}.$$

### O.3.68 2. Shared and inherited axes and thresholds

**Inherited structure.** Theories in  $K_9$  provide objects;  $K_{10}$  provides morphisms:

$$\text{Models}(K_9) \rightarrow \text{Functors}(K_{10}).$$

Symbolic structures, logic and methodologies from  $K_9$  become elements in the categorical architecture of  $K_{10}$ .

**New axes in  $K_{10}$ .** These include:

- $A_{\text{functor}}$  — types of model-transformations,
- $A_{\text{meta}}$  — meta-logical and meta-ontological distinctions,
- $A_{\text{compat}}$  — degrees of cross-theory compatibility,
- $A_{\text{reflect}}$  — axes for self-description.

**Meta-thresholds.**  $\Theta^{10}$  contains:

- **coherence threshold** — categorical diagrams must commute,
  - **compatibility threshold** — functors must preserve structure between theories,
  - **meta-logical threshold** — avoiding paradox or inconsistency,
  - **expressivity threshold** — minimal ability to represent distinctions across all lower levels.
- Violation of  $\Theta^9$  prevents formation of  $K_{10}$ .

### O.3.69 3. Cross-level flows, cycles, and tensions

**Flows.** Flows in  $K_{10}$  extend theoretical flows:

$$J^{10} = (J_{\text{interpret}}, J_{\text{translate}}, J_{\text{reconstruct}}, J_{\text{lift}}),$$

where:

- $J_{\text{interpret}}$  — reinterpretation of a theory in a new frame,
- $J_{\text{translate}}$  — mappings between formalisms,
- $J_{\text{reconstruct}}$  — rebuilding theory structure,
- $J_{\text{lift}}$  — representing a theory as part of a higher category or meta-model.

**Cycles.** Meta-cycles  $C^{10}$  include:

$$C_{\text{meta}} : \text{model} \rightarrow \text{interpretation} \rightarrow \text{translation} \rightarrow \text{refinement} \rightarrow \text{model}.$$

These cycles ensure stability of the meta-continuum.

**Tension.** Cross-level tension:

$$T_{9,10} = f(\Theta^{10} - P^{10}, A^{10} - A^9, J^{10} - J^9, \Theta^9 - \Theta^{10}).$$

If  $T_{9,10}$  exceeds its dimensional threshold,  $K_{10}$  collapses into a non-meta theoretical regime.

### O.3.70 4. Birth and death conditions across the levels

**Birth of  $K_{10}$ .** The metatheoretical continuum emerges when:

$$T_9 > \Theta_{9,\text{dim}} \quad \text{and} \quad A^{10} \in M_{10} \setminus A(K_9).$$

Expansion:

$$\Omega(K_{10}) = \Omega(K_9) \cup \Delta\Omega_{\text{meta}}.$$

**Death propagation.** If theoretical stability fails ( $\Theta^9$ ),  $K_{10}$  cannot form.

If meta-thresholds  $\Theta^{10}$  fail (functorial incompatibility, meta-logical paradox),  $K_{10}$  collapses but  $K_9$  persists.

### O.3.71 5. Conceptual examples

**Comparing theories.**  $K_{10}$  enables structured comparison of theories via functors:

$$\mathcal{F} : \text{Mod}(T_1) \rightarrow \text{Mod}(T_2).$$

**Interpreting paradigms.** Incompatible theories in  $K_9$  can be reconciled by lifting them into a meta-framework in  $K_{10}$ .

**Avoiding paradox.** If a meta-framework violates the meta-logical threshold  $\Theta_{\text{coh}}^{10}$ , the entire meta-continuum collapses.

Thus the  $K_9 \rightarrow K_{10}$  transition can be summarised in bounded terms as the emergence of a meta-layer capable of comparing, transforming and reconstructing entire theoretical systems, including itself.

### O.3.72 K10–K11 cross-level structure

The transition  $K_{10} \rightarrow K_{11}$  marks the emergence of a **trans–metatheoretical continuum** from the metatheoretical continuum. While  $K_{10}$  provides functorial mappings, meta-logic and categorical structures that relate theoretical systems,  $K_{11}$  introduces:

- universal operator classes acting across multiple meta-frameworks,
- higher-order compatibility conditions for meta-models,
- global thresholds for trans-meta coherence,
- flows that restructure entire families of meta-models,
- cycles that regulate meta-level evolution itself,
- a constraint layer ensuring that model evolution across all levels  $K_0 \dots K_{10}$  remains consistent within a higher envelope.

In Core 1.3,  $K_{11}$  is treated as an active upper-level doctrine rather than as an unresolved placeholder. This section summarises its structural role within the promoted cross-level architecture.

### O.3.73 1. Levels involved and their roles

**K10 (metatheoretical continuum).**  $K_{10}$  provides:

- axes  $A^{10}$  (functors, meta-logic, compatibility categories),
- potentials  $P^{10}$  (meta-coherence, expressivity, universality),
- thresholds  $\Theta^{10}$  (diagrammatic consistency, meta-logical well-formedness),
- flows  $J^{10}$  (interpretation, translation, reconstruction),
- cycles  $C^{10}$  (meta-stability, refinement loops).

**K11 (trans–metatheoretical continuum).**  $K_{11}$ , in its Core 1.3 formulation, introduces:

- axes  $A^{11}$  describing relations between entire classes of meta-frameworks,
- potentials  $P^{11}$  quantifying global coherence across meta-levels,
- thresholds  $\Theta^{11}$  for trans-meta consistency and compatibility,
- flows  $J^{11}$  restructuring families of metamodels,
- cycles  $C^{11}$  regulating the evolution of  $K_{10}$  structures,
- a measure  $k(K_{11})$  for universal structural coherence.

$K_{11}$  serves as a constraint layer for all lower modelling levels.

### O.3.74 2. Shared / inherited axes and thresholds

**Inherited structure.** Metamodels in  $K_{10}$  become objects;  $K_{11}$  introduces morphisms between meta-morphisms:

$$\text{MetaFunctors}(K_{10}) \rightarrow \text{TransFunctors}(K_{11}).$$

**New axes.** Axes in  $A^{11}$  include:

- $A_{\text{ultra}}$  — operator classes acting on categories of meta-models,
- $A_{\text{compat}}^{(11)}$  — trans-meta compatibility relations,
- $A_{\text{reflect}}^{(11)}$  — higher-order self-reflection axes,
- $A_{\text{limit}}$  — constraints from universal consistency.

**New thresholds.**  $\Theta^{11}$  includes:

- **trans-meta coherence threshold** — failure collapses  $K_{11}$ ,
  - **compatibility threshold** — incompatible meta-frameworks cannot coexist,
  - **universal consistency threshold** — prohibits paradox at the trans-meta level,
  - **expressivity threshold** — minimal representational sufficiency across  $K_0 \dots K_{10}$ .
- Violation of  $\Theta^{10}$  prevents the emergence of  $K_{11}$ .

### O.3.75 3. Cross-level flows, cycles, and tensions

**Flows.** Flows in  $K_{11}$  extend meta-flows:

$$J^{11} = (J_{\text{ultra}}, J_{\text{crossmeta}}, J_{\text{lift2}}, J_{\text{constraint}}),$$

where:

- $J_{\text{ultra}}$  — restructuring of categories of metamodels,
- $J_{\text{crossmeta}}$  — translation between incompatible meta-levels,
- $J_{\text{lift2}}$  — lifting meta-frameworks into  $K_{11}$ ,
- $J_{\text{constraint}}$  — enforcing global structural limits.

**Cycles.**  $C^{11}$  includes higher-order cycles such as:

$$C_{\text{ultra}} : \text{metamodel} \rightarrow \text{lift} \rightarrow \text{constraint} \rightarrow \text{refinement} \rightarrow \text{metamodel}.$$

These cycles regulate evolution of the meta-continuum.

**Tension.** Cross-level tension:

$$T_{10,11} = f(\Theta^{11} - P^{11}, A^{11} - A^{10}, J^{11} - J^{10}, \Theta^{10} - \Theta^{11}).$$

Excess tension collapses  $K_{11}$  to a simpler metatheoretical regime.

### O.3.76 4. Birth and death conditions across the levels

**Birth of  $K_{11}$ .**  $K_{11}$  emerges when:

$$T_{10} > \Theta_{10,\text{dim}} \quad \text{and} \quad A^{11} \in M_{11} \setminus A(K_{10}),$$

i.e. when distinctions between entire meta-framework classes become necessary.

Expansion:

$$\Omega(K_{11}) = \Omega(K_{10}) \cup \Delta\Omega_{\text{ultra}}.$$

**Death propagation.** If  $\Theta^{10}$  fails, the trans-meta layer cannot exist.

If  $\Theta^{11}$  fails (incompatibility, meta-paradox),  $K_{11}$  collapses but  $K_{10}$  persists:

$$\Omega(K_{11}) \rightarrow \emptyset.$$

### O.3.77 5. Conceptual examples

**Reconciling incompatible meta-frameworks.**  $K_{11}$  can provide a higher-level compatibility operator that relates meta-frameworks which cannot directly map into one another in  $K_{10}$ .

**Universal constraints.**  $K_{11}$  enforces global consistency across all modelling levels:

$$K_{11} : \{K_0, \dots, K_{10}\} \rightarrow \text{coherent envelope}.$$

**Collapse case.** If a trans-meta inconsistency arises:

$$T_{10,11} > \Theta_{\text{coh}}^{11},$$

$K_{11}$  collapses into  $K_{10}$ .

Thus the  $K_{10} \rightarrow K_{11}$  transition can be summarised in bounded terms as the emergence of universal operators and constraints capable of relating and regulating entire classes of metatheoretical frameworks.

### O.3.78 K11–K12 cross-level structure

The transition  $K_{11} \rightarrow K_{12}$  is the terminal cross-level step in Core 1.3. While  $K_{11}$  provides a trans–metatheoretical continuum with universal operators acting on entire classes of meta-frameworks,  $K_{12}$  binds those upper-level structures into the final unified-science envelope in which cross-domain compatibility and global integrability are assessed.

The level remains abstract, but it is no longer a placeholder. Its role is to state the admissibility conditions under which the promoted hierarchy can be treated as one coherent scientific stack.

### O.3.79 1. Levels involved and their roles

**K11 (trans–metatheoretical continuum).**  $K_{11}$  provides:

- axes  $A^{11}$  relating classes of meta-frameworks,
- potentials  $P^{11}$  expressing trans-meta coherence,
- thresholds  $\Theta^{11}$  for universal consistency and compatibility,
- flows  $J^{11}$  restructuring families of metamodels,
- cycles  $C^{11}$  regulating evolution of meta-level structures.

**K12 (formal upper continuum).** In Core 1.3,  $K_{12}$  is defined through:

- the existence of an ambient space  $M_{12}$  of possible axes,
- the possibility of new distinctions not representable in  $A^{11}$ ,
- abstract thresholds  $\Theta^{12}$  bounding structural consistency,
- generic flows  $J^{12}$  acting on classes of  $K_{11}$  objects,
- potential cycles  $C^{12}$  stabilising new high-level structures.

$K_{12}$  is not a concrete domain model. It is the global integrability layer that constrains how the closed hierarchy may be unified without contradiction.

### O.3.80 2. Shared / inherited axes and thresholds

**Inheritance.** All axes, potentials and thresholds of  $K_{11}$  are embeddable into the ambient space of  $K_{12}$ :

$$A^{11} \subseteq M_{12}, \quad \Theta^{11} \subseteq \Theta^{12}.$$

**New distinctions.** If  $K_{12}$  exists, its axes must satisfy:

$$A^{12} \in M_{12} \setminus A(K_{11}),$$

consistent with Theorem 4 (new axes cannot be self-generated by  $K_{11}$ ).

**Threshold extension.**  $\Theta^{12}$  may introduce:

- **extended coherence thresholds** involving entire chains  $K_0 \dots K_{11}$ ,
- **upper-bound constraints** (limits imposed by  $M_{12}$ ),
- **compatibility thresholds** for structures beyond meta-levels.

### O.3.81 3. Cross-level flows, cycles, and tensions

**Flows.** The abstract form of flows is sufficient here:

$$J^{12} = (J_{\text{lift3}}, J_{\text{ultra2}}, J_{\text{constraint2}}, J_{\text{embed}}),$$

meaning that  $K_{12}$  would operate on  $K_{11}$  structures the same way  $K_{11}$  operates on  $K_{10}$ , but at a higher-order level.

**Cycles.** Potential  $C^{12}$  cycles generalise trans-meta cycles:

$$C_{\text{limit}} : \text{K11-structure} \rightarrow \text{lift} \rightarrow \text{constraint} \rightarrow \text{refinement} \rightarrow \text{K11-structure}.$$

**Tension.** Cross-level tension obeys the usual form:

$$T_{11,12} = f(\Theta^{12} - P^{12}, A^{12} - A^{11}, J^{12} - J^{11}, \Theta^{11} - \Theta^{12}).$$

Excess tension collapses  $K_{12}$  into  $K_{11}$ .

### O.3.82 4. Birth and death conditions across the levels

**Birth of  $K_{12}$ .** In the promoted formalisation:

$$T_{11} > \Theta_{11,\text{dim}} \quad \text{and} \quad A^{12} \in M_{12},$$

i.e. new distinctions must exist in the ambient space  $M_{12}$  that exceed the representational capacity of  $K_{11}$ .

State-space expansion:

$$\Omega(K_{12}) = \Omega(K_{11}) \cup \Delta\Omega_{\text{limit}}.$$



**Death propagation.** If  $\Theta^{11}$  fails, no  $K_{12}$  structure can be formed.

If  $\Theta^{12}$  fails (extended incompatibility or paradox), then:

$$\Omega(K_{12}) \rightarrow \emptyset,$$

while  $K_{11}$  remains intact.

### O.3.83 5. Conceptual examples

**Meta-limit extension.**  $K_{12}$  could express properties relating entire chains of mappings across levels  $K_0 \dots K_{11}$ .

**Ultra-consistency.** A universal constraint might require coherence of the entire model under transformations not representable in  $K_{11}$ .

**Collapse case.** Any violation of extended coherence:

$$T_{11,12} > \Theta_{\text{coh}}^{12},$$

collapses the hypothetical  $K_{12}$  back into the fully specified  $K_{11}$ .

Thus, within Core 1.3, the  $K_{11} \rightarrow K_{12}$  transition is summarised in bounded terms as a formal upper envelope for cross-domain compatibility and global integrability, ensuring that the promoted hierarchy remains scientifically coherent while respecting the continuity and dimensional axioms.

## O.4 Cycle Structure in the Ontology of Continua

This master section provides the unified formal framework for cycle structure in the Ontology of Continua (OC). Cycles constitute the central mechanism of structural persistence: a continuum remains alive if and only if it maintains at least one structurally stable cycle that stays within its admissible state space  $\Omega(K)$  and maintains positive continuumness  $k(K, t)$ . The material below consolidates all general results on cycles across levels  $K_0$ – $K_{12}$ , independent of domain-specific interpretations.

### O.4.1 Definition of Cycles

A *cycle* of a continuum  $K$  is a closed trajectory

$$C : [0, \tau] \rightarrow \Omega(K), \quad C(0) = C(\tau),$$

such that:

1.  $C(t)$  remains strictly inside the admissible domain:  $\text{dist}(C(t), \partial\Omega(K)) > 0$ ;
2. the flow field  $J(t)$  supports the trajectory:  $\dot{C}(t) = J(C(t), t)$ ;
3. the cycle forms a stable attractor under the evolution operator  $E : K(t) \rightarrow K(t + dt)$ .

A cycle is *structurally stable* if perturbations of potentials, flows or thresholds do not destroy its closed form: there exists  $\varepsilon > 0$  such that all trajectories with initial distance  $< \varepsilon$  return to the cycle within bounded time.

### O.4.2 Role of Cycles in Continuum Persistence

In OC, cycles are not dynamical ornaments but structural necessities. They serve several essential functions:

- **Maintain distance from the boundary  $\partial\Omega(K)$ .** Without stable cycles, trajectories generically reach the boundary, where thresholds saturate and the continuum collapses.
- **Stabilise potentials and flows.** Cycles maintain oscillatory, periodic, quasi-periodic or recurrent regimes that preserve the structural viability of  $K$ .
- **Carry the measure of continuumness.** The contribution of cycles enters explicitly into the global measure  $k(K, t)$  through their stability, diversity and distance to  $\partial\Omega(K)$ .
- **Ensure structural identity.** The presence of characteristic cycle complexes allows a continuum to preserve its structural type across time.

As a consequence, the disappearance of all structurally stable cycles coincides with the death of the continuum.

### O.4.3 Cycle Complexes

For most continua, cycles do not appear in isolation but in interconnected clusters called *cycle complexes*. A cycle complex  $C_{\text{cx}}$  is a finite or countably infinite collection of cycles  $\{C_i\}$  satisfying:

1. shared supporting flows;
2. structural compatibility under perturbations of thresholds;
3. existence of transition paths between cycles that do not reach  $\partial\Omega(K)$ .

The *maximal cycle complex*  $C_{\text{max}}(K)$  is the set of all cycles that remain stable under the full threshold landscape  $\Theta(K)$ .

Theorem 7 (OC Core) states that

$$C_{\text{max}}(K) = \emptyset \iff \Omega(K) = \emptyset,$$

i.e. death is equivalent to the disappearance of the maximal cycle complex.

### O.4.4 Metrics on Cycles

OC defines several universal metrics on cycles, needed for expressing continuumness, structural tension and stability criteria.

**Length.** The geometric or functional length of a cycle is defined by

$$L(C) = \int_0^\tau \|\dot{C}(t)\| dt.$$

**Efficiency.** Cycle efficiency measures the degree to which flows align with axes:

$$C_{\text{eff}}(C) = \frac{1}{L(C)} \int_0^\tau J(C(t), t) \cdot dA(C(t)).$$

**Stability.** The distance of a cycle to the boundary is

$$S(C) = \min_{t \in [0, \tau]} d(C(t), \partial\Omega(K)).$$

This quantity plays a fundamental role:

$$S(C) = 0 \iff C \text{ is not structurally viable.}$$

**Weight.** A structural weight  $w(C)$  encodes relative contribution of the cycle to global viability and is defined as an aggregate of length, efficiency, stability and embedding-compatibility criteria.

#### O.4.5 Cycle Dynamics and Operator $Q$

The operator  $Q$  defines the time-evolution of cycle structure:

$$\frac{dC}{dt} = Q(C, P, J, \Theta).$$

Core behaviours captured by  $Q$  include:

- **Cycle formation:** birth of new cycles when flows become recurrent and thresholds stabilise oscillations.
- **Cycle stabilisation:** increase of  $S(C)$  and reduction of oscillatory distance to the attractor.
- **Cycle deformation:** changes in length, shape or period due to modifications in potentials or flows.
- **Cycle destruction:** collapse when  $S(C) \rightarrow 0$  or flows fail to support recurrence.

This operator interacts with all other structural operators ( $F, G, H, R, S, U$ ) and contributes directly to the derivative of continuumness:

$$\frac{dk}{dt} = U(\Omega, A, P, J, \Theta, C, \partial\Omega).$$

#### O.4.6 Cycles Across the K-Hierarchy

Cycles appear at every level of the continuum hierarchy, though their interpretations differ:

- $K_1$ : periodic field or geometric oscillations.
- $K_2$ : coherence cycles, topological defects, phase loops.
- $K_3$ : catalytic cycles in RAF networks.
- $K_4$ : metabolic and membrane-maintenance cycles.
- $K_5$ : excitation–repolarisation cycles (proto-spikes).
- $K_6$ : cognitive feedback loops and prediction cycles.
- $K_7$ : institutional cycles and cooperation loops.
- $K_8$ : infrastructural and energy-logistics cycles.
- $K_9, K_{10}$ : theoretical and meta-theoretical cycles.

Despite these differences, all such cycles share: closure inside  $\Omega(K)$ , positive distance to  $\partial\Omega$ , and stabilising interaction with thresholds.

#### O.4.7 Cycles and Dimensional Transitions

Dimensional birth ( $K_x \rightarrow K_{x+1}$ ) requires that cycles at level  $K_x$  approach the dimensional threshold  $\Theta_{\dim}(K_x)$  in such a way that their representational capacity becomes insufficient. Loss of cycle stability is one of the earliest indicators of an impending dimensional transition.

#### O.4.8 Summary

Cycles encode the persistence, identity and structural viability of continua across all domains. Their stability, diversity and organisation form a key component of continuumness  $k(K, t)$ . The disappearance of all structurally stable cycles corresponds to the death of a continuum, while deformation and emergence of new cycles signal structural reorganisation or dimensional transition.

#### O.4.9 Cycles on $K_0$

Level  $K_0$  represents the meta-ontological background of the Ontology of Continua. It possesses no geometry, no time, no admissible state space in the sense of  $\Omega(K)$ , no flows  $J$ , no thresholds  $\Theta$ , and no operators of evolution. For this reason, the notion of a cycle cannot be defined on  $K_0$ .

#### O.4.10 Absence of State Space and Time

Cycles require:

1. a non-empty state space  $\Omega(K)$ ;
2. a notion of time or parametrisation  $t \in [0, \tau]$ ;
3. a flow field  $J$  generating closed trajectories.

None of these structures exist on  $K_0$ . Formally:

$$\Omega(K_0) = \emptyset, \quad \partial\Omega(K_0) \text{ undefined}, \quad J_{K_0} \text{ undefined},$$

and no parameter  $t$  exists that could index a trajectory.

Therefore, a cycle

$$C : [0, \tau] \rightarrow \Omega(K), \quad C(0) = C(\tau),$$

cannot be formulated.

#### O.4.11 Structural Reason

Level  $K_0$  is not a continuum but a logical condition of possibility for continua of levels  $K_1$  and above. It establishes no dynamical or geometric structure from which cycles could emerge. All cycle-related notions—length, stability, distance to boundary, cycle complexes, or contributions to continuumness—are strictly inapplicable.

#### O.4.12 Implication for Higher Levels

The absence of cycles on  $K_0$  implies:

- the concept of time  $\tau(K)$  begins only at  $K_1$ , where a continuous axis is first defined;
- the first non-trivial cycles appear at  $K_1$  as trivial or degenerate oscillatory structures of a one-dimensional field;
- structural cycles in the full sense (stability, distance to  $\partial\Omega$ , interaction with thresholds) emerge only at levels  $K_2$  and above;
- the birth of cycles is itself a signature of the phase transition  $K_0 \rightarrow K_1$ .

#### O.4.13 Summary

No cycles exist on  $K_0$ . Cycle structure becomes meaningful only beginning with  $K_1$ , where time, state space and flows first appear. Accordingly, the file `cycles_k0.tex` records a formal *non-existence statement*, required for the internal consistency of the cycle taxonomy across the entire hierarchy of continua.

#### O.4.14 Cycles on $K_1$

Level  $K_1$  is the first genuine continuum in the Ontology of Continua. It is defined by a single continuous axis

$$A_1(x) = x, \quad x \in I = (a, b),$$

and a non-empty state space

$$\Omega(K_1) = C^0(T, H^1(I, V)) \cap C^1(T, L^2(I, V)),$$

as established in the formal construction of  $K_1$ .

Time  $\tau(K_1)$  is first definable here, enabling the introduction of cycles. Therefore  $K_1$  is the minimal level at which cycles exist.

#### O.4.15 Definition of Cycles on $K_1$

Let  $s(t) \in \Omega(K_1)$  denote the state of the continuum. A cycle is a mapping

$$C : [0, \tau] \rightarrow \Omega(K_1), \quad C(0) = C(\tau),$$

generated by the evolution operator

$$\frac{dA_1}{dt} = F_1(A_1, P_1, J_1, \Theta_1),$$

subject to the admissibility conditions of  $\Omega(K_1)$ .

Because  $K_1$  has only a single axis and no internal thresholds beyond  $\Theta_1$ , the dynamics is one-dimensional and cannot create non-trivial topological or structural cycle complexes. All cycles on  $K_1$  are therefore “trivial” in the thermodynamic and structural sense.

#### O.4.16 Characteristics of Cycles on $K_1$

Cycles at this level have several characteristic features:

- **Topology:** The cycle lives on a 1D manifold; no loop nesting or cycle complexes exist.
- **Degeneracy:** The cycle is typically a minimal oscillation or a closed 1-parameter trajectory of the field  $A_1(x, t)$ .
- **No critical thresholds:** The threshold vector  $\Theta_1$  contains only the basic admissibility bound arising from the construction of  $K_1$ . There are no phase transitions associated with  $C$ .
- **No structural tension:** The structural tension  $T_1(t)$  is defined but cannot generate complex bifurcations or instabilities because  $K_1$  admits no incompatible differences.
- **Boundary behaviour:** The boundary  $\partial\Omega(K_1)$  is trivial; cycles cannot meaningfully approach or cross it.
- **Energy consistency:** The action functional

$$S[A_1] = \int_0^\tau E(A_1, P_1, J_1) dt$$

determines the dynamical feasibility of a cycle but does not create higher-order constraints.

#### O.4.17 Metrics for $K_1$ Cycles

From the universal cycle formalism developed in Run 9, cycles on  $K_1$  have:

$$L(C) = \int_0^\tau \|\dot{A}_1(t)\| dt,$$

$$C_{\text{eff}} = \frac{1}{L(C)} \int_0^\tau J_1(t) \cdot dA_1(t),$$

and a trivial stability

$$S(C) = \min_{t \in [0, \tau]} d(\Omega(K_1), \partial\Omega(K_1)).$$

Since the boundary is trivial and unattainable,  $S(C)$  is always maximal for admissible states:

$$S(C) = S_{\text{max}}.$$

#### O.4.18 Role of $K_1$ Cycles in the Hierarchy

The existence of cycles on  $K_1$  is essential for the higher levels:

- They establish the first non-zero temporal structure.
- They serve as the “seed” cycles from which  $K_2$  inherits periodicity and dynamical recurrence.
- They provide the baseline from which non-trivial structural cycles (e.g. percolation cycles, metabolic cycles, spike cycles) arise at  $K_2$ – $K_5$ .

Thus, although cycles on  $K_1$  are dynamically simple, they are structurally indispensable for the continuity of the hierarchy.

#### O.4.19 Summary

$K_1$  supports the simplest possible cycles — closed trajectories of a 1D continuum with no criticality, no bifurcations and no internal structure. They mark the birth of time and dynamical recurrence, enabling all higher-order cycles in the Ontology of Continua.

#### O.4.20 Cycles on $K_2$

Level  $K_2$  corresponds to the continuum of connectivity. Its state space is represented by a percolation-like configuration

$$\Omega(K_2) \cong \{0, 1\}^{|E|},$$

where  $E$  is the set of edges, and the structural variable is the occupation probability  $p$ . The fundamental phenomenon at this level is the emergence or destruction of global connectivity.

Cycles on  $K_2$  arise due to oscillations in cluster structure, connectivity measures, and the effective time  $\tau(K_2)$  associated with correlation dynamics.

#### O.4.21 Definition of Cycles on $K_2$

A cycle on  $K_2$  is a dynamical trajectory

$$C : [0, \tau] \rightarrow \Omega(K_2)$$

such that:

$$C(0) = C(\tau),$$

and its evolution is governed by the operator

$$\frac{dp}{dt} = F_2(p, A_2, P_2, J_2, \Theta_2),$$

where:

- $A_2$  are connectivity axes (local “occupied/unoccupied” states),
- $P_2$  are potentials derived from cluster sizes, correlation length, or energy-like parameters,
- $J_2$  are flows of connectivity changes,
- $\Theta_2$  contains the critical percolation threshold  $p_c$ .

Thus a cycle is a closed recurrence of connectivity configurations.

#### O.4.22 Types of Cycles on $K_2$

Three major classes of cycles exist.

**(1) Cluster–Rearrangement Cycles.** Local edge-flip dynamics generates oscillatory behaviour in cluster structure. These cycles do not approach criticality and have bounded correlation length.

**(2) Subcritical Connectivity Cycles.** For  $p < p_c$ , the system may undergo recurrent behaviour in the size distribution of finite clusters:

$$C : N(s, t) \rightarrow N(s, t + \tau),$$

where  $N(s)$  is the number of clusters of size  $s$ . These cycles are structurally stable due to the absence of a giant component.

**(3) Critical–Near-Critical Cycles.** For  $p \approx p_c$ , oscillations may bring the system repeatedly close to the critical percolation threshold. These cycles have:

- long correlation lengths  $\xi(p(t))$ ,
- high structural tension  $T_2(t)$ ,
- sensitivity to thresholds  $\Theta_2$ ,
- potential for phase-transition-like excursions.

This is the first level in the hierarchy where non-trivial structural cycles appear.

#### O.4.23 Structural Tension and Threshold Interaction

Structural tension on  $K_2$  is given by:

$$T_2(p) = f(\xi(p), \text{Conn}(p), |p - p_c|),$$

where  $f$  grows as the system approaches criticality.

A cycle can be periodic only if:

$$T_2(t) < \Theta_{\text{death}}$$

for all  $t$ . If the loop crosses the phase boundary

$$p = p_c,$$

the system undergoes a structural transition, and the cycle cannot close smoothly unless the dynamics returns  $p$  across the same boundary.

This leads to the possibility of **\*\*critical cycles\*\***, the first instance in OC of loops that interact with phase transitions.

#### O.4.24 Metrics of Cycles on $K_2$

Consistent with Run 9, cycles are measured via:

$$\begin{aligned} L(C) &= \int_0^\tau \|\dot{p}(t)\| dt, \\ C_{\text{eff}} &= \frac{1}{L(C)} \int_0^\tau J_2(t) \cdot dA_2(t), \\ S(C) &= \min_{t \in [0, \tau]} d(\Omega(K_2), \partial\Omega(K_2)), \end{aligned}$$

where  $d$  depends on the proximity to criticality.

For near-critical cycles:

$$S(C) \rightarrow 0 \quad \text{as} \quad p \rightarrow p_c.$$

This is the origin of “critical slowing down” and “critical cycle degeneration.”

#### O.4.25 Birth of Time and Non-Trivial Cycles

According to the formal definition of  $\tau(K_2)$  (memory block C.1–7), time on  $K_2$  is determined by the recurrence period of correlation structures:

$$\tau(K_2) \sim \frac{1}{\omega_{\text{corr}}}.$$

Thus cycles on  $K_2$  are directly tied to:

- correlation length evolution  $\xi(t)$ ,
- cluster rearrangement timescales,
- occupation-flow dynamics  $dp/dt$ .

This establishes  $K_2$  as the first level with **\*\*dynamic temporal complexity\*\***.

#### O.4.26 Role of $K_2$ Cycles in the Hierarchy

Cycles on  $K_2$  are the precursor to:

- chemical reaction cycles on  $K_3$ ;
- membrane and metabolic cycles on  $K_4$ ;
- spike cycles on  $K_5$ ;
- cognitive recurrence loops on  $K_6$ ;
- institutional cycles on  $K_7$ ;
- civilisation-level reproduction cycles on  $K_8$ .

The essential mechanism introduced at  $K_2$  is *critical recurrence* — the possibility of interacting with thresholds in a cyclical manner.

Thus  $K_2$  is the minimal level where the Ontology of Continua admits non-trivial, threshold-sensitive, structurally complex cycles.



### O.4.27 Cycles on $K_3$

Level  $K_3$  corresponds to the RAF–chemical continuum: a self-sustaining reaction network enabled by autocatalysis and F-closure. Its state space is given by

$$\Omega(K_3) = \{(M, R, C, F) \mid (M', R') \in \text{RAF}(G_2)\},$$

where  $G_2$  is the connectivity graph inherited from  $K_2$ , and the RAF condition enforces both autocatalytic support and reachability of all reactants from the food-set  $F$ . Cycles on  $K_3$  emerge as recurrent chemical transformations involving reaction pathways, catalytic flows, and closure restoration.

### O.4.28 Fundamental Types of Cycles on $K_3$

Three structural classes of cycles define the dynamical complexity of the RAF–continuum.

**(1) Autocatalytic Reaction Cycles.** These are closed paths in reaction space:

$$m_{i_1} \xrightarrow{r_1} m_{i_2} \xrightarrow{r_2} \dots \xrightarrow{r_k} m_{i_1},$$

where each reaction  $r_j$  is catalysed by some  $m \in M'$ . Such cycles maintain concentrations, control production ratios, and propagate RAF stability. They represent the minimal dynamical substrate of  $K_3$ .

**(2) F-closure Restoration Cycles.** The RAF property requires:

$$\text{all reactants in } R' \text{ must be reachable from } F.$$

Perturbations (e.g. deficit of a molecule) can temporarily break F-closure. Cycles restoring F-closure satisfy:

$$F \rightarrow M_{\text{lost}} \rightarrow M_{\text{restored}} \rightarrow F$$

and are essential for the viability of  $K_3$ . They interact directly with the threshold  $\Theta_{\text{kas}}$ .

**(3) Energetic and Redox Cycles.** Following the extended chemical block (memory #52), chemical potentials include:

$$P_{\text{chem}}(t), P_{\text{redox}}(t), P_{\text{grad}}(t).$$

Correspondingly,  $K_3$  supports energetic cycles:

$$C_{\text{energy}} : P_{\text{chem}} \rightarrow P_{\text{redox}} \rightarrow P_{\text{grad}} \rightarrow P_{\text{chem}},$$

which prefigure the metabolic and gradient cycles of  $K_4$ .

These processes keep the RAF system within the energy-viable region of state space.

### O.4.29 Structural Tension and Thresholds

The structural tension  $T_3(t)$  is determined by:

$$T_3 = f(\text{deficit}_{\text{catalysis}}, 1 - \rho_{\text{closure}}, \text{energy deficit}, \text{redox imbalance}),$$

where  $\rho_{\text{closure}}$  is the closure metric defined by:

$$\rho_{\text{closure}} = \frac{\text{number of reachable reactants from } F}{\text{total reactants in } R'}.$$

Cycle viability requires:

$$T_3(t) < \Theta_{\text{death}}^{(3)},$$

and RAF existence requires:

$$\rho_{\text{cat}} \cdot \rho_{\text{closure}} \geq \Theta_{\text{kas}}.$$

Thus cycles on  $K_3$  exist inside a sharply defined viability region.

### O.4.30 Metrics of Cycles on $K_3$

Consistent with the universal metrics (Run 9):

**Length.**

$$L(C) = \int_0^\tau \|J_{\text{chem}}(t)\| dt,$$

where  $J_{\text{chem}}$  is the reaction–flow vector.

**Efficiency.**

$$C_{\text{eff}} = \frac{1}{L(C)} \int_0^\tau J_{\text{chem}}(t) \cdot dA_{\text{chem}}(t),$$

where  $A_{\text{chem}}$  are chemical-state axes (active/inactive reaction, presence/absence of species).

**Stability.**

$$S(C) = \min_{t \in [0, \tau]} d(\Omega(K_3), \partial\Omega(K_3)),$$

which corresponds to proximity to breaking either autocatalysis or closure.

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), \text{energy turnover}).$$

### O.4.31 Time and Cycle Periods

The chemical time on  $K_3$  is defined via:

$$\tau(K_3) = \int \frac{ds}{v_{\text{chem}}},$$

where  $v_{\text{chem}}$  is the effective reaction velocity. For RAF cycles, the characteristic recurrence period is:

$$\tau_{\text{RAF}} \sim \frac{1}{k_{\text{cat}}}$$

where  $k_{\text{cat}}$  is the effective catalytic rate.

Time itself is thus a function of recurring reaction structure.

### O.4.32 Birth of Higher-Level Cycles

Cycles at  $K_3$  serve as the precursors to  $K_4$  cycles:

- membrane-maintenance cycles,
- energy–gradient cycles across a boundary,
- proto-metabolic loops,
- pre-excitability cycles (via ion or proton gradients).

The continuity map  $K_3 \rightarrow K_4$  (memory #55, #62, #69, #70) shows that cycles of:

$$(P_{\text{chem}}, P_{\text{grad}}, J_{\text{chem}})$$

gradually become:

$$(P_{\text{mem}}, P_{\text{grad}}, J_{\text{in/out}})$$

as a membrane  $\partial\Omega$  forms.

Thus cycles on  $K_3$  encode the earliest organisational structure. Biological cycles (metabolic, membrane, excitability) emerge only later from that basis.

### O.4.33 Cycles on $K_4$

Level  $K_4$  corresponds to the protocellular continuum: a chemically self-organised domain enclosed by a semi-permeable membrane  $\partial\Omega(K_4)$ . Cycles define the persistent dynamic structure of the protocell: maintenance of gradients, metabolic turnover, membrane repair, energy conversion, replication accuracy, and early proto-excitability. They determine the viability of the protocell and the possibility of transition to the excitability level  $K_5$ .

### O.4.34 Classes of Cycles on $K_4$

Following the Biology U0.3b formalisation,  $K_4$  supports six fundamental families of cycles.

#### O.4.35 1. Membrane-Maintenance Cycles $C_{\text{mem}}$

These cycles repair, stabilise and reshape the membrane. They involve:

$$J_{\text{lipid}}, J_{\text{in/out}}, P_{\text{mem}}, \gamma_{\text{edge}}, \kappa_{\text{bend}},$$

where  $\gamma_{\text{edge}}$  is the line tension of membrane pores and  $\kappa$  is the bending modulus (memory #57).

The cycle structure is:

$$\partial\Omega \rightarrow \text{stretching} \rightarrow \text{repair} \rightarrow \partial\Omega.$$

Stability requires:

$$T_{\text{mem}}(t) < \Theta_{\text{mem}},$$

where  $T_{\text{mem}}$  measures osmotically and mechanically induced tension.

#### O.4.36 2. Gradient-Maintenance Cycles $C_{\text{grad}}$

The membrane introduces new axes:

$$A_{\text{in/out}}, A_{\text{perm}}, A_{\text{grad}}.$$

Correspondingly, gradient cycles regulate:

$$P_{\text{grad}} = (\Delta\text{pH}, \Delta\text{ion}, \Delta\text{redox}),$$

with flows:

$$J_{\text{pump}}, J_{\text{leak}}, J_{\text{channel}}.$$

The cycle has the form:

$$\Delta G \rightarrow \text{pump} \rightarrow \text{leak} \rightarrow \text{dissipation} \rightarrow \Delta G,$$

where  $P_{\text{grad}}$  must remain in the viable domain:

$$|P_{\text{grad}}| < \Theta_{\text{grad}}.$$

These cycles are precursors of the electrochemical structure of  $K_5$ .

### O.4.37 3. Energetic and Redox Cycles $C_{\text{energy}}$

From memory #67, the energetic architecture includes:

$$P_{\text{energy}}, P_{\text{redox}}, P_{\text{grad-energy}},$$

and flows:

$$J_{\text{energy}}, J_{\text{redox}}, J_{\text{grad}}.$$

The protometabolic cycle is:

$$\Delta G_{\text{in}} \rightarrow \text{activation} \rightarrow \text{transfer} \rightarrow \text{dissipation} \rightarrow \Delta G_{\text{in}}.$$

Existence requires:

$$T_{\text{cycle-energy}} < \Theta_{\text{cycle-energy}}.$$

These cycles stabilise the internal environment and support gradient-maintenance cycles.

### O.4.38 4. Metabolic Turnover Cycles $C_{\text{metabolic}}$

Building on the RAF precursor structure from  $K_3$ ,  $K_4$  forms localised protometabolic loops embedded inside the membrane:

$$M_{\text{in}} \xrightarrow{r_1} M_{\text{act}} \xrightarrow{r_2} M_{\text{out}} \xrightarrow{r_3} M_{\text{in}}.$$

They contribute to:

$$P_{\text{in}}, P_{\text{chem}}, P_{\text{stability}},$$

and maintain viable concentration ranges.

Disruption:

$$T_{\text{metabolic}} > \Theta_{\text{stability}} \Rightarrow \Omega(K_4) = \emptyset.$$

### O.4.39 5. Information-Stability Cycles $C_{\text{info}}$

From the replication accuracy block (memory #68), information-bearing polymers must satisfy:

$$P_{\text{copy}} = (1 - \varepsilon)^L \geq P_{\text{min}}.$$

The mutation–correction–selection loop:

$$\text{template} \rightarrow \text{copy} \rightarrow \text{mutation} \rightarrow \text{correction} \rightarrow \text{selection} \rightarrow \text{template}$$

forms a genuine cycle.

Its viability is equivalent to:

$$\varepsilon < \Theta_{\text{error}}.$$

Above the error threshold, the protocell’s informational structure collapses.

### O.4.40 6. Proto-Excitability Cycles $C_{\text{exc}}$

Based on memory #75 and #76, early channels and gradients produce ignition–front cycles (“proto–action potentials”):

$$P_{\text{grad}} \rightarrow \text{local ignition} \rightarrow \text{front propagation} \rightarrow \text{recovery} \rightarrow P_{\text{grad}}.$$

Ignition requires:

$$G_{\text{lat}} > \Theta_{\text{lat}}^{\text{crit}} \quad \text{and} \quad C_{\text{mem}} > C_{\text{mem}}^{\text{crit}}.$$

These cycles define the soft transition  $K_4 \rightarrow K_5$  (Polish W3 memory).

#### O.4.41 Metrics of Cycles at $K_4$

**Length.**

$$L(C) = \oint d\Omega,$$

where  $\Omega$  is the state space of internal composition.

**Efficiency.**

$$C_{\text{eff}} = \frac{\int J \cdot dA}{L(C)},$$

with flows  $J$  across:

$$A_{\text{mem}}, A_{\text{grad}}, A_{\text{metabolic}}.$$

**Stability.**

$$S(C) = \min_{t \in C} d(\Omega(K_4), \partial\Omega(K_4)).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), \text{energy turnover}, P_{\text{grad}}).$$

#### O.4.42 Time and Recurrence Structure

The defining timescales of  $K_4$  cycles include:

$$\tau_{\text{pump}}, \tau_{\text{leak}}, \tau_{\text{repair}}, \tau_{\text{metabolic}}, \tau_{\text{exc}}.$$

The continuum possesses time if these satisfy stable recurrence:

$$\tau_i < \infty \quad \forall i.$$

#### O.4.43 Collapse and Viability of $C(K_4)$

The death of  $K_4$  (memory #69) occurs if any of:

$$\Theta_{\text{mem}}, \Theta_{\text{grad}}, \Theta_{\text{cycle-energy}}, \Theta_{\text{error}}, \Theta_{\text{stability}}$$

is violated.

All cycles break simultaneously:

$$C_j \rightarrow \emptyset, \quad \Omega(K_4) = \emptyset.$$

#### O.4.44 Transition Toward $K_5$

Cycles  $C_{\text{exc}}$  and  $C_{\text{grad}}$  gradually stiffen into temporal–electrical loops, matching the proto–spike structure of  $K_5$ :

$$C_{\text{exc}} \longrightarrow C_{\text{spike}}.$$

This transition is continuous ( $C^1$ ), as ensured by the Polish W3 soft–threshold formulation.

#### O.4.45 Cycles on $K_5$

Level  $K_5$  corresponds to the early neural (bioelectrical) continuum. It is defined by the presence of an excitable membrane, ion-channel architecture, threshold-governed spikes, recovery dynamics, and electrical–chemical coupling. Cycles on  $K_5$  represent the fundamental recurrent processes that endow the system with excitability, signalling, rhythmicity and proto-network coordination.

#### O.4.46 Overview of Cycle Structure on $K_5$

According to the representability theorem for neurons (memory #26), the state space  $\Omega(K_5)$  includes:

$$V_m(t), P_{\text{ion}}(t), \{g_i(t)\}, \{w_{ij}(t)\}, \text{Ca}^{2+}(t), \text{channel states},$$

with thresholds:

$$\Theta_{\text{spike}}, \Theta_{\text{exc}}, \Theta_{\text{noise—elec}}, \Theta_{\text{front}}, \Theta_{\text{refrac}}, \Theta_{\text{spatial}}.$$

Cycles arise when flows  $J(t)$ , potentials  $P(t)$ , and axes  $A(t)$  return to an equivalent configuration after a finite time  $\tau$ . Time exists on  $K_5$  only because these cycles are stable and recurrent.

The core families of cycles are:

- action-potential cycles  $C_{\text{spike}}$ ,
- subthreshold oscillatory cycles  $C_{\text{sub}}$ ,
- refractory-reset cycles  $C_{\text{refrac}}$ ,
- channel-gating cycles  $C_{\text{gate}}$ ,
- calcium/second-messenger cycles  $C_{\text{Ca}}$ ,
- synaptic potentiation/depression cycles  $C_{\text{syn}}$ ,
- network rhythmic cycles  $C_{\text{net}}$  forming the extended continuum  $K'_5$ .

#### O.4.47 1. Action-Potential Cycles $C_{\text{spike}}$

The spike is a phase transition caused by:

$$T(t) > \Theta_{\text{spike}},$$

as stated in memory #26.

A complete cycle consists of:

rest  $\rightarrow$  depolarisation  $\rightarrow$  overshoot  $\rightarrow$  repolarisation  $\rightarrow$  hyperpolarisation  $\rightarrow$  rest.

Formally:

$$C_{\text{spike}} = \{(V_m(t), g_i(t), P_{\text{ion}}(t))\}_{t_0}^{t_0 + \tau_{\text{spike}}},$$

with period  $\tau_{\text{spike}} < \infty$ .

The spike cycle is the canonical electrical cycle on  $K_5$  and the central element of its structural identity.

#### O.4.48 2. Subthreshold Oscillatory Cycles $C_{\text{sub}}$

When the membrane potential oscillates without crossing  $\Theta_{\text{spike}}$  but approaches the excitable regime:

$$|V_m(t) - V_{\text{rest}}| < \Theta_{\text{exc}},$$

recurrent oscillations emerge:

$$V_m(t) = V_{\text{rest}} + A \sin(\omega t).$$

These cycles are precursors to rhythmogenesis in  $K'_5$ .

#### O.4.49 3. Refractory-Reset Cycles $C_{\text{refrac}}$

Following a spike, thresholds shift (memory #26, #77):

$$\Theta_{\text{spike}}(t + \delta t) > \Theta_{\text{spike}}(t).$$

The refractory cycle is:

spike  $\rightarrow$  absolute refractory  $\rightarrow$  relative refractory  $\rightarrow$  threshold reset.

Its period  $\tau_{\text{refrac}}$  determines the maximal firing rate.

#### O.4.50 4. Channel-Gating Cycles $C_{\text{gate}}$

Ion channels undergo cyclic transitions among states:

$$S_i \in \{\text{open, closed, inactive}\}.$$

The gating cycle is driven by:

$$g_i(t) = g_{\text{max}} m(t)^p h(t)^q,$$

where  $m, h$  satisfy their own recurrence equations.

The full cycle:

$$S_i^{(\text{open})} \rightarrow S_i^{(\text{inactive})} \rightarrow S_i^{(\text{closed})} \rightarrow S_i^{(\text{open})}.$$

Gating cycles stabilise  $C_{\text{spike}}$  and  $C_{\text{sub}}$ .

#### O.4.51 5. Calcium and Second-Messenger Cycles $C_{\text{Ca}}$

Calcium concentration dynamics:

$$\text{Ca}^{2+}(t) \rightleftharpoons \text{buffered Ca, pumped Ca,}$$

form cycles involving release, sequestration and pumping.

Stability requires:

$$T_{\text{Ca}} < \Theta_{\text{noise-elec}}.$$

These cycles modulate both excitability and synaptic change.

#### O.4.52 6. Synaptic Potentiation/Depression Cycles $C_{\text{syn}}$

Sustained activity leads to cyclic modulation of synaptic weight:

$$w_{ij}(t + \tau) = w_{ij}(t),$$

driven by:

$$J_{\text{pre}}, J_{\text{post}}, [\text{Ca}^{2+}],$$

and threshold conditions for plasticity:

$$T_{\text{plastic}} > \Theta_{\text{plastic}}.$$

These cycles enable memory and persistent patterns.

#### O.4.53 7. Network-Level Cycles $C_{\text{net}}$ (Continuum $K'_5$ )

Using the representability theorem for neural networks (memory #26), a network forms an extended continuum  $K'_5$  with cycles:

$$C_{\text{rhythm}}, \quad C_{\text{sync}}, \quad C_{\text{burst}}.$$

They arise when collective activation fields  $p_i(t)$  satisfy the recurrence:

$$p_i(t + \tau) = p_i(t),$$

and threshold conditions:

$$T_{\text{sync}} < \Theta_{\text{front}}, \quad T_{\text{burst}} < \Theta_{\text{spatial}}.$$

These cycles are the basis of organized oscillations and proto-cognition in  $K'_5$ .

#### O.4.54 Metrics of Cycles on $K_5$

Using the universal cycle metrics (memory #41):

**Length.**

$$L(C) = \oint d\Omega(K_5).$$

**Efficiency.**

$$C_{\text{eff}} = \frac{\oint J \cdot dA}{L(C)},$$

where  $A$  includes:

$$A_{\text{exc}}, A_{\text{ion}}, A_{\text{syn}}, A_{\text{Ca}}.$$

**Stability.**

$$S(C) = \min_{t \in C} d(\Omega(K_5), \partial\Omega(K_5)).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), P_{\text{ion}}, P_{\text{Ca}}, w_{ij}).$$

#### O.4.55 Collapse of $C(K_5)$

According to memory #69, collapse occurs if:

$$\Theta_{\text{exc}}, \Theta_{\text{spike}}, \Theta_{\text{noise-elec}}, \Theta_{\text{front}}, \Theta_{\text{spatial}}, \Theta_{\text{refrac}}$$

are violated.

Then:

$$C_j \rightarrow \emptyset, \quad \Omega(K_5) = \emptyset, \quad k_5 \rightarrow 0.$$

#### O.4.56 Continuity from $K_4$ to $K_5$

From Polish W3 (memory #74), the transition is  $C^1$ -continuous:

$$C_{\text{exc}}^{(K_4)} \longrightarrow C_{\text{spike}}^{(K_5)},$$

with smooth emergence of:

$$A_{\text{exc}}, \Theta_{\text{spike}}, C_{\text{refrac}}.$$

This soft transition ensures the viability of the first neural cells.

#### O.4.57 Cycles on $K_6$

The level  $K_6$  is the cognitive continuum: a system with representational axes, predictive dynamics, binding mechanisms, memory integration and structured internal models. Cycles on  $K_6$  correspond to recurrent cognitive processes that stabilise representations, maintain coherence, and permit prediction and learning.

The structure of cycles on  $K_6$  is governed by the cognitive axes

$$A_{\text{rep}}, A_{\text{bind}}, A_{\text{pred}}, A_{\text{mem}}, A_{\text{attn}},$$

the potentials

$$P_{\text{rep}}, P_{\text{pred}}, P_{\text{error}}, P_{\text{salience}}, P_{\text{stability}},$$

flows  $J_6$  (memory #44), and thresholds:

$$\Theta_{\text{PE}}, \Theta_{\text{bind}}, \Theta_{\text{coh}}, \Theta_{\text{expressivity}}, \Theta_{\text{noise}}.$$



#### O.4.58 Overview of Cognition-Level Cycles

Cognitive cycles arise because cognitive states must be actively maintained against noise, instability and representational decay. Every cycle on  $K_6$  expresses a closed-loop regime of:

perception  $\rightarrow$  prediction  $\rightarrow$  comparison  $\rightarrow$  error evaluation  $\rightarrow$  update  $\rightarrow$  re – stabilisation.

These cycles provide the structural basis of cognition.

#### O.4.59 1. Predictive-Processing Cycle $C_{PE}$

The core cognitive cycle arises from predictive processing, governed by the flow  $J_{PE}$  (memory #44):

$$C_{PE} = \{(P_{pred}(t), P_{rep}(t), P_{error}(t))\}_{t_0}^{t_0 + \tau_{PE}}.$$

The cycle exists when the prediction–error loop is stable:

$$P_{error}(t + \tau_{PE}) = P_{error}(t),$$

and cognitive tension satisfies:

$$T_{PE} < \Theta_{PE}.$$

Violation of  $\Theta_{PE}$  generates cognitive collapse or chaotic prediction error.

#### O.4.60 2. Binding Cycle $C_{bind}$

Cognitive representations have to bind their components across representational axes  $A_{bind}$ .

The binding cycle:

feature extraction  $\rightarrow$  coherence formation  $\rightarrow$  integration  $\rightarrow$  comparison  $\rightarrow$  re – binding.

Formally:

$$C_{bind} = \{A_{bind}(t), P_{rep}(t)\}_{t_0}^{t_0 + \tau_{bind}}.$$

Existence requires:

$$T_{bind} < \Theta_{bind}.$$

This cycle corresponds to perceptual and conceptual coherence.

#### O.4.61 3. Attention–Salience Cycle $C_{attn}$

Attention modulates the salience potential  $P_{salience}$  and selectively routes flows  $J_6$ .

The cycle is:

salience detection  $\rightarrow$  attentional shift  $\rightarrow$  enhancement  $\rightarrow$  integration  $\rightarrow$  decay.

The recurrence condition:

$$P_{salience}(t + \tau_{attn}) = P_{salience}(t).$$

Stability requires:

$$T_{attn} < \Theta_{coh}.$$

#### O.4.62 4. Working-Memory Cycle $C_{\text{wm}}$

Working memory involves recurrent maintenance of representational states. The classical loop:

load  $\rightarrow$  maintain  $\rightarrow$  transform  $\rightarrow$  offload.

Formally:

$$C_{\text{wm}} = \{P_{\text{mem}}(t), A_{\text{mem}}(t)\}_{t_0}^{t_0 + \tau_{\text{wm}}}.$$

Existence requires:

$$T_{\text{wm}} < \Theta_{\text{expressivity}}.$$

This threshold embodies the minimal expressive capacity of  $K_6$ .

#### O.4.63 5. Conceptual Cycle $C_{\text{concept}}$

A concept is stabilised through recurrence across transformation axes:

$$C_{\text{concept}} = \{A_{\text{rep}}(t), P_{\text{rep}}(t), P_{\text{stability}}(t)\}.$$

It is defined by:

$$P_{\text{stability}}(t + \tau) = P_{\text{stability}}(t),$$

and coherence threshold:

$$T_{\text{rep}} < \Theta_{\text{coh}}.$$

This cycle generalises symbolic stability without requiring language.

#### O.4.64 6. Cognitive Rhythm Cycle $C_{\text{cog-rhythm}}$

Cognition often relies on rhythmic coordination (precursor to the neural rhythms of  $K'_5$  but now internal to representation).

The cognitive rhythm cycle:

$$P_{\text{rep}}(t) = P_0 + A \sin(\omega t), \quad A < \Theta_{\text{noise}}.$$

This cycle maintains coherence of representational frames.

#### O.4.65 7. Multi-Scale Integration Cycle $C_{\text{integrate}}$

Cognition integrates across timescales:

fast prediction  $\leftrightarrow$  slow model update.

Formally a slow-fast cycle:

$$C_{\text{integrate}} = \{(P_{\text{pred}}^{\text{fast}}(t), P_{\text{rep}}^{\text{slow}}(t))\}.$$

The cycle is viable only when:

$$T_{\text{multi}} < \min(\Theta_{\text{PE}}, \Theta_{\text{expressivity}}).$$

#### O.4.66 8. Meta-Cognitive Cycle $C_{\text{meta}}$

Meta-cognition corresponds to cycles in which:

$$\text{model} \rightarrow \text{self} - \text{monitoring} \rightarrow \text{evaluation} \rightarrow \text{correction} \rightarrow \text{model}.$$

This is the earliest precursor of the meta-theoretical continua  $K_9$ .

Stability condition:

$$T_{\text{meta}} < \Theta_{\text{coh}}.$$

#### O.4.67 Metrics of Cognitive Cycles

Using the universal cycle metrics (memory #41):

**Length.**

$$L(C) = \oint d\Omega(K_6).$$

**Efficiency.**

$$C_{\text{eff}} = \frac{\oint J_6 \cdot dA}{L(C)}.$$

**Stability.**

$$S(C) = \min_{t \in C} d(\Omega(K_6), \partial\Omega(K_6)).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), P_{\text{pred}}, P_{\text{mem}}, P_{\text{error}}).$$

#### O.4.68 Collapse of $K_6$ Cycles

Cognitive collapse (memory #69) occurs when:

$$T_{\text{PE}} > \Theta_{\text{PE}}, \quad T_{\text{bind}} > \Theta_{\text{bind}}, \quad T_{\text{coh}} > \Theta_{\text{coh}}, \quad T_{\text{expressivity}} > \Theta_{\text{expressivity}}.$$

Then:

$$C_j \rightarrow \emptyset, \quad \Omega(K_6) = \emptyset, \quad k_6 \rightarrow 0.$$

#### O.4.69 Continuity from $K_5$ to $K_6$

The transition  $K'_5 \rightarrow K_6$  (memory #26, #76) corresponds to the rise of:

$$A_{\text{rep}}, A_{\text{bind}}, A_{\text{pred}}, A_{\text{mem}},$$

together with new thresholds:

$$\Theta_{\text{PE}}, \Theta_{\text{bind}}, \Theta_{\text{coh}}, \Theta_{\text{expressivity}}.$$

This is a dimensional transition governed by universal conditions:

$$T > \Theta_{\text{dim}}.$$

The emergent cycles on  $K_6$  complete the formation of a fully cognitive continuum.

#### O.4.70 Cycles on $K_7$

The level  $K_7$  represents the social continuum: a structured domain of roles, norms, communicative flows, trust dynamics, and institutional stabilisation. Cycles on  $K_7$  formalise recurrent social processes that maintain the persistence of social groups, institutions, and collective behaviour.

Axes on  $K_7$  include:

$$A_{\text{role}}, A_{\text{norm}}, A_{\text{trust}}, A_{\text{comm}}, A_{\text{coop}},$$

with potentials:

$$P_{\text{trust}}, P_{\text{coh}}, P_{\text{normative}}, P_{\text{stability}},$$

and flows (memory #44, #45):

$$J_{\text{comm}}, J_{\text{coop}}, J_{\text{stab}}.$$

Thresholds determining cycle viability:

$$\Theta_{\text{trust}}, \Theta_{\text{norm}}, \Theta_{\text{coh-soc}}, \Theta_{\text{embed}}, \Theta_{\text{fragility}}.$$

#### O.4.71 Overview

A social continuum persists not because its components are static, but because they are dynamically maintained through recurrent cycles of communication, cooperation, norm enforcement, trust regeneration, and institutional reproduction. This section formalises these cycles in the language of the OC framework.

##### O.4.72 1. Communication Cycle $C_{\text{comm}}$

Communication is the core process establishing shared state within the social continuum.

The cycle:

$$\text{signal} \rightarrow \text{interpretation} \rightarrow \text{feedback} \rightarrow \text{alignment} \rightarrow \text{signal}.$$

Formally:

$$C_{\text{comm}} = \{A_{\text{comm}}(t), P_{\text{coh}}(t), J_{\text{comm}}(t)\}.$$

Stability requires:

$$T_{\text{comm}} < \Theta_{\text{coh-soc}}.$$

##### O.4.73 2. Trust-Regeneration Cycle $C_{\text{trust}}$

Trust is a dynamic potential  $P_{\text{trust}}$  subject to fluctuations, erosion, and restoration.

Cycle structure:

$$\text{cooperation} \rightarrow \text{verification} \rightarrow \text{reinforcement} \rightarrow \text{renewal} \rightarrow \text{cooperation}.$$

Formally:

$$C_{\text{trust}} = \{P_{\text{trust}}(t), J_{\text{coop}}(t), A_{\text{trust}}(t)\}.$$

Existence condition:

$$P_{\text{trust}}(t + \tau_{\text{trust}}) = P_{\text{trust}}(t).$$

Threshold:

$$T_{\text{trust}} < \Theta_{\text{trust}}.$$

#### O.4.74 3. Norm Enforcement Cycle $C_{\text{norm}}$

Social norms regulate expectations and behaviour.

Cycle:

activation  $\rightarrow$  monitoring  $\rightarrow$  sanction  $\rightarrow$  repair  $\rightarrow$  activation.

Form:

$$C_{\text{norm}} = \{A_{\text{norm}}(t), P_{\text{normative}}(t), J_{\text{stab}}(t)\}.$$

Threshold condition:

$$T_{\text{norm}} < \Theta_{\text{norm}}.$$

#### O.4.75 4. Role-Structure Cycle $C_{\text{role}}$

Social roles must be continuously reproduced to sustain institutional order.

Cycle:

role learning  $\rightarrow$  performance  $\rightarrow$  evaluation  $\rightarrow$  adjustment  $\rightarrow$  role learning.

Formally:

$$C_{\text{role}} = \{A_{\text{role}}(t), P_{\text{stability}}(t), J_{\text{comm}}(t)\}.$$

Threshold:

$$T_{\text{role}} < \Theta_{\text{coh-soc}}.$$

#### O.4.76 5. Cooperation Cycle $C_{\text{coop}}$

Cooperation, sustained by  $J_{\text{coop}}$ , is a recurrent structure:

proposal  $\rightarrow$  coordination  $\rightarrow$  execution  $\rightarrow$  reward  $\rightarrow$  renewal.

Formally:

$$C_{\text{coop}} = \{J_{\text{coop}}(t), P_{\text{trust}}(t), A_{\text{coop}}(t)\}.$$

Stability condition:

$$T_{\text{coop}} < \Theta_{\text{trust}}.$$

#### O.4.77 6. Institutional Reproduction Cycle $C_{\text{inst}}$

Institutions are persistent social structures defined by:

codification  $\rightarrow$  implementation  $\rightarrow$  monitoring  $\rightarrow$  repair  $\rightarrow$  codification.

Institutional reproduction depends on:

$$C_{\text{inst}} = \{A_{\text{norm}}(t), A_{\text{role}}(t), P_{\text{stability}}(t), J_{\text{stab}}(t)\}.$$

Threshold:

$$T_{\text{inst}} < \Theta_{\text{embed}}.$$

The embedding-space constraint encodes the dependence of  $K_7$  on  $M_7$ .

#### O.4.78 7. Social-Stability Cycle $C_{\text{stab}}$

This is a multi-cycle integration:

$$C_{\text{stab}} = C_{\text{comm}} \cup C_{\text{trust}} \cup C_{\text{norm}} \cup C_{\text{role}} \cup C_{\text{coop}}.$$

Stability requires:

$$T_{\text{soc}} < \min(\Theta_{\text{trust}}, \Theta_{\text{norm}}, \Theta_{\text{coh-soc}}, \Theta_{\text{embed}}).$$

This cycle corresponds to the existence of a coherent social continuum.

#### O.4.79 Cycle Metrics on $K_7$

**Length.**

$$L(C) = \oint d\Omega(K_7).$$

**Efficiency.**

$$C_{\text{eff}} = \frac{\oint J_7 \cdot dA}{L(C)}.$$

**Stability.**

$$S(C) = \min_{t \in C} d(\Omega(K_7), \partial\Omega(K_7)).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), P_{\text{trust}}, P_{\text{normative}}, P_{\text{coh}}).$$

#### O.4.80 Collapse of $K_7$ Cycles

Collapse of social cycles occurs when:

$$T_{\text{trust}} > \Theta_{\text{trust}}, \quad T_{\text{norm}} > \Theta_{\text{norm}}, \quad T_{\text{comm}} > \Theta_{\text{coh-soc}},$$

or when embedding-space constraints fail:

$$M_7 \not\subseteq \Omega(K_7).$$

Collapse implies:

$$C_j \rightarrow \emptyset, \quad \Omega(K_7) = \emptyset, \quad k_7 \rightarrow 0.$$

#### O.4.81 Continuity from $K_6$ to $K_7$

The transition  $K_6 \rightarrow K_7$  (memory #49) occurs when:

$$T_{\text{comm}} > \Theta_{\text{dim}},$$

and new axes appear:

$$A_{\text{role}}, A_{\text{norm}}, A_{\text{trust}}, A_{\text{coop}}.$$

This generates new thresholds and permits the existence of social cycles with no analogue at the cognitive level  $K_6$ .

#### O.4.82 Cycles on $K_8$

The level  $K_8$  represents the civilizational continuum: a large-scale, multi-component system integrating population, infrastructures, energy base, logistics, information systems and economic mechanisms of reproduction. Cycles on  $K_8$  formalise the recurrent macro-processes that allow a civilization to sustain itself over historical time.

The state of  $K_8$  is:

$$s(t) = (\text{Pop}(t), \text{Infra}(t), \text{Tech}(t), \text{Energy}(t), \text{Info}(t), \text{Econ}(t)).$$

The fundamental flows (memory #28) are:

$$J^{(8)} = \begin{pmatrix} J_{\text{energy}}, J_{\text{material}}, J_{\text{logistics}}, \\ J_{\text{info}}, J_{\text{population}} \end{pmatrix}.$$

Critical thresholds:

$$\Theta_E, \Theta_L, \Theta_I, \Theta_M, \Theta_R.$$

Structural tension:

$$T_8 = w_E(E_{\text{crit}} - E_{\text{density}}) + w_L(L_{\text{demand}} - L_{\text{capacity}}) \\ + w_M(M_{\text{demand}} - M_{\text{repair}}) + w_I(I_{\text{complexity}} - I_{\text{coherence}}) \\ + w_R(R_{\text{demand}} - R_{\text{supply}})$$

The existence of a civilizational continuum requires:

$$T_8 < 0, \quad k_8 > 0.$$

#### O.4.83 Overview

Civilizational cycles describe the macro-recurrent processes through which societies maintain material reproduction, energy flows, knowledge coherence, logistics, infrastructure sustainability and demographic stability. These cycles integrate the dynamic interactions between subsystems and ensure the persistence of  $\Omega(K_8)$ .

##### O.4.84 1. Energy Reproduction Cycle $C_{\text{energy}}$

Energy density is the primary determinant of civilizational viability.

Extraction  $\rightarrow$  Conversion  $\rightarrow$  Distribution  $\rightarrow$  Use  $\rightarrow$  Maintenance  $\rightarrow$  Extraction.

Formally:

$$C_{\text{energy}} = \{J_{\text{energy}}(t), E_{\text{density}}(t), \text{Infra}_{\text{energy}}(t)\}.$$

Threshold:

$$E_{\text{density}} > \Theta_E.$$

##### O.4.85 2. Material Production–Logistics Cycle $C_{\text{mat/log}}$

The macro-cycle linking production, transport and consumption.

Production  $\rightarrow$  Transport  $\rightarrow$  Use  $\rightarrow$  Waste processing  $\rightarrow$  Repair  $\rightarrow$  Production.

Form:

$$C_{\text{mat/log}} = \left\{ J_{\text{material}}(t), J_{\text{logistics}}(t), \right. \\ \left. L_{\text{capacity}}(t), \text{Infra}_{\text{industrial}}(t) \right\}.$$

Threshold:

$$L_{\text{capacity}} > \Theta_L.$$

### O.4.86 3. Infrastructure Maintenance Cycle $C_{\text{infra}}$

Civilizational stability depends on maintaining complex infrastructures.

Inspection  $\rightarrow$  Repair  $\rightarrow$  Upgrade  $\rightarrow$  Integration  $\rightarrow$  Inspection.

Form:

$$C_{\text{infra}} = \left\{ \begin{array}{l} M_{\text{repair}}(t), \text{ Infra}(t), \\ J_{\text{material}}(t) \end{array} \right\}.$$

Threshold:

$$M_{\text{repair}} > \Theta_M.$$

### O.4.87 4. Information Coherence Cycle $C_{\text{info}}$

A civilization requires consistent information-processing capacity.

Generation  $\rightarrow$  Transmission  $\rightarrow$  Integration  $\rightarrow$  Coherence checking  $\rightarrow$  Update  $\rightarrow$  Generation.

Form:

$$C_{\text{info}} = \left\{ \begin{array}{l} J_{\text{info}}(t), I_{\text{coherence}}(t), \\ I_{\text{complexity}}(t) \end{array} \right\}.$$

Threshold:

$$I_{\text{coherence}} > \Theta_I.$$

### O.4.88 5. Demographic Reproduction Cycle $C_{\text{pop}}$

Demographic stability is necessary for sustaining labor, knowledge transfer, military capacity and inter-generational reproduction.

Birth  $\rightarrow$  Growth  $\rightarrow$  Education  $\rightarrow$  Labour  $\rightarrow$  Aging  $\rightarrow$  Birth.

Form:

$$C_{\text{pop}} = \left\{ \begin{array}{l} J_{\text{population}}(t), R_{\text{supply}}(t), \\ R_{\text{demand}}(t) \end{array} \right\}.$$

Threshold:

$$R_{\text{supply}} > \Theta_R.$$

### O.4.89 6. Macro-Economic Reproduction Cycle $C_{\text{econ}}$

The economic cycle:

Investment  $\rightarrow$  Production  $\rightarrow$  Distribution  $\rightarrow$  Consumption  $\rightarrow$  Reinvestment.

Formally:

$$C_{\text{econ}} = \left\{ \begin{array}{l} \text{Econ}(t), J_{\text{material}}(t), \\ J_{\text{energy}}(t), J_{\text{info}}(t) \end{array} \right\}.$$

Stability requires:

$$T_{\text{econ}} < \min(\Theta_L, \Theta_E).$$



#### O.4.90 7. Integrated Civilizational Cycle $C_{\text{civ}}$

The civilizational continuum persists only if all major reproduction cycles remain within stable thresholds.

$$C_{\text{civ}} = C_{\text{energy}} \cup C_{\text{mat/log}} \cup C_{\text{infra}} \cup C_{\text{info}} \cup C_{\text{pop}} \cup C_{\text{econ}}$$

Stability requires:

$$T_8 < \min(\Theta_E, \Theta_L, \Theta_I, \Theta_M, \Theta_R).$$

If any subthreshold fails,  $\Omega(K_8)$  undergoes collapse.

#### O.4.91 Cycle Metrics on $K_8$

**Cycle length.**

$$L(C) = \oint d\Omega(K_8).$$

**Efficiency.**

$$C_{\text{eff}} = \frac{\oint J^{(8)} \cdot dA}{L(C)}.$$

**Stability.**

$$S(C) = \min_{t \in C} d(\Omega(K_8), \partial\Omega(K_8)).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), E_{\text{density}}, L_{\text{capacity}}, I_{\text{coherence}}).$$

#### O.4.92 Collapse of $K_8$ Cycles

Collapse occurs when:

$$E_{\text{density}} < \Theta_E, \quad L_{\text{capacity}} < \Theta_L, \quad I_{\text{coherence}} < \Theta_I, \quad M_{\text{repair}} < \Theta_M, \quad R_{\text{supply}} < \Theta_R.$$

Under collapse:

$$C_j \rightarrow \emptyset, \quad \Omega(K_8) = \emptyset, \quad k_8 \rightarrow 0.$$

This corresponds to full civilizational breakdown.

#### O.4.93 Continuity from $K_7$ to $K_8$

The transition  $K_7 \rightarrow K_8$  occurs when:

$$T_{\text{soc}} > \Theta_{\text{dim}},$$

and new axes appear:

$$A_{\text{energy}}, A_{\text{logistics}}, A_{\text{infra}}, A_{\text{population}}, A_{\text{econ}}.$$

This introduces macro-reproduction cycles with no analogue at the social level  $K_7$ .

#### O.4.94 Cycles on $K_9$

The level  $K_9$  represents the meta-theoretical continuum: the domain of theories, paradigms, logics, ontologies and methodological frameworks. Cycles on  $K_9$  describe the recurrent structural processes through which scientific knowledge evolves, stabilises, reorganises and sometimes collapses.

The state space is

$$\Omega(K_9) = \{\text{theories, logics, paradigms, ontologies, models, interpretations}\}.$$

The axes  $A^9(t)$  include:

$$A^9 = \{\text{logical distinctions, ontological distinctions, methodological distinctions, interpretative dimensions}\}.$$

Potentials  $P^9(t)$ :

$$P^9(t) = \{\text{explanatory strength, predictive power, simplicity, systematicity, coherence}\}.$$

Flows  $J^9(t)$ :

$$J^9(t) = \{\text{argumentation, proofs, interpretations, criticism, theory transitions}\}.$$

Critical thresholds:

$$\Theta_{\text{logic}}^9, \quad \Theta_{\text{emp}}^9, \quad \Theta_{\text{heur}}^9, \quad \Theta_{\text{axiom}}^9.$$

A meta-theoretical continuum exists if

$$k(K_9) > 0, \quad T_9 < \min(\Theta^9).$$

#### O.4.95 Overview

Cycles on  $K_9$  encode the dynamics of theory formation, stabilization, evaluation, correction, and revolutionary replacement. They formalise the systemic processes underlying scientific knowledge and its evolution. These cycles operate on the structural space  $\Omega(K_9)$  and are driven by logical, empirical and methodological pressures expressed through  $T_9$ .

#### O.4.96 1. The Normal Science Cycle $C_{\text{norm}}$

This cycle corresponds to stable periods of theory-driven research.

$$\begin{aligned} &\text{Puzzle selection} \rightarrow \text{Method application} \rightarrow \text{Result integration} \\ &\rightarrow \text{Coherence increase} \rightarrow \text{Puzzle selection.} \end{aligned}$$

Formally:

$$C_{\text{norm}} = \{J_{\text{arg}}^9, P_{\text{coherence}}^9, P_{\text{systematicity}}^9\},$$

where  $J_{\text{arg}}^9$  is the argumentative flow supporting integration.

Stability requires:

$$\Theta_{\text{logic}}^9 < P_{\text{coherence}}^9.$$

#### O.4.97 2. The Proof-Refutation Cycle $C_{\text{proof/ref}}$

The foundational logical cycle in science.

$$\begin{aligned} &\text{Hypothesis} \rightarrow \text{Derivation} \rightarrow \text{Test} \rightarrow \text{Refutation or Confirmation} \\ &\rightarrow \text{Refinement} \rightarrow \text{Hypothesis.} \end{aligned}$$

Form:

$$C_{\text{proof/ref}} = \{J_{\text{proof}}^9, J_{\text{crit}}^9, P_{\text{predictive}}^9, \Theta_{\text{emp}}^9\}.$$

Threshold:

$$P_{\text{predictive}}^9 > \Theta_{\text{emp}}^9.$$

### O.4.98 3. The Interpretation Cycle $C_{\text{interp}}$

Interpretative cycles reorganize theories without altering empirical content.

Reformulation  $\rightarrow$  Conceptual mapping  $\rightarrow$  Reduction or extension  $\rightarrow$  Unification  $\rightarrow$  Reformulation.

Form:

$$C_{\text{interp}} = \{A_{\text{interpretation}}^9, P_{\text{simplicity}}^9, J_{\text{reinterpretation}}^9\}.$$

Stability condition:

$$P_{\text{simplicity}}^9 > \Theta_{\text{heur}}^9.$$

### O.4.99 4. The Paradigm Cycle $C_{\text{paradigm}}$

This cycle describes long-term evolution of entire scientific paradigms.

Model development  $\rightarrow$  Problem solving  $\rightarrow$  Anomaly accumulation  
 $\rightarrow$  Crisis  $\rightarrow$  Reconstruction  $\rightarrow$  Model development.

Form:

$$C_{\text{paradigm}} = \{P_{\text{explanatory}}^9, J_{\text{crit}}^9, A_{\text{ontology}}^9, \Theta_{\text{axiom}}^9\}.$$

Threshold (Kuhn-type):

$$T_9 > \Theta_{\text{logic}}^9 \Rightarrow \text{Paradigm instability.}$$

### O.4.100 5. The Scientific Revolution Cycle $C_{\text{rev}}$

The rapid, high-tension transition where a new paradigm replaces the old.

Crisis  $\rightarrow$  Competing frameworks  $\rightarrow$  Selection  $\rightarrow$  Reinterpretation of past  $\rightarrow$  Stabilisation  $\rightarrow$  Crisis.

Formally:

$$C_{\text{rev}} = \{J_{\text{transition}}^9, A_{\text{methodology}}^9, P_{\text{explanatory}}^9, P_{\text{predictive}}^9\}.$$

The transition condition (meta-theoretical phase transition):

$$T_9 = \Theta_{\text{dim}}^9.$$

Cycle reinstates a new structure of  $\Omega(K_9)$ .

### O.4.101 6. The Logical Consistency Cycle $C_{\text{logic}}$

Based on internal logic and axiomatics.

Axiom selection  $\rightarrow$  Derivation  $\rightarrow$  Consistency check  $\rightarrow$  Revision  $\rightarrow$  Axiom selection.

Form:

$$C_{\text{logic}} = \{A_{\text{logic}}^9, \Theta_{\text{axiom}}^9, J_{\text{proof}}^9\}.$$

Linked to Gödel-type limits:

$$P_{\text{coherence}}^9 > \Theta_{\text{axiom}}^9 \text{ for viability.}$$

#### O.4.102 7. The Meta-Theory Evolution Cycle $C_{\text{meta}}$

This cycle governs transitions between entire meta-frameworks and logics.

Framework  $\rightarrow$  Cross-comparison  $\rightarrow$  Meta-critique  $\rightarrow$  Higher-order reconstruction  $\rightarrow$  Framework.

Form:

$$C_{\text{meta}} = \{A_{\text{paradigm}}^9, A_{\text{methodology}}^9, J_{\text{meta}}^9, \Theta_{\text{heur}}^9\}.$$

Condition:

$$k(K_9) > 0 \Leftrightarrow C_{\text{meta}} \text{ remains stable.}$$

#### O.4.103 Cycle Metrics on $K_9$

Using the general definitions (memory #41):

**Cycle length.**

$$L(C) = \oint d\Omega(K_9).$$

**Efficiency.**

$$C_{\text{eff}} = \frac{\oint J^9 \cdot dA^9}{L(C)}.$$

**Stability.**

$$S(C) = \min_{t \in C} d(\Omega(K_9), \partial\Omega(K_9)).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), P_{\text{coherence}}^9, P_{\text{predictive}}^9).$$

#### O.4.104 Collapse of $K_9$ Cycles

Collapse occurs when:

$$P_{\text{coherence}}^9 < \Theta_{\text{logic}}^9, \quad P_{\text{predictive}}^9 < \Theta_{\text{emp}}^9, \quad P_{\text{simplicity}}^9 < \Theta_{\text{heur}}^9.$$

Then:

$$C_j \rightarrow \emptyset, \quad \Omega(K_9) = \emptyset, \quad k(K_9) \rightarrow 0.$$

This corresponds to the death of a paradigm or meta-framework.

#### O.4.105 Continuity from $K_8$ to $K_9$

The transition  $K_8 \rightarrow K_9$  occurs when:

$$T_8 > \Theta_{\text{dim}}^9,$$

and new reflective axes appear:

$$A_{\text{logic}}^9, A_{\text{ontology}}^9, \\ A_{\text{methodology}}^9, A_{\text{interpretation}}^9.$$

This yields a new meta-theoretical continuum with its own cycles and stability conditions.

#### O.4.106 Cycles on $K_{10}$

The level  $K_{10}$  represents the meta-model continuum: the space of higher-order theoretical structures capable of describing, evaluating and transforming entire meta-theories ( $K_9$ ), including their logics, axiomatics, methodological frameworks, and the mechanisms of self-description. Cycles on  $K_{10}$  correspond to recurrent processes in the evolution of meta-frameworks, meta-logics and higher-order operators.

The state space:

$$\Omega(K_{10}) = \{\text{meta-models, meta-logics, meta-ontologies, higher-order operators, self-descriptions}\}.$$

Axes:

$$A^{10} = \{\text{meta-logical distinctions, meta-ontological distinctions, meta-methodological distinctions, cross-theory abstraction axes}\}.$$

Potentials:

$$P^{10} = \{\text{meta-coherence, meta-explanatory power, meta-predictivity, consistency of higher-order operators}\}.$$

Flows:

$$J^{10} = \{\text{meta-transformation flows, operator evolution, cross-level abstraction flows, self-referential reasoning}\}.$$

Thresholds:

$$\Theta_{10} = \{\Theta_{\text{coh}}^{10}, \Theta_{\text{cons}}^{10}, \Theta_{\text{meta}}^{10}, \Theta_{\text{dim}}^{10}\}.$$

The continuum exists if:

$$k(K_{10}) > 0, \quad T_{10} < \min(\Theta_{10}).$$

#### O.4.107 Overview

Cycles on  $K_{10}$  orchestrate the dynamic evolution of meta-models, meta-logical systems and higher-order inferential frameworks. They regulate how theories about theories develop, how self-descriptions stabilize, how operator systems evolve, and under which conditions the entire structure of scientific knowledge transitions to a higher-order organisation.

These cycles operate on  $\Omega(K_{10})$  and lie above the  $K_9$  cycles of paradigms and meta-theories. They are responsible for the creation of universal operators, the structuring of meta-spaces, and the emergence of self-referential stability.

#### O.4.108 1. The Meta-Operator Cycle $C_{\text{op}}$

This cycle governs the evolution of higher-order operators such as  $\Psi$ ,  $\Phi$ ,  $\Lambda$ ,  $U$ , and  $X$ .

Operator definition  $\rightarrow$  Application  $\rightarrow$  Evaluation  $\rightarrow$  Revision  $\rightarrow$  Operator definition.

Formally:

$$C_{\text{op}} = \{J_{\text{op}}^{10}, P_{\text{cons}}^{10}, \Theta_{\text{coh}}^{10}\}.$$

Stability condition:

$$P_{\text{cons}}^{10} > \Theta_{\text{coh}}^{10}.$$

#### O.4.109 2. The Higher-Order Logic Cycle $C_{\text{logic}}^{(10)}$

This cycle pertains to the formation and refinement of meta-logics capable of describing and constraining logical systems on  $K_9$ .

Meta – axiom selection  $\rightarrow$  Higher – order inference  $\rightarrow$  Consistency analysis  
 $\rightarrow$  Meta – revision  $\rightarrow$  Meta – axiom selection.

Form:

$$C_{\text{logic}}^{(10)} = \{ A_{\text{meta-logic}}^{10}, J_{\text{meta-proof}}^{10}, \Theta_{\text{cons}}^{10} \}.$$

Gödel-type constraint at meta-level:

$$P_{\text{meta-coherence}}^{10} > \Theta_{\text{cons}}^{10}.$$

#### O.4.110 3. The Meta-Theory Unification Cycle $C_{\text{unif}}$

This cycle unifies diverse meta-theories and organizes cross-theory abstractions.

Abstract extraction  $\rightarrow$  Generalisation  $\rightarrow$  Unification  
 $\rightarrow$  Constraint derivation  $\rightarrow$  Abstract extraction.

Form:

$$C_{\text{unif}} = \{ A_{\text{abstraction}}^{10}, P_{\text{meta-explanatory}}^{10}, J_{\text{cross}}^{10} \}.$$

Unification involves the emergence of cross-level invariants.

#### O.4.111 4. The Self-Description Cycle $C_{\text{self}}$

This cycle is fundamental for  $K_{10}$  and has no analogue at lower levels. It formalises the process by which a meta-model describes its own structure.

Self – mapping  $\rightarrow$  Meta – analysis  $\rightarrow$  Correction  
 $\rightarrow$  Re – mapping  $\rightarrow$  Self – mapping.

Form:

$$C_{\text{self}} = \{ A_{\text{self-ref}}^{10}, J_{\text{self}}^{10}, P_{\text{coh}}^{10}, \Theta_{\text{meta}}^{10} \}.$$

Stability condition:

$$T_{10} < \Theta_{\text{meta}}^{10}.$$

#### O.4.112 5. The Meta-Framework Evolution Cycle $C_{\text{meta}}^{(10)}$

This cycle manages transitions between entire meta-frameworks, extending the  $K_9$  meta-cycle into higher-order organisation.

Framework  $\rightarrow$  Cross – evaluation  $\rightarrow$  Meta – critique  
 $\rightarrow$  Higher – order reconstruction  $\rightarrow$  Framework.

Form:

$$C_{\text{meta}}^{(10)} = \{ A_{\text{methodology}}^{10}, J_{\text{meta}}^{10}, P_{\text{predictive}}^{10}, \Theta_{\text{dim}}^{10} \}.$$

Transition condition:

$$T_{10} = \Theta_{\text{dim}}^{10}$$

corresponds to the birth of new meta-axes and a higher meta-space.

#### O.4.113 6. The Cross-Level Abstraction Cycle $C_{\text{cross}}$

This cycle links  $K_{10}$  with lower levels by abstracting over multiple structural continua.

Data integration  $\rightarrow$  Multi-level pattern extraction  $\rightarrow$  Abstraction  
 $\rightarrow$  Constraint projection  $\rightarrow$  Data integration.

Form:

$$C_{\text{cross}} = \{ J_{\text{cross}}^{10}, A_{\text{abstraction}}^{10}, P_{\text{meta-predictive}}^{10} \}.$$

#### O.4.114 7. The Universal Operator Cycle $C_{\text{univ}}$

This cycle generates and stabilises universal operators such as the universal evolution operator  $U$ , the dimensional operator  $\Psi$ , the collapse operator  $X$ , and others.

Operator construction  $\rightarrow$  Universality test  $\rightarrow$  Generalisation  
 $\rightarrow$  Stabilisation  $\rightarrow$  Operator construction.

Form:

$$C_{\text{univ}} = \{ A_{\text{meta-logic}}^{10}, P_{\text{consistency}}^{10}, J_{\text{meta-evolution}}^{10} \}.$$

#### O.4.115 Cycle Metrics on $K_{10}$

Using the general definitions:

**Length.**

$$L(C) = \oint d\Omega(K_{10}).$$

**Efficiency.**

$$C_{\text{eff}} = \frac{\oint J^{10} \cdot dA^{10}}{L(C)}.$$

**Stability.**

$$S(C) = \min_{t \in C} d(\Omega(K_{10}), \partial\Omega(K_{10})).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), P_{\text{meta-coherence}}^{10}, P_{\text{meta-predictive}}^{10}).$$

#### O.4.116 Collapse of $K_{10}$ Cycles

Collapse occurs when:

$$P_{\text{meta-coherence}}^{10} < \Theta_{\text{coh}}^{10}, \quad P_{\text{consistency}}^{10} < \Theta_{\text{cons}}^{10}, \quad P_{\text{meta-predictive}}^{10} < \Theta_{\text{meta}}^{10}.$$

Then:

$$C_j \rightarrow \emptyset, \quad \Omega(K_{10}) = \emptyset, \quad k(K_{10}) \rightarrow 0.$$

This corresponds to collapse of a meta-logical framework or an inconsistency cascade.

#### O.4.117 Continuity from $K_9$ to $K_{10}$

The transition  $K_9 \rightarrow K_{10}$  requires:

$$T_9 > \Theta_{\text{dim}}^{10},$$

leading to emergence of new abstraction and meta-logical axes:

$$A_{\text{meta-logic}}^{10}, A_{\text{meta-ontology}}^{10}, A_{\text{self-ref}}^{10}, A_{\text{abstraction}}^{10}.$$

This yields a higher-order continuum with recursive self-description and universal operators.

#### O.4.118 Cycles on $K_{11}$

The continuum  $K_{11}$  corresponds to the level of *meta-evolution*: the domain where the evolution of higher-order operators, meta-logical frameworks, meta-ontologies and meta-spaces becomes itself a structured and dynamical process. If  $K_{10}$  describes meta-models and universal operators, then  $K_{11}$  describes the *evolution of the mechanisms* that govern their transformation.

The state space:

$$\Omega(K_{11}) = \{\text{meta-evolution rules, evolution of universal operators, meta-transformations of meta-logics, evolution of meta-spaces, adaptive higher-order constraints}\}.$$

Axes:

$$A^{11} = \{\text{evolutionary axes over operators, axes of meta-evolution of logics, axes of meta-space adaptation, axes of constraint evolution}\}.$$

Potentials:

$$P^{11} = \{P_{\text{meta-evol}}^{11}, P_{\text{adapt}}^{11}, P_{\text{universality}}^{11}, P_{\text{stability}}^{11}\}.$$

Flows:

$$J^{11} = \{J_{\text{evol}}^{11}, J_{\text{operator}}^{11}, J_{\text{meta-space}}^{11}, J_{\text{constraint}}^{11}, J_{\text{recursive}}^{11}\}.$$

Thresholds:

$$\Theta_{11} = \{\Theta_{\text{evol}}^{11}, \Theta_{\text{meta}}^{11}, \Theta_{\text{univ}}^{11}, \Theta_{\text{dim}}^{11}\}.$$

The continuum exists if:

$$T_{11} < \min(\Theta_{11}), \quad k(K_{11}) > 0.$$

#### O.4.119 Overview

Cycles on  $K_{11}$  represent higher-order dynamical systems that govern how universal operators evolve, how meta-logics adapt to new meta-levels, how meta-spaces grow, and how constraints reorganise under the pressure of structural tension  $T_{11}$ . They form the backbone of the “meta-evolutionary architecture” of the Ontology of Continua.

The defining characteristic of  $K_{11}$  cycles is that they describe *evolution of the mechanisms of evolution* themselves.

This introduces a recursive depth not present in  $K_{10}$ , where operators and meta-logics evolve but their evolutionary rules are fixed. In  $K_{11}$  the rules themselves become variable and dynamic.



#### O.4.120 1. The Meta-Evolutionary Operator Cycle $C_{\text{evol-op}}$

This cycle governs the evolution of universal operators ( $U, \Psi, \Phi, \Lambda, X$ ) at the level of their transformation rules.

Rule definition  $\rightarrow$  Operator evolution  $\rightarrow$  Meta-evaluation  
 $\rightarrow$  Rule revision  $\rightarrow$  Rule definition.

Form:

$$C_{\text{evol-op}} = \{A_{\text{operator}}^{11}, J_{\text{operator}}^{11}, P_{\text{universality}}^{11}, \Theta_{\text{univ}}^{11}\}.$$

Stability:

$$P_{\text{universality}}^{11} > \Theta_{\text{univ}}^{11}.$$

#### O.4.121 2. The Meta-Logic Evolution Cycle $C_{\text{meta-logic}}^{(11)}$

This cycle evolves the rules that govern meta-logics themselves.

Meta-logic generation  $\rightarrow$  Meta-inference dynamics  $\rightarrow$  Meta-constraint evaluation  
 $\rightarrow$  Meta-revision of rules  $\rightarrow$  Meta-logic generation.

Form:

$$C_{\text{meta-logic}}^{(11)} = \{A_{\text{meta-logic}}^{11}, J_{\text{recursive}}^{11}, P_{\text{stability}}^{11}, \Theta_{\text{meta}}^{11}\}.$$

Condition:

$$T_{11} < \Theta_{\text{meta}}^{11}.$$

#### O.4.122 3. The Meta-Space Evolution Cycle $C_{\text{meta-space}}$

This cycle describes the dynamics of meta-spaces  $M_x$  and their structural adaptation to new dimensional pressures.

Meta-space formation  $\rightarrow$  Dimensional interaction  $\rightarrow$  Meta-space constraint evaluation  
 $\rightarrow$  Meta-space adaptation  $\rightarrow$  Meta-space formation.

Form:

$$C_{\text{meta-space}} = \{A_{\text{meta-space}}^{11}, J_{\text{meta-space}}^{11}, P_{\text{adapt}}^{11}, \Theta_{\text{dim}}^{11}\}.$$

#### O.4.123 4. The Constraint Evolution Cycle $C_{\text{constraint}}$

This cycle updates and reorganises higher-order constraints that act as bridges between meta-logics, operators and meta-spaces.

Constraint derivation  $\rightarrow$  Constraint interaction  $\rightarrow$  Constraint evaluation  
 $\rightarrow$  Constraint adaptation  $\rightarrow$  Constraint derivation.

Form:

$$C_{\text{constraint}} = \{A_{\text{constraint}}^{11}, J_{\text{constraint}}^{11}, P_{\text{meta-evol}}^{11}\}.$$

#### O.4.124 5. The Higher-Order Adaptation Cycle $C_{\text{adapt}}$

This cycle orchestrates adaptation of the entire meta-evolutionary system.

Detection of meta—pressures  $\rightarrow$  Structural adjustment  $\rightarrow$  Meta—interpretation  
 $\rightarrow$  Recalibration  $\rightarrow$  Detection of meta—pressures.

Form:

$$C_{\text{adapt}} = \{J_{\text{recursive}}^{11}, P_{\text{adapt}}^{11}, \Theta_{\text{evol}}^{11}\}.$$

Stability:

$$P_{\text{adapt}}^{11} > \Theta_{\text{evol}}^{11}.$$

#### O.4.125 Cycle Metrics on $K_{11}$

**Length.**

$$L(C) = \oint d\Omega(K_{11}).$$

**Efficiency.**

$$C_{\text{eff}} = \frac{\oint J^{11} \cdot dA^{11}}{L(C)}.$$

**Stability.**

$$S(C) = \min_{t \in C} d(\Omega(K_{11}), \partial\Omega(K_{11})).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), P_{\text{meta-evol}}^{11}, P_{\text{universality}}^{11}).$$

#### O.4.126 Collapse of $K_{11}$ Cycles

Collapse occurs when meta-evolutionary stability fails:

$$P_{\text{adapt}}^{11} < \Theta_{\text{evol}}^{11}, \quad P_{\text{universality}}^{11} < \Theta_{\text{univ}}^{11}, \quad P_{\text{stability}}^{11} < \Theta_{\text{meta}}^{11}.$$

Then:

$$C_j \rightarrow \emptyset, \quad \Omega(K_{11}) = \emptyset, \quad k(K_{11}) \rightarrow 0.$$

#### O.4.127 Continuity from $K_{10}$ to $K_{11}$

The transition requires:

$$T_{10} > \Theta_{\text{dim}}^{11},$$

which forces new axes associated with meta-evolution of operators and meta-spaces:

$$A_{\text{meta-logic}}^{11}, A_{\text{meta-space}}^{11}, \\ A_{\text{constraint}}^{11}, A_{\text{operator}}^{11}.$$

The result is a continuum of evolving meta-rules.

#### O.4.128 Cycles on $K_{12}$

The continuum  $K_{12}$  represents the highest-order domain in the vertical hierarchy of continua. It is the level at which the evolution of the *meta-evolutionary system itself* becomes a structured, dynamic object. While  $K_{11}$  governs the evolution of mechanisms of evolution,  $K_{12}$  governs the evolution of the *space of possible meta-evolutionary mechanisms* and the global coherence of the entire hierarchy  $K_0 \rightarrow K_{12}$ .

Thus, cycles on  $K_{12}$  represent the most abstract class of dynamical structures: they ensure the viability of the full ontological chain by maintaining the meta-continuity of meta-spaces, the meta-integrity of operators, and the global recursive coherence of the Ontology of Continua.

State space:

$$\Omega(K_{12}) = \{\text{meta-meta-evolution rules, global recursive consistency conditions, meta-universal constraints, meta-space generative dynamics, global K-chain coherence structures}\}.$$

Axes:

$$A^{12} = \{\text{axes of meta-meta-evolution, axes of global recursive coherence, axes of universal constraint evolution, axes of meta-space generation}\}.$$

Potentials:

$$P^{12} = \{P_{\text{coherence}}^{12}, P_{\text{meta-universality}}^{12}, P_{\text{global-stability}}^{12}, P_{\text{recursive}}^{12}\}.$$

Flows:

$$J^{12} = \{J_{\text{recursive}}^{12}, J_{\text{meta-space}}^{12}, J_{\text{universal}}^{12}, J_{\text{coherence}}^{12}\}.$$

Thresholds:

$$\Theta_{12} = \{\Theta_{\text{coherence}}^{12}, \Theta_{\text{meta-universality}}^{12}, \Theta_{\text{recursive}}^{12}, \Theta_{\text{dim}}^{12}\}.$$

Existence:

$$T_{12} < \min(\Theta_{12}), \quad k(K_{12}) > 0.$$

#### O.4.129 1. Global Recursive Coherence Cycle $C_{\text{rec}}^{(12)}$

This cycle maintains the recursive consistency of the entire hierarchy  $K_0 \rightarrow K_{12}$ .

Hierarchy scan  $\rightarrow$  Recursive evaluation  $\rightarrow$  Global coherence adjustment  
 $\rightarrow$  Constraint reintegration  $\rightarrow$  Hierarchy scan.

Form:

$$C_{\text{rec}}^{(12)} = \{A_{\text{recursive}}^{12}, J_{\text{recursive}}^{12}, P_{\text{coherence}}^{12}, \Theta_{\text{coherence}}^{12}\}.$$

Condition:

$$P_{\text{coherence}}^{12} > \Theta_{\text{coherence}}^{12}.$$

#### O.4.130 2. Meta-Space Generative Cycle $C_{\text{meta-space}}^{(12)}$

This cycle evolves the generative architecture of meta-spaces  $M_x$ .

Meta-space generation  $\rightarrow$  Dimensional extension  $\rightarrow$  Global constraint evaluation  
 $\rightarrow$  Meta-space revision  $\rightarrow$  Meta-space generation.

Form:

$$C_{\text{meta-space}}^{(12)} = \{A_{\text{meta-space}}^{12}, J_{\text{meta-space}}^{12}, P_{\text{global-stability}}^{12}, \Theta_{\text{dim}}^{12}\}.$$

### O.4.131 3. Universal Constraint Evolution Cycle $C_{\text{univ}}^{(12)}$

This cycle regulates the evolution of highest-order constraints that bind all levels  $K_x$  and all meta-spaces  $M_x$  into a single coherent ontological structure.

Universal constraint definition  $\rightarrow$  Constraint interaction across levels  $\rightarrow$  Global evaluation  
 $\rightarrow$  Constraint adaptation  $\rightarrow$  Universal constraint definition.

Form:

$$C_{\text{univ}}^{(12)} = \{A_{\text{constraint}}^{12}, J_{\text{universal}}^{12}, P_{\text{meta-universality}}^{12}, \Theta_{\text{meta-universality}}^{12}\}.$$

### O.4.132 4. Meta-Meta-Evolution Cycle $C_{\text{mme}}$

This cycle governs the evolution of the evolution rules themselves – the highest recursion depth appearing in OC.

Meta–meta–rule generation  $\rightarrow$  Meta–meta–evaluation  $\rightarrow$  Meta–meta–adaptation  
 $\rightarrow$  Reintegration  $\rightarrow$  Meta–meta–rule generation.

Form:

$$C_{\text{mme}} = \{A_{\text{meta-evol}}^{12}, J_{\text{recursive}}^{12}, P_{\text{recursive}}^{12}\}.$$

### O.4.133 Cycle Metrics on $K_{12}$

**Length.**

$$L(C) = \oint d\Omega(K_{12}).$$

**Efficiency.**

$$C_{\text{eff}} = \frac{\oint J^{12} \cdot dA^{12}}{L(C)}.$$

**Stability.**

$$S(C) = \min d(\Omega(K_{12}), \partial\Omega(K_{12})).$$

**Weight.**

$$w(C) = F(C_{\text{eff}}, S(C), P_{\text{coherence}}^{12}, P_{\text{meta-universality}}^{12}).$$

### O.4.134 Collapse of $K_{12}$ Cycles

Collapse occurs when highest-order coherence cannot be maintained:

$$P_{\text{coherence}}^{12} < \Theta_{\text{coherence}}^{12}, \quad P_{\text{meta-universality}}^{12} < \Theta_{\text{meta-universality}}^{12}.$$

Then:

$$C_j \rightarrow \emptyset, \quad \Omega(K_{12}) = \emptyset, \quad k(K_{12}) \rightarrow 0.$$

#### O.4.135 Continuity from $K_{11}$ to $K_{12}$

The birth  $\Psi_{11 \rightarrow 12}$  is triggered when:

$$T_{11} > \Theta_{\text{dim}}^{12},$$

forcing the creation of axes supporting highest-order recursive and meta-universal structures:

$$\begin{aligned} &A_{\text{recursive}}^{12}, A_{\text{meta-space}}^{12}, \\ &A_{\text{constraint}}^{12}, A_{\text{meta-evol}}^{12}. \end{aligned}$$

Thus  $K_{12}$  becomes the universal coherence regulator of the full hierarchy of continua.

### O.5 Structural Jets

The jet formalism introduced in this section provides a unified, dimension-agnostic description of *infinitesimal structural propagation* across all continua in the Ontology of Continua (OC). While the operators  $E$ ,  $\Psi$ ,  $\Phi$  and  $U$  describe macroscopic evolution, embedding, deformation of admissible regions and global continuumness, the jet hierarchy captures the *local differential structure* governing how continua deform along their axes, potentials, flows, thresholds, and boundary gradients.

In classical mathematics, jets represent truncated Taylor expansions of smooth functions. In OC, jets generalise this idea: a jet is a local, structurally admissible fragment of continuum evolution, encoding *all allowed derivatives of all structural components* up to a specified order. This includes derivatives of:

- axes  $A(t)$  and their deformation modes,
- potentials  $P(t)$  and their gradients  $\partial P / \partial A$ ,
- thresholds  $\Theta(t)$  and their rate of change,
- flows  $J(t)$ , including cross-level flows and boundary exchange,
- cycles  $C(t)$  and their infinitesimal stability conditions,
- admissible regions  $\Omega(K)$  and their boundary geometry  $\partial\Omega(K)$ ,
- continuumness  $k(t)$  and its differential response.

Thus, a jet represents the *infinitesimal content* of the universal evolution operator:

$$E(K(t), M(t)) = K(t + \Delta t),$$

providing a differential expansion that allows OC to connect macroscopic evolution to local geometric constraints.

#### O.5.1 Definition of Structural Jets

For any continuum  $K_x$ , we define its  $n$ -th jet  $J^n(K_x)$  as:

$$J^n(K_x)(t) = \left\{ \frac{d^m A}{dt^m}, \frac{d^m P}{dt^m}, \frac{d^m \Theta}{dt^m}, \frac{d^m J}{dt^m}, \frac{d^m C}{dt^m}, \frac{d^m \partial\Omega}{dt^m}, \frac{d^m k}{dt^m} \mid 0 \leq m \leq n \right\}. \quad (\text{O.1})$$

A jet exists if and only if:

1. all derivatives up to order  $n$  respect the admissibility conditions of the continuum;
2. the deformation does not push  $K_x$  outside  $\Omega(M_x)$ ;
3. thresholds at every order remain finite; and

4. the structural tension at each order satisfies:

$$T^{(m)}(t) < \Theta_{\text{collapse}}^{(m)}.$$

Thus, jets encode differential consistency conditions across all levels of OC.

### O.5.2 Role of Jets in Universal Dynamics

Jets serve four structural functions:

**(1) Local generator of global evolution.** While  $E$  provides the global update, jets describe its infinitesimal expansion:

$$E(K(t)) = K(t) + dt \cdot J^1(K) + dt^2 \cdot J^2(K) + \dots$$

**(2) Compatibility conditions for  $\Psi$  and  $\Phi$ .** The embedding operator  $\Psi$  and the evolution of the ambient space  $\Phi$  must satisfy jet-consistency constraints, ensuring that:

$$J^n(K_x) \subseteq J^n(M_x).$$

**(3) Local structure of phase transitions.** Every dimensional transition (e.g.,  $K_x \rightarrow K_{x+1}$ ) begins as a jet instability: an  $n$ th-order divergence along a new axis.

**(4) Diagnosis of collapse modes.** Death of a continuum corresponds to jet-level singularities:

$$\left| \frac{d^m X}{dt^m} \right| \rightarrow \infty,$$

for some structural component  $X$ .

### O.5.3 Jets Across the Levels $K_0 \rightarrow K_{12}$

Every level  $K_x$  has a characteristic jet structure:

- $K_0$ : trivial jet (no derivatives, no axes).
- $K_1$ : geometric jets on  $(X, \tau)$ , first-order boundary motion.
- $K_2$ : physical jets (fields, gradients, variational structure).
- $K_3$ : chemical jets (activation energies, reaction profiles).
- $K_4$ : biological jets (membrane curvature, gradient kinetics).
- $K_5$ : electrical jets (ion flux derivatives, excitability).
- $K_6$ : cognitive jets (prediction-error gradients).
- $K_7$ : social jets (resource flows, institutional rigidity).
- $K_8$ : civilizational jets (symbolic/technological propagation).
- $K_9$ : theoretical jets (paradigm deformation, model flux).
- $K_{10}$ : meta-theoretical jets (functorial consistency).
- $K_{11}$ : hyper-theoretical jets (modal universe curvature).
- $K_{12}$ : universal jets (derivatives of structural laws).

All specialised jet files extend this master definition.

### O.5.4 Boundary and Threshold Constraints

For any jet to exist, the boundary and threshold conditions must remain admissible:

$$\partial\Omega(K_x)(t + dt) = \partial\Omega(K_x)(t) + dt \cdot \frac{d}{dt}\partial\Omega(K_x) + O(dt^2), \quad (\text{O.2})$$

$$\Theta_x(t + dt) = \Theta_x(t) + dt \cdot \frac{d\Theta_x}{dt} + O(dt^2). \quad (\text{O.3})$$

A jet exists if and only if:

$$\partial\Omega(K_x)(t + dt) \subseteq \Omega(M_x)(t + dt), \quad \Theta_x^{(m)} < \Theta_{\max}^{(m)}.$$

Failures of these conditions correspond to structural collapse.

### O.5.5 Summary

This master section establishes jets as the infinitesimal, level-agnostic formalism linking all structural components of OC. It provides the foundation for each specialised jets-file that follows, ensuring consistency across  $K_0 \rightarrow K_{12}$  and integration with the universal operators  $E$ ,  $\Psi$ ,  $\Phi$ , and  $U$ .

### O.5.6 Jets on $K_0$

The level  $K_0$  represents the degenerate, pre-geometric continuum of the Ontology of Continua (OC). It contains no axes, no potentials, no thresholds beyond the minimal existence threshold  $\Theta_0 = \varepsilon > 0$ , and no temporal evolution. Consequently, the jet structure on  $K_0$  is both trivial and foundational: it provides the limiting case against which all higher-order jets are defined.

Formally, the structural content of  $K_0$  is given by:

$$K_0 = (S, \Delta, \mathcal{C}),$$

where  $S$  is an undifferentiated structural substrate,  $\Delta$  is the minimal distinction required for the existence of a continuum, and  $\mathcal{C}$  denotes the admissible compositions of distinctions. No metric, topological, temporal, or dynamical structure exists at this level.

Thus  $K_0$  admits only the *zero jet*, representing the fact that no derivatives of any structural quantity can be defined.

### O.5.7 Definition of Jets on $K_0$

Since  $K_0$  possesses no time parameter, no axes, and no dynamical fields, the general definition of  $n$ -jets collapses to the trivial object:

$$J^n(K_0) = \{0\} \quad \text{for all } n \geq 0.$$

This means:

- no quantities can vary with time (time does not yet exist),
- no gradients can be defined (no axes exist),
- no flows can be defined (no admissible transitions),
- no thresholds evolve (only the existence threshold  $\Theta_0$ ),
- $\partial\Omega(K_0)$  is static and degenerate.

The jet formalism therefore records the essential fact that  $K_0$  exhibits *zero differential structure*. All higher jet structures  $J^n(K_x)$  for  $x \geq 1$  must reduce to  $J^n(K_0)$  in the appropriate singular limit.

### O.5.8 Jet Constraints and Admissibility

The existence of jets at  $K_0$  reduces to the requirement that the minimal threshold of existence is respected:

$$T_0 < \Theta_0 = \varepsilon.$$

If this condition fails, the continuum does not exist at all:

$$K_0 = \emptyset.$$

Since  $K_0$  has no time evolution, the inequality is structural rather than dynamical; it represents the minimal consistency condition of the pre-continuum. Any attempt to introduce derivative-like behaviour would violate admissibility, since no such behaviour is defined.

### O.5.9 Transition to $K_1$ : Jets and the Birth of Time

The transition  $K_0 \rightarrow K_1$  is governed by the operator  $\Psi_{0 \rightarrow 1}$ , which introduces:

- the first axis  $A_1$ ,
- a topological substrate  $(X, \tau)$ ,
- classical admissible configurations  $\Omega_{\text{cl}}(K_1)$ ,
- the first boundary structure  $\partial\Omega_{\text{cl}}$ ,
- the notion of time as the parameter of evolution.

In terms of jets, this transition is described as:

$$J^0(K_0) = \{0\} \longrightarrow J^1(K_1) = \left\{ \frac{dA_1}{dt}, \frac{d}{dt}\partial\Omega_{\text{cl}}, \frac{dE}{dt}, \frac{dT_1}{dt}, \dots \right\}. \quad (\text{O.4})$$

Thus, the birth of  $K_1$  is the birth of nontrivial jet structure. All higher continua inherit and then enrich this basic infinitesimal geometry.

The jet structure therefore provides a rigorous mathematical interpretation of the emergence of time and geometry from the degenerate pre-continuum  $K_0$ .

### O.5.10 Summary

Jets on  $K_0$  are trivial but foundational:

- all jets reduce to the zero jet,
- no derivatives exist,
- no flows, gradients, or potentials exist,
- the only admissibility condition is structural existence,
- the first nontrivial jet appears only at  $K_1$ .

These properties ensure that  $K_0$  acts as the unique jet-free anchor of the entire continuum hierarchy, providing the reference point for the emergence of infinitesimal structure at every higher level.

### O.5.11 Jets on $K_1$

The level  $K_1$  is the first continuum in the OC hierarchy that admits a genuine infinitesimal structure. It introduces a topological space  $(X, \tau)$ , a single axis  $A_1$ , a configuration space  $\Omega(K_1) = C^0(X, V)$ , and a classical energy functional  $E$  satisfying axioms E1–E3. Time  $t$  appears as the parameter of evolution of configurations, making the jet formalism  $J^n(K_1)$  nontrivial for the first time.

This section defines and analyses jets on  $K_1$ , their admissibility, their interaction with the classical boundary  $\partial\Omega_{\text{cl}}$ , and their role in the transition  $K_1 \rightarrow K_2$ .



### O.5.12 Jet Structure on $K_1$

Let  $\phi(x, t) \in \Omega(K_1)$  denote an admissible classical configuration. Since  $(X, \tau)$  carries no geometry beyond continuity, the only derivatives available at  $K_1$  are *time derivatives*. Spatial derivatives arise only at  $K_2$ , where the continuum gains a geometric substrate.

Thus the  $n$ -jet of  $\phi$  on  $K_1$  has the form:

$$J^n(\phi) = \left\{ \frac{d^k \phi}{dt^k}(x, t) \mid 0 \leq k \leq n \right\}.$$

No derivatives with respect to  $x \in X$  exist; no connection or metric structure is available. Jets at  $K_1$  describe purely temporal deformations of admissible configurations.

### O.5.13 Energy and Action Jets

The classical energy functional on  $K_1$  is:

$$E[\phi] = \int_X \mathcal{E}(\phi(x, t)) d\mu_1,$$

where  $d\mu_1$  is the minimal admissible measure (arising from the minimal measurability axiom). Since  $\mathcal{E}$  contains no gradient terms, its jet structure reduces to:

$$J^1(E) = \left\{ \frac{dE}{dt} \right\} = \left\{ \int_X \frac{\partial \mathcal{E}}{\partial \phi} \frac{d\phi}{dt} d\mu_1 \right\}.$$

The action functional  $S[\phi]$ —when defined—inherits the same temporal jet structure:

$$J^1(S) = \left\{ \frac{dS}{dt} \right\}.$$

This makes  $K_1$  the minimal continuum where variational principles can be meaningfully formulated, albeit without spatial structure.

### O.5.14 Jets of the Boundary $\partial\Omega_{\text{cl}}$

The classical boundary  $\partial\Omega_{\text{cl}}(K_1)$  is defined as the set of configurations violating the admissibility imposed by the energy and threshold conditions of  $K_1$ :

$$\partial\Omega_{\text{cl}} = \{ \phi \in \Omega(K_1) \mid E[\phi] = \Theta_1 \}.$$

The jet of the boundary is:

$$J^1(\partial\Omega_{\text{cl}}) = \left\{ \frac{d}{dt} (E[\phi(t)] - \Theta_1) \right\}.$$

Since  $\Theta_1$  is structural and constant on  $K_1$ , the evolution of the boundary is governed entirely by  $\frac{dE}{dt}$ :

$$\frac{d}{dt} \partial\Omega_{\text{cl}} \iff \frac{dE}{dt}.$$

A configuration exits  $K_1$  when:

$$\frac{dE}{dt} > 0 \quad \text{and} \quad E(t) \rightarrow \Theta_1.$$

This prepares the ground for the emergence of gradient energy at  $K_2$ .

### O.5.15 Jets of Structural Quantities

Jets also apply to the structural components of  $K_1$ :

- **Axis:**

$$J^1(A_1) = \left\{ \frac{dA_1}{dt} \right\},$$

describing its admissible temporal deformation.

- **Tension:**

$$J^1(T_1) = \left\{ \frac{dT_1}{dt} \right\},$$

where  $T_1$  is induced from the distribution of potential values and boundary proximity.

- **Thresholds:**

$$J^1(\Theta_1) = \{0\},$$

since thresholds at  $K_1$  are structural and time-invariant.

- **Continuum Measure:**

$$J^1(k_1) = \left\{ \frac{dk_1}{dt} \right\},$$

linking infinitesimal change in continuumness to the tension and the evolution of  $\Omega(K_1)$ .

### O.5.16 Admissibility and Jet Constraints

The general admissibility condition for jets on  $K_1$  is:

$$T_1(t) < \Theta_1.$$

A jet  $J^n(\phi)$  is admissible iff all derivatives preserve this condition. For example:

$$\frac{d\phi}{dt} \text{ is admissible} \iff \frac{dE}{dt} < \frac{d\Theta_1}{dt} = 0.$$

If the jet violates admissibility, the configuration leaves the continuum:

$$\Omega(K_1) \rightarrow \emptyset, \quad k_1 \rightarrow 0.$$

This is the earliest level where collapse due to jet divergence can be defined.

### O.5.17 Role in the Transition $K_1 \rightarrow K_2$

The emergence of  $K_2$  requires:

- a new axis  $A_{\text{geom}}$  not expressible as a deformation of  $A_1$ ,
- a nonzero gradient term in the energy functional,
- a deformation of the boundary  $\partial\Omega$  indicating the need for spatial structure.

The jet interpretation of the transition is:

$$\frac{d^2\phi}{dt^2} \text{ and } \frac{dT_1}{dt} \text{ grow large} \implies \exists A_{\text{geom}} \notin \text{span}\{A_1\}.$$

This implements the formal operator  $F_{1 \rightarrow 2}$  and initiates the first genuine geometric continuum.

### O.5.18 Summary

Jets on  $K_1$  are the first nontrivial jets in the continuum hierarchy. They describe:

- temporal deformation of configurations,
- temporal variation of energy and tension,
- evolution of the boundary  $\partial\Omega_{\text{cl}}$ ,
- constraints from the threshold  $\Theta_1$ ,
- first stability analyses via jet admissibility.

They also provide the mathematical language for the transition to  $K_2$ , where spatial structure and gradient jets first appear.

### O.5.19 Jets on $K_2$

The continuum  $K_2$  is the first level in the OC hierarchy that admits a genuine geometric substrate. Unlike  $K_1$ , where only temporal jets exist, the emergent axis  $A_{\text{geom}}$  at  $K_2$  supports spatial structure, spatial gradients, and geometric contributions to the energy and tension. Jets on  $K_2$  therefore include both temporal and spatial components. They encode the fine-scale behaviour of fields near percolation thresholds, cluster boundaries, and structural transitions that enable the emergence of higher-order continua such as  $K_3$ .

### O.5.20 Jet Space and Geometric Axis

The emergence of  $K_2$  occurs through the operator  $F_{1 \rightarrow 2}$ , which introduces a new axis

$$A_{\text{geom}} \notin \text{span}\{A_1\}.$$

This axis supports spatial variation and connectedness structures.

A configuration on  $K_2$  is a field

$$\phi : X \rightarrow V,$$

where  $X$  now carries minimal geometric structure sufficient to define adjacency, neighbourhoods, and spatial differences.

The  $(m, n)$ -jet of  $\phi$  at  $K_2$  consists of all mixed temporal and spatial derivatives up to order  $m$  in time and  $n$  in space:

$$J^{m,n}(\phi) = \left\{ \partial_t^k \partial_x^\ell \phi \mid 0 \leq k \leq m, 0 \leq \ell \leq n \right\}.$$

Spatial jets make sense because  $K_2$  possesses:

- a geometric adjacency structure (arising from percolation),
- definable local neighbourhoods,
- minimal differentiability of admissible configurations.

### O.5.21 Jets and Percolation Structure

The structural hallmark of  $K_2$  is the presence of clusters and connected components arising from percolation. A configuration  $\phi$  near the percolation threshold  $p_c$  has large correlation lengths and strong sensitivity to small perturbations.

Let  $\mathcal{C}(\phi)$  denote the connected cluster containing a reference point  $x$ . The jet of the cluster indicator is:

$$J^1(\chi_{\mathcal{C}}) = \left\{ \frac{d}{dt} \chi_{\mathcal{C}}(x, t), \partial_x \chi_{\mathcal{C}}(x, t) \right\}.$$

Spatial jets capture how the local connectivity changes under perturbations. For example:

$$\partial_x \chi_C(x) \neq 0 \iff x \text{ lies on a cluster boundary.}$$

Temporal jets capture the rate at which a configuration approaches or moves away from the percolation threshold. These jets are essential in determining structural tension  $T_2$ .

### O.5.22 Jets of the Energy Functional

The energy functional at  $K_2$  includes spatial-derivative terms:

$$E[\phi] = \int_X (\mathcal{E}_0(\phi) + \alpha |\nabla \phi|^2) d\mu_2,$$

where  $\alpha > 0$  quantifies the geometric stiffness of the continuum.

The jet of the energy is:

$$J^1(E) = \left\{ \frac{dE}{dt} = \int_X \left( \frac{\partial \mathcal{E}_0}{\partial \phi} \partial_t \phi + 2\alpha \nabla \phi \cdot \nabla (\partial_t \phi) \right) d\mu_2 \right\}.$$

Spatial jets appear even in the first temporal jet:

$$\nabla \phi, \quad \nabla (\partial_t \phi).$$

They quantify how spatial structure contributes to the energy flow.

### O.5.23 Mass as a Jet Derivative of Tension

The mass operator derived in the OC framework is

$$m = \partial_{A_{\text{geom}}} T_2,$$

the structural derivative of the tension with respect to the geometric axis.

Jets of tension take the form:

$$J^{1,1}(T_2) = \left\{ \frac{dT_2}{dt}, \partial_x T_2 \right\}.$$

The presence of  $\partial_x T_2$  is a purely  $K_2$  phenomenon; it does not exist on  $K_1$ .

Mass appears as a first-order spatial jet of tension. Thus:

$$m(x) = \partial_x T_2(x).$$

This interpretation reproduces the dependence of mass on geometric structure, including emergent mass from cluster boundaries, curvature of field configurations, and the energy of spatial gradients.

### O.5.24 Jets of the Boundary $\partial\Omega(K_2)$

The admissible region  $\Omega(K_2)$  is determined by the combined constraints of:

- geometric connectivity,
- bounded spatial gradients,
- bounded tension  $T_2 < \Theta_2$ .

The boundary  $\partial\Omega(K_2)$  therefore has a nontrivial jet structure:

$$J^{1,1}(\partial\Omega) = \left\{ \partial_t (E[\phi] - \Theta_2), \partial_x (E[\phi] - \Theta_2) \right\}.$$

The spatial component indicates where local geometric instabilities arise.

A configuration approaches collapse when:

$$T_2(x, t) \rightarrow \Theta_2 \quad \text{and/or} \quad |\nabla \phi| \rightarrow \infty.$$

The latter has no analogue at  $K_1$ .

### O.5.25 Critical Jets Near the Percolation Threshold

Near the percolation threshold  $p_c$ , the continuum exhibits critical behaviour:

- correlation length  $\xi \rightarrow \infty$ ,
- fluctuations become scale-free,
- jet magnitudes diverge.

The jet signature of criticality is:

$$|\nabla\phi| \sim \xi^{-1} \rightarrow 0, \quad |\nabla^2\phi| \sim \xi^{-2} \rightarrow 0,$$

but higher-order statistical jets (variance, covariance of jets) grow without bound.

Critical jets mark the onset of dimension-increase readiness:

$$T_2 \rightarrow \Theta_{\text{dim}} \implies A_{\text{chem}} \text{ becomes admissible (transition to } K_3).$$

### O.5.26 Admissibility of Jets on $K_2$

A jet  $J^{m,n}(\phi)$  is admissible iff:

$$T_2 < \Theta_2, \quad |\nabla\phi| < G_{\text{max}}, \quad \text{and} \quad \mathcal{C}(\phi) \text{ percolates.}$$

The explicit gradient bound  $G_{\text{max}}$  appears for the first time at  $K_2$ , since  $K_1$  had no gradient energy.

The violation of admissibility results in:

$$\Omega(K_2) \rightarrow \emptyset, \quad k_2 \rightarrow 0.$$

### O.5.27 Role in the Transition $K_2 \rightarrow K_3$

Jets provide the analytic mechanism for the emergence of the chemical continuum  $K_3$ .

The transition occurs when:

- spatial jets become sufficiently structured,
- tension gradients localize to form stable motifs,
- energy jets favour formation of discrete interaction sites.

Formally:

$$\partial_x T_2 \text{ large and stable} \implies A_{\text{chem}} \in A(M_2) \setminus A(K_2),$$

which is precisely the jet signature of the birth of chemical interaction axes.

This is the geometric-to-chemical phase transition encoded in the operator  $F_{2 \rightarrow 3}$ .

### O.5.28 Summary

Jets on  $K_2$  are the first jets with full mixed temporal and spatial structure. They encode:

- geometric variation,
- percolation behaviour,
- spatial gradients of tension (mass),
- boundary geometry of  $\partial\Omega(K_2)$ ,
- critical phenomena near  $p_c$ ,
- admissibility constraints for stable continua,

- the mechanism for emergence of  $K_3$ .

They form the mathematical backbone of the physical continuum and the bridge toward chemical organisation.

### O.5.29 Jets on $K_3$

The continuum  $K_3$  introduces chemical organisation on top of the geometric substrate of  $K_2$ . Jets on  $K_3$  describe how chemical potentials, reaction pathways, activation energies, and configurational structures vary under infinitesimal changes of state. They provide the local analytic machinery for chemical kinetics, catalytic effects, RAF-network formation, membrane precursor dynamics, and the emergence of prebiotic compartments that form the bridge toward  $K_4$ .

### O.5.30 Emergence of Chemical Jet Structure

The transition  $K_2 \rightarrow K_3$  is driven by localisation of tension gradients from  $K_2$ , which creates stable interaction sites. The operator  $F_{2 \rightarrow 3}$  introduces a new axis

$$A_{\text{chem}} \in A(M_2) \setminus A(K_2),$$

corresponding to chemical state transitions (bond formation, breaking, reaction coordinates).

A configuration on  $K_3$  takes the form:

$$\phi = (\{\text{atoms}\}, \{\text{bonds}\}, \{\text{molecules}\}),$$

with chemical potentials

$$P_{\text{chem}} = (\mu_i, E_{\text{act}}, E_{\text{conf}}, E_{\text{bond}})$$

defined on molecular states.

Jets encode variations of all these quantities:

$$J^{m,n}(\phi) = \left\{ \partial_t^k \partial_{A_{\text{chem}}}^\ell \phi \right\}_{0 \leq k \leq m, 0 \leq \ell \leq n}.$$

### O.5.31 Jets of the Chemical Potential

The chemical potential  $\mu_i$  for species  $i$  is a function of concentration, temperature, and molecular configuration. Its first jet is:

$$J^1(\mu_i) = \left\{ \frac{d\mu_i}{dt}, \partial_{A_{\text{chem}}} \mu_i \right\}.$$

The second component expresses sensitivity to infinitesimal displacement along the reaction coordinate.

Explicitly:

$$\partial_{A_{\text{chem}}} \mu_i = \frac{\partial \mu_i}{\partial \xi}, \quad \xi \text{ reaction coordinate.}$$

This derivative governs:

- direction of spontaneous reaction flow,
- local curvature of the energy landscape,
- stability of transient intermediates.

### O.5.32 Jets of Activation Energy and Reaction Pathways

Each chemical reaction has an activation barrier  $E_{\text{act}}$ . Jets describe how this barrier changes under perturbations of state:

$$J^1(E_{\text{act}}) = \left\{ \frac{dE_{\text{act}}}{dt}, \partial_{A_{\text{chem}}} E_{\text{act}} \right\}.$$

The spatial derivative along the chemical axis determines how the reaction accelerates or decelerates locally.

For a reaction path  $\gamma$  in state space,

$$\gamma : [0, 1] \rightarrow \Omega(K_3),$$

the jet of the energy profile  $E(\gamma(s))$  satisfies:

$$\partial_s E = \nabla E \cdot \dot{\gamma}, \quad \partial_s^2 E = H_E(\dot{\gamma}, \dot{\gamma}),$$

where  $H_E$  is the Hessian of the chemical energy.

These jets determine:

- saddle points (transition states),
- local minima (stable molecules),
- curvature of reaction funnels.

### O.5.33 Jets of Bond Structure and Configurational Energy

The bond network  $B$  of a molecular configuration changes under the chemical jet flow:

$$\partial_{A_{\text{chem}}} B \quad \text{encodes infinitesimal bond weakening/strengthening.}$$

Configurational energy  $E_{\text{conf}}$  has the jet:

$$J^1(E_{\text{conf}}) = \left\{ \frac{dE_{\text{conf}}}{dt}, \partial_{A_{\text{chem}}} E_{\text{conf}} \right\}.$$

The latter derivative determines:

- rotational barriers,
- torsional stiffness,
- conformational transitions,
- emergence of proto-membrane lipids and amphiphiles.

### O.5.34 Jets and Catalysis (RAF Networks)

Catalysis lowers energy barriers without altering final states. In OC formalism, catalysis introduces a new reaction path supported by a sub-continuum:

$$K_{\text{path}}(C),$$

whose jets capture intermediate configurations induced by the catalyst.

The catalytic jet signature is:

$$\partial_{A_{\text{chem}}} E_{\text{act}}(C) < \partial_{A_{\text{chem}}} E_{\text{act}},$$

consistent with the structural theorem of catalysis in the OC core.

RAF (Reflexively Autocatalytic and Food-generated) networks exhibit collective jet coherence:

$$J^1(E_{\text{act}}^{\text{network}}) \ll J^1(E_{\text{act}}^{\text{isolated}}),$$

which is the analytic expression of their stabilising effect.

Jets quantify how catalytic pathways reshape  $\partial\Omega(K_3)$  by lowering local thresholds.

### O.5.35 Jets of the Boundary $\partial\Omega(K_3)$

The admissible chemical region is:

$$\Omega(K_3) = \{\text{configurations with bounded potential, finite gradients, and } T_3 < \Theta_{\text{chem}}\}.$$

The boundary carries a jet structure:

$$J^1(\partial\Omega) = \{\partial_t(E_{\text{act}} - \Theta_{\text{chem}}), \partial_{A_{\text{chem}}}(E_{\text{act}} - \Theta_{\text{chem}})\}.$$

Crossing the boundary corresponds to chemical collapse:

- runaway reactions,
- unbounded radical formation,
- breakdown of stable molecular species.

### O.5.36 Jets of Chemical Oscillations and Early Networks

Many prebiotic systems exhibit oscillatory or autocatalytic dynamics. Jets provide their analytic characterisation.

For a chemical cycle with state variable  $x$ :

$$\dot{x} = f(x), \quad J^1(x) = \{\dot{x}\}, \quad J^2(x) = \{\ddot{x}\}.$$

Oscillatory behaviour requires:

$$\ddot{x} < 0 \text{ at turning points.}$$

Jets identify:

- proto-metabolic cycles,
- oscillatory redox systems,
- lipid-assembly/dissolution cycles,
- proto-feedback networks.

### O.5.37 Jets and Prebiotic Compartment Formation

The transition  $K_3 \rightarrow K_4$  requires the formation of compartments with semi-stable boundaries. Jets capture:

$$\partial_{A_{\text{chem}}} E_{\text{conf}} \text{ large and negative} \implies \text{spontaneous aggregation (micelles, vesicles).}$$

Reaction jets determine:

- stability of amphiphilic assemblies,
- membrane curvature formation,
- emergence of proto-boundaries  $\partial\Omega(K_4)$ ,
- redox-gradient localisation.

These jet signatures produce the \*closure conditions\* that define early compartments.



### O.5.38 Admissibility of Jets on $K_3$

A jet on  $K_3$  is admissible iff:

$$T_3 < \Theta_{\text{chem}}, \quad E_{\text{act}} < E_{\text{max}}, \quad |\partial_{A_{\text{chem}}} \phi| < G_{\text{chem}}.$$

Violation implies:

$$\Omega(K_3) \rightarrow \emptyset, \quad k_3 \rightarrow 0.$$

### O.5.39 Role in Transition $K_3 \rightarrow K_4$

Jets provide the analytic mechanism for the shift from purely chemical reaction networks to bounded biochemical systems.

The transition requires:

- localisation of gradients,
- formation of stable amphiphilic structures,
- reduction of activation barriers for boundary-forming reactions,
- temporal coherence of chemical oscillations.

Formally:

$$\partial_{A_{\text{chem}}} E_{\text{conf}} \text{ stable and negative} \implies A_{\text{comp}} (\text{the boundary axis}) \in A(M_3).$$

This jet signature marks the emergence of the membrane continuum  $K_4$ .

### O.5.40 Summary

Jets on  $K_3$  encode:

- fine-scale variation of potentials along reaction coordinates,
- sensitivity of activation barriers,
- bond network dynamics,
- catalytic modification of reaction pathways,
- structural deformation of  $\partial\Omega(K_3)$ ,
- oscillatory and autocatalytic phenomena,
- emergence of compartments and preparation for  $K_4$ .

They form the analytic core of chemical organisation and the essential bridge to the biological continuum.

### O.5.41 Jets on $K_4$

The continuum  $K_4$  introduces biological organisation grounded in semi-stable boundaries, compartments, osmotic regulation, ionic gradients, and metabolic pre-cycles. Jets on  $K_4$  describe the infinitesimal variations of membrane geometry, permeability, gradients, fluxes, and boundary potentials that determine the viability and stability of protocells. They constitute the analytic engine of the transition from chemical collective dynamics ( $K_3$ ) to biological autonomy ( $K_5$ ).

### O.5.42 Emergence of the Boundary Jet Structure

The defining feature of  $K_4$  is the existence of a physical boundary  $\partial\Omega(K_4)$ , generated by amphiphilic self-assembly. The new boundary axis

$$A_{\text{boundary}} \in A(M_3) \setminus A(K_3)$$

controls the degrees of freedom associated with curvature, permeability, hydration, and membrane tension.

A membrane configuration is represented as:

$$\phi = (\text{lipid composition, curvature field } C(x), \text{ permeability profile } P_{\text{perm}}(x), \Delta V, \nabla\mu_i),$$

with membrane potentials:

$$P_{\text{mem}} = (\Delta V, P_{\text{grad}}, P_{\text{osm}}, P_{\text{charge}}).$$

Jets describe infinitesimal variations of these fields under shifts along the boundary axis and time:

$$J^{m,n}(\phi) = \left\{ \partial_t^k \partial_{A_{\text{boundary}}}^\ell \phi \right\}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n.$$

### O.5.43 Jets of Membrane Curvature and Shape

Let  $C(x)$  denote the local curvature of the boundary. Its jet is:

$$J^1(C) = \left\{ \frac{dC}{dt}, \partial_{A_{\text{boundary}}} C \right\}.$$

The directional derivative reflects the membrane's geometric response to pressure differentials and gradients.

Key biological consequences of curvature jets:

- initiation of budding or vesicle division,
- stabilisation of micelle-to-vesicle transitions,
- curvature-driven concentration of catalysts and reactants,
- localisation of redox and proton gradients.

In the patch-model:

$$\partial_{A_{\text{boundary}}} C_i \text{ large} \quad \Rightarrow \quad \text{local instability or budding.}$$

### O.5.44 Jets of Permeability and Transport Thresholds

Permeability  $P_{\text{perm}}$  varies across the membrane. Jets encode its sensitivity:

$$J^1(P_{\text{perm}}) = \{ \dot{P}_{\text{perm}}, \partial_{A_{\text{boundary}}} P_{\text{perm}} \}.$$

This determines how transport thresholds change:

$$\Theta_{\text{perm}}(t) \sim \frac{1}{P_{\text{perm}}(t)}, \quad \partial_{A_{\text{boundary}}} \Theta_{\text{perm}} \propto - \frac{\partial_{A_{\text{boundary}}} P_{\text{perm}}}{P_{\text{perm}}^2}.$$

Consequences:

- membrane collapse (when  $\Theta_{\text{perm}}$  diverges),
- transition to selective permeability (proto-channels),
- asymmetric transport leading to proto-metabolic cycles.

### O.5.45 Jets of Osmotic Potential and Volume Regulation

Osmotic potential:

$$P_{\text{osm}} = \Pi_{\text{out}} - \Pi_{\text{in}},$$

with jet:

$$J^1(P_{\text{osm}}) = \{\dot{P}_{\text{osm}}, \partial_{A_{\text{boundary}}} P_{\text{osm}}\}.$$

Volume dynamics depend on:

$$\dot{V} = P_{\text{perm}} P_{\text{osm}}.$$

Jets determine:

- thresholds of osmotic collapse,
- stabilisation of volume under active or buffered conditions,
- oscillatory swelling–shrinking cycles.

Small perturbations near critical osmotic points yield:

$$\partial_{A_{\text{boundary}}} \dot{V} = \partial_{A_{\text{boundary}}} (P_{\text{perm}} P_{\text{osm}}) = (\partial_{A_{\text{boundary}}} P_{\text{perm}}) P_{\text{osm}} + P_{\text{perm}} \partial_{A_{\text{boundary}}} P_{\text{osm}}.$$

### O.5.46 Jets of Chemical and Ionic Gradients

Gradients:

$$P_{\text{grad}} = \nabla \mu_i, \quad P_{\text{charge}} = \Delta V.$$

Jets:

$$J^1(P_{\text{grad}}) = \{\dot{P}_{\text{grad}}, \partial_{A_{\text{boundary}}} P_{\text{grad}}\}, \quad J^1(\Delta V) = \{\Delta \dot{V}, \partial_{A_{\text{boundary}}} \Delta V\}.$$

These derivatives govern:

- initiation of proto-pumps (patterned flux),
- coupling of chemical and electrical potentials,
- emergence of spatially structured reaction zones,
- proto-information gradients.

### O.5.47 Jets of Passive vs Active Fluxes

Total flux:

$$J_{\text{total}} = J_{\text{diff}} + J_{\text{osm}} + J_{\text{active}}.$$

First jets:

$$J^1(J_{\text{diff}}) = \{\dot{J}_{\text{diff}}, \partial_{A_{\text{boundary}}} J_{\text{diff}}\},$$

$$J^1(J_{\text{active}}) = \{\dot{J}_{\text{active}}, \partial_{A_{\text{boundary}}} J_{\text{active}}\}.$$

Active flux appears when metabolic sub-cycles in  $K_4$  produce energetic potentials (proton gradients, redox potentials).

Jets reveal:

- onset of energy-driven asymmetry,
- bifurcation into stable vs unstable transport regimes,
- thresholds for appearance of proto-pumps.

### O.5.48 Patch Model Jets and Local Instabilities

In the patch discretisation of  $\partial\Omega(K_4)$ , each patch  $i$  has:

$$(C_i, P_{\text{perm},i}, \Delta V_i, P_{\text{grad},i}, P_{\text{osm},i})$$

Jets:

$$J^1(X_i) = \{\dot{X}_i, \partial_{A_{\text{boundary}}} X_i\}.$$

Local instabilities arise when:

$$|\partial_{A_{\text{boundary}}} X_i| > \Theta_{X,i},$$

where  $\Theta_{X,i}$  is the local threshold for curvature, permeability, or charge variation.

Consequences:

- flicker instability (rapid permeability switching),
- burst permeability (loss of compartment integrity),
- curvature-driven vesicle budding,
- proto-spike appearance ( $K_4 \rightarrow K_5$  pathway).

### O.5.49 Jets and Proto-Excitability ( $K_4 \rightarrow$ Early $K_5$ )

The earliest electrical axis arises when jets of charge and permeability produce a threshold-crossing event:

$$\partial_{A_{\text{boundary}}} \Delta V > \Theta_{\text{exc}}.$$

This creates the proto-excitability cycle:

$$C_{\text{exc}} = (\Delta V \uparrow, P_{\text{perm}} \uparrow, J_{\text{ion}} \uparrow, \Delta V \downarrow),$$

where jets capture the derivative structure at each phase.

Necessary jet conditions:

$$\begin{array}{ll} \partial_{A_{\text{boundary}}} \Delta V > 0 & \text{(charge accumulation),} \\ \partial_{A_{\text{boundary}}} P_{\text{perm}} > 0 & \text{(channel-like opening),} \\ \partial_{A_{\text{boundary}}} J_{\text{ion}} > 0 & \text{(ion influx),} \\ \partial_{A_{\text{boundary}}} \Delta V < 0 & \text{(recovery).} \end{array}$$

These jets provide the formal analytic signature of the birth of the electrical axis  $A_{\text{exc}}$ , distinguishing  $K_5$  from  $K_4$ .

### O.5.50 Jets of Boundary Tension and Stability

Boundary tension:

$$T_{\text{mem}} = f(C, P_{\text{perm}}, P_{\text{osm}}, \Delta V, P_{\text{grad}}).$$

Jet:

$$J^1(T_{\text{mem}}) = \{\dot{T}_{\text{mem}}, \partial_{A_{\text{boundary}}} T_{\text{mem}}\}.$$

Critical condition for stability:

$$\partial_{A_{\text{boundary}}} T_{\text{mem}} < \Theta_{\text{mem}}.$$

Exceeding this jet threshold induces:

- membrane collapse,
- permeabilisation bursts,
- loss of volume control,
- death of  $K_4$  ( $\Omega \rightarrow \emptyset$ ).

### O.5.51 Jets and $\partial\Omega(K_4)$ Dynamics

The boundary of admissible configurations is controlled by:

$$E_{\text{conf}}, \quad T_{\text{mem}}, \quad P_{\text{osm}}, \quad P_{\text{grad}}, \quad P_{\text{perm}}.$$

Jets determine the deformation of this boundary:

$$\partial_{A_{\text{boundary}}}(E_{\text{conf}} - \Theta_{\text{closure}}), \quad \partial_{A_{\text{boundary}}}(T_{\text{mem}} - \Theta_{\text{mem}}),$$

where  $\Theta_{\text{closure}}$  is the threshold for vesicle closure.

Crossing  $\partial\Omega(K_4)$  implies death or transition.

### O.5.52 Summary

Jets on  $K_4$  encode the infinitesimal structure of:

- curvature and shape dynamics of membranes,
- permeability and osmotic response,
- chemical and electrical gradients,
- passive and active fluxes,
- patch-level instabilities,
- proto-excitability cycles (birth of  $K_5$ ),
- stability of membrane tension,
- deformation of  $\partial\Omega(K_4)$ .

These jets provide the analytic foundation for biological autonomy and the emergence of the neural continuum  $K_5$ .

### O.5.53 Jets on $K_5$

The continuum  $K_5$  is characterised by the emergence of an electrical dimension enabled by ion gradients, membrane potential, channel-mediated flows, and proto-spike dynamics. Jets on  $K_5$  formalise the infinitesimal variations of electrical, ionic, and permeability fields, their coupling to membrane geometry, stochastic noise, and excitation–recovery cycles that define the viability of early neural systems.

### O.5.54 Birth of the Electrical Jet Structure

The defining axis of  $K_5$  is the electrical axis

$$A_{\text{exc}} \in A(M_4) \setminus A(K_4),$$

created when membrane-bound charge separation becomes dynamically coupled to selective ion permeation.

The core field is the membrane potential:

$$\Delta V = V_{\text{in}} - V_{\text{out}},$$

with associated ionic reversal potentials  $E_{\text{ion}}$ , conductances  $g_{\text{ion}}$ , and channel states

$$s \in \{\text{open, closed, blocked, leaky}\}.$$

Jets capture infinitesimal variations:

$$J^{m,n}(\Delta V) = \{\partial_t^k \partial_{A_{\text{exc}}}^\ell \Delta V\}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n.$$

### O.5.55 Jets of Ion Channel States and Conductance

A channel is defined by:

$$J_{\text{ion}} = g_{\text{channel}}(\Delta V - E_{\text{ion}}),$$

where  $g_{\text{channel}}$  depends on the state  $s$ .

The jet structure includes:

$$J^1(g_{\text{channel}}) = \{\dot{g}_{\text{channel}}, \partial_{A_{\text{exc}}} g_{\text{channel}}\}.$$

For discrete channel states, jets describe the transition rates between them:

$$\lambda_{s \rightarrow s'} = \partial_t p(s') = M_{s \rightarrow s'}(t) p(s),$$

with stochastic operator  $M$  from Fix.W2.

Key consequences:

- sensitivity of excitation thresholds to channel kinetics,
- onset of proto-spike oscillations,
- channel noise amplification when approaching criticality.

### O.5.56 Jets of the Membrane Potential $\Delta V$

The temporal derivative:

$$\dot{\Delta V} = - \sum_{\text{ion}} g_{\text{ion}}(\Delta V - E_{\text{ion}}) + J_{\text{pump}} + J_{\text{leak}},$$

Jets:

$$J^1(\Delta V) = \{\dot{\Delta V}, \partial_{A_{\text{exc}}} \Delta V\}.$$

Spatial or structural derivatives appear via boundary patches:

$$\partial_{A_{\text{exc}}} \Delta V_i = \Delta V'_i \quad (\text{local change under electrical axis variation}).$$

These jets govern:

- spike initiation,
- spike propagation across patches,
- local failures of excitability,
- refractory behaviour.

### O.5.57 Jets of Ionic Fluxes and Pump–Leak Balance

Ion flux:

$$J_{\text{ion}} = g_{\text{ion}}(\Delta V - E_{\text{ion}}).$$

Jets:

$$J^1(J_{\text{ion}}) = \{\dot{J}_{\text{ion}}, \partial_{A_{\text{exc}}} J_{\text{ion}}\}.$$

Active pumping:

$$J_{\text{pump}} = \alpha P_{\text{energy}}, \quad \dot{J}_{\text{pump}} \propto \dot{P}_{\text{energy}}.$$

Leak:

$$J_{\text{leak}} = L(\Delta V - E_{\text{env}}).$$

Jets reveal:

- emergence of pump-dominated regimes,
- instability when leak > pump (collapse of  $\Delta V$ ),
- oscillatory behaviour in overcompensated regimes,
- transitions to stable excitability cycles.

### O.5.58 Jets of Excitation Threshold and Recovery Threshold

The excitation threshold:

$$\Theta_{\text{exc}} = \Theta(\Delta V, g_{\text{channel}}, J_{\text{ion}}, \eta_{\text{noise}}),$$

with noise  $\eta_{\text{noise}}$  from Fix.W2.

Jets:

$$J^1(\Theta_{\text{exc}}) = \{\dot{\Theta}_{\text{exc}}, \partial_{A_{\text{exc}}} \Theta_{\text{exc}}\}.$$

Similarly for recovery threshold  $\Theta_{\text{rec}}$ .

Their jet derivatives determine:

- existence and shape of spike cycles,
- refractory period duration,
- sensitivity to noise and channel density,
- boundary of the admissible excitability region  $\partial\Omega(K_5)$ .

### O.5.59 Jets of Channel Noise and Stochastic Logic

Stochastic logic (Fix.W2) formalises membrane state transitions with probability vector  $p(t)$  evolving via:

$$p(t + \Delta t) = M(t)p(t), \quad \eta_{\text{noise}} = \text{stochastic component.}$$

Noise jets:

$$J^1(\eta_{\text{noise}}) = \{\dot{\eta}, \partial_{A_{\text{exc}}} \eta\}.$$

Biological effects:

- stochastic opening of channels near threshold,
- noise-driven proto-spikes,
- noise-suppression cycles (stability control),
- emergence of proto-computational logic gates on  $K_5$ .

### O.5.60 Patch Jets and Proto-Spike Propagation

Each membrane patch  $i$  has variables:

$$(\Delta V_i, g_{\text{ion},i}, P_{\text{open},i}, \eta_i).$$

Jets:

$$J^1(\Delta V_i) = \{\dot{\Delta V}_i, \partial_{A_{\text{exc}}} \Delta V_i\}, \quad J^1(P_{\text{open},i}) = \{\dot{P}_{\text{open},i}, \partial_{A_{\text{exc}}} P_{\text{open},i}\}.$$

Propagation condition:

$$\partial_{A_{\text{exc}}} \Delta V_i > \Theta_{\text{prop}},$$

where  $\Theta_{\text{prop}}$  is the minimal local depolarisation needed to activate neighbouring patch  $i + 1$ .

Consequences:

- wave-like proto-spike propagation,
- failure modes (local quenching),
- splitting of excitation waves,
- basis for one-dimensional neural conduction.

### O.5.61 Jets of Refractory Dynamics

Refractory behaviour arises when channel states become transiently inactivated. Let  $s_{\text{inact}}$  be an inactivated state.

Jet condition:

$$\partial_{A_{\text{exc}}} p(s_{\text{inact}}) > \Theta_{\text{ref}}.$$

This ensures:

- suppression of immediate reactivation,
- control of spike frequency,
- temporal separation of signals,
- stability of  $C_{\text{spike}}$  cycles.

### O.5.62 Jets of Boundary Coupling (K4→K5)

Although  $K_5$  possesses a new electrical axis, it remains constrained by the membrane structure inherited from  $K_4$ :

$$T_{\text{mem}}(t), \quad C_i(t), \quad P_{\text{perm}}(t).$$

Jets of electrical variables couple to membrane jets through:

$$\partial_{A_{\text{exc}}} g_{\text{channel}} \propto \partial_{A_{\text{boundary}}} P_{\text{perm}}, \quad \partial_{A_{\text{exc}}} \Delta V \propto \partial_{A_{\text{boundary}}} C.$$

These relations encode the residual dependence of neural excitability on biophysical structure.

### O.5.63 Jets and $\partial\Omega(K_5)$ Stability

The boundary of admissibility for  $K_5$  is shaped by:

- excitability thresholds,
- pump–leak balance,
- channel density,
- noise amplitude,
- membrane tension inherited from  $K_4$ .

Jets determine when the system approaches violation of admissibility:

$$\partial_{A_{\text{exc}}} (\Delta V - \Theta_{\text{exc}}) < 0 \quad \Rightarrow \quad \text{collapse of excitability.}$$

Collapse modes include:

- permanent depolarisation,
- permanent hyperpolarisation,
- noise-dominated failure,
- loss of pump dominance.



### O.5.64 Jets and the Full Proto-Spike Cycle

The spike cycle on  $K_5$ :

$$C_{\text{spike}} = (\text{rest} \rightarrow \text{rise} \rightarrow \text{peak} \rightarrow \text{fall} \rightarrow \text{recovery}),$$

has a jet representation:

$$J^1(C_{\text{spike}}) = \{\partial_t \Delta V, \partial_{A_{\text{exc}}} \Delta V, \partial_t g_{\text{ion}}, \partial_{A_{\text{exc}}} g_{\text{ion}}, \partial_t p(s), \partial_{A_{\text{exc}}} p(s)\}.$$

These jets encode:

- spike amplitude,
- spike width,
- phase durations,
- stability under perturbations,
- susceptibility to noise and membrane deformation.

### O.5.65 Summary

Jets on  $K_5$  provide the formal infinitesimal structure of neural excitability:

- membrane potential dynamics and electrical axis derivatives,
- channel state kinetics and conductance jets,
- ionic flow and pump–leak balance,
- excitation and recovery thresholds,
- stochastic noise and logic gates,
- patch-level propagation,
- refractory structure,
- coupling to  $K_4$  membrane jets,
- stability of  $\partial\Omega(K_5)$ ,
- analytic structure of the full spike cycle.

These jets constitute the mathematical foundation of the transition to higher neural organisation in  $K_6$ .

### O.5.66 Jets on $K_6$

The continuum level  $K_6$  describes cognitive organisation: axes of representation, prediction, attention and control, structured by potentials (prediction error, salience, valence), flows (selection, binding, model-updates) and thresholds for coherence and stability. Jets on  $K_6$  capture *infinitesimal variations* of cognitive states and models along these axes and provide a differential view on how small perturbations in inputs, internal parameters or structural constraints propagate through the cognitive continuum.

We denote the cognitive continuum by

$$K_6 = (\Omega(K_6), A_6, P_6, \Theta_6, J_6, C_6, k_6(t)),$$

where  $\Omega(K_6)$  is the space of admissible cognitive states,  $A_6$  the set of representational and control axes,  $P_6$  the relevant potentials (prediction error, salience, value, ...),  $\Theta_6$  the thresholds (coherence, complexity,

error, ...),  $J_6$  the cognitive flows,  $C_6$  the cycles (attention, learning, consolidation, ...), and  $k_6(t)$  the measure of cognitive continuum strength at time  $t$ .

Jets on  $K_6$  describe derivatives of these objects with respect to the underlying axes and parameters and therefore encode how micro-perturbations in  $A_6$ ,  $P_6$  or  $\Theta_6$  deform  $\Omega(K_6)$  and the flows  $J_6$ .

### O.5.67 State fields and cognitive coordinates

A cognitive state at time  $t$  can be written as a field

$$x : A_6 \rightarrow \mathbb{R}^n,$$

assigning to each representational axis  $a \in A_6$  a finite vector of activations or parameters. In practice,  $A_6$  may contain axes for sensory features, abstract concepts, task variables, latent factors, or decision-relevant dimensions.

We write  $x(a)$  for the local state along axis  $a$  and collect these values into a global state  $x \in \Omega(K_6)$ . A change of state under a small perturbation  $\delta a$  of an axis is described by the jet

$$j^1 x(a) \simeq x(a) + \partial_a x(a) \delta a,$$

where  $\partial_a x(a)$  is the derivative of  $x$  with respect to the axis  $a$ . Higher-order jets capture more complex dependencies (e.g. curvature, mixed derivatives across axes).

### O.5.68 Potentials and their jets

Central potentials on  $K_6$  include prediction error, salience, and value-like quantities. Let

$$P_{\text{pred}}, P_{\text{sal}}, P_{\text{val}}$$

denote prediction error, salience and value potentials, respectively. Given a cognitive state  $x$  and a model  $m$ , we can write

$$P_{\text{pred}} = P_{\text{pred}}(x, m), \quad P_{\text{sal}} = P_{\text{sal}}(x, m), \quad P_{\text{val}} = P_{\text{val}}(x, m).$$

Jets of these potentials describe how small changes in state, model or axes change the underlying cognitive pressures. For example, for a representational axis  $a \in A_6$ , the first jet of prediction error along  $a$  is

$$j^1 P_{\text{pred}}(a) \simeq P_{\text{pred}}(x, m) + \partial_a P_{\text{pred}}(x, m) \delta a.$$

Large values of  $\partial_a P_{\text{pred}}$  identify axes along which the system is particularly sensitive to perturbations.

### O.5.69 Jets of flows and cycles

Cognitive flows  $J_6$  describe time evolution within  $\Omega(K_6)$ : selection, binding, compression, prediction and model updates. We can write generically

$$\frac{dx}{dt} = J_6(x; P_6, \Theta_6).$$

Jets of flows encode how these dynamics change under small perturbations of axes, thresholds or external inputs. For a parameter  $\lambda$  (e.g. a control gain, noise level, or resource constraint) we obtain the first jet

$$j^1 J_6(x; \lambda) \simeq J_6(x; \lambda) + \partial_\lambda J_6(x; \lambda) \delta \lambda.$$

For a cycle  $C \in C_6$  (e.g. an attention–prediction–update cycle or a learning–consolidation cycle) with period  $\tau_C$ , jets characterise stability under perturbations of parameters and axes:

$$j^1 \tau_C(\lambda) \simeq \tau_C(\lambda) + \partial_\lambda \tau_C(\lambda) \delta \lambda,$$

and similarly for measures of cycle amplitude or coherence.

### O.5.70 Jets at the interface with $K_5$

The coupling between  $K_5$  (bioelectrical excitable continua) and  $K_6$  (cognitive continua) is mediated by jets that connect electrical variables (membrane potentials, spiking statistics) to cognitive axes and potentials.

Let  $\Delta V$  denote membrane potential and  $A_{\text{exc}}$  the electrical excitability axis on  $K_5$ . A first jet of  $\Delta V$  along  $A_{\text{exc}}$  is

$$j^1 \Delta V(A_{\text{exc}}) \simeq \Delta V + \partial_{A_{\text{exc}}}(\Delta V) \delta A_{\text{exc}},$$

where *no extra brace* appears in the derivative term. This jet captures how small changes in the excitability axis (e.g. effective channel density, gating kinetics, or patch composition) alter the local membrane potential.

On the  $K_6$  side, an axis of proto-representation  $A_{\text{rep}}$  may be driven by spike statistics (rates, synchrony, patterns). A jet of a representational coordinate  $x_{\text{rep}}$  with respect to a spiking feature  $s$  can be written as

$$j^1 x_{\text{rep}}(s) \simeq x_{\text{rep}}(s) + \partial_s x_{\text{rep}}(s) \delta s.$$

Non-zero  $\partial_s x_{\text{rep}}$  indicates an effective projection from spike-space to representational space and thus supports the emergence of cognitive axes from underlying electrical dynamics.

### O.5.71 Jets of thresholds and boundary deformations

Thresholds on  $K_6$  (e.g. prediction-error threshold  $\Theta_{\text{pred}}$ , complexity threshold  $\Theta_{\text{comp}}$ , coherence thresholds) define the boundary  $\partial\Omega(K_6)$  of admissible cognitive states. Jets of thresholds describe how boundary conditions deform under structural or parametric changes.

For a threshold  $\Theta_i \in \Theta_6$  depending on a parameter  $\lambda$  we have

$$j^1 \Theta_i(\lambda) \simeq \Theta_i(\lambda) + \partial_\lambda \Theta_i(\lambda) \delta \lambda.$$

Similarly, jets of the boundary itself can be written in terms of normal variations along  $\partial\Omega(K_6)$ , capturing how regions of admissible cognition expand, contract or change shape under slow structural evolution.

### O.5.72 Jets and the evolution of $k_6(t)$

The measure of cognitive continuum strength  $k_6(t)$  depends on the structure of  $\Omega(K_6)$ , the richness and stability of cycles  $C_6$ , and the balance of flows  $J_6$ . Jets of  $k_6(t)$  with respect to parameters or axes quantify how small changes in structure influence overall cognitive viability.

For a parameter  $\lambda$  we write

$$j^1 k_6(t; \lambda) \simeq k_6(t; \lambda) + \partial_\lambda k_6(t; \lambda) \delta \lambda.$$

If  $\partial_\lambda k_6 > 0$  in some direction, then small perturbations along  $\lambda$  increase the strength of the cognitive continuum; if  $\partial_\lambda k_6 < 0$  they weaken it. Jets therefore provide a local, differential tool for mapping directions of stabilisation, destabilisation and possible transition paths towards higher levels (e.g.  $K_7$ ) or towards collapse of  $K_6$ .

### O.5.73 Summary

Jets on  $K_6$  provide a differential calculus for cognitive continua. They describe:

- infinitesimal changes of state fields along cognitive axes,
- sensitivity of prediction, salience and value potentials,
- local stability of cognitive flows and cycles,
- coupling between electrical excitability on  $K_5$  and representational axes on  $K_6$ ,

- deformations of thresholds and boundaries,
- and local directions of growth or decay of  $k_6(t)$ .

This jet structure connects the qualitative description of cognitive dynamics in  $K_6$  with precise local variations, providing the bridge to more formal differential or variational formulations of cognitive evolution.

#### O.5.74 Jets on $K_7$

The level  $K_7$  introduces social continua: multi-agent structures with collective states, norms, roles, distributed information, and institutional constraints. Jets on  $K_7$  provide the infinitesimal analysis of social potentials, flows, coordination mechanisms, and instability thresholds governing cooperation, conflict, and institutional persistence.

#### O.5.75 State Space and Jet Coordinates of $K_7$

A social-state  $\omega_7$  consists of

$$\omega_7 = (G, R, S, N, P, J),$$

where:

- $G$  — group-structure configuration (membership, topology),
- $R$  — role assignments and behavioural repertoires,
- $S$  — status and authority distributions,
- $N$  — norms and institutional constraints,
- $P$  — social potentials (trust, cohesion, sanction pressure),
- $J$  — social flows (information, resources, influence).

The  $m$ - $n$  jet bundle of  $K_7$ :

$$J^{m,n}(K_7) = \{\partial_t^k \partial_{A_7}^\ell (G, R, S, N, P, J)\}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n,$$

where  $A_7$  denotes the set of social axes (coordination, imitation, status-gradient, normative force, institutional rigidity, network connectivity).

#### O.5.76 Jets of Group Structure $G$

Group structure evolves by:

$$\dot{G} = F_G(G, S, P, J_{\text{comm}}).$$

Jets:

$$J^1(G) = \{\dot{G}, \partial_{A_7} G\}.$$

Interpretation:

- changes in group topology (coalitions, fragmentation),
- sensitivity to trust potentials and influence gradients,
- onset of coordination or polarisation,
- conditions for network bifurcations.

### O.5.77 Jets of Roles $R$

Roles formalise stable behavioural channels:

$$\dot{R} = F_R(R, N, S, P).$$

Jets:

$$\partial_{A_7} R = \partial_S R \partial_{A_7} S + \partial_N R \partial_{A_7} N + \partial_P R \partial_{A_7} P.$$

They encode:

- role-switch dynamics,
- emergence of functional differentiation,
- instability when norms degrade,
- coupling between status and role coherence.

### O.5.78 Jets of Status Distribution $S$

Status field:

$$\dot{S} = F_S(S, G, P, J_{\text{influence}}).$$

Jet:

$$J^1(S) = \{\dot{S}, \partial_{A_7} S\}.$$

Interpretation:

- status competition and stabilisation,
- influence propagation,
- transition from hierarchical to flat structures,
- bifurcations at institutional thresholds  $\Theta_{\text{inst}}$ .

### O.5.79 Jets of Norms and Institutional Constraints $N$

Norm dynamics:

$$\dot{N} = F_N(N, P, S, J_{\text{sanction}}).$$

Jets:

$$\partial_{A_7} N = \partial_P N \partial_{A_7} P + \partial_S N \partial_{A_7} S.$$

Norm jets capture:

- stability of rule-sets,
- emergence of local conventions,
- institutional drift,
- conditions of breakdown when normative force falls below  $\Theta_{\text{norm}}$ .

### O.5.80 Jets of Social Potentials $P_7$

Social potentials include trust, cohesion, sanction pressure, expectation alignment, and cooperative incentive structure:

$$P = (P_{\text{trust}}, P_{\text{cohesion}}, P_{\text{sanction}}, P_{\text{coord}}, P_{\text{inst}}).$$

Their jets:

$$J^1(P) = \{\dot{P}, \partial_{A_7} P\}.$$

These derivatives quantify:

- erosion or reinforcement of trust,
- formation of cooperative equilibria,
- phase transitions between coordination regimes,
- strengthening or weakening of institutions.

### O.5.81 Jets of Social Flows $J_7$

Flows in  $K_7$  include:

$$J_{\text{comm}}, J_{\text{resource}}, J_{\text{influence}}, J_{\text{sanction}}, J_{\text{coord}}.$$

Dynamics:

$$\dot{J}_7 = F_J(J_7, P, G, S, N).$$

Jet:

$$J^1(J_7) = \{\dot{J}_7, \partial_{A_7} J_7\}.$$

Interpretation:

- stability of communication networks,
- propagation of influence and norms,
- coordination efficiency,
- onset of systemic conflict or alignment.

### O.5.82 Jets of Social Thresholds $\Theta_7$

Key thresholds:

$$\Theta_7 = \{\Theta_{\text{trust}}, \Theta_{\text{cohesion}}, \Theta_{\text{norm}}, \Theta_{\text{inst}}, \Theta_{\text{coord}}\}.$$

Their jets:

$$J^1(\Theta_7) = \{\dot{\Theta}_7, \partial_{A_7} \Theta_7\}.$$

Threshold jets describe:

- proximity to institutional collapse,
- formation of new stable norms,
- collective action failure,
- condensation of coordinated behaviour.

### O.5.83 Jets of Social Cycles $C_7$

The social continuum contains characteristic cycles:

$$C_{\text{coord}}, C_{\text{norm}}, C_{\text{status}}, C_{\text{resource}}, C_{\text{influence}}, C_{\text{institution}}.$$

A jet of a cycle  $C_j$  is:

$$J^1(C_j) = \{\partial_t X_j, \partial_{A_7} X_j\}, \quad X_j \in \{G, R, S, N, P, J\}.$$

Cycle jets diagnose:

- coordination oscillations and coherence domains,
- status rivalry / stabilisation,
- normative reinforcement loops,

- resource redistribution dynamics,
- institutional drift and lock-in,
- conditions for emergence of  $K_8$ -level structures.

#### O.5.84 Jets and Boundary Stability $\partial\Omega(K_7)$

The social continuum becomes unstable when:

$$\partial_{A_7}(P_{\text{trust}} - \Theta_{\text{trust}}) < 0,$$

or

$$\partial_{A_7}(P_{\text{inst}} - \Theta_{\text{inst}}) < 0,$$

indicating breakdown of trust or institutional authority.

Similarly, coordination collapse occurs if

$$\partial_{A_7}(P_{\text{coord}} - \Theta_{\text{coord}}) < 0.$$

Jets quantify distances to these boundaries and predict which axes are responsible for approaching  $\partial\Omega$ .

#### O.5.85 Jets and the Transition $K_6 \rightarrow K_7$

The emergence of the social continuum arises when cognitive models become aligned across agents:

$$\partial_{A_7}G \propto \partial_{A_6}M_{\text{shared}}, \quad \partial_{A_7}N \propto \partial_{A_6}\Pi_{\text{collective}},$$

$$\partial_{A_7}P_{\text{trust}} \propto \partial_{A_6}C_{\text{model}}.$$

Thus, jets of  $K_6$  (model coherence, attention, prediction cycles) stabilise and extend into the social domain, giving rise to coordinated groups, norms, and institutions.

#### O.5.86 Summary

Jets on  $K_7$  formalise the infinitesimal structure of social continua:

- group topology and coordination,
- role and status differentiation,
- norm dynamics and institutional strength,
- social potentials (trust, cohesion, sanction),
- information and resource flows,
- social thresholds and boundary stability,
- characteristic coordination and norm cycles,
- emergence of collective structures from  $K_6$  cognition.

They form the transition layer toward  $K_8$ , where symbolic and civilisational structures emerge.

#### O.5.87 Jets on $K_8$

The level  $K_8$  corresponds to civilisational continua: large-scale aggregates of agents, institutions, symbolic structures, technologies, and energy flows. Jets on  $K_8$  allow us to analyse the infinitesimal evolution of infrastructures, symbolic coherence, systemic fragility, and long-term stability of complex societies.

### O.5.88 State Space and Jet Coordinates of $K_8$

A civilisational state  $\omega_8$  is represented as

$$\omega_8 = (I, T, S, P, J),$$

where

- $I$  — infrastructure configurations (transport, energy, logistics),
- $T$  — technological systems and production networks,
- $S$  — symbolic structures (language, media, cultural codes),
- $P$  — macro-potentials (energy capacity, institutional tension, symbolic coherence, systemic stress),
- $J$  — flows (information, energy, materials, finance).

Jet coordinates:

$$J^{m,n}(K_8) = \{\partial_t^k \partial_{A_8}^\ell (I, T, S, P, J)\}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n,$$

where  $A_8$  includes infrastructural, technological, symbolic, institutional, and macro-energetic axes.

### O.5.89 Jets of Infrastructure $I$

Infrastructure dynamics:

$$\dot{I} = F_I(I, T, P_{\text{energy}}, J_{\text{material}}).$$

Jet:

$$J^1(I) = \{\dot{I}, \partial_{A_8} I\}.$$

Interpretation:

- robustness and redundancy of networks,
- sensitivity to energy constraints,
- propagation of failures (cascades),
- infrastructure-driven phase transitions.

### O.5.90 Jets of Technological Systems $T$

Technological systems evolve via:

$$\dot{T} = F_T(T, I, S, P_{\text{innovation}}, J_{\text{info}}).$$

Jet:

$$\partial_{A_8} T = \partial_I T \partial_{A_8} I + \partial_S T \partial_{A_8} S + \partial_P T \partial_{A_8} P.$$

This captures:

- innovation dynamics,
- technological lock-in,
- dependence on symbolic structures (education, language),
- transitions between technological regimes.



### O.5.91 Jets of Symbolic Structures $S$

Symbolic structures include shared languages, media, narratives and cultural systems:

$$\dot{S} = F_S(S, T, P_{\text{coherence}}, J_{\text{info}}).$$

Jet:

$$J^1(S) = \{\dot{S}, \partial_{A_8} S\}.$$

Interpretation:

- symbolic coherence and fragmentation,
- emergence of shared narratives,
- media-driven instability,
- cultural drift and codification.

### O.5.92 Jets of Civilisational Potentials $P_8$

Macro-potentials include:

$$P_8 = (P_{\text{energy}}, P_{\text{coherence}}, P_{\text{inst}}, P_{\text{stress}}, P_{\text{innovation}}).$$

Jets:

$$J^1(P_8) = \{\dot{P}_8, \partial_{A_8} P_8\}.$$

Potentials encode:

- available energy reserves and conversion efficiency,
- symbolic cohesion,
- institutional tension and legitimacy,
- systemic stress accumulation,
- capacity for technological advance.

### O.5.93 Jets of Civilisational Flows $J_8$

Flows contain:

$$J_8 = (J_{\text{energy}}, J_{\text{material}}, J_{\text{info}}, J_{\text{finance}}, J_{\text{institution}}).$$

Dynamics:

$$\dot{J}_8 = F_J(J_8, I, T, S, P_8).$$

Jet:

$$J^1(J_8) = \{\dot{J}_8, \partial_{A_8} J_8\}.$$

Interpretation:

- global coupling of subsystems,
- congestion and diffusion regimes,
- propagation of shocks,
- formation of large-scale attractors.

#### O.5.94 Jets of Thresholds $\Theta_8$

Key civilisational thresholds:

$$\Theta_8 = \{\Theta_{\text{energy}}, \Theta_{\text{infrastructure}}, \Theta_{\text{coherence}}, \Theta_{\text{inst}}, \Theta_{\text{fragility}}\}.$$

Jets:

$$J^1(\Theta_8) = \{\dot{\Theta}_8, \partial_{A_8} \Theta_8\}.$$

Interpretation:

- proximity to infrastructural collapse,
- symbolic fragmentation boundary,
- institutional instability,
- energy constraints on growth,
- systemic fragility amplification.

#### O.5.95 Jets of Civilisational Cycles $C_8$

Typical cycles of  $K_8$ :

$$C_{\text{infra}}, C_{\text{energy}}, C_{\text{symbolic}}, C_{\text{tech}}, C_{\text{inst}}, C_{\text{stress}}.$$

A jet of a cycle  $C_j$ :

$$J^1(C_j) = \{\partial_t X_j, \partial_{A_8} X_j\}, \quad X_j \in \{I, T, S, P, J\}.$$

Cycle jets reveal:

- long-term civilisational rhythms,
- shocks and recovery processes,
- positive feedback loops,
- transitions from expansion to stagnation,
- systemic crises (collapse attractors),
- pathways to new civilisational phases.

#### O.5.96 Jets and Boundary Stability $\partial\Omega(K_8)$

Instability occurs when:

$$\partial_{A_8}(P_{\text{energy}} - \Theta_{\text{energy}}) < 0,$$

or

$$\partial_{A_8}(P_{\text{coherence}} - \Theta_{\text{coherence}}) < 0,$$

or

$$\partial_{A_8}(P_{\text{inst}} - \Theta_{\text{inst}}) < 0.$$

These inequalities indicate:

- resource exhaustion,
- symbolic fragmentation,
- institutional collapse.

Jets quantify the rates and directions by which the system approaches  $\partial\Omega(K_8)$ .

### O.5.97 Jets and the Transition $K_7 \rightarrow K_8$

The rise of the civilisational continuum corresponds to:

$$\partial_{A_8} S \propto \partial_{A_7} C_{\text{coord}},$$

$$\partial_{A_8} I \propto \partial_{A_7} J_{\text{resource}},$$

$$\partial_{A_8} P_{\text{coherence}} \propto \partial_{A_7} P_{\text{trust}}.$$

Thus stable symbolic structures at  $K_7$  become enlarged and codified into long-range infrastructures, institutional systems, and technological networks characteristic of  $K_8$ .

### O.5.98 Summary

Jets on  $K_8$  formalise the infinitesimal dynamics of civilisational continua:

- infrastructure robustness and propagation of failures,
- technological regimes and innovation dynamics,
- symbolic coherence and cultural drift,
- macro-potentials governing stability and growth,
- large-scale flows across energy, information and materials,
- civilisational thresholds and systemic fragility,
- long-term cycles and collapse dynamics,
- emergence from  $K_7$  social coordination.

They provide the analytical basis for transitions to higher-level meta-structural continua  $K_9$  and  $K_{10}$ .

### O.5.99 Jets on $K_9$

The level  $K_9$  corresponds to scientific paradigms: structured systems of theories, models, inference rules, evidential practices and epistemic constraints. Jets on  $K_9$  describe infinitesimal changes in the configuration of paradigms, enabling a differential analysis of conceptual evolution, epistemic tension, evidential shocks and paradigm transition.

### O.5.100 State Space and Jet Coordinates of $K_9$

A paradigm state is given by

$$\omega_9 = (\mathcal{T}, \mathcal{M}, E, P_9, J_9),$$

where

- $\mathcal{T}$  — theoretical structures (axioms, constructs, inference rules),
- $\mathcal{M}$  — class of admissible models and representations,
- $E$  — evidential corpus (data, experiments, observations),
- $P_9$  — epistemic potentials (coherence, explanatory power, epistemic tension, uncertainty),
- $J_9$  — epistemic flows (data influx, model updates, inferential propagation, anomaly accumulation).

Jet coordinates:

$$J^{m,n}(K_9) = \{\partial_t^k \partial_{A_9}^\ell (\mathcal{T}, \mathcal{M}, E, P_9, J_9)\}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n.$$

### O.5.101 Jets of Theoretical Structures $\mathcal{T}$

Theoretical constructions evolve via

$$\dot{\mathcal{T}} = F_{\mathcal{T}}(\mathcal{T}, \mathcal{M}, P_9, E).$$

Jet:

$$J^1(\mathcal{T}) = \{\dot{\mathcal{T}}, \partial_{A_9} \mathcal{T}\}.$$

Interpretation:

- structural refinement of theories,
- shifts in inferential norms,
- reduction of internal contradiction,
- expansion or restriction of conceptual domains.

### O.5.102 Jets of Model Spaces $\mathcal{M}$

Models evolve under evidential pressure and theoretical adjustment:

$$\dot{\mathcal{M}} = F_{\mathcal{M}}(\mathcal{M}, \mathcal{T}, E, P_9).$$

Jet:

$$\partial_{A_9} \mathcal{M} = \partial_{\mathcal{T}} \mathcal{M} \partial_{A_9} \mathcal{T} + \partial_E \mathcal{M} \partial_{A_9} E + \partial_{P_9} \mathcal{M} \partial_{A_9} P_9.$$

Interpretation:

- change of admissible solution classes,
- refinement of parameters and structures,
- extension to new domains,
- collapse of obsolete representations.

### O.5.103 Jets of the Evidential Corpus $E$

Evidence dynamics:

$$\dot{E} = F_E(E, J_9, \mathcal{M}, \mathcal{T}).$$

Jet:

$$J^1(E) = \{\dot{E}, \partial_{A_9} E\}.$$

Interpretation:

- accumulation of new data,
- emergence of anomalies,
- evidential saturation,
- domain shifts produced by new observations.

#### O.5.104 Jets of Epistemic Potentials $P_9$

Epistemic potentials include:

$$P_9 = (P_{\text{coh}}, P_{\text{exp}}, P_{\text{tens}}, P_{\text{unc}}),$$

corresponding to coherence, explanatory power, epistemic tension and uncertainty.

Jet:

$$J^1(P_9) = \{\dot{P}_9, \partial_{A_9} P_9\}.$$

Interpretation:

- tracking epistemic stress,
- assessing coherence gradients,
- quantifying paradigm vulnerability,
- measuring informational load.

#### O.5.105 Jets of Epistemic Flows $J_9$

Epistemic flows:

$$J_9 = (J_{\text{data}}, J_{\text{infer}}, J_{\text{model}}, J_{\text{anomaly}}).$$

Dynamics:

$$\dot{J}_9 = F_J(J_9, E, \mathcal{T}, \mathcal{M}, P_9).$$

Jet:

$$J^1(J_9) = \{\dot{J}_9, \partial_{A_9} J_9\}.$$

Interpretation:

- propagation of inferential consequences,
- anomaly diffusion,
- evidential feedback loops,
- destabilisation of paradigmatic structures.

#### O.5.106 Jets of Thresholds $\Theta_9$

Thresholds at  $K_9$  include:

$$\Theta_9 = \{\Theta_{\text{coh}}, \Theta_{\text{exp}}, \Theta_{\text{tens}}, \Theta_{\text{unc}}, \Theta_{\text{cons}}\},$$

corresponding to coherence collapse, loss of explanatory power, epistemic tension exceeding sustainable limits, runaway uncertainty and logical inconsistency.

Jets:

$$J^1(\Theta_9) = \{\dot{\Theta}_9, \partial_{A_9} \Theta_9\}.$$

Interpretation:

- onset of paradigm crisis,
- tightening or loosening of admissible theories,
- proximity to conceptual breakdown,
- transition into new theoretical regimes.

### O.5.107 Jets of Paradigm Cycles $C_9$

Typical cycles include:

$$C_{\text{Popper}}, \quad C_{\text{Kuhn}}, \quad C_{\text{evid}}, \quad C_{\text{stress}}.$$

A jet of cycle  $C_j$ :

$$J^1(C_j) = \{\partial_t X_j, \partial_{A_9} X_j\}, \quad X_j \in \{\mathcal{T}, \mathcal{M}, E, P_9, J_9\}.$$

Cycle jets reveal:

- oscillations between normal science and crisis,
- evidential accumulation vs. anomaly saturation,
- shifts in theoretical dominance,
- stability windows and tipping points,
- pathways to paradigm replacement.

### O.5.108 Jets and Boundary Stability $\partial\Omega(K_9)$

Instability conditions:

$$\begin{aligned} \partial_{A_9}(P_{\text{coh}} - \Theta_{\text{coh}}) &< 0, & \partial_{A_9}(P_{\text{exp}} - \Theta_{\text{exp}}) &< 0, \\ \partial_{A_9}(P_{\text{tens}} - \Theta_{\text{tens}}) &> 0. \end{aligned}$$

These indicate:

- breakdown of coherence,
- loss of explanatory adequacy,
- runaway epistemic tension.

Jets allow us to track the rate and direction of approach to the paradigm boundary  $\partial\Omega(K_9)$ .

### O.5.109 Jets and the Transition $K_8 \rightarrow K_9$

Transition arises when symbolic structures of  $K_8$  become abstracted and codified into explicit formal systems:

$$\partial_{A_9} \mathcal{T} \propto \partial_{A_8} S, \quad \partial_{A_9} E \propto \partial_{A_8} J_{\text{info}}.$$

Thus:

- long-range cultural-symbolic coherence becomes formal theory,
- infrastructural flows become structured evidential flows,
- civilisational potentials become epistemic potentials.

### O.5.110 Summary

Jets on  $K_9$  characterise the differential evolution of scientific paradigms:

- transformation of theories and inferential rules,
- evolution of model classes,
- evidential shocks and anomaly accumulation,
- epistemic potentials and tension gradients,

- flows of data, inference and conceptual change,
- dissolution of paradigms and emergence of new ones.

They provide the formal link between civilisational symbolic systems ( $K_8$ ) and the meta-theoretical continua characteristic of  $K_{10}$ .

#### O.5.111 Jets on $K_{10}$

The level  $K_{10}$  represents meta-theoretical continua: coherent systems of modelling frameworks, categories of theories, functors between them, modal spaces and operators of second-order differences. Jets on  $K_{10}$  capture infinitesimal changes in these meta-structures, enabling a differential analysis of how modelling capacities, functorial stability, modal reach and reflexive depth evolve over time.

#### O.5.112 State Space and Jet Coordinates of $K_{10}$

A meta-theoretical state is given by

$$\omega_{10} = (\mathcal{F}, \mathcal{C}, \Lambda, \mathfrak{D}),$$

where

- $\mathcal{F}$  — meta-modelling operators;
- $\mathcal{C}$  — categories of models and functors;
- $\Lambda$  — modal spaces and possible-world structures;
- $\mathfrak{D}$  — second-order difference operators (rules for constructing axes and ontologies).

Jets on  $K_{10}$  are defined over partial derivatives with respect to meta-axes  $A_{10}$ :

$$J^{m,n}(K_{10}) = \{\partial_t^k \partial_{A_{10}}^\ell (\mathcal{F}, \mathcal{C}, \Lambda, \mathfrak{D})\}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n.$$

#### O.5.113 Jets of Meta-Modelling Operators $\mathcal{F}$

Dynamics:

$$\dot{\mathcal{F}} = F_{\mathcal{F}}(\mathcal{F}, \mathcal{C}, \Lambda, \mathfrak{D}).$$

Jet:

$$J^1(\mathcal{F}) = \{\dot{\mathcal{F}}, \partial_{A_{10}} \mathcal{F}\}.$$

Interpretation:

- refinement of transformation rules for theories,
- evolution of constraints on models and representations,
- emergence of new meta-syntactic operations,
- restructuring of admissible mappings between modelling domains.

#### O.5.114 Jets of Categories of Models $\mathcal{C}$

Categories evolve through changes in admissible objects and functors, driven by meta-level pressure from  $\mathcal{F}$  and  $\mathfrak{D}$ :

$$\dot{\mathcal{C}} = F_{\mathcal{C}}(\mathcal{C}, \mathcal{F}, \Lambda, \mathfrak{D}).$$

Jet:

$$\partial_{A_{10}} \mathcal{C} = \partial_{\mathcal{F}} \mathcal{C} \partial_{A_{10}} \mathcal{F} + \partial_{\Lambda} \mathcal{C} \partial_{A_{10}} \Lambda + \partial_{\mathfrak{D}} \mathcal{C} \partial_{A_{10}} \mathfrak{D}.$$

Interpretation:

- transformation of object classes,

- emergence or collapse of functors,
- changes in compositional rules,
- destabilisation of categorical coherence.

### O.5.115 Jets of Modal Spaces $\Lambda$

Modal spaces store the structure of admissible possibilities and counterfactuals for theories.

Dynamics:

$$\dot{\Lambda} = F_{\Lambda}(\Lambda, \mathcal{F}, \mathcal{C}, \mathfrak{D}).$$

Jet:

$$J^1(\Lambda) = \{\dot{\Lambda}, \partial_{A_{10}}\Lambda\}.$$

Interpretation:

- expansion or contraction of modal reach,
- refinement of accessibility relations,
- restructuring of possible-worlds geometry,
- shifts in the dimensionality of modality.

### O.5.116 Jets of Difference Operators $\mathfrak{D}$

Difference operators encode the rules for constructing new axes, ontologies and structural distinctions.

Dynamics:

$$\dot{\mathfrak{D}} = F_{\mathfrak{D}}(\mathfrak{D}, \mathcal{F}, \mathcal{C}, \Lambda).$$

Jet:

$$J^1(\mathfrak{D}) = \{\dot{\mathfrak{D}}, \partial_{A_{10}}\mathfrak{D}\}.$$

Interpretation:

- emergence of higher-order distinctions,
- refinement of ontological construction rules,
- destabilisation of self-referential structures,
- birth of new modelling dimensions.

### O.5.117 Jets of Thresholds $\Theta_{10}$

Thresholds at  $K_{10}$ :

$$\Theta_{10} = (\Theta_{\text{self}}, \Theta_{\text{meta}}, \Theta_{\text{mod}}, \Theta_{\text{functor}}),$$

where each component represents:

- $\Theta_{\text{self}}$ : limit of self-reflexive depth;
- $\Theta_{\text{meta}}$ : maximal admissible meta-complexity;
- $\Theta_{\text{mod}}$ : boundary on modal dimensionality;
- $\Theta_{\text{functor}}$ : stability region for functors.

Jet:

$$J^1(\Theta_{10}) = \{\dot{\Theta}_{10}, \partial_{A_{10}}\Theta_{10}\}.$$

These jets identify approach to meta-theoretical collapse:

$$T_{10} = w_{\text{self}}(d_{\text{reflex}} - \Theta_{\text{self}}) + w_{\text{meta}}(C_{\text{meta}} - \Theta_{\text{meta}}) + w_{\text{mod}}(D_{\text{mod}} - \Theta_{\text{mod}}) + w_{\text{fun}}(S_{\text{functor}} - \Theta_{\text{functor}}).$$



### O.5.118 Jets of Meta-Theoretical Flows $J^{(10)}$

Flows:

$$J^{(10)} = \left( \frac{d\mathcal{F}}{dt}, \frac{d\mathcal{C}}{dt}, \frac{d\Lambda}{dt}, \frac{d\mathfrak{D}}{dt} \right).$$

Jet:

$$J^1(J^{(10)}) = \left\{ \partial_t^2(\mathcal{F}, \mathcal{C}, \Lambda, \mathfrak{D}), \partial_{A_{10}} \partial_t(\mathcal{F}, \mathcal{C}, \Lambda, \mathfrak{D}) \right\}.$$

Interpretation:

- acceleration/deceleration of conceptual evolution,
- meta-level resonance phenomena,
- divergence of modelling branches,
- onset of functorial instability.

### O.5.119 Jets of Meta-Theoretical Cycles $C_{10}$

Cycles are defined through closed sequences:

$$C_j : \text{metatheory} \rightarrow \text{metamodel} \rightarrow \text{model} \rightarrow \text{updated metatheory}.$$

A jet of cycle:

$$J^1(C_j) = \{ \partial_t X_j, \partial_{A_{10}} X_j \}, \quad X_j \in \{ \mathcal{F}, \mathcal{C}, \Lambda, \mathfrak{D} \}.$$

Jets reveal:

- coherence of meta-theoretical evolution,
- existence of stable meta-cycles,
- approach to collapse when  $\oint d\mathcal{F} \neq 0$ ,
- emergence of higher-order modelling structures.

### O.5.120 Jets and Boundary Stability $\partial\Omega(K_{10})$

Instability arises when:

$$\partial_{A_{10}}(d_{\text{reflex}} - \Theta_{\text{self}}) > 0, \quad \partial_{A_{10}}(C_{\text{meta}} - \Theta_{\text{meta}}) > 0,$$

$$\partial_{A_{10}}(D_{\text{mod}} - \Theta_{\text{mod}}) > 0.$$

Interpretation:

- runaway self-reflexivity,
- meta-theoretical overload,
- modal explosion,
- functorial incoherence.

Approach to  $\partial\Omega(K_{10})$  predicts meta-theoretical death:

$$\Omega(K_{10}) = \emptyset.$$

### O.5.121 Jets and the Transition $K_9 \rightarrow K_{10}$

Transition occurs when structured scientific paradigms ( $K_9$ ) acquire:

- second-order difference operators,
- formalised relations between theories (functors),
- modal operators describing possibility spaces,
- meta-rules for generating new modelling axes.

Jet relation:

$$\partial_{A_{10}}(\mathcal{F}, \mathcal{C}, \Lambda, \mathfrak{D}) \propto \partial_{A_9}(\mathcal{T}, \mathcal{M}, E, P_9, J_9).$$

### O.5.122 Summary

Jets on  $K_{10}$  provide a differential description of how meta-theoretical continua evolve:

- transformation of meta-modelling operators,
- restructuring of categories and functors,
- shifts in modal dimensionality,
- refinement of higher-order difference operators,
- destabilisation and collapse through meta-level thresholds,
- transition from paradigm dynamics to meta-theoretical dynamics.

They constitute the necessary bridge from  $K_9$  (scientific paradigms) to the full higher-order modelling structures characteristic of advanced meta-theories.

### O.5.123 Jets on $K_{11}$

The level  $K_{11}$  represents fully formalised ontological systems: closed logical structures equipped with admissible rules of inference, operators on differences of arbitrary order, and a rigorously defined boundary of coherent ontology. Jets on  $K_{11}$  describe infinitesimal changes in these formal systems and capture the onset of instability, collapse or restructuring of the deepest representational layer of a continuum.

### O.5.124 State Space and Jet Coordinates of $K_{11}$

A state of  $K_{11}$  is a formal ontological structure:

$$\omega_{11} = (\mathcal{O}, \mathcal{R}, \Delta, \Pi),$$

where:

- $\mathcal{O}$  — the set of admissible ontological objects;
- $\mathcal{R}$  — formal relations and inference rules;
- $\Delta$  — higher-order difference operators (beyond  $\mathfrak{D}$  of  $K_{10}$ );
- $\Pi$  — admissible proof, consistency and closure schemes.

Jets on  $K_{11}$  are defined as:

$$J^{m,n}(K_{11}) = \left\{ \partial_t^k \partial_{A_{11}}^\ell (\mathcal{O}, \mathcal{R}, \Delta, \Pi) \right\}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n,$$

where  $A_{11}$  is the axis of formal-ontological variation.

### O.5.125 Jets of Ontological Objects $\mathcal{O}$

Dynamics:

$$\dot{\mathcal{O}} = F_{\mathcal{O}}(\mathcal{O}, \mathcal{R}, \Delta, \Pi).$$

Jet:

$$J^1(\mathcal{O}) = \{\dot{\mathcal{O}}, \partial_{A_{11}} \mathcal{O}\}.$$

Interpretation:

- refinement or restriction of admissible objects;
- birth of new permissible ontological entities;
- contraction of  $\mathcal{O}$  as boundaries approach collapse;
- elimination of objects violating consistency schemes.

### O.5.126 Jets of Ontological Relations $\mathcal{R}$

Relations and inference rules evolve as constraints from  $\Delta$  and  $\Pi$  shift.

Dynamics:

$$\dot{\mathcal{R}} = F_{\mathcal{R}}(\mathcal{R}, \mathcal{O}, \Delta, \Pi).$$

Jet:

$$\partial_{A_{11}} \mathcal{R} = \partial_{\mathcal{O}} \mathcal{R} \partial_{A_{11}} \mathcal{O} + \partial_{\Delta} \mathcal{R} \partial_{A_{11}} \Delta + \partial_{\Pi} \mathcal{R} \partial_{A_{11}} \Pi.$$

Interpretation:

- modifications of admissible inference patterns,
- structural shifts in relation schemas,
- axiomatic transformations,
- collapse of relations under inconsistency.

### O.5.127 Jets of Higher-Order Difference Operators $\Delta$

In  $K_{11}$ , difference operators become fully formal: they act on distinctions of arbitrary order.

Dynamics:

$$\dot{\Delta} = F_{\Delta}(\Delta, \mathcal{O}, \mathcal{R}, \Pi).$$

Jet:

$$J^1(\Delta) = \{\dot{\Delta}, \partial_{A_{11}} \Delta\}.$$

Interpretation:

- emergence of new orders of distinction,
- restructuring of difference hierarchies,
- incompatibility cascades leading to collapse,
- sharpening of ontological axes.

### O.5.128 Jets of Consistency and Closure Schemes $\Pi$

$\Pi$  encodes formal admissibility, proof systems, soundness, closure under inference and anti-paradox constraints.

Dynamics:

$$\dot{\Pi} = F_{\Pi}(\Pi, \mathcal{O}, \mathcal{R}, \Delta).$$

Jet:

$$J^1(\Pi) = \{\dot{\Pi}, \partial_{A_{11}} \Pi\}.$$

Interpretation:

- strengthening or weakening of consistency boundaries,
- refinement of admissible derivational schemes,
- meta-logical phase transitions,
- approach to inconsistency collapse.

### O.5.129 Jets of Thresholds $\Theta_{11}$

$K_{11}$  inherits and amplifies the meta-theoretical thresholds:

$$\Theta_{11} = (\Theta_{\text{consist}}, \Theta_{\text{closure}}, \Theta_{\text{order}}, \Theta_{\text{reflex}}),$$

with interpretations:

- $\Theta_{\text{consist}}$  — limit of consistency load;
- $\Theta_{\text{closure}}$  — maximal admissible closure depth;
- $\Theta_{\text{order}}$  — boundary on order of permissible distinctions;
- $\Theta_{\text{reflex}}$  — limit of self-referential admissibility.

Jet:

$$J^1(\Theta_{11}) = \{\dot{\Theta}_{11}, \partial_{A_{11}} \Theta_{11}\}.$$

Instability signals:

$$T_{11} = w_c(C_{\text{consist}} - \Theta_{\text{consist}}) + w_o(O_{\text{order}} - \Theta_{\text{order}}) + w_r(R_{\text{reflex}} - \Theta_{\text{reflex}}),$$

with  $T_{11} > \Theta_{\text{death}}$  implying  $\Omega(K_{11}) = \emptyset$ .

### O.5.130 Jets of Formal Flows $J^{(11)}$

Flows:

$$J^{(11)} = \left( \frac{d\mathcal{O}}{dt}, \frac{d\mathcal{R}}{dt}, \frac{d\Delta}{dt}, \frac{d\Pi}{dt} \right).$$

Jet:

$$J^1(J^{(11)}) = \{\partial_t^2(\mathcal{O}, \mathcal{R}, \Delta, \Pi), \partial_{A_{11}} \partial_t(\mathcal{O}, \mathcal{R}, \Delta, \Pi)\}.$$

Interpretation:

- acceleration of formal structural change,
- divergence of inference schemes,
- formation of inconsistency fronts,
- transition to global ontological collapse.

### O.5.131 Jets of Formal Cycles $C_{11}$

Cycles appear only in highly stable  $K_{11}$  regimes:

$$C_j : \text{ontology} \rightarrow \text{inference} \rightarrow \text{difference hierarchy} \rightarrow \text{closure} \rightarrow \text{ontology}.$$

Jet of cycle:

$$J^1(C_j) = \{\partial_t X_j, \partial_{A_{11}} X_j\}, \quad X_j \in \{\mathcal{O}, \mathcal{R}, \Delta, \Pi\}.$$

Interpretation:

- existence of stable logical-ontological equilibrium,
- capacity of the system to regenerate formal coherence,
- sensitivity to perturbations in higher-order distinctions,
- precursor signals of meta-logical collapse.

### O.5.132 Jets and Boundary Geometry $\partial\Omega(K_{11})$

Approach to the boundary occurs when:

$$\begin{aligned} \partial_{A_{11}}(C_{\text{consist}} - \Theta_{\text{consist}}) &> 0, \quad \partial_{A_{11}}(O_{\text{order}} - \Theta_{\text{order}}) > 0, \\ \partial_{A_{11}}(R_{\text{reflex}} - \Theta_{\text{reflex}}) &> 0. \end{aligned}$$

Interpretation:

- rise of super-critical self-referential depth,
- runaway escalation of distinction orders,
- destabilisation of proof/closure systems,
- collapse of formal ontological space.

### O.5.133 Relation to $K_{10} \rightarrow K_{11}$ Transition

Transition is triggered when:

- meta-theoretical constructs become strictly formalised,
- modal operators acquire consistency-sensitive interpretation,
- functorial and category-theoretic structures become axiomatic,
- higher-order differences become admissible inference operators.

The jet relation:

$$\partial_{A_{11}}(\mathcal{O}, \mathcal{R}, \Delta, \Pi) \propto \partial_{A_{10}}(\mathcal{F}, \mathcal{C}, \Lambda, \mathfrak{D})$$

marks the embedding of meta-theory into fully formal ontology.

### O.5.134 Summary

Jets on  $K_{11}$  describe:

- infinitesimal changes in ontological objects, relations and proof systems;
- restructuring of difference hierarchies and admissible distinctions;
- behaviour of formal thresholds and reflexive limits;
- dynamics of formal cycles and precursors of collapse;
- transition from meta-theoretical continua to fully formal ontological continua.

They provide a differential machinery for analysing the birth, instability and collapse of the highest-order representational structures.

### O.5.135 Jets on $K_{12}$

Level  $K_{12}$  represents the highest-order meta-ontological stratum: a continuum whose states are admissible ontological universes, together with their axes, potentials, thresholds, flows, cycles, and evolution rules. Jets on  $K_{12}$  capture infinitesimal changes in the space of all possible continua, describing micro-perturbations in the architecture of reality itself.

### O.5.136 State Space of $K_{12}$

A state of  $K_{12}$  is:

$$\omega_{12} = (\mathfrak{K}, \mathfrak{A}, \mathfrak{P}, \mathfrak{f}, \mathfrak{J}, \mathfrak{C}, \mathfrak{E}),$$

where:

- $\mathfrak{K}$  — admissible continua (the class of  $K_0 \rightarrow K_{11}$  structures),
- $\mathfrak{A}$  — allowed axes applicable to those continua,
- $\mathfrak{P}$  — meta-potentials governing entire ontological spaces,
- $\mathfrak{f}$  — meta-thresholds determining possibility of continua,
- $\mathfrak{J}$  — admissible classes of flows,
- $\mathfrak{C}$  — allowed cycle types across continuum categories,
- $\mathfrak{E}$  — admissible evolution operators (e.g.,  $E, \Psi, \Phi, U$ ).

Jets on  $K_{12}$  quantify infinitesimal variations of these meta-structures.

### O.5.137 Jet Coordinates of $K_{12}$

$$J^{m,n}(K_{12}) = \left\{ \partial_t^k \partial_{\mathcal{X}}^\ell (\mathfrak{K}, \mathfrak{A}, \mathfrak{P}, \mathfrak{f}, \mathfrak{J}, \mathfrak{C}, \mathfrak{E}) \right\},$$

where:

$$\mathcal{X} \in A_{12}$$

is the \*\*axis of meta-level variation\*\*, controlling how the space of all admissible continua deforms.

### O.5.138 Jets of Admissible Continua $\mathfrak{K}$

Dynamics:

$$\dot{\mathfrak{K}} = F_{\mathfrak{K}}(\mathfrak{K}, \mathfrak{A}, \mathfrak{P}, \mathfrak{f}).$$

Jet:

$$J^1(\mathfrak{K}) = \{\dot{\mathfrak{K}}, \partial_{A_{12}} \mathfrak{K}\}.$$

Interpretation:

- emergence of new admissible continuum types,
- extinction of unstable continuum families,
- deformation of allowable K-level hierarchies,
- restriction or expansion of existence conditions for  $K_x$ .

### O.5.139 Jets of Meta-Axes $\mathfrak{A}$

Axes define which transformations are possible at any level.

Dynamics:

$$\dot{\mathfrak{A}} = F_{\mathfrak{A}}(\mathfrak{K}, \mathfrak{A}, \mathfrak{f}).$$

Jet:

$$J^1(\mathfrak{A}) = \{\dot{\mathfrak{A}}, \partial_{A_{12}}\mathfrak{A}\}.$$

Interpretation:

- appearance of new ontological axes,
- removal of axes that violate meta-threshold conditions,
- shifts in the dimensional growth rules across continua.

### O.5.140 Jets of Meta-Potentials $\mathfrak{P}$

Meta-potentials control the general "pressure landscape" shaping which continua exist and how they evolve.

Dynamics:

$$\dot{\mathfrak{P}} = F_{\mathfrak{P}}(\mathfrak{P}, \mathfrak{f}, \mathfrak{K}, \mathfrak{A}).$$

Jet:

$$J^1(\mathfrak{P}) = \{\dot{\mathfrak{P}}, \partial_{A_{12}}\mathfrak{P}\}.$$

Interpretation:

- global shifts in meta-stability,
- modifications of inter-level compatibility,
- changes in the universal potential landscape for continua.

### O.5.141 Jets of Meta-Thresholds $\mathfrak{f}$

Meta-thresholds determine \*which continua can exist at all\*.

Dynamics:

$$\dot{\mathfrak{f}} = F_{\mathfrak{f}}(\mathfrak{f}, \mathfrak{P}, \mathfrak{K}).$$

Jet:

$$J^1(\mathfrak{f}) = \{\dot{\mathfrak{f}}, \partial_{A_{12}}\mathfrak{f}\}.$$

Interpretation:

- tightening or loosening existence conditions for  $K_x$ ,
- modification of the dimension-birth thresholds,
- global shifts in compatibility conditions between  $K$ -levels.

### O.5.142 Jets of Meta-Flows $\mathfrak{J}$

Meta-flows represent permissible transformation patterns between continua or between their structural components.

Dynamics:

$$\dot{\mathfrak{J}} = F_{\mathfrak{J}}(\mathfrak{J}, \mathfrak{K}, \mathfrak{A}, \mathfrak{P}).$$

Jet:

$$J^1(\mathfrak{J}) = \{\dot{\mathfrak{J}}, \partial_{A_{12}}\mathfrak{J}\}.$$

Interpretation:

- birth of new universal transformation principles,
- suppression of flows causing cross-level inconsistency,
- emergence of novel meta-evolutionary patterns.

#### O.5.143 Jets of Meta-Cycles $\mathfrak{C}$

Cycles at  $K_{12}$  encode stable patterns of universe-level structure.

Dynamics:

$$\dot{\mathfrak{C}} = F_{\mathfrak{C}}(\mathfrak{C}, \mathfrak{K}, \mathfrak{A}, \mathfrak{P}, \mathfrak{f}).$$

Jet:

$$J^1(\mathfrak{C}) = \{\dot{\mathfrak{C}}, \partial_{A_{12}} \mathfrak{C}\}.$$

Interpretation:

- stability or fragility of meta-level recurrence,
- conditions for regeneration of continuum families,
- catastrophic breakage of universal structural cycles.

#### O.5.144 Jets of Meta-Evolution Operators $\mathfrak{E}$

Operators include  $E, \Psi, \Phi, U$  and any future higher-order evolution schemes.

Dynamics:

$$\dot{\mathfrak{E}} = F_{\mathfrak{E}}(\mathfrak{E}, \mathfrak{K}, \mathfrak{A}, \mathfrak{P}, \mathfrak{f}).$$

Jet:

$$J^1(\mathfrak{E}) = \{\dot{\mathfrak{E}}, \partial_{A_{12}} \mathfrak{E}\}.$$

Interpretation:

- modifications of universe-level evolution laws,
- changes in operators that govern dimension birth or collapse,
- restructuring of compatibility conditions across ontological strata.

#### O.5.145 Jets and Boundary Geometry $\partial\Omega(K_{12})$

Approach to the boundary occurs when:

$$\begin{aligned} \partial_{A_{12}}(\mathfrak{f}_{\text{exist}} - L_{\text{min}}) &> 0, \\ \partial_{A_{12}}(\mathfrak{P}_{\text{global}} - P_{\text{crit}}) &> 0, \\ \partial_{A_{12}}(\text{Compat}(K_x, K_y) - \Theta_{\text{compat}}) &> 0. \end{aligned}$$

Interpretation:

- rising global instability of universe-level architecture,
- failure of cross-level compatibility conditions,
- collapse of entire families of continua.



### O.5.146 Relation to $K_{11} \rightarrow K_{12}$ Transition

Transition occurs when:

- formal ontologies ( $K_{11}$ ) become variables of a higher meta-structure;
- thresholds and axes themselves become deformable entities;
- the evolution operator must operate on entire ontology classes, not singular structures;
- consistency becomes global rather than internal.

In jet form:

$$\partial_{A_{12}}(\mathcal{O}, \mathcal{R}, \Delta, \Pi) \subset \partial_{A_{12}}(\mathfrak{K}, \mathfrak{A}, \mathfrak{P}, f).$$

### O.5.147 Summary

Jets on  $K_{12}$  describe:

- infinitesimal variations in the space of possible universes;
- the birth or extinction of entire continuum families;
- global deformations of thresholds, potentials and axes;
- micro-dynamics of universe-level flows and cycles;
- evolution of evolution operators themselves;
- meta-stability, compatibility and catastrophic transitions.

They provide the differential structure required to understand how the very possibility space of reality shifts, stabilises or collapses.

## O.6 Processes

Processes constitute the dynamic layer of the Ontology of Continua (OC). A *process* is defined as any structured transformation

$$\mathcal{P} : \Omega(K) \longrightarrow \Omega(K)$$

regulated by the internal axes  $A(K)$ , potentials  $P(K)$ , flows  $J(K)$ , thresholds  $\Theta(K)$ , structural tension  $T(K)$ , and the operators  $\Psi, \Phi, \Lambda, U, X$ .

A process exists only if the continuum  $K$  maintains non-zero continuumness:

$$k(K, t) > 0,$$

and the transformation remains within the admissible region:

$$\mathcal{P}(s) \in \Omega(K) \quad \forall s \in \Omega(K).$$

This master module provides the unified framework for all process-descriptions in OC and applies to every level  $K_0$ – $K_{12}$ .

### O.6.1 Definition of a Process

A process on a continuum  $K$  is a tuple:

$$\Pi(K) = (\Omega(K), A(K), P(K), J(K), \Theta(K), C(K), T(K), k(K), \mathcal{O}),$$

where  $\mathcal{O} = \{\Psi, \Phi, \Lambda, U, X\}$  is the set of universal OC operators.

Formally, a process is a map:

$$\mathcal{P}(t) : \Omega(K, t) \rightarrow \Omega(K, t + \Delta t)$$

satisfying:

1. **Admissibility:**  $\mathcal{P}(t)(\Omega(K, t)) \subseteq \Omega(K, t + \Delta t)$ ;
2. **Threshold compliance:**  $f_{\Theta}(s) \leq 0$  for all evolved states  $s$ ;
3. **Flow coherence:**  $J(K, t)$  must permit the transition enforced by  $\mathcal{P}$ ;
4. **Cycle compatibility:**  $\mathcal{P}$  either preserves or transforms the existing cycles  $C(K)$  without destroying the structural coherence of  $K$ ;
5. **Continuity condition:**  $k(K, t + \Delta t) \geq 0$ , and death is defined by  $k \rightarrow 0$ .

### O.6.2 Operators and Processes

Processes arise through the universal operators:

- $\Psi$ : generative operator (birth of distinctions, axes, and new states);
- $\Phi$ : structural operator (reconfiguration, alignment, mapping);
- $\Lambda$ : boundary operator (modifies  $\partial\Omega$ , creates new regions);
- $U$ : evolution operator (global update of  $k, T, J, P, \Theta$ );
- $X$ : collapse operator (extinction, fragmentation, death).

Every process is expressible as a composition of these:

$$\mathcal{P} = \Lambda^{\alpha} \circ \Phi^{\beta} \circ \Psi^{\gamma} \circ U^{\delta} \circ X^{\epsilon},$$

with exponents indicating activation or multiplicity.

### O.6.3 Classes of Processes

We classify processes into six universal families:

1. **Generative processes** ( $\Psi$ -dominated): birth of axes, dimensions, states, or entire continua.
2. **Structural processes** ( $\Phi$ -dominated): rearrangement of  $A, P, J$ , or internal geometry of  $\Omega(K)$ .
3. **Boundary processes** ( $\Lambda$ -dominated): modifications of  $\partial\Omega(K)$ , creation or destruction of admissible regions.
4. **Evolutionary processes** ( $U$ -dominated): updates of tension, thresholds, flows, and continuumness.
5. **Cyclic processes** ( $C(K)$ -dominated): self-sustaining loops generating stability and coherence.
6. **Collapse processes** ( $X$ -dominated): destruction, degeneration, fragmentation, or death of the continuum.

Every specific continuum  $K$  exhibits its own instantiation of these universal classes.

### O.6.4 Conditions for the Existence of Processes

For a process to exist at time  $t$ :

$$k(K, t) > 0,$$

$$T(K, t) < \Theta_{\text{death}},$$

$$\Omega(K, t) \neq \emptyset,$$

and

$J(K, t)$  supports the transformation.

Violating any of these conditions induces a collapse trajectory under  $X$ .

### O.6.5 Birth of a Process

A process is born when:

- new distinctions emerge (via  $\Psi$ ),
- $A(K)$  gains a new axis,
- $P(K)$  changes beyond a generative threshold  $\Theta_{\text{gen}}$ ,
- a new region of  $\Omega(K)$  becomes accessible,
- a cycle  $C(K)$  begins to self-close,
- structural tension  $T$  crosses a metastable boundary.

Birth is always accompanied by a local increase in  $k(K)$ .

### O.6.6 Evolution of a Process

A process evolves according to:

$$\frac{d}{dt}\mathcal{P}(t) = U(P, A, \Theta, J, T, k),$$

where the evolution operator  $U$  updates:

$$A(t), P(t), \Theta(t), J(t), T(t), k(t).$$

The evolution is coherent if:

$$C(K, t) \neq \emptyset \quad \text{and} \quad f_{\Theta}(s) \leq 0.$$

### O.6.7 Death of a Process

A process dies when any of the following occur:

$$k(K, t) \rightarrow 0,$$

$$T(K, t) > \Theta_{\text{death}},$$

$\partial\Omega(K)$  undergoes fragmentation,

$$J_{\text{support}} < J_{\text{critical}},$$

$C(K, t)$  breaks (cycle collapse).

Death may be localized (partial collapse) or global ( $K$  itself collapses).

### O.6.8 Universal Schema of a Process

Every process in OC follows the structure:

birth  $\xrightarrow{\Psi}$  structural alignment  $\xrightarrow{\Phi}$  boundary update  $\xrightarrow{\Lambda}$  evolution  $\xrightarrow{U}$  stability or collapse  $\xrightarrow{X}$ .

This schema is invariant from  $K_0$  to  $K_{12}$ .

### O.6.9 Processes Across Levels $K_0$ – $K_{12}$

The nature of processes varies with level:

- $K_0$ : pre-continuum proto-processes (formation of distinctions).
- $K_1$ : geometric and energetic processes.
- $K_2$ : dynamical and temporal processes.
- $K_3$ : chemical reactions, bond formation/breaking.
- $K_4$ : metabolic, osmotic, excitability processes.
- $K_5$ : network, signaling, proto-cognitive processes.
- $K_6$ : cognitive, representational, inferential processes.
- $K_7$ : social, normative, institutional processes.
- $K_8$ : civilizational, infrastructural, systemic processes.
- $K_9$ : theoretical processes (model formation).
- $K_{10}$ : formal, logical, recursive processes.
- $K_{11}$ : cross-model, intertheoretic processes.
- $K_{12}$ : meta-dynamic processes among model-spaces.

At each level, processes inherit the universal schema but operate on different structures of  $\Omega(K)$  and  $\partial\Omega(K)$ .

### O.6.10 Processes and M-Spaces

In M-spaces, processes are governed by:

$$\Theta^M, A^M, \partial\Omega(M), J^M, C^M,$$

and represent transformations between entire continua.

A process in an M-space may induce:

- birth of new continua,
- branching of K-levels,
- dimensional shifts,
- cross-level fusions,
- global reconfiguration of  $\Omega$ .

M-processes obey the same universal schema but take place in the superstructure where  $K_0$ – $K_{12}$  reside as objects.

### O.6.11 Summary

This module provides the universal definition and classification of processes across all levels of the Ontology of Continua. It establishes the formal conditions for birth, evolution, stability, and death of processes, and clarifies the role of universal operators. It serves as the foundation for all level-specific process modules in the subsequent files.

### O.6.12 Processes on $K_0$

Processes on  $K_0$  differ fundamentally from all higher-level processes: they occur in a pre-continuum substrate where no axes, no flows, no potentials, and no temporal structure yet exist. All subsequent continuum dynamics originate from these proto-processes.

$K_0$  processes are not “physical” or “temporal”; they are structural transformations internal to the primitive configuration

$$K_0 = (S, \Delta, \mathcal{C}),$$

where  $S$  is the set of primitive distinctions,  $\Delta$  is the structural relation, and  $\mathcal{C}$  is the composition operator.

### O.6.13 Nature of $K_0$ Processes

A process on  $K_0$  is defined as a transformation

$$\mathcal{P}_0 : (S, \Delta, \mathcal{C}) \longrightarrow (S', \Delta', \mathcal{C}')$$

that respects the  $K_0$  axioms and maintains the minimal non-zero degree of continuumness:

$$k_0 > \Theta_0 = \varepsilon > 0.$$

Because  $K_0$  lacks axes, potentials, time, flows, or thresholds of higher levels, a  $K_0$  process operates exclusively by:

1. generating distinctions in  $S$ ;
2. modifying the adjacency or relational pattern  $\Delta$ ;
3. forming or dissolving compositional structures  $\mathcal{C}$ .

No coordinate, metric, or temporal interpretation exists at this level.

### O.6.14 The Three Fundamental Proto-Processes

There are exactly three types of processes admissible on  $K_0$ , corresponding to its three structural components.

**1. Distinction-Generation Process ( $\Psi_0$ ).** This is the primordial operation:

$$\Psi_0 : S \rightarrow S \cup \{s_{\text{new}}\}.$$

A new distinction arises when the internal structural tension in  $(S, \Delta)$  exceeds the primitive generative threshold:

$$T_0 > \Theta_{\text{gen}}^0.$$

This is the only way new “elements” of the proto-continuum can appear.

**2. Relational Reconfiguration Process ( $\Phi_0$ ).** A transformation of the primordial relation:

$$\Phi_0 : \Delta \mapsto \Delta'.$$

This process alters which distinctions can be composed, compared, or juxtaposed. It prepares the conditions for the emergence of geometric structure in  $K_1$ .

**3. Compositional Assembly Process ( $\Lambda_0$ ).** A modification of the compositional operator:

$$\Lambda_0 : \mathcal{C} \mapsto \mathcal{C}'.$$

This process forms proto-structures that become the precursors of connected regions in  $\Omega(K_1)$ .

### O.6.15 Absence of Time and Flows

On  $K_0$  there is no time:

$$\tau(K_0) \text{ does not exist.}$$

Thus, a  $K_0$  process is not temporally ordered. Its “sequence” is purely structural and becomes temporal only after the birth of the time-axis in  $K_1$ .

Similarly:

$$J(K_0) = \emptyset, \quad P(K_0) = \emptyset.$$

There are no flows, because flows require axes and potentials.

### O.6.16 Structural Tension and Thresholds on $K_0$

$K_0$  admits only one non-trivial threshold:

$$\Theta_0 = \varepsilon > 0,$$

the minimal requirement for existence.

Structural tension  $T_0$  is defined as:

$$T_0 = f(S, \Delta),$$

a measure of incompatibility among distinctions relative to the structural relation  $\Delta$ .

Generative transformations occur when:

$$T_0 > \Theta_{\text{gen}}^0.$$

Collapse occurs if:

$$T_0 > \Theta_{\text{collapse}}^0.$$

Because  $K_0$  is minimal, collapse means complete extinction:

$$\Omega(K_0) \rightarrow \emptyset.$$

### O.6.17 Birth of the First Axis and the Transition to $K_1$

The most important  $K_0$  process is the birth of the first axis, which marks the transition  $K_0 \rightarrow K_1$ .

When the interplay of  $\Psi_0$ ,  $\Phi_0$ , and  $\Lambda_0$  produces a stable structured pattern that cannot be described within the one-dimensional structure of  $(S, \Delta, \mathcal{C})$ , a new axis  $A_1$  is forced into existence.

Formally:

$$\exists S', \Delta', \mathcal{C}' \text{ such that the dimension constraint } \dim(S', \Delta', \mathcal{C}') > 0.$$

This yields:

$$A_1 = \text{first axis in } K_1.$$

Temporal structure  $\tau$  emerges simultaneously as the minimal ordering needed to track transformations of  $A_1$ .

This process is governed by the operator:

$$\Psi_{0 \rightarrow 1} : K_0 \rightarrow K_1.$$

#### O.6.18 The Universal Schema for $K_0$ Processes

$K_0$  processes instantiate a truncated form of the universal process schema (valid for all  $K$ ):

$$\begin{aligned} \text{distinction-generation} &\xrightarrow{\Psi_0} \text{relational reconfiguration} \\ &\xrightarrow{\Phi_0} \text{compositional assembly} \\ &\xrightarrow{\Lambda_0} \text{birth of the first axis.} \end{aligned}$$

No further structural operators exist at this level.

#### O.6.19 Death of $K_0$ Processes

A  $K_0$  process dies when:

$$T_0 > \Theta_{\text{collapse}}^0,$$

or

$$k_0 \rightarrow 0.$$

Because  $K_0$  has no internal redundancy or cycles, any collapse is total.

#### O.6.20 Summary

Processes on  $K_0$  are purely structural transformations of the primitive triplet  $(S, \Delta, \mathcal{C})$ . They generate distinctions, modify relations, assemble the first coherent structures, and enable the birth of  $K_1$ .

All higher-level dynamics in the Ontology of Continua originate from these proto-processes.

#### O.6.21 Processes on $K_1$

Processes on  $K_1$  constitute the first fully dynamical class of processes in the Ontology of Continua. Unlike  $K_0$ , which admits only structural transformations of  $(S, \Delta, \mathcal{C})$ , the continuum  $K_1$  possesses:

- a state space  $\Omega(K_1) = C^0(X, V)$ ,
- a boundary  $\partial\Omega(K_1)$ ,
- a continuous axis (the first axis)  $A_1$ ,
- time  $\tau(K_1)$  as an ordering parameter,
- flows  $J_1$  along the axis,
- an energy functional  $E[\phi]$ ,
- and an action functional  $S[\phi]$ .

Thus, processes on  $K_1$  are the first in the hierarchy that behave like physical dynamical systems.

### O.6.22 Definition of a $K_1$ Process

A process on  $K_1$  is any evolution of a field-like configuration:

$$\phi : X \times \tau \rightarrow V,$$

driven by admissible flows  $J_1$ , constrained by the boundary  $\partial\Omega(K_1)$ , and regulated by the energy and action functionals.

Formally:

$$\mathcal{P}_1 : \phi(t) \longrightarrow \phi(t + dt) \quad \text{with} \quad (\phi, \partial_x \phi, \partial_t \phi) \in \Omega(K_1).$$

All  $K_1$  processes satisfy the Sobolev regularity conditions:

$$\phi \in H^1(X, V), \quad \partial_x \phi, \partial_t \phi \in L^2(X, V).$$

### O.6.23 Types of Processes on $K_1$

There are four canonical classes of processes at this level.

**1. Evolution Through Flows ( $J_1$ ).** Flows are the primary mechanism of change in  $K_1$ :

$$J_1 = f(\phi, \partial_x \phi, \partial_t \phi).$$

These flows govern the redistribution of quantities along the axis  $A_1$ .

The universal evolution equation is:

$$\partial_t \phi = \Phi_1(\phi, \partial_x \phi, J_1),$$

where  $\Phi_1$  is the level-1 instance of the general evolutionary operator.

**2. Energy-Driven Processes.** Processes can minimize or redistribute the  $K_1$  energy:

$$E[\phi] = \int_X \left( \frac{1}{2} |\partial_x \phi|^2 + U(\phi) \right) dx.$$

A gradient-flow process satisfies:

$$\partial_t \phi = - \frac{\delta E}{\delta \phi}.$$

This captures diffusion-like behaviour, smoothing, and relaxation.

**3. Action-Driven (Variational) Processes.** When driven by the action functional:

$$S[\phi] = \int_\tau \int_X \mathcal{L}(\phi, \partial_t \phi, \partial_x \phi) dx dt,$$

processes follow Euler–Lagrange dynamics:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_t \left( \frac{\partial \mathcal{L}}{\partial (\partial_t \phi)} \right) - \partial_x \left( \frac{\partial \mathcal{L}}{\partial (\partial_x \phi)} \right) = 0.$$

This is the first emergence of variational physics in the OC framework.

**4. Boundary-Driven Processes.** Changes in  $\partial\Omega(K_1)$  induce boundary conditions:

$$\phi|_{\partial\Omega} = g(t), \quad \partial_x \phi|_{\partial\Omega} = h(t),$$

which can serve as sources, sinks, or constraints initiating new flows.



### O.6.24 Role of Operators on $K_1$

$K_1$  processes are governed by the action of the general operators:

- $\Psi$  — generative operator (creates new modes or patterns),
- $\Phi$  — evolutionary operator (drives temporal change),
- $\Lambda$  — compositional operator (combines fields/structures),
- $U$  — universal operator governing  $k(t)$  evolution,
- $X$  — boundary and constraint operator.

Their  $K_1$  forms reduce to:

$$\Psi_1 : \text{mode creation} \Rightarrow \phi \mapsto \phi + \delta\phi_{\text{mode}},$$

$$\Phi_1 : \text{flow-based evolution} \Rightarrow \partial_t \phi = \Phi_1(\phi, J_1),$$

$$\Lambda_1 : \text{composition} \Rightarrow \phi = \phi_1 \oplus \phi_2,$$

$$U_1 : \text{update of continuumness} \Rightarrow k_1(t + dt) = U_1(k_1, \phi, J_1, T_1, \Theta_1),$$

$$X_1 : \text{boundary enforcement.}$$

### O.6.25 Continuumness Evolution

$K_1$  admits a non-trivial evolution of the measure of continuumness:

$$k_1(t) = F_k(\Omega(K_1), J_1, T_1, \Theta_1).$$

$k_1$  increases under coherent flows and decreases under irregular, high-gradient, or threshold-violating behaviour.

If

$$k_1(t) \rightarrow 0,$$

the continuum internally collapses into a  $K_0$ -like state.

### O.6.26 Structural Tension and Threshold-Crossing Processes

$K_1$  includes:

$$T_1 = f(\phi, \partial_x \phi, \partial_t \phi),$$

with thresholds:

- $\Theta_{\text{stab}}$  — stability threshold,
- $\Theta_{\text{crit}}$  — critical tension (phase change),
- $\Theta_{\text{death}}$  — death of continuum.

Processes exhibit:

- **stability regime:**  $T_1 < \Theta_{\text{stab}}$ ,
- **critical regime:**  $T_1 = \Theta_{\text{crit}}$ ,
- **phase transition:**  $T_1 > \Theta_{\text{crit}}$  with new structural modes arising,
- **collapse:**  $T_1 > \Theta_{\text{death}}$ .

### O.6.27 Emergence of $K_2$ Processes

$K_1$  processes generate  $K_2$  when the flows and relational structure induce non-trivial interactions that cannot be encoded in one axis.

Formally, the transition is triggered when:

$$\exists \phi, J_1 \text{ such that } \dim(\Omega(K_1)) < \dim(\Omega(K_2)).$$

This is mediated by the operator:

$$\Psi_{1 \rightarrow 2} : K_1 \rightarrow K_2.$$

The hallmark of this process is the birth of percolation-like structures and causal networks.

### O.6.28 Death of $K_1$ Processes

A  $K_1$  process terminates when:

$$T_1 > \Theta_{\text{death}}, \quad k_1 \rightarrow 0, \quad \text{or} \quad \partial\Omega(K_1) \rightarrow \emptyset.$$

Because  $K_1$  has no redundancy of cycles (unlike  $K_2$  and above), collapse is typically global.

### O.6.29 Summary

Processes on  $K_1$  introduce:

- temporal evolution,
- field-like dynamics,
- flows and gradients,
- variational behaviour,
- threshold-induced transitions,
- and the first emergence of physical-like laws.

$K_1$  is the first level where continuum physics appears, and all higher levels inherit its dynamical schema.

### O.6.30 Processes on $K_2$

Processes on  $K_2$  describe the first genuinely spatial and topological dynamics in the Ontology of Continua. While  $K_1$  supports temporal evolution of a single-axis field,  $K_2$  introduces:

- a network-like state space  $\Omega(K_2)$ ,
- a topology of connectivity and clusters,
- percolation processes with thresholds,
- causal propagation,
- multi-axis interactions,
- and emergent geometrical structure.

Thus, processes at  $K_2$  correspond to the birth of spatial organisation and the first form of physical locality.

### O.6.31 Structure of a $K_2$ Process

A process on  $K_2$  is an evolution of:

$$G(t) = (V, E(t)),$$

where  $V$  is the inherited set of nodes (states at  $K_1$ ), and  $E(t)$  the time-dependent set of edges representing interactions, adjacency, or causal relations.

Each edge  $e_{ij} \in E(t)$  has:

$$p_{ij}(t) \in [0, 1], \quad J_{ij}(t) \in \mathbb{R}, \quad \omega_{ij}(t) \in \mathcal{W},$$

representing:

- probability of connectivity,
- flow along the link,
- weight or metric contribution.

The evolution of  $E(t)$  follows:

$$\partial_t p_{ij} = \Phi_2(p_{ij}, J_{ij}, T_2, \Theta_2),$$

which generalises  $K_1$  flows to relational structure.

### O.6.32 Percolation Processes

The hallmark of  $K_2$  is the percolation process.

Define the mean connectivity:

$$p = \frac{1}{|E_{\max}|} \sum_{i < j} p_{ij}.$$

There exists a critical threshold  $p_c$  such that:

$$\begin{cases} p < p_c & \Rightarrow \text{only small clusters,} \\ p = p_c & \Rightarrow \text{critical state,} \\ p > p_c & \Rightarrow \text{giant connected component.} \end{cases}$$

A  $K_2$  percolation process is any evolution where:

$$\partial_t p \neq 0,$$

or, more generally, when cluster connectivity undergoes transitions.

Near  $p_c$ , processes show:

- critical slowing down of  $\tau$ ,
- emergence of long-range correlations,
- divergence of cluster size variance,
- onset of spatial dimensionality.

This is the fundamental mechanism underpinning the birth of “space”. Spatial axes  $A_2$  emerge precisely when percolation produces stable pathways across the graph.

### O.6.33 Causal Processes

Once percolation produces extended clusters, flows  $J_{ij}(t)$  propagate across them, giving rise to causal dynamics.

A causal process on  $K_2$  is defined by:

$$\partial_t \phi_i = \sum_{j \in \mathcal{N}(i,t)} J_{ij}(t),$$

where  $\mathcal{N}(i, t)$  is the dynamic neighbourhood induced by  $E(t)$ .

Causality is not fundamental but emergent: it arises when flows become constrained by stable connectivity and finite transmission time.

### O.6.34 Metric-Emergence Processes

Edges carry weights  $\omega_{ij}(t)$  that define a proto-metric:

$$d(i, j) = \min_{\text{paths } P} \sum_{(k\ell) \in P} \omega_{k\ell}.$$

A process is metric-forming if:

$$\partial_t d(i, j) \neq 0.$$

Stable metrics (constant or stationary  $d$ ) correspond to emergence of geometric structure and spatial axes.

This is the first point in the OC hierarchy where geometry becomes a process in itself.

### O.6.35 Topological Processes

Processes that change the qualitative structure of  $\Omega(K_2)$  include:

- edge formation / deletion,
- cluster merging / splitting,
- loop creation,
- change of homology groups,
- emergence of percolating structures,
- collapse of connectivity.

Formally, a topological transition occurs when:

$$\pi_n(G(t^-)) \not\cong \pi_n(G(t^+)).$$

Such transitions often correspond to threshold-crossing events:

$$T_2 = \Theta_2^{\text{top}},$$

where structural tension reaches the critical value associated with topological instability.

### O.6.36 Processes Driven by Operators

The operators act on  $K_2$  as follows:

**Generative Operator  $\Psi_2$ .**

$\Psi_2$  : create or modify edges, clusters, or weights.

This operator generates:

- new connectivity,
- new metric contributions,
- new interaction channels.

**Evolution Operator  $\Phi_2$ .**

$$\partial_t p_{ij} = \Phi_2(p_{ij}, J_{ij}, T_2, \Theta_2).$$

This is the canonical form of  $K_2$  evolution.

**Compositional Operator  $\Lambda_2$ .** Merges clusters, composes subnetworks, or lifts patterns from  $K_1$  into  $K_2$  relational structure.

**Universal Operator  $U_2$ .** Updates continuumness:

$$k_2(t + dt) = U_2(k_2, G(t), J_2, T_2, \Theta_2).$$

**Constraint Operator  $X_2$ .** Enforces boundary constraints on:

- allowable edges,
- maximum degree,
- metric bounds,
- interaction locality.

**O.6.37 Structural Tension and Thresholds**

Structural tension for a graph-like  $K_2$  is:

$$T_2 = f(p_{ij}, J_{ij}, \omega_{ij}, \text{cluster structure}).$$

Critical thresholds include:

- $\Theta_2^{\text{perc}}$  — percolation threshold,
- $\Theta_2^{\text{top}}$  — topology-changing threshold,
- $\Theta_2^{\text{metric}}$  — metric-instability threshold,
- $\Theta_2^{\text{death}}$  — collapse of connectivity.

Crossing thresholds induces discrete changes in the geometry or topology of the continuum.

### O.6.38 Cycles on $K_2$

Cycles  $C(K_2)$  correspond to:

- loop flows,
- circulation in connected components,
- recurrence patterns in connectivity,
- and re-emerging metric configurations.

A cycle process satisfies:

$$G(t + T) \simeq G(t),$$

where  $T$  is the cycle period.

Cycles stabilise  $k_2$  and define early “laws” akin to conservation through recurrence.

### O.6.39 Emergence of $K_3$ Processes

Processes on  $K_2$  generate  $K_3$  when chemical-like structure emerges:

$$\exists \text{ stable subgraphs } H \subseteq G \text{ with } \Phi_2(H) \approx 0.$$

Stable, bounded, non-trivial relational clusters behave as molecules.

The transition operator:

$$\Psi_{2 \rightarrow 3} : K_2 \rightarrow K_3$$

activates when:

$$T_2 > \Theta_2^{\text{chem}},$$

producing chemically interpretable binding relations.

### O.6.40 Collapse and Death of $K_2$

A  $K_2$  continuum collapses when:

- $p \rightarrow 0$  (loss of connectivity),
- cluster sizes remain  $O(1)$  for all time,
- metric degenerates ( $d \rightarrow \infty$  or 0 everywhere),
- structural tension exceeds  $\Theta_2^{\text{death}}$ ,
- cycles collapse and  $k_2 \rightarrow 0$ .

In collapse,  $K_2$  decays into multiple disconnected  $K_1$ -like subcontinua.

### O.6.41 Summary

Processes on  $K_2$  introduce:

- percolation and emergence of space,
- causal propagation constrained by connectivity,
- proto-metric formation,
- topological transitions,

- multi-axis interactions,
- stable loops and early “laws”,
- and the generative pathway to  $K_3$ .

$K_2$  thus forms the bridge between purely dynamical continua ( $K_1$ ) and materially structured continua ( $K_3$ ).

#### O.6.42 Processes on $K_3$

Processes on  $K_3$  describe the first materially-structured dynamics in the hierarchy of continua. While  $K_2$  provides a relational and topological substrate,  $K_3$  introduces:

- chemically meaningful state configurations,
- binding and dissociation processes,
- activation barriers and energy thresholds,
- reaction pathways and fluxes,
- stable and metastable molecular structures,
- and the first robust “objects” with internal organisation.

Thus,  $K_3$  is the level where physical connectivity becomes chemical structure and reactive dynamics.

#### O.6.43 Structure of a $K_3$ Process

The state space  $\Omega(K_3)$  contains configurations:

$$X = (\{A_i\}, \{\text{bonds}_{ij}\}, \{\rho(\mathbf{r})\}, P_{\text{chem}}),$$

where:

- $A_i$  are atomic-like units inherited from  $K_2$  clusters,
- $\text{bonds}_{ij}$  encode binding potentials,
- $\rho(\mathbf{r})$  is the electronic or charge-density analogue,
- $P_{\text{chem}}$  is the chemical potential landscape.

A process on  $K_3$  is the evolution:

$$\partial_t X = \Phi_3(X, J_3, T_3, \Theta_3),$$

where the operator  $\Phi_3$  encodes kinetics, energy landscapes, threshold conditions, and topological transformations of molecular structure.

#### O.6.44 Binding and Dissociation Processes

The defining feature of  $K_3$  is the existence of chemically meaningful binding energies  $E_{\text{bond}}$ .

A bond between units  $A_i$  and  $A_j$  exists if:

$$E < \Theta_{\text{bond}} = E_{\text{bond}},$$

where  $E$  is the local energy.

A *binding process* is any trajectory where:

$$E(t) \downarrow E_{\text{bond}} \Rightarrow \text{bond}_{ij}(t + dt) = 1.$$

A *dissociation process* occurs when:

$$E(t) > E_{\text{bond}} \quad \Rightarrow \quad \text{bond}_{ij}(t + dt) = 0.$$

Dissociation is the canonical form of “death” for a local subsystem of  $K_3$ ; binding is a local birth event.

These transitions constitute the simplest non-trivial topological updates of molecular graphs.

### O.6.45 Activation Processes and Energy Barriers

Many transformations in  $K_3$  require surpassing an activation energy:

$$E_{\text{act}}.$$

The canonical activated process:

$$\text{State } X \rightarrow X' \quad \text{requires} \quad E \geq E_{\text{act}}.$$

The rate is typically:

$$J_{\text{act}} = J_0 \exp \left[ -\frac{E_{\text{act}} - E}{k_B T} \right],$$

expressed symbolically in OC notation as:

$$J_3 = f(P_{\text{chem}}, T_3, \Theta_3^{\text{act}}).$$

Activation processes enable:

- isomerisations,
- bond rearrangements,
- energy-driven structural transitions,
- catalytic pathways (via reduced  $\Theta_3^{\text{act}}$ ).

### O.6.46 Reaction Pathway Processes

A reaction pathway is a sequence:

$$X_0 \xrightarrow{J_3} X_1 \xrightarrow{J_3} \dots \xrightarrow{J_3} X_n,$$

with intermediates and transition states lying in  $\Omega(K_3)$ .

Reaction dynamics follow:

$$\partial_t P(X) = \sum_{X'} \left( J_{X' \rightarrow X} P(X') - J_{X \rightarrow X'} P(X) \right).$$

A  $K_3$  reaction process is characterised by:

- fluxes  $J_3$  along reaction channels,
- transitions over energy landscapes,
- threshold-controlled jumps at saddle configurations,
- and stable minima representing molecules.

This is the minimal structure necessary for chemical kinetics.



### O.6.47 Processes of Potential Landscape Deformation

Chemical potentials  $P_{\text{chem}}$  evolve under external conditions (temperature, pressure, pH-like parameters, electromagnetic fields).

A deformation process is defined by:

$$\partial_t P_{\text{chem}} \neq 0.$$

Such processes change:

- depths of minima (stability of molecules),
- heights of barriers (reaction rates),
- accessible channels (topology of  $\Omega(K_3)$ ),
- positions of transition states.

They are the origin of context-dependence in chemical reactivity.

### O.6.48 Processes of Concentration and Diffusive Motion

At  $K_3$  one may define concentration-like coordinates  $c_i(t)$  for species or clusters.

A concentration process satisfies:

$$\partial_t c_i = D_i \nabla^2 c_i - \sum_j J_{ij}(c),$$

with diffusion coefficients  $D_i$  inherited from the connectivity structure of  $K_2$ .

These processes encode:

- spatial mixing,
- reaction-diffusion waves,
- gradients that later drive  $K_4$  metabolism.

### O.6.49 Processes Driven by Operators

**Generative Operator  $\Psi_3$ .** Creates or removes bonds, introduces new atomic clusters, or modifies potential wells:

$$\Psi_3 : \Omega(K_3) \rightarrow \Omega(K_3).$$

**Evolution Operator  $\Phi_3$ .** Governs chemical kinetics:

$$\partial_t X = \Phi_3(\{E\}, P_{\text{chem}}, J_3, \Theta_3).$$

**Compositional Operator  $\Lambda_3$ .** Combines substructures into larger molecules or fragments:

$$\Lambda_3(H_1, H_2) = H_1 \cup H_2 \quad \text{if } E_{\text{bond}}^{\text{new}} < \Theta_3.$$

**Universal Operator  $U_3$ .** Updates continuumness:

$$k_3(t + dt) = U_3(k_3, J_3, \Omega(K_3), \Theta_3).$$

**Constraint Operator  $X_3$ .** Imposes:

- valence limits,
- geometric constraints,
- conservation rules,
- bounding of energy surfaces.

### O.6.50 Threshold Processes in $K_3$

Key thresholds of  $K_3$  include:

- $\Theta_3^{\text{bond}}$  — bond formation threshold,
- $\Theta_3^{\text{diss}}$  — dissociation threshold,
- $\Theta_3^{\text{act}}$  — activation barrier,
- $\Theta_3^{\text{stab}}$  — stability threshold for molecules,
- $\Theta_3^{\text{chem}}$  — transition to  $K_4$  metabolic structure.

Crossing these thresholds induces discrete structural transformations in the chemical graph.

### O.6.51 Cycles on $K_3$

Cycles  $C(K_3)$  include:

- catalytic cycles,
- closed reaction loops,
- oscillatory chemical networks,
- periodic bond rearrangements,
- relaxation oscillations.

A cycle process satisfies:

$$X(t + T) \simeq X(t),$$

where  $T$  is determined by the interplay of  $J_3$ , barrier heights, and dissipation.

Cycles stabilise  $k_3$  and mark the emergence of chemical “laws” analogous to periodicity and conservation.

### O.6.52 Emergence of $K_4$ Processes

A  $K_3$  process generates  $K_4$  when reaction networks become autocatalytic and produce membranes or boundary-like structures.

Formally, a transition occurs when:

$$\exists H \subseteq \Omega(K_3) \text{ such that } \Phi_3(H) \approx 0 \text{ and } J_3(H) > J_{\text{crit}}.$$

This marks the formation of:

- stable autocatalytic sets,
- metabolic-like fluxes,
- boundary-forming amphiphilic molecules.

The operator  $\Psi_{3 \rightarrow 4}$  activates once:

$$T_3 > \Theta_3^{\text{metabolic}},$$

producing structures that behave as protocellular precursors.

### O.6.53 Collapse and Death of $K_3$

A  $K_3$  continuum collapses when:

- all bonds dissociate ( $\text{bond}_{ij} \rightarrow 0$ ),
- no stable minima remain in  $P_{\text{chem}}$ ,
- activation barriers prevent reaction fluxes ( $J_3 \rightarrow 0$ ),
- energy surfaces flatten to noise,
- structural tension  $T_3$  exceeds  $\Theta_3^{\text{death}}$ .

In collapse,  $K_3$  degenerates to disconnected reactive fragments, behaving like  $K_2$  clusters without chemical identity.

### O.6.54 Summary

Processes on  $K_3$  include:

- formation and breaking of chemical bonds,
- activation-driven transitions,
- kinetic reaction pathways,
- diffusion and concentration dynamics,
- catalytic and oscillatory cycles,
- potential landscape deformation,
- and the emergence of metabolic structure (precursor to  $K_4$ ).

$K_3$  thus bridges the relational topology of  $K_2$  and the biological continuity of  $K_4$ .

### O.6.55 Processes on $K_4$

Processes on  $K_4$  describe the dynamics of protocellular continua: systems in which chemical networks, gradients, and boundary structures form a unified and self-maintaining whole.  $K_4$  marks the transition from chemistry ( $K_3$ ) to early biology, where membranes, flux balance, and metabolic closure become the dominant organising principles.

A process on  $K_4$  is an evolution of the tuple:

$$X_4 = \begin{pmatrix} \Omega(K_4), \partial\Omega(K_4), A_{\text{mem}}, A_{\text{grad}}, A_{\text{flux}}, \\ P_{\text{mem}}, P_{\text{chem}}, P_{\text{ion}}, J_{\text{met}}, J_{\text{grad}}, \\ J_{\text{ion}}, J_{\text{leak}}, C_{\text{met}}, C_{\text{buffer}}, C_{\text{pump}}, \\ \Theta_{\text{mem}}, \Theta_{\text{grad}}, \Theta_{\text{cycle}} \end{pmatrix},$$

updated by the evolution operator  $\Phi_4$ :

$$\partial_t X_4 = \Phi_4(X_4, J_4, T_4, \Theta_4).$$

### O.6.56 Membrane Formation, Deformation and Maintenance

The membrane is the defining structural feature of  $K_4$ . A membrane process is governed by:

$$\partial_t \partial\Omega(K_4) = f(\gamma, \kappa, \Delta P, J_{\text{lipid}}, T_4),$$

where:

- $\gamma$  is surface tension,
- $\kappa$  is bending rigidity,
- $\Delta P$  is osmotic or hydrostatic pressure difference,
- $J_{\text{lipid}}$  is lipid flux across or along the membrane,
- $T_4$  is structural tension.

Membrane processes include:

- vesicle growth and shrinkage,
- budding, fusion and fission,
- curvature-driven deformation,
- flickering fluctuations (near-critical  $\kappa$  regime),
- thickness and permeability changes.

A membrane is stable iff:

$$T_4 < \Theta_{\text{mem}} \quad \text{and} \quad |\Delta P| < \Theta_{\text{osm}}.$$

### O.6.57 Osmotic, Ionic and Chemical Gradient Processes

Gradients are the main driving potentials of  $K_4$ . We define concentration, pH, redox and ionic axes:

$$A_{\text{grad}} = \{[S_i], \Delta\text{pH}, \Delta V, \Delta\text{redox}\}.$$

A gradient-evolution process satisfies:

$$\partial_t P_{\text{grad}} = J_{\text{pump}} - J_{\text{leak}} + J_{\text{chem}}.$$

There are three principal classes:

#### (1) Osmotic Gradient Processes

$$J_{\text{osm}} = f(\Delta[S], \Theta_{\text{osm}}, P_{\text{mem}}).$$

These regulate swelling, shrinking and mechanical stress on the boundary.

#### (2) Ionic Gradient Processes

$$J_{\text{ion}} = g(\Delta V, g_{\text{channel}}, \Theta_{\text{ion}}).$$

Channels, pores, and non-specific leakage govern  $\Delta V$ .

#### (3) Chemical / pH / Redox Gradient Processes

$$\partial_t(\Delta\text{pH}) = J_{\text{acid/base}} - J_{\text{buffer}},$$

$$\partial_t(\Delta\text{redox}) = J_{\text{redox}}^{\text{met}} - J_{\text{redox}}^{\text{env}}.$$

Gradients drive metabolism and structure; collapse of gradients often triggers death of  $K_4$ .

### O.6.58 Metabolic and Autocatalytic Processes

A key innovation of  $K_4$  is the formation of metabolic cycles  $C_{\text{met}}$ , which include:

- substrate uptake,
- internal transformation,
- energy production,
- waste generation and export.

Let  $X_i$  be metabolite concentrations. A general metabolic process is:

$$\partial_t X_i = \sum_j J_{j \rightarrow i}^{\text{react}} - \sum_k J_{i \rightarrow k}^{\text{react}} + J_{\text{import}} - J_{\text{export}}.$$

Autocatalytic closure occurs when:

$$\exists C \subset \Omega(K_4) \quad \text{such that} \quad \Lambda_4(C) = C \quad \text{and} \quad J_{\text{met}}(C) > 0.$$

This condition marks the completion of a RAF-like set: Reflexively Autocatalytic and F-generated.

### O.6.59 Waste–Pressure–Tension Loop Processes

Waste accumulation is a uniquely destabilising  $K_4$  process.

Define:

$$W(t) = \text{total waste concentration.}$$

Waste dynamics:

$$\partial_t W = J_{\text{waste,prod}} - J_{\text{waste,export}}.$$

Pressure response:

$$\Delta P = f(W, \Theta_{\text{perm}}, \Theta_{\text{osm}}).$$

Tension increase:

$$T_4 = T_4^0 + \alpha \Delta P.$$

If:

$$T_4 > \Theta_{\text{mem}},$$

the membrane undergoes rupture or uncontrolled leak — a canonical death mode for  $K_4$ .

### O.6.60 Processes of Permeability, Leakage, and Ion Balance

Membrane permeability defines the stability window for  $K_4$ .

Leak flux:

$$J_{\text{leak}} = g_{\text{leak}}(P_{\text{mem}}, \Delta V, \Delta[S]).$$

Channels (primitive pores) contribute:

$$J_{\text{channel}} = g_{\text{channel}}(\Delta V, \Theta_{\text{channel}}).$$

High leak destroys gradients and collapses  $k_4$ .

### O.6.61 Pump-Driven Processes and Proto-Energy Handling

Primitive pumps (chemical or electrochemical) update gradients:

$$J_{\text{pump}} = f(E_{\text{met}}, \Delta G, \Theta_{\text{pump}}).$$

Their existence is a defining step toward  $K_5$  excitability.

Energy cycles include:

$$C_{\text{energy}} : E_{\text{nutrient}} \rightarrow E_{\text{usable}} \rightarrow E_{\text{gradient}}.$$

These processes regulate:

- maintenance of  $\Delta V$ ,
- maintenance of pH and redox gradients,
- activation of metabolic pathways,
- proto-excitability transitions.

### O.6.62 Processes of Boundary Excitability (Proto-AP)

The LUX Biology run identified the ignition condition for proto-action potentials (proto-AP), a key transition to early  $K_5$ .

Define excitability variable:

$$E_{\text{exc}}(t).$$

Ignition condition:

$$E_{\text{exc}} > \Theta_{\text{exc}} \Rightarrow \text{activation of a propagating boundary front.}$$

Front propagation:

$$v_{\text{front}} = f(g_{\text{channel}}, \Delta V, \Theta_{\text{curv}}, \Theta_{\text{perm}}, J_{\text{ion}}).$$

A proto-AP process is a boundary-localised transient that:

- couples membrane curvature,
- couples ion flux,
- propagates laterally along the membrane,
- increases local metabolic demand.

This is the precursor of spiking dynamics at  $K_5$ .

### O.6.63 Processes of Buffering and Homeostatic Regulation

Buffers regulate pH, redox balance, and metabolite concentration.

A buffering process satisfies:

$$\partial_t X_i = -k_{\text{bind}} X_i + k_{\text{release}} B_i,$$

with buffer states  $B_i$ .

Homeostatic regulation emerges when:

$$\partial_t P_{\text{grad}} \approx 0 \quad \text{while} \quad J_{\text{met}} > 0.$$

Such plateaus define the long-living operational zone of  $K_4$ .

#### O.6.64 Processes of Spatial Compartmentalisation

$K_4$  is the first level capable of producing internal compartments.

Compartment formation requires:

$$T_4 < \Theta_{\text{curv}} \quad \text{and} \quad J_{\text{lipid}} > J_{\text{crit}}.$$

Sub-compartments modify:

- reaction rates,
- metabolic flux patterns,
- spatial separation of incompatible reactions.

This is a precursor to organelle-like structure at  $K_5/K_6$ .

#### O.6.65 Processes Driving $K_4 \rightarrow K_5$ Transition

The following conditions collectively mark the emergence of  $K_5$ :

1. Persistent electrical gradients ( $\Delta V$ ) maintained by pumps.
2. Appearance of ignition-capable excitability variable  $E_{\text{exc}}$ .
3. Formation of proto-AP fronts.
4. Increase of information-carrying boundary modes.
5. Local spiking cycles coupled to metabolic cycles.

When:

$$\Delta V > \Theta_{\text{exc}} \quad \text{and} \quad C_{\text{met}} \text{ supports } C_{\text{spike}},$$

a new axis  $A_{\text{exc}}$  emerges, giving birth to  $K_5$ .

#### O.6.66 Collapse and Death of $K_4$

A  $K_4$  continuum collapses when at least one of the following holds:

- **Membrane rupture:**

$$T_4 > \Theta_{\text{mem}}.$$

- **Osmotic explosion or implosion:**

$$|\Delta P| > \Theta_{\text{osm}}.$$

- **Gradient collapse:**

$$\Delta V, \Delta \text{pH}, \Delta \text{redox} \rightarrow 0.$$

- **Metabolic failure:**

$$J_{\text{met}} \rightarrow 0.$$

- **Uncontrolled permeability:**

$$J_{\text{leak}} \gg J_{\text{pump}}.$$

Collapse removes boundary integrity and reduces the system to  $K_3$  chemical fragments.

### O.6.67 Summary

Processes on  $K_4$  comprise:

- membrane formation, deformation, stability and rupture,
- osmotic, ionic and chemical gradient dynamics,
- metabolic and autocatalytic flux cycles,
- waste–pressure–tension instability loops,
- permeability and leak processes,
- pump-driven proto-energy handling,
- proto-excitability and boundary front propagation,
- buffering and homeostasis,
- spatial compartmentalisation,
- and transitions toward full excitability at  $K_5$ .

$K_4$  is therefore the minimal level of biological organisation: a self-maintaining protocell-like continuum capable of gradients, cycles, and boundary-mediated information flow.

### O.6.68 Processes on $K_5$

$K_5$  is the first level at which excitability, signalling, and propagating activation modes become stable components of the continuum. While  $K_4$  is defined by membrane–gradient–metabolism closure,  $K_5$  introduces a new structural axis:

$$A_{\text{exc}} = \{E_{\text{exc}}, \Delta V, g_{\text{channel}}, C_{\text{spike}}\},$$

supporting fast, nonlinear, threshold-driven processes. A  $K_5$  process is an evolution of:

$$X_5 = \begin{pmatrix} \Omega(K_5), \partial\Omega(K_5), A_{\text{exc}}, A_{\text{grad}}, A_{\text{ion}}, \\ P_{\text{exc}}, P_{\text{ion}}, P_{\text{chem}}, J_{\text{ion}}, J_{\text{pump}}, \\ J_{\text{leak}}, J_{\text{met}}, C_{\text{spike}}, C_{\text{recovery}}, C_{\text{energy}}, \\ \Theta_{\text{exc}}, \Theta_{\text{ion}}, \Theta_{\text{perm}} \end{pmatrix},$$

evolved by the operator  $\Phi_5$ :

$$\partial_t X_5 = \Phi_5(X_5, J_5, T_5, \Theta_5).$$

### O.6.69 Excitability and Ignition Processes

Excitability is the defining new capacity of  $K_5$ .

The excitability variable obeys:

$$\partial_t E_{\text{exc}} = f(\Delta V, P_{\text{ion}}, g_{\text{channel}}, C_{\text{energy}}) - h(E_{\text{exc}}),$$

where  $h$  encodes refractoriness and recovery.

Ignition occurs when:

$$E_{\text{exc}} > \Theta_{\text{exc}}.$$

This triggers a rapid influx/outflux of ions:

$$J_{\text{ion}}^{\text{exc}} = g_{\text{channel}}(\Delta V - \Delta V_{\text{rev}}),$$

producing a local spike.

Spike are not yet “neuronal”; they are generic, membrane-bound activation events coupling:



- ion flows,
- curvature changes,
- metabolic demand,
- mechanical tension.

### O.6.70 Action-Potential-Like Front Processes

When ignition propagates across  $\partial\Omega(K_5)$ , a proto-action-potential front forms.

Front velocity:

$$v_{\text{front}} = F(g_{\text{channel}}, \Delta V, \Theta_{\text{curv}}, \Theta_{\text{perm}}, J_{\text{ion}}, J_{\text{pump}}).$$

A propagating front exists iff:

$$g_{\text{channel}} > g_{\text{crit}} \quad \text{and} \quad \Delta V > \Theta_{\text{exc}}.$$

Propagation is stabilised by:

- high membrane integrity ( $\Theta_{\text{mem}}$  not exceeded),
- limited leakage ( $J_{\text{leak}} < J_{\text{pump}}$ ),
- local metabolic support,
- adequate channel density and distribution.

A front process is a self-sustaining cycle:

$$C_{\text{spike}} : \text{rest} \rightarrow \text{depolarisation} \rightarrow \text{peak} \rightarrow \text{repolarisation} \rightarrow \text{recovery}.$$

### O.6.71 Ion-Flux and Channel-Gating Processes

Ion fluxes now follow nonlinear gating laws. Let  $m(t)$  be a channel activation variable:

$$\partial_t m = \alpha(\Delta V)(1 - m) - \beta(\Delta V)m.$$

Ion current:

$$J_{\text{ion}} = g_{\text{max}} m^p (\Delta V - \Delta V_{\text{rev}}).$$

This introduces:

- channel state transitions,
- voltage-dependent dynamics,
- metastable gating regimes,
- fast-slow variable splits.

These processes allow  $K_5$  to support signal initiation, amplification and suppression.

### O.6.72 Recovery, Refractory and Reset Processes

After each spike, the system undergoes recovery governed by:

$$\partial_t R = r_1(E_{\text{exc}}) - r_2 R,$$

where  $R$  is a recovery variable.

Refractoriness arises when:

$$R < \Theta_{\text{ref}}.$$

A refractory process ensures unidirectional front propagation, prevents re-ignition and stabilises oscillatory cycles.

### O.6.73 Pump-Driven Restoration Processes

To sustain excitability, pumps restore gradients depleted during spikes.

Pump flux:

$$J_{\text{pump}} = f(E_{\text{ATP}}, \Delta G, \Theta_{\text{pump}}).$$

Pumps compensate:

$$J_{\text{ion}}^{\text{exc}} + J_{\text{leak}} \longrightarrow \text{restored } \Delta V.$$

A pump-deficient state collapses excitability:

$$\partial_t(\Delta V) \rightarrow 0 \quad \Rightarrow \quad E_{\text{exc}} \rightarrow 0.$$

### O.6.74 Energy Cycle Processes Supporting Excitability

$K_5$  couples energy and signalling tightly.

Energy cycle:

$$C_{\text{energy}} : \quad \text{nutrient} \rightarrow \text{ATP-like potential} \rightarrow \text{pump flux} \rightarrow \Delta V.$$

The cycle's stability determines:

- firing frequency,
- refractory period,
- maximum signal length,
- tolerance to noise.

Breakdown of  $C_{\text{energy}}$  leads to:

$$\text{spike failure} \quad \text{and eventually} \quad K_5 \rightarrow K_4.$$

### O.6.75 Oscillatory and Pattern-Formation Processes

$K_5$  supports autonomous oscillations. Let  $x$  be an excitability–chemical coupled variable.

Generic oscillatory system:

$$\partial_t x = F(x, E_{\text{exc}}, \Delta V),$$

$$\partial_t E_{\text{exc}} = G(x, \Delta V),$$

with nullcline geometry producing limit cycles.

Pattern formation emerges when:

$$D_{\text{ion}} \nabla^2(\Delta V) + F(E_{\text{exc}}) = 0,$$

yielding:

- traveling waves,
- standing patterns,
- spiral waves (depending on curvature and leakage),
- multi-front interactions.

These processes generalise beyond biology (chemical, mechanical waves).

### O.6.76 Compartmental and Network-Coupling Processes

$K_5$  supports coupling of multiple excitability domains.

Coupling flux:

$$J_{\text{couple}} = g_{\text{junction}}(\Delta V_1 - \Delta V_2).$$

When many compartments interact, the system supports:

- synchronisation,
- phase locking,
- wave interference,
- distributed signalling.

This is the precursor of neural networks at  $K_6$ .

### O.6.77 Noise-Driven and Stochastic Activation Processes

Thermal/chemical noise may trigger:

$$E_{\text{exc}} + \eta(t) > \Theta_{\text{exc}}.$$

Noise produces:

- spontaneous spikes,
- subthreshold oscillations,
- channel flicker,
- stochastic resonance enhancing weak signals.

$K_5$  therefore acts as a sensitive excitable medium.

### O.6.78 Processes of Information-Carrying Capacity

Because spikes carry distinguishable temporal and spatial signatures,  $K_5$  supports primitive information dynamics.

Information process:

$$I(t) = \mathcal{F}(C_{\text{spike}}, v_{\text{front}}, \Delta V(t)),$$

where  $\mathcal{F}$  maps spike patterns to informational states.

This is the first level at which the continuum's state can encode and transmit structured information beyond chemical gradients.

### O.6.79 Processes Driving the Transition $K_5 \rightarrow K_6$

A continuum transitions to  $K_6$  when:

1. stable networks of excitability domains form,
2. excitability couples to internal state variables (proto-neural),
3. information transmission becomes multi-step and referential,
4. temporal integration processes arise,
5. an emergent internal model begins to form.

Formally, a new axis appears:

$$A_{\text{cog}} = \{\text{patterns of spikes, state integration, } J_{\text{info}}\},$$

marking the birth of cognition.

### O.6.80 Collapse and Death of $K_5$

$K_5$  collapses if any of the following conditions hold:

#### 1. Gradient loss

$$\Delta V \rightarrow 0 \quad \Rightarrow \quad E_{\text{exc}} \rightarrow 0.$$

#### 2. Pump failure

$$C_{\text{energy}} \text{ collapses.}$$

#### 3. Excessive leakage

$$J_{\text{leak}} \gg J_{\text{pump}}.$$

#### 4. Ion imbalance

$$|P_{\text{ion}} - P_{\text{ion}}^{\text{stable}}| > \Theta_{\text{ion}}.$$

**5. Structural failure of  $\partial\Omega$**  When membrane tension exceeds  $\Theta_{\text{mem}}$ , the entire excitability layer is destroyed.

The system reverts to  $K_4$  or fragments to  $K_3$  chemical subsystems.

### O.6.81 Summary

Processes on  $K_5$  include:

- excitability and ignition,
- spike and wavefront propagation,
- channel gating and ion-flux control,
- refractory and recovery dynamics,
- pump-driven gradient restoration,
- metabolic–electrical coupling,
- oscillations and pattern formation,
- stochastic activation,
- information-carrying spike dynamics,
- compartment coupling and emergent networks,
- transitions to early cognition ( $K_6$ ).

$K_5$  is therefore the first truly information-bearing excitable continuum, bridging protocells and proto-neurons, chemistry and computation.

### O.6.82 Processes on $K_6$

$K_6$  is the first level at which excitability becomes cognitively structured. A  $K_6$  continuum supports stable attractors, multi-step information flows, temporal integration, pattern-based state updates, and the minimal architecture of proto-representation. The defining new axis is

$$A_{\text{cog}} = \{\text{pattern space, integration window, internal state variables, proto-model variables}\},$$

and the defining operator is the information-flow operator:

$$J_{\text{info}} : \Omega(K_6) \rightarrow \Omega(K_6).$$

Processes on  $K_6$  evolve the system

$$X_6 = (\Omega_c, A_{\text{cog}}, A_{\text{exc}}, P_{\text{cog}}, P_{\text{info}}, J_{\text{exc}}, J_{\text{info}}, C_{\text{pattern}}, C_{\text{integration}}, C_{\text{predict}}, \Theta_{\text{cog}}, \Theta_{\text{cons}}, T_6),$$

under the operator  $\Phi_6$ :

$$\partial_t X_6 = \Phi_6(X_6, J_6, T_6, \Theta_6).$$

### O.6.83 Pattern Formation and Stabilisation Processes

$K_6$  supports autonomous formation of information-bearing patterns.

Let  $S(t)$  be the instantaneous spike-pattern vector derived from  $K_5$ :

$$S(t) = \mathcal{E}(\Delta V, E_{\text{exc}}, \text{spike events}).$$

Patterns stabilise when they enter an attractor basin:

$$\partial_t S = F(S) + J_{\text{info}}(S) \Rightarrow S \rightarrow S^*,$$

where  $S^*$  is a stable pattern.

A process of pattern stabilisation includes:

- formation of recurrent motifs,
- decay of noise-driven fluctuations,
- establishment of spatio-temporal correlational structure.

Pattern stability is the first necessary condition for memory.

### O.6.84 Attractor Dynamics and State Convergence

Let  $x(t)$  be an internal cognitive variable.

State evolution:

$$\partial_t x = G(x, S(t), P_{\text{cog}}).$$

An attractor  $x^*$  satisfies:

$$G(x^*, S(t), P_{\text{cog}}) = 0.$$

Processes:

1. **Formation of attractor basins:** structure emerges from network coupling and stability of feedback loops.
2. **Convergence:** initial states collapse into low-dimensional manifolds.
3. **Cycle attractors:** recurrent thought-like loops (proto-inference).

These processes convert raw excitation patterns into stable internal states.

### O.6.85 Temporal Integration Processes

$K_6$  introduces explicit temporal integration:

$$I(t) = \int_{t-\tau}^t S(\xi) w(t - \xi) d\xi,$$

where  $\tau$  is the integration window and  $w$  is a weighting kernel.

Integration processes:

- smoothing,

- prediction,
- temporal binding,
- context construction.

Temporal integration is necessary for recognition and proto-prediction.

### O.6.86 Multi-Step Information Flow Processes

Information at  $K_6$  is not single-step (as in  $K_5$ ), but iterative:

$$S(t) \xrightarrow{J_{\text{info}}} I(t) \xrightarrow{\Psi_6} x(t) \xrightarrow{\Phi_6} S(t + \Delta t).$$

This defines:

- recurrent inference loops,
- internal updating cycles,
- formation of internal models.

The system becomes capable of representing and modifying its own state.

### O.6.87 Consistency Maintenance Processes

Internal states must satisfy a cognitive consistency threshold  $\Theta_{\text{cons}}$ .

Let  $x_i$  be a set of interacting internal variables.

Consistency metric:

$$C_{\text{cons}} = \sum_{i,j} W_{ij} |x_i - f_{ij}(x_j)|.$$

A consistency-maintenance process minimises:

$$\partial_t C_{\text{cons}} = -\gamma C_{\text{cons}},$$

unless external input forces reconfiguration.

Excessive inconsistency leads to tension spikes:

$$T_6 > \Theta_{\text{cog}} \Rightarrow \text{cognitive collapse or restructuring.}$$

### O.6.88 State-Update, Comparison, and Error-Correction Processes

Let  $x_{\text{old}}$  be the previous internal state and  $x_{\text{new}}$  the proposed updated state.

Error estimate:

$$\epsilon = D(x_{\text{new}}, x_{\text{old}}).$$

Processes include:

- comparison of predicted and observed patterns,
- error-driven adjustments,
- propagation of correction across internal variables,
- reweighting of pattern associations.

This is the formal analogue of learning in the  $K_6$  continuum.

### O.6.89 S-Cell (S-Unit) Meaning-Formation Processes

The S-module gives a universal minimal unit of meaning:

$$A \rightarrow \text{fix} \rightarrow \text{expect } B \rightarrow C \rightarrow \text{compare} \rightarrow \text{update}.$$

Processes at this level:

1. **Fixation:** selecting a pattern as a reference.
2. **Expectation:** predicting the next state.
3. **Comparison:** evaluating deviation from prediction.
4. **Update:** adjusting internal variables or thresholds.

Meaning is thus the result of dynamic alignment between prediction, observation and internal model.

### O.6.90 Information Routing and Selective Attention Processes

Routing is controlled by  $J_{\text{info}}$  acting on selected subspaces.

Let  $\Pi$  be a projection operator selecting relevant channels.

Selective routing:

$$J_{\text{info}}^{(\Pi)} = \Pi \circ J_{\text{info}}.$$

This creates processes:

- dynamic reallocation of processing resources,
- selective enhancement of patterns,
- suppression of irrelevant information,
- stabilisation of attention loops.

This emerges naturally from thresholds and attractor geometry.

### O.6.91 Stability, Conflict and Cognitive Tension Processes

Structural tension at  $K_6$ :

$$T_6 = F(\epsilon, C_{\text{cons}}, \Theta_{\text{cog}}, J_{\text{info}}, P_{\text{cog}}).$$

Processes:

- tension accumulation (prediction error, inconsistency),
- local correction (partial updates),
- global reorganisation (attractor reshaping),
- breakdown (collapse to  $K_5$  dynamics).

High  $T_6$  triggers qualitative shifts in cognitive structure.

### O.6.92 Memory Formation and Retrieval Processes

Memory is a stable attractor or pattern stored in  $\Omega(K_6)$ .

Encoding:

$$\partial_t x = \Phi_6(x, S(t)).$$

Strengthening:

$$\partial_t W_{ij} = H(S_i, S_j, x).$$

Retrieval:

$$S_{\text{cue}} \rightarrow x^* \rightarrow S_{\text{restored}}.$$

Memory is thus a dynamical process, not an object.

### O.6.93 Proto-Representation and Internal Modelling Processes

A representation occurs when:

$$x_r(t) \approx \mathcal{M}(S(t)),$$

with  $\mathcal{M}$  an internal model mapping external patterns into internal variables.

Processes:

- construction of latent variables,
- predictive modelling,
- generation of counterfactual patterns,
- simulation-like internal cycles.

This is not full symbolic reasoning, but structured proto-modelling.

### O.6.94 Decision and Action-Preparation Processes

When multiple attractors compete, decision processes select one:

$$x_{\text{selected}} = \arg \min_x E_{\text{tension}}(x).$$

Action preparation occurs when internal states generate predictive signals:

$$J_{\text{prep}} = \Psi_6(x_{\text{selected}}).$$

This is still internal; external motor action appears only at  $K_7$ .

### O.6.95 Noise-Driven Cognitive Fluctuation Processes

Noise  $\eta(t)$  acts on internal variables:

$$\partial_t x = G(x) + \eta(t).$$

Processes include:

- spontaneous reconfiguration,
- escape from shallow attractors,
- creativity-like divergence,
- noise-induced transitions between states.

Noise becomes a functional exploration tool.

### O.6.96 Transition Processes $K_6 \rightarrow K_7$

A transition occurs when:

1. multiple  $K_6$  continua synchronise,
2. information flows become inter-agent,
3. shared symbols or norms stabilise,
4. externalised patterns (signals/actions) form coupling loops.

This produces:

$$A_{\text{soc}}, J_{\text{soc}}, C_{\text{norm}}, \Theta_{\text{legit}},$$

signalling the emergence of  $K_7$ .



### O.6.97 Collapse and Death of $K_6$

$K_6$  collapses when:

1. excitability support ( $K_5$  substrate) fails,
2. attractors destabilise catastrophically,
3. inconsistency exceeds  $\Theta_{\text{cog}}$ ,
4. information routing breaks down,
5. tension  $T_6$  diverges.

The result is reversion to  $K_5$  dynamics or fragmentation into disconnected subsystems.

### O.6.98 Summary

Processes at  $K_6$  include:

- pattern stabilisation and attractor formation,
- temporal integration,
- multi-step inference loops,
- cognitive consistency maintenance,
- prediction–error correction,
- meaning-formation via S-cells,
- selective attention and routing,
- tension dynamics and reorganisation,
- memory encoding and retrieval,
- proto-representation and internal modelling,
- early decision processes,
- noise-driven exploration,
- transition dynamics to  $K_7$ .

$K_6$  is the first level supporting genuine cognition: a continuum able to integrate patterns, maintain internal models, predict, correct itself and generate meaning.

### O.6.99 Processes on $K_7$

$K_7$  is the social continuum: a structured space of norms, roles, institutions, sanctions, expectations, and collective information flows. A  $K_7$  continuum emerges when multiple  $K_6$  cognitive continua synchronise their internal variables through communication and stabilise shared patterns in a common social space  $\Omega(K_7)$ .

Processes on  $K_7$  evolve the system:

$$X_7 = (\Omega^s, A^s, P^s, J^s, C^s, \Theta^s, T_7, k_7),$$

under the social operator  $\Phi_7$ :

$$\partial_t X_7 = \Phi_7(X_7, J^s, \Theta^s, T_7).$$

We describe the major classes of processes constitutive of  $K_7$ .

### O.6.100 Norm Formation and Stabilisation Processes

Norms are stable attractors in the space of shared expectations. Let  $N(t)$  be a normative pattern represented by a distribution over expected behaviours.

Norm formation occurs when:

$$J_{\text{comm}}^s(i, j) > \Theta_{\text{sync}}^s \Rightarrow N_i(t) \approx N_j(t).$$

Processes include:

- **Convergence of expectations:** cognitive  $K_6$  models align through communication.
- **Pattern generalisation:** local rules scale to group-wide norms.
- **Stabilisation:** norms become fixed points under repeated social cycles  $C_{\text{norm}}^s$ .

Norms survive as long as:

$$T_7(N) < \Theta_{\text{norm-stability}}^s.$$

### O.6.101 Role Construction and Assignment Processes

Roles  $R$  are social axes  $A_{\text{role}}^s$  assigning differential expectations.

Processes include:

- **Role emergence:** patterns of behaviour differentiate into persistent categories.
- **Role allocation:** agents adopt roles due to competence, negotiation, or constraint.
- **Role stabilisation:** deviations are corrected by sanction cycles.

Mathematically:

$$\partial_t R_i = F(R_i, P_{\text{status}}^s, J_{\text{comm}}^s, \Theta_{\text{role}}^s).$$

### O.6.102 Institution Formation and Maintenance Processes

Institutions are higher-order attractors regulating multiple norms.

Let  $I$  be an institutional configuration.

Formation:

$$C_{\text{norm}}^s \circ C_{\text{sanction}}^s \rightarrow I.$$

Maintenance processes:

1. integration of heterogeneous norms,
2. enforcement cycles,
3. symbolic reinforcement,
4. boundary maintenance ( $\partial\Omega^s$  construction).

Stability conditions:

$$T_7(I) < \Theta_{\text{inst-collapse}}^s.$$

Institutions collapse when trust, compliance, or shared expectations fall below critical thresholds.

### O.6.103 Trust Dynamics and Social Cohesion Processes

Trust  $P_{\text{trust}}^s$  is a central potential of  $K_7$ .

Evolution:

$$\partial_t P_{\text{trust}}^s = G(P_{\text{trust}}^s, J_{\text{interaction}}^s, E_{\text{expect}}).$$

Processes:

- **Trust accumulation** through repeated successful interactions.
- **Trust erosion** under unpredictability or norm violation.
- **Trust repair** via institutional reinforcement.
- **Cohesion formation** when trust networks percolate and form a giant cluster.

The emergence of a connected trust graph is the hallmark of a unified community:

$$p_{\text{trust}} > p_c^{\text{soc}} \Rightarrow k_7 \text{ increases.}$$

### O.6.104 Sanction, Reward, and Enforcement Processes

Sanctioning is represented by a flow:

$$J_{\text{sanction}}^s : \Omega^s \rightarrow \Omega^s.$$

Processes include:

- negative feedback (punishment),
- positive feedback (reward),
- stabilising feedback (norm reinforcement),
- suppressive feedback (role regulation).

Sanction cycles ensure:

$$N(t + \Delta t) = N(t) + F_{\text{sanction}}(T_7, \Theta_{\text{norm}}^s).$$

Sanction failure is a precursor to normative collapse.

### O.6.105 Collective Attention and Information Routing Processes

Collective attention is social-scale selective routing:

$$J_{\text{focus}}^s = \Pi^s \circ J^s,$$

where  $\Pi^s$  selects relevant topics, symbols, or events.

Processes:

- focusing of the group on specific issues,
- amplification of signals,
- suppression of noise,
- coordination of group behaviour.

Collective attention enables synchronised decision making.

### O.6.106 Symbol Formation and Shared Meaning Processes

Symbols  $S_{\text{sym}}$  arise when cognitive S-cells of multiple agents converge to shared mappings:

$$A \xrightarrow{\text{fix}} B \quad \text{across agents.}$$

Processes:

1. **Cross-agent fixation:** selecting common referents.
2. **Stabilisation of meaning:** meaning becomes invariant under communication cycles.
3. **Symbolic expansion:** symbols anchor roles, norms, and institutions.

Symbol systems form the substrate for  $K_8$ .

### O.6.107 Collective Decision-Making Processes

Decisions  $D$  emerge from the aggregation of internal cognitive models into a unified social state.

Let  $x_i$  be individual cognitive states.

Collective decision rule:

$$D = \arg \min_x \sum_i W_i E_{\text{soc}}(x, x_i).$$

Processes include:

- consensus formation,
- conflict resolution,
- authority selection,
- policy emergence.

Decision processes shape the evolution of  $\Omega(K_7)$ .

### O.6.108 Conflict, Social Tension, and Reorganisation Processes

Structural social tension:

$$T_7 = F(\Theta^s, J^s, P^s, C^s).$$

Processes:

- **tension accumulation:** norm conflict, institutional overload, divergent roles, inconsistent expectations.
- **local correction:** targeted sanctions, small-scale norm shifts.
- **global reorganisation:** institutional restructuring, role redistribution.
- **collapse:** if  $T_7 > \Theta_{\text{collapse}}^s$ , the social order fragments.

High tension expands  $\partial\Omega^s$  and decreases  $k_7$ .

### O.6.109 Collective Memory and Tradition Processes

Collective memory is a stable attractor in the inter-agent pattern space.

Formation:

$$S_{\text{event}} \rightarrow S_{\text{symbol}} \rightarrow M_{\text{collective}}.$$

Processes:

- encoding of shared events,

- ritualisation and symbolic reinforcement,
- intergenerational transmission,
- condensation into narratives and traditions.

Collective memory increases stability but may inhibit adaptation.

#### O.6.110 Social Learning Processes

Learners acquire norms, symbols, and roles through:

1. imitation,
2. correction (sanction-based),
3. guided interaction,
4. narrative transmission.

Formally:

$$\partial_t x_i^{(\text{soc})} = \Phi_7(S_{\text{social}}, x_i^{(\text{cog})}, J_{\text{input}}^s).$$

Social learning increases integration of newcomers into  $K_7$ .

#### O.6.111 Network Growth, Percolation, and Connectivity Processes

Let  $G(t)$  be the social graph.

Processes:

- node addition (birth, migration),
- link formation (interaction),
- link deletion (conflict, isolation),
- cluster merging (cohesion),
- fragmentation (collapse).

Percolation threshold:

$$p_{\text{connect}} > p_c^{\text{soc}} \Rightarrow \text{unified social continuum, } k_7 > 0.$$

Fragmentation corresponds to social death:

$$\Omega(K_7) \rightarrow \emptyset.$$

#### O.6.112 Transition Processes $K_7 \rightarrow K_8$

Transition occurs when:

1. institutions become platforms for technological organisation,
2. formal roles induce economic specialisation,
3. symbol systems evolve into symbolic knowledge systems,
4. flows  $J^s$  support complex external infrastructure,
5. cycles  $C_{\text{institution}}^s$  stabilise logistics, resource flows,
6. thresholds allow scalable coordination.

This produces:

$$A^8, P^8, J^8, C^8, \Theta^8,$$

marking the rise of  $K_8$  (civilisational continuum).

### O.6.113 Collapse and Death of $K_7$

Collapse occurs when:

- trust networks fracture,
- sanction systems fail,
- institutional thresholds are exceeded,
- norms lose stability,
- tension  $T_7$  diverges.

Death of  $K_7$  yields:

$$\Omega(K_7) \rightarrow \emptyset, \quad \text{agents revert to isolated } K_6 \text{ modes.}$$

### O.6.114 Summary

Key processes at  $K_7$  include:

- norm formation and stabilisation,
- role construction,
- institutional emergence and maintenance,
- trust dynamics and cohesion,
- sanction and enforcement,
- collective attention and information routing,
- symbol formation and meaning synchronisation,
- collective decision-making,
- tension, conflict, and reorganisation,
- collective memory and tradition,
- social learning,
- network growth and percolation,
- transitions to civilisation-level dynamics at  $K_8$ .

$K_7$  is the first continuum that generates external, shared, structurally stable social reality.

### O.6.115 Processes on $K_8$

$K_8$  is the civilisational continuum. It extends the social dynamics of  $K_7$  into material infrastructures, economic cycles, technological systems, cities, markets, long-range logistics, and distributed knowledge structures.

The system is defined by:

$$X_8 = (\Omega^8, A^8, P^8, J^8, C^8, \Theta^8, T_8, k_8),$$

with operators inherited from  $K_7$  but now applied to physical and technological environments.

Processes on  $K_8$  describe how civilisations grow, stabilise, adapt, industrialise, expand their spatial domains, accumulate complexity, and undergo collapse.

### O.6.116 Infrastructure Formation and Expansion Processes

Let  $I_{\text{infra}}$  denote infrastructural states: roads, energy grids, communication systems, water supply, sewage, storage, transportation networks.

Formation:

$$\partial_t I_{\text{infra}} = F(I_{\text{infra}}, J_{\text{labor}}^8, J_{\text{resource}}^8, \Theta_{\text{build}}^8).$$

Processes include:

- construction and expansion of physical networks,
- growth of capacity and throughput,
- integration of specialised subsystems,
- maintenance and repair cycles,
- redundancy creation for resilience.

Infrastructure expands  $\Omega(K_8)$  and raises  $k_8$ .

### O.6.117 Economic Production and Resource Allocation Processes

Economic activity is described by flows:

$$J_{\text{prod}}^8, \quad J_{\text{exchange}}^8, \quad J_{\text{consumption}}^8, \quad J_{\text{resource}}^8.$$

Processes:

1. **production:** transformation of raw materials into goods,
2. **exchange:** markets, prices, trade networks,
3. **distribution:** logistics and allocation,
4. **consumption:** satisfying demand,
5. **recycling:** reintroducing materials into the cycle.

Stability requires:

$$T_8^{\text{econ}} < \Theta_{\text{market-collapse}}^8.$$

Economic collapse corresponds to a breakdown of  $J_{\text{prod}}^8$  or a divergence of resource tension.

### O.6.118 Technological Innovation and Knowledge Accumulation Processes

Let  $K_{\text{tech}}(t)$  represent the technological knowledge set.

Processes include:

- invention (creation of new techniques),
- refinement and optimisation,
- institutionalisation of expertise,
- codification into formal knowledge,
- diffusion across domains and agents.

Formal dynamics:

$$\partial_t K_{\text{tech}} = \Psi_8(K_{\text{tech}}, J_{\text{research}}^8, J_{\text{learning}}^8).$$

Technological innovation expands  $A^8$ , producing new axes of operation.

Knowledge accumulation increases:

$$k_8 \quad \text{and} \quad |\Omega(K_8)|.$$

### O.6.119 Urbanisation and Spatial Organisation Processes

Cities are dense attractors in  $\Omega(K_8)$  providing:

- high interaction rates,
- knowledge clusters,
- labour concentration,
- infrastructural efficiency,
- rapid innovation.

Urbanisation occurs when:

$$J_{\text{migration}}^8 > \Theta_{\text{urban-threshold}}^8.$$

Spatial organisation processes include:

- cluster growth,
- network densification,
- zoning and functional specialisation,
- megaregional integration,
- resource hinterlands and supply chains.

Urban failure (e.g., water system breakdown, sanitation collapse) can trigger systemic  $K_8$  destabilisation.

### O.6.120 Civilisational Memory, Archives, and Knowledge Institutions

Memory at  $K_8$  becomes externalised:

$$M_8 = S_{\text{symbols}} + S_{\text{records}} + S_{\text{institutions}}.$$

Processes:

1. creation of written, digital, architectural, and legal records,
2. formation of educational institutions,
3. intergenerational transmission of knowledge,
4. preservation and restoration cycles,
5. protection of memory during crises.

Collapse of memory institutions shrinks  $\Omega(K_8)$  and reduces  $k_8$ .

### O.6.121 Macro-Institutional Governance Processes

$K_8$  introduces large-scale decision-making through:

$$A_{\text{governance}}^8, \quad J_{\text{authority}}^8, \quad C_{\text{policy}}^8.$$

Processes:

- legislative cycles,
- executive coordination,
- judicial stabilisation,



- taxation and resource reallocation,
- crisis management,
- regulation of conflict.

Governance stabilises  $K_8$  if:

$$T_8^{\text{gov}} < \Theta_{\text{state-collapse}}^8.$$

State failure is a partial death of  $K_8$  but not necessarily collapse of the whole civilisation.

### O.6.122 Logistics, Supply Chains, and Energy Flow Processes

Civilisations depend on long-range flows:

$$J_{\text{energy}}^8, \quad J_{\text{food}}^8, \quad J_{\text{materials}}^8, \quad J_{\text{transport}}^8.$$

Processes include:

- extraction,
- conversion (e.g., fuel  $\rightarrow$  electricity),
- transportation along networks,
- distribution to demand centers,
- accumulation and storage,
- system redundancy,
- emergency rerouting.

Energy crisis corresponds to:

$$J_{\text{energy}}^8 < \Theta_{\text{min-energy}}^8.$$

Such failures propagate tension  $T_8$  through all other processes.

### O.6.123 Specialisation and Functional Differentiation Processes

Let  $S_i$  denote specialised subsystems: agriculture, metallurgy, finance, education, military, health, computation, etc.

Processes:

1. emergence of new specialisations,
2. deepening of division of labour,
3. interdependence of subsystems,
4. formation of supply, knowledge, and governance hierarchies,
5. failure propagation between sectors.

Differentiation scales  $\Omega(K_8)$  but increases fragility.

### O.6.124 Long-Range Coordination and Synchronisation Processes

Civilisational behaviour requires global patterns:

$$C_{\text{coord}}^8 : \Omega^8 \rightarrow \Omega^8.$$

Processes include:

- synchronised production cycles,
- temporal standardisation,
- global communication protocols,
- shared currencies and metrics,
- supranational governance.

Failure of synchronisation lowers  $k_8$ .

### O.6.125 Environmental Interaction and Ecological Feedback Processes

$K_8$  interacts with  $M$ -spaces (natural environment, climate, biomes).

Processes:

1. resource extraction,
2. pollution and waste cycles,
3. land-use transformation,
4. agricultural systems,
5. ecological collapse feedback,
6. climate-induced destabilisation.

Ecological thresholds:

$$\Theta_{\text{eco}}^8 : P_{\text{env-damage}}^8 < \Theta_{\text{ecological-collapse}}^8.$$

Crossing them destabilises  $K_8$ .

### O.6.126 Cultural Production, Media, and Symbolic Layer Processes

Symbols at  $K_8$  become mass-propagated through:

$$J_{\text{media}}^8, C_{\text{culture}}^8.$$

Processes:

- mass communication,
- cultural production and reinforcement,
- symbolic innovation,
- ideological propagation,
- memetic spread,
- cultural conflict and alignment.

The symbolic layer serves as the scaffolding for  $K_9$ .

### O.6.127 Crisis, Degeneration, and Collapse Processes

Civilisational collapse corresponds to:

$$\Omega(K_8) \rightarrow \emptyset, \quad k_8 \rightarrow 0.$$

Collapse drivers:

- energy breakdown,
- ecological overshoot,
- institutional decay,
- supply chain fragmentation,
- technological regress,
- cultural disintegration,
- warfare,
- pandemic-propagated failure,
- systemic tension divergence:  $T_8 > \Theta_{\text{collapse}}^8$ .

Collapse may be local (partial) or total (death of  $K_8$ ).

### O.6.128 Transition Processes $K_8 \rightarrow K_9$

Transition to  $K_9$  (the theoretical/ideational megastructure continuum) occurs when:

1. symbolic systems become formalised into explicit theories,
2. knowledge institutions stabilise abstract reasoning,
3. technological and scientific practices generate formalism,
4. meta-symbolic cycles  $C^9$  appear,
5. new axes  $A^9$  (logic, theory, abstraction) arise.

This produces:

$$(\Omega^9, A^9, P^9, J^9, C^9, \Theta^9),$$

marking the birth of  $K_9$ .

### O.6.129 Summary

Processes on  $K_8$  include:

- infrastructural growth,
- economic production cycles,
- technological innovation,
- urbanisation,
- civilisational memory formation,
- macro-institutional governance,
- supply-chain and energy dynamics,

- specialisation and coordination,
- environmental interaction and ecological feedback,
- cultural production,
- systemic crisis and collapse,
- transition toward  $K_9$ .

$K_8$  is the first continuum that creates persistent, large-scale technological environments that extend far beyond the cognitive or social scales of  $K_6$ – $K_7$ .

#### O.6.130 Processes on $K_9$

$K_9$  is the theoretical–ideational continuum. It emerges when civilisational symbolic systems ( $K_8$ ) become internally structured, recursively self-referential and capable of generating stable bodies of theory, logic, mathematics, scientific frameworks, and formal reasoning.

The formal structure of  $K_9$  is:

$$X_9 = (\Omega^9, A^9, P^9, J^9, C^9, \Theta^9, T_9, k_9),$$

with processes defined over symbolic, conceptual and epistemic causality.

Processes on  $K_9$  describe how theories form, evolve, stabilise, compete, collapse, and generate new abstract axes.

#### O.6.131 Concept Formation and Semantic Stabilisation

Let  $C_{\text{concept}}$  denote conceptual structures.

Concept formation:

$$\partial_t C_{\text{concept}} = \Psi_9(J_{\text{inference}}^9, J_{\text{abstraction}}^9, P_{\text{coherence}}^9, \Theta_{\text{semantic}}^9)$$

Processes:

- extraction of invariant patterns from symbolic material ( $K_8$ ),
- creation of semantic primitives and categories,
- stabilisation under repeated use,
- rejection of unstable or contradictory formations,
- emergence of conceptual networks.

Stable concepts increase  $|\Omega(K_9)|$  and raise  $k_9$ .

#### O.6.132 Model Building and Theorisation Processes

Let  $M_{\text{theory}}$  be the set of theoretical models.

Model-building processes include:

1. identifying variables and structures,
2. constructing internal relations,
3. formalising rules (axioms, equations, semantics),
4. validating internal coherence,
5. extending explanatory reach.

Formal dynamics:

$$\partial_t M_{\text{theory}} = \Phi_9(M_{\text{theory}}, P_{\text{rigor}}^9, J_{\text{derivation}}^9, \Theta_{\text{consistency}}^9)$$

Theories increase  $A^9$  by creating new axes of abstraction.

### O.6.133 Inference, Deduction, and Proof Processes

Inference is the primary flow at  $K_9$ :

$$J_{\text{infer}}^9, \quad J_{\text{deduce}}^9, \quad J_{\text{compute}}^9, \quad J_{\text{verify}}^9.$$

Processes:

- deductive reasoning using established rules,
- logical inference and derivation chains,
- algorithmic computation,
- proof construction and verification,
- detection of contradictions.

$K_9$  is the first continuum where inference becomes a physically realised operator on abstract space.

Failure of inference coherence pushes

$$T_9^{\text{logic}} > \Theta_{\text{inconsistency}}^9,$$

triggering collapse of the affected subtheory.

### O.6.134 Paradigm Formation and Shift Cycles

Paradigms  $\Pi$  are high-level, self-stabilising structures integrating:

$$\Pi = (A_{\Pi}^9, P_{\Pi}^9, \Theta_{\Pi}^9, C_{\Pi}^9).$$

Processes:

1. paradigm construction,
2. problem-solving cycles,
3. anomaly accumulation,
4. tension buildup,
5. paradigm shift if  $T_9 > \Theta_{\text{shift}}^9$ .

Paradigm cycles are the central  $C^9$  dynamics.

### O.6.135 Abstraction, Generalisation, and Compression Processes

$K_9$  systems compress lower-level structures ( $K_6$ – $K_8$ ) into:

- laws,
- universal relations,
- mathematical structures,
- meta-models.

Compression dynamics:

$$J_{\text{compress}}^9 : \Omega(K_{\leq 8}) \rightarrow A^9$$

Generalisation occurs when:

$$P_{\text{general}}^9 > \Theta_{\text{generalisation}}^9$$

Excessive compression can create epistemic blind spots:

$$T_9^{\text{loss}} > \Theta_{\text{semantic-retention}}^9.$$

### O.6.136 Cross-Theory Fusion and Integration Processes

Integration takes the form:

$$\Psi_{\text{fusion}}(T_1, T_2) \rightarrow T^*.$$

Processes:

- mapping between conceptual axes,
- translating models across domains,
- merging compatible rule systems,
- resolving incompatibilities,
- constructing unified frameworks.

Fusion increases the dimensionality of  $A^9$ .

Failure of integration produces incoherent hybrids that reduce  $k_9$ .

### O.6.137 Formalisation and Mathematisation

Formalisation transforms informal symbolic content into:

1. axioms,
2. definitions,
3. typed structures,
4. formal languages,
5. rigorous proof systems.

Dynamics:

$$\partial_t P_{\text{formal}}^9 = J_{\text{precision}}^9 - J_{\text{ambiguity}}^9$$

Mathematisation is a key  $K_9$  route to the birth of  $K_{10}$ .

### O.6.138 Epistemic Governance, Methodology, and Scientific Cycles

Scientific cycles  $C_{\text{science}}^9$  include:

- hypothesis formation,
- model building,
- prediction generation,
- empirical testing (via  $K_8$  subsystems),
- theory revision.

Methodological governance sets:

$$\Theta_{\text{validity}}^9, \quad \Theta_{\text{falsifiability}}^9, \quad \Theta_{\text{rigour}}^9.$$

Processes ensure stability and reliability of  $K_9$  reasoning.

### O.6.139 Internal Collapse, Degeneration, and Paradigm Death

Death of a theoretical subsystem occurs when:

$$T_9 > \Theta_{\text{collapse}}^9.$$

Causes:

- contradictions,
- inability to solve core problems,
- accumulation of anomalies,
- empirical falsification at  $K_8$ ,
- loss of internal coherence,
- breakdown of inferential flows.

Collapse shrinks  $\Omega(K_9)$  but may produce new  $\Pi$  (via  $\Psi_9$ ).

### O.6.140 Meta-Processes and Emergence of $K_{10}$

Transition to  $K_{10}$  requires:

1. recursive formalisation of theories,
2. explicit construction of meta-languages,
3. stable proof systems,
4. appearance of recursion operators,
5. identification of formal limits (Gödel, Turing),
6. differentiation of syntax–semantics axes.

The operator:

$$\Lambda_{9 \rightarrow 10} : K_9 \rightarrow K_{10}$$

marks the birth of formal recursion and the death of purely semantic theoretical forms.

$K_{10}$  begins when theory becomes self-computable.

### O.6.141 Summary

Processes on  $K_9$  include:

- concept formation and semantic stabilisation,
- theory construction and model-building,
- inference, deduction, and proof,
- paradigm cycles and shifts,
- abstraction, compression, generalisation,
- cross-theory integration,
- formalisation and mathematisation,
- epistemic governance and scientific cycles,
- collapse of degenerating theories,
- meta-level transition toward  $K_{10}$ .

$K_9$  is the first continuum whose internal processes are entirely epistemic, symbolic, and inferential, setting the stage for the recursive formal structures of  $K_{10}$ .

### O.6.142 Processes on $K_{10}$

$K_{10}$  is the formal-recursive continuum. It arises when theoretical structures of  $K_9$  become:

- fully formalised,
- axiomatically specified,
- recursively computable,
- capable of defining and analysing themselves,
- subject to limits of consistency, decidability and computability.

The structure of  $K_{10}$  is:

$$X_{10} = (\Omega^{10}, A^{10}, P^{10}, J^{10}, C^{10}, \Theta^{10}, T_{10}, k_{10}),$$

with processes acting on symbolic, formal, computational and meta-formal flows.

Processes on  $K_{10}$  govern the dynamics of axioms, rules, proofs, recursions, reductions, and meta-theoretical operators.

### O.6.143 Axiomatisation and Rule Formation

Let  $\mathcal{A}$  denote the set of axioms; let  $\mathcal{R}$  be rules.

Axiomatisation is the process:

$$\partial_t \mathcal{A} = \Psi_{10}(P_{\text{expressive}}^{10}, \Theta_{\text{consistency}}^{10}, J_{\text{formalisation}}^{10})$$

Processes:

- extraction of primitive statements,
- specification of inference rules,
- construction of formal grammars,
- elimination of ambiguity,
- stabilisation of a formal system.

The quality and expressive power of  $\mathcal{A}$  raises  $k_{10}$ .

### O.6.144 Proof Construction, Verification, and Proof-Flows

Let  $J_{\text{proof}}^{10}$  be the proof-flow.

Proof processes:

$$J_{\text{proof}}^{10} = (\text{deduction, inference, verification, reduction})$$

Dynamics:

$$\partial_t \Pi_{\text{proof}} = \Phi_{10}(J_{\text{proof}}^{10}, P_{\text{rigor}}^{10}, \Theta_{\text{proof-sound}}^{10})$$

Proof-flows generate:

- derivations,
- soundness/consistency checks,
- refutations,
- algorithmic or mechanised proofs,
- proof-minimisation and formal optimisation.

Breakdown of the proof-flow raises:

$$T_{10}^{\text{proof}} > \Theta_{\text{inconsistency}}^{10},$$

triggering collapse of that subsystem.



### O.6.145 Recursive Definition, Fixed Points, and Self-Reference

Central to  $K_{10}$  is recursion.

Let  $\mathcal{F}$  be the class of recursively defined functions.

Processes:

$$\mathcal{F}(x) = \Lambda_{10}(\mathcal{F})(x),$$

where  $\Lambda_{10}$  is the recursion operator (with fixed points).

Processes include:

- recursive definitions of functions,
- construction of fixed-point combinators,
- meta-theoretical self-reference,
- generation of Gödel coding,
- emergence of undecidability phenomena.

Self-reference increases  $A^{10}$  by enabling meta-level axes.

### O.6.146 Computability, Reduction, and Algorithmic Flows

Let  $J_{\text{compute}}^{10}$  denote computational flows.

Computability processes:

$$\partial_t \Omega_{\text{computable}}^{10} = J_{\text{compute}}^{10} - J_{\text{noncompute}}^{10}$$

Processes include:

- execution of algorithms,
- reduction to canonical forms,
- decision procedures (when they exist),
- simulation of machines,
- complexity growth and collapse.

A system collapses if placed beyond its own algorithmic capacity:

$$T_{10}^{\text{load}} > \Theta_{\text{computability}}^{10}.$$

### O.6.147 Interpretation, Translation, and Meta-Formal Flows

Interpretation processes map one formal system to another:

$$\Psi_{\text{interp}}^{10} : (\mathcal{A}, \mathcal{R})_1 \rightarrow (\mathcal{A}, \mathcal{R})_2.$$

Key processes:

- encoding and decoding formal languages,
- semantic interpretation of syntax,
- relative consistency proofs,
- inter-theory embeddings,
- categorical or structural translations.

Interpretation flows create a graph structure between formal systems:

$$J_{\text{interp}}^{10} : \Omega^{10} \rightarrow \Omega^{10}.$$

### O.6.148 Soundness, Completeness, and Meta-Theoretic Cycles

Let  $C_{SC}^{10}$  denote the soundness–completeness cycle.

Processes:

1. proving soundness,
2. proving completeness,
3. discovering incompleteness,
4. establishing independence of axioms,
5. quantifying expressiveness.

Cycle dynamics:

$$C_{SC}^{10} = (J_{\text{sound}}^{10}, J_{\text{complete}}^{10}, J_{\text{incomplete}}^{10})$$

These cycles shape  $\Theta^{10}$ , including Gödelian thresholds:

$$\Theta_{\text{Gödel}}^{10}, \Theta_{\text{Turing}}^{10}, \Theta_{\text{undecidable}}^{10}.$$

### O.6.149 Evolution, Expansion, and Collapse of Formal Systems

A formal system evolves when:

$$P_{\text{expressive}}^{10} \uparrow, \quad J_{\text{proof}}^{10} \uparrow, \quad \Theta_{\text{consistency}}^{10} > 0$$

Expansion processes:

- addition of axioms,
- extension to stronger logics,
- incorporation of new types or operators,
- refinement of inference rules.

Collapse occurs if:

$$T_{10} > \Theta_{\text{collapse}}^{10},$$

e.g. due to:

- inconsistency,
- unbounded recursion without halting,
- violation of computability,
- paradox generation,
- failure of interpretability.

Collapse shrinks  $\Omega(K_{10})$  but may create new consistent fragments.

### O.6.150 Relation to $K_9$ and Emergence Toward $K_{11}$

Upward direction:

Transition to  $K_{11}$  requires:

- higher-order recursion,
- layered meta-levels,
- stratified types,

- cross-system interpretability,
- abstraction over formal languages themselves.

Operator:

$$\Lambda_{10 \rightarrow 11} : K_{10} \rightarrow K_{11}$$

The emergence of  $K_{11}$  marks the point where formal systems cease to be merely computable and become meta-architectural.

Downward direction:

$K_{10}$  governs and constrains:

- all theories ( $K_9$ ),
- all symbolic systems ( $K_8$ ),
- all cognitive models ( $K_6$ ),
- all processes depending on logic or rules.

### O.6.151 Summary

Processes on  $K_{10}$  include:

- axiomatisation and rule formation,
- proof construction, verification, and proof-flows,
- recursion, fixed points, and self-reference,
- computability, reduction, and algorithmic flows,
- interpretation and translation of formal systems,
- soundness–completeness–incompleteness cycles,
- evolution and collapse of formal systems,
- meta-theoretic ascent toward  $K_{11}$ .

$K_{10}$  is the first continuum with inherent limits of formalisation and computation. Its processes define the boundary of what any system—biological, cognitive, social, or technological—can ever formalise or prove.

### O.6.152 Processes on $K_{11}$

$K_{11}$  is the meta-architectural continuum that arises when the recursive–formal structures of  $K_{10}$  become:

- multi-layered,
- type-stratified,
- architecturally managed,
- capable of organising entire families of formal systems,
- governed by meta-rules, meta-consistency thresholds, and meta-interpretation operators.

Formally:

$$X_{11} = (\Omega^{11}, A^{11}, P^{11}, J^{11}, C^{11}, \Theta^{11}, T_{11}, k_{11})$$

where processes act on *meta-structures of formal systems* rather than on systems themselves.

The hallmark processes of  $K_{11}$  are:

- multi-level recursion and stratification,
- meta-interpretation and architecture formation,

- control over families of logics and formalisms,
- categorical and higher-type organisation,
- emergence of meta-cycles and universal structural constraints.

### O.6.153 Stratification and Multi-Level Recursion

$K_{11}$  extends recursion from  $K_{10}$  to multiple—possibly transfinite—levels.

Let

$$\mathcal{R}_0, \mathcal{R}_1, \dots, \mathcal{R}_n$$

be layers of recursive rules.

Processes:

$$\partial_t \mathcal{R}_k = \Lambda^{11}(\mathcal{R}_{k-1}, P_{\text{meta-recursion}}^{11}, \Theta_{\text{layer-consistency}}^{11})$$

They include:

- construction of layered recursion schemas,
- definition of types across levels,
- resolution of cross-level dependencies,
- control of infinite or transfinite recursion chains,
- regularisation of meta-self-reference.

Breakdown occurs if:

$$T_{11}^{\text{strat}} > \Theta_{\text{strat-collapse}}^{11}.$$

### O.6.154 Meta-Interpretation and Inter-System Architecture

Central to  $K_{11}$  is the architecture governing entire families of systems.

Let  $\mathbb{S}$  denote the space of formal systems in  $K_{10}$ .

A meta-interpretation operator:

$$\Psi_{\text{meta}}^{11} : \mathbb{S} \rightarrow \mathbb{S}$$

Processes:

- formation of architectures of theories,
- meta-selection of axiomatic bases,
- regulation of expressiveness,
- identification of universal schemas across systems,
- maintenance of coherence between heterogeneous logics.

These processes create a higher-order organisational geometry over  $\Omega^{10}$ .

### O.6.155 Categorical Organisation and Higher-Type Structure

Processes in  $K_{11}$  include categorical abstraction:

Let  $\mathbf{C}$  be a category of theories, functors, or proofs.

Meta-categorical processes:

$$\partial_t \mathbf{C} = \Phi^{11}(J_{\text{functor}}^{11}, P_{\text{abstraction}}^{11}, \Theta_{\text{categorical-coherence}}^{11})$$

Components:

- construction of functors between formal systems,

- natural transformations as regulatory flows,
- higher-type structures (e.g. 2-categories,  $\infty$ -categories),
- abstraction over types, rules, or meanings,
- identification of universal constructions.

Stability of categorical organisation contributes to  $k_{11}$ .

#### O.6.156 Meta-Thesholds, Meta-Consistency, and Structural Stability

Where  $K_{10}$  includes consistency thresholds for a single system,  $K_{11}$  involves thresholds for *families* of systems.

Let:

$$\Theta^{11} = (\Theta_{\text{meta-consistency}}^{11}, \Theta_{\text{functorial-coherence}}^{11}, \Theta_{\text{type-stability}}^{11}, \Theta_{\text{universal-validity}}^{11}).$$

Processes:

- maintaining coherence across levels of recursion,
- enforcing cross-theory compatibility,
- preventing contradictions due to heterogeneous logics,
- managing global meta-stability.

Violation leads to architectural collapse:

$$T_{11} > \Theta_{\text{meta-collapse}}^{11}.$$

#### O.6.157 Architectural Flows and Meta-Level Control Operators

Define meta-architectural flows:

$$J_{\text{arch}}^{11} = (\text{layer-control}, \text{meta-selection}, \text{schema-regulation}, \text{consistency-balancing})$$

Processes include:

- selection of stable sub-theories,
- suppression of inconsistent branches,
- amplification of coherent formalisms,
- construction of layered architectures,
- control of global expressive power.

These flows serve as the control mechanism for

$$\Phi^{10}, \Psi^{10}, U_{10}, \Lambda_{10}.$$

Thus,  $K_{11}$  is the operator-of-operators layer.

#### O.6.158 Meta-Cycles: Universality, Extension, Collapse

Cycles on  $K_{11}$ :

$$C^{11} = \{C_{\text{universality}}, C_{\text{extension}}, C_{\text{meta-stability}}, C_{\text{collapse}}\}$$

Each cycle corresponds to a recurrent global process:

- **Universality cycle** — identification of universal patterns, logics, and structures.
- **Extension cycle** — expansion of architecture by adding new layers or functors.

- **Meta-stability cycle** — balancing coherence across multiple levels.
- **Collapse cycle** — pruning or restructuring unstable branches.

Cycles define long-term evolution of formal architectures.

### O.6.159 Emergence Toward $K_{12}$

Transition to  $K_{12}$  requires:

$$P_{\text{universality}}^{11} \uparrow, \quad A_{\text{meta-abstraction}}^{11} \uparrow, \quad C_{\text{unification}}^{11} \text{ sustained.}$$

Processes driving the emergence:

- synthesis of meta-architectures into universal operators,
- abstraction beyond any specific family of systems,
- construction of structural invariants at all levels,
- convergence of functorial and recursive frameworks.

The operator:

$$\Xi_{11 \rightarrow 12} : K_{11} \rightarrow K_{12}$$

creates a universal meta-space where architectural constraints become absolute rather than relative.

### O.6.160 Summary

Processes characteristic of  $K_{11}$  include:

- multi-level recursive stratification,
- formation of architectures of formal systems,
- meta-interpretation between entire families of theories,
- categorical and higher-type organisation,
- enforcement of meta-consistency across levels,
- architectural flows regulating expressiveness and coherence,
- universal, extension, stability, and collapse meta-cycles,
- emergence of universal structures leading to  $K_{12}$ .

$K_{11}$  is the first continuum where *structures themselves become subjects of global architecture*. It governs not proofs or theories, but the organisation of all possible families of formal systems.

### O.6.161 Processes on $K_{12}$

$K_{12}$  is the highest continuum definable within the Core. It represents a *universal meta-space* in which all lower continua  $K_0$ – $K_{11}$  appear as internal substructures, limits, or projections. Processes in  $K_{12}$  act not on theories, nor on families of theories, nor on architectures, but on the *space of possible architectures* and on the universal invariants connecting them.

Where  $K_{11}$  governs meta-architectural organisation,  $K_{12}$  governs the *constraints that any architecture must satisfy*.

Formally:

$$X_{12} = (\Omega^{12}, A^{12}, P^{12}, J^{12}, C^{12}, \Theta^{12}, T_{12}, k_{12})$$

with  $\Omega^{12}$  containing all permissible structural configurations consistent with the axioms of the Ontology of Continua.

### O.6.162 Universal Structural Invariants and Global Constraints

The defining process of  $K_{12}$  is the enforcement of *universal invariants* across all lower-level architectures.

Let  $\mathfrak{A}$  be the class of admissible architectures (from  $K_{11}$ ).

There exists a universal operator:

$$\Upsilon_{12} : \mathfrak{A} \rightarrow \mathfrak{A}$$

such that:

$$\Upsilon_{12}(A) = A \iff A \text{ respects all universal constraints (laws) of } K_{12}.$$

Processes include:

- identification of absolute structural invariants,
- projection of constraints onto all  $K_n$ ,
- elimination of forbidden or inconsistent architectures,
- stabilisation of the universal structural manifold.

These constraints represent the “laws of laws” within the OC model.

### O.6.163 Trans-Architectural Integration and Completion

Given a family of architectures

$$\{\mathcal{A}_i\}_{i \in I} \subset K_{11},$$

$K_{12}$  performs a process of *trans-architectural integration*:

$$\mathcal{A}_{\text{unified}} = \Xi(\{\mathcal{A}_i\}, P^{12}, J^{12}, \Theta^{12})$$

Processes:

- formation of universal envelopes for entire architecture families,
- closure under meta-operations (composition, recursion, abstraction),
- emergence of canonical forms of architecture,
- convergence to a unified structural manifold.

This is the point where the classification of all possible structures becomes itself a structure.

### O.6.164 Global Meta-Operators and Structural Absolutes

While  $K_{11}$  controls operators of  $K_{10}$ ,  $K_{12}$  controls the *space of all such operators*.

A global meta-operator:

$$\mathcal{O}_{12} : \{\Psi, \Phi, U, \Lambda, \mathbf{X}\} \rightarrow \{\Psi, \Phi, U, \Lambda, \mathbf{X}\}$$

acts on the class of structural operators and enforces:

- invariance under admissible transformations,
- preservation of universal constraints,
- elimination of degenerate or contradictory operator families,
- canonical normalisation of operator hierarchies.

Thus  $K_{12}$  is the domain of *operator universality*.

### O.6.165 Universal Thresholds and the Absolute Boundary

$K_{12}$  introduces threshold conditions that bound the entire ontology from above.

Universal thresholds:

$$\Theta^{12} = (\Theta_{\text{univ-consistency}}, \Theta_{\text{meta-unification}}, \Theta_{\text{absolute-coherence}}, \Theta_{\text{structure-existence}}).$$

Processes ensure:

- impossibility of exceeding structural dimensionality,
- consistency across all possible continua,
- global closure under recursion and abstraction,
- preservation of the universal boundary  $\partial\Omega^{12}$ .

If

$$T_{12} > \Theta_{\text{existence}}^{12},$$

the continuum collapses – no structure beyond  $K_{12}$  is definable.

This is consistent with the theorem of impossibility of self-generated dimension increase beyond the meta-limit.

### O.6.166 Flows on $K_{12}$ : Universal Balancing and Projection

Flows in  $K_{12}$  mediate between all structure levels.

Define:

$$J^{12} = (J_{\text{projection}}, J_{\text{unification}}, J_{\text{constraint-flow}}, J_{\text{abstraction}}).$$

Processes:

- projection of universal constraints onto  $K_0$ – $K_{11}$ ,
- upward integration of structural information,
- balancing tensions between incompatible architectures,
- generating canonical forms via universal flows.

These flows guarantee the coherence of the entire ontology.

### O.6.167 Cycles of Universality, Closure, and Completion

Cycles in  $K_{12}$  are not cycles of evolution of systems, but cycles of finalisation and universality.

$$C^{12} = \{C_{\text{universality}}, C_{\text{completion}}, C_{\text{constraint-stabilisation}}, C_{\text{absolute-collapse}}\}.$$

Meaning:

- **Universality cycle** – discovery and enforcement of universal laws.
- **Completion cycle** – closure of architecture families into universal structures.
- **Constraint-stabilisation cycle** – maintaining global consistency across all continua.
- **Absolute-collapse cycle** – destruction of any meta-structure violating universal limits.

Cycles define the static-dynamic nature of the highest level.



### O.6.168 Continuumness $k_{12}$ and Universal Stability

The measure  $k_{12}$  describes whether the entire ontological stack  $K_0$ – $K_{11}$  is coherent with universal invariants.

$$k_{12} = F_{12}(\Omega^{12}, J^{12}, \Theta^{12}, T_{12})$$

High  $k_{12}$  corresponds to:

- strict satisfaction of universal invariants,
- stability across all structure levels,
- minimal meta-tension,
- robust global closure.

If  $k_{12} \rightarrow 0$ , no consistent ontology exists – the system collapses beyond definability.

### O.6.169 Summary

Processes on  $K_{12}$  include:

- enforcement of universal structural invariants,
- integration of entire families of architectures,
- regulation of all lower-level structural operators,
- meta-unification and closure under abstraction,
- universal threshold control and absolute boundary maintenance,
- projection and balancing flows across all continua,
- universality, completion, stabilisation, and collapse cycles,
- definition of the global limit of the Ontology of Continua.

$K_{12}$  is the final definable continuum: the space of all possible structural laws that any  $K$ -level must obey. Beyond  $K_{12}$ , no structurally meaningful continuum exists within the theory.

## O.7 Universal Evolution Operators

The evolution of a continuum  $K(t)$  is governed by a family of operators

$$F, G, H, Q, R, S, U,$$

which determine the dynamics of axes, potentials, thresholds, flows, boundary, cycles, and continuumness.

For brevity we write

$$K(t) = (\Omega(t), \partial\Omega(t), A(t), P(t), \Theta(t), J(t), C(t), \tau(K), k(t)),$$

and omit the explicit  $t$ -dependence when no confusion arises. Structural tension is denoted by  $T(t)$  and is understood as a functional of  $\Omega, A, P, \Theta, J$ .

### O.7.1 Operator $F$ : dynamics of axes $A$

**Definition.** The operator  $F$  governs the evolution of the axis set  $A(t)$ :

$$\frac{dA_i}{dt} = F_i(A(t), P(t), \Theta(t), J(t), T(t), k(t)),$$

for each axis component  $A_i \in A$ . Here  $T(t)$  is the structural tension and  $k(t)$  the continuumness.

**Domain and codomain.** Formally,

$$F : A \times P \times \Theta \times J \times \mathbb{R}_{\geq 0} \times [0, 1] \longrightarrow T_A,$$

where  $T_A$  is the tangent space to the manifold of admissible axis configurations.

**Output.** The vector  $\frac{dA}{dt} \in T_A$  encodes:

- birth of new axes (via the dimensional operator  $\Psi$ );
- deactivation or effective collapse of redundant axes;
- smooth deformation of existing axes.

**Relation to components of  $K$ .** The operator  $F$  is sensitive to:

- gradients and distributions of potentials (through  $P$  and  $J$ );
- structural tension  $T$  and its approach to  $\Theta_{\text{dim}}$ ;
- current continuumness  $k$ , suppressing axis changes when the continuum is close to death.

### O.7.2 Operator $G$ : dynamics of potentials $P$

**Definition.** The operator  $G$  governs the evolution of potentials:

$$\frac{dP_i}{dt} = G_i(A(t), J(t), \nabla P(t), \Theta(t)),$$

for each potential component  $P_i$ .

**Domain and codomain.** We write

$$G : A \times J \times \mathcal{F}(P) \times \Theta \longrightarrow T_P,$$

where  $\mathcal{F}(P)$  is the space of potential gradients and  $T_P$  is the tangent space of the potential space.

**Output.** The vector  $\frac{dP}{dt} \in T_P$  is aligned with the effective gradients of the potentials under threshold constraints  $\Theta$ .

**Relation to components of  $K$ .**

- $G$  shapes the energetic landscape over  $\Omega$ .
- It determines which regions of  $\Omega$  are preferred and which are unstable.
- It contributes directly to structural tension  $T(t)$  and to the triggering of phase transitions.

### O.7.3 Operator $H$ : dynamics of thresholds $\Theta$

**Definition.** The operator  $H$  governs the evolution of thresholds:

$$\frac{d\Theta_k}{dt} = H_k(A(t), P(t), J(t), C(t)),$$

for each threshold component  $\Theta_k$ .

**Domain and codomain.**

$$H : A \times P \times J \times C \longrightarrow T_\Theta,$$

where  $T_\Theta$  is the tangent space of the threshold space.

**Output.** The vector  $\frac{d\Theta}{dt} \in T_\Theta$  encodes:

- adaptation of the continuum to environmental and internal conditions;
- tightening or relaxation of admissibility constraints;
- motion toward or away from  $\Theta_{\text{crit}}, \Theta_{\text{dim}}, \Theta_{\text{death}}$ .

**Relation to components of  $K$ .**

- Changes in  $\Theta$  deform the admissible region  $\Omega \subset \Omega(M)$ .
- Through  $\Theta$ , phase transitions and the onset of death are regulated.

#### O.7.4 Operator $Q$ : dynamics of flows $J$

**Definition.** The operator  $Q$  governs the evolution of flows:

$$\frac{dJ_i}{dt} = Q_i(A(t), P(t), \partial\Omega(t), \Theta(t)),$$

for each flow component  $J_i$ .

**Domain and codomain.**

$$Q : A \times P \times \partial\Omega \times \Theta \longrightarrow T_J,$$

where  $T_J$  is the tangent space of the flow space.

**Output.** The vector  $\frac{dJ}{dt} \in T_J$  describes:

- redistribution of flows within  $\Omega$ ;
- modification of flows across the boundary  $\partial\Omega$ ;
- creation or annihilation of transport channels.

**Relation to components of  $K$ .**

- Flows constrain which cycles  $C_j$  can exist and remain stable.
- Changes in flows influence energetic stability and thus  $k(t)$ .

#### O.7.5 Operator $R$ : dynamics of the boundary $\partial\Omega$

**Definition.** The operator  $R$  governs the evolution of the boundary:

$$\frac{d(\partial\Omega)}{dt} = R(A(t), P(t), J(t), \Theta(t)).$$

**Domain and codomain.**

$$R : A \times P \times J \times \Theta \longrightarrow T_{\partial\Omega},$$

where  $T_{\partial\Omega}$  is the space of boundary deformations.

**Output.** The quantity  $\frac{d(\partial\Omega)}{dt} \in T_{\partial\Omega}$  describes changes of shape and permeability of the boundary, including:

- expansion or contraction of  $\Omega$ ;
- modification of permeability for different flows;
- birth of new internal interfaces within the continuum.

**Relation to components of  $K$ .**

- The boundary determines the separation between continuum and environment.
- The geometry and topology of  $\partial\Omega$  shape the structure of flows and cycles.

**O.7.6 Operator  $S$ : dynamics of cycles  $C$** 

**Definition.** The operator  $S$  governs the evolution of cycles:

$$\frac{dC_j}{dt} = S_j(A(t), J(t), \Theta(t), P(t)),$$

for each cycle  $C_j$ .

**Domain and codomain.**

$$S : A \times J \times \Theta \times P \longrightarrow T_C,$$

where  $T_C$  is the space of deformations and transformations of cycles.

**Output.** The quantity  $\frac{dC}{dt} \in T_C$  captures changes in the cycle system:

- birth of new cycles from combinations of flows;
- breaking of existing cycles;
- restructuring of cycle periods and amplitudes.

**Relation to components of  $K$ .**

- Cycles support the temporal rhythm encoded in  $\tau(K)$ .
- Destruction of critical cycles typically reduces  $k(t)$  and may trigger death.

**O.7.7 Operator  $U$ : dynamics of continuumness  $k(t)$** 

**Definition.** The operator  $U$  governs the evolution of continuumness:

$$\frac{dk}{dt} = U(\Omega(t), \partial\Omega(t), A(t), P(t), \Theta(t), J(t), C(t)).$$

**Domain and codomain.**

$$U : \mathcal{P}(\Omega) \times \partial\Omega \times A \times P \times \Theta \times J \times C \longrightarrow \mathbb{R},$$

where  $\mathcal{P}(\Omega)$  denotes the space of admissible deformations of the state region.

**Output.** The scalar  $\frac{dk}{dt}$  determines:

- growth of  $k(t)$  when structural coherence and cycle support increase;
- decay of  $k(t)$  when cycles are destroyed, thresholds are violated, or axes become inconsistent;
- stabilization of  $k(t)$  when gains and losses of structure balance.

**Relation to components of  $K$ .**

- $U$  aggregates the effects of all other operators on the single scalar order parameter  $k(t)$ .
- The sign of  $U$  determines whether the continuum is moving toward higher stability or toward collapse.

### O.7.8 Joint Evolution Operator $E$

**Definition.** The combined evolution of the continuum is given by the operator

$$E : K(t) \mapsto K(t + dt),$$

which, in differential form, is realized by the system:

$$\begin{aligned}\frac{dA}{dt} &= F(A, P, \Theta, J, T, k), \\ \frac{dP}{dt} &= G(A, J, \nabla P, \Theta), \\ \frac{d\Theta}{dt} &= H(A, P, J, C), \\ \frac{dJ}{dt} &= Q(A, P, \partial\Omega, \Theta), \\ \frac{d(\partial\Omega)}{dt} &= R(A, P, J, \Theta), \\ \frac{dC}{dt} &= S(A, J, \Theta, P), \\ \frac{dk}{dt} &= U(\Omega, \partial\Omega, A, P, \Theta, J, C).\end{aligned}$$

**Compatibility.** The operator  $E$  is compatible with the Level-0 axioms and the axioms of continua  $K$ :

- it does not drive the system outside  $\Omega(M)$  while the continuum is alive ( $k > 0$ );
- it respects the refinement relation between  $\Theta$  and  $\Theta^M$ ;
- it preserves the logical constraints required by the meta-domain evolution operator  $\Phi$ .

## O.8 Master Theorems of Core 1.2

This section collects the key theorems of Core 1.2, based on the Level-0 axioms, the axioms of continua  $K$ , and the universal evolution operators.

### O.8.1 Representability Theorems for Levels $K_2, K_4, K_8$

**Theorem 1 (Representability of  $K_2$ ).** *Any Level- $K_2$  system described by local fields, threshold interactions, and quantized excitations can be represented as a continuum*

$$K_2 = (\Omega_2, \partial\Omega_2, A_2, P_2, \Theta_2, J_2, C_2, \tau(K_2), k_2)$$

satisfying the axioms of Core 1.2.

*Proof sketch.* Local fields and their excitations define axes  $A_2$ ; threshold interactions are encoded in  $\Theta_2$ ; admissible field configurations form  $\Omega_2$ ; the field dynamics induce flows  $J_2$  and cycles  $C_2$  (e.g. periodic or quasi-periodic configurations). By construction, the structural tuple satisfies Axioms K.1–K.7. The joint evolution operator  $E$  can be chosen to coincide with the local field dynamics, ensuring compatibility with the Level-0 axioms and preserving admissibility in  $\Omega(M)$ .

**Theorem 2 (Representability of  $K_4$ ).** *Any autocatalytic chemical system with closed metabolic cycles and a stable boundary can be represented as a continuum  $K_4$ .*

*Proof sketch.* Concentrations and reaction configurations span  $\Omega_4$ ; the membrane or compartment defines the boundary  $\partial\Omega_4$ ; stoichiometric directions define axes  $A_4$ ; chemical potentials provide  $P_4$ ; kinetic constraints (e.g. saturation, inhibition, energetics) define thresholds  $\Theta_4$ ; material fluxes define flows  $J_4$ ; metabolic cycles correspond to cycles  $C_4$ . As long as at least one cycle closes and the boundary prevents uncontrolled leakage, we obtain  $k_4 > 0$ . All structural elements map directly to the axioms of a continuum, so  $K_4$  satisfies Core 1.2.

**Theorem 3 (Representability of  $K_8$ ).** *Any civilizational–technological system with stable infrastructure, norms, and information flows can be represented as a continuum  $K_8$ .*

*Proof sketch.* Agents and their states form  $\Omega_8$ ; inter-system boundaries (political, economic, infrastructural) define  $\partial\Omega_8$ ; technological and symbolic codes define axes  $A_8$ ; resources and indicators (energy, capital, capacity, knowledge) define potentials  $P_8$ ; institutional and technological constraints define thresholds  $\Theta_8$ ; material and informational streams define flows  $J_8$ ; institutional and infrastructural feedback loops define cycles  $C_8$ . If the underlying interaction graph is connected enough and key cycles are sustained, then  $k_8 > 0$ . Thus the system is representable as a continuum of Level  $K_8$ .

### O.8.2 Compatibility Theorem $K \leftrightarrow M$

**Theorem 4 (Necessary and sufficient conditions for compatibility  $K \leftrightarrow M$ ).** *A continuum  $K$  can evolve under the joint operator  $E$  without destruction if and only if the following conditions hold:*

1.  $\Omega(K) \subseteq \Omega(M)$ ;
2.  $A(K) \subseteq A(M)$ ;
3. thresholds  $\Theta$  are refinements of the environmental thresholds  $\Theta^M$ ;
4. the dynamics  $E$  are consistent with the meta-domain evolution  $\Phi$ .

*Proof sketch.* Necessity: violation of (1) drives the trajectory  $K(t)$  outside  $\Omega(M)$ ; violation of (2) or (3) leads to axes or thresholds unsupported by the meta-domain, violating Axioms 0.2, 0.6, and  $\Phi.2$ . In each case, Axioms D.1–D.3 imply that either  $\Omega$  becomes empty or death thresholds are crossed. Sufficiency: if (1)–(4) hold, then trajectories generated by  $E$  remain inside  $\Omega(M)$ , with thresholds compatible by design. No forced violation of death thresholds occurs, so destruction is not imposed by the environment.

### O.8.3 Theorem on Impossibility of Evolution Outside $M$

**Theorem 5 (Impossibility of evolution outside the meta-domain).** *Let  $K \subset M$  be a continuum satisfying the axioms of Core 1.2. If there exists a time  $t^*$  and a state  $s^* \notin \Omega(M)$ , then there is no valid evolution operator  $E$  that extends the evolution of  $K$  through  $t^*$  without destruction.*

*Proof sketch.* States outside  $\Omega(M)$  violate the meta-thresholds  $\Theta^M$ , contradicting Axiom 0.6. Any operator  $E$  mapping to such a state breaks Axiom  $\Phi.2$  and the compatibility conditions of Theorem 4. The only way to keep the axioms consistent is that  $\Omega(K)$  becomes empty (death) before reaching  $s^*$ . Thus no non-destructive evolution beyond  $t^*$  exists.

### O.8.4 Theorem on Birth of a New Dimension via $\Psi$

**Theorem 6 (Birth of a new dimension).** *Let  $K \subset M$  be a living continuum with structural tension  $T(t)$  monotonically increasing. If there exists a time  $t^*$  such that  $T(t^*) > \Theta_{\dim}$  and there is an axis  $a_{\text{new}} \in A(M) \setminus A(K)$ , then there exists and is unique (up to axis equivalence) a continuum  $K' = \Psi(K, M)$  with  $\dim K' = \dim K + 1$ .*

*Proof sketch.* By Axiom  $\Theta.4$ , crossing  $\Theta_{\dim}$  requires dimensional growth. By Axiom  $\Psi.1$ , any new axis must come from  $A(M) \setminus A(K)$ . By Axiom 0.9, local structural complexity is finite, so the set of candidate axes is finite. Identifying axes that are equivalent with respect to their effect on  $\Omega$  and  $\Theta$  yields a unique equivalence class of admissible new axes. Thus  $K'$  is unique up to this equivalence and satisfies  $\dim K' = \dim K + 1$ .

### O.8.5 Theorem on Death of a Continuum

**Theorem 7 (Equivalence of death conditions).** *For a continuum  $K$ , the following conditions are equivalent:*

1.  $\Omega(t^*) = \emptyset$ ;
2.  $k(t) \rightarrow 0$  as  $t \rightarrow t^*$ ;
3.  $\tau_{\text{cycle}} \rightarrow \infty$  and there exist no new cycles with finite frequency;
4. at least one threshold in  $\Theta_{\text{death}}$  is violated.

*Proof sketch.* (1) $\Rightarrow$ (2): by Axiom K.2, a living continuum requires  $\Omega \neq \emptyset$ ; if  $\Omega = \emptyset$ , then  $k = 0$ . (2) $\Rightarrow$ (3): by Axiom C.2 and k.1, non-zero continuumness requires at least one supporting cycle with finite period;  $k \rightarrow 0$  implies all such cycles degrade to  $\tau_{\text{cycle}} \rightarrow \infty$ . (3) $\Rightarrow$ (4): absence of supporting cycles and diverging periods violate stability thresholds, which are part of  $\Theta_{\text{death}}$ . (4) $\Rightarrow$ (1): by Axiom D.3, violation of any death threshold forces  $\Omega = \emptyset$ . Thus all four conditions are equivalent characterizations of death.

### O.8.6 Theorem on Phase Transition

**Theorem 8 (Phase transition upon crossing  $\Theta_{\text{crit}}$ ).** *Let  $K$  be a living continuum with structural tension  $T(t)$ . If there exists a time  $t^*$  such that*

$$\Theta_{\text{crit}} < T(t^*) < \min\{\Theta_{\text{dim}}, \Theta_{\text{death}}\},$$

*then:*

1. the admissible region  $\Omega(t)$  undergoes a qualitative restructuring (splitting into new components or merging of old ones);
2. the continuum remains alive ( $k(t^*) > 0$ );
3. the dimension  $\dim K$  is preserved.

*Proof sketch.* By definition of the critical threshold  $\Theta_{\text{crit}}$  (Axiom  $\Theta.3$ ), crossing it induces a structural reorganization of  $\Omega$  without forcing death or dimensional growth. The inequalities  $T < \Theta_{\text{dim}}, \Theta_{\text{death}}$  guarantee that neither dimensional expansion nor death is triggered. Therefore the system undergoes a phase transition within fixed dimension and non-zero continuumness.

### O.8.7 Theorem on Impossibility of Degradational Evolution

**Theorem 9 (No “backward” dimensional reduction for a living continuum).** *If there exists a time  $t^*$  such that*

$$\dim K(t^* + dt) < \dim K(t^*),$$

*then the continuum  $K$  is dead at time  $t^* + dt$ .*

*Proof sketch.* By Axiom  $\Psi.3$ , a living continuum cannot decrease its dimension; any formal reduction of dimension implies that the admissible region collapses to  $\Omega = \emptyset$  and  $k = 0$ . Thus, if the dimension is strictly smaller at  $t^* + dt$ , the resulting configuration must be dead, and the change cannot be interpreted as a genuine evolution of a living continuum.

## O.9 Complexity Metric $S(K)$

### O.9.1 Definition of Complexity Components

The complexity of a continuum  $K$  measures the structural richness of its state space  $\Omega$ , axis set  $A$ , cycle system  $C$ , threshold structure  $\Theta$ , and flows  $J$ . We decompose it into five components:

$$S(K) = w_{\Omega}S_{\Omega} + w_AS_A + w_CS_C + w_{\Theta}S_{\Theta} + w_JS_J,$$

where  $w_{\bullet} \geq 0$  are weight coefficients.

**Component  $S_\Omega$  (state space).** Let  $\mu(\Omega)$  be a normalized measure of the admissible state region  $\Omega$  inside  $\Omega(M)$ , with  $0 \leq \mu(\Omega) \leq 1$ . Then we define

$$S_\Omega(K) = \log(1 + \alpha_\Omega \mu(\Omega)),$$

where  $\alpha_\Omega > 0$  is a scale parameter.

**Component  $S_A$  (axes).** Let  $\dim K = |A|$  be the number of independent axes. Then

$$S_A(K) = \log(1 + \alpha_A \dim K),$$

with  $\alpha_A > 0$  a scale parameter.

**Component  $S_C$  (cycles).** Let  $|C|$  be the number of independent cycles and  $\overline{\tau_C}$  the mean period of the stable cycles. We define

$$S_C(K) = \log(1 + \alpha_C |C|) \cdot \log(1 + \beta_C / \overline{\tau_C}),$$

where  $\alpha_C, \beta_C > 0$ . Thus the contribution of cycles grows with both their number and their frequency.

**Component  $S_\Theta$  (threshold structure).** Let  $N_{\text{eff}}(\Theta)$  be the effective number of independent thresholds, i.e. the minimal number of threshold functions  $f_k$  sufficient to define the admissible region  $\Omega$ . Then

$$S_\Theta(K) = \log(1 + \alpha_\Theta N_{\text{eff}}(\Theta)),$$

with  $\alpha_\Theta > 0$ .

**Component  $S_J$  (flows).** Let  $J$  be the set of flows and  $\|J\|$  some aggregated norm (for example, the time-averaged sum of absolute values of all flows). We set

$$S_J(K) = \log(1 + \alpha_J \|J\|),$$

with  $\alpha_J > 0$ .

## O.9.2 Unified Complexity Formula

Combining the components, we obtain

$$S(K) = w_\Omega \log(1 + \alpha_\Omega \mu(\Omega)) + w_A \log(1 + \alpha_A \dim K) + w_C \log(1 + \alpha_C |C|) \log(1 + \beta_C / \overline{\tau_C}) \quad (\text{O.5}) \\ + w_\Theta \log(1 + \alpha_\Theta N_{\text{eff}}(\Theta)) + w_J \log(1 + \alpha_J \|J\|).$$

This formula defines a scalar complexity metric  $S(K) \geq 0$  that increases monotonically with each component, assuming the others are fixed.

## O.9.3 Relation of $S(K)$ to $k(t)$ , $T(t)$ , and $\tau(K)$

**Relation to continuumness  $k(t)$ .** Continuumness  $k(t)$  reflects the *quality* of coherence and stability, whereas  $S(K)$  reflects the *amount* of structure. In general:

- low  $S(K)$  restricts the achievable maximal value of  $k(t)$  because there is not enough structure to support rich dynamics;
- excessively high  $S(K)$  at fixed resources can reduce  $k(t)$  due to overload of thresholds  $\Theta$  and insufficient support by cycles.

Functionally, the universal operator  $U$  governing  $dk/dt$  depends on  $S(K)$ :

$$\frac{dk}{dt} = U(\dots, S(K), \dots),$$

with the contribution of  $S(K)$  to  $U$  being typically positive for moderate complexity and negative for excessive complexity.



**Relation to structural tension  $T(t)$ .** Structural tension  $T(t)$  tends to grow when:

- $\mu(\Omega)$  increases without a corresponding increase in the number of axes  $A$ ;
- $N_{\text{eff}}(\Theta)$  increases at fixed axes  $A$ ;
- flows  $J$  and cycles  $C$  become misaligned or incoherent.

Thus  $T(t)$  can be expressed as a functional of the components  $S_\Omega, S_A, S_C, S_\Theta, S_J$ . The phase thresholds  $\Theta_{\text{crit}}$ ,  $\Theta_{\text{dim}}$ , and  $\Theta_{\text{death}}$  appear as critical values for specific combinations of these components.

**Relation to the temporal structure  $\tau(K)$ .** The temporal structure  $\tau(K)$  captures how quickly the continuum can process its complexity:

- for fixed  $S(K)$ , decreasing  $\tau_{\text{response}}$  and  $\tau_{\text{regen}}$  increases the probability of maintaining  $k(t) > 0$ ;
- if  $S(K)$  grows faster than the continuum can adapt its temporal constants, the system approaches a collapse threshold  $\Theta_{\text{collapse}}$ .

Formally, the lifetime functional

$$\tau_{\text{life}}(K) = \int_{t_0}^{t_{\text{death}}} k(t) dt$$

is determined by the trajectory  $S(K(t))$  together with the temporal parameters  $\tau(K)$ .

#### O.9.4 Interpretation of the Complexity Metric

**Interpretation of  $S_\Omega$ .**  $S_\Omega$  captures the volume of admissible states. An increase in  $\mu(\Omega)$  without an accompanying growth in axes raises structural tension.

**Interpretation of  $S_A$ .**  $S_A$  captures the dimensionality and the number of independent directions of difference. Growth of  $\dim K$  via  $\Psi$  allows tension to be redistributed but increases the cost of maintaining structure.

**Interpretation of  $S_C$ .**  $S_C$  measures the number and speed of cycles that support the life of the continuum. Too few cycles make the system fragile; too many cycles at fixed resources can lead to competition for flows and a drop in  $k(t)$ .

**Interpretation of  $S_\Theta$ .**  $S_\Theta$  captures the complexity of the threshold landscape. A large number of independent thresholds makes  $\Omega$  narrow and increases the risk of crossing death thresholds  $\Theta_{\text{death}}$ .

**Interpretation of  $S_J$ .**  $S_J$  measures the intensity of motion inside the continuum. Insufficient flows make cycles impossible; excessive flows can erode the boundary  $\partial\Omega$  and violate thresholds  $\Theta$ .

Taken together, the components  $S_\Omega, S_A, S_C, S_\Theta, S_J$  provide a quantitative measure of complexity  $S(K)$  that supports the formal description of growth, phase transitions, and death of continua within Core 1.2.

## P Reference and Benchmark Atlas

This appendix restores the full recursive reference subtree for experiments, falsifiability, and predictions. These materials remain inside the flagship volume as inspectable benchmark and operational surfaces, but they are staged here as reference atlases rather than breaking the teaching flow of the main monograph.

## P.1 Structural Experiments

The experimental programme of the Ontology of Continua (OC) provides the empirical and computational foundation for testing structural predictions across all continuum levels  $K_0$ – $K_{12}$ . Unlike domain-specific experiments contained in the per-level files (`experiments_kX.tex`), the present master module outlines the general methodology, validation strategies, and cross-domain principles that govern the design and interpretation of experiments within OC.

Experiments in OC serve several structural functions:

- verifying the existence and saturation of thresholds  $\Theta$ ;
- tracking the behaviour of flows  $J$  under controlled perturbations;
- measuring cycle stability  $C_{\text{stab}}$  and collapse dynamics;
- identifying birth conditions for new continua via  $\Psi_{x \rightarrow x+1}$ ;
- testing cross-level invariants that hold across physics, chemistry, biology, cognition, social systems, and meta-theoretical continua.

Experimental design spans physical, chemical, biological, cognitive, institutional, and computational modalities. Across those domains, every experiment still receives one structural interpretation in terms of  $(\Omega, A, P, J, \Theta, \partial\Omega, C, k)$ .

### P.1.1 Purpose of Structural Experiments

Structural experiments aim to test not domain-specific phenomenology, but the general principles of continuum dynamics. Every experiment targets at least one of the following claims:

1. **Thresholds govern qualitative change.** Phase transitions, regime shifts, collapse events, and dimensional births occur when structural tension  $T$  saturates a threshold  $\Theta_{\text{crit}}$  or  $\Theta_{\text{dim}}$ .
2. **Cycles maintain persistence.** Stable cycles  $C$  underpin viability; their destruction predicts or marks collapse.
3. **Flows drive evolution.** The balance between supporting and destructive flows determines system stability.
4. **Birth of new continua requires tension.** Dimensional transitions arise only when  $T > \Theta_{\text{dim}}$ .
5. **Collapse is threshold-driven and irreversible.** Violating  $\Theta_{\text{death}}$  eliminates the admissible region  $\Omega$ , forcing  $k \rightarrow 0$ .
6. **Cross-domain invariance.** The same structural phenomena appear in  $K_2$ – $K_4$  experiments (physical and chemical), in  $K_4$ – $K_5$  experiments (biological excitability), in  $K_7$ – $K_8$  (social collapse), and in  $K_9$ – $K_{10}$  (paradigm coherence and theoretical collapse).

### P.1.2 Types of Experiments

OC distinguishes three major categories of experiments:

**1. Physical / Chemical Experiments.** These include tests of percolation thresholds, BKT transitions, catalytic closure in RAF networks, vesicle stability under osmotic stress, curvature thresholds, and the emergence of proto-excitability.

**2. Biological / Cognitive Experiments.** Tests include ion channel gating statistics, proto-spike generation, membrane recovery cycles, binding limits in cognitive tasks, and prediction thresholds in representational systems.

**3. Social / Institutional / Theoretical Experiments.** These evaluate trust thresholds, institutional stability, infrastructure cycles, and logical or model–theoretic coherence thresholds in  $K_9$ – $K_{10}$ .

### P.1.3 Universal Experiment Structure

Every experiment is represented by the tuple:

$$E = (\text{initial state, perturbation, measurement,} \\ \text{threshold observation, cycle tracking, collapse / recovery criteria}).$$

This generic structure applies across all levels, where:

- perturbations drive flows  $J$ ;
- measurements track potentials  $P$  and axes  $A$ ;
- threshold observations determine whether  $\Theta$  is approached or crossed;
- cycle tracking identifies the stability of  $C$  over time;
- collapse/recovery criteria map the behaviour of  $k(t)$ .

### P.1.4 Measurement Protocols

OC introduces universal measurement protocols applicable to any continuum:

1. **Threshold Tracking.** Measure  $P$ ,  $J$ , or  $A$  as a function of controlled perturbation to locate  $\Theta_{\text{crit}}$ ,  $\Theta_{\text{stab}}$ ,  $\Theta_{\text{dim}}$ , or  $\Theta_{\text{death}}$ .
2. **Cycle Stability Analysis.** Observe long–range coherence of  $C$ , compute:

$$S(C) = \min d(\Omega, \partial\Omega), \quad C_{\text{eff}} = \frac{\oint J \cdot dA}{L(C)}.$$

3. **Collapse Mapping.** Identify divergence of structural tension:

$$T \rightarrow \infty \quad \Rightarrow \quad \Omega \rightarrow \emptyset, \quad k \rightarrow 0.$$

4. **Birth Condition Evaluation.** Detect the emergence of new axes when:

$$T > \Theta_{\text{dim}}.$$

5. **Cross–Domain Comparisons.** Validate invariants across continuum levels.

### P.1.5 Computational Experiments

Many structural predictions of OC are most readily tested via computational experiments, including:

- percolation simulations for  $K_2$ ;
- stochastic RAF network generation for  $K_3$ ;
- membrane patch models and ion channel dynamics for  $K_4$ – $K_5$ ;
- predictive coding / binding models for  $K_6$ ;
- agent–based models of trust collapse for  $K_7$ ;
- infrastructure–energy co–evolution for  $K_8$ ;
- model–theoretic consistency checkers for  $K_9$ – $K_{10}$ .

Computational experiments allow testing structural conditions that are difficult or impossible to measure directly in natural systems.

### P.1.6 Experimental Validation Pipeline

The OC validation pipeline contains four universal stages:

1. **Design:** identify the structural prediction and the relevant K-level.
2. **Perturbation:** apply controlled changes to  $P$ ,  $A$ , or  $J$ .
3. **Measurement:** measure  $T$ ,  $\Theta$ ,  $C$ , and  $k(t)$ .
4. **Interpretation:** evaluate structural behaviour relative to expected threshold saturation or cycle dynamics.

### P.1.7 Relation to Falsifiability

All experiments in OC support falsifiability by targeting:

- prediction violations;
- threshold misalignment;
- cycle mismatch;
- incorrect dimensional birth conditions;
- unexpected collapse behaviour.

The master falsifiability module (Section 15) provides the system-level criteria, while the present module organizes the experimental methodology.

### P.1.8 Cross-Level Generality

The experimental framework is intentionally cross-domain and cross-level. This universality is a defining feature of OC and a central criterion for falsifiable theory construction: a structural prediction validated in physics ( $K_2$ ) must have a corresponding instantiation in chemistry ( $K_3$ – $K_4$ ), biology ( $K_4$ – $K_5$ ), cognition ( $K_6$ ), social systems ( $K_7$ – $K_8$ ), and theoretical continua ( $K_9$ – $K_{10}$ ), unless an explicit domain-specific break is justified by the operator structure.

### P.1.9 Structure of the Per-Level Experiment Files

The following files (generated from `master_core_structure.yaml`) contain domain-specific experimental designs:

- `experiments_k0.tex` — substrate-level tests;
- `experiments_k1.tex` — one-dimensional continua;
- `experiments_k2.tex` — physical continua;
- `experiments_k3.tex` — chemical RAF/network experiments;
- `experiments_k4.tex` — membrane thresholds & cycles;
- `experiments_k5.tex` — early excitability & proto-spikes;
- `experiments_k6.tex` — cognitive thresholds;
- `experiments_k7.tex` — institutional tests;
- `experiments_k8.tex` — civilizational dynamics;
- `experiments_k9.tex` — theoretical stability;

- `experiments_10.tex` — meta-theoretical coherence;
- `experiments_11.tex` — evolution-of-evolution;
- `experiments_12.tex` — meta-recursive experiments.

These specialised modules implement the universal methodology presented here.

### P.1.10 Experiments for $K_0$

The continuum level  $K_0$  represents the meta-ontological substrate of the Ontology of Continua. It does not possess geometry, time, energy, flows, thresholds, or dynamics. Consequently, no physical or empirical experiments can be performed on  $K_0$  in the usual sense. Instead, the validation of  $K_0$  proceeds through *structural* and *logical* experiments: tests of consistency, definability, and well-posedness of the transition operator  $\Psi_{0 \rightarrow 1}$ .

$K_0$ -experiments therefore evaluate the *conditions of possibility* for any continuum to exist, rather than the behaviour of a continuum itself.

### P.1.11 Nature of $K_0$ Experiments

Because  $K_0$  lacks:

- a state space  $\Omega(K_0)$ ,
- axes  $A$ ,
- potentials  $P$ ,
- flows  $J$ ,
- thresholds  $\Theta$ ,
- cycles  $C$ ,
- and measurable time  $t$ ,

experiments reduce to formal tests of the logical structure underlying continuum formation.

These tests target the following principles:

1. **Non-contradiction of the ontological substrate.** The foundational axioms governing  $K_0$  must not generate paradoxes.
2. **Minimal definability of a proto-parameter.** A single real-valued proto-parameter  $p_1 : S \rightarrow \mathbb{R}$  must be definable as a precondition for  $K_1$  (Axiom  $A_p$ ).
3. **Connectedness of the proto-domain.** The closure of  $p_1(S)$  must define a single connected interval  $(a, b)$  (Axiom  $A_{\text{conn}}$ ), ensuring that the resulting  $K_1$  continuum is one-dimensional and connected.
4. **Absence of illicit structure.** No geometry, metric, energy, or temporal order may be smuggled into  $K_0$ . Any such appearance signals a violation of the axioms.
5. **Well-posedness of  $\Psi_{0 \rightarrow 1}$ .** The transition operator must produce a valid continuum  $K_1$  with:

$$\Omega(K_1), A_1, P_1, J_1, \Theta_1, C_1, k_1(t)$$

in full agreement with the formal definition established in Core 1.3.

### P.1.12 Type I Logical Experiments: Axiom Consistency Checks

These experiments verify that the axioms governing  $K_0$  form a coherent and non-contradictory set. The tests include:

- consistency proofs for the absence of geometry;
- confirmation that no structure of dimension  $\geq 1$  is forced at  $K_0$ ;
- syntactic and semantic validation of the proto-parameter axioms;
- ensuring that no implicit metric or temporal order emerges from the axiom set.

Although not empirical, these experiments are essential: if  $K_0$  were inconsistent, the entire vertical chain  $K_1$ – $K_{12}$  would collapse.

### P.1.13 Type II Domain of Definition Experiments

These experiments validate the minimal domain  $S$  from which the proto-parameter  $p_1$  is defined. The goal is to ensure that:

1.  $S$  contains no structure exceeding the axioms;
2.  $p_1(S)$  is definable without prior geometry;
3. the mapping  $p_1$  does not introduce forbidden structure.

A successful experiment demonstrates that the raw substrate does not pre-encode any continuum and is compatible with the birth of  $K_1$ .

### P.1.14 Type III Transition Experiments for $\Psi_{0 \rightarrow 1}$

The most important  $K_0$  experiments evaluate the correctness of the transition operator  $\Psi_{0 \rightarrow 1}$ .

These experiments ask:

- Does  $\Psi_{0 \rightarrow 1}$  produce a state space  $\Omega(K_1)$  that is well-defined as a smooth 1D manifold?
- Does it produce the correct initial axis  $A_1(x) = x$ ?
- Does the resulting continuum satisfy all  $K_1$  axioms, including the definition of flows  $J_1$ , energy  $E_1$ , tension  $T_1$ , and the trivial cycle  $C_{\text{triv}}$ ?
- Does the mapping avoid creating multiple connected components?
- Are the boundary conditions encoded by  $\partial\Omega(K_1)$  consistent with the source axioms at  $K_0$ ?

A failure of any of these tests indicates a structural error in the foundation of the entire model.

### P.1.15 Type IV Meta-Consistency Experiments

Finally,  $K_0$  supports meta-theoretical consistency tests that ensure the entire vertical chain  $K_0 \rightarrow K_1 \rightarrow K_2 \rightarrow \dots \rightarrow K_{12}$  can be defined without contradiction.

These experiments include:

- verifying that  $K_0$  places no constraints incompatible with higher continua (e.g., physics or biology);
- checking that the transition maps  $\Psi_{x \rightarrow x+1}$  all have a well-defined domain containing  $K_0$  as substrate;
- confirming that no cyclic contradictions are introduced at the highest meta-levels ( $K_{10}$ – $K_{12}$ ).

Because  $K_0$  is outside time, these experiments are purely structural and serve as the logical foundation for the falsifiability pipeline.

### P.1.16 Summary

Experiments on  $K_0$  are not empirical but *foundational*. They verify:

- logical coherence of the axioms,
- definability of the proto-parameter,
- absence of illicit structure,
- well-posedness of  $\Psi_{0 \rightarrow 1}$ ,
- compatibility with the entire K-hierarchy.

All higher experiments depend on the correctness of these structural tests.

### P.1.17 Experiments for $K_1$

The continuum level  $K_1$  represents the first fully realized continuum in the Ontology of Continua. It is defined as a smooth one-dimensional manifold  $X = (a, b)$  equipped with:

- a state space  $\Omega(K_1)$  satisfying  $C^0(T, H^1(X, V)) \cap C^1(T, L^2(X, V))$ ;
- the axis  $A_1(x) = x$ ;
- potentials  $P_1(t, x)$ ;
- flows  $J_1(t, x)$ ;
- structural tension  $T_1(t)$ ;
- a stability threshold  $\Theta_1$ ;
- the trivial cycle  $C_{\text{triv}}$ ;
- a well-defined time parameter  $\tau(K_1)$ .

Because  $K_1$  is the lowest continuum that possesses genuine dynamics, its experimental programme establishes the basic empirical and analytic tests that validate the continuum formalism and its evolution operator  $E(K_1)$ .

### P.1.18 Objectives of $K_1$ Experiments

Experiments at the level of  $K_1$  aim to verify:

1. **Existence and smoothness of the continuum.** Confirming the manifold structure of  $X = (a, b)$  and the smoothness requirements on fields.
2. **Correctness of the axis  $A_1$ .** The axis must correspond to the real coordinate of the manifold and must remain stable under the evolution operator.
3. **Continuity and differentiability of flows.** Flows  $J_1(t, x)$  must satisfy the regularity imposed by the space  $\Omega(K_1)$  and produce finite energy and tension.
4. **Existence of a trivial cycle.** The cycle  $C_{\text{triv}}$  must be realizable in practice: a constant or periodic configuration with vanishing net flow.
5. **Threshold behaviour.** Experimental perturbations must reproduce the critical condition  $T_1 = \Theta_1$  marking the onset of instability.
6. **Reproducibility of the evolution operator.** Numerical and analytical experiments must validate the operator  $E(K_1)$  as defined in Core 1.3.

### P.1.19 Type I Experiments: Continuity and Smoothness

These experiments verify that fields inhabiting  $\Omega(K_1)$  satisfy the required regularity:

- $P_1(t, x)$  and  $J_1(t, x)$  must be continuous in  $t$  and  $x$ ;
- spatial derivatives must lie in  $L^2(X)$ ;
- temporal derivatives must lie in  $L^2(X)$  in the sense of distributions.

Typical experiments:

1. Approximating fields by smooth test functions and evaluating convergence in  $H^1$ .
2. Numerical integration of flows with imposed small perturbations.
3. Regularity checks via discrete Sobolev norms.

### P.1.20 Type II Experiments: Flow Stability and Tension

The structural tension  $T_1(t)$  is measurable through the deviation of flows from stable configurations. Experiments focus on:

- establishing a stable flow  $J_1^{\text{stable}}(x)$  with minimal tension;
- perturbing it and measuring  $T_1(t)$  as the system relaxes;
- identifying the threshold  $\Theta_1$  at which relaxation fails.

Such tests validate the definition of stability encoded in the Core.

### P.1.21 Type III Experiments: Existence of the Trivial Cycle

The trivial cycle  $C_{\text{triv}}$  is defined as a configuration that remains invariant under evolution:

$$E(K_1)[C_{\text{triv}}] = C_{\text{triv}}.$$

Experiments include:

1. constructing constant fields  $P_1(x) = \text{const}$ ,  $J_1(x) = 0$ ;
2. evolving them numerically or analytically;
3. verifying invariance and stability for small perturbations.

These experiments establish the existence of time at  $K_1$  in the precise sense of a closed minimal cycle.

### P.1.22 Type IV Experiments: Threshold Dynamics

Experiments probe the behaviour near the critical threshold  $\Theta_1$ :

- applying increasing perturbations to a stable configuration;
- measuring tension  $T_1$  as a function of perturbation amplitude;
- detecting the point where  $T_1 = \Theta_1$ ;
- observing the onset of instability or collapse.

These tests confirm the role of  $\Theta_1$  as an existence and stability threshold.



### P.1.23 Type V Experiments: Validation of $E(K_1)$

The evolution operator  $E(K_1)$  must satisfy:

$$K_1(t + \Delta t) = E(K_1(t)).$$

Experiments validate this by:

1. integrating the fields forward in time using numerical schemes;
2. checking conservation or decay properties of energy  $E_1$ ;
3. verifying tension evolution  $dT_1/dt$  against theoretical predictions derived from the Core equations;
4. benchmarking analytical solutions against the evolution operator.

These experiments are the first point where the general evolution operator from the Core becomes empirically testable.

### P.1.24 Summary

Experiments on  $K_1$  establish the foundations of physical dynamics in the Ontology of Continua:

- continuity,
- smoothness,
- flow regularity,
- existence of cycles,
- threshold behaviour,
- validity of evolution.

All higher levels depend on the correctness of these baseline tests.

### P.1.25 Experiments for $K_2$

The continuum level  $K_2$  is the first level at which physical connectivity, percolation phenomena, spatial axes and energetic thresholds appear. Its experimental programme therefore focuses on empirically testing the structure of  $\Omega(K_2)$ , the behaviour of the connectivity operator, the emergence of continuous symmetries, and the critical thresholds that govern the birth and death of physical continua.

Experiments at  $K_2$  correspond to tests of:

- percolation transitions,
- connectivity and cluster formation,
- threshold behaviour ( $\Theta_{\text{phys}}$ ),
- emergence and stability of the physical axis  $A_{\text{phys}}$ ,
- BKT-type transitions,
- coherence collapse and restoration,
- structural tension  $T_2(t)$  and its critical scaling.

This section provides the canonical experimental programme validating the  $K_1 \rightarrow K_2$  transition and the structure of physical continua.

### P.1.26 Objectives of $K_2$ Experiments

The experimental programme aims to test the following principles:

1. **Emergence of connectivity.** Verification of the connectivity function  $\text{Conn}(p)$ , the critical percolation threshold  $p_c$ , and the transition from disconnected to globally connected configurations.
2. **Existence of a physical axis.** Emergence of  $A_{\text{phys}}$  from the underlying connectivity structure, including its stability under perturbations.
3. **Threshold-induced structural transitions.** Measurement of the physical threshold  $\Theta_{\text{phys}}$  at which the system undergoes collapse or a phase transition.
4. **Coherence and decoherence behaviour.** Testing the coherence radius, correlation decay, and collapse under critical tension.
5. **Critical scaling laws.** Measurement of scaling exponents associated with  $K_2$  thresholds and phase transitions.
6. **Validation of the operator  $F_{1 \rightarrow 2}$ .** The transition from  $K_1$  to  $K_2$  must exhibit the predicted deformation of  $\partial\Omega$ , growth of dimension, and birth of the new axis.

### P.1.27 Type I Experiments: Percolation and Connectivity

These experiments test the structure of  $\Omega(K_2)$  through percolation.

The system is initialized on a discrete lattice or graph with occupation probability  $p$ , and the connectivity function  $\text{Conn}(p)$  is measured.

Key observables:

- size of the largest cluster,
- mean cluster size,
- susceptibility,
- correlation length,
- probability of global connectivity.

Experiments:

1. Scan  $p$  across the full range  $[0, 1]$  and measure  $\text{Conn}(p)$ .
2. Estimate the critical threshold  $p_c$ .
3. Validate the monotonicity of the connectivity operator:

$$\text{Conn}(p_1) \leq \text{Conn}(p_2) \quad \text{whenever} \quad p_1 < p_2.$$

4. Observe the divergence of the correlation length at  $p = p_c$ .

### P.1.28 Type II Experiments: Emergence of the Physical Axis

The physical axis  $A_{\text{phys}}$  is defined as the continuous limit of coarse-grained connectivity structures.

Experiments include:

1. coarse-graining occupied clusters at progressively larger length scales  $s$ ;
2. demonstrating convergence of the coarse-grained embedding to a smooth axis;

3. verifying that the emergent axis remains stable across multiple realizations;
4. measuring the variance of the reconstructed coordinate.

The goal is to show that  $A_{\text{phys}}$  is not imposed but emerges from the structure of  $\Omega(K_2)$ .

### **P.1.29 Type III Experiments: Energetic Thresholds $\Theta_{\text{phys}}$**

$K_2$  is the first level where energy enters as a structural property of the continuum. The threshold  $\Theta_{\text{phys}}$  determines:

- whether global connectivity persists,
- whether the physical axis remains stable,
- whether the continuum survives or collapses.

Experimental procedure:

1. prepare a connected configuration near the percolation threshold;
2. apply energy injections or deformations (random or structured);
3. measure tension  $T_2(t)$  as a function of applied perturbation;
4. determine the critical point where  $T_2(t) = \Theta_{\text{phys}}$ ;
5. observe collapse of global connectivity for  $T_2 > \Theta_{\text{phys}}$ .

These experiments validate the role of energetic thresholds in the Core model.

### **P.1.30 Type IV Experiments: BKT-Type Transitions**

The Berezinskii–Kosterlitz–Thouless (BKT) transition is a canonical  $K_2$ -level phenomenon associated with topological excitations and correlation decay.

Experiments:

- simulate a 2D XY-model or equivalent system;
- identify vortex–antivortex unbinding at the critical temperature;
- measure the universal jump in the helicity modulus;
- compare the behaviour with the predicted threshold structure in the Core (C.1-8: birth of the phase axis).

This experiment validates the mechanism of threshold-induced dimension change.

### **P.1.31 Type V Experiments: Coherence Collapse**

Coherence collapse experiments test:

- coherence radius,
- decay of correlation functions,
- loss of global order under tension.

Procedure:

1. initialize a near-ordered configuration;

2. introduce controlled randomness or noise;
3. track the decay of the correlation function  $G(r)$ ;
4. identify the critical noise amplitude at which coherence collapses;
5. compare behaviour to the tension threshold  $\Theta_{\text{phys}}$ .

### P.1.32 Type VI Experiments: Validation of the Operator $F_{1 \rightarrow 2}$

The operator  $F_{1 \rightarrow 2}$  describes the transition from  $K_1$  to  $K_2$ :

$$K_1 \xrightarrow{F_{1 \rightarrow 2}} K_2.$$

Experiments validate:

- deformation of  $\partial\Omega(K_1)$  into  $\partial\Omega(K_2)$ ;
- the appearance of the connectivity structure;
- the birth of the physical axis;
- the increase of dimension;
- the existence of a nontrivial threshold.

These experiments provide direct empirical grounding for the dimension-birth theorems.

### P.1.33 Summary

The  $K_2$  experimental programme establishes:

- the emergence of connectivity,
- the behaviour of critical thresholds,
- the birth of the physical axis,
- critical phenomena including BKT transitions,
- coherence behaviour,
- the validity of the  $K_1 \rightarrow K_2$  operator.

Because  $K_2$  is the foundation for all physical continua, its experimental programme is central to the empirical testing of the Ontology of Continua.

### P.1.34 Experiments for $K_3$

The continuum level  $K_3$  corresponds to the chemical domain: the space of atomic and molecular configurations, chemical bonds, reaction pathways, catalytic networks, energetic landscapes, and the first appearance of autocatalytic organisation (RAF structures). The experimental programme for  $K_3$  tests the emergence, stability, and transformation of chemical continua and validates the theoretical predictions regarding thresholds, flows, and structure of  $\Omega(K_3)$  derived in the Chemistry Block (U0.2–Final).

The goal of  $K_3$  experiments is to verify:

- the structure of the configuration space  $\Omega(K_3)$ ,
- stability and collapse of chemical continua,
- threshold behaviour (activation thresholds  $\Theta_{\text{chem}}$ ),
- the formation and closure of RAF networks,

- the behaviour of reaction flows  $J_{\text{chem}}$ ,
- constraints imposed by the boundary  $\partial\Omega(K_3)$ ,
- empirical support for the operator  $F_{2\rightarrow 3}$ .

### P.1.35 Objectives of $K_3$ Experiments

The experiments for  $K_3$  pursue the following main objectives:

1. **Validate the structure of  $\Omega(K_3)$  as the space of chemical configurations.** Demonstrate that only configurations compatible with valence rules, spatial geometry, and energy constraints belong to the accessible region of  $\Omega(K_3)$ .
2. **Measure activation thresholds  $\Theta_{\text{chem}}$ .** These thresholds govern whether reactions can proceed, whether complexes remain stable, and whether catalytic cycles can close.
3. **Verify the existence of chemical flows  $J_{\text{chem}}$ .** Establish that flows through reaction channels obey:

$$J = f(P, \Theta, T)$$

as predicted by the Core.

4. **Test emergence and stability of RAF networks.** Demonstrate the catalytic closure and persistence of reaction sets with sufficient autocatalytic structure.
5. **Validate the operator  $F_{2\rightarrow 3}$ .** Confirm the transition from purely physical continua ( $K_2$ ) to chemical continua ( $K_3$ ) via the birth of new axes and the appearance of chemical thresholds.

### P.1.36 Type I Experiments: Structure of $\Omega(K_3)$

These experiments aim to reconstruct the domain of chemically allowed states.

Procedure:

1. Enumerate or simulate atomic configurations under:
  - valence constraints,
  - bond-angle constraints,
  - steric hindrance,
  - quantum chemical stability conditions.
2. Identify stable and metastable regions of  $\Omega(K_3)$ .
3. Map the boundary  $\partial\Omega(K_3)$  as the set of configurations with:

$$E > \Theta_{\text{break}}$$

or impossible geometric/energetic constraints.

4. Compute connectivity within  $\Omega(K_3)$ : clusters of configurations connected via feasible reaction paths.

Expected results:  $\Omega(K_3)$  must exhibit distinct basins of attraction, reflecting chemical families and reaction pathways.

### P.1.37 Type II Experiments: Activation Thresholds $\Theta_{\text{chem}}$

Activation thresholds determine the feasibility of reactions:

$$\Theta_{\text{chem}} = E_{\text{activation}}.$$

Experiments:

1. Measure activation energies for controlled reactions (e.g., bond formation, bond breaking, polymerisation).
2. Validate the existence of a minimum energy input required for reaction pathways to open.
3. Test prediction:

$$J_{\text{chem}} = 0 \quad \text{for} \quad E < \Theta_{\text{chem}}.$$

These experiments test the Core prediction that thresholds restrict the accessible region of  $\Omega(K_3)$  and determine the structure of chemical dynamics.

### P.1.38 Type III Experiments: Reaction Flows $J_{\text{chem}}$

Chemical flows trace the movement of configurations along reaction pathways.

Experiments include:

- time-resolved spectroscopy,
- microfluidic reaction monitoring,
- concentration dynamics in batch and flow reactors,
- rate measurements for elementary reactions.

Goals:

1. Quantify  $J_{\text{chem}}$  as a function of potentials (chemical potentials  $\mu$ ), temperature, and thresholds.
2. Verify tension-driven regime changes predicted by Core:

$$T_{\text{chem}} > \Theta_{\text{route}} \Rightarrow \text{reaction pathway collapse.}$$

3. Confirm existence of alternative low-threshold pathways (catalysis).

### P.1.39 Type IV Experiments: Catalysis and Lowering of Thresholds

Catalysis is the hallmark of  $K_3$  and one of its defining mechanisms.

The Core predicts:

- unchanged  $\Omega(\text{reactants})$  and  $\Omega(\text{products})$ ,
- reduced threshold  $\Theta_{\text{chem}}$  in the presence of a catalyst,
- emergence of a catalytic sub-continuum  $K_{\text{path}}$ ,
- increased number of accessible trajectories,
- unchanged final state.

Experiments validate:

1. threshold lowering for catalysed vs. uncatalysed reactions,
2. existence of intermediate catalytic states,
3. expanded set of feasible reaction paths.

#### P.1.40 Type V Experiments: Autocatalytic Sets (RAF Networks)

Experiments to test the existence and stability of RAF networks include:

- constructing minimal experimental RAF systems,
- monitoring self-sustaining production cycles,
- applying perturbations to test closure conditions:

$$\text{RAF closed} \Leftrightarrow \forall r \in R_{\text{RAF}} : \text{reactants produced within RAF.}$$

- validating the existence of catalytic feedback loops,
- identifying collapse thresholds of RAF sets.

Expected outcomes: RAF networks exhibit sharp thresholds of existence and collapse, in line with  $K_3$  predictions.

#### P.1.41 Type VI Experiments: Failure and Collapse of Chemical Continua

Experiments analyse breakdown scenarios:

- temperature-induced decomposition,
- pH-driven instability,
- oxidation and reduction breakdown,
- photochemical destruction.

Core prediction: Collapse occurs when chemical tension exceeds the structural threshold:

$$T_{\text{chem}} > \Theta_{\text{collapse}}.$$

Experiments should map the threshold landscape for various molecular families.

#### P.1.42 Type VII Experiments: Validation of the Operator $F_{2 \rightarrow 3}$

The transition from  $K_2$  to  $K_3$  involves:

- birth of chemical axes (bond states, electron configurations),
- deformation of  $\partial\Omega(K_2)$  into  $\partial\Omega(K_3)$ ,
- appearance of activation thresholds,
- creation of structured reaction pathways,
- emergence of catalytic cycles.

Experiments validating  $F_{2 \rightarrow 3}$  include:

1. reconstruction of low-dimensional potential surfaces,
2. identification of emergent minima corresponding to stable molecules,
3. confirmation of chemical axis birth through stability diagrams,
4. demonstration that connectivity now reflects chemical similarity, not spatial adjacency.

### P.1.43 Summary

The experimental programme for  $K_3$  confirms:

- structure of chemical configuration space,
- activation thresholds and reaction feasibility constraints,
- behaviour of chemical flows,
- catalytic lowering of thresholds,
- existence and stability of RAF networks,
- mechanisms of chemical collapse,
- the validity of  $K_2 \rightarrow K_3$  transition.

These experiments ground the chemical level of the Ontology of Continua and provide the empirical foundation for the emergence of  $K_4$ .

### P.1.44 Experiments for $K_4$

The continuum level  $K_4$  corresponds to the domain of protocellular biological organisation: membranes, osmotic and electrochemical gradients, transport processes, early metabolic potentials, regulatory precursors, and the first stable biological cycles. This level marks the emergence of a genuine internal–external boundary  $\partial\Omega(K_4)$  and the construction of a structured bioenergetic landscape. The experimental programme for  $K_4$  serves to test the predictions of the Chemistry–Biology transition formalised in Biology U0.3b.

The central goal is to empirically validate:

- the birth and stability of biological boundaries,
- the appearance and maintenance of gradients  $P_{\text{bio}}$ ,
- the existence of transport flows  $J_{\text{in/out}}$ ,
- the behaviour of membrane patches ( $L_\alpha$ ,  $L_\beta$ ,  $L_o$ ),
- the structure of bioenergetic cycles,
- threshold conditions for excitability and regulation,
- collapse phenomena of proto-biological continua.

### P.1.45 Objectives of the $K_4$ Experimental Programme

Experimental tests for  $K_4$  pursue the following aims:

1. **Validate the emergence of the membrane boundary  $\partial\Omega(K_4)$ .** Reproduce self-assembly regimes where amphiphiles form vesicles, protocells, or patch-stabilised surfaces.
2. **Verify the existence of internal–external gradients  $P_{\text{bio}}$ .** Demonstrate stable pH, ion, redox, and concentration differences supported by membrane integrity.
3. **Measure biological thresholds  $\Theta_{\text{grad}}$ ,  $\Theta_{\text{osm}}$ ,  $\Theta_{\text{perm}}$ ,  $\Theta_{\text{charge}}$ .**
4. **Validate the core flows  $J_{\text{in}}$ ,  $J_{\text{out}}$ ,  $J_{\text{pump}}$ ,  $J_{\text{redox}}$ ,  $J_{\text{buffer}}$ .**
5. **Test patch-level dynamical behaviour.** Demonstrate spatial heterogeneity and graininess of  $\partial\Omega$ .
6. **Confirm conditions for proto-excitability (early  $A_{\text{exc}}$ ).** Validate pre-spike dynamics and early electrical instabilities.
7. **Empirically test collapse conditions** for  $K_4$  continua.



### P.1.46 Type I Experiments: Self-Assembly and Birth of $\partial\Omega(K_4)$

These experiments reconstruct the transition from  $K_3$  chemical aggregates to membrane-bounded  $K_4$  protocells.

#### Procedures:

1. Assemble fatty-acid, phospholipid, or mixed amphiphile vesicles under variable pH, salt concentration, and temperature.
2. Characterise vesicle formation using:
  - fluorescence microscopy,
  - cryo-EM,
  - scattering techniques,
  - microfluidic encapsulation.
3. Quantify membrane tension  $T_{\text{mem}}$  and bending modulus.

#### Core predictions validated:

- Existence of a stable boundary region  $\partial\Omega(K_4)$ .
- Window of stability for membrane closure ( $\Theta_{\text{closure}}$ ).
- Transition from disordered aggregates to coherent compartments.

### P.1.47 Type II Experiments: Appearance and Stability of Gradients

Gradients  $P_{\text{grad}}$ ,  $P_{\text{ion}}$ ,  $P_{\text{redox}}$ ,  $P_{\text{pH}}$  are defining potentials of  $K_4$ .

#### Experimental programme:

1. Introduce controlled pH/ion gradients across protocell membranes.
2. Measure:
  - membrane potential  $\Delta V$ ,
  - redox potential differences,
  - osmotic pressure  $\Delta\pi$ ,
  - proton gradients.
3. Determine leakage rates and effective permeability.

#### Predictions tested:

- Thresholds  $\Theta_{\text{grad}}$ ,  $\Theta_{\text{osm}}$ ,  $\Theta_{\text{charge}}$  define gradient survival.
- Collapse of gradients occurs when:

$$J_{\text{leak}} > J_{\text{pump}}.$$

- Stability of gradients requires:

$$T_{\text{mem}} < \Theta_{\text{perm}}.$$

### P.1.48 Type III Experiments: Transport Flows $J_{\text{in/out}}$

Transport through primitive membranes is central for  $K_4$  dynamics.

**Experimental procedures:**

- Measure passive diffusion rates for ions and small molecules.
- Perform experiments on facilitated diffusion via early channel analogues (peptides, pores, mineral surfaces).
- Characterise active and pseudo-active transport using:
  - chemical pumps,
  - redox-driven transport,
  - pH-driven uptake mechanisms.

**Core predictions:**

- Existence of flow regimes ( $J_{\text{diffusion}}$ ,  $J_{\text{pump}}$ ,  $J_{\text{leak}}$ ) constrained by thresholds.
- Membrane patch heterogeneity produces spatially variable transport.
- Breakdown occurs when  $J_{\text{leak}}$  exceeds local patch stability thresholds  $\Theta_{\text{perm}}$ .

**P.1.49 Type IV Experiments: Patch Dynamics and Spatial Graininess**

The membrane boundary is not uniform but patch-structured ( $L\alpha$ ,  $L\beta$ ,  $L_o$  phases).

**Experimental programme:**

1. Construct vesicles with controlled lipid heterogeneity.
2. Image and map dynamic patches using:
  - STED or super-resolution microscopy,
  - AFM force mapping,
  - fluorescence lifetime imaging.
3. Track patch transitions and flickering.

**Predicted outcomes:**

- Emergence of local thresholds  $\Theta_i$  for distinct patches.
- Existence of localised collapse events (bursting, flicker instability).
- Patch-dependent regulation of proto-excitability.

**P.1.50 Type V Experiments: Bioenergetic and Buffering Cycles**

Core bioenergetic cycles, predicted by  $K_4$ , must be detectable:

- metabolic drive cycles ( $C_{\text{energy}}$ ),
- buffer cycles ( $C_{\text{buffer}}$ ),
- proton and ion recycling ( $C_{\text{pump/leak}}$ ).

**Experimental validation:**

1. Construct minimal systems supporting sustained redox or pH gradients.
2. Demonstrate periodic or quasi-periodic flow cycles.
3. Investigate threshold behaviour for cycle persistence.

### P.1.51 Type VI Experiments: Pre-Regulation and Signal Networks

Early regulatory networks must satisfy:

- thresholding behaviour ( $\Theta_{\text{reg}}$ ),
- inhibition/activation cycles ( $C_{\text{reg}}$ ),
- stochastic logic structure defined by  $A_{\text{logic}}$ .

#### Experimental approaches:

1. Construct ligand–membrane or ligand–peptide signalling systems.
2. Measure probabilistic switching behaviour.
3. Apply noise to test stability of logic transitions.

### P.1.52 Type VII Experiments: Proto-Excitability and Early Electrical Dynamics

The birth of the electrical axis ( $A_{\text{exc}}$ ) and proto-spike behaviour is the defining sign of  $K_4 \rightarrow K_5$  transition.

#### Procedures:

- Introduce cation/anion gradients and measure  $\Delta V$  evolution.
- Test primitive channels (peptide pores, mineral pores).
- Detect transient voltage spikes (proto-spikes).

#### Predictions validated:

- A minimal excitability threshold  $\Theta_{\text{exc}}$  exists.
- The spike regime emerges when:
$$T_{\text{elec}} > \Theta_{\text{exc}}.$$
- Patch dynamics critically modulate proto-spike behaviour.

### P.1.53 Type VIII Experiments: Collapse of $K_4$ Continua

Breakdown mechanisms include:

- osmotic bursting,
- leakage-induced collapse,
- pH-driven disintegration,
- loss of redox balance,
- membrane tension runaway.

#### Core prediction:

$$T_{\text{total}}(K_4) > \Theta_{\text{collapse}} \Rightarrow \Omega(K_4) = \emptyset.$$

Experiments must map collapse thresholds as functions of membrane composition, patch distribution, and metabolic drive.

### P.1.54 Summary

The experiments for  $K_4$  validate:

- emergence of biological boundaries,
- stability of gradients and internal potentials,
- structure of transport flows,
- patch-graininess of  $\partial\Omega(K_4)$ ,
- operation of early bioenergetic and regulatory cycles,
- birth of excitability,
- collapse mechanisms of the first biological continua.

These empirical results anchor the  $K_4$  level and establish the experimental foundation for the emergence of  $K_5$ .

### P.1.55 Experiments for $K_5$

The continuum level  $K_5$  marks the emergence of bioelectrical excitability: stable ion gradients, selective ion channels, membrane potentials  $\Delta V$ , proto-spike dynamics, patch-mediated gating and the first excitation–recovery cycles. Experiments at this level test the transition from  $K_4$  protocells to early neural continua and validate the full structure of the electrical axis  $A_{\text{exc}}$ , flows  $J_{\text{ion}}$ , thresholds  $\Theta_{\text{exc}}$ , membrane noise, and proto-circuit formation.

This experimental programme operationalises the predictions of Biology U0.3b and  $K_5$  Run 6, covering excitability, regulation, cycles, collapse, and the birth of proto-networks.

### P.1.56 Objectives of the $K_5$ Experimental Programme

Experiments aim to validate:

- formation and stability of membrane potentials  $\Delta V$ ,
- dynamics of ion channels and gating transitions,
- existence of proto-spikes and thresholds  $\Theta_{\text{exc}}$ ,
- structure of excitation–recovery cycles  $C_{\text{exc}}$ ,
- role of membrane patchiness in gating behaviour,
- emergence of stochastic logic  $A_{\text{logic}}$ ,
- collapse mechanisms of early excitable systems.

### P.1.57 Type I Experiments: Emergence of Membrane Potential $\Delta V$

Experiments test whether minimal biological systems can maintain stable voltage differences across membranes.

#### Procedures:

1. Prepare protocell or lipid vesicle systems with asymmetrical distributions of ions ( $\text{K}^+$ ,  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{Ca}^{2+}$ ).
2. Introduce primitive ion channels or pores (peptide pores, mineral channels).
3. Measure  $\Delta V$  using:

- microelectrode techniques,
  - voltage-sensitive dyes,
  - patch-clamp on GUVs.
4. Vary environmental parameters: pH, ionic strength, temperature.

**Predictions validated:**

- Stable  $\Delta V$  emerges when
 
$$J_{\text{pump}} > J_{\text{leak}}.$$
- Existence of a minimal stability threshold  $\Theta_{\text{charge}}$ .
- Membrane potential depends on patch composition.

**P.1.58 Type II Experiments: Ion Channel Opening, Closing and Gating**

Ion channels are the structural heart of  $K_5$ .

**Experimental programme:**

1. Reconstitute primitive ion channels in lipid bilayers.
2. Perform patch-clamp recordings at:
  - single-channel resolution,
  - whole-vesicle configuration.
3. Measure:
  - conductance  $g_{\text{channel}}$ ,
  - open/closed dwell times  $\tau_{\text{open}}$ ,  $\tau_{\text{close}}$ ,
  - selectivity  $S_{\text{selectivity}}$ ,
  - noise levels  $\eta_{\text{noise}}$ .

**Predictions validated:**

- Distinct gating states (open/closed/leaky/blocked) exist.
- Channels obey threshold conditions  $\Theta_{\text{open}}$ ,  $\Theta_{\text{close}}$ ,  $\Theta_{\text{leak}}$ .
- Noise-induced transitions follow predictions of the stochastic logic model  $A_{\text{logic}}$ .

**P.1.59 Type III Experiments: Proto-Spike Dynamics**

The proto-spike is the defining event of  $K_5$ : a transient depolarisation–repolarisation cycle.

**Procedures:**

1. Induce proto-spikes by controlled changes in ionic gradients.
2. Use high-speed voltage imaging and patch-clamp.
3. Characterise spike parameters:
  - amplitude,
  - duration,
  - rise/decay time,
  - refractory period.

**Core predictions:**

- Existence of a sharp excitability threshold  $\Theta_{\text{exc}}$ :

$$T_{\text{elec}} > \Theta_{\text{exc}}.$$

- Spike morphology depends on membrane patch regime ( $L_\alpha$ ,  $L_\beta$ ,  $L_0$ ).
- Recovery dynamics produce a genuine  $C_{\text{exc}}$  cycle: excitation  $\rightarrow$  depolarisation  $\rightarrow$  recovery  $\rightarrow$  rest.

**P.1.60 Type IV Experiments: Patch-Dependent Electrical Dynamics**

Patch structure deeply modulates excitability.

**Experimental programme:**

1. Construct vesicles with controlled patch heterogeneity.
2. Use super-resolution techniques to map local  $\Delta V_i$ .
3. Measure spatially resolved channel behaviour across patches.

**Predictions validated:**

- Local thresholds  $\Theta_{\text{exc},i}$  differ between patches.
- Proto-spike propagation may halt or amplify at patch boundaries.
- Patch flickering produces stochastic excitability windows.

**P.1.61 Type V Experiments: Excitation–Recovery Cycles**

$K_5$  predicts the existence of structured cyclic dynamics underlying the first excitable continua.

**Procedures:**

1. Trigger repeatable proto-spikes.
2. Measure:
  - inter-spike intervals,
  - refractory periods,
  - amplitude adaptation.
3. Identify oscillatory or quasi-periodic regimes.

**Predictions validated:**

- Existence of stable or metastable  $C_{\text{exc}}$  cycles.
- Appearance of recovery cycles  $C_{\text{recovery}}$ .
- Breakdown of cycles when thresholds shift (temperature, lipid composition, noise).

**P.1.62 Type VI Experiments: Stochastic Logic and Early Regulatory Behaviour**

As predicted by Biology U0.3b, early membranes implement probabilistic logic.

**Experimental programme:**

1. Quantify probability of channel opening under variable inputs.
2. Characterise switching curves  $p_{\text{open}}(V)$ .
3. Perturb channels with oscillatory and noisy stimuli.

**Predictions validated:**

- Logic gates emerge from channel behaviour (AND-like, OR-like).
- Threshold  $\Theta_{\text{logic}}$  defines stable logic.
- Collapse occurs when logic entropy  $S_{\text{logic}}$  exceeds  $S_{\text{max}}$ .

**P.1.63 Type VII Experiments: Proto-Networks and Early Connectivity**

The final prediction of  $K_5$  is the appearance of structured electrical interactions between protocells.

**Procedures:**

1. Arrange protocells in microfluidic arrays or gels.
2. Introduce ionic bridges or primitive gap-junction analogues.
3. Measure correlated  $\Delta V(t)$  dynamics across multiple cells.

**Predictions validated:**

- Emergence of primitive network motifs (chains, bifurcations).
- Synchronisation and entrainment of proto-spikes.
- Existence of a minimal connectivity threshold  $\Theta_{\text{conn}}$  for network behaviour.

**P.1.64 Type VIII Experiments: Collapse of  $K_5$  Continua**

Collapse at level  $K_5$  occurs when the electrical axis cannot be stabilised:

- runaway depolarisation,
- loss of  $\Delta V$  maintenance,
- destructive noise in channel behaviour,
- breakdown of recovery cycles,
- patch-driven instabilities.

**Prediction:**

$$T_{\text{elec}}(K_5) > \Theta_{\text{collapse}} \Rightarrow \Omega(K_5) = \emptyset.$$

Mapping these thresholds establishes the survivability range of early excitable systems and identifies the stable region for the transition  $K_5 \rightarrow K_6$ .

### P.1.65 Summary

Experiments for  $K_5$  validate the structural core of early excitability:

- membrane potentials and ion gradients,
- ion channel gating and stochasticity,
- proto-spike morphology and thresholds,
- patch-modulated excitability,
- excitation–recovery cycles,
- proto-network formation,
- collapse phenomenology.

Together, these experimental results confirm that  $K_5$  is a genuine excitable continuum and provide the empirical foundation for the emergence of  $K_6$  cognitive dynamics.

### P.1.66 Experiments for $K_6$

Continuum level  $K_6$  marks the emergence of cognitive dynamics: stable internal representations, binding operations, prediction, model coherence, and memory stabilisation. Experiments for this level must therefore probe structural, computational and behavioural manifestations of the cognitive cycles  $C_{\text{model}}$ ,  $C_{\text{bind}}$ ,  $C_{\text{pred}}$ , and  $C_{\text{mem}}$ . They operationalise predictions of the  $K_6$  Run, including the representational–predictive architecture, cognitive thresholds, flow patterns  $J_6$ , and the collapse conditions of cognitive continua.

### P.1.67 Objectives of the $K_6$ Experimental Programme

The goals of the experimental programme are to validate:

- the existence of structured representational axes  $A_{\text{rep}}$  and their corresponding state space,
- binding operations and multi-feature integration,
- prediction and prediction-error dynamics,
- model stability and collapse thresholds,
- memory formation and maintenance cycles,
- structural tension  $T_6$  in cognitive tasks,
- flow-based organisation of cognitive processes  $J_6$ .

### P.1.68 Type I Experiments: Representational Capacity and Axes

These experiments test the basic structure of cognitive representations.

#### Procedures:

1. Use high-dimensional stimulus sets (visual, auditory, multisensory) to map representational manifolds.
2. Measure neural population activity (fMRI, MEG, high-density EEG, multi-unit recordings in model organisms).
3. Apply dimensionality reduction and manifold learning (PCA, t-SNE, UMAP, Laplacian eigenmaps).



**Predictions validated:**

- Existence of stable representational axes  $A_{\text{rep}}$ .
- Representational geometry follows  $\partial\Omega(K_6)$  constraints (smooth low-dimensional manifolds).
- Breakdown when  $\Theta_{\text{rep}}$  is exceeded (noise, overload, fragmentation).

**P.1.69 Type II Experiments: Binding and Multi-Feature Integration**

Binding is the characteristic operation of  $K_6$ .

**Experimental programme:**

1. Use classical binding paradigms: colour–shape binding, location–identity binding, auditory–visual binding.
2. Measure behavioural accuracy and reaction times.
3. Perform neural recordings to detect synchrony or phase-locking patterns.
4. Manipulate task complexity to push the system towards the binding threshold  $\Theta_{\text{bind}}$ .

**Predictions validated:**

- A sharp capacity limit for binding (bounded representational conjunctions).
- Emergence of synchronisation patterns supporting  $C_{\text{bind}}$  cycles.
- Collapse of binding when tension  $T_6$  exceeds  $\Theta_{\text{bind}}$ .

**P.1.70 Type III Experiments: Prediction and Prediction-Error Dynamics**

Prediction is a central mechanism of  $K_6$ .

**Procedures:**

1. Use sequence learning, next-item prediction, or sensory prediction tasks.
2. Measure neural correlates of prediction error: mismatch negativity (MMN), temporal response functions, error-related potentials.
3. Modulate uncertainty or volatility of stimuli.

**Predictions validated:**

- Existence of a prediction threshold  $\Theta_{\text{pred}}$  below which reliable prediction occurs.
- Structured prediction cycles  $C_{\text{pred}}$ .
- Collapse when prediction error saturates the threshold:

$$T_{\text{pred}} > \Theta_{\text{pred}} \Rightarrow C_{\text{pred}} \text{ breaks.}$$

**P.1.71 Type IV Experiments: Internal Models and Coherence Testing**

These experiments probe the structural integrity of cognitive models.

**Experimental programme:**

1. Conduct tasks requiring internal scene construction, inference, or hierarchical reasoning.
2. Use perturbations including:
  - inconsistent stimuli,
  - contradictory evidence,
  - rapid reversals of contingencies.
3. Measure model coherence indicators: consistency of beliefs, updating speed, stability of latent-state estimates.

**Predictions validated:**

- A coherence threshold  $\Theta_{\text{model}}$  exists.
- When exceeded, model collapse occurs: fragmentation of latent representations, inconsistent inference, runaway prediction error.
- Stable internal cycles  $C_{\text{model}}$  support ongoing coherence.

**P.1.72 Type V Experiments: Memory Encoding, Storage and Retrieval**

Memory cycles distinguish  $K_6$  from purely excitable continua.

**Procedures:**

1. Use working memory, episodic memory or associative memory tasks.
2. Apply high-density EEG, MEG, calcium imaging or single-unit electrophysiology.
3. Track neural signatures of encoding, consolidation and retrieval.

**Predictions validated:**

- Existence of memory cycles  $C_{\text{mem}}$ .
- Memory stability depends on thresholds  $\Theta_{\text{mem}}^{\min}$  and  $\Theta_{\text{mem}}^{\max}$ .
- Collapse when memory interference raises tension above  $\Theta_{\text{mem}}$ .

**P.1.73 Type VI Experiments: Flow Dynamics  $J_6$  and Cognitive Load**

Flow-based organisation is essential to the  $K_6$  framework.

**Experimental programme:**

1. Vary cognitive load across tasks (dual-task paradigms, high-complexity reasoning, sustained attention).
2. Measure variability of information flow via transfer entropy, Granger causality, directed connectivity measures.
3. Determine load thresholds where flows become unstable.

**Predictions validated:**

- Cognitive flows  $J_6$  follow structural constraints of  $A_6$ ,  $P_6$  and  $\Theta_6$ .
- Overload produces a rise in tension  $T_6$  and flow collapse.

**P.1.74 Type VII Experiments: Cognitive Collapse and Recovery**

Cognitive collapse is the analogue of excitability breakdown at  $K_5$ , now expressed in representational and predictive terms.

**Procedures:**

1. Introduce high-noise, high-volatility or contradictory tasks.
2. Track breakdown of coherence, binding, prediction and memory.
3. Apply recovery protocols: rest, structured cues, controlled re-exposure, process-level hints.

**Predictions validated:**

- Collapse corresponds to:

$$\Omega(K_6) \rightarrow \emptyset, \quad k_6 \rightarrow 0.$$

- Collapse is preceded by runaway increase in tension  $T_6$ .
- Recovery depends on rebuilding cycles  $C_{\text{model}}$  and  $C_{\text{bind}}$ .

**P.1.75 Summary**

Experiments for  $K_6$  validate the core architecture of cognitive continua:

- representational axes and geometry,
- binding dynamics and capacity limits,
- prediction and prediction-error mechanisms,
- model coherence and collapse thresholds,
- memory cycles and stability conditions,
- information flows  $J_6$  under load,
- collapse and recovery dynamics.

Together, these results provide the empirical and computational foundation for the transition  $K_6 \rightarrow K_7$  and the emergence of social continua.

**P.1.76 Experiments for  $K_7$** 

Continuum level  $K_7$  corresponds to social systems: groups, institutions, communication networks, normative structures, resource flows, and trust dynamics. Experiments for this level probe the existence and stability of social cycles  $C_{\text{inst}}$ ,  $C_{\text{norm}}$ ,  $C_{\text{comm}}$ , the threshold landscape  $\Theta_7 = \{\Theta_{\text{trust}}, \Theta_{\text{coh}}, \Theta_{\text{stab}}, \Theta_{\text{frag}}\}$ , and the flows  $J_7$  (communication, resources, influence, norms).

Because  $K_7$  systems operate at population scale, the experimental programme combines behavioural laboratory methods, large-scale data analysis, computational simulations, and institutional stress testing.

### **P.1.77 Objectives of the $K_7$ Experimental Programme**

Experiments at this level aim to validate:

- the existence of structured social axes  $A_7$  (trust, hierarchy, group identity, communication channels),
- social potentials  $P_{\text{soc}}$  (resource gradients, normative pressures, influence fields),
- the threshold landscape governing stability and fragility,
- stable social cycles (institutional, communicative, normative),
- tension-driven collapse mechanisms,
- transitions between group states under controlled perturbations.

### **P.1.78 Type I Experiments: Trust Dynamics and Thresholds**

Trust is the primary axis of  $K_7$ . Testing the trust threshold  $\Theta_{\text{trust}}$  is essential.

#### **Experimental procedures:**

1. Classical trust games, repeated games, coordination games.
2. Varying uncertainty, payoff structures, noise, partner switching.
3. Measuring:
  - trust formation,
  - trust decay,
  - trust restoration.
4. Manipulate misinformation or communication clarity.

#### **Predictions validated:**

- Nonlinear trust response near  $\Theta_{\text{trust}}$ .
- Collapse of coordination when trust falls below threshold.
- Stabilisation when supportive communication flows  $J_{\text{comm}}$  increase.

### **P.1.79 Type II Experiments: Normative Cycles and Social Coherence**

Norms form a structural potential  $P_{\text{norm}}$  and generate stabilising cycles  $C_{\text{norm}}$ .

#### **Experimental programme:**

1. Rule-following tasks with internal conflicts.
2. Social dilemmas (public goods, commons problems).
3. Manipulating:
  - normative salience,
  - group identity,
  - enforcement strength.
4. Measuring norm internalisation and breakdown.

**Predictions validated:**

- Normative coherence occurs when  $T_{\text{norm}} < \Theta_{\text{coh}}$ .
- Normative collapse produces fragmentation of  $A_7$ .
- Cycles  $C_{\text{norm}}$  maintain stability.

**P.1.80 Type III Experiments: Communication Networks and Flow Stability**

Communication flows  $J_{\text{comm}}$  shape the structure of  $K_7$ .

**Procedures:**

1. Controlled communication network experiments (laboratory micro-societies, online platforms).
2. Vary network topology (centralised, decentralised, modular).
3. Introduce noise, latency, bottlenecks, misinformation.
4. Track information propagation, consensus formation, and network resilience.

**Predictions validated:**

- Stable  $J_{\text{comm}}$  flows require  $T_7$  below  $\Theta_{\text{flow}}$ .
- Network bottlenecks push tension above critical levels, producing fragmentation.
- Consensus cycles  $C_{\text{comm}}$  emerge under low noise.

**P.1.81 Type IV Experiments: Institutional Stability and Collapse**

Institutions instantiate the social cycles  $C_{\text{inst}}$ .

**Experimental programme:**

1. Stress testing of institutional processes (governance labs, voting simulations, procedural perturbations).
2. Controlled failures of subcomponents (resource shocks, procedural inconsistencies, overload).
3. Measuring:
  - response time,
  - coherence,
  - recovery,
  - fragmentation.

**Predictions validated:**

- A stability threshold  $\Theta_{\text{stab}}$  exists.
- Exceeding stability threshold triggers institutional collapse:

$$\Omega(K_7) \rightarrow \emptyset.$$

- Recovery requires re-establishing  $C_{\text{inst}}$  cycles.

### **P.1.82 Type V Experiments: Resource Flows and Social Gradients**

Resource potentials  $P_{\text{res}}$  and flows  $J_{\text{res}}$  govern large-scale social tension.

#### **Procedures:**

1. Simulate resource inequalities in laboratory or online groups.
2. Introduce controlled shocks (redistribution, scarcity).
3. Measure systemic tension  $T_{\text{res}}$  and group stability.

#### **Predictions validated:**

- High resource gradients destabilise  $A_7$ .
- A redistribution threshold  $\Theta_{\text{res}}$  defines whether inequality reduces or increases tension.
- Collapse occurs when resource flows cease (network freeze).

### **P.1.83 Type VI Experiments: Social Identity and Fragmentation**

Group identity modulates  $\Omega(K_7)$  and influences all thresholds.

#### **Experimental programme:**

1. Minimal group paradigms.
2. Manipulation of identity salience.
3. Introduce cross-cutting identities and conflicting signals.
4. Measure cooperation, polarisation, and fragmentation.

#### **Predictions validated:**

- Identity alignment lowers tension  $T_7$ .
- Identity conflict pushes tension above  $\Theta_{\text{frag}}$ .
- Fragmentation corresponds to collapse of  $C_{\text{comm}}$  and  $C_{\text{norm}}$ .

### **P.1.84 Type VII Experiments: Collapse and Reorganisation of Social Continua**

Collapse at  $K_7$  is the social analogue of cognitive breakdown at  $K_6$ .

#### **Procedures:**

1. Induce high-tension conditions: misinformation cascades, resource interruption, governance overload, institutional mismatch.
2. Track:
  - cycle degradation,
  - tension divergence,
  - collapse of coherence,
  - spontaneous reorganisation.
3. Probe the transition to  $K_8$  (civilisational-level cycles).

**Predictions validated:**

- Collapse corresponds to:

$$k_7 \rightarrow 0, \quad \Omega(K_7) \rightarrow \emptyset.$$

- Reorganisation requires formation of new  $C_{\text{inst}}$  or transition to emergent  $K_8$  axes.

**P.1.85 Summary**

Experiments for  $K_7$  empirically validate the structural theory of social continua:

- trust dynamics and thresholds,
- normative cycles and coherence,
- communication flows and stability,
- institutional robustness and collapse,
- resource gradients and social tension,
- identity-driven fragmentation,
- collapse, recovery and transition to civilisational dynamics.

Together, these experiments form the empirical foundation for the transition  $K_7 \rightarrow K_8$  and the emergence of civilisational continua.

**P.1.86 Experiments for  $K_8$** 

Continuum level  $K_8$  corresponds to civilisational systems: large-scale infrastructures, written symbolic systems, scientific and technological structures, long-range institutions, and cross-generational flows of information, energy, resources, and norms. Experiments at this level validate the structure, thresholds, cycles, and collapse mechanisms of civilisational continua.

Because  $K_8$  spans centuries and large populations, the experimental programme consists of a combination of:

- historical reconstructions,
- computational simulations,
- comparative civilisational studies,
- large-scale data analysis,
- controlled stress-tests of modern infrastructures,
- detection of universal invariant patterns.

**P.1.87 Objectives of the  $K_8$  Experimental Programme**

The experimental goals are to validate the formal structure:

- axes  $A_8$  (infrastructure, symbolic systems, scientific paradigms),
- potentials  $P_8$  (symbolic energy, technological potential, information density, institutional load),
- threshold landscape  $\Theta_8$  (infrastructure collapse thresholds, information coherence thresholds, resource and demographic thresholds),
- civilisational cycles  $C_8$  (science, bureaucracy, taxation, literacy, infrastructure maintenance),
- collapse and reorganisation dynamics,
- transition conditions for  $K_8 \rightarrow K_9$  (meta-theoretical consolidation).

### **P.1.88 Type I Experiments: Written Symbolic Systems and $\Theta_{\text{sym}}$**

Civilisational stability requires stable written symbolic systems. The symbolic potential  $P_{\text{sym}}$  supports coherence and long-term memory.

#### **Experimental programme:**

1. Historical analysis of script emergence, degradation, and replacement.
2. Quantitative modelling of literacy spread (percolation models).
3. Laboratory micro-societies using constructed scripts.
4. Measuring symbol retention, mutation, compression, and decay.

#### **Predictions validated:**

- A critical symbolic threshold  $\Theta_{\text{sym}}$  is required for civilisational memory persistence.
- Below  $\Theta_{\text{sym}}$ , symbolic collapse occurs and  $k_8$  drops.
- Symbolic cycles  $C_{\text{sym}}$  stabilise knowledge across generations.

### **P.1.89 Type II Experiments: Infrastructure Dynamics and Collapse Thresholds**

Infrastructure axes include transport, energy, communication, water, sanitation, and administrative systems.

#### **Procedures:**

1. Infrastructure stress simulations (load, failure propagation, maintenance decay, redundancy variation).
2. Historical reconstructions of infrastructural collapse (Rome, Maya, Bronze Age, Angkor, modern blackouts).
3. Quantifying dependency networks and vulnerability.
4. Controlled perturbations in real infrastructures (limited scale, e.g. grid stress-tests, redundancy switching).

#### **Predictions validated:**

- Infrastructure collapse occurs when  $T_{\text{infra}} > \Theta_{\text{infra}}$ .
- Redundancy lowers structural tension.
- Infrastructure cycles  $C_{\text{infra}}$  extend civilisational lifespan.

### **P.1.90 Type III Experiments: Scientific Cycles and Paradigm Transitions**

Science generates the civilisational potential  $P_{\text{sci}}$ .

#### **Experimental procedures:**

1. Scientometric analysis of paradigm formation, expansion, and collapse.
2. Modelling of knowledge networks and innovation percolation.
3. Studying growth and saturation cycles  $C_{\text{sci}}$ .
4. Controlled experiments on epistemic communities (collaborative reasoning, distributed inference).



**Predictions validated:**

- Paradigm transitions occur when epistemic tension exceeds  $\Theta_{\text{sci}}$ .
- Knowledge systems exhibit universal growth–collapse–renewal cycles.
- The transition  $K_8 \rightarrow K_9$  requires coherent meta-theoretical integration.

**P.1.91 Type IV Experiments: Technological Acceleration and Stability**

Technological potential  $P_{\text{tech}}$  can destabilise  $K_8$  if its growth rate exceeds civilisational capacity.

**Procedures:**

1. Technological acceleration curve analysis.
2. Stress tests for technology–infrastructure mismatch.
3. Simulation of automation, digitisation, and AI shocks.
4. Multi-agent models of labour, knowledge, and skill redistribution.

**Predictions validated:**

- A critical threshold  $\Theta_{\text{tech}}$  governs technological sustainability.
- Overshooting leads to runaway tension and fragmentation.
- Balanced cycles  $C_{\text{tech}}$  stabilise growth.

**P.1.92 Type V Experiments: Information Density, Cohesion, and Collapse**

Information density and coherence shape the potential  $P_{\text{info}}$ .

**Experimental programme:**

1. Measuring fragmentation in media ecosystems.
2. Modelling information cascades, echo chambers, and global coherence.
3. Historical analysis of breakdowns in knowledge systems.
4. Controlled micro-society experiments with communication overload.

**Predictions validated:**

- Excess information tension  $T_{\text{info}} > \Theta_{\text{coh}}$  generates fragmentation.
- Civilisations collapse into localised  $K_7$  clusters when global coherence is lost.
- Cohesion cycles  $C_{\text{info}}$  are necessary for  $k_8 > 0$ .

**P.1.93 Type VI Experiments: Demography, Resources, and Long-Range Flows**

Civilisations depend on sustainable resource and demographic flows  $J_8$ .

**Procedures:**

1. Large-scale demographic modelling.
2. Historical reconstructions of resource pathway collapse.
3. Controlled simulations of migration and trade disruptions.
4. Multi-region agent-based civilisational models.

**Predictions validated:**

- Civilisational collapse occurs when long-range flows drop below  $\Theta_{\text{flow}}$ .
- Demographic gradients produce structural tension.
- Stable  $J_8$  flows maintain  $k_8$  and prevent fragmentation.

**P.1.94 Type VII Experiments: Civilisational Collapse and Reorganisation**

Civilisational collapse is a structural phenomenon, not a historical accident.

**Experimental programme:**

1. Reconstruction of collapse signatures across civilisations.
2. Comparative analysis of failure modes (infrastructure collapse, symbolic collapse, overextension, ecological collapse).
3. Simulation of collapse cascades on interconnected networks.
4. Study of spontaneous reorganisation and new axis formation.

**Predictions validated:**

- Collapse corresponds to  $k_8 \rightarrow 0, \quad \Omega(K_8) \rightarrow \emptyset.$
- Reorganisation requires new stable cycles  $C_8$  or transition to  $K_9$  (meta-theoretical consolidation).

**P.1.95 Summary**

Experiments for  $K_8$  establish civilisational-scale validation for the Ontology of Continua:

- symbolic thresholds and stability,
- infrastructure cycles and collapse dynamics,
- scientific and technological cycles,
- information density and fragmentation,
- resource and demographic flows,
- universal signatures of collapse and reorganisation.

These results form the empirical foundation for the transition  $K_8 \rightarrow K_9$  and validate the structural unity of civilisational continua.

### P.1.96 Experiments for $K_9$

Continuum level  $K_9$  describes scientific paradigms, meta-models, and meta-theoretical structures governing the production, organisation, and evolution of knowledge. Experiments for  $K_9$  validate the structure of meta-theories, their thresholds, cycles, and transitions, including the conditions for the emergence of  $K_{10}$ .

Because meta-theories operate on knowledge systems rather than physical matter, experiments require a combination of:

- scientometric measurements,
- reconstruction of paradigm dynamics,
- controlled reasoning experiments,
- logic-based stress tests,
- meta-model simulations,
- cross-domain comparative studies.

### P.1.97 Objectives of the $K_9$ Experimental Programme

The aim is to empirically validate the  $K_9$  structure:

- axes  $A_9$  (paradigms, logical frameworks, model categories),
- potentials  $P_9$  (epistemic tension, coherence, explanatory power),
- thresholds  $\Theta_9$  (coherence limits, contradiction accumulation, paradigm collapse thresholds),
- flows  $J_9$  (information, inference steps, proofs, inter-model mappings),
- cycles  $C_9$  (paradigm formation, saturation, crisis, replacement, consolidation),
- conditions for the transition  $K_9 \rightarrow K_{10}$ .

### P.1.98 Type I Experiments: Paradigm Reconstruction and Coherence Thresholds

Scientific paradigms emerge, stabilise, and collapse in predictable patterns.

#### Experimental programme:

1. Reconstruction of paradigm histories across disciplines.
2. Quantifying coherence via contradiction density and model-error.
3. Historical detection of collapse points for  $\Theta_{\text{coh}}^{(9)}$ .
4. Network analysis of conceptual dependencies.

#### Predictions validated:

- Paradigms collapse when
$$T_{\text{epistemic}} > \Theta_{\text{coh}}^{(9)}$$
- Coherence decay follows universal patterns across sciences.
- Paradigm cycles  $C_{\text{paradigm}}$  are structurally invariant.

### **P.1.99 Type II Experiments: Logical Framework Stress Tests**

Meta-theories rely on formal logics with finite stability.

#### **Procedures:**

1. Stress-testing logical systems with paradox-generating inputs.
2. Measuring stability of deduction under noise.
3. Comparing resilience of different logical frameworks (classical, intuitionistic, type-theoretic, categorical).
4. Examining threshold behaviour in proof networks.

#### **Predictions validated:**

- Each logical framework has a measurable stability threshold  $\Theta_{\text{logic}}$ .
- Exceeding this threshold induces inconsistency cascades.
- More expressive systems produce higher epistemic tension.

### **P.1.100 Type III Experiments: Meta-Model Dynamics and Explanatory Power**

Meta-theories coordinate families of models, not individual datasets.

#### **Experimental programme:**

1. Tracking growth of explanatory power in emerging meta-models.
2. Reconstruction of model-category evolution (functors, adjunctions).
3. Computation of representational tension in multi-model systems.
4. Modelling inter-theory translations and loss of information.

#### **Predictions validated:**

- Meta-models saturate and enter crisis when explanatory load exceeds  $\Theta_{\text{load}}$ .
- Functorial collapse patterns are universal across disciplines.
- Successful meta-models minimise epistemic tension  $T_{\text{epistemic}}$  across model families.

### **P.1.101 Type IV Experiments: Inconsistency Cascades and $K_9$ Collapse**

A defining prediction of OC is that meta-theories exhibit catastrophic failure when inconsistency flows exceed stability limits.

#### **Procedures:**

1. Simulation of inconsistency propagation in proof networks.
2. Measuring cascade rates in highly interdependent model systems.
3. Historical reconstruction of paradigm collapse via contradiction chains.
4. Logical percolation studies using distributed reasoning agents.

**Predictions validated:**

- Inconsistency cascades exhibit critical behaviour similar to physical percolation.
- Collapse corresponds to

$$\Omega(K_9) \rightarrow \emptyset, \quad k_9 \rightarrow 0.$$

- Reorganisation requires formation of a new coherent axis  $A'_9$ .

**P.1.102 Type V Experiments: Cross-Disciplinary Meta-Theoretical Integration**

The transition  $K_9 \rightarrow K_{10}$  requires coherence across entire families of meta-theories.

**Experimental programme:**

1. Studying emergence of unified explanatory frameworks.
2. Mapping inter-theoretic coherence and contradiction flows.
3. Detecting conditions for convergent unification.
4. Empirical testing of the limit  $\Theta_{\text{meta}}$ , beyond which no stable meta-theory can exist.

**Predictions validated:**

- Meta-theories unify when epistemic tension gradients align:

$$\nabla P_9 \approx 0.$$

- Excessive meta-complexity destroys global coherence and prevents formation of  $K_{10}$ .
- A stable  $K_{10}$  requires saturation of  $C_{\text{meta}}$  cycles.

**P.1.103 Type VI Experiments: Agent-Based Reasoning and Collective Cognition**

Meta-theoretical structures depend on distributed reasoning.

**Procedures:**

1. Agent-based models of collective scientific inference.
2. Measuring group reasoning bias and convergence.
3. Stimulating epistemic crises via controlled misinformation.
4. Testing resilience of scientific communities under load.

**Predictions validated:**

- Distributed reasoning increases stability up to a limit  $\Theta_{\text{collective}}$ .
- Above this limit, collective cognition becomes chaotic.
- Emergence of meta-theoretical order corresponds to stabilisation of group inference cycles  $C_{\text{group}}$ .

### P.1.104 Summary

Experiments for  $K_9$  validate the structural unity of scientific and meta-theoretical continua:

- coherence thresholds and paradigm collapse,
- stress tests for logical frameworks,
- dynamics of explanatory power and model categories,
- inconsistency cascades and epistemic percolation,
- cross-disciplinary unification and the formation of  $K_{10}$ ,
- collective reasoning and epistemic stability.

These experimental lines establish the empirical basis for the existence and stability of  $K_{10}$  and confirm the structural predictions of the Ontology of Continua at meta-theoretical scale.

### P.1.105 Experiments for $K_{10}$

Level  $K_{10}$  describes meta-model continua: coherent systems of model categories, functorial structures, modal spaces, and second-order difference operators. Experiments for  $K_{10}$  are conceptual, mathematical, and epistemic-structural: they validate the stability, thresholds, and dynamics of meta-model continua, including conditions under which  $K_{10}$  collapses or transitions toward higher-order structures.

The goal is to empirically and formally verify:

- the existence and stability of  $\Omega(K_{10})$ ,
- thresholds  $\Theta_{10}$  (self-reference, meta-complexity, modal dimensionality, functor stress),
- flows  $J^{(10)}$  (changes in functor sets, modal transitions, category deformation),
- cycles  $C^{(10)}$  (meta-theory  $\rightarrow$  meta-model  $\rightarrow$  model  $\rightarrow$  paradigm update  $\rightarrow$  meta-theory),
- structural tension  $T_{10}$ ,
- collapse behaviour  $\Omega(K_{10}) \rightarrow \emptyset$ ,
- conditions for the emergence of  $K_{11}$  (if any).

### P.1.106 Type I Experiments: Coherence and Self-Reference Thresholds

Self-reference depth and meta-theoretical recursion are fundamental components of  $K_{10}$ . Experiments aim to measure the depth at which a meta-model becomes unstable.

#### Procedures:

1. Construct meta-model chains (categories-of-categories, functors-between-functors).
2. Measure coherence loss as recursion depth increases.
3. Quantify self-referential tension:

$$T_{\text{self}} = w_{\text{self}}(\text{ReflexDepth} - \Theta_{\text{self}}).$$

4. Simulate self-reference loops in proof assistants and dependent-type frameworks (Coq, Lean, Agda).

**Predictions validated:**

- There exists a finite threshold  $\Theta_{\text{self}}$  beyond which no stable meta-model can exist.
- Exceeding  $\Theta_{\text{self}}$  collapses  $\Omega(K_{10})$  regardless of the specific formal system.
- Systems with higher expressive power reach instability faster.

**P.1.107 Type II Experiments: Functor Stress Tests and Category Dynamics**

Functorial structures define the skeleton of  $K_{10}$ .

**Experimental programme:**

1. Construct multi-level categorical structures: categories, functors, natural transformations, adjunction networks.
2. Introduce controlled deformations in objects/morphisms.
3. Measure functor stress (failure of naturality, loss of adjunctions).
4. Run simulations of category evolution under structural load.

**Predictions validated:**

- Functorial stability collapses when
 
$$T_{\text{functor}} > \Theta_{\text{functor}}.$$
- Loss of adjunctions is the dominant early-warning signal.
- Categorical collapse always precedes global  $K_{10}$  collapse.

**P.1.108 Type III Experiments: Modal Spaces and Dimensionality Thresholds**

Modal spaces  $\Lambda$  encode possible worlds, model variants, and counterfactuals.

**Procedures:**

1. Construct modal spaces of increasing dimension.
2. Measure combinatorial explosion of possible transitions.
3. Compute modal tension:
 
$$T_{\text{mod}} = w_{\text{mod}}(\text{ModDim} - \Theta_{\text{mod}}).$$
4. Simulate accessibility relations and modal collapse.

**Predictions validated:**

- There exists a finite threshold  $\Theta_{\text{mod}}$  for modal dimensionality.
- Exceeding  $\Theta_{\text{mod}}$  induces modal collapse: indistinguishability of worlds.
- Modal collapse triggers breakdown of second-order difference operators.

**P.1.109 Type IV Experiments: Operator Stability and Difference Dynamics**

Operators  $\mathfrak{D}$  generate second-order differences (models of models, distinctions of distinctions).

**Experimental programme:**

1. Construct operator chains  $\mathfrak{D}_1, \mathfrak{D}_2, \dots$ .
2. Introduce perturbations in input theories.
3. Measure stability of operator outputs.
4. Analyse breakdown patterns under noise.

**Predictions validated:**

- Second-order difference operators have finite stability regions.
- Noise amplification follows universal patterns across formalisms.
- Collapse occurs when operators are no longer contractive mappings over meta-model space.

**P.1.110 Type V Experiments: Cycle Stability and Meta-Theoretical Recurrence**

The fundamental dynamical loop of  $K_{10}$  is:

$$\text{meta-theory} \xrightarrow{\mathfrak{D}} \text{meta-model} \xrightarrow{\Phi} \text{model} \xrightarrow{\Psi} \text{updated meta-theory}.$$

**Procedures:**

1. Empirical reconstruction of real scientific cycles (e.g., physics: classical  $\rightarrow$  quantum  $\rightarrow$  QFT  $\rightarrow$  category-theoretic formalisms).
2. Simulation of recurrence cycles using agent-based meta-logical systems.
3. Measurement of cycle stability and attractor size.

**Predictions validated:**

- Cycles  $C^{(10)}$  stabilise only when tension gradients align.
- Excess meta-complexity collapses cycles to fixed points or chaos.
- Stable  $K_{10}$  requires a nonzero cycle measure  $|C_{\max}|$ .

**P.1.111 Type VI Experiments: Collapse of  $K_{10}$** 

Collapse occurs when no meta-model remains coherent:

$$\Omega(K_{10}) = \emptyset, \quad k_{10} = 0.$$

**Procedures:**

1. Simulate extreme self-reference and functor stress.
2. Analyse inconsistencies in hierarchical categorical systems.
3. Trigger modal overload in high-dimensional  $\Lambda$  spaces.
4. Model cascading failure across the  $K_{10}$  operators.



**Predictions validated:**

- Collapse follows a universal multi-operator instability pattern.
- Pre-collapse tension profile is detectable in  $\{T_{\text{self}}, T_{\text{mod}}, T_{\text{functor}}\}$ .
- Collapse forces reinitialisation of meta-theoretical axes  $A'_{10}$ .

**P.1.112 Type VII Experiments: Transition to  $K_{11}$** 

If  $K_{11}$  exists, its emergence requires:

- a new axis  $A_{11}$  not representable as a deformation of  $A_{10}$ ,
- a meta-stable configuration of operators,
- modal dimensionality below the collapse threshold,
- stabilised cycles  $C^{(10)}$ ,
- non-zero continuumness  $k_{10} > 0$ .

**Experimental programme:**

1. Search for formal systems with strictly higher representational power.
2. Analyse if second-order operators can yield consistent third-order structures.
3. Determine whether a new boundary  $\partial\Omega(K_{11})$  can be formally defined.

**Predictions validated:**

- Transition  $K_{10} \rightarrow K_{11}$  is possible only under strict tension bounds and stabilised cycles.
- Most known formalisms reach collapse before achieving  $K_{11}$  conditions.
- Evidence of  $K_{11}$  is falsifiable via *stability-of-operators tests*.

**P.1.113 Summary**

Experiments for  $K_{10}$  validate the structural and mathematical predictions of meta-model continua:

- self-reference and meta-complexity thresholds,
- functor stability and adjunction breakdown,
- modal dimensionality limits,
- second-order difference operator stability,
- recurrence cycles of meta-theoretical evolution,
- collapse signatures of  $K_{10}$ ,
- potential pathways toward  $K_{11}$ .

These experiments collectively confirm the structural form of the meta-model continuum and provide empirical–formal criteria for the existence and stability of higher-order continua.

#### **P.1.114 Experiments for $K_{11}$**

Level  $K_{11}$  represents a hypothetical higher-order continuum beyond the meta-model layer  $K_{10}$ . If it exists,  $K_{11}$  would encompass third-order operators, hyper-modal structures, and recursive architectures that exceed the representational capacities of  $K_{10}$ . The purpose of experiments for  $K_{11}$  is to identify, formalise, and validate the minimal structural conditions required for such a continuum to exist, and to determine whether any known or constructible formal system satisfies them.

Experiments at this level are necessarily structural, axiomatic–logical, and meta-representational. They test limits of recursion, category iteration, modal explosion, and operator stability.

#### **P.1.115 Type I Experiments: Detecting New Axes $A_{11}$**

A necessary condition for  $K_{11}$  is the existence of a new axis  $A_{11}$  not derivable from the operator set of  $K_{10}$  and not representable as a deformation of any axis in  $A(K_{10})$ .

##### **Procedures:**

1. Analyse formal systems with representational power exceeding second-order meta-models.
2. Attempt construction of operators  $\mathfrak{D}^{(3)}$  that produce distinctions between second-order distinctions.
3. Test whether resulting structures form a coherent space of states  $\Omega(K_{11})$ .
4. Evaluate independence of the candidate axis  $A_{11}$  from existing axes using:
  - failure of representability,
  - functor non-deformability,
  - modal non-reducibility.

##### **Predictions validated:**

- $A_{11}$  exists only if operator depth becomes strictly richer than  $\mathfrak{D}^{(2)}$ .
- Most known logical and categorical formalisms fail to produce non-deformable axes.

#### **P.1.116 Type II Experiments: Hyper-Modal Spaces and Dimensionality**

A defining property of  $K_{11}$  is the presence of hyper-modal spaces  $\Lambda^{(11)}$  with modal relations that cannot be embedded into the modal structure of  $K_{10}$ .

##### **Experimental programme:**

1. Construct multi-layer modal spaces (modalities over modalities over modalities).
2. Measure growth of modal dimension and determine whether it surpasses  $\Theta_{\text{mod}}^{(10)}$  without collapse.
3. Simulate accessibility relations with meta-hyper-modal branching.
4. Test stability of hyper-modal transitions under noise.

##### **Predictions validated:**

- Hyper-modal spaces quickly approach modal overload unless new stabilisation mechanisms exist.
- Stability of  $\Lambda^{(11)}$  is evidence for a distinct structural regime beyond  $K_{10}$ .

### P.1.117 Type III Experiments: Third-Order Operator Stability

Operators for  $K_{11}$  must enable stable transitions among hyper-modal, hyper-categorical, and third-order meta-structures.

#### Procedures:

1. Attempt construction of operators  $\mathcal{O}^{(11)}$  such that:

$$\mathcal{O}^{(11)} : (\text{second-order distinctions}) \longrightarrow (\text{third-order distinctions}).$$

2. Evaluate stability of such operators under perturbations.
3. Test for contractivity or divergence under iteration.
4. Compare with collapse behaviour known for  $\mathfrak{D}^{(2)}$  on  $K_{10}$ .

#### Predictions validated:

- Stable third-order operators are extremely rare or may not exist in known formalisms.
- Divergence of operator chains signals impossibility of  $K_{11}$  inside the tested formalism.

### P.1.118 Type IV Experiments: Categorical Tower Construction

$K_{11}$  requires coherent categorical structures above categories-of-categories (i.e., beyond 2-categories,  $n$ -categories, and infinity-categories if they remain reducible).

#### Procedures:

1. Build towers of categorical abstraction:

$$\text{Cat} \rightarrow 2\text{-Cat} \rightarrow n\text{-Cat} \rightarrow \dots$$

2. Attempt to construct a non-reducible layer above all known  $n$ -categorical systems.
3. Test whether new compositional laws remain coherent.
4. Evaluate functorial stress at each level of the tower.

#### Predictions validated:

- The vast majority of attempts collapse to lower-level categorical structures.
- Failure of coherence at high categorical levels is a primary obstacle to  $K_{11}$ .

### P.1.119 Type V Experiments: Hyper-Recurrence and Cycle Stability

Cycles  $C^{(11)}$  represent recurrence dynamics of third-order meta-theoretical systems.

**Experimental programme:**

1. Define hypothetical cycles of the form:

$$K_{10} \longrightarrow K_{11} \longrightarrow K'_{10} \longrightarrow K'_{11} \longrightarrow \dots$$

2. Simulate cycle dynamics by recursively transforming theories, meta-theories, and meta-meta-theories.
3. Measure cycle attractor size and stability.
4. Characterise collapse channels unique to  $K_{11}$ :
  - infinite regress,
  - operator explosion,
  - hyper-modal collapse.

**Predictions validated:**

- Stable  $K_{11}$  cycles require strict control of operator growth.
- Cycles diverge rapidly unless new constraints or structural invariants exist.

**P.1.120 Type VI Experiments: Collapse Modes of  $K_{11}$** 

$K_{11}$ , if constructible, is expected to be highly unstable. Experiments focus on identifying universal collapse signatures.

**Procedures:**

1. Introduce maximal recursion depth in third-order operators.
2. Amplify hyper-modal branching until instability.
3. Stress-test the categorical tower beyond  $n$ -levels.
4. Compute tension components:

$$T_{11} = w_{\text{self}}T_{\text{self}}^{(11)} + w_{\text{mod}}T_{\text{mod}}^{(11)} + w_{\text{functor}}T_{\text{functor}}^{(11)}.$$

**Predictions validated:**

- Collapse occurs when any major component of  $T_{11}$  exceeds its corresponding threshold.
- Collapse is generally unavoidable in most tested formalisms.

**P.1.121 Type VII Experiments: Detectability of  $K_{11}$** 

Even if  $K_{11}$  is not constructible, its absence is falsifiable. These experiments determine the detectability criteria.

**Experimental programme:**

1. Evaluate whether extension of  $K_{10}$  operators produces coherent third-order objects.
2. Attempt to construct new boundaries  $\partial\Omega(K_{11})$  from the deformation of existing modal/categorical spaces.
3. Test whether any stable, non-reducible categorical or modal regime emerges under extreme conditions.

**Predictions validated:**

- If no coherent third-order operators can be constructed,  $K_{11}$  is falsified in the tested framework.
- Evidence for  $K_{11}$  requires:
  - non-zero  $k_{11}$ ,
  - well-formed  $\Omega(K_{11})$ ,
  - at least one stable axis  $A_{11}$ .

**P.1.122 Summary**

Experiments for  $K_{11}$  aim to test the theoretical upper bounds of representability in the Ontology of Continua. They analyse:

- existence and independence of new axes  $A_{11}$ ,
- hyper-modal stability and dimensionality,
- operator regimes beyond second-order,
- construction of higher categorical towers,
- recurrence and cycle stability at third-order depth,
- collapse behaviour under meta-structural tension,
- detectability and falsification criteria for  $K_{11}$ .

These experiments provide the first systematic programme for detecting or falsifying the existence of  $K_{11}$  continua.

**P.1.123 Experiments for  $K_{12}$** 

The level  $K_{12}$  represents the hypothetical maximal extension of the continuum hierarchy consistent with the formal principles of the Ontology of Continua. While  $K_{10}$  captures meta-theory and  $K_{11}$  explores third-order recursive structures,  $K_{12}$  would represent a radically higher-order continuum characterised by:

- hyper-transfinite modal spaces,
- fourth-order distinction operators  $\mathfrak{D}^{(4)}$ ,
- unbounded categorical towers with new coherence laws,
- global structural recursion beyond all known meta-levels,
- extreme tension landscapes ( $T_{12}$ ) with new threshold types.

Experiments for  $K_{12}$  therefore constitute an investigation of the absolute limits of structural representation. They serve two goals:

1. to determine whether a  $K_{12}$ -continuum is theoretically possible within the axioms of OC,
2. to establish falsifiability criteria if such a continuum cannot be realised.

**P.1.124 Type I Experiments: Existence of Fourth-Order Axes  $A_{12}$** 

A necessary and non-negotiable condition for  $K_{12}$  is the emergence of at least one new axis  $A_{12}$  that:

- is not reducible to any axis of  $K_0$ – $K_{11}$ ,
- cannot be generated via deformation of  $\mathfrak{D}^{(1)}$ ,  $\mathfrak{D}^{(2)}$ ,  $\mathfrak{D}^{(3)}$ ,
- requires a genuinely new distinction operator  $\mathfrak{D}^{(4)}$ .

**Experimental programme:**

1. Attempt construction of formal systems with representational power exceeding third-order meta-systems.
2. Analyse whether any structure supports distinctions between third-order distinctions in a non-degenerate way.
3. Evaluate the independence of candidate axes by:
  - functorial independence tests,
  - modal non-embeddability tests,
  - non-reducibility to composition of lower-order operators.

**Predictions validated:**

- Most known formalisms cannot generate  $A_{12}$ .
- Existence of  $A_{12}$  would imply a fundamentally new class of mathematical structures.

**P.1.125 Type II Experiments: Hyper-Transfinite Modal Spaces  $\Lambda^{(12)}$** 

$K_{12}$  demands modal spaces whose dimensions grow faster than those available even in hyper-modal  $K_{11}$  structures.

These spaces must exceed the classical and constructive limits of accessibility relations, enabling forms of modal branching that cannot be embedded into any  $\Lambda^{(x)}$  for  $x \leq 11$ .

**Procedures:**

1. Construct families of modal spaces with transfinite or inaccessible cardinal branching.
2. Benchmark modal complexity against  $\Theta_{\text{mod}}^{(11)}$ .
3. Stress-test candidate hyper-transfinite structures under:
  - perturbations,
  - modal collapse scenarios,
  - functorial stress tests.
4. Detect fixed points or attractors in hyper-modal transitions.

**Predictions validated:**

- Most candidate hyper-transfinite spaces collapse into lower modal regimes under perturbation.
- Existence of stable  $\Lambda^{(12)}$  requires entirely new structural invariants.

**P.1.126 Type III Experiments: Fourth-Order Operator Viability**

The operator family for  $K_{12}$  must extend the known operators  $F, G, H, Q, R, S, U$ , and in particular the distinction operators  $\mathfrak{D}^{(n)}$ , beyond  $n = 3$ .

**Procedures:**

1. Attempt to define  $\mathfrak{D}^{(4)}$  such that

$$\mathfrak{D}^{(4)} : (\text{third-order distinctions}) \longrightarrow (\text{fourth-order distinctions})$$

is well-defined and non-degenerate.

2. Test for:
  - contractivity,
  - explosion,
  - self-consistency,
  - coherence with existing operators.
3. Evaluate dynamic behaviour under iteration.
4. Compare collapse behaviour with known limits for  $\mathfrak{D}^{(3)}$  on  $K_{11}$ .

**Predictions validated:**

- Most  $\mathfrak{D}^{(4)}$  candidates diverge rapidly.
- Convergence would constitute strong evidence for  $K_{12}$ 's theoretical viability.

**P.1.127 Type IV Experiments: Tower-of-Theories Construction**

In  $K_9$  and  $K_{10}$ , theories and meta-theories already form categories and functorial layers. In  $K_{11}$ , categorical towers extend further.  $K_{12}$  requires the construction of theory towers that exceed all known  $n$ -categorical and  $\infty$ -categorical structures.

**Procedures:**

1. Examine the adjoint behaviour of towers extending beyond known  $n$ -categories.
2. Attempt to build families of theory transformations with non-reducible, non-collapse-prone compositional laws.
3. Evaluate functorial stress  $\Theta_{\text{functor}}^{(11)}$  at each stage.
4. Quantify instability signals:
  - coherence failures,
  - compositional divergence,
  - collapse into  $K_{11}$ -like structures.

**Predictions validated:**

- The majority of theoretical towers collapse below  $K_{12}$ .
- Any demonstration of a stable higher-order tower is evidence for  $K_{12}$ .

**P.1.128 Type V Experiments: Hyper-Recursive Cycle Dynamics  $C_{12}$** 

Cycles  $C_{12}$  represent recurrence dynamics of fourth-order meta-theoretical transformations.

**Experimental programme:**

1. Construct hypothetical hyper-recursive chains:

$$K_{10} \rightarrow K_{11} \rightarrow K_{12} \rightarrow K'_{11} \rightarrow K'_{10} \rightarrow \dots$$

2. Model cycle attractors in hyper-transfinite state spaces.
3. Quantify cycle stability and divergence speed.
4. Identify special collapse signatures:
  - recursion explosion,
  - transfinite fixed-point loss,
  - functorial meltdown.

**Predictions validated:**

- Stable hyper-recursive cycles are extremely unlikely.
- Existence of a stable cycle  $C_{12}$  would suggest a dramatically new form of structural order.

**P.1.129 Type VI Experiments: Collapse Signatures of  $K_{12}$** 

The collapse conditions for  $K_{12}$  follow the general theory:

$$\Omega(K_{12}) = \emptyset, \quad k_{12} \rightarrow 0, \quad T_{12} > \Theta_{\text{crit}}^{(12)}.$$

**Procedures:**

1. Amplify hyper-transfinite modal branching until thresholds are exceeded.
2. Probe instability of  $\mathfrak{D}^{(4)}$  under recursion.
3. Push categorical towers past coherence failure.
4. Compute the composite tension:

$$T_{12} = w_1 T_{\text{mod}}^{(12)} + w_2 T_{\text{self}}^{(12)} + w_3 T_{\text{functor}}^{(12)}.$$

5. Identify universal collapse signatures, including:
  - modal implosion,
  - operator explosion,
  - category tower collapse.

**Predictions validated:**

- Most candidate structures collapse before reaching  $K_{12}$  thresholds.
- Survival implies existence of exotic stabilisation mechanisms not seen in lower continua.

**P.1.130 Type VII Experiments: Detectability and Falsification**

Even if  $K_{12}$  cannot be constructed, its falsification requires specific tests.



**Conditions for non-existence:**

- No stable  $\mathfrak{D}^{(4)}$  can be built,
- all candidate  $\Lambda^{(12)}$  collapse,
- categorical towers remain reducible,
- no independent axis  $A_{12}$  emerges,
- $k_{12}$  cannot remain  $> 0$  under any perturbation.

**Conditions for possible existence:**

- Discovery of at least one stable hyper-transfinite modal structure,
- identification of a non-deformable fourth-order axis,
- evidence of fourth-order operators with finite tension,
- demonstration of coherent transfinite cycle dynamics.

**P.1.131 Summary**

Experiments for  $K_{12}$  explore the absolute theoretical limits of representable continua. They test:

- existence of new axes and operators,
- viability of hyper-transfinite modal spaces,
- stability of fourth-order distinctions,
- construction of theory towers beyond all known frameworks,
- hyper-recursive cycles,
- universal collapse signatures,
- detectability or falsification of  $K_{12}$ .

If  $K_{12}$  exists, it represents the furthest possible extension of the continuum hierarchy consistent with the axioms of the Ontology of Continua.

**P.2 Falsifiability**

Falsifiability in the Ontology of Continua (OC) differs fundamentally from standard empirical or phenomenological falsifiability. Because OC is a structural theory, falsification proceeds through *violations of structural constraints* rather than mismatches between numerical predictions and measurements. A continuum exists only if it satisfies the axioms governing:

$$(\Omega, A, P, J, \Theta, \partial\Omega, C, k).$$

Thus, any inconsistency between these components—or any failure of their dynamic evolution—is sufficient to falsify a given instantiation of the theory.

This master file summarises the universal falsifiability programme for all continua  $K_0$ – $K_{12}$  and connects it to the general operators  $F, G, H, Q, R, S, U$ , the threshold landscape, and the embedding-space constraints of  $M_x$ .

### P.2.1 Structural Meaning of Falsifiability

A continuum  $K_x$  is falsified when the structure assigned to it cannot satisfy the OC axioms or when its dynamic evolution contradicts the universal operators. The following structural criteria serve as the foundation for falsifiability across all levels:

1. **Violation of existence thresholds ( $\Theta_{\text{exist}}$ ):** If no configuration of  $(\Omega, A, P, J)$  can be made compatible with  $\Theta_{\text{exist}}$ , the continuum cannot exist.
2. **Boundary inconsistency:** If  $\partial\Omega$  cannot be defined such that it separates admissible from inadmissible states, structural incoherence arises.
3. **Operator incompatibility:** If  $F, G, H, Q, R, S, U$  cannot be consistently defined for the proposed continuum, it is structurally impossible.
4. **Cycle impossibility:** If no non-trivial cycles  $C$  can be maintained, the continuum has  $k = 0$  and collapses.
5. **Embedding-space violations:** If  $\Omega(K_x) \not\subseteq \Omega(M_x)$ , the continuum is falsified by Theorem 0.0 (embedding constraint).
6. **Threshold contradictions:** If  $\Theta_{\text{crit}}$  or  $\Theta_{\text{dim}}$  cannot be satisfied, structural transitions become impossible.
7. **Failure of monotonic dimension theorem:** If  $\dim K_x$  decreases without collapse, OC is falsified.

These structural criteria are directly tied to the core OC axioms and therefore constitute universal falsifiability conditions.

### P.2.2 Universal Falsifiability Criteria

Across all continua, the following universal conditions provide a complete falsification toolkit:

**(F1) Incoherent State Space  $\Omega$ .** If  $\Omega$  cannot form a connected, non-empty, admissible region under the given axes and potentials, the continuum is impossible.

**(F2) Impossible Axes  $A$ .** If an axis cannot support at least two incompatible states (Theorem 3: minimality of axes), the continuum collapses.

**(F3) Potentials leaving admissible ranges ( $P_i \notin \text{Dom}(P)$ ).** Violations of potential bounds force  $T \rightarrow \infty$  or collapse.

**(F4) Threshold incompatibility.** If thresholds cannot be simultaneously satisfied:

$$\Theta_{\text{exist}} < \Theta_{\text{stab}} < \Theta_{\text{crit}},$$

the continuum cannot stabilise.

**(F5) Flow impossibility.** If no supporting flows  $J_{\text{support}}$  exist,  $k(t)$  cannot remain positive.

**(F6) Boundary collapse.** If  $\partial\Omega$  cannot remain well-defined under  $R$ , collapse is inevitable.

**(F7) Operator inconsistency.** If the evolution operator

$$K(t + \Delta t) = E(K(t))$$

cannot be defined, the continuum is ruled out.

**(F8) Violation of embedding constraints.** If the embedding space  $M_x$  cannot host the continuum, falsification is immediate.

### P.2.3 Cross-Level Falsifiability Structure

Falsifiability strengthens at higher  $K$ -levels because each continuum inherits constraints from all preceding levels. Thus:

- $K_2$  must satisfy all constraints of  $K_0$  and  $K_1$ .
- $K_5$  must satisfy all constraints of  $K_0$ – $K_4$ .
- $K_{10}$  must satisfy all constraints of  $K_0$ – $K_9$ .

This inheritance principle ensures:

1. structural upward compatibility,
2. impossibility of “shortcut continua” skipping levels,
3. monotonic tightening of thresholds at each level,
4. cumulative constraints on dynamics and stability.

Cross-level falsifiability also allows identifying the level at which a model fails to satisfy OC, making structural diagnosis possible.

### P.2.4 Falsifiability via Dynamics and Collapse

The dynamics of collapse provide additional, stringent falsification criteria.

**(F9) Diverging structural tension  $T$ .** If  $T(t)$  grows without bound and no axis, boundary, or potential reconfiguration reduces it, collapse is guaranteed.

**(F10) Cycle breakdown.** If essential cycles  $C$  disappear, the continuum cannot persist.

**(F11) Boundary failure.** If local threshold interactions cause patch collapse in  $\partial\Omega$ , the entire continuum becomes unphysical.

**(F12) Dimensional stagnation.** If  $T > \Theta_{\text{dim}}$  but no new axis emerges, the continuum cannot support the required dimensional transition and collapses.

These conditions ground falsifiability in the dynamical behaviour of the continuum as it evolves.

### P.2.5 Falsifiability via Embedding Spaces

The embedding spaces  $M_x$  impose external constraints:

$$\Omega(K_x) \subseteq \Omega(M_x).$$

Thus falsifiability can be tested by:

1. expanding  $\Omega(K_x)$  until it attempts to leave  $\Omega(M_x)$ ,
2. varying potentials  $P$  to explore the edges of  $M_x$ ,
3. perturbing axes  $A(K_x)$  to test if admissible values remain inside  $M_x$ ,
4. detecting incompatible states whose realisation would violate  $M_x$  structure.

If any such situation occurs, the continuum cannot exist within its embedding space.

### P.2.6 Falsifiability in Practice: Multi-Domain Implementation

OC provides concrete falsifiability conditions for:

- physics (coherence thresholds, phase transitions, percolation),
- chemistry (RAF closure, membrane thresholds),
- early life (osmotic/curvature collapse),
- biology (excitation thresholds, membrane failure),
- cognition (binding failure, predictive collapse),
- society (trust breakdown, institutional failure),
- civilisation (infrastructure collapse, system-wide tension),
- theory and meta-theory (coherence thresholds, recursion failure).

In each case, falsifiability follows the same structural pattern:

$$\text{violate thresholds} \Rightarrow T \rightarrow \infty \Rightarrow \partial\Omega \text{ destabilises} \Rightarrow k \rightarrow 0 \Rightarrow \Omega = \emptyset.$$

This universality is one reason the falsifiability programme remains structurally uniform across domains.

### P.2.7 Summary

The universal falsifiability programme for the Ontology of Continua is structural, not phenomenological. It tests continua by probing their ability to satisfy the OC axioms, threshold hierarchy, and dynamic operator constraints. Any breakdown of:

- admissible state space,
- axes,
- potentials,
- flows,
- thresholds,
- cycles,

- boundaries,
- operators,
- embedding constraints,

immediately falsifies the proposed continuum.

This framework provides a precise scientific basis for evaluating all levels  $K_0$ – $K_{12}$  and their domain-specific realisations.

### P.2.8 Falsifiability of $K_0$

The continuum  $K_0$  represents the minimal structural substrate of the Ontology of Continua. It is defined as a pre-dynamic object

$$K_0 = (S, \Delta, \mathcal{C}),$$

with no time, no flows, no potentials, no thresholds beyond the minimal existence bound  $\Theta_0 = \varepsilon > 0$ , and a trivial cycle  $C_{\text{triv}}$ . Falsifiability of  $K_0$  therefore tests the logical and structural consistency of the most primitive form of continuumhood.

Despite its simplicity,  $K_0$  imposes strict constraints. Any violation of its axioms is sufficient to falsify not only  $K_0$  but also all higher continua, because the full hierarchy  $K_1$ – $K_{12}$  is built upon the existence of a well-defined  $K_0$ .

### P.2.9 Fundamental Falsifiability Principles for $K_0$

The falsification of  $K_0$  occurs when the structural triple  $(S, \Delta, \mathcal{C})$  fails to satisfy the defining axioms:

- **Axioms D1–D4:** governing the set  $S$  of primitive distinctions and the minimal structural properties of  $\Delta$ .
- **Axioms C1–C4':** governing the construction and compatibility of the composition operator  $\mathcal{C}$ .
- **Axiom 0.1 (Existence):** which provides the minimal condition for a continuum to exist.
- **Minimal threshold:**  $\Theta_0 = \varepsilon > 0$ , ensuring non-degeneracy.

Thus,  $K_0$  is falsified whenever the primitive combinatorial structure fails to meet these requirements.

### P.2.10 Structural Falsifiability Conditions

Several structural incompatibilities can invalidate  $K_0$ :

**(F0.1) Undefined or incoherent  $S$ .** If the primitive set  $S$  of distinctions is empty or logically inconsistent, no continuum can be constructed.

**(F0.2) Breakdown of  $\Delta$ .** If the difference relation  $\Delta$  cannot be defined such that it respects axioms D1–D4 (irreflexivity, minimal consistency, closure under composition), then  $K_0$  is impossible.

**(F0.3) Inconsistent composition  $\mathcal{C}$ .** If  $\mathcal{C}$  cannot produce valid composite distinctions or violates C1–C4', structural coherence is lost.

**(F0.4) Violation of the minimal existence threshold.** If under the primitive structure the threshold

$$\Theta_0 = \varepsilon > 0$$

cannot be maintained (i.e. the structure collapses to the empty set),  $K_0$  cannot exist.

**(F0.5) Lack of a trivial cycle.** Even at  $K_0$ , the trivial cycle  $C_{\text{triv}}$  must exist. A failure to define this cycle violates the minimal continuity condition.

**(F0.6) Contradiction in structural pairs.** If the structural pair  $(S, \Delta)$  cannot exist without contradiction (violating the “structural pair axiom”),  $K_0$  is falsified.

### P.2.11 Boundary and Region Falsifiability

Although  $K_0$  has no dynamical boundary in the usual sense, it still requires a well-defined primitive boundary of admissible configurations:

$$\partial\Omega(K_0) = \{\text{distinctions permitted by } S, \Delta, \mathcal{C}\}.$$

$K_0$  is falsified if:

- $\partial\Omega$  cannot be defined (the primitive region has no consistent boundary),
- $\Omega(K_0) = \emptyset$ ,
- the boundary admits contradictions in distinction composition.

In other words, if the primitive “space of distinctions” cannot be constructed, no continuum can exist at all.

### P.2.12 Collapse and Non-Existence at $K_0$

Since  $K_0$  has no dynamics, collapse is strictly structural rather than temporal. The continuum  $K_0$  fails when it cannot satisfy its definition in a single static configuration.

This occurs when:

1.  $S$  degenerates into a contradictory or empty set;
2.  $\Delta$  cannot be defined coherently;
3.  $\mathcal{C}$  produces contradictions or cannot operate;
4.  $\Theta_0$  cannot be respected;
5.  $C_{\text{triv}}$  cannot be defined.

Any such violation implies the total non-existence of  $K_0$  and therefore renders all higher continua impossible.

### P.2.13 Embedding-Space Constraints for $K_0$

Although  $K_0$  has no parent continuum, it is still embedded in the minimal admissible meta-space that supports primitive distinction structures. Falsifiability via embedding arises when:

- the meta-space cannot host any non-degenerate distinction set,
- the assumed primitive distinctions require an impossible background structure,
- $\Omega(K_0)$  cannot be embedded into this minimal meta-space.

These conditions guarantee that  $K_0$  is not merely a logical construction but must be realisable within an admissible structural environment.

### P.2.14 Summary

$K_0$  is falsified purely by structural contradictions. Because  $K_0$  contains no time, no potentials, no flows, and no dynamics, every falsification is immediate and global. A single violation of:

- the primitive distinction set  $S$ ,
- the difference relation  $\Delta$ ,
- the composition operation  $\mathcal{C}$ ,
- the minimal threshold  $\Theta_0$ ,
- the existence of  $C_{\text{triv}}$ ,
- or the coherence of  $\partial\Omega(K_0)$

invalidates  $K_0$  entirely.

Because every higher continuum  $K_x$  requires a valid  $K_0$ , falsifying  $K_0$  collapses the entire hierarchy.

### P.2.15 Falsifiability of $K_1$

The continuum  $K_1$  is the first non-trivial continuum in the hierarchy of the Ontology of Continua. It introduces a one-dimensional axis of admissible variation, typically interpreted as the minimal temporal or parametric axis  $\tau$ , together with a space of fields on this axis:

$$K_1 = (X, \tau, A_1, V, \Omega, E, \varphi_0).$$

Unlike  $K_0$ , which is purely structural and pre-dynamic,  $K_1$  already possesses:

- a topological line  $(X, \tau)$ ,
- a field space  $V$ ,
- an admissible configuration space  $\Omega = C^0(X, V)$ ,
- an energy functional  $E$  obeying axioms E1–E3,
- a non-trivial admissibility region with boundary  $\partial\Omega$ ,
- a minimal threshold  $\Theta_1$  for allowed configurations,
- the trivial cycle  $C_{\text{triv}}$  defined over the axis.

Falsifiability of  $K_1$  concerns the possibility that any of these structures fail to meet the defining axioms of the continuum.

### P.2.16 Structural and Topological Falsifiability

The continuum  $K_1$  is falsified if its minimal topological and field structures cannot be coherently defined. The key conditions include:

**(F1.1) Breakdown of the axis  $(X, \tau)$ .**  $K_1$  requires a one-dimensional continuum with a valid topology. If  $(X, \tau)$  cannot be defined, is inconsistent, or collapses into a disconnected or discrete structure incompatible with the axioms, the continuum fails.

**(F1.2) Invalid or ill-defined field space  $V$ .** The field space must be a well-defined vector or configuration space; if  $V$  admits contradictions or cannot support continuous fields,  $\Omega$  becomes ill-defined.

**(F1.3) Non-existence of the admissible configuration space  $\Omega = C^0(X, V)$ .** If continuity cannot be defined on  $(X, \tau)$  or if  $V$  does not support continuous maps, then  $\Omega = \emptyset$  and  $K_1$  is impossible.

**(F1.4) Inconsistent admissibility region.** The admissible region  $\Omega_{\text{adm}}$  must satisfy the  $K_1$  constraints. If it cannot be defined or is immediately empty, the continuum fails.

### P.2.17 Energy Functional and Threshold Falsifiability

The energy functional

$$E : \Omega \rightarrow \mathbb{R}$$

is essential for defining admissibility at  $K_1$ , together with its axioms (E1: boundedness below, E2: locality, E3: stability).

$K_1$  is falsified when:

**(F1.5)  $E$  violates boundedness below (E1).** If  $E$  has no lower bound, the continuum becomes dynamically unstable.

**(F1.6)  $E$  violates locality (E2).** Non-local energy at  $K_1$  contradicts the definition of the continuum and invalidates its structure.

**(F1.7)  $E$  violates stability (E3).** If small perturbations lead to divergent energy without a stabilising mechanism,  $K_1$  collapses.

**(F1.8) Violation of the threshold  $\Theta_1$ .** If no configuration satisfies

$$E(\varphi) \leq \Theta_1,$$

or if  $\Theta_1$  cannot be defined consistently,  $K_1$  is impossible.

### P.2.18 Boundary and Region Falsifiability

The admissible region  $\Omega$  has a boundary  $\partial\Omega$  defined by energy and continuity constraints.  $K_1$  is falsified if:

- $\partial\Omega$  cannot be defined,
- $\partial\Omega$  contradicts the topology  $(X, \tau)$ ,
- $\Omega$  collapses to the empty set,
- admissible configurations violate the  $K_1$  axioms.

The boundary is essential for the continuum's existence; without a valid boundary, the continuum has no admissible states.

### P.2.19 Cycle and Evolution Falsifiability

Even though  $K_1$  is minimally dynamic, it must support the trivial cycle  $C_{\text{triv}}$  and the minimal action of the operator  $F_1$ , defined by

$$K_1(t + dt) = F_1(K_1(t)).$$

The continuum is falsified if:

**(F1.9) The trivial cycle cannot be defined.** If  $C_{\text{triv}}$  fails to exist, continuity along the axis is impossible.

**(F1.10) The operator  $F_1$  cannot act consistently.** If evolution under  $F_1$  breaks admissibility or contradicts the axioms,  $K_1$  collapses.



**(F1.11) No stable configuration  $\varphi_0$ .** If there exists no baseline configuration  $\varphi_0$  satisfying continuity and threshold conditions, the continuum is inconsistent.

### P.2.20 Embedding-Space and Transition Falsifiability

The transition from  $K_0$  to  $K_1$  is governed by the operator  $\Psi_{0 \rightarrow 1}$  and requires the existence of the one-dimensional axis as a new admissible dimension.

$K_1$  is falsified when:

**(F1.12)  $\Psi_{0 \rightarrow 1}$  cannot act.** If the embedding space of  $K_0$  cannot support a one-dimensional axis, no  $K_1$  can arise.

**(F1.13) Dimensional threshold failure.** If the structural tension at  $K_0$  does not reach

$$T_0 > \Theta_{\text{dim}},$$

the new axis cannot emerge.

**(F1.14) Embedding contradiction.** If  $K_1$  cannot be embedded within the admissible meta-space specified by  $M_1$ , the continuum is impossible.

### P.2.21 Summary

$K_1$  is falsified whenever:

- its minimal topology  $(X, \tau)$  is inconsistent,
- its field space  $V$  is invalid or contradictory,
- the configuration space  $\Omega$  cannot be defined,
- the energy functional  $E$  violates axioms E1–E3,
- the threshold  $\Theta_1$  cannot be satisfied,
- the boundary  $\partial\Omega$  breaks consistency,
- the trivial cycle  $C_{\text{triv}}$  cannot exist,
- the evolution operator  $F_1$  cannot act coherently,
- the transition from  $K_0$  to  $K_1$  fails structurally.

Any such violation renders the continuum  $K_1$  impossible and blocks the entire emergence of higher continua  $K_2$ – $K_{12}$ .

### P.2.22 Falsifiability of $K_2$

The continuum  $K_2$  is the first fully physical continuum in the OC hierarchy. It is characterised by:

- the emergence of a new axis  $A_{\text{phys}}$  not contained in  $\text{span}_{\text{str}}(A_1)$ ,
- the birth of physical configuration space  $\Omega(K_2)$  defined by connectivity, percolation and local interaction,
- the existence of a critical threshold  $p_c$  determining global connectedness,
- the appearance of physical potentials and flows  $(P_{\text{phys}}, J_{\text{phys}})$ ,
- physically meaningful boundaries  $\partial\Omega(K_2)$  defined by field constraints,
- non-trivial structural tension  $T_2$ .

Falsifiability of  $K_2$  concerns the possibility that any of these structures fail.

### P.2.23 Topological and Connectivity Falsifiability

$K_2$  crucially depends on global connectivity. The physical continuum is defined by a connected region emerging after a percolation-like transition.

It is falsified if:

**(F2.1) No global connected component exists.** If the system remains below the percolation threshold  $p_c$  for all configurations, then

$$\text{Conn}(\Omega(K_2)) = 0$$

and the continuum cannot exist.

**(F2.2) Percolation theory predicts no admissible connected phase.** If the structure of  $K_1$  or the geometry of admissible states forbids a transition through  $p_c$ ,  $K_2$  cannot arise.

**(F2.3) Connectivity changes violate stability.** If connected components appear and disappear without a stable regime,  $K_2$  becomes dynamically inconsistent.

**(F2.4) Breakdown of the physical axis  $A_{\text{phys}}$ .**  $K_2$  requires a new axis with distinct admissible states (e.g. spatial dimension, phase angle, field intensity). If an axis with  $|A| < 2$  cannot be defined, the continuum fails.

### P.2.24 Percolation and Threshold Falsifiability

The fundamental transition from  $K_1$  to  $K_2$  is governed by a percolation threshold  $p_c$  and the dimensional threshold  $\Theta_{\text{dim}}$ .

$K_2$  is falsified if:

**(F2.5) The critical threshold  $p_c$  cannot be defined.** If no mathematically coherent condition separates connected and disconnected phases,  $K_2$  loses its defining structure.

**(F2.6)  $p_c$  is never attainable.** If the admissible configurations of  $K_1$  lie permanently in  $p < p_c$ ,  $K_2$  cannot emerge.

**(F2.7) The system overshoots the threshold incoherently.** If crossing  $p_c$  causes divergence or collapse of  $\Omega$  instead of forming a connected phase, the continuum fails.

**(F2.8) Dimensional threshold violation.** If the structural tension of  $K_1$  does not reach

$$T_1 > \Theta_{\text{dim}},$$

no new axis can form, and  $K_2$  cannot be born.

**(F2.9) Threshold landscape contradiction.** If the physical threshold set  $\Theta_{\text{phys}}$  cannot be consistently defined, physical interactions cannot stabilise.

### P.2.25 Boundary and Admissibility Falsifiability

The physical continuum requires a non-empty admissible region  $\Omega(K_2)$  with a well-defined boundary  $\partial\Omega(K_2)$ .

$K_2$  is falsified when:

**(F2.10)**  $\Omega(K_2) = \emptyset$ . If all physically admissible configurations violate thresholds, admissibility collapses.

**(F2.11)**  $\partial\Omega(K_2)$  **cannot be defined**. Boundaries must be set by physical fields, gradients, and potentials. If these contradict each other,  $\partial\Omega$  becomes undefined.

**(F2.12)**  $\partial\Omega(K_2)$  **contradicts continuity**. If the boundary cannot be embedded in  $\Omega(K_1)$ , the transition is invalid.

**(F2.13)** **Non-physical configurations leak into the admissible set**. If configurations violating interaction rules or connectivity are still admissible, the continuum becomes inconsistent.

### P.2.26 Potentials, Flows and Field Falsifiability

At  $K_2$  physical potentials  $P_{\text{phys}}$  and flows  $J_{\text{phys}}$  appear for the first time.

$K_2$  is falsified if:

**(F2.14)** **Physical potentials cannot be defined**. If no consistent scalar or vector potentials can be assigned, physical interaction is impossible.

**(F2.15)** **Potentials violate admissibility ranges**. If  $P_{\text{phys}}$  cannot be kept within the allowable region defined by  $\Theta_{\text{phys}}$ , the continuum collapses.

**(F2.16)** **Physical flows  $J_{\text{phys}}$  cannot be formed**. Flows must be definable from gradients and potentials. If no such dynamics exist,  $K_2$  fails.

**(F2.17)** **Flow divergence destroys connectivity**. If flows necessarily break the percolated structure, the continuum cannot maintain a global state.

### P.2.27 Dynamic and Evolutionary Falsifiability

The continuum  $K_2$  evolves under its physical operator:

$$K_2(t + dt) = F_2(K_2(t)),$$

where  $F_2$  incorporates physical potentials, flows, and connectivity constraints.

It is falsified if:

**(F2.18)**  $F_2$  **cannot be defined**. If no lawful evolution exists, the continuum is invalid.

**(F2.19)**  $F_2$  **breaks admissibility**. If evolution forces the system outside  $\Omega(K_2)$  or below  $p_c$ , stability is lost.

**(F2.20)**  $F_2$  **destroys the new axis**. If evolution eliminates the admissible states of  $A_{\text{phys}}$ , the continuum collapses.

**(F2.21)** **No stable attractors exist**. Physical continuum requires attractors or stable regions. If all trajectories diverge,  $K_2$  is falsified.

### P.2.28 Transition and Embedding Falsifiability

The transition  $K_1 \rightarrow K_2$  is governed by the operator  $F_{1 \rightarrow 2}$  and requires a coherent embedding in the meta-space  $M_2$ .

$K_2$  is falsified if:

**(F2.22)  $F_{1 \rightarrow 2}$  cannot act.** If  $K_1$  cannot generate the new axis or connectivity, the transition is impossible.

**(F2.23) Embedding inconsistency.** If  $K_2$  cannot be embedded in the admissible region of  $M_2$ , it cannot exist.

**(F2.24) Conflict with physical laws encoded in  $M_2$ .** If physical interactions required by  $K_2$  contradict the structure of  $M_2$ , the continuum fails.

### P.2.29 Summary

$K_2$  is falsified whenever:

- no global connected component forms,
- the percolation threshold  $p_c$  is unattainable or undefined,
- the physical axis  $A_{\text{phys}}$  cannot emerge,
- $\Omega(K_2)$  or its boundary  $\partial\Omega(K_2)$  cannot be defined,
- physical potentials and flows cannot be introduced,
- evolution under  $F_2$  is impossible or destroys stability,
- the transition  $K_1 \rightarrow K_2$  fails structurally or dynamically.

Failure of any of these conditions renders the physical continuum  $K_2$  impossible and blocks the entire physical branch of the OC hierarchy.

### P.2.30 Falsifiability of $K_3$

The continuum  $K_3$  corresponds to chemistry: the domain where configurations of atoms, molecules, and reaction networks form a coherent connected continuum governed by chemical thresholds, reaction pathways, and catalytic structure. It arises from the physical continuum  $K_2$  once stable molecular structures become admissible states and reaction networks form self-consistent cycles. Falsifiability of  $K_3$  concerns the conditions under which a chemical continuum fails to exist or loses stability.

The core components required for  $K_3$  are:

- a non-empty chemical configuration space  $\Omega(K_3)$ ,
- chemical axes  $A_3$  (bond states, conformations, reaction coordinates),
- admissible potentials  $P_3$  (activation energies, bond energies, chemical potentials),
- chemical thresholds  $\Theta_3$  (bond stability, catalytic thresholds, closure thresholds),
- flows  $J_3$  (reaction fluxes, diffusion, activation–decay processes),
- reaction cycles  $C_3$  (RAF networks, catalytic loops),
- stable boundaries  $\partial\Omega(K_3)$  separating chemically allowed and forbidden configurations.

Failure of any of these components falsifies  $K_3$ .

### P.2.31 Falsifiability via Chemical Configuration

The continuum  $K_3$  requires a coherent molecular configuration space.

$K_3$  is falsified if:

**(F3.1)  $\Omega(K_3)$  is empty.** If no set of molecular configurations satisfies all bond thresholds  $\Theta_{\text{bond}}$ , chemical geometry cannot form a continuum.

**(F3.2) Bonding contradictions.** If bonding rules (valence, orbital geometry, electron sharing) cannot be consistently defined, the admissible configuration space collapses.

**(F3.3) Instability of all molecular states.** If every molecular configuration violates stability thresholds  $(\Theta_{\text{bond}}, \Theta_{\text{conf}})$ , then

$$\Omega(K_3) = \emptyset,$$

and chemistry cannot arise.

**(F3.4) No connected component in chemical configuration space.** If molecular states exist but do not form any connected set under allowed transformations, the continuum loses coherence.

**(F3.5) Disconnected reaction pathways.** If transitions between configurations are forbidden or inconsistent,  $\Omega(K_3)$  loses its path-connectedness.

### P.2.32 Falsifiability via Reaction Networks

The defining feature of  $K_3$  is the emergence of reaction networks and chemical flows  $J_3$ .

The continuum is falsified when:

**(F3.6) No reaction flows can be defined.** If activation energies, reaction coordinates, or intermediate states cannot be consistently specified, chemical transitions cannot occur.

**(F3.7) Reaction flows violate admissibility.** If  $J_3$  forces the system into forbidden regions of  $\Omega(K_3)$  (e.g. energetically impossible intermediates), continuity fails.

**(F3.8) Absence of catalytic pathways.** If no catalytic enhancements can lower activation thresholds,

$$\Theta_{\text{act}} = \Theta_{\text{act}}^{\min}$$

remains too high for reaction cycles to form.

**(F3.9) Contradictory reaction topology.** If reaction graphs necessarily produce broken or inconsistent pathways, chemical cycles cannot stabilise.

### P.2.33 Falsifiability via RAF Networks and Closure

$K_3$  includes RAF structures: reflexively autocatalytic and food-generated networks that support self-sustaining chemical dynamics.

The continuum fails if:

**(F3.10) RAF closure is impossible.** If no subset of reactions satisfies RAF conditions simultaneously, chemical self-maintenance cannot emerge.

**(F3.11) Catalytic inconsistency.** If catalytic rules violate activation or bonding thresholds, the network cannot close.

**(F3.12) Food-set inconsistency.** If the set of basic molecules cannot generate the full RAF system, closure collapses.

**(F3.13) Unstable intermediary states.** If intermediates required for RAF structure violate  $\Theta_3$ , the network cannot stabilise.

**(F3.14) RAF cycles conflict with  $\partial\Omega(K_3)$ .** If catalytic loops require forbidden configurations outside the admissible domain, the continuum becomes inconsistent.

### P.2.34 Falsifiability via Chemical Potentials and Thresholds

The chemical continuum requires well-defined potentials  $P_3$  and chemical thresholds  $\Theta_3$ .

$K_3$  is falsified if:

**(F3.15) Chemical potentials cannot be defined.** If free energies, activation energies, or chemical potentials contradict each other or cannot be assigned consistently,  $K_3$  collapses.

**(F3.16) Threshold contradictions.** If the set of thresholds  $\Theta_3$  (bond, activation, catalytic, network) fails to admit any consistent configuration, admissibility fails.

**(F3.17) Threshold–flow inconsistency.** If physical flows (diffusion, reaction rates) require surpassing forbidden threshold regions, the continuum breaks.

**(F3.18) No stable energetic minima.** If potentials admit no attractors or local minima, chemical cycles cannot settle and the continuum diverges.

### P.2.35 Falsifiability via Boundaries and Admissibility

Chemical admissibility is defined by  $\partial\Omega(K_3)$ , separating chemically allowed and forbidden regions.

$K_3$  is falsified if:

**(F3.19) Boundary inconsistency.** If the admissible boundary cannot be defined using chemical potentials or bonding rules,  $\Omega(K_3)$  collapses.

**(F3.20) Forbidden leakage.** If configurations violating  $\Theta_3$  appear as accessible, the continuum becomes ill-defined.

**(F3.21) Loss of chemical domain.** If environmental fluctuations (temperature, pressure, radiation) regularly force the system outside  $\Omega(K_3)$ , stability is impossible.

**(F3.22) No embedding into  $K_2$ .** If chemical configurations cannot be embedded in physical constraints ( $K_2$ ), the transition fails.

### P.2.36 Falsifiability via Dynamics and Evolution

The evolution of chemical systems is governed by the operator:

$$K_3(t + dt) = F_3(K_3(t)),$$

which includes reaction dynamics, diffusion, catalytic enhancement, thresholds, and energy landscapes.

$K_3$  is falsified when:

**(F3.23)  $F_3$  is undefined.** If no coherent dynamical rules exist, the continuum collapses.

**(F3.24)  $F_3$  violates admissibility.** If evolution drives the system outside  $\Omega(K_3)$ , chemical coherence fails.

**(F3.25) No stable cycles form.** If all reaction cycles decay or diverge, no persistent chemical dynamics exist.

**(F3.26) Reaction blow-up.** If flows  $J_3$  produce runaway reactions that exceed all thresholds,  $K_3$  collapses.

**(F3.27) Loss of catalytic support.** If regulatory or catalytic effects destabilise over time, RAF networks break.

### P.2.37 Falsifiability via Transition $K_2 \rightarrow K_3$

The transition from physics to chemistry is mediated by the operator  $F_{2 \rightarrow 3}$ , which introduces molecular axes, chemical potentials and reaction pathways.

The continuum is falsified if:

**(F3.28)  $F_{2 \rightarrow 3}$  cannot act.** If molecular configurations cannot be generated from  $K_2$ , chemistry fails to arise.

**(F3.29) Incompatibility with physical potentials.** If chemical potentials require physical states forbidden by  $K_2$ , embedding fails.

**(F3.30) Failure of new axis.** If the reaction coordinate or bonding axis has  $|A_3| < 2$ , no chemical continuum exists.

**(F3.31) Dimensional threshold violation.** If structural tension at  $K_2$  never reaches  $\Theta_{\text{dim}}$ , the new chemical axis cannot be born.

### P.2.38 Summary

$K_3$  is falsified if any of the following fail:

- the chemical configuration space  $\Omega(K_3)$ ,
- reaction flows  $J_3$  and RAF cycles  $C_3$ ,
- chemical potentials and thresholds  $\Theta_3$ ,
- stability of boundaries  $\partial\Omega(K_3)$ ,
- dynamical operator  $F_3$ ,

- the transition  $K_2 \rightarrow K_3$ .

If any of these structures cannot be defined or sustained, the chemical continuum collapses and the pathway toward  $K_4$  becomes impossible.

### P.2.39 Falsifiability of $K_4$

The continuum  $K_4$  represents the biological proto-cellular domain: systems with a stable boundary  $\partial\Omega(K_4)$  (membrane or equivalent), internal-external potential separation, metabolic and redox flows, incipient regulatory logic, and the first appearance of an internal energetic landscape. Falsifiability of  $K_4$  concerns the structural, dynamical, and energetic conditions under which such a proto-biological continuum cannot arise or cannot persist.

A valid  $K_4$  requires the following:

- a non-empty state space  $\Omega(K_4)$  of compartmentalised, gradient-supported molecular systems;
- a physically stable boundary  $\partial\Omega(K_4)$  (lipid-like compartments, porous mineral compartments, vesicles or early protocells);
- a set of biological axes  $A_4$ , including gradient axes, charge separation, redox axes, permeability axes, and boundary curvature;
- internal and external potentials  $P_4$ :  $P_{\text{grad}}, P_{\text{ion}}, P_{\text{redox}}, P_{\text{membrane}}, P_{\text{env}}$ ;
- biological thresholds  $\Theta_4$  (closure threshold, permeability threshold, osmotic threshold, redox stability, minimal gradient stability);
- flows  $J_4$  (diffusion, active transport, redox fluxes, osmosis, proto-metabolic fluxes);
- regulatory and feedback cycles  $C_4$  (buffering, pump-leak cycles, gradient-maintenance cycles);
- positive continuum measure  $k_4 > 0$ , supported by stable internal gradients and boundary integrity.

If any of these structures fail, the biological continuum  $K_4$  is falsified.

### P.2.40 Falsifiability via Compartment Formation

A core requirement for  $K_4$  is the existence of a stable compartment with a well-defined boundary.

The continuum collapses if:

**(F4.1) No stable boundary can form.** If amphiphilic or mineral structures cannot create a persistent  $\partial\Omega(K_4)$ , compartmentalisation is impossible.

**(F4.2) Boundary violates closure threshold  $\Theta_{\text{closure}}$ .** If the membrane does not maintain enclosure under mechanical or chemical perturbations,  $K_4$  cannot arise.

**(F4.3) Excessive permeability ( $\Theta_{\text{perm}}$  not satisfied).** If solutes, ions or redox species cross the membrane too freely, no stable gradients can be maintained.

**(F4.4) Zero permeability (no exchange).** If  $\partial\Omega(K_4)$  prevents all flows  $J_4$ , metabolic or redox cycles cannot operate, and the proto-cell becomes inert.

**(F4.5) Boundary geometry instability.** If curvature fluctuations exceed  $\Theta_{\text{curv}}$ , the compartment collapses or fragments.

**(F4.6) Patch-level instability.** If local membrane patches have incompatible phase states ( $L_0, L_\alpha, L_\beta$ ) or violate local thresholds  $\Theta_{\text{mem},i}$ , then  $\partial\Omega(K_4)$  is globally unstable.



#### P.2.41 Falsifiability via Biological Potentials $P_4$

$K_4$  requires a distinct separation of internal and external potentials.

The continuum is falsified when:

**(F4.7) No internal–external potential difference exists.** If  $P_{\text{in}} = P_{\text{out}}$  for all axes (ion gradients, redox potential, pH), the proto–cell lacks biological structure.

**(F4.8) Gradient instability.** If gradients cannot be maintained due to leaks or insufficient transport,  $\partial P / \partial A$  collapses.

**(F4.9) Redox potential inconsistency.** If  $P_{\text{redox}}$  cannot be defined or violates  $\Theta_{\text{redox}}$ , metabolic cycles cannot function.

**(F4.10) Osmotic imbalance.** If osmotic pressure exceeds  $\Theta_{\text{osm}}$ , the membrane bursts.

**(F4.11) Energetic flatness.** If the internal potential landscape lacks minima or metastable basins, the system cannot sustain cycles  $C_4$ .

#### P.2.42 Falsifiability via Flows $J_4$

Flows  $J_4$  support metabolism-like behaviour and gradient maintenance.

$K_4$  is falsified if:

**(F4.12) No flows exist.** If all flows vanish ( $J_4 = 0$ ), the compartment becomes chemically inert.

**(F4.13) Uncontrolled leak flux.** If passive leaks exceed pump capacity or buffering potential, gradients collapse.

**(F4.14) Redox runaway.** If redox flux pushes reaction intermediates beyond  $\Theta_{\text{redox-max}}$ , structural collapse occurs.

**(F4.15) Osmotic runaway.** If inflow of water exceeds membrane elasticity limits, compartment destruction follows.

**(F4.16) Counter-flow inconsistency.** If opposing flows violate conservation or potential gradients, the system cannot settle into a self-maintaining regime.

**(F4.17) No balancing cycles.** If pump–leak cycles fail to stabilise  $J_4$ , gradients vanish and  $k_4 \rightarrow 0$ .

#### P.2.43 Falsifiability via Cycles $C_4$

Biological cycles at  $K_4$  include:

- buffer cycles, - redox cycles, - proton gradient cycles, - pump–leak cycles, - primitive regulatory cycles.

The continuum is falsified if:

**(F4.18) No stabilising cycles  $C_4$  exist.** If all cycles decay, gradients and potentials collapse.

**(F4.19) Cycles violate thresholds.** If a cycle requires states outside  $\Omega(K_4)$  or exceeding  $\Theta_4$ , it cannot run.

**(F4.20) Feedback divergence.** If a feedback loop amplifies fluctuations without damping, structural tension  $T_4$  diverges.

**(F4.21) Buffering failure.** If pH or ion-buffer cycles cannot stabilise internal chemistry,  $P_{\text{in}}$  becomes undefined.

**(F4.22) Cycle incompatibility.** If cycles impose contradictory requirements on gradients or potentials, no coherent  $C_4$  exists.

#### **P.2.44 Falsifiability via Membrane-Bound Information and Regulation**

The early regulatory axis  $A_{\text{logic}}$  emerges at  $K_4$  as stochastic logic (Fix.W2), forming probabilistic switches regulating flows.

The continuum is falsified when:

**(F4.23) No logical switching.** If the switching probability matrix  $M(t)$  cannot be defined or  $\Theta_{\text{logic}}$  is not met, regulation is impossible.

**(F4.24) Noise-dominated logic.** If noise exceeds  $\Theta_{\text{noise}}$ , switching becomes random and control collapses.

**(F4.25) Logical entropy divergence.** If  $S_{\text{logic}} > S_{\text{max}}$  (observed range 1.0–1.4), stable regulation cannot be maintained.

**(F4.26) Regulatory inconsistency.** If regulatory cycles conflict with flows or potentials,  $C_4$  fails to stabilise the system.

#### **P.2.45 Falsifiability via Structural Tension $T_4$**

Structural tension combines membrane, chemical, and electrical stresses:

$$T_4 = T_{\text{chem}} + T_{\text{mem}} + T_{\text{elec}}.$$

The continuum is falsified if:

**(F4.27)  $T_4$  exceeds collapse threshold  $\Theta_{\text{collapse}}$ .** If structural tension surpasses membrane strength or chemical compatibility, the system ruptures.

**(F4.28) Persistent local overstress.** Local patch tension  $T_i$  exceeding  $\Theta_{\text{patch}}$  leads to nucleation of collapse.

**(F4.29) Electric instability.** If membrane potential  $\Delta V$  oscillates beyond  $\Theta_{\text{exc-pre}}$  without recovery cycles, the boundary loses integrity.

**(F4.30) No stress-relief pathways.** If the system lacks compensatory cycles, tension accumulates without bounds.

#### P.2.46 Falsifiability via Transition $K_3 \rightarrow K_4$

The transition is governed by the operator  $\Psi_{3 \rightarrow 4}$ , requiring:

- stable compartment formation, - gradient separation, - emergence of  $A_{\text{grad}}$  and  $A_{\text{charge}}$ , - partial redox/metabolic cycles.

$K_4$  is falsified if:

**(F4.31)  $\Psi_{3 \rightarrow 4}$  cannot be defined.** If no compartment–gradient system can be constructed from chemistry, the transition fails.

**(F4.32) Inconsistency with  $K_3$  potentials.** If chemical potentials conflict with gradient/boundary requirements, the embedding fails.

**(F4.33) No new axis with  $|A_4| \geq 2$ .** If the biological axes (gradient, charge, redox, permeability) cannot instantiate incompatible states, dimensional birth fails.

**(F4.34) Threshold mismatch.** If  $\Theta_{\text{closure}}$ ,  $\Theta_{\text{grad}}$ ,  $\Theta_{\text{perm}}$  cannot be met simultaneously,  $\Omega(K_4)$  is empty.

**(F4.35) No viable  $\partial\Omega(K_4)$ .** If membrane formation is impossible, no biological domain arises.

#### P.2.47 Summary

The biological proto–cell continuum  $K_4$  is falsified when any of the following structural elements cannot be defined or maintained:

- stable boundary  $\partial\Omega(K_4)$ ;
- internal–external potentials  $P_4$  and gradients;
- biologically relevant flows  $J_4$ ;
- stabilising cycles  $C_4$ ;
- regulatory switching dynamics;
- bounded structural tension  $T_4$ ;
- a viable transition  $\Psi_{3 \rightarrow 4}$  from chemistry.

If these conditions fail,  $K_4$  collapses, and the biological lineage toward  $K_5$  cannot emerge.

#### P.2.48 Falsifiability of $K_5$

The continuum  $K_5$  formalises early neuronal excitability: stable membrane-bound electrical potentials, ion-selective channels, proto–spikes, controlled pump–leak cycles, and emerging electrical signalling. Falsifiability of  $K_5$  concerns the necessary structural and dynamical conditions under which excitability cannot arise or cannot persist.

A valid  $K_5$  requires:

- a stable membrane boundary  $\partial\Omega(K_5)$  capable of maintaining an electrical potential difference;
- a set of electrical axes  $A_5$  including  $A_{\text{exc}}$ ,  $A_{\text{channel}}$ ,  $A_{\text{perm}}$ , and early stochastic logic  $A_{\text{logic}}$ ;
- electrical potentials  $P_5$ : membrane potential  $\Delta V$ , Nernst potentials  $E_{\text{ion}}$ , redox-driven energetic potentials, and channel gating energies;
- biological thresholds  $\Theta_5$ : excitation threshold  $\Theta_{\text{exc}}$ , recovery threshold  $\Theta_{\text{rec}}$ , leak threshold  $\Theta_{\text{leak}}$ , noise threshold  $\Theta_{\text{noise}}$ ;

- flows  $J_5$ : ion fluxes through channels, leaks, pumps, excitatory flows, recovery flows;
- cycles  $C_5$ : excitation–recovery cycles, leak–pump balance cycles, electrical buffering cycles;
- $k_5 > 0$  supported by stable excitability and recovery.

If any of these structures fails, the neuronal continuum  $K_5$  collapses and cannot exist.

#### P.2.49 Falsifiability via Membrane and Electrical Boundary

A stable membrane is a prerequisite for  $K_5$ . The continuum is falsified when:

**(F5.1) No stable membrane potential.** If  $\Delta V = 0$  for all time and cannot be induced by ionic concentrations or pumps, excitability cannot arise.

**(F5.2) Excessive leakage.** If passive ion leaks exceed pump capacity (violating  $\Theta_{\text{leak}}$ ),  $\Delta V$  cannot be maintained.

**(F5.3) Patch instability.** If membrane patches fluctuate beyond local thresholds  $\Theta_{\text{mem},i}$  (LUX patch model), global excitability fails.

**(F5.4) No channel selectivity.** If channels cannot discriminate ions,  $E_{\text{ion}}$  is undefined, so  $\Delta V$  becomes incoherent.

**(F5.5) Osmotic/electrical incompatibility.** If osmotic forces violate curvature or tension thresholds,  $\partial\Omega(K_5)$  collapses.

#### P.2.50 Falsifiability via Electrical Potentials $P_5$

Neuronal excitability depends on:

-  $\Delta V$  — membrane potential; -  $E_{\text{ion}}$  — equilibrium potentials; - gating potentials for channel opening/closing.

The continuum is falsified when:

**(F5.6) Undefined Nernst potentials.** If ion gradients collapse ( $P_{\text{grad}}$  fails),  $E_{\text{ion}}$  cannot be defined.

**(F5.7) Flattened electrical landscape.** If gating energies cannot produce stable open/closed states, channels cannot support excitability.

**(F5.8) Divergent  $\Delta V$ .** If  $\Delta V$  exceeds  $\Theta_{\text{exc},\text{max}}$  without recovery, structural tension  $T_5$  diverges.

**(F5.9) No recovery potentials.** If the system lacks a path back to baseline ( $P_{\text{rec}}$  undefined), cycles  $C_5$  cannot close.

**(F5.10) Redox instability.** If redox energy driving pumps violates  $\Theta_{\text{redox}}$ , electrical potentials cannot be maintained.

#### P.2.51 Falsifiability via Ion Flows $J_5$

Flows constitute the dynamical basis of excitability.

$K_5$  is falsified if:

**(F5.11) No ion channels.** If  $J_{\text{channel}} = 0$  for all states, excitation dynamics cannot occur.

**(F5.12) Channel gating incompatibility.** If channel transitions (open/closed/refrac/noisy) violate

$$J_{\text{ion}} = g_{\text{channel}}(\Delta V - E_{\text{ion}}),$$

excitability fails.

**(F5.13) Leak runaway.** If leak currents exceed  $\Theta_{\text{leak}}$ , the system collapses to a depolarised state.

**(F5.14) Pump failure.** If metabolic pumps cannot compensate leaks,  $\Delta V$  vanishes and  $k_5 \rightarrow 0$ .

**(F5.15) No refractory flow.** If  $J_{\text{refrac}}$  cannot restore channel states, spike cycles cannot repeat.

**(F5.16) Counter-flow conflict.** If fluxes violate conservation or create contradictory  $P_5$ , the continuum destabilises.

### P.2.52 Falsifiability via Neural Cycles $C_5$

The early neuronal continuum requires several stabilising cycles:

- the excitation cycle  $C_{\text{exc}}$ , - the recovery cycle  $C_{\text{rec}}$ , - the leak–pump balance cycle  $C_{\text{leak}}$ , - early electrical buffering cycles.

The continuum is falsified when:

**(F5.17) No excitation cycle.** If an initial excitation cannot propagate along  $A_{\text{exc}}$  or fails to reach threshold  $\Theta_{\text{exc}}$ , action-potential–like dynamics do not exist.

**(F5.18) Failure to reach recovery.** If the system cannot return to baseline, the excitation–recovery cycle cannot close.

**(F5.19) Divergent recovery.** If recovery overshoots beyond stability thresholds, oscillations destroy boundary integrity.

**(F5.20) Leak–pump imbalance.** If  $C_{\text{leak}}$  cannot offset leaks, stable  $\Delta V$  is impossible.

**(F5.21) Cycle incompatibility.** If cycles impose conflicting demands on  $\Delta V$  or  $E_{\text{ion}}$ , no coherent excitability regime exists.

### P.2.53 Falsifiability via Stochastic Logic and Information Flow

At  $K_5$  the stochastic logic axis  $A_{\text{logic}}$  becomes functional (Fix.W2). Switching dynamics are governed by:

$$p(t + \Delta t) = M(t)p(t),$$

where  $M(t)$  depends on potentials, flows and thresholds.

The continuum collapses when:

**(F5.22) No switching matrix  $M(t)$ .** If logical transitions cannot be defined, regulatory control is absent.

**(F5.23) Excessive noise.** If noise exceeds  $\Theta_{\text{noise}}$ , switching becomes random, destroying coordinated excitability.

**(F5.24) Logical entropy divergence.** If the logical entropy  $S_{\text{logic}} > S_{\text{max}}$ , stable information processing cannot occur.

**(F5.25) Incompatible regulatory cycles.** If regulatory logic conflicts with cycles  $C_5$  or flows  $J_5$ , the continuum destabilises.

#### P.2.54 Falsifiability via Structural Tension $T_5$

Electrical activity induces tension:

$$T_5 = T_{\text{chem}} + T_{\text{mem}} + T_{\text{elec}}.$$

$K_5$  is falsified when:

**(F5.26)  $T_5$  exceeds collapse threshold  $\Theta_{\text{collapse}}$ .** Excitability leads to membrane rupture or ionic overload.

**(F5.27) Persistent local overstress.** Local over stresses  $T_i$  (patch-level) induce failure cascades.

**(F5.28) Refractory instability.** If refractory processes cannot reduce tension, excitability cannot be cyclic.

**(F5.29) Electric runaway.** If sequential spikes exceed tension bounds, the system transitions into uncontrollable oscillations.

#### P.2.55 Falsifiability of Transition $K_4 \rightarrow K_5$

The transition is governed by the operator  $\Psi_{4 \rightarrow 5}$ , requiring:

- stable  $\partial\Omega(K_4)$ , - persistent  $\Delta V$ , - ion channels, - gating dynamics, - excitation and recovery within bounds.

Transition fails when:

**(F5.30)  $\Psi_{4 \rightarrow 5}$  is undefined.** If no electrical axis  $A_{\text{exc}}$  can emerge,  $K_5$  cannot be born.

**(F5.31) No incompatible states in  $A_{\text{channel}}$ .** If channels have fewer than two incompatible gating states, dimension growth fails (Theorem 5).

**(F5.32) Threshold mismatch.** If  $\Theta_{\text{exc}}$ ,  $\Theta_{\text{rec}}$ ,  $\Theta_{\text{leak}}$  cannot be simultaneously satisfied,  $\Omega(K_5)$  is empty.

**(F5.33) Pump–energy incompatibility.** If energetic potentials from  $K_4$  cannot support pumping,  $\Delta V$  vanishes.

**(F5.34) No viable recovery path.** If cycles cannot restore the system to baseline,  $K_5$  cannot stabilise.

### P.2.56 Summary

The neuronal proto-excitable continuum  $K_5$  is falsified if any of:

- membrane potential  $\Delta V$  cannot be maintained;
- ion flows  $J_5$  cannot create or sustain excitability;
- cycles  $C_5$  cannot close;
- stochastic logic becomes noise-dominated;
- structural tension  $T_5$  exceeds thresholds;
- the transition  $\Psi_{4 \rightarrow 5}$  is not viable.

If these conditions fail, the electrical axis collapses, and no neuronal lineage toward  $K_6$  can arise.

### P.2.57 Falsifiability of $K_6$

The continuum  $K_6$  formalises cognitive organisation: stable representations, binding dynamics, predictive processing, memory, model coherence and informational flows. Falsifiability of  $K_6$  identifies the structural and dynamical conditions under which cognition cannot arise or cannot persist.

A valid  $K_6$  requires:

- representational axes  $A_6$  (binding, comparison, prediction, memory);
- cognitive potentials  $P_6$ : representational energy, expected value, prediction error, model confidence;
- stable thresholds  $\Theta_6$ : binding threshold  $\Theta_{\text{bind}}$ , prediction threshold  $\Theta_{\text{pred}}$ , memory stability threshold  $\Theta_{\text{mem}}$ ;
- information flows  $J_6$ : prediction flows, comparison flows, model-update flows, memory flows;
- cognitive cycles  $C_6$ : selection, comparison, binding, prediction, model formation, memory;
- representational state space  $\Omega(K_6)$  with stable boundary  $\partial\Omega(K_6)$ ;
- structural tension  $T_6$  within bounds;
- positive continuumness  $k_6 > 0$ .

If any of these conditions fails, cognitive organisation collapses and the system cannot maintain  $K_6$ .

### P.2.58 Falsifiability via Representational Structure

A cognitive continuum requires coherent representational geometry.

$K_6$  is falsified when:

**(F6.1) No representational axes.** If representational dimensions (features, concepts, percepts) do not form a stable axis system  $A_6$ , cognition cannot arise.

**(F6.2) Unstable representational potentials.** If representational energy diverges or cannot stabilise,  $\Omega_6$  becomes fragmented.

**(F6.3) Boundary incoherence.** If representational boundaries  $\partial\Omega_6$  fluctuate beyond  $\Theta_{\text{bound}}$ , the space of meanings collapses.

**(F6.4) Degenerate representation.** If all representational states collapse to a single fixed point, no discrimination or inference is possible.

**(F6.5) No representational gradients.** If  $\partial P_6 / \partial A_6 = 0$ , there is no informational drive for cognition (prediction, comparison, binding).

### P.2.59 Falsifiability via Binding Dynamics

Binding is essential for constructing compound representations.

$K_6$  is falsified when:

**(F6.6) Binding threshold not reached.** If  $\Theta_{\text{bind}}$  cannot be satisfied, binding cycles  $C_{\text{bind}}$  cannot close.

**(F6.7) Overbinding.** If binding is too strong (super-saturation), representations fuse uncontrollably, destroying structure.

**(F6.8) Underbinding.** If binding energy is too weak, complex representations cannot form.

**(F6.9) Incompatible bindings.** If binding requirements conflict across axes,  $T_6$  diverges and  $\Omega_6$  becomes inconsistent.

**(F6.10) Binding incoherence across cycles.** If binding cannot synchronise with selection or comparison cycles, the representational hierarchy collapses.

### P.2.60 Falsifiability via Predictive Processing

Prediction is a defining axis of  $K_6$ .

The continuum is falsified when:

**(F6.11) Prediction threshold not met.** If prediction error cannot be driven below  $\Theta_{\text{pred}}$ , predictive cycles cannot stabilise.

**(F6.12) Divergent prediction error.** If prediction error grows without bounds, model coherence collapses.

**(F6.13) No prediction flows.** If  $J_{\text{pred}} = 0$ , the system cannot update or test models.

**(F6.14) Negative model confidence.** If  $P_{\text{model}}$  falls below stability threshold, the system cannot sustain internal models.

**(F6.15) Prediction–binding conflict.** If prediction dynamics contradict binding constraints, no integrated cognition exists.

### P.2.61 Falsifiability via Comparison and Selection

Cognition requires comparing representations and selecting relevant ones.

$K_6$  is falsified when:

**(F6.16) Failed comparison cycles.** If comparison flows cannot discriminate representations, selection and inference fail.

**(F6.17) Saturated comparison.** If all comparison gradients collapse (all-to-one similarity), no reasoning is possible.



**(F6.18) No selection dynamics.** If cycles  $C_{\text{sel}}$  cannot produce stable selection, the system cannot form meaningful internal states.

**(F6.19) Selection–prediction incompatibility.** If selection contradicts predictive requirements, the continuum destabilises.

### **P.2.62 Falsifiability via Memory and Stability**

Memory provides temporal continuity for cognition.

$K_6$  is falsified when:

**(F6.20) No stable memory.** If memory traces decay too quickly,  $\Theta_{\text{mem}}$  is violated.

**(F6.21) Memory overflow.** If memory accumulates without pruning, tension  $T_6$  becomes unsustainable.

**(F6.22) Inconsistent memory cycles.** If memory update cycles  $C_{\text{mem}}$  conflict with prediction or binding cycles, representational collapse occurs.

**(F6.23) Memory structural failure.** If storage axes cannot maintain consistent states, long-term consistency breaks.

**(F6.24) Memory–boundary incoherence.** If stored representations exceed  $\partial\Omega_6$ , semantic collapse results.

### **P.2.63 Falsifiability via Cognitive Flows $J_6$**

Flows govern dynamics of inference, prediction, and model update.

The continuum collapses when:

**(F6.25) No information flow.** If  $J_6 = 0$ , no cognitive process can evolve.

**(F6.26) Flow saturation.** If flows become too weak or too strong, cycles cannot coordinate.

**(F6.27) Contradictory flows.** If flows impose incompatible demands on  $P_6$ , the system becomes unstable.

**(F6.28) Destructive flows.** If  $J_{\text{kill}} > J_{\text{critical}}$ , cognition collapses entirely.

### **P.2.64 Falsifiability via Cognitive Cycles $C_6$**

There are six primary cognitive cycles:

$$C_6 = \{C_{\text{sel}}, C_{\text{cmp}}, C_{\text{bind}}, C_{\text{pred}}, C_{\text{model}}, C_{\text{mem}}\}.$$

$K_6$  is falsified when any stabilising cycle breaks:

**(F6.29) Broken selection cycle.** No stable criteria for relevance.

**(F6.30) Broken comparison cycle.** No resolvable representational differences.

**(F6.31) Broken binding cycle.** Compound representations cannot form.

**(F6.32) Broken prediction cycle.** Models cannot be evaluated or refined.

**(F6.33) Broken model cycle.** The system cannot maintain a coherent internal world-model.

**(F6.34) Broken memory cycle.** Traces cannot be stored or recalled.

### **P.2.65 Falsifiability via Structural Tension $T_6$**

Cognitive tension includes representational, predictive, and memory-related components.  $K_6$  collapses when:

**(F6.35)  $T_6$  exceeds cognitive collapse threshold.** Overload destroys representational integrity.

**(F6.36) Persistent prediction–error stress.** If prediction error stays high for long periods, the system cannot stabilise its models.

**(F6.37) Memory–tension feedback loop.** If memory accumulation increases  $T_6$  beyond thresholds, runaway fragmentation occurs.

**(F6.38) Boundary tension divergence.** If  $\partial\Omega_6$  is destabilised by representational conflict, the continuum dies.

### **P.2.66 Falsifiability of Transition $K_5 \rightarrow K_6$**

Transition operator  $\Psi_{5 \rightarrow 6}$  requires:

- stable excitability and proto-processing at  $K_5$ , - binding-capable representational axes, - predictive and comparison capacities, - memory stability.

Transition is falsified when:

**(F6.39) No representational birth.** If no new representational axis emerges, cognition cannot arise.

**(F6.40) No predictive axis.** If prediction flows cannot form, the system remains at  $K_5$ .

**(F6.41) Binding dimension failure.** If binding cannot reach  $\Theta_{\text{bind}}$ , no composite representations exist.

**(F6.42) Model-collapse at birth.** If early models cannot stabilise,  $\Omega(K_6)$  is empty.

**(F6.43) No temporal integration.** If memory axes cannot form, the system cannot maintain persistent states over time.

### **P.2.67 Summary**

The cognitive continuum  $K_6$  is falsified if any of the following hold:

- representational geometry is unstable or degenerate;
- binding, prediction, comparison or memory cycles fail;
- cognitive flows  $J_6$  are inconsistent or destructive;

- structural tension  $T_6$  exceeds thresholds;
- the transition  $\Psi_{5 \rightarrow 6}$  cannot produce a stable representational axis.

If these conditions fail, coherent cognition is impossible and the continuum collapses to lower organisational levels.

### P.2.68 Falsifiability of $K_7$

The continuum  $K_7$  formalises social organisation: communication, coordination, role-dynamics, normative structures, collective behaviour and institutional coherence.

A valid  $K_7$  requires:

- social axes  $A_7$ : communication, roles, norms, coordination, cooperation;
- social potentials  $P_7$ : trust, status, resources, normative pressure, coordination cost, cohesion potential;
- thresholds  $\Theta_7$ : trust threshold  $\Theta_{\text{trust}}$ , cohesion threshold  $\Theta_{\text{coh}}$ , coordination threshold  $\Theta_{\text{coord}}$ , conflict threshold  $\Theta_{\text{conf}}$ ;
- flows  $J_7$ : communication flows, coordination flows, resource flows, normative flows;
- cycles  $C_7$ : communication cycle, coordination cycle, norm-formation cycle, institutionalisation cycle, conflict-resolution cycle;
- stable social state space  $\Omega(K_7)$  and boundary  $\partial\Omega(K_7)$ ;
- bounded structural tension  $T_7$ ;
- non-zero continuumness  $k_7 > 0$ .

Violation of any of these conditions falsifies  $K_7$ .

### P.2.69 Falsifiability via Communication Structure

Communication is the fundamental axis of  $K_7$ .

$K_7$  is falsified when:

**(F7.1) No communication coherence.** If communication signals cannot propagate or be interpreted, social organisation cannot arise.

**(F7.2) Breakdown of communication axes.** If  $A_{\text{comm}}$  becomes unstable or incoherent, collective states cannot be formed.

**(F7.3) Communication noise beyond threshold.** If noise exceeds  $\Theta_{\text{comm-noise}}$ ,  $\Omega_7$  loses connectivity.

**(F7.4) Non-reciprocal communication.** If communication flows collapse into unilateral signalling, coordination cycles fail.

**(F7.5) Divergent communication tension.** If communicative demands exceed  $T_7$ -bounds, social structure fragments.

### P.2.70 Falsifiability via Trust and Cohesion

Trust is a structural social potential.

$K_7$  is falsified when:

**(F7.6) Trust threshold not met.** If  $\Theta_{\text{trust}}$  cannot be achieved, stable cooperation cannot form.

**(F7.7) Trust collapse.** If trust drops below critical threshold, roles and coordination axes disintegrate.

**(F7.8) Cohesion instability.** If cohesion potential oscillates beyond  $\Theta_{\text{coh}}$ , the group fragments.

**(F7.9) Negative cohesion gradients.** If  $\partial P_{\text{coh}}/\partial A_7 < 0$ , collective behaviour becomes impossible.

**(F7.10) Cohesion–conflict inversion.** If conflict potentials dominate cohesion potentials,  $\Omega_7$  becomes unstable.

### **P.2.71 Falsifiability via Roles and Normative Structure**

Roles and norms enable predictable collective behaviour.

$K_7$  is falsified when:

**(F7.11) No stable role structure.** If role boundaries cannot stabilise, coordination and communication collapse.

**(F7.12) Norm instability.** If norms fluctuate beyond  $\Theta_{\text{norm}}$ , social cycles cannot close.

**(F7.13) Role–norm inconsistency.** If role expectations contradict normative constraints,  $T_7$  diverges.

**(F7.14) Norm saturation or overload.** If normative demands exceed cognitive bounds of individuals ( $K_6 \rightarrow K_7$  mismatch), collective stability collapses.

**(F7.15) Failed institutional coherence.** If institutions cannot maintain stable normative flows, no long-lived social organisation can emerge.

### **P.2.72 Falsifiability via Coordination Dynamics**

Coordination is a defining axis of social continua.

The continuum is falsified when:

**(F7.16) Coordination threshold not met.** If  $\Theta_{\text{coord}}$  cannot be achieved, collective action cannot stabilise.

**(F7.17) Coordination breakdown.** If individuals cannot align actions, flows  $J_{\text{coord}}$  become inconsistent.

**(F7.18) Coordination–communication mismatch.** If coordination demands contradict communication capacities, cycles break.

**(F7.19) Divergence of coordination cost.** If coordination cost potential  $P_{\text{coord}}$  grows unbounded, collective behaviour collapses.

**(F7.20) Coordination asymmetry.** Large asymmetries in coordination flows lead to instability of roles.

### P.2.73 Falsifiability via Resource Flows $J_7$

Resource exchange is a structural element of  $K_7$ .

$K_7$  is falsified when:

(F7.21) **No resource flows.** If  $J_{\text{res}} = 0$ , no sustained social structure is possible.

(F7.22) **Resource inequality beyond threshold.** If inequality exceeds  $\Theta_{\text{ineq}}$ , social cohesion collapses.

(F7.23) **Resource–trust breakdown.** If resource flows destroy trust gradients, the system becomes unstable.

(F7.24) **Destructive flows.** If  $J_{\text{kill}} > J_{\text{critical}}$ , group disintegrates.

(F7.25) **Resource saturation.** If flows overshoot capacity of  $\Omega_7$ , structural tension  $T_7$  diverges.

### P.2.74 Falsifiability via Social Cycles $C_7$

$K_7$  includes cycles:

$$C_7 = \{C_{\text{comm}}, C_{\text{coord}}, C_{\text{norm}}, C_{\text{inst}}, C_{\text{conflict}}\}.$$

The continuum collapses when:

(F7.26) **Communication cycle fails.** No stable exchange of information.

(F7.27) **Coordination cycle fails.** Collective behaviour cannot stabilise.

(F7.28) **Norm-formation cycle fails.** Norms cannot emerge or stabilise.

(F7.29) **Institutional cycle fails.** Institutions cannot maintain structure over time.

(F7.30) **Conflict-resolution cycle fails.** Conflicts accumulate until  $\Theta_{\text{conf}}$  is exceeded, destroying  $\Omega_7$ .

### P.2.75 Falsifiability via Structural Tension $T_7$

Social tension  $T_7$  includes communication tension, coordination tension, normative tension, and conflict tension.

$K_7$  is falsified when:

(F7.31)  **$T_7$  exceeds collapse threshold.** The group loses structural integrity.

(F7.32) **Communication–coordination divergence.** If demands from communication and coordination axes conflict,  $T_7$  rises until cycles break.

(F7.33) **Normative overload.** If normative tension exceeds sustainable bounds, social collapse occurs.

(F7.34) **Conflict accumulation.** If conflict grows beyond resolution capacity,  $\Omega_7$  becomes empty.

(F7.35) **Boundary tension.** If  $\partial\Omega_7$  destabilises (e.g. through fragmentation), the continuum dies.

### P.2.76 Falsifiability of Transition $K_6 \rightarrow K_7$

The transition  $\Psi_{6 \rightarrow 7}$  requires:

- stable cognition at  $K_6$ , - communication axes emergent from cognitive signalling, - trust and coordination potentials, - early norm-formation capability.

Transition is falsified when:

**(F7.36) No communicative birth.** If communication axes do not emerge, no social organisation is possible.

**(F7.37) Failed trust formation.** If initial trust gradients cannot stabilise, cooperation cannot emerge.

**(F7.38) Coordination impossibility.** If  $J_{\text{coord}}$  cannot form, the system remains at  $K_6$ .

**(F7.39) Normative incoherence at birth.** If norms cannot stabilise,  $\Omega_7$  cannot form.

**(F7.40) Cognitive–social incompatibility.** If capacities of  $K_6$  cannot support demands of  $K_7$ , the transition fails.

### P.2.77 Summary

The social continuum  $K_7$  is falsified if any of the following fail:

- communication structure,
- trust, cohesion, roles or norms,
- coordination dynamics or resource flows,
- social cycles,
- structural tension bounds,
- the transition  $\Psi_{6 \rightarrow 7}$ .

If these conditions are violated, collective organisation collapses and  $K_7$  cannot exist.

### P.2.78 Falsifiability of $K_8$

The continuum  $K_8$  describes civilisation-level organisation: symbolic systems, cumulative knowledge, institutional memory, technological infrastructure, written communication, and long-range coordination across large populations and long timescales.

A valid  $K_8$  requires:

- axes  $A_8$ : writing, symbolic systems, knowledge structures, large-scale institutions, technology, cultural reproduction;
- potentials  $P_8$ : epistemic potential, symbolic-energy potential, technological potential, institutional stability, cultural coherence, long-range coordination potential;
- thresholds  $\Theta_8$ : epistemic threshold, cultural-coherence threshold, institutional-stability threshold, technological-sufficiency threshold, complexity-control threshold;
- flows  $J_8$ : knowledge flows, symbolic flows, technological flows, institutional flows, long-range communication flows;
- cycles  $C_8$ : knowledge-cycle, technological-cycle, institutional-cycle, cultural-cycle, innovation-cycle, archival-cycle;

- stable civilisation space  $\Omega(K_8)$  and boundary  $\partial\Omega(K_8)$ ;
- bounded structural tension  $T_8$ ;
- non-zero continuumness  $k_8 > 0$ .

Violation of any of these leads to falsification of  $K_8$ .

### P.2.79 Falsifiability via Symbolic Systems and Writing

Symbolic systems and writing are defining axes of  $K_8$ . They enable cumulative culture, long-range memory, and expansion of social organisation beyond individual cognition.

$K_8$  is falsified when:

**(F8.1) No stable symbolic system.** If symbols cannot stabilise or be interpreted consistently, cumulative knowledge cannot form.

**(F8.2) Writing instability.** If written records degrade faster than they are reproduced, the civilisation cannot preserve information.

**(F8.3) Loss of symbolic coherence.** If symbolic systems fragment beyond  $\Theta_{\text{symbol}}$ ,  $\Omega_8$  loses connect-  
edness.

**(F8.4) Symbolic overload.** If symbolic complexity grows beyond  $\Theta_{\text{symbolic\_load}}$ , interpretability col-  
lapses.

**(F8.5) Divergent symbolic tension.** If symbolic demands exceed capacity of institutions and cognition,  $T_8$  diverges, destroying civilisation-level structure.

### P.2.80 Falsifiability via Knowledge Systems and Epistemic Structure

Knowledge systems define civilisation-level memory.

$K_8$  is falsified when:

**(F8.6) Epistemic threshold not met.** If epistemic potentials are insufficient to encode, preserve, and reproduce complex knowledge, civilisation cannot emerge.

**(F8.7) Epistemic collapse.** If knowledge decays faster than it accumulates, the system reverts to  $K_7$ .

**(F8.8) Epistemic fragmentation.** If incompatible knowledge clusters exceed  $\Theta_{\text{epistemic-frag}}$ , global co-  
herence collapses.

**(F8.9) Breakdown of scientific/institutional memory.** Failure of science, archives, schools, universi-  
ties, or epistemic institutions destroys  $\Omega_8$ .

**(F8.10) Cognitive–epistemic mismatch.** If cognitive capacities of individuals ( $K_6$ ) cannot support epistemic demands of  $K_8$ , civilisation becomes unstable.

### P.2.81 Falsifiability via Institutions and Governance

Civilisations require stable large-scale institutions.

$K_8$  is falsified when:

**(F8.11) Institutional instability.** If institutions oscillate beyond  $\Theta_{\text{instab}}$ , civilisation-scale coherence cannot hold.

**(F8.12) Collapse of coordination institutions.** If coordination, enforcement, or governance institutions fail, long-range organisation breaks down.

**(F8.13) Institutional–cultural mismatch.** If institutions contradict cultural norms beyond tolerance,  $T_8$  diverges.

**(F8.14) Bureaucratic overload.** If administrative complexity exceeds  $\Theta_{\text{bureaucratic}}$ , institutional flows  $J_8$  stall.

**(F8.15) Institutional memory decay.** If institutions lose continuity, civilisation loses stability.

### **P.2.82 Falsifiability via Technology and Infrastructure**

Technological systems in  $K_8$  create new degrees of freedom for resource flow, communication, and symbolic reproduction.

$K_8$  is falsified when:

**(F8.16) Technological sufficiency threshold not met.** If technologies required for knowledge reproduction do not exist, civilisation cannot persist.

**(F8.17) Infrastructure collapse.** If communication, energy, or transport infrastructure collapses, global cycles  $C_8$  break.

**(F8.18) Tech–institution mismatch.** If institutions cannot regulate or integrate technological change, stability fails.

**(F8.19) Negative technological externalities.** If technological side-effects exceed  $\Theta_{\text{tech\_damage}}$ , structure becomes unstable.

**(F8.20) Overacceleration.** If technological development grows faster than the adaptation capacity of institutions and culture, civilisation collapses.

### **P.2.83 Falsifiability via Cultural Dynamics**

Culture ensures long-term reproduction of civilisation patterns.

$K_8$  is falsified when:

**(F8.21) Cultural incoherence.** If cultural potentials diverge beyond  $\Theta_{\text{culture}}$ , civilisation loses identity and stability.

**(F8.22) Cultural stagnation.** If cultural reproduction fails (loss of traditions, language, norms), institutional stability collapses.

**(F8.23) Cultural fragmentation.** If groups split into incompatible cultural clusters,  $\Omega_8$  becomes disconnected.



**(F8.24) Cultural–symbolic divergence.** If cultural norms contradict symbolic systems (writing, science, law), the system becomes unstable.

**(F8.25) Cultural entropy.** Excessive randomness in cultural reproduction destroys large-scale coherence.

#### **P.2.84 Falsifiability via Civilisational Flows $J_8$**

Civilisation requires stable flows of information, resources, knowledge, symbols, and institutional authority.

$K_8$  is falsified when:

**(F8.26) Breakdown of long-range communication.** If communication networks fail, civilisation-scale organisation collapses.

**(F8.27) Resource flow collapse.** If energy, food, or material flows collapse, institutions break down.

**(F8.28) Knowledge-flow bottlenecks.** If knowledge cannot propagate, the innovation cycle fails.

**(F8.29) Symbolic-flow imbalance.** If symbolic flows overload cultural or institutional systems,  $T_8$  increases beyond stability.

**(F8.30) Institutional-flow disruption.** If authority or governance flows collapse,  $\Omega_8$  disintegrates.

#### **P.2.85 Falsifiability via Civilisational Cycles $C_8$**

$K_8$  depends on cycles:

$$C_8 = \{C_{\text{knowledge}}, C_{\text{technology}}, C_{\text{institution}}, C_{\text{culture}}, C_{\text{innovation}}, C_{\text{archive}}\}.$$

$K_8$  is falsified when:

**(F8.31) Knowledge-cycle collapse.** If knowledge cannot be preserved, tested, and transmitted, civilisation regresses to  $K_7$ .

**(F8.32) Institutional-cycle failure.** If institutions cannot reproduce themselves, long-term coherence is lost.

**(F8.33) Innovation-cycle breakdown.** If innovation cannot be sustained, technological stagnation collapses  $P_8$ .

**(F8.34) Archival-cycle failure.** If archives cannot preserve symbolic systems,  $\Omega_8$  collapses.

**(F8.35) Culture-cycle failure.** If culture cannot reproduce itself, civilisation dissolves.

#### **P.2.86 Falsifiability via Structural Tension $T_8$**

Structural tension at  $K_8$  includes:

- epistemic tension, - symbolic tension, - institutional tension, - cultural tension, - technological tension.

$K_8$  is falsified when:

**(F8.36)  $T_8$  exceeds collapse threshold.** Civilisation cannot stabilise under excessive internal stress.

**(F8.37) Multi-axis tension divergence.** If epistemic, institutional, and technological tensions diverge simultaneously, collapse is unavoidable.

**(F8.38) Boundary instability.** If  $\partial\Omega_8$  (geographical, cultural, or epistemic boundaries) destabilises, the continuum dies.

**(F8.39) Complexity overload.** If systemic complexity exceeds  $\Theta_{\text{complexity}}$ , institutions and culture cannot control it.

**(F8.40) System-wide resonance.** Synchronised failures across  $A_8$  axes destroy  $k_8$ .

### **P.2.87 Falsifiability of Transition $K_7 \rightarrow K_8$**

The transition  $\Psi_{7 \rightarrow 8}$  requires:

- emergence of stable writing and symbolic systems,
- scaling of institutions from group-level to civilisation-level,
- long-range communication and resource flows,
- cumulative culture and knowledge storage,
- technological and epistemic support.

The transition fails when:

**(F8.41) No writing emergence.** Without writing or symbolic storage, civilisation cannot form.

**(F8.42) Failed scaling of institutions.** If institutions cannot scale beyond group-level,  $K_8$  cannot stabilise.

**(F8.43) Insufficient technological base.** If technologies enabling symbolic reproduction do not exist, the transition stalls.

**(F8.44) Cultural incoherence at birth.** If culture cannot support symbolic systems and institutions,  $\Omega_8$  cannot form.

**(F8.45) Collapse of long-range flows.** If long-distance flows  $J_8$  cannot stabilise, the system remains at  $K_7$ .

### **P.2.88 Summary**

$K_8$  is falsified if civilisation-level symbolic, epistemic, institutional, technological, cultural, or flow structures cannot stabilise or maintain bounded tension.

If any of these conditions fail, the civilisation continuum collapses and  $K_8$  ceases to exist.

### P.2.89 Falsifiability of $K_9$

The continuum  $K_9$  corresponds to the level of scientific paradigms, formal theories, representational frameworks, and high-order epistemic structures. It formalises how civilisations generate, preserve, revise, and replace scientific world-models.

A functioning  $K_9$  requires:

- axes  $A_9$ : paradigms, models, formal languages, inferential frameworks, methodological rules, meta-epistemic structures;
- potentials  $P_9$ : epistemic potential, coherence potential, explanatory potential, predictive potential, paradigm-stability potential;
- thresholds  $\Theta_9$ : coherence, consistency, explanatory, model-complexity, and methodological-stability thresholds;
- flows  $J_9$ : flows of scientific information, paradigmatic flows, inferential flows, methodological flows;
- cycles  $C_9$ : theory-cycle, paradigm-cycle, model-validation cycle, anomaly-resolution cycle, replication cycle;
- stable paradigm space  $\Omega(K_9)$  and boundary  $\partial\Omega(K_9)$ ;
- bounded structural tension  $T_9$ ;
- non-zero continuumness  $k_9 > 0$ .

Violation of any of these conditions yields falsification of  $K_9$ .

### P.2.90 Falsifiability via Coherence and Logical Consistency

Scientific paradigms require consistency, coherence, and interpretability. Failures here collapse the epistemic continuum.

$K_9$  is falsified when:

**(F9.1) Logical inconsistency.** If the core paradigms become inconsistent (beyond  $\Theta_{\text{logic}}$ ), they cease to define valid models.

**(F9.2) Semantic incoherence.** If the meanings of fundamental concepts cannot be stabilised or aligned,  $\Omega_9$  breaks into incompatible fragments.

**(F9.3) Collapse of representational capacity.** If models cannot represent empirical or theoretical structures, explanatory potential falls below viability.

**(F9.4) Category-theoretical breakdown.** If mappings between models fail to form coherent categories or functors, meta-theoretical structure collapses.

**(F9.5) Reflexive inconsistency.** If meta-theoretical claims contradict the lower-level theories they regulate,  $T_9$  diverges and the continuum collapses.

### P.2.91 Falsifiability via Paradigm Dynamics

Paradigm transitions constitute the central mechanism of  $K_9$ .

$K_9$  is falsified when:

**(F9.6) Paradigm stagnation.** If anomalies accumulate without triggering paradigm revision, the system freezes into an incoherent state.

**(F9.7) Paradigm fragmentation.** If multiple incompatible paradigms coexist without integration,  $\Omega_9$  loses connectedness.

**(F9.8) Excessive paradigm volatility.** If paradigms change too rapidly, coherence is lost due to insufficient stability.

**(F9.9) Absence of Kuhnian cycles.** If science does not exhibit the predicted normal science  $\rightarrow$  anomalies  $\rightarrow$  crisis  $\rightarrow$  paradigm shift cycle, the continuum-level description is falsified.

**(F9.10) Paradigm collapse via cognitive overload.** If models exceed cognitive capacities of  $K_6$ , cascade failure propagates to  $K_9$ .

## **P.2.92 Falsifiability via Epistemic Potentials and Explanatory Power**

A scientific paradigm must provide explanation and prediction.

$K_9$  is falsified when:

**(F9.11) Explanatory collapse.** If a paradigm fails to explain known empirical structures, its explanatory potential falls below  $\Theta_{\text{exp}}$ .

**(F9.12) Predictive failure.** If predictions systematically fail, epistemic potential collapses.

**(F9.13) Degeneration of theoretical programmes.** If auxiliary hypotheses proliferate to patch anomalies,  $T_9$  diverges.

**(F9.14) Lack of compressive power.** If paradigms cannot compress empirical regularities, model complexity exceeds thresholds.

**(F9.15) Inability to integrate cross-domain evidence.** If the paradigm cannot map higher- or lower-level continua (physics, biology, cognition, society), it loses coherence in the  $K_9$  space.

## **P.2.93 Falsifiability via Methodological Structure**

Methodological rules regulate theory construction, validation, and replacement.

$K_9$  is falsified when:

**(F9.16) Methodological incoherence.** If rules of inference or validation contradict each other, models lose structural integrity.

**(F9.17) Replication crisis.** If empirical studies cannot be replicated, the model-validation cycle collapses.

**(F9.18) Method collapse under paradigm shift.** If methodological rules cannot adapt when paradigms change, transition cycles break.

**(F9.19) Methodological overfitting.** If methods become too narrow or specialised, cross-domain integration fails.

**(F9.20) Methodological–epistemic mismatch.** If methods cannot support epistemic goals, the continuum destabilises.

#### P.2.94 Falsifiability via Scientific Flows $J_9$

Flows of scientific information ensure the circulation of knowledge, correction of errors, and formation of stable epistemic structures.

$K_9$  is falsified when:

**(F9.21) Disruption of inferential flows.** If inferential steps cannot propagate through the community, science becomes non-functional.

**(F9.22) Failure of paradigmatic flows.** If paradigms cannot spread or stabilise, the system becomes fragmented.

**(F9.23) Breakdown of replication flows.** If replication data does not circulate, paradigm testing is impossible.

**(F9.24) Information bottlenecks.** If knowledge cannot propagate across institutions,  $\Omega_9$  becomes disconnected.

**(F9.25) Flow–capacity mismatch.** If scientific flows exceed institutional or cognitive capacities,  $T_9$  diverges.

#### P.2.95 Falsifiability via Scientific Cycles $C_9$

The  $K_9$  continuum depends on robust cycles:

$$C_9 = \{C_{\text{paradigm}}, C_{\text{theory}}, C_{\text{model}}, C_{\text{replication}}, C_{\text{validation}}, C_{\text{anomaly}}\}.$$

$K_9$  is falsified when:

**(F9.26) Breakdown of the paradigm cycle.** If anomalies do not trigger shifts or synthesis, the system loses adaptability.

**(F9.27) Breakdown of the theory cycle.** If theories cannot evolve or be replaced, science stagnates.

**(F9.28) Breakdown of the model cycle.** If models cannot be refined or tested, explanatory power collapses.

**(F9.29) Breakdown of replication cycle.** If knowledge cannot be validated, the paradigm collapses.

**(F9.30) Breakdown of anomaly cycle.** If anomalies cannot be identified or integrated, the theory loses self-correction.

#### P.2.96 Falsifiability via Structural Tension $T_9$

Structural tension at  $K_9$  includes:

- logical tension, - semantic tension, - methodological tension, - paradigmatic tension, - representational tension.

$K_9$  is falsified when:

**(F9.31)  $T_9$  exceeds collapse threshold.** If tension from anomalies, inconsistencies, or cross-paradigm conflicts exceeds  $\Theta_{\text{collapse}}$ , science destabilises.

**(F9.32) Multi-axis tension divergence.** If logical, methodological, and semantic tensions diverge simultaneously, the continuum collapses.

**(F9.33) Boundary instability.** If  $\partial\Omega_9$  (the boundary of conceivable scientific models) shifts uncontrollably, representational collapse follows.

**(F9.34) Complexity overload.** If theory-space complexity exceeds  $\Theta_{\text{complexity}}$ , the continuum becomes unstable.

**(F9.35) Reflexive overload.** If meta-theories grow faster than foundational theories, reflexive instability destroys coherence.

### **P.2.97 Falsifiability of the Transition $K_8 \rightarrow K_9$**

The transition  $\Psi_{8 \rightarrow 9}$  requires:

- stable civilisation-level symbolic and epistemic systems,
- mature institutions supporting science,
- technological systems enabling cumulative empirical work,
- coherent cultural and symbolic frameworks,
- sufficient cognitive capacity from  $K_6$ ,
- stable  $K_8$  cycles and flows.

The transition is falsified when:

**(F9.36) No emergence of formal models.** If symbolic systems cannot evolve into formal theories,  $K_9$  cannot form.

**(F9.37) Insufficient institutional support.** If science cannot be organised institutionally, paradigm dynamics remain at  $K_8$ .

**(F9.38) Cognitive–conceptual mismatch.** If conceptual demands of  $K_9$  exceed cognitive capacities, the transition cannot occur.

**(F9.39) Collapse of empirical infrastructure.** If empirical data cannot be collected or reproduced, theory space cannot stabilise.

**(F9.40) Epistemic incoherence at birth.** If early theories contradict each other beyond tolerance,  $\Omega_9$  fails to form.

### **P.2.98 Summary**

$K_9$  is falsified if scientific paradigms, theories, formal languages, methodological rules, or scientific flows cannot stabilise or maintain bounded structural tension. Falsification manifests as loss of coherence, representational collapse, or failure of scientific cycles. If these conditions fail, the  $K_9$  continuum collapses and science regresses to lower levels.

### P.2.99 Falsifiability of $K_{10}$

The continuum  $K_{10}$  corresponds to the level of meta-theoretical structures: systems that operate on theories themselves, construct and transform model categories, generate modal spaces, regulate functorial mappings, and maintain the coherence of high-order conceptual frameworks.

$K_{10}$  emerges only when:

- meta-theoretic axes  $A_{10}$  exist (modal axes, category axes, second-order difference operators, functorial axes);
- potentials  $P_{10}$  are stable (meta-coherence, functorial stability, modal potential, reflexive potential);
- thresholds  $\Theta_{10}$  are satisfied ( $\Theta_{\text{self}}$ ,  $\Theta_{\text{meta}}$ ,  $\Theta_{\text{mod}}$ ,  $\Theta_{\text{functor}}$ );
- flows  $J^{(10)}$  of meta-transformations remain within bounds;
- cycles  $C_{10}$  are stable (theory  $\rightarrow$  meta-theory  $\rightarrow$  meta-model  $\rightarrow$  theory);
- structural tension  $T_{10}$  remains bounded;
- the modal and functorial space  $\Omega(K_{10})$  is non-empty.

Any violation falsifies  $K_{10}$ .

### P.2.100 Falsifiability via Reflexive and Meta-Theoretical Coherence

$K_{10}$  requires consistency not only at the level of theories, but at the level of operations on theories. Failures here directly collapse the continuum.

$K_{10}$  is falsified when:

**(F10.1) Reflexive inconsistency.** If self-referential or self-transformative structures contradict each other, the reflexive potential exceeds  $\Theta_{\text{self}}$ , and  $\Omega(K_{10})$  collapses.

**(F10.2) Meta-theoretical incoherence.** If the meta-framework generates incompatible or contradictory transformations across theories, meta-coherence fails.

**(F10.3) Failure of second-order difference operators.** If operators of differences-of-differences ( $\mathfrak{D}$ ) yield inconsistent or ill-defined chains, the continuum becomes undefined.

**(F10.4) Incoherence between levels.** If meta-theories contradict the theories they regulate,  $T_{10}$  diverges.

**(F10.5) Breakdown of reflexive hierarchy.** If meta-levels collapse into a single level, dimensionality is lost.

### P.2.101 Falsifiability via Category-Theoretical and Functorial Structure

$K_{10}$  requires that theories, models, and transformations form coherent categories with stable, composable functors.

$K_{10}$  is falsified when:

**(F10.6) Failure of category formation.** If models cannot form objects and morphisms of a coherent category, the categorical structure of  $K_{10}$  does not exist.

**(F10.7) Functorial inconsistency.** If functors between categories of theories fail to preserve identity and composition,  $\Theta_{\text{functor}}$  is violated.

**(F10.8) Breakdown of naturality conditions.** If natural transformations cannot be defined or become unstable, functorial structure collapses.

**(F10.9) Collapse of higher-order functors.** If functors between functors (2-functors, adjunctions, monads) exceed modal/structural capacity, the continuum becomes inconsistent.

**(F10.10) Loss of coherence in categorical diagrams.** If commutative diagrams fail to commute, meta-theoretic structure collapses.

### **P.2.102 Falsifiability via Modal Spaces and Modal Dimensions**

A defining property of  $K_{10}$  is the explicit construction and operation on possible-world or modal spaces  $\Lambda_t$ .

$K_{10}$  is falsified when:

**(F10.11) Modal explosion.** If the modal space grows beyond  $\Theta_{\text{mod}}$ , representational capacity collapses.

**(F10.12) Absence of modal structure.** If no structured modal space can be constructed,  $K_{10}$  cannot exist.

**(F10.13) Modal inconsistency.** If possible worlds contradict structural rules of  $K_9$ , modal structure becomes incoherent.

**(F10.14) Cross-modal mapping failure.** If mappings between modal worlds cannot be defined,  $\Omega(K_{10})$  loses connectivity.

**(F10.15) Collapse of modal hierarchy.** If modal levels merge or collapse into fewer dimensions, the continuum loses structural degrees of freedom.

### **P.2.103 Falsifiability via Meta-Level Potentials $P_{10}$**

Meta-potentials regulate structural stability of theoretical ecosystems.

$K_{10}$  is falsified when:

**(F10.16) Meta-coherence potential collapse.** If  $P_{\text{meta}}$  falls below  $\Theta_{\text{meta}}$ , the system cannot maintain stable transformations.

**(F10.17) Reflexive potential divergence.** If reflexive operations accumulate without stabilising,  $T_{10}$  diverges.

**(F10.18) Functorial potential collapse.** If stable functorial mappings cannot form, the system loses compositional integrity.

**(F10.19) Modal potential collapse.** If modal transitions cannot be represented by operators in  $\mathcal{F}_t$ , modal structure becomes ill-defined.

**(F10.20) Epistemic potential mismatch.** If meta-theories cannot increase or preserve epistemic power, continuity of  $K_{10}$  is violated.



### P.2.104 Falsifiability via Meta-Flows $J^{(10)}$

Flows  $J^{(10)}$  describe how theories evolve, how meta-operators act, and how conceptual structures propagate across levels.

$K_{10}$  is falsified when:

**(F10.21) Breakdown of meta-transformative flows.** If transformations between theories cannot propagate, meta-dynamics becomes impossible.

**(F10.22) Divergence of flow intensity.** If meta-flows exceed capacity,  $T_{10}$  exceeds collapse thresholds.

**(F10.23) Semantic bottleneck.** If transformations cannot preserve interpretability, meta-theory collapses.

**(F10.24) Reflexive flow inversion.** If flows invert their meaning due to contradiction, coherence is destroyed.

**(F10.25) Flow–potential mismatch.** If flows cannot be supported by available potentials (e.g., modal, functorial, epistemic), the system destabilises.

### P.2.105 Falsifiability via Meta-Cycles $C_{10}$

$C_{10}$  includes cycles such as:

$$C_{10} = \{C_{\text{theory} \rightarrow \text{meta} \rightarrow \text{theory}}, C_{\text{model} \rightarrow \text{meta} \rightarrow \text{model}}, C_{\text{paradigm} \rightarrow \text{meta} \rightarrow \text{paradigm}}, C_{\text{functor} \rightarrow \text{meta} \rightarrow \text{functor}}\}.$$

$K_{10}$  is falsified when:

**(F10.26) Breakdown of the theory–meta-theory cycle.** If  $\text{theory} \rightarrow \text{meta-theory} \rightarrow \text{refined theory}$  does not converge, continuity breaks.

**(F10.27) Meta-model cycle collapse.** If model spaces cannot be lifted to meta-model spaces and back,  $\Omega(K_{10})$  loses structure.

**(F10.28) Functor cycle failure.** If functors cannot be refined or stabilised, categorical architecture collapses.

**(F10.29) Reflexive cycle instability.** If reflexive cycles diverge (due to too deep self-reference),  $\Theta_{\text{self}}$  is violated.

**(F10.30) Epistemic cycle degeneration.** If meta-cycles do not improve explanations or predictions, the system regresses to  $K_9$ .

### P.2.106 Falsifiability via Structural Tension $T_{10}$

Structural tension  $T_{10}$  combines:

- reflexive tension,
- modal tension,
- functorial tension,

- categorical tension,
- epistemic tension.

$K_{10}$  is falsified when:

**(F10.31) Divergence of reflexive tension.** If self-referential structures generate unbounded sequences, the continuum collapses.

**(F10.32) Modal–categorical inconsistency.** If modal mappings cannot be expressed functorially,  $T_{10}$  diverges.

**(F10.33) Breakdown of boundary stability.** If  $\partial\Omega(K_{10})$  shifts uncontrollably (due to conceptual drift), stability of meta-spaces is lost.

**(F10.34) Meta-complexity overload.** If conceptual complexity exceeds  $\Theta_{\text{meta complexity}}$ , the system becomes unmanageable.

**(F10.35) Cross-level tension feedback.** If tension from  $K_9$  propagates uncontrollably into  $K_{10}$ , meta-structure collapses.

#### **P.2.107 Falsifiability of the Transition $K_9 \rightarrow K_{10}$**

$K_{10}$  emerges only when:

- stable paradigms exist at  $K_9$ ,
- scientific cycles operate reliably,
- formal modelling tools are sufficiently expressive,
- meta-theoretical questions become necessary and stable,
- modal and categorical abstractions become essential,
- flows between theories require higher-order operators.

The transition is falsified when:

**(F10.36) Insufficient paradigm stability at  $K_9$ .** If  $K_9$  collapses, meta-theory cannot form above it.

**(F10.37) Lack of meta-coherence.** If early meta-theories contradict themselves,  $\Omega(K_{10})$  cannot be born.

**(F10.38) Failure to form modal spaces.** If new axes (modal axes) do not emerge,  $K_{10}$  cannot stabilise.

**(F10.39) Failure to form functorial structure.** If no functorial mappings are needed or possible, meta-theory cannot exceed  $K_9$ .

**(F10.40) Collapse of representational capacity.** If meta-level constructions cannot improve coherence, the continuum fails at birth.

### P.2.108 Summary

$K_{10}$  is falsified when meta-theoretical frameworks, modal spaces, functorial structures, or reflexive operators fail to stabilise, when structural tension exceeds thresholds, or when meta-cycles diverge. If these conditions fail, the continuum collapses into  $K_9$  or lower levels.

### P.2.109 Falsifiability of $K_{11}$

The continuum  $K_{11}$  corresponds to the level of *meta-meta structures*, or third-order conceptual organisation: a domain in which entire meta-theoretical frameworks can be compared, transformed, classified, and embedded into one another. While  $K_{10}$  requires stable meta-theories, modal spaces, and functorial mappings,  $K_{11}$  introduces a new dimension: **operators on families of meta-theories**, global constraints on admissible meta-spaces, and higher-order consistency conditions.

Because this is the highest structured level defined before  $K_{12}$ , its falsifiability is governed by strict conditions on:

- global stability of families of meta-theories,
- higher-order modal and functorial coherence,
- meta-meta-potentials and their thresholds,
- ultra-high-level conceptual flows  $J^{(11)}$ ,
- higher-order cycles  $C_{11}$ ,
- stability of boundaries  $\partial\Omega(K_{11})$ ,
- structural tension  $T_{11}$  across nested conceptual hierarchies.

Any violation implies immediate collapse of  $K_{11}$  into  $K_{10}$ .

### P.2.110 Falsifiability via Meta-Meta Coherence

$K_{11}$  carries the strongest possible coherence requirements within the Core: meta-theories themselves must be embedded into a coherent structure of transformations, adjunctions, modal lifts, and higher-order categories.

$K_{11}$  is falsified when:

**(F11.1) Collapse of meta-meta consistency.** If two or more meta-theories cannot be jointly embedded into a coherent structural space,  $\Omega(K_{11})$  becomes inconsistent.

**(F11.2) Divergence of second-order reflexivity.** If reflexive operators at  $K_{10}$  explode when lifted to  $K_{11}$ , the reflexive potential exceeds  $\Theta_{\text{self}}^{(11)}$ .

**(F11.3) Higher-order contradiction.** If meta-theories compatible at level  $K_{10}$  contradict one another once compared under  $K_{11}$ 's global constraints, the continuum collapses.

**(F11.4) Instability of meta-meta inference rules.** If inference rules governing transformations between meta-theories cannot remain stable, the level becomes undefined.

**(F11.5) Incompatibility of meta-frameworks.** If no unifying operator  $\Lambda_{11}$  can simultaneously classify or embed the relevant families of meta-theories,  $K_{11}$  fails.

### P.2.111 Falsifiability via Higher-Order Categorical Structures

$K_{11}$  demands coherence at the level of *categories of meta-theoretical categories*: a 3-level hierarchy requiring stability of transformations between categories, functors, and natural transformations of functors.

$K_{11}$  is falsified when:

**(F11.6) Breakdown of 2-categorical or 3-categorical structure.** If meta-theoretical categories fail to form a stable 2-category or 3-category, no  $K_{11}$  structure exists.

**(F11.7) Failure of higher-order functoriality.** If functors between functors (2-functors), or transformations between 2-functors cannot remain coherent,  $\Theta_{\text{functor}}^{(11)}$  is exceeded.

**(F11.8) Collapse of naturality at higher orders.** If commutative diagrams at the meta-meta level do not commute, the categorical boundary  $\partial\Omega(K_{11})$  collapses.

**(F11.9) Breakdown of adjoint or monadic structure.** If adjunctions or monads between meta-theories cannot be defined, the structural continuum becomes incomplete.

**(F11.10) Failure of global coherence theorems.** If coherence conditions for higher-order categories (e.g., Mac Lane-style global coherence) break down,  $K_{11}$  collapses to  $K_{10}$ .

### P.2.112 Falsifiability via Higher-Order Modal Spaces

$K_{11}$  introduces yet a higher modal operator: modalities of modal spaces, i.e., global conditions governing families of possible-world structures.

$K_{11}$  is falsified when:

**(F11.11) Modal hyper-explosion.** If the modal dimension exceeds  $\Theta_{\text{mod}}^{(11)}$ , representability collapses.

**(F11.12) Incoherence of cross-modal embeddings.** If embeddings between different modal systems become inconsistent, meta-modal coherence breaks.

**(F11.13) Violation of second-order modal laws.** If the rules governing modal transitions at  $K_{10}$  cannot be lifted without contradiction, the higher modal operator fails.

**(F11.14) Boundary drift in modal hyper-spaces.** If  $\partial\Omega(K_{11})$  changes uncontrollably due to modal spillover, stability collapses.

**(F11.15) Collapse of modal stratification.** If modal levels merge or lose definability, dimension of  $K_{11}$  becomes zero, implying collapse.

### P.2.113 Falsifiability via Potentials $P_{11}$

Potentials at  $K_{11}$  quantify the stability of hyper-theoretical integration.

The continuum is falsified when:

**(F11.16) Collapse of meta-meta coherence potential.** If  $P_{\text{meta}}^{(11)}$  falls below  $\Theta_{\text{meta}}^{(11)}$ , coherence across families of meta-theories disappears.

**(F11.17) Divergence of reflexive potential.** If global reflexive operations accumulate without stabilisation,  $T_{11}$  diverges.

**(F11.18) Failure of integrative potential.** If meta-theories cannot be jointly embedded or synchronised,  $P_{\text{integrative}}^{(11)}$  becomes insufficient.

**(F11.19) Violation of epistemic potential.** If hyper-level transformations do not improve epistemic structure,  $K_{11}$  cannot generate stable complexity.

**(F11.20) Functorial potential mismatch.** If higher-order functors require more regularity than the system can support, the continuum collapses.

#### **P.2.114 Falsifiability via Flows $J^{(11)}$**

Flows in  $K_{11}$  propagate transformations between entire meta-theoretical landscapes.

The continuum collapses when:

**(F11.21) Breakdown of hyper-flows.** If transformations between families of meta-theories cannot propagate stably,  $J^{(11)}$  is insufficient.

**(F11.22) Divergence of flow intensity.** If hyper-flows exceed allowable thresholds, structural tension  $T_{11}$  grows unbounded.

**(F11.23) Semantic inversion of hyper-transformations.** If hyper-level transformations invert their meaning, the system loses interpretability.

**(F11.24) Flow–potential inconsistency.** If flows require potentials that cannot be provided, the meta-meta structure collapses.

**(F11.25) Global obstruction in theoretical migrations.** If theories cannot be moved, lifted, or compared in the higher meta-space,  $\Omega(K_{11})$  becomes disconnected.

#### **P.2.115 Falsifiability via Cycles $C_{11}$**

Canonical cycles for  $K_{11}$  include integrations and comparisons between meta-theories, meta-models, and transformation logics.

$K_{11}$  is falsified when:

**(F11.26) Breakdown of meta-meta integration cycles.** If cycles that integrate several meta-theories fail to converge, hyper-coherence is impossible.

**(F11.27) Failure of global paradigm alignment.** If paradigms cannot be synchronised across meta-theories,  $C_{11}$  cannot stabilise.

**(F11.28) Collapse of hyper-reflexive cycles.** If cycles of self-application at order 3 diverge,  $\Theta_{\text{self}}^{(11)}$  is violated.

**(F11.29) Incompatibility of category-theoretic foundations.** If cycles of categorical refinement break, hyper-level morphisms become undefined.

**(F11.30) Breakdown of epistemic lift cycles.** If higher-order cycles cannot improve explanatory power, the continuum regresses to  $K_{10}$ .

### **P.2.116 Falsifiability via Structural Tension $T_{11}$**

$T_{11}$  quantifies tensions between entire theoretical ecosystems.

$K_{11}$  is falsified when:

**(F11.31) Reflexive overflow.** If higher-order reflexive chains become unbounded,  $T_{11}$  diverges.

**(F11.32) Modal–categorical incompatibility.** If modal spaces and categorical structures cannot coexist coherently,  $T_{11}$  exceeds stability thresholds.

**(F11.33) Boundary instability.** If the hyper-boundary of  $\Omega(K_{11})$  cannot remain stable, the continuum collapses.

**(F11.34) Global complexity overload.** If the conceptual complexity of the system surpasses  $\Theta_{\text{complex}}^{(11)}$ , stability is impossible.

**(F11.35) Cascading tension from lower levels.** If excess tension from  $K_{10}$  propagates uncontrollably,  $K_{11}$  collapses.

### **P.2.117 Falsifiability of the Transition $K_{10} \rightarrow K_{11}$**

A transition to  $K_{11}$  requires:

- stable families of meta-theories at  $K_{10}$ ,
- coherent modal and categorical architectures,
- existence of higher-order operators (operators on sets of meta-theories),
- sufficient epistemic and structural potential,
- positive dimensionality of the new axis  $A_{11}$ .

The transition is falsified when:

**(F11.36) Failure of meta-theoretical diversity.** If  $K_{10}$  does not provide multiple compatible meta-theories, no higher-order comparison can be formed.

**(F11.37) Failure to generate new dimensionality.** If no new axis  $A_{11}$  appears, the transition cannot occur.

**(F11.38) Meta-theoretical incoherence at scale.** If contradictions appear only once large collections of meta-theories are considered,  $K_{11}$  collapses at birth.

**(F11.39) Lack of integrative potential.** If new embeddings and comparisons cannot be supported,  $\Omega(K_{11})$  is empty.

**(F11.40) Collapse of structural tension gradients.** If  $T_{11}$  exceeds  $\Theta_{\text{dim}}^{(11)}$  during birth, the transition fails.

### P.2.118 Summary

$K_{11}$  is falsified when higher-order coherence, higher-order modal and functorial structures, meta-meta potentials, hyper-flows, or complex integration cycles fail to stabilise. Because  $K_{11}$  is a global meta-ecosystem, its collapse is typically immediate and total, returning the system to  $K_{10}$  or lower.

### P.2.119 Falsifiability of $K_{12}$

The continuum  $K_{12}$  represents the highest definable level in the current Ontology of Continua: the space of *universal structural laws governing all possible meta-theoretical systems*, including their birth, interaction, collapse, and representability. While  $K_{11}$  governs hyper-theories and meta-meta structures,  $K_{12}$  captures the conditions under which *any structurally admissible universe of continua can exist*. As such, falsifiability of  $K_{12}$  is expressed through absolute constraints: violations result not in collapse to  $K_{11}$ , but in the impossibility of coherent structural organisation altogether.

Because of its universal role, falsifiability of  $K_{12}$  is determined by:

- global structural laws and their consistency,
- admissibility of all lower-level continua  $K_0 \rightarrow K_{11}$ ,
- coherence of universal operators,
- existence and definability of  $\Omega(K_{12})$ ,
- universal thresholds  $\Theta_{12}$  limiting all structure.

If any criterion is violated,  $K_{12}$  cannot exist even momentarily.

### P.2.120 Falsifiability via Universal Structural Consistency

$K_{12}$  unifies structural constraints across all possible continua. It is falsified when global consistency cannot be maintained.

**(F12.1) Violation of universal coherence.** If the structural laws governing continua contradict each other, no  $\Omega(K_{12})$  can be defined.

**(F12.2) Inconsistency across levels.** If any lower level  $K_x$  (for  $0 \leq x \leq 11$ ) is incompatible with universal structural laws,  $K_{12}$  becomes undefined.

**(F12.3) Incompleteness of universal rules.** If the universal rule set fails to determine behaviour of continua in general,  $K_{12}$  cannot exist as a formal system.

**(F12.4) Collapse of universal representability.** If some continua cannot be represented in the universal structural language, representability fails.

**(F12.5) Violation of monotonicity of dimension.** If dimension can decrease without collapse, or behave inconsistently with the general axioms,  $K_{12}$  is falsified.

### P.2.121 Falsifiability via Universal Modal and Categorical Structure

$K_{12}$  requires coherence not only among meta-theories but across all modal, categorical, and higher-order structures.

**(F12.6) Modal universes become inconsistent.** If modal spaces across continua cannot be unified,  $\Theta_{\text{mod}}^{(12)}$  is exceeded.

**(F12.7) Categorical universality fails.** If higher categories required for embedding all continua do not exist or contradict, global structure collapses.

**(F12.8) Breakdown of global adjunctions.** If adjunctions or universal categorical constructions cannot be maintained across arbitrary continua,  $K_{12}$  becomes incoherent.

**(F12.9) Failure of universal naturality.** If diagrams governing continua-level transformations fail to commute universally, structural coherence is destroyed.

**(F12.10) Collapse of trans-universal functoriality.** If functorial mappings between families of continua cannot be defined, no operator algebra exists for  $K_{12}$ .

#### **P.2.122 Falsifiability via Universal Potentials $P_{12}$**

Universal potentials  $P_{12}$  describe the global conditions permitting the existence of continua.

$K_{12}$  is falsified when:

**(F12.11) Inadequate global coherence potential.** If  $P_{\text{coh}}^{(12)} < \Theta_{\text{coh}}^{(12)}$ , universal structure cannot stabilise.

**(F12.12) Divergence of universal tension.** If universal tension  $T_{12}$  becomes unbounded, no continuum can exist consistently.

**(F12.13) Insufficient generative potential.** If the system cannot generate continua of arbitrary dimension, universality is impossible.

**(F12.14) Collapse of integrability potential.** If continua cannot be jointly embedded into a universal structure,  $\Omega(K_{12})$  is empty.

**(F12.15) Violation of global energetic constraints.** If interaction of potentials across continua violates structural conservation laws,  $K_{12}$  collapses.

#### **P.2.123 Falsifiability via Universal Flows $J^{(12)}$**

Flows at  $K_{12}$  quantify how continua transform across universes.

The continuum is falsified when:

**(F12.16) Non-existence of universal flows.** If  $J^{(12)}$  cannot be defined for arbitrary continua, the system cannot describe transformations at all.

**(F12.17) Divergence of transformation intensities.** If hyper-transformations grow beyond universal thresholds, structural integrity fails.

**(F12.18) Incompatibility of universal flow laws.** If flow rules contradict across continua, no unifying operator  $F_{12}$  exists.

**(F12.19) Loss of interpretability of universal transitions.** If universal transitions become semantically inconsistent,  $K_{12}$  cannot sustain meaning.



**(F12.20) Breakdown of cross-universal communication.** If continua cannot exchange structure through  $J^{(12)}$ , universality collapses.

#### **P.2.124 Falsifiability via Universal Cycles $C_{12}$**

Cycles  $C_{12}$  describe closed transformations among entire universes of continua.

$K_{12}$  is falsified when:

**(F12.21) Collapse of universal integration cycles.** If cycles integrating classes of continua cannot converge, universal coherence is impossible.

**(F12.22) Breakdown of universal stabilisation cycles.** If universal processes cannot stabilise structural laws,  $C_{12}$  diverges.

**(F12.23) Failure of universal creation cycles.** If cycles responsible for structural birth of continua fail, universality breaks.

**(F12.24) Breakdown of meta-paradigm cycles.** If cycles mapping between paradigm families become unstable, no global organisation can persist.

**(F12.25) Collapse of conservation cycles.** If conserved structural quantities (tension, dimensionality, reachability) cannot be globally maintained,  $K_{12}$  falters.

#### **P.2.125 Falsifiability via Universal Boundaries $\partial\Omega(K_{12})$**

Universal boundaries define the admissible region for all continua. Falsification occurs when:

**(F12.26) Boundary indeterminacy.** If  $\partial\Omega(K_{12})$  cannot be defined, universality fails.

**(F12.27) Boundary inconsistency.** If the universal boundary contradicts any lower-level boundary, structure collapses.

**(F12.28) Boundary leakage.** If continua require states outside  $\Omega(K_{12})$ , universal admissibility is violated.

**(F12.29) Dimensional inconsistency.** If boundary conditions depend on dimension in a contradictory way,  $\Theta_{\text{dim}}^{(12)}$  is exceeded.

**(F12.30) Breakdown of universal reachability.** If reachability of continua within the universal domain is violated, no global structure can exist.

#### **P.2.126 Falsifiability of $K_{11} \rightarrow K_{12}$ Transition**

The transition requires:

- existence of multiple coherent higher-meta-theories at  $K_{11}$ ,
- stability of higher-order modal and categorical universes,
- emergence of a universal axis  $A_{12}$ ,
- stabilisation of global hyper-coherence potential,

- definability of universal structural operators.

The transition is falsified when:

**(F12.31) Insufficient meta-meta diversity.** If  $K_{11}$  does not provide enough complexity, universality cannot emerge.

**(F12.32) Failure to generate a universal axis  $A_{12}$ .** If no new dimension appears,  $K_{12}$  cannot exist.

**(F12.33) Collapse of global coherence scaling.** If coherence cannot scale from  $K_{11}$  to the universal level, the transition fails.

**(F12.34) Universal tension singularity.** If  $T_{12}$  becomes singular during emergence, the birth of  $K_{12}$  is impossible.

**(F12.35) Structural underdetermination.** If universal operators cannot be uniquely defined,  $\Omega(K_{12})$  cannot stabilise.

### P.2.127 Summary

$K_{12}$  is falsified by violations of universal structural laws, categorical and modal universality, global potentials, universal flows, cycles, or boundary constraints. As the highest definable continuum, its collapse implies the impossibility of any coherent universe of continua.

## P.3 Predictions

Predictions in the Ontology of Continua (OC) are structural consequences of the formal definition of a continuum

$$K = (\Omega, A, P, J, \Theta, C, k)$$

and of the compatibility requirement with the corresponding meta-space  $M$ :

$$\Omega(K) \subseteq \Omega(M).$$

OC does not generate phenomenological predictions directly. Instead, it yields *structural invariants*, *threshold relations*, and *dynamical constraints* that must hold in any admissible real physical, chemical, biological, cognitive, social or meta-theoretical instantiation.

The predictions are therefore:

1. **universal** (independent of domain);
2. **structural** (arising from the architecture of continua);
3. **threshold-based** (arising from  $\Theta$ -conditions);
4. **dimensional** (arising from emergence of new axes);
5. **cycle-dependent** (linked to  $C_j$  existence/viability);
6. **meta-constrained** (arising from compatibility with  $M$ ).

The framework of predictions is uniform across all levels  $K_0$ – $K_{12}$ .

### P.3.1 Prediction Types

A prediction  $P_i$  belongs to one of five classes.

**(1) Invariance Predictions.** Any structure that appears in  $\Omega(K)$  must satisfy invariants implied by axes  $A$ , thresholds  $\Theta$ , and admissible flows  $J$ . Example pattern:

If  $A_{\text{new}} \notin A(M)$ , then  $A_{\text{new}}$  cannot appear in  $K$ .

**(2) Threshold Predictions.** Whenever a potential crosses a threshold, a phase transition or loss of existence occurs:

$$P(t) > \Theta \Rightarrow \Omega(K) \rightarrow \Omega'(K).$$

**(3) Dimensional Predictions.** A new dimension may emerge only if the meta-space admits it and the system reaches the required structural tension:

$$T > \Theta_{\text{dim}} \Rightarrow \dim(K) \rightarrow \dim(K) + 1.$$

**(4) Cycle Predictions.** Life/operation/continuity requires at least one stabilising cycle:

$$C_j \text{ exists} \Rightarrow k(t) > 0.$$

Loss of all such cycles implies collapse.

**(5) Collapse Predictions.** A continuum dies when compatibility is lost:

$$\Omega(K) \not\subseteq \Omega(M) \Rightarrow k(K) = 0.$$

### P.3.2 Universal Prediction Constraints

All predictions across  $K_0$ – $K_{12}$  must satisfy:

**1. Meta-space admissibility:**

$$P_i \text{ valid} \implies \exists M : \Omega(K) \subseteq \Omega(M).$$

**2. Structural monotonicity:** Predictions must not violate the monotonicity of axes, thresholds or dimensionality:

$$\dim(K_{x+1}) \geq \dim(K_x).$$

**3. Threshold consistency:** All phenomena must emerge at threshold crossings definable in terms of  $P$ ,  $J$ ,  $\Theta$ .

**4. Continuity of  $\Omega$ :** Predictions must respect

$$\partial\Omega(K_x) \subseteq \partial\Omega(M_{x+1}).$$

**5. No ad hoc parameters:** Predictions arise only from operators already defined:  $\Psi$ ,  $\Phi$ ,  $\Lambda$ ,  $U$ ,  $X$ .

### P.3.3 Prediction Template for Each K-level

A prediction for level  $K_x$  has the canonical form:

$$P_i^{(x)} = (\text{Assumptions on } A_x, P_x, J_x, \Theta_x) \Rightarrow (\text{Invariant / Transition / Collapse}).$$

Examples of generic prediction templates implemented throughout the module:

• **Invariant Template**

$$A_x^{(i)} = \text{const} \Rightarrow T_x(t) \text{ monotonic.}$$

- **Threshold Template**

$$P_x^{(j)} > \Theta_x^{(j)} \Rightarrow \text{Phase transition in } \Omega(K_x).$$

- **Dimensional Template**

$$T_x(t) > \Theta_{\dim, x} \Rightarrow \dim(K_x) \rightarrow \dim(K_x) + 1.$$

- **Cycle Template**

$$C_x^{(k)} \text{ broken} \Rightarrow k_x(t) \rightarrow 0.$$

- **Collapse Template**

$$\Omega(K_x) \not\subseteq \Omega(M_x) \Rightarrow K_x \text{ ceases to exist.}$$

These templates guarantee uniformity across domains and allow each prediction module  $P(K_0)$ – $P(K_{12})$  to be derived transparently from the formal structure.

### P.3.4 Role of Predictions Within Core 1.3

Predictions serve three purposes in OC:

**(i) Structural falsifiability.** Every prediction is paired with a falsifiability criterion defined in the corresponding module:

$$P_i \text{ false} \Rightarrow \text{the relevant bounded OC claim must be revised.}$$

**(ii) Cross-domain coherence.** A prediction derived at  $K_x$  must not contradict predictions at lower or higher levels  $K_y$  whenever their domains overlap.

**(iii) Empirical anchoring.** Predictions relating to physical, chemical, biological, cognitive or social continua become empirically testable via the experiments defined in the corresponding  $K_x$  experimental modules.

### P.3.5 Interaction with Meta-Spaces

Predictions are constrained by meta-spaces  $M_x$ :

$$P_i^{(x)} \text{ valid} \Leftrightarrow P_i^{(x)} \in \Omega(M_x) \text{ and does not violate } \Theta(M_x).$$

This ensures:

- existence of  $K_x$ ,
- admissibility of transitions,
- compatibility with higher-level spaces,
- logical consistency across the hierarchy.

### P.3.6 Summary

The master prediction module provides:

- a universal formalism for deriving predictions at all K-levels,
- structural templates linking potentials, flows and thresholds,
- a unified theory of dimensional transitions,
- compatibility rules with meta-spaces  $M_x$ ,
- a foundation for empirical testing in accompanying modules.

This framework ensures that the predictive content of OC is mathematically grounded, operationalisable, falsifiable and consistent across the entire K/M hierarchy.

### P.3.7 Predictions for $K_0$

Predictions at level  $K_0$  follow from the minimal structure

$$K_0 = (S, \Delta, \mathcal{C})$$

equipped with the axioms of locality, regularity, admissible compositions and the minimal existence threshold  $\Theta_0 = \varepsilon > 0$ .

Because  $K_0$  contains no axes, no potentials, no flows and no cycles, all predictions are structural and concern the necessary behaviour of *any* proto-continuum that may give rise to higher levels  $K_1, K_2, \dots$

The predictions of  $K_0$  fall into five groups.

### P.3.8 P1: Predictions Concerning Existence

**(P1.1) Minimal Extent Prediction.** If a proto-continuum exists, its configuration set  $\Delta$  must satisfy

$$|\Delta| \geq 1,$$

otherwise no continuum exists, even in degenerate form.

**(P1.2) Non-zero Structural Threshold.** The minimal threshold

$$\Theta_0 = \varepsilon > 0$$

predicts that no continuum can exist with exact zero structural slack. Any structure with  $\Theta_0 = 0$  is forbidden:

$$\Theta_0 = 0 \Rightarrow K_0 \text{ impossible.}$$

**(P1.3) No Perfect Determinacy.** Because admissible configurations form  $\Delta$  rather than a single element,

$$|\Delta| = 1 \Rightarrow K_0 \text{ collapses,}$$

i.e. a proto-continuum must have at least some configurational latitude.

### P.3.9 P2: Predictions Concerning Structure and Composition

**(P2.1) Compositional Closure Prediction.** From axioms of  $\mathcal{C}$ :

$$\mathcal{C}(x, y) \text{ defined} \Rightarrow \mathcal{C}(y, x) \text{ not required.}$$

Thus OC predicts that  $K_0$  composition is not necessarily symmetric, supporting the later emergence of directional dynamics ( $K_1$ ).

**(P2.2) No Global Composition Prediction.**  $K_0$  predicts:

$$\neg \exists \mathcal{C}_{\text{global}} : S \times S \rightarrow S.$$

Global closure is impossible; only partial compositions are admissible.

This matches the observed emergence of local-to-global structure only at  $K_1$ .

**(P2.3) Partiality is Necessary, not Contingent.** A full composition table is forbidden:

$$\forall x, y \in S, \mathcal{C}(x, y) \text{ defined} \Rightarrow K_0 \text{ collapses.}$$

### P3.10 P3: Predictions Concerning Transitions and Dimensionality

**(P3.1) No Dimensional Self-Generation.** OC predicts:

$$K_0 \not\Rightarrow K_1$$

unless the meta-space  $M_0$  provides an admissible axis  $A_1 \in M_0$ . Thus spontaneous dimensionality increase is forbidden.

**(P3.2) Transition Requires an Axis Seed.** A transition

$$K_0 \longrightarrow K_1$$

is possible only if an operator  $\Psi_{0 \rightarrow 1}$  is defined on  $M_0$ ,

mapping  $(S, \Delta, \mathcal{C})$  to a field  $\phi(x)$  over a newly born axis. Therefore:

$$\neg \exists A_1 \in \Omega(M_0) \Rightarrow K_0 \not\rightarrow K_1.$$

**(P3.3) No Temporal Structure Prediction.** Since  $K_0$  contains no time axis,

$$\tau \notin A(K_0).$$

Thus  $K_0$  predicts that no temporal evolution is definable at this level. All dynamics are emergent properties of  $K_1$ .

### P3.11 P4: Predictions Concerning Tension and Collapse

**(P4.1) Zero Tension is Necessary.** Since  $T(K_0)$  is undefined (no potentials, no axes), the only admissible value is structural quasi-zero:

$$T_0 = 0.$$

Thus  $K_0$  predicts that no internal tension may accumulate.

**(P4.2) Collapse Trigger Prediction.** If compositional inconsistencies accumulate (violation of C1–C4'),

$$\exists x, y, z : \mathcal{C}(x, y), \mathcal{C}(y, z), \text{ but } \mathcal{C}(x, z) \text{ undefined}$$

and this violates minimal coherence constraints, then:

$$K_0 \rightarrow \emptyset.$$

**(P4.3) No Partial Collapse.** Predicted:

$$K_0 \text{ exists} \Rightarrow \text{fully coherent};$$

there is no notion of partial existence because  $k$  is undefined at this level.

### P.3.12 P5: Predictions Concerning Compatibility with Meta-Spaces

**(P5.1) Meta-space Incompleteness Prediction.** OC predicts:

$$\Omega(M_0) \subsetneq \Omega(M_1)$$

by the Theorem of Incomplete Meta-spaces. Thus  $M_0$  cannot be maximal.

**(P5.2) Forbidden Full Closure.**  $M_0$  cannot contain the full structure of  $K_1$ :

$$A_1 \notin M_0, \quad \Delta_1 \notin M_0.$$

**(P5.3) Boundary Prediction.** Since boundaries are undefined in  $K_0$ ,

$$\partial\Omega(K_0) = \emptyset,$$

predicting that boundaries only emerge at  $K_1$  with the birth of the first axis.

### P.3.13 Summary

The predictions for  $K_0$  state that:

- proto-continua must possess nonzero structural slack  $\Theta_0 > 0$ ;
- compositional structure must be partial and cannot be globally defined;
- no dimensionality, no time and no dynamics can emerge from  $K_0$  internally;
- transitions to  $K_1$  require admissibility from  $M_0$ ;
- collapse occurs whenever minimal coherence fails;
- boundaries and axes are impossible at this level.

These predictions provide the invariant conditions under which the emergence, coherence and eventual transition of proto-continua are permitted within the OC framework.

### P.3.14 Predictions for $K_1$

Predictions for  $K_1$  follow from the formal structure established in the Core: the birth of the first axis, the emergence of time  $\tau$ , the space of admissible fields

$$\Omega(K_1) = C^0(X, V) \subset H^1(X, V),$$

the presence of a boundary class  $\partial\Omega_{\text{cl}}$ , an energy functional  $E[\phi]$  and an action  $S[\phi]$ , with minimal threshold  $\Theta_1$  determining the stability of  $K_1$ .

The predictions divide into six categories.

### P.3.15 P1: Predictions Concerning Axes and Time

**(P1.1) Existence of a Single Axis.**  $K_1$  predicts that exactly one axis is definable:

$$\dim A(K_1) = 1.$$

Any structure exhibiting two independent axes contradicts  $K_1$  and belongs instead to  $K_2$ .

**(P1.2) Time Emerges Uniquely.** The first axis must behave as a temporal axis:

$$A_1 \equiv \tau.$$

Thus  $K_1$  predicts that the earliest continuum with dynamics must possess a monotonic ordering of configurations:

$$\tau : X \rightarrow \mathbb{R}, \quad \frac{d\tau}{ds} \geq 0.$$

**(P1.3) No Spatial Degrees of Freedom.**  $K_1$  predicts:

$$\neg \exists x \in \text{spatial manifold} \quad \text{at this level.}$$

All spatial structure must be emergent in  $K_2$ .

### P3.16 P2: Predictions Concerning Fields and Regularity

**(P2.1) Minimal Regularity Prediction.** Fields  $\phi(\tau)$  must satisfy

$$\phi \in C^0(\tau), \quad \partial_\tau \phi \in L^2,$$

i.e.  $K_1$  predicts Sobolev regularity  $H^1$  as the \*minimal coherent structure\* of any evolving continuum. Any dynamics requiring higher derivatives (e.g., curvature) cannot arise at this level.

**(P2.2) No Higher-Order Terms.** Energy and action must be first-order in  $\partial_\tau \phi$ :

$$E[\phi] = \int f(\phi(\tau), \partial_\tau \phi(\tau)) d\tau.$$

$K_1$  predicts the absence of second-order spatial or temporal operators.

**(P2.3) No Local Interactions.** All interactions must be global or integral, not local in space, because no spatial neighbourhood exists.

### P3.17 P3: Predictions Concerning Dynamics and Evolution

**(P3.1) One-Dimensional Causality.** Causality reduces to ordering:

$$\tau_1 < \tau_2 \quad \Rightarrow \quad \phi(\tau_1) \text{ influences } \phi(\tau_2).$$

No multi-directional causal cones exist at  $K_1$ .

**(P3.2) Universal Relaxation Prediction.** Because the only motion permitted is along  $\tau$ ,  $K_1$  predicts that all stable solutions exhibit relaxation to fixed points:

$$\partial_\tau \phi = 0 \text{ at equilibrium.}$$

**(P3.3) No Oscillatory Dynamics.**  $K_1$  forbids periodic solutions:

$$\phi(\tau + T) = \phi(\tau) \quad \Rightarrow \quad \text{collapse of } K_1.$$

Oscillations require a second axis  $\rightarrow K_2$ .



**(P3.4) First-Order Evolution Equation.** Predicted form:

$$\partial_\tau \phi = -\frac{\delta E}{\delta \phi},$$

i.e., gradient-like evolution is the only admissible dynamic.

### **P3.18 P4: Predictions Concerning Thresholds and Tension**

**(P4.1) Single Stability Threshold.**  $K_1$  predicts a single effective threshold:

$$T_1(\tau) < \Theta_1 \Rightarrow \text{stable.}$$

**(P4.2) Critical Slowing Prediction.** As  $T_1 \rightarrow \Theta_1$ , the relaxation time diverges:

$$\tau_{\text{relax}} \rightarrow \infty.$$

This provides a direct empirical signature of  $K_1$  dynamics.

**(P4.3) No Stress Propagation.** Because no spatial axes exist:

stress cannot propagate or diffuse.

Any observed propagation indicates  $K_2$ .

### **P3.19 P5: Predictions Concerning Collapse and Boundaries**

**(P5.1) Collapse Criterion.**  $K_1$  collapses when

$$E[\phi] \notin \mathbb{R}, \quad \partial_\tau \phi \notin L^2, \quad T_1 > \Theta_1,$$

or when  $\phi$  exits  $\Omega(K_1)$ .

**(P5.2) Boundary Prediction.**  $K_1$  predicts the existence of a boundary class:

$$\partial\Omega_{\text{cl}} = \{\phi : \partial_\tau \phi \in L^2 \text{ but } \phi \notin C^0\}.$$

Crossing this boundary signals the onset of  $K_0$ -like behaviour (degenerate continuum death).

**(P5.3) No Partial Collapse.** Collapse is binary because  $k_1$  has no spatial decomposition:

$$k_1 = 0 \quad \text{or} \quad k_1 > 0.$$

No domainwise collapse exists at this level.

### **P3.20 P6: Predictions Concerning Transitions and M-Spaces**

**(P6.1) Necessary Condition for  $K_1 \rightarrow K_2$ .** A transition requires an admissible spatial axis in  $M_1$ :

$$A_2 \in \Omega(M_1) \Rightarrow K_1 \rightarrow K_2.$$

**(P6.2) Forbidden Self-Generation of Space.**  $K_1$  predicts:

$$\neg \exists \text{ internal process generating a second axis.}$$

Spatial structure cannot emerge from temporal dynamics alone.

**(P6.3) Dimensional Tension Prediction.** If the system accumulates structural tension  $T_1$  that cannot be resolved within the single axis, then either:

$$K_1 \rightarrow K_2 \quad \text{or} \quad K_1 \rightarrow \emptyset.$$

### P.3.21 Summary

$K_1$  predicts that:

- exactly one axis exists and it must behave as time;
- fields exhibit  $C^0$  regularity with  $L^2$  derivatives;
- dynamics are first-order, causal, and purely relaxational;
- oscillations, waves and spatial propagation are impossible;
- stability is governed by a single threshold  $\Theta_1$ ;
- collapse is global and binary;
- transition to  $K_2$  requires a spatial axis provided by  $M_1$ .

These predictions represent the minimal empirically testable signatures of one-dimensional continua and define the necessary and sufficient conditions for the birth of classical dynamical structure at  $K_2$ .

### P.3.22 Predictions for $K_2$

Level  $K_2$  corresponds to the first continuum possessing a spatial axis, nontrivial topology, and a connected state space that supports extended dynamics. Formally,  $K_2$  emerges when the axis structure expands from the single temporal axis of  $K_1$  to a two-dimensional configuration:

$$A(K_2) = \{\tau, x\},$$

where  $x$  denotes the first spatial coordinate. The key structural ingredient is the onset of percolation and the formation of connected spatial domains.

Predictions for  $K_2$  follow from the structure of  $\Omega(K_2)$ , the percolation threshold  $p_c$ , the tension threshold  $\Theta_2$ , and the dynamics induced by the universal evolution operator  $U$ .

The predictions fall into six major groups.

### P.3.23 P1: Predictions Concerning Axes and Geometry

**(P2.1) Existence of a Single Spatial Axis.**  $K_2$  predicts:

$$\dim A(K_2) = 2, \quad A(K_2) = \{\tau, x\}.$$

Any continuum with more than one spatial axis belongs to  $K_3$  or higher.

**(P2.2) Prediction of Coordinate Continuity.** Fields  $\phi(x, \tau)$  must be continuous in both coordinates:

$$\phi \in C^0(X \times \tau).$$

**(P2.3) Prediction of Finite Spatial Neighbourhoods.** The geometry of  $K_2$  must admit neighbourhoods  $U_\epsilon(x)$ , enabling:

local interactions, diffusion, propagation.

Such locality is impossible at  $K_1$ .

### P.3.24 P2: Predictions Concerning Percolation and Connectivity

**(P2.4) Existence of a Critical Percolation Threshold.**  $K_2$  predicts a structural phase transition at

$$p = p_c,$$

where  $p$  denotes the effective occupation/connectivity probability. For  $p < p_c$ ,  $\Omega(K_2)$  fragments; for  $p \geq p_c$ , a giant connected component exists:

$$C_{\max} \sim O(|X|).$$

**(P2.5) Uniqueness of the Giant Cluster.** Above  $p_c$ ,  $K_2$  predicts:

there exists exactly one macroscopic connected component.

**(P2.6) Scaling Laws Near  $p_c$ .** The geometry must exhibit critical exponents (dimension-dependent):

$$\xi \sim (p - p_c)^{-\nu}, \quad C_{\max} \sim (p - p_c)^\beta.$$

Any violation implies the continuum is not  $K_2$ .

### P.3.25 P3: Predictions Concerning Dynamics

**(P3.1) Existence of Wave-Like Propagation.** Because a spatial axis exists,  $K_2$  predicts:

$$\partial_\tau^2 \phi, \quad \partial_x^2 \phi$$

may appear in  $U$ . Thus wave propagation and diffusion are admissible and must occur.

**(P3.2) Diffusion as a Universal Process.** The universal form of  $K_2$  dynamics predicts:

$$\partial_\tau \phi = D \partial_x^2 \phi + \text{lower-order terms.}$$

**(P3.3) Finite-Range Causality.** Because spatial neighbourhoods exist, disturbances satisfy a causal bound:

$$v_{\text{prop}} < \infty,$$

where  $v_{\text{prop}}$  is the propagation velocity determined by  $U$ .

**(P3.4) Critical Slowing Down Near  $p_c$ .** As  $p \rightarrow p_c^+$ ,

$$\tau_{\text{relax}} \sim \xi^z \rightarrow \infty,$$

with  $\xi$  the correlation length.

### P.3.26 P4: Predictions Concerning Thresholds and Tension

**(P4.1) Spatial Tension Threshold.**  $K_2$  predicts a new class of thresholds:

$$\Theta_2 = \{\Theta_{\text{conn}}, \Theta_{\text{dim}}, \Theta_{\text{stab}}\},$$

with  $\Theta_{\text{conn}}$  the percolation threshold.

**(P4.2) Stress Propagation.** Unlike  $K_1$ , stress can propagate:

$$\partial_\tau T = D_T \partial_x^2 T + \dots$$

**(P4.3) Local Collapse Criterion.** Because space is extended, collapse may be local:

$$k_2(x, \tau) = 0 \quad \text{for some } x.$$

Partial failure is a prediction unique to  $K_2$  and above.

### P.3.27 P5: Predictions Concerning Boundary, Collapse and Phase Transitions

**(P5.1) Boundary Geometry.**  $K_2$  predicts:

$$\partial\Omega(K_2) = \text{sets of fields where connectivity breaks.}$$

**(P5.2) Collapse via Connectivity Loss.** Collapse happens when:

$$p < p_c,$$

even if local dynamics remain smooth.

**(P5.3) Topological Death.**  $K_2$  predicts a form of death absent at  $K_1$ :

$$\Omega(K_2) \rightarrow \emptyset \text{ if the space fragments beyond recovery.}$$

### P.3.28 P6: Predictions Concerning Transitions to Higher Levels

**(P6.1) Necessary Condition for  $K_2 \rightarrow K_3$ .** Transition requires:

$$\exists y \in A(K_3), \quad \text{independent of } x.$$

I.e., a *second spatial axis* must be present in  $M_2$ .

**(P6.2) Emergence of Two-Dimensional Geometry.**  $K_2$  predicts that in the presence of sufficient tension

$$T_2 > \Theta_{\text{dim}},$$

the system must either transition to  $K_3$  or collapse.

**(P6.3) Critical Tension as Predictor of Dimensional Growth.** Dimensionality growth is predicted when:

$$\frac{\partial T}{\partial x} \text{ cannot be balanced within 1D space.}$$

### P.3.29 Summary

$K_2$  makes the following core predictions:

- existence of one spatial axis and emergent geometry;
- existence of a percolation threshold  $p_c$  and a giant cluster;
- wave propagation, diffusion and finite-range causality;
- spatial stress propagation and local collapse;
- divergence of relaxation times near  $p_c$ ;
- transitions to  $K_3$  driven by dimensional tension.

These predictions constitute the empirically testable signature of the first extended spatial continuum.

### P.3.30 Predictions for $K_3$

Level  $K_3$  is the first continuum with a genuinely two-dimensional spatial structure. Whereas  $K_2$  supports a single spatial axis  $x$ ,  $K_3$  supports a pair of independent axes:

$$A(K_3) = \{\tau, x, y\},$$

giving rise to two-dimensional geometry, extended connectivity, and new topological phenomena that cannot exist at lower levels.

Predictions for  $K_3$  rest on the expanded state space  $\Omega(K_3)$  and on the geometry of two-dimensional neighbourhoods. They also use the threshold structure  $\Theta_3$  and the evolution operator  $U$  acting on fields  $\phi(x, y, \tau)$ .

We group the predictions into six families.

### P.3.31 P1: Predictions Concerning Geometry and Dimensionality

**(P3.1) Two-Dimensional Spatial Geometry.**  $K_3$  predicts:

$$\dim A(K_3) = 3, \quad A(K_3) = \{\tau, x, y\}.$$

**(P3.2) Existence of 2D Neighbourhoods.** The continuum must admit open sets of the form:

$$U_\epsilon(x, y) \subset \mathbb{R}^2.$$

**(P3.3) Dimensional Independence.**  $x$  and  $y$  are independent:

$$\partial_x \not\propto \partial_y.$$

**(P3.4) Curvature Emergence.**  $K_3$  predicts the possibility of nonzero Gaussian curvature:

$$K(x, y) \neq 0,$$

whereas  $K_2$  cannot express curvature at all.

### P.3.32 P2: Predictions Concerning Connectivity and Percolation

**(P3.5) 2D Percolation Thresholds.**  $K_3$  predicts distinct percolation behaviour from  $K_2$ :

$$p_c^{(2D)} < p_c^{(1D)},$$

and the existence of 2D critical exponents:

$$\xi \sim |p - p_c|^{-\nu_{2D}}, \quad C_{\max} \sim |p - p_c|^{\beta_{2D}}.$$

**(P3.6) Loop-Dominated Connectivity.** Connectivity is mediated not only by paths but also by closed cycles. Thus  $K_3$  predicts:

$$\exists \text{ nontrivial loops in } \Omega(K_3).$$

**(P3.7) Boundary Percolation.** The continuum supports boundary-driven transitions:

$$\partial\Omega(K_3) \text{ may undergo percolation independently of bulk.}$$

### P.3.33 P3: Predictions Concerning Dynamics

**(P3.8) 2D Wave and Diffusion Dynamics.** The universal evolution operator must contain:

$$\partial_x^2 + \partial_y^2,$$

predicting:

- 2D diffusion,
- 2D wave propagation,
- anisotropic or isotropic dynamics depending on  $U$ .

**(P3.9) Vortex Formation.** Because  $K_3$  supports circulation and closed loops,  $K_3$  predicts:

$$\oint_{\gamma} \nabla \phi \cdot dl \neq 0$$

for at least some fields. Vortex-like phenomena are impossible at  $K_2$ .

**(P3.10) Long-Range 2D Correlations.** Near  $p_c^{(2D)}$ :

$$\tau_{\text{relax}} \sim \xi^{z_{2D}} \rightarrow \infty.$$

**(P3.11) Edge Modes.**  $K_3$  predicts excitations confined to boundaries:

$$\phi(x, y, \tau)|_{\partial\Omega} \text{ supports distinct modes.}$$

### P.3.34 P4: Predictions Concerning Thresholds and Tension

**(P3.12) Dimensional Threshold for  $K_2 \rightarrow K_3$ .** The transition requires:

$$T_2 > \Theta_{\text{dim}}^{(2 \rightarrow 3)},$$

where  $\Theta_{\text{dim}}^{(2 \rightarrow 3)}$  is the minimal tension that cannot be resolved in 1D geometry.

**(P3.13) Shear Stress.**  $K_3$  predicts new stress components:

$$T_{xy} \neq 0,$$

absent at  $K_2$ , which only has scalar tension.

**(P3.14) Topological Stability Threshold.** Two-dimensional geometry admits stability thresholds associated with:

vorticity, winding number, defect creation/annihilation.

**(P3.15) Multi-Point Collapse.**  $K_3$  predicts the possibility of nontrivial collapse patterns:

$$k_3(x_i, y_i) = 0 \quad \text{for multiple points } i,$$

leading to complex fragmentation.

### P.3.35 P5: Predictions Concerning Boundary, Collapse and Topological Effects

**(P3.16) Boundary Curvature Transitions.**  $\partial\Omega(K_3)$  may change topology under tension, predicting:

boundary folds, cusps, detachment.

**(P3.17) Existence of Topological Defects.**  $K_3$  predicts:

$$\exists \text{ point defects } D(x_0, y_0)$$

analogous to vortices or disclinations.

**(P3.18) Collapse via Curvature Blow-Up.** A direct geometric prediction:

$$K(x, y) \rightarrow \infty \Rightarrow \Omega(K_3) \rightarrow \emptyset.$$

**(P3.19) Topological Death.**  $K_3$  can die via:

destruction of the fundamental group  $\pi_1(\Omega)$ .

### P3.36 P6: Predictions Concerning Transitions to Higher Levels

**(P3.20) Necessary Condition for  $K_3 \rightarrow K_4$ .** Transition requires:

appearance of internal degrees of freedom (chemical axes).

**(P3.21) Prediction of Chemical Reactivity.**  $K_3$  predicts the emergence of:

$$\phi(x, y, \tau) \Rightarrow \{\phi_i\} \text{ with interaction rules.}$$

**(P3.22) Dimensional Saturation.**  $K_3$  predicts that spatial dimensionality must *saturate* at 2 unless:

$$\exists A_{\text{chem}} \in M_3.$$

**(P3.23) Threshold for Catalyst-Like Behaviour.**  $K_3$  predicts that catalytic behaviour becomes possible only when:

$$J_{\text{react}} \neq 0,$$

an indicator of transition toward  $K_4$ .

### P3.37 Summary

$K_3$  predicts the following structural and empirical features:

- Two-dimensional geometry with curvature and independent axes;
- 2D percolation with distinct exponents and loop-driven connectivity;
- Vortex formation, topological defects and edge modes;
- Novel tension components (shear) and dimensional thresholds;
- Complex collapse patterns involving curvature singularities;
- Conditions for chemical structure and the transition to  $K_4$ .

These predictions uniquely characterize the first fully two-dimensional continuum and distinguish  $K_3$  from both  $K_2$  and  $K_4$ .

### P.3.38 Predictions for $K_4$

Level  $K_4$  is the first biological continuum. It is defined by the existence of a semi-permeable membrane  $\partial\Omega(K_4)$ , chemical gradients across the membrane, metabolic flow cycles, and internal reaction networks capable of maintaining non-equilibrium structure.

Predictions for  $K_4$  follow from:

- membrane thresholds  $\Theta_{\text{mem}}$  (stretch, curvature, osmosis, charge);
- gradient axes  $A_{\text{grad}}, A_{\text{ion}}, A_{\text{pH}}, A_{\text{redox}}$ ;
- metabolic potentials  $P_{\text{energy}}, P_{\text{redox}}, P_{\text{grad}}$ ;
- flows  $J_{\text{pump}}, J_{\text{redox}}, J_{\text{metabolic}}$ ;
- cycles  $C_{\text{energy}}, C_{\text{buffering}}, C_{\text{redox}}$ ;
- vesicle flickering regimes and instability thresholds;
- collapse modes discovered in Chemistry Run (pH-collapse, osmotic burst, waste-pressure loop).

The predictions are grouped into seven families.

### P.3.39 P1: Predictions Concerning Membrane Structure and Stability

**(P4.1) Existence of a Stable Semi-Permeable Boundary.**  $K_4$  predicts that the membrane must satisfy:

$$\Theta_{\text{stretch}} > T_{\text{mem}}(t), \quad \Theta_{\text{curv}} > |H(x, t)|,$$

where  $H$  is mean curvature. Otherwise  $\partial\Omega(K_4)$  ruptures and  $K_4$  dies.

**(P4.2) Flickering Regime.**  $K_4$  predicts a regime of membrane flickering:

$$C_{\text{flickering}} \neq 0$$

near critical osmotic and curvature thresholds. This regime is a universal near-critical phenomenon absent in  $K_3$ .

**(P4.3) Osmotic Equilibrium Constraint.** The osmotic potential must satisfy:

$$|\Delta\Pi| < \Theta_{\text{osm}},$$

predicting a maximum sustainable solute difference before collapse.

**(P4.4) Charge Threshold.** A membrane charge threshold:

$$\Theta_{\text{charge}}$$

limits electric potential before destabilisation.

### P.3.40 P2: Predictions Concerning Internal Chemical Organization

**(P4.5) Reaction Network Closure.**  $K_4$  predicts the existence of at least one internally closed metabolic loop:

$$C_{\text{core}} = A \rightarrow B \rightarrow C \rightarrow A.$$

**(P4.6) Threshold for Waste Accumulation.** There exists a metabolic waste threshold:

$$\Theta_{\text{waste}},$$

where exceeding it generates internal pressure and induces collapse.



**(P4.7) pH Buffering Cycle.**  $K_4$  predicts:

$$C_{\text{buffering}} \neq 0,$$

a cycle regulating internal pH and protecting against the pH-collapse mode identified in Chemistry Run.

**(P4.8) Redox Potential Gradient.** A redox-driven potential difference must exist:

$$P_{\text{redox}}(\text{in}) \neq P_{\text{redox}}(\text{out}),$$

predicting metabolic energy production even in early protocells.

### **P.3.41 P3: Predictions Concerning Flows and Transport**

**(P4.9) Active Transport.**  $K_4$  predicts that passive transport is insufficient for stability:

$$J_{\text{active}} > 0$$

must compensate losses and maintain gradients.

**(P4.10) Gradient-Driven Coupling.** Flows satisfy:

$$J_{\text{ion}} \propto \nabla P_{\text{grad}}, \quad J_{\text{metabolic}} \propto P_{\text{energy}}.$$

**(P4.11) Osmotic Inflow Instability.** There exists a regime where:

$$\frac{dV}{dt} > 0, \quad \frac{d^2V}{dt^2} > 0,$$

predicting runaway swelling until membrane rupture.

### **P.3.42 P4: Predictions Concerning Cycles and Non-Equilibrium Dynamics**

**(P4.12) Existence of Energy Cycle.**  $K_4$  predicts:

$$C_{\text{energy}} \neq \emptyset,$$

minimal for maintaining  $k_4(t) > 0$ .

**(P4.13) Turnover Cycle.** A turnover cycle:

$$C_{\text{turnover}} = \{\text{production} \rightarrow \text{consumption} \rightarrow \text{waste export}\}$$

must exist and determines lifetime of  $K_4$ .

**(P4.14) Redox Cycle Oscillations.** Small oscillations in redox potential are predicted:

$$P_{\text{redox}}(t) = P_0 + \delta P(t).$$

**(P4.15) Metabolic Bottleneck Threshold.** If:

$$T_{\text{metabolic}} > \Theta_{\text{path}},$$

then collapse via pathway exhaustion occurs.

### **P.3.43 P5: Predictions Concerning Collapse, Death and Threshold Behaviour**

**(P4.16) pH-Collapse (Golden Test 1).**  $K_4$  predicts catastrophic loss of continuity if:

$$P_{H^+}(t) > \Theta_{pH},$$

causing membrane destabilisation and enzymatic inactivity.

**(P4.17) Osmotic Burst (Golden Test 2).** If:

$$\Delta\Pi(t) > \Theta_{osm},$$

then the membrane ruptures and  $\Omega(K_4)$  disappears.

**(P4.18) Waste–Pressure Loop (Golden Test 3).** A positive feedback loop is predicted:

$$\text{waste} \uparrow \Rightarrow P_{int} \uparrow \Rightarrow k_4(t) \downarrow.$$

**(P4.19) Critical Flicker Collapse.** Near:

$$T_{mem} \approx \Theta_{curv},$$

flickering frequency diverges, predicting structural failure.

**(P4.20) Charge-Induced Death.** Excess membrane potential:

$$|\Delta V| > \Theta_{charge}$$

destabilises  $\partial\Omega(K_4)$ .

### **P.3.44 P6: Predictions Concerning Transition to $K_5$**

**(P4.21) Threshold for Electrical Excitability.**  $K_4$  predicts:

$$\exists \Theta_{exc} \text{ such that } |\Delta V| > \Theta_{exc} \Rightarrow \text{proto-action potential.}$$

**(P4.22) Birth of a New Axis  $A_{exc}$ .** When membrane depolarisation becomes cyclic:

$$C_{spike} \neq 0,$$

a new dynamical axis of excitability emerges — signature of  $K_5$ .

**(P4.23) Transition via Ion Channel Specialisation.** The transition  $K_4 \rightarrow K_5$  requires:

$$g_{channel} > \Theta_{channel-open},$$

predicting evolution of early ion channels.

**(P4.24) Redox-to-Electrical Coupling.** The early form of electrophysiological behaviour is predicted:

$$J_{redox} \rightarrow J_{ion} \rightarrow \Delta V.$$

**(P4.25) Spatially Propagating Excitation (proto-AP).** If:

$$G_{lat} > G_{crit},$$

a travelling excitation front emerges — defining property of  $K_5$ .

### P.3.45 P7: Predictions Concerning Evolution and Higher-Level Structure

**(P4.26) Prediction of Autocatalytic Complexity Growth.**  $K_4$  predicts that metabolic complexity grows if:

$$\frac{dC_{\text{core}}}{dt} > 0,$$

under stable membrane conditions.

**(P4.27) Prediction of Genetic Precursors.** Template-like molecular structures must appear if:

$$J_{\text{ligation}} - J_{\text{fragmentation}} > 0,$$

predicting the onset of informational continuity.

**(P4.28) Prediction of Spatial Heterogeneity.**  $K_4$  predicts emergence of:

$$\nabla P_{\text{grad}}(x, y) \neq 0,$$

leading to compartmentalisation, precursor to  $K_5$  morphology.

**(P4.29) Evolutionary Boundary Stability.** Systems evolve toward:

$$\Theta_{\text{mem}} \uparrow, \Theta_{\text{osm}} \uparrow,$$

predicting progressive robustness of biological membranes.

**(P4.30) Prediction of Multi-Vesicle Aggregation.** At low tension:

$$T_{\text{mem}} < \Theta_{\text{adhesion}},$$

vesicles fuse or aggregate, precursor to multicellular scaffolds.

### P.3.46 Summary

Level  $K_4$  predicts:

- stable semi-permeable boundaries with curvature, stretch and osmotic thresholds;
- membrane flickering as a universal near-critical regime;
- internal metabolic loops, redox cycles and buffering systems;
- active transport and gradient maintenance;
- three universal collapse modes: pH-collapse, osmotic burst, waste–pressure loop;
- conditions for emergence of excitability and the transition to  $K_5$ ;
- early evolutionary dynamics: complexity growth, proto-genetic systems, vesicle aggregation.

These predictions uniquely define  $K_4$  as the first biological continuum and distinguish it sharply from both chemical systems ( $K_3$ ) and neural-excitable systems ( $K_5$ ).

### P.3.47 Predictions for $K_5$

Level  $K_5$  is the continuum of *elementary excitability*: a membrane-bound system capable of generating, propagating, and regulating electrical excitation fronts driven by ion fluxes and membrane potential dynamics.  $K_5$  emerges from  $K_4$  when a new axis  $A_{\text{exc}}$  appears and when the system becomes capable of sustaining:

$$C_{\text{spike}} \neq 0, \quad \Delta V(t) \text{ self-amplifying for a finite interval.}$$

Predictions for  $K_5$  follow from:

- excitability thresholds  $\Theta_{\text{exc}}, \Theta_{\text{channel}}$ ;
- membrane ionic conductances  $g_{\text{ion}}$  and their nonlinearities;
- lateral coupling and conductivity  $G_{\text{lat}}$ ;
- ignition condition and critical membrane dynamics (L2.3);
- flows  $J_{\text{ion}}, J_{\text{pump}}, J_{\text{leak}}$ ;
- cycles  $C_{\text{spike}}, C_{\text{refractory}}, C_{\text{reset}}$ ;
- metabolic potentials  $P_{\text{grad}}, P_{\text{energy}}$ ;
- redox–electrical coupling from late  $K_4$ .

Predictions are grouped into seven categories.

### P.3.48 P1: Predictions Concerning the Birth of Excitability

**(P5.1) Threshold of Electrical Excitability  $\Theta_{\text{exc}}$ .**  $K_5$  predicts the existence of a membrane potential threshold:

$$|\Delta V| > \Theta_{\text{exc}} \Rightarrow \text{self-amplifying excitation event (proto-spike).}$$

**(P5.2) Emergence of the Excitability Axis.** A new dynamical axis  $A_{\text{exc}}$  is predicted when  $\Delta V$  acquires its own internal dynamics:

$$\frac{d\Delta V}{dt} = F(\Delta V, J_{\text{ion}}, J_{\text{pump}}),$$

with positive feedback for short time intervals.

**(P5.3) Necessity of Voltage-Gated Behaviour.**  $K_5$  predicts that purely passive membranes cannot sustain excitability; thus:

$$g_{\text{channel}}(\Delta V) \text{ must be nonlinear.}$$

**(P5.4) Existence of Ion Channel Threshold  $\Theta_{\text{channel}}$ .** Excitation requires:

$$g_{\text{channel}} > \Theta_{\text{channel-open}},$$

identifying a minimal conductance for spike initiation.

### P.3.49 P2: Predictions Concerning Spatial Propagation

**(P5.5) Ignition Condition for Proto-AP Fronts (L2.3).** A spatially propagating excitation front exists iff:

$$G_{\text{lat}} > G_{\text{crit}}, \quad C_{\text{mem}} > C_{\text{mem}}^{\text{crit}},$$

where  $G_{\text{lat}}$  is lateral conductivity, and  $C_{\text{mem}}$  is membrane capacitance.

**(P5.6) Minimal Spatial Extent for Propagation.** There exists a critical cluster size:

$$C_{\text{critical}} \sim O(10) \text{ units,}$$

below which propagation cannot be sustained.

**(P5.7) Conduction Velocity Scaling.**  $K_5$  predicts:

$$v_{\text{cond}} \propto \sqrt{G_{\text{lat}}}.$$

### **P.3.50 P3: Predictions Concerning Spike Dynamics**

**(P5.8) Existence of Spike Cycle  $C_{\text{spike}}$ .** The spike cycle consists of:

$$\text{activation} \rightarrow \text{peak} \rightarrow \text{inactivation} \rightarrow \text{reset},$$

and must be closed for stability of  $K_5$ .

**(P5.9) Refractory Cycle  $C_{\text{refractory}}$ .**  $K_5$  predicts a refractory period:

$$T_{\text{ref}} > 0,$$

arising from the interplay of channel inactivation and pump recovery.

**(P5.10) Reset Cycle  $C_{\text{reset}}$ .** A return to baseline potential requires:

$$J_{\text{pump}} - J_{\text{leak}} > 0,$$

predicting pump-dominant recovery.

**(P5.11) Spike Amplitude Threshold.** Amplitude satisfies:

$$A_{\text{spike}} > \Theta_{\text{amp}},$$

a necessary condition for propagation.

### **P.3.51 P4: Predictions Concerning Ion Channels and Conductance**

**(P5.12) Evolutionary Prediction: Specialisation of Ion Channels.**  $K_5$  predicts differentiation of:

$$g_{\text{Na}}, g_{\text{K}}, g_{\text{Ca}},$$

arising from selection pressures for excitability robustness.

**(P5.13) Conductance Time-Scale Separation.** For sustained spiking:

$$\tau_{\text{open}} \ll \tau_{\text{close}},$$

i.e. opening must be fast and closing must be slow.

**(P5.14) Metabolic Cost Constraint.**  $K_5$  predicts an energy requirement:

$$P_{\text{grad}} \downarrow \Rightarrow A_{\text{spike}} \downarrow \quad \text{or} \quad \text{failure.}$$

**(P5.15) Noise-Induced Spiking Threshold.** Thermal noise may induce:

$$\Delta V + \eta(t) > \Theta_{\text{exc}},$$

predicting spontaneous spikes near threshold.

### **P.3.52 P5: Predictions Concerning Collapse, Death and Threshold Phenomena**

**(P5.16) Excitability Collapse via Ion Exhaustion.**  $K_5$  predicts collapse when:

$$P_{\text{grad}}(t) < \Theta_{\text{grad-min}},$$

causing failure to reach  $\Theta_{\text{exc}}$ .

**(P5.17) Refractory Overload.** If:

$$T_{\text{ref}} \rightarrow \infty,$$

the spike cycle collapses and  $K_5$  loses continuity.

**(P5.18) Blow-Up Regime (Runaway Excitation).** If positive feedback dominates:

$$\frac{d\Delta V}{dt} > \Theta_{\text{blowup}},$$

leading to membrane destruction or channel burnout.

**(P5.19) Oscillatory Collapse (Near-Hopf).**  $K_5$  predicts:

$$\text{limit cycle} \rightarrow \text{unstable cycle} \rightarrow \text{collapse},$$

under near-critical  $G_{\text{lat}}$  or  $P_{\text{grad}}$  conditions.

**(P5.20) Electro-Osmotic Death Mode.** Large  $\Delta V$  induces:

$$J_{\text{osmotic}} \gg 0,$$

swelling and rupture (coupling  $K_5$  collapse to  $K_4$  death modes).

### **P.3.53 P6: Predictions Concerning the Transition to $K_6$**

**(P5.21) Emergence of Patterned Excitation.** When excitation becomes structured:

$$C_{\text{pattern}} \neq 0,$$

a new axis  $A_{\text{pattern}}$  emerges, precursor to cognition.

**(P5.22) Multi-Spike Integration.**  $K_5$  predicts:

$$\text{summation of spikes} \Rightarrow \text{pattern formation}.$$

**(P5.23) Threshold for Binding Capacity.**  $K_5$  predicts a minimal number of simultaneously active units:

$$N_{\text{bind}} > \Theta_{\text{bind}},$$

required to form coherent patterns.

**(P5.24) Emergence of Attractors.** If:

$$G_{\text{lat}} \text{ sufficiently high,}$$

then stable attractor-like patterns appear — signature of  $K_6$ .

**(P5.25) Coding via Spike Timing.**  $K_5$  predicts:

$$t_{\text{spike}} \mapsto \text{information,}$$

a precursor of temporal coding in  $K_6$  cognitive continua.

### **P.3.54 P7: Predictions Concerning Evolution and Higher-Level Behaviour**

**(P5.26) Prediction of Axonal Conductance Scaling.**  $K_5$  predicts that selection favours:

$$G_{\text{lat}} \uparrow \Rightarrow v_{\text{cond}} \uparrow,$$

consistent with the emergence of elongated processes.

**(P5.27) Prediction of Myelination Precursors.** Energy efficiency predicts:

$$C_{\text{mem}} \downarrow \Rightarrow v_{\text{cond}} \uparrow,$$

leading to insulating structures.

**(P5.28) Prediction of Channel Diversity Increase.** Evolution moves toward:

$$N_{\text{channel-types}} \uparrow,$$

increasing dynamical richness.

**(P5.29) Prediction of Metabolic Coupling.** Excitability requires:

$$C_{\text{energy}} \neq 0,$$

predicting stronger metabolic integration compared to  $K_4$ .

**(P5.30) Prediction of Early Neural Networks.** Aggregations with strong lateral coupling evolve into:

$$K_5 \rightarrow K_6,$$

giving rise to proto-neural structures.

### **P.3.55 Summary**

$K_5$  predicts:

- a strict excitability threshold  $\Theta_{\text{exc}}$  and channel threshold  $\Theta_{\text{channel}}$ ;
- the emergence of a new axis  $A_{\text{exc}}$  and spike cycle  $C_{\text{spike}}$ ;
- existence of spatially propagating excitation fronts and an ignition condition ( $G_{\text{lat}} > G_{\text{crit}}$ );
- structured spike dynamics: refractory cycle, reset cycle, and amplitude threshold;
- several collapse modes: ion exhaustion, refractory lock, runaway excitation, oscillatory collapse, electro-osmotic destruction;

- evolutionary predictions: channel diversification, proto-axonal conduction, insulating precursors, metabolic–electrical coupling;
- high-level predictions for the transition into  $K_6$ : pattern formation, attractors, binding capacity threshold, temporal coding.

These predictions uniquely define  $K_5$  as the continuum of elementary excitability and distinguish it sharply from chemical-membrane systems ( $K_4$ ) and cognitive continua ( $K_6$ ).

### P.3.56 Predictions for $K_6$

Level  $K_6$  is the continuum of *cognition*: the domain in which neural excitability ( $K_5$ ) becomes structured into stable, reproducible, self-organising patterns capable of classification, comparison, inference, and error-driven update. Its defining features are:

$$A_{\text{cog}} \neq 0, \quad C_{\text{cog}} \neq 0, \quad k_6(t) > 0,$$

where  $A_{\text{cog}}$  are the cognitive axes (pattern, comparison, error), and  $C_{\text{cog}}$  are cognitive cycles (observation, fixation, comparison, update).

Predictions for  $K_6$  arise from:

- the S-module (S.1–S.5),
- pattern–attractor formation from structured  $K_5$  activity,
- thresholds  $\Theta_{\text{cog}}$  for coherence, contradiction, and stability,
- flows  $J_{\text{info}}$ ,  $J_{\text{comparison}}$ ,  $J_{\text{update}}$ ,
- structural tension  $T_{\text{cog}}$  and its resolution cycles,
- the  $K_6$  Stress Test: 7 falsifiable predictions about cognitive structure.

We group predictions in eight categories.

### P.3.57 P1: Predictions Concerning the Birth of Cognition

**(P6.1) Threshold for Pattern Stability  $\Theta_{\text{pattern}}$ .**  $K_6$  predicts that cognition begins when neural patterns satisfy:

$$\frac{d}{dt}P_{\text{pattern}} = 0 \quad \text{for a finite interval,}$$

i.e. a stable attractor emerges.

**(P6.2) Existence of the Cognitive Axes  $A_{\text{cog}}$ .** A new family of axes appears:

$$A_{\text{pattern}}, A_{\text{comparison}}, A_{\text{error}},$$

predicting classification, similarity estimation, and contradiction detection.

**(P6.3) Minimal Binding Capacity.** Cognition requires:

$$N_{\text{bind}} > \Theta_{\text{bind}},$$

i.e. a minimal number of mutually coherent active units.

**(P6.4) Emergence of Representational Coherence.**  $K_6$  predicts that stable patterns encode:

$$\text{relations, categories, similarities,}$$

not just excitation magnitudes.



### **P3.58 P2: Predictions from the S-module (Meaning Formation)**

**(P6.5) Existence of the S-cell (meaning cell) structure.** A minimal cognitive process must contain:

$$A \rightarrow \text{fixation} \rightarrow \text{expectation} \rightarrow B \rightarrow C \rightarrow \text{comparison} \rightarrow \text{update},$$

predicting that any biological or artificial  $K_6$ -system will implement these operations.

**(P6.6) Emergence of Expectation.**  $K_6$  predicts:

$$\text{fixation of pattern } A \Rightarrow \text{generation of expected pattern } B_*.$$

**(P6.7) Error Signal as an Axial Value.** Error is not a scalar but an axis:

$$e = A - B,$$

predicting that cognitive update behaviours universally seek to minimise  $e$ .

**(P6.8) Update Cycle Necessity.** If:

$$C_{\text{update}} = 0,$$

then meaning cannot form and  $K_6$  collapses to  $K_5$  behaviour.

### **P3.59 P3: Predictions Concerning Attractors and Pattern Dynamics**

**(P6.9) Attractor Stability Threshold.** Patterns must satisfy:

$$\lambda_{\max} < 0,$$

predicting negative Lyapunov exponents in stable cognitive states.

**(P6.10) Multi-Attractor Organisation.**  $K_6$  predicts:

multiple semi-stable attractors

corresponding to distinct concepts or pattern classes.

**(P6.11) Attractor Competition and Selection.** If two attractors compete:

$$T_{\text{cog}} \uparrow,$$

the system predicts winner-take-all resolution or coexistence via splitting.

### **P3.60 P4: Predictions Concerning Comparison, Similarity and Distance**

**(P6.12) Existence of Intrinsic Cognitive Distance.**  $K_6$  predicts a metric:

$$d_{\text{cog}}(A, B),$$

arising from comparison flows  $J_{\text{comparison}}$ .

**(P6.13) Prediction of Similarity-Based Generalisation.** Patterns close under  $d_{\text{cog}}$  generalise:

$$A \sim B \Rightarrow \text{shared activation basin.}$$

**(P6.14) Prediction of Prototype Formation.**  $K_6$  predicts:

$$\text{prototype} = \arg \min_X \sum_i d_{\text{cog}}(X, A_i).$$

### **P.3.61 P5: Predictions Concerning Structural Tension and Contradiction**

**(P6.15) Cognitive Tension as a Measurable Quantity.** Contradictory patterns produce:

$$T_{\text{cog}} > 0,$$

predicting tension-driven reorganisation.

**(P6.16) Contradiction Threshold  $\Theta_{\text{contradiction}}$ .** When:

$$T_{\text{cog}} > \Theta_{\text{contradiction}},$$

a structural update or collapse of the current pattern is predicted.

**(P6.17) Resolution via Update Cycles.**  $K_6$  predicts:

$$C_{\text{update}} \text{ acts to reduce } T_{\text{cog}}.$$

**(P6.18) Prediction of Cognitive Drift.** If contradictions remain unresolved:

$$\frac{d}{dt} P_{\text{pattern}} \neq 0,$$

leading to drift or restructuring.

### **P.3.62 P6: Predictions Concerning Memory and Stability**

**(P6.19) Existence of Cognitive Memory Attractors.**  $K_6$  predicts stable long-lived attractors storing:

past patterns.

**(P6.20) Memory Capacity Threshold.** There exists:

$$C_{\text{mem-capacity}} > 0$$

such that exceeding it leads to attractor interference.

**(P6.21) Prediction of Catastrophic Forgetting.** If:

$$N_{\text{attractors}} \uparrow \quad \text{and} \quad d_{\text{cog}} \downarrow,$$

then earlier attractors collapse — a falsifiable prediction.

### **P.3.63 P7: Predictions Concerning Information Flow and Computation**

**(P6.22) Directed Information Flow  $J_{\text{info}}$ .** Cognition predicts:

$$J_{\text{info}} \text{ has preferred directions,}$$

not isotropic flow as in  $K_5$ .

**(P6.23) Existence of Computation via Pattern Dynamics.**  $K_6$  predicts:

computation emerges by transitions between attractors.

**(P6.24) Prediction of Hierarchical Pattern Composition.** Patterns combine:

$$A \oplus B \rightarrow C,$$

forming higher-level structures.

**(P6.25) Time-Scale Separation.**  $K_6$  predicts:

$$\tau_{\text{fast}} (\text{excitation}) \ll \tau_{\text{slow}} (\text{cognition}),$$

a biologically universal constraint.

### P.3.64 P8: Predictions Concerning Collapse and Transitions

**(P6.26) Cognitive Collapse via Overload.** If:

$$T_{\text{cog}} \gg 0,$$

patterns become unstable and collapse.

**(P6.27) Collapse via Noise.** Noise can push:

$$d_{\text{cog}}(A, B) < \Theta_{\text{blur}},$$

producing ambiguous or fused patterns.

**(P6.28) Collapse via Excessive Binding.** If:

$$N_{\text{bind}} \gg \Theta_{\text{bind}},$$

patterns merge into a non-functional cluster.

**(P6.29) Transition to  $K_7$  (Social Continuum).**  $K_6$  predicts:

$$\text{shared attractors} \Rightarrow K_7,$$

when multiple  $K_6$  systems synchronise via communication.

### P.3.65 Summary

$K_6$  predicts the emergence of:

- stable attractors, cognitive axes, and the S-cell structure;
- comparison, expectation, error, contradiction resolution;
- memory attractors, capacity limits, and catastrophic forgetting;
- structured information flows and pattern computation;
- multiple collapse modes: contradiction overload, noise blur, excessive binding, and attractor instability;
- the transition to  $K_7$  through synchronisation of attractor structures.

These predictions sharply distinguish  $K_6$  from mere excitability ( $K_5$ ) and ensure theoretical continuity toward  $K_7$ .

### P.3.66 Predictions for $K_7$

Level  $K_7$  describes *social continua*: structured collectives of cognitive agents ( $K_6$ -systems) whose patterns, norms, trust cycles, and institutions form a higher-level continuum with its own state space, axes, thresholds, flows, cycles, and collapse conditions.

A social continuum exists when:

$$k(K_7) > 0, \quad C^s \neq 0, \quad J^s \neq 0, \quad \partial\Omega^s \text{ defines institutional boundaries.}$$

Predictions for  $K_7$  arise from:

- Theorem of Representability 19 (social system  $\rightarrow K_7$ ),
- social axes  $A^s$  (normative, status, institutional, role),
- social potentials  $P^s$  (trust, authority, legitimacy, capital),
- thresholds  $\Theta^s$  (legitimacy, stability, collapse),
- flows  $J^s$  (communication, sanctions, resources, migration),
- cycles  $C^s$  (norm cycle, trust cycle, institutional cycle, power cycle),
- transition mechanism  $K_6 \rightarrow K_7$  via synchronisation of attractors.

We present predictions in nine categories.

### P.3.67 P1: Predictions Concerning the Birth of a Social Continuum

**(P7.1) Synchronised attractors as the necessary condition.** A social continuum emerges when multiple  $K_6$  systems satisfy:

$$A_i^{\text{cog}} \approx A_j^{\text{cog}}, \quad C^s \neq 0,$$

i.e. their cognitive attractors synchronise through communication flows  $J^s$ .

**(P7.2) Collective Norm Formation.**  $K_7$  predicts:

$$\exists A_{\text{norm}}^s \text{ iff shared attractor} \rightarrow \text{normative stability.}$$

**(P7.3) Boundary Formation.** A social boundary  $\partial\Omega(K_7)$  forms when:

$$T_{\text{soc}} < \Theta_{\text{boundary}},$$

i.e. when common norms and flows stabilise a recognisable group.

**(P7.4) Institutional Emergence Threshold.** Institutions arise when:

$$C_{\text{norm}}^s \text{ closes,}$$

predicting that repeated norm cycles create durable constraints.

### P.3.68 P2: Predictions Concerning Social Axes and Structure

**(P7.5) Existence of Social Axes  $A^s$ .**  $K_7$  predicts the following minimal axes:

$$A_{\text{norm}}, A_{\text{status}}, A_{\text{role}}, A_{\text{institution}}.$$

**(P7.6) Status Gradient Prediction.** Any  $K_7$  system will exhibit:

$$P_{\text{status}}(x) \text{ is not uniform,}$$

predicting intrinsic and measurable status gradients.

**(P7.7) Role Differentiation.** The system predicts the emergence of:

$$A_{\text{role}}^s \neq 0$$

whenever flows  $J^s$  exceed the norm-cycle capacity.

**(P7.8) Symbolic Capital as a Potential.** Social potential:

$$P_{\text{symbolic}}$$

is predicted to behave analogously to  $P_{\text{energy}}$  in  $K_4$ : it accumulates, flows, and dissipates.

### P.3.69 P3: Predictions Concerning Trust and Legitimacy

**(P7.9) Trust as a Social Potential.** Trust  $P_{\text{trust}}$  satisfies:

$$\frac{d}{dt}P_{\text{trust}} = f(J_{\text{info}}, C_{\text{trust}}, \Theta_{\text{trust}})$$

predicting measurable trust cycles.

**(P7.10) Legitimacy Threshold  $\Theta_{\text{legit}}$ .** Institutions remain stable only when:

$$P_{\text{legit}} > \Theta_{\text{legit}}.$$

**(P7.11) Predictable Collapse of Illegitimate Systems.** If:

$$P_{\text{legit}} < \Theta_{\text{legit}},$$

collapse of the institutional attractor is inevitable.

**(P7.12) Trust–Norm Feedback Loop.**  $K_7$  predicts:

$$P_{\text{trust}} \uparrow \Rightarrow \Theta_{\text{norm}} \downarrow,$$

i.e. high trust lowers the threshold for adopting new norms.

### P.3.70 P4: Predictions About Social Flows

**(P7.13) Directed Information Flow  $J_{\text{comm}}$ .** Social communication is anisotropic:

$$J_{\text{comm}}(x \rightarrow y) \neq J_{\text{comm}}(y \rightarrow x),$$

predicting asymmetric influence networks.

**(P7.14) Resource Flow Hierarchies.**  $K_7$  predicts:

$$J_{\text{resource}} \text{ forms hierarchical channels.}$$

**(P7.15) Sanction Flow Necessity.** Norms cannot be stable without:

$$J_{\text{sanction}} \neq 0.$$

**(P7.16) Migration as Boundary Pressure.** Large migration flow:

$$J_{\text{mig}} \gg 0$$

increases structural tension  $T_{\text{soc}}$  and predicts boundary adaptation or collapse.

### P.3.71 P5: Predictions Concerning Social Cycles

**(P7.17) Existence of the Trust Cycle  $C_{\text{trust}}^s$ .**  $K_7$  predicts a recurrent loop:

expectation  $\rightarrow$  interaction  $\rightarrow$  outcome  $\rightarrow$  update.

**(P7.18) Norm Cycle Closure as a Stability Condition.** Norms are stable iff:

$C_{\text{norm}}^s$  is closed and repeated.

**(P7.19) Institutional Cycle.**  $K_7$  predicts a long-scale cycle:

legitimation  $\rightarrow$  formalisation  $\rightarrow$  codification  $\rightarrow$  adaptation.

**(P7.20) Power Cycle.** Status and authority evolve through:

$$C_{\text{power}} : P_{\text{status}} \rightarrow J_{\text{resource}} \rightarrow P'_{\text{status}}.$$

### P.3.72 P6: Predictions Concerning Stability and Continuumness

**(P7.21) Social Coherence Threshold.** A social continuum exists only when:

$$k(K_7) > 0 \iff \text{norms, flows, and cycles satisfy } \Theta^s.$$

**(P7.22) Predictable Response to Stress.** If structural tension:

$$T_{\text{soc}} > \Theta_{\text{stress}},$$

the system predicts branching or institutional mutation.

**(P7.23) Predictable Polarisation.** If:

$$\Theta_{\text{norm}} \uparrow \quad \text{and} \quad P_{\text{trust}} \downarrow,$$

polarisation appears as two competing social attractors.

### P.3.73 P7: Predictions Concerning Collapse of K7

**(P7.24) Collapse via Broken Norm Cycle.** If:

$$C_{\text{norm}}^s = 0,$$

the social continuum collapses.

**(P7.25) Collapse via Legitimacy Failure.**

$$P_{\text{legit}} < \Theta_{\text{legit}} \Rightarrow \Omega(K_7) \rightarrow \emptyset.$$

**(P7.26) Collapse via Trust Decay.** Trust decay obeys:

$$\frac{d}{dt} P_{\text{trust}} < 0 \text{ for extended time} \Rightarrow k(K_7) \downarrow.$$

**(P7.27) Collapse via Flow Disruption.** When critical flows vanish:

$$J_{\text{comm}} = 0 \quad \text{or} \quad J_{\text{resource}} = 0,$$

the continuum disintegrates.

**P.3.74 P8: Predictions Concerning Transition to  $K_8$**

**(P7.28) Infrastructure Attractor Formation.**  $K_7$  predicts formation of:

technical, economic, legal attractors

which define the birth of  $K_8$ .

**(P7.29) Institutional Over-stability  $\rightarrow K_8$ .** If institutional cycles:

$$C_{\text{institution}}^s$$

become rigid and self-supporting, a transition to  $K_8$  occurs.

**(P7.30) Threshold for Macro-structural Emergence.** When:

$$P_{\text{infrastructure}} > \Theta_{\text{infra}},$$

the continuum expands to  $K_8$ .

**P.3.75 Summary**

$K_7$  predicts:

- synchronisation of cognitive attractors forming social norms,
- emergence of roles, status gradients, and symbolic capital,
- trust and legitimacy as measurable social potentials,
- directional communication and resource flows,
- self-sustaining cycles: norms, trust, institutions, power,
- polarisation, stabilisation, and collapse via specific thresholds,
- transition to  $K_8$  through infrastructure attractors.

These predictions distinguish  $K_7$  sharply from  $K_6$  and form the bridge to the complex structural domains of  $K_8$ .

### P.3.76 Predictions for $K_8$

Level  $K_8$  describes *civilizational-infrastructure continua*: large-scale systems whose stability depends on infrastructure, economic flows, technological bases, regulatory architectures, collective memory, and long-range coordination mechanisms. The defining structural objects of  $K_8$  are:

$$\Omega(K_8), \quad A^8, \quad P^8, \quad \Theta^8, \quad J^8, \quad C^8, \quad k(K_8), \quad T_{\text{infra}}.$$

A system qualifies as a  $K_8$  continuum when:

$$\begin{aligned} C^8 &\neq 0, \\ J_{\text{infra}}^8 &\neq 0, \\ k(K_8) &> 0, \\ \partial\Omega^8 &\text{ defines macro-level institutional-infrastructure boundaries.} \end{aligned}$$

Below are the predictions grouped by structural class.

### P.3.77 P1: Predictions on the Birth of $K_8$

**(P8.1) Infrastructure Attractor Formation.**  $K_8$  emerges when an attractor of infrastructure states appears:

$$A_{\text{infra}}^8 \neq 0 \iff \text{persistent technical-economic-legal cycles form.}$$

**(P8.2) Threshold for Macro-structural Emergence.** Let  $P_{\text{infra}}$  denote the infrastructure potential. A necessary condition:

$$P_{\text{infra}} > \Theta_{\text{infra}},$$

predicting a detectable phase transition from social continuity ( $K_7$ ) to infrastructural invariants ( $K_8$ ).

**(P8.3) Persistence Condition.** An emerging  $K_8$  continuum persists only if long-range flows satisfy:

$$J_{\text{logistics}} > J_{\text{min}}, \quad J_{\text{regulation}} > 0, \quad J_{\text{information}} \text{ remains non-fragmented.}$$

**(P8.4) Civilizational Memory Formation.**  $K_8$  predicts that once collective memory becomes an axis:

$$A_{\text{memory}}^8 \neq 0,$$

the continuum gains temporal depth (historical inertia).

### P.3.78 P2: Predictions about Axes and Structural Geometry

**(P8.5) Minimal Axes at  $K_8$ .**  $K_8$  necessarily exhibits the following axes:

$$A_{\text{infra}}, \quad A_{\text{tech}}, \quad A_{\text{econ}}, \quad A_{\text{reg}}, \quad A_{\text{memory}}.$$

**(P8.6) Separation of Scales.**  $K_8$  predicts strong multi-scale separation:

$$\tau_{\text{tech}} \ll \tau_{\text{institution}} \ll \tau_{\text{civilizational}}.$$

**(P8.7) Existence of Regulatory Geometry.** The geometry of constraints (laws, standards) behaves as:

$$\partial\Omega_{\text{reg}} \text{ acts as a rigid boundary reducing } T_{\text{infra}}.$$



### P.3.79 P3: Predictions for Potentials $P^8$

**(P8.8) Economic Potential as Stabilizer.**  $P_{\text{econ}}$  satisfies:

$$\frac{d}{dt}k(K_8) = F(J_{\text{trade}}, C_{\text{production}}, \Theta_{\text{econ}}).$$

**(P8.9) Technological Acceleration.** If:

$$P_{\text{tech}} \uparrow,$$

then:

$$\Theta_{\text{infra}} \downarrow,$$

predicting easier formation of new infrastructural attractors.

**(P8.10) Collective Memory Inertia.** Potential  $P_{\text{memory}}$  increases stability but slows adaptation:

$$P_{\text{memory}} \uparrow \Rightarrow \frac{d}{dt}\Theta_{\text{institution}} \uparrow.$$

**(P8.11) Regulatory Potential  $P_{\text{reg}}$ .**  $K_8$  predicts that:

$$P_{\text{reg}} > \Theta_{\text{reg}}$$

is required for large-scale coordination.

### P.3.80 P4: Predictions for Infrastructural and Economic Flows

**(P8.12) Asymmetric Infrastructure Flows.**  $K_8$  predicts intrinsic directionality:

$$J_{\text{infra}}(x \rightarrow y) \neq J_{\text{infra}}(y \rightarrow x),$$

leading to core–periphery structures.

**(P8.13) Logistic Thresholds.** Critical flows:

$$J_{\text{logistics}} > \Theta_{\text{logistics}}$$

are necessary for the survival of the continuum.

**(P8.14) Regulatory Flow Closure.** A functioning  $K_8$  always satisfies:

$$J_{\text{regulation}} : \text{policy} \rightarrow \text{implementation} \rightarrow \text{feedback loop}.$$

**(P8.15) Energy and Resource Scaling Laws.** Civilizational energy flow:

$$J_{\text{energy}} \sim \Omega_{\text{population}}^\alpha, \quad \alpha > 1,$$

predicting superlinear scaling.

### P.3.81 P5: Predictions for Cycles $C^8$

**(P8.16) Production–Distribution–Consumption Cycle.**  $K_8$  predicts a self-sustaining cycle:

$$C_{\text{econ}} : \text{production} \rightarrow \text{distribution} \rightarrow \text{consumption} \rightarrow \text{renewal}.$$

**(P8.17) Institutional Reproduction Cycle.** Institutions must follow:

$$C_{\text{institution}}^8 = \text{codification} \rightarrow \text{enforcement} \rightarrow \text{adaptation}.$$

**(P8.18) Innovation Cycle.**  $K_8$  predicts:

$$C_{\text{innovation}} : P_{\text{tech}} \rightarrow \text{application} \rightarrow J_{\text{econ}} \rightarrow P'_{\text{tech}},$$

forming positive feedback.

**(P8.19) Memory Cycle.** Collective memory evolves through:

$$\text{selection} \rightarrow \text{institutional embedding} \rightarrow \text{reproduction}.$$

### P.3.82 P6: Predictions for Stability and $k(K_8)$

**(P8.20) Multi-Threshold Stability Condition.**

$$k(K_8) > 0 \iff \forall k : P^8 \in \mathcal{D}_k^{(8)},$$

i.e. all infrastructural, economic, regulatory and memory thresholds must hold.

**(P8.21) Stress Response.** If infrastructural tension:

$$T_{\text{infra}} > \Theta_{\text{stress}}^{(8)},$$

the system predicts either:

- infrastructure branching,
- regulatory mutation,
- collapse of logistic flows.

**(P8.22) Predictable Failure Modes.**  $K_8$  collapses when:

$$J_{\text{energy}} = 0, \quad \text{or} \quad J_{\text{logistics}} = 0, \quad \text{or} \quad J_{\text{regulation}} = 0.$$

**(P8.23) Predictable Polarisation at Civilizational Scale.** If:

$$P_{\text{memory}} \uparrow \text{ and } J_{\text{information}} \downarrow,$$

then:

polarisation attractors emerge.

### P.3.83 P7: Predictions Concerning Collapse of $K_8$

**(P8.24) Collapse via Infrastructure Decay.** If:

$$C_{\text{infra}}^8 = 0,$$

the continuum loses its boundary and collapses.

**(P8.25) Collapse via Economic Cycle Failure.** If:

$$C_{\text{econ}} = 0,$$

the system enters  $k(K_8) \rightarrow 0$ .

**(P8.26) Collapse via Memory Fragmentation.** If:

$$A_{\text{memory}}^8 \text{ splits into incompatible subspaces,}$$

the continuum fragments into sub- $K_8$  entities.

**(P8.27) Collapse via Regulatory Failure.** If:

$$P_{\text{reg}} < \Theta_{\text{reg}},$$

long-range coordination fails and  $\Omega(K_8)$  dissolves.

### P.3.84 P8: Predictions for Transition to $K_9$

**(P8.28) Meta-structural Layer Emergence.** Transition to  $K_9$  begins when:

$$A_{\text{meta}}^9 \neq 0,$$

i.e. when the system constructs theories, models, and meta-representations of itself.

**(P8.29) Threshold of Abstract Coherence.**  $K_8 \rightarrow K_9$  transition requires:

$$P_{\text{abstraction}} > \Theta_{\text{meta}},$$

predicting emergence of philosophical, scientific, mathematical structures.

**(P8.30) Recursive Institutionalisation.** When:

$$C_{\text{institution}}^8 \text{ includes self-description,}$$

a new dimension opens and  $K_9$  is born.

### P.3.85 Summary

The predictions for  $K_8$  include:

- formation of infrastructural attractors,
- multi-axis macro-organization (technical, economic, regulatory, memory),
- non-trivial potentials governing civilizational stability,
- superlinear energy scaling laws,
- closure of economic, institutional, innovation and memory cycles,
- predictable stress responses, polarisation, and collapse modes,
- emergence of meta-structures signalling the transition to  $K_9$ .

### P.3.86 Predictions for $K_9$

Level  $K_9$  describes theory-producing continua: systems whose dynamics arise from symbolic representation, conceptual distinction, methodological structuring, and deliberate construction of abstract models of reality. The defining structural components are:

$$\Omega(K_9), A^9, P^9, \Theta^9, J^9, C^9, k(K_9), T_{\text{theory}}.$$

A continuum qualifies as  $K_9$  when:

$$A_{\text{concept}}^9 \neq 0, \quad A_{\text{logic}}^9 \neq 0, \quad C_{\text{theory}}^9 \neq 0, \quad P_{\text{abstraction}} > \Theta_{\text{cognitive}},$$

i.e. it develops stable structures of conceptualization, reasoning, and meta-representation.

Below are structural predictions deduced from the Core formalism.

### P.3.87 P1: Predictions on the Birth of $K_9$

**(P9.1) Emergence of a Conceptual Axis.**  $K_9$  appears when a system extends

$$A_{\text{semantic}}^8 \rightarrow A_{\text{concept}}^9,$$

creating conceptual distinctions irreducible to  $K_8$ .

**(P9.2) Abstraction Threshold.** Transition from  $K_8$  to  $K_9$  requires:

$$P_{\text{abstraction}} > \Theta_{\text{meta}},$$

predicting abrupt appearance of theory-like structures.

**(P9.3) Birth of Meta-cycles.** If:

$$C_{\text{institution}}^8 \text{ begins to include self-description,}$$

then:

$$C_{\text{theory}}^9 \neq 0,$$

marking the birth of theoretical reflexivity.

**(P9.4) Propagation of Logical Constraints.**  $K_9$  predicts the inevitability of:

$$\partial\Omega_{\text{logic}}^9 \subseteq \partial\Omega(K_{10}),$$

thus anticipating the need for stricter formal systems ( $K_{10}$ ).

### P.3.88 P2: Predictions Concerning Axes $A^9$

**(P9.5) Minimal Necessary Axes.**  $K_9$  necessarily possesses:

$$A_{\text{concept}}, A_{\text{logic}}, A_{\text{semantics}}, A_{\text{method}}, A_{\text{representation}}.$$

**(P9.6) Dimensional Expansion Law.** An increase in conceptual distinctions implies:

$$\dim \Omega(K_9) \uparrow \Rightarrow \Theta_{\text{coherence}} \uparrow,$$

predicting higher coherence requirements in richer theories.

**(P9.7) Multi-Layered Semantic Geometry.** Semantic axes form a stratified geometry:

$$A_{\text{object}} < A_{\text{model}} < A_{\text{meta}}.$$

### P.3.89 P3: Predictions for Potentials $P^9$

**(P9.8) Expressive Power Tends to Grow.** For sufficiently stable  $K_9$  continua:

$$\frac{d}{dt} P_{\text{expressive}} > 0,$$

predicting expansion of symbolic resources (languages, formalisms).

**(P9.9) Conceptual Energy Minimization.**  $K_9$  theories evolve toward reducing conceptual tension:

$$T_{\text{theory}} \downarrow \iff \text{emergence of unifying abstractions.}$$

**(P9.10) Convergence Toward Formalization.** If:

$$P_{\text{precision}} \uparrow,$$

then:

$$K_9 \rightarrow K_{10}$$

becomes inevitable.

**(P9.11) Abstraction Saturation.** When:

$$P_{\text{abstraction}} > \Theta_{\text{saturation}},$$

a new meta-level ( $M$ -space) becomes required (predicting extension to  $M_1, M_2, \dots$ ).

### P.3.90 P4: Predictions for Flows $J^9$

**(P9.12) Proof-like Flows Are Emergent.** Even without formal logic,  $K_9$  predicts:

$$J_{\text{reason}}^9 \sim \text{proto-deductive chains.}$$

**(P9.13) Interpretative Translation Is Necessary.** Flows between theories satisfy:

$$J_{\text{translation}}(T_i \rightarrow T_j) \neq 0 \iff k(K_9) > 0.$$

**(P9.14) Flow Saturation Leads to Paradigm Shift.** If:

$$J_{\text{contradiction}}^9 > \Theta_{\text{critical}},$$

a new  $\Omega(K_9)$  region appears  $\rightarrow$  theoretical revolution.

**(P9.15) Information Compression.**  $K_9$  predicts signatures of compression:

$$J_{\text{compression}} \uparrow \Rightarrow A_{\text{unification}} \uparrow.$$

### P.3.91 P5: Predictions for Cycles $C^9$

**(P9.16) Universal Theory Cycle.** All  $K_9$  systems exhibit:

$$C^9 = (\text{model construction}) \rightarrow (\text{evaluation}) \rightarrow (\text{revision}) \rightarrow (\text{reintegration}).$$

**(P9.17) Meta-stability Through Iteration.**  $K_9$  predicts:

$$\#(C^9) \uparrow \Rightarrow k(K_9) \uparrow.$$

**(P9.18) Divergence Condition.** If:

$$C_{\text{revision}}^9 \text{ diverges,}$$

then:

$$\Omega(K_9) \rightarrow \emptyset,$$

i.e. the theory collapses.

**(P9.19) Cross-Disciplinary Fusion.** Cycles of theory tend to merge when:

$$J_{\text{translation}} \uparrow, \quad P_{\text{unification}} > \Theta_{\text{unif.}}$$

### P.3.92 P6: Predictions for Stability and $k(K_9)$

#### (P9.20) Consistency Threshold.

$$k(K_9) > 0 \iff P_{\text{coherence}} > \Theta_{\text{inconsistency}}.$$

#### (P9.21) Theoretical Stress Response. If theoretical tension:

$$T_{\text{theory}} > \Theta_{\text{critical}}^{(9)},$$

the system undergoes:

- conceptual unification,
- fragmentation into sub-theories,
- formalization (transition to  $K_{10}$ ).

#### (P9.22) Predictable Modes of Collapse. Collapse occurs when:

$$C_{\text{theory}} = 0, \quad \text{or} \quad J_{\text{reason}} = 0.$$

#### (P9.23) Predictable Over-Expansion Failure. If:

$$\dim A^9 \gg P_{\text{coherence}},$$

the continuum becomes unstable  $\rightarrow$  fragmentation.

### P.3.93 P7: Predictions for Transition to $K_{10}$

#### (P9.24) Formalization Trigger. Transition begins when:

$A_{\text{logic}}^9$  stabilizes into discrete syntactic categories.

#### (P9.25) Recursion Threshold. If:

$$P_{\text{recursion}} > \Theta_{\text{rec}},$$

the system necessarily moves to  $K_{10}$ .

#### (P9.26) Birth of Proof-Flow Attractors. As:

$$J_{\text{reason}}^9 \rightarrow J_{\text{proof}}^{10},$$

the structural geometry of  $K_9$  collapses into formal logic of  $K_{10}$ .

#### (P9.27) Meta-theoretical Stabilization. If:

$$C_{\text{meta}}^9 \neq 0,$$

then the system builds stable structures of type-theory, category-theory, formal semantics, etc.  $\rightarrow$  clear  $K_9 \rightarrow K_{10}$  transition.

### P.3.94 Summary

The structural predictions for  $K_9$  include:

- emergence of conceptual, logical, semantic and methodological axes,
- appearance of theory-producing cycles and meta-reflexive structures,
- growth of expressive and abstract potentials,
- proof-like and translation flows between theories,
- universal patterns of theory evolution, collapse and unification,
- thresholds for coherence, abstraction, recursion and meta-stability,
- well-defined triggers for the transition from  $K_9$  to the formal  $K_{10}$  continuum.

### P.3.95 Predictions for $K_{10}$

Level  $K_{10}$  describes formal–recursive continua: systems whose states are symbolic configurations with well-defined syntax, semantics, recursion principles, and proof or computation flows. The structural components are:

$$\Omega(K_{10}), A^{10}, P^{10}, \Theta^{10}, J^{10}, C^{10}, k(K_{10}), T_{\text{formal}}.$$

A continuum qualifies as  $K_{10}$  when:

$$A_{\text{logic}}^9 \rightarrow A_{\text{formal}}^{10}, \quad P_{\text{recursion}} > \Theta_{\text{rec}}, \quad J_{\text{proof}}^{10} \neq 0.$$

Below are the predictions derived from the Core.

### P.3.96 P1: Predictions About the Birth of Formality

**(P10.1) Recursion as a Dimensional Trigger.**  $K_{10}$  emerges when the system constructs a stable operator:

$$\text{Fix}(f) \Rightarrow A_{\text{rec}}^{10} \neq 0,$$

predicting that recursion is the minimal requirement for formalization.

**(P10.2) Syntactic Solidification.** Transition from  $K_9$  to  $K_{10}$  occurs when:

$$\partial\Omega_{\text{syntax}}^{10} = \text{discrete},$$

i.e. symbolic categories acquire rigid combinatorial boundaries.

**(P10.3) Proof-Flow Emergence.** If:

$$J_{\text{reason}}^9 = O(1),$$

and abstraction is stable, then:

$$J_{\text{proof}}^{10} = O(n),$$

predicting increasing determinacy of inferential steps.

**(P10.4) Collapse of Semantic Ambiguity.** When:

$$P_{\text{precision}} > \Theta_{\text{semantic}},$$

semantic latitude collapses into formal interpretability.

### P.3.97 P2: Predictions Concerning Axes $A^{10}$

**(P10.5) Necessary Axes for Any Formal System.** Each  $K_{10}$  continuum necessarily includes:

$$A_{\text{syntax}}, A_{\text{semantics}}, A_{\text{recursion}}, A_{\text{type}}, A_{\text{proof}}, A_{\text{model}}.$$

**(P10.6) Growth of Logical Dimensionality.** Increasing the expressive power implies:

$$\dim A_{\text{type}}^{10} \uparrow \Rightarrow \Theta_{\text{consistency}} \uparrow,$$

predicting larger fragility of rich formalisms.

**(P10.7) Hierarchical Layering.** Formal axes form predictable strata:

$$A_{\text{term}} < A_{\text{type}} < A_{\text{meta}}.$$

### P.3.98 P3: Predictions About Potentials $P^{10}$

**(P10.8) Expressive Power Exhibits Phase Transitions.** There exist critical transitions:

$$P_{\text{expressive}} = \Theta_{\text{Turing}}, P_{\text{expressive}} = \Theta_{\text{2nd-order}},$$

predicting threshold jumps in computational ability.

**(P10.9) Stability Requires Coherence Energy.** Formal stability requires:

$$P_{\text{coherence}} > \Theta_{\text{Gödel}},$$

predicting inevitable incompleteness at high expressive power.

**(P10.10) Recursion Increases Structural Tension.** The presence of unbounded recursion implies:

$$\frac{d}{dt} T_{\text{formal}} > 0,$$

until new meta-levels are introduced.

**(P10.11) Completeness Is Structurally Limited.** For sufficiently expressive continua:

$$P_{\text{expressive}} > \Theta_{\text{Gödel}} \Rightarrow C_{\text{completeness}}^{10} = 0.$$

### P.3.99 P4: Predictions for Proof/Computation Flows $J^{10}$

**(P10.12) Proof Flows Have Attractor Structure.**  $K_{10}$  predicts that:

$$J_{\text{proof}}^{10} \rightarrow \text{canonical normal forms},$$

whenever reduction is confluent.

**(P10.13) Computation Saturation.** If:

$$J_{\text{compute}} \rightarrow \infty,$$

the continuum tends toward undecidability regions in  $\Omega(K_{10})$ .



**(P10.14) Interpretation Flows Are Monotonic.** For any interpretation:

$$T_i \rightarrow T_j, \quad \text{monotone increase in } P_{\text{structure}}.$$

**(P10.15) Explosion Threshold.** If contradiction flows satisfy:

$$J_{\text{contradiction}}^{10} > \Theta_{\text{explosion}},$$

then:

$$\Omega(K_{10}) = \emptyset.$$

**P.3.100 P5: Predictions for Cycles  $C^{10}$**

**(P10.16) Universal Cycle of Formal Systems.** All  $K_{10}$  systems obey:

$$C^{10} = (\text{axioms}) \rightarrow (\text{derivations}) \rightarrow (\text{reductions}) \rightarrow (\text{normal forms}) \rightarrow (\text{meta-analysis}).$$

**(P10.17) Meta-stability Requires Finite Cycles.** Stability requires:

$$|C_{\text{meta}}| < \infty,$$

predicting collapse when meta-levels proliferate without bound.

**(P10.18) Canonical Collapse Forms.** Collapse of a formal system follows one of:

1. inconsistency,
2. triviality,
3. undecidability saturation,
4. infinite regress of meta-levels.

**(P10.19) Criterion for Formal Robustness.** Robustness increases with:

$$J_{\text{normalization}}^{10} \uparrow, \quad P_{\text{redundancy}} \uparrow.$$

**P.3.101 P6: Predictions for Stability and  $k(K_{10})$**

**(P10.20) Nontriviality Threshold.**

$$k(K_{10}) > 0 \iff P_{\text{consistency}} > \Theta_{\text{collapse}}.$$

**(P10.21) Predictable Growth of Incompleteness.** As expressivity increases:

$$k(K_{10}) \downarrow, \quad T_{\text{Gödel}} \uparrow.$$

**(P10.22) Formal Stress Modes.** Stress is relieved only by:

- restriction of expressive power,
- stratification into type levels,
- transition to an  $M$ -space.

### P.3.102 P7: Predictions for Transitions Out of $K_{10}$

(P10.23) Transition to  $M$ -Spaces. If:

$$A_{\text{meta}}^{10} \text{ becomes unstable,}$$

the continuum ascends:

$$K_{10} \rightarrow M_x.$$

(P10.24) Birth of Metalogic. When:

$$P_{\text{reflection}} > \Theta_{\text{reflect}},$$

the system must generate:

$$\Omega(M_1),$$

i.e. formal reflection on the formal system itself.

(P10.25) Predictable Expansion of Expressive Hierarchy.  $K_{10}$  predicts:

$$\exists \text{ infinite hierarchy of reflective levels } (M_0, M_1, M_2, \dots).$$

(P10.26) Necessity of Category-Level Abstractions. If:

$$\dim A_{\text{structure}}^{10} \rightarrow \infty,$$

then category-theoretic, type-theoretic or topos-theoretic abstractions must appear.

### P.3.103 Summary

The core predictions for  $K_{10}$  include:

- necessity of recursion, syntax, semantics, and proof flows,
- existence of phase transitions in expressivity,
- unavoidable incompleteness at sufficient expressiveness,
- attractor structure of proof and computation,
- canonical forms of collapse and robustness,
- strict thresholds for consistency, decidability, and recursion,
- predictable emergence of meta-logical and  $M$ -space structures.

### P.3.104 Predictions for $K_{11}$

Level  $K_{11}$  describes *model-space continua*: meta-structures whose states are classes of models, interpretations, functors, and transformations between formal systems. While  $K_{10}$  operates on symbolic and recursive structure,  $K_{11}$  operates on the *space of all structures* associated with a given formal system or hierarchy of systems.

The defining components are:

$$\Omega(K_{11}), A^{11}, P^{11}, \Theta^{11}, J^{11}, C^{11}, k(K_{11}), T_{11},$$

where  $\Omega(K_{11})$  is a model-space (or category) and  $J^{11}$  describes transformations and adjunction flows.

Below are the structural predictions derived from the Core.

### P.3.105 P1: Predictions About the Birth of Model-Spaces

**(P11.1) Necessity of Multiple Valid Interpretations.** A transition  $K_{10} \rightarrow K_{11}$  occurs when:

$$|\text{Mod}(T)| > 1,$$

predicting that *non-categorical semantics* forces the birth of a model-space continuum.

**(P11.2) Model-Space Dimensional Trigger.** If:

$$P_{\text{expressive}}^{10} > \Theta_{\text{Gödel}},$$

then  $K_{10}$  cannot remain self-contained and must generate:

$$A_{\text{model}}^{11} \neq 0.$$

**(P11.3) Functorial Coherence Threshold.** Model-spaces arise when:

$$J_{\text{interpretation}}^{10} \text{ becomes functorial,}$$

predicting categorical organization of interpretations.

**(P11.4) Collapse of Pure Syntax.** Whenever:

$$P_{\text{sem}}^{10} > \Theta_{\text{syntax}},$$

semantics can no longer be reduced to syntax, forcing the continuum to expand to  $K_{11}$ .

### P.3.106 P2: Predictions Concerning Axes $A^{11}$

**(P11.5) Universal Axes of Model-Spaces.** Every  $K_{11}$  continuum contains axes corresponding to:

$$A_{\text{models}}, A_{\text{interpretations}}, A_{\text{functors}}, A_{\text{equivalences}}, A_{\text{topologies}}, A_{\text{universality}}.$$

**(P11.6) Existence of Model-Topologies.** Each  $K_{11}$  continuum must admit a Grothendieck-like topology:

$$A_{\text{covers}}^{11} \neq \emptyset,$$

thereby enabling sheaf-like reconstruction of semantics.

**(P11.7) Equivalence Collapse Threshold.** If:

$$P_{\text{structure}} \uparrow,$$

then:

$$\dim A_{\text{equiv}}^{11} \downarrow,$$

predicting collapse to fewer equivalence classes of models.

**(P11.8) Predictable Emergence of Adjoint Axes.** Whenever two model dimensions grow monotonically:

$$A_i^{11} \uparrow, A_j^{11} \uparrow,$$

the system predicts the appearance of an adjunction:

$$A_{i \dashv j}^{11}.$$

### P.3.107 P3: Predictions About Potentials $P^{11}$

**(P11.9) Semantic Stability Requires Higher-Order Coherence.** The continuum predicts:

$$P_{\text{coherence}}^{11} > \Theta_{\text{coh}},$$

is necessary for stable interpretation classes.

**(P11.10) Universality Potential Appears Naturally.** If the space of models grows without bound:

$$|\Omega(K_{11})| \rightarrow \infty,$$

then:

$$P_{\text{universality}}^{11} > 0$$

predicting emergence of universal models or universal morphisms.

**(P11.11) Locality–Globality Transition.** There exists a threshold  $\Theta_{\text{local}}$  such that:

$$P_{\text{local}} < \Theta_{\text{local}} \Rightarrow \text{global structures emerge.}$$

**(P11.12) Necessity of Higher Adjointness.** As meta-complexity grows:

$$P_{\text{adjunction}} \uparrow,$$

and adjoint functors must appear to maintain stability.

### P.3.108 P4: Predictions for Transformation Flows $J^{11}$

**(P11.13) Functorial Flow Directionality.** Every  $K_{11}$  continuum predicts:

$$J_{\text{interpretation}}^{11} \text{ is } \text{Funct}(T \rightarrow \mathcal{M}),$$

i.e. interpretation flows become strictly functorial.

**(P11.14) Natural Transformation Attractors.** Flows between models admit attractors in the form of:

$$\eta : F \Rightarrow G, \quad \text{and} \quad \varepsilon : G \Rightarrow F,$$

predicting stabilization via natural transformations.

**(P11.15) Collapse of Non-Coherent Flows.** If:

$$J_{\text{model}}^{11} \text{ fails coherence constraints,}$$

then transitions collapse to:

$$\Omega(K_{11})_{\text{coh}}.$$

**(P11.16) Universality as Flow Fixed Point.** Every universal morphism:

$$u : X \rightarrow U,$$

is predicted to be a fixed point of the flow:

$$J_{\text{model}}^{11}.$$

### P.3.109 P5: Predictions for Cycles $C^{11}$

(P11.17) **The Model-Theoretic Cycle.** All  $K_{11}$  continua follow the cycle:

$$(\text{syntax}) \rightarrow (\text{semantics}) \rightarrow (\text{models}) \rightarrow (\text{functors}) \rightarrow (\text{universality}) \rightarrow (\text{reflection}).$$

(P11.18) **Existence of Global–Local Cycles.** Whenever covers exist:

$$C_{\text{local/global}}^{11} \text{ forms a commuting diagram.}$$

(P11.19) **Meta-Stability Requires Triangulated Cycles.** Stability predicts:

$$C^{11} \text{ contains a triangulated subcycle,}$$

analogous to:

$$X \rightarrow Y \rightarrow Z \rightarrow X[1].$$

(P11.20) **Collapse Conditions.** Collapse occurs when:

1. there is no coherent interpretation class,
2. universality cannot be established,
3. equivalences proliferate without convergence,
4. adjunction cycles fail to close.

### P.3.110 P6: Predictions for Continuumness $k(K_{11})$

(P11.21) **Model-Space Fragmentation Reduces  $k(K_{11})$ .**

$$k(K_{11}) \downarrow \quad \text{if} \quad |\pi_0(\Omega(K_{11}))| \uparrow,$$

predicting fragmentation as a sign of collapse risk.

(P11.22) **Universality Increases  $k(K_{11})$ .** The measure of universality contributes positively:

$$k(K_{11}) \propto P_{\text{universality}}^{11}.$$

(P11.23) **Adjoint Stability Criterion.** If adjunctions exist:

$$F \dashv G,$$

then:

$$k(K_{11}) \uparrow.$$

(P11.24) **Topological Cohesion Threshold.** A necessary condition for  $k(K_{11}) > 0$ :

$$\Theta_{\text{cover}} < P_{\text{coverage}},$$

i.e. the model-space must admit adequate topologies.

### P.3.111 P7: Predictions for Transition to $K_{12}$

(P11.25) Emergence of Meta-Dynamics. If:

$$J_{\text{model}}^{11} \text{ develops dynamical degrees,}$$

then:

$$K_{11} \rightarrow K_{12}.$$

(P11.26) Necessity of Dynamic Universality. Transition requires:

$$P_{\text{universality}}^{11} > \Theta_{\text{dynamic}},$$

predicting activation of dynamic model-spaces.

(P11.27) Predictable Growth of Meta-Structure. As:

$$\dim A^{11} \uparrow,$$

the system must generate:

$$A_{\text{dynamics}}^{12}.$$

(P11.28) Categorical Collapse as Transition Trigger. When equivalence classes collapse too aggressively:

$$A_{\text{equiv}}^{11} \Downarrow\Downarrow,$$

the continuum must move to:

$$K_{12},$$

where dynamics replaces static structural comparison.

### P.3.112 Summary

The predictions for  $K_{11}$  include:

- birth of model-spaces when interpretations proliferate,
- functorial and adjoint structure as structural invariants,
- threshold conditions for coherence, universality and locality,
- attractor behaviour via natural transformations,
- canonical global–local cycles and triangulated cycles,
- collapse modes tied to failure of equivalence convergence,
- universality and adjointness as stabilizing forces,
- predictable emergence of  $K_{12}$  when dynamics intrudes.

### P.3.113 Predictions for $K_{12}$

Level  $K_{12}$  describes *dynamic model-spaces*: meta-continua whose states are not individual models or classes of models (as in  $K_{11}$ ), but *dynamical laws governing the evolution of model-spaces themselves*. Thus  $K_{12}$  is the level at which the continuum can reorganize, reparameterize or transform entire semantic–formal hierarchies.

A  $K_{12}$  continuum is characterized by:

$$\Omega(K_{12}), A^{12}, P^{12}, \Theta^{12}, J^{12}, C^{12}, k(K_{12}), T_{12},$$

with  $J^{12}$  representing dynamic flows between model-spaces.

Below we list the structural predictions.

### P.3.114 P1: Predictions About the Birth of $K_{12}$

**(P12.1) Dynamization Threshold of Model-Spaces.** A transition  $K_{11} \rightarrow K_{12}$  occurs when:

$J_{\text{model}}^{11}$  acquires an intrinsic temporal or dynamical parameter.

**(P12.2) Necessity of Dynamic Universality.** If:

$$P_{\text{universality}}^{11} > \Theta_{\text{dynamic}},$$

then static model-spaces cannot remain stable and must expand to  $K_{12}$ .

**(P12.3) Collapse of Static Equivalence Classes.** Whenever equivalence classes in  $K_{11}$  collapse or proliferate uncontrollably:

$$A_{\text{equiv}}^{11} \downarrow \quad \text{or} \quad A_{\text{equiv}}^{11} \rightarrow \infty,$$

the continuum is predicted to generate dynamic structure.

**(P12.4) Trigger by Meta-Adjunction.** If adjunctions in  $K_{11}$  become unstable:

$$F \dashv G \quad \text{fails to persist,}$$

then  $K_{12}$  emerges to absorb the instability.

### P.3.115 P2: Predictions About Axes $A^{12}$

**(P12.5) Existence of Dynamic Axes.** Every  $K_{12}$  continuum contains axes of the form:

$$A_{\text{flow}}, A_{\text{reconfiguration}}, A_{\text{meta-time}}, A_{\text{meta-dimension}},$$

representing directions in which model-spaces can evolve.

**(P12.6) Emergence of 2- and 3-categorical Axes.** The continuum predicts:

$$A_{2\text{cat}}^{12} \neq 0,$$

i.e. dynamic transformations between functors themselves.

**(P12.7) Structural Expansion Threshold.** If:

$$\dim A^{11} \rightarrow \infty,$$

then  $A^{12}$  must appear to preserve coherence.

**(P12.8) Predictable Birth of Meta-Dimensionality.** A hallmark:

$$A_{\text{dim}}^{12} > 0,$$

indicating reconfigurable dimensional structure.

### P.3.116 P3: Predictions About Potentials $P^{12}$

**(P12.9) Instability of Static Semantics.**  $K_{12}$  arises when:

$$P_{\text{semantic}}^{11} > \Theta_{\text{static}},$$

making semantics impossible without dynamic restructuring.

**(P12.10) Existence of Meta-Energy.** Prediction:

$$E^{12} = E^{11} + \delta E_{\text{dynamic}},$$

i.e. energy-like potentials emerge to regulate dynamic transformations.

**(P12.11) Growth of Dynamic Expressivity.** If:

$$P_{\text{expressive}}^{11} \uparrow,$$

then:

$$P_{\text{dynamic}}^{12} \uparrow,$$

predicting the rise of dynamic meta-models.

**(P12.12) Dynamic Universality Potential.** The continuum expects:

$$P_{\text{universal-flow}}^{12} > 0,$$

representing the possibility of flows with universal properties.

**P.3.117 P4: Predictions for Transformation Flows  $J^{12}$**

**(P12.13) Dynamic Functoriality.**  $J^{12}$  is predicted to become:

$$J^{12} : \mathcal{M}_1(t) \rightarrow \mathcal{M}_2(t'),$$

i.e. functorial in both domain and codomain that themselves evolve.

**(P12.14) Existence of Flow-Equivalences.** There exist morphisms:

$$\Phi_{t,t'} : \Omega(K_{12})(t) \rightarrow \Omega(K_{12})(t'),$$

indicating equivalence classes of flows.

**(P12.15) Higher-Order Natural Transformations.** Flows admit dynamic natural transformations:

$$\alpha(t) : F_t \Rightarrow G_t.$$

**(P12.16) Meta-Stable Fixed Points.** Predicts existence of fixed points:

$$J^{12}(X) = X,$$

interpretable as dynamically invariant theories.

**P.3.118 P5: Predictions for Cycles  $C^{12}$**

**(P12.17) Dynamic Reconstruction Cycle.** K12 predicts a canonical cycle:

$$(\text{model-spaces}) \rightarrow (\text{flows}) \rightarrow (\text{reconfigurations}) \rightarrow (\text{meta-stability}) \rightarrow (\text{new model-spaces}),$$

resembling a bootstrapping mechanism.

**(P12.18) Existence of Multi-Level Cycles.** Cycles span multiple categorical levels:

$$C^{12} : 0 \rightarrow 1 \rightarrow 2 \rightarrow 3\text{-level},$$

predicting stable structures only if all commute.



**(P12.19) Threshold for Dynamic Collapse.** Collapse occurs when:

$$J^{12} \text{ overwhelms } \Theta_{\text{stability}},$$

leading to:

$$\Omega(K_{12}) \rightarrow \emptyset.$$

**(P12.20) Meta-Temporal Loop Formation.** Predicts:

$$C_{\text{meta-time}}^{12} \text{ forms closed trajectories,}$$

giving rise to re-emerging structural configurations.

**P.3.119 P6: Predictions for Continuumness**  $k(K_{12})$

**(P12.21) Dynamic Cohesion Requirement.**

$$k(K_{12}) > 0 \quad \Leftrightarrow \quad \exists C_{\text{coherent}}^{12}.$$

**(P12.22) Meta-Dimensional Stability.** If:

$$\dim A^{12} \text{ remains bounded,}$$

then:

$$k(K_{12}) \uparrow .$$

**(P12.23) Growth of Dynamic Universality.**

$$k(K_{12}) \propto P_{\text{universal-flow}}^{12}.$$

**(P12.24) Fragmentation of Model-Dynamics Reduces**  $k(K_{12})$ . If:

$$|\pi_0(J^{12})| \uparrow,$$

then:

$$k(K_{12}) \downarrow .$$

**P.3.120 P7: Predictions for Transition Beyond**  $K_{12}$

**(P12.25) Existence of Hyper-Dynamic Structures.** Predicts:

$$\exists K_{13} \text{ only if } P_{\text{dynamic}}^{12} > \Theta_{\text{hyper}}.$$

**(P12.26) No Stable Termination of Model-Dynamics.** The continuum predicts that  $K_{12}$  cannot serve as a final level:

$$T_{12} \not\rightarrow 0.$$

**(P12.27) Necessity of Meta-Stable Attractors.** Transition requires:

$$J^{12} \text{ has attractors with } k > 0.$$

**(P12.28) Dimensional Overflow Condition.** If:

$$\sum_{i=1}^{12} \dim A^i \rightarrow \infty,$$

a new meta-level is required.

### P.3.121 Summary

The predictions for  $K_{12}$  include:

- dynamic restructuring of entire model-spaces,
- emergence of meta-time and reconfiguration axes,
- higher-order functorial flows and dynamic natural transformations,
- cyclic meta-dynamics enabling self-reconstruction,
- stability conditions based on dynamic universality and bounded dimensionality,
- collapse mechanisms tied to instability of model-dynamics,
- predictable transitions toward hypothetical  $K_{13}$ .

## Q Unified Synthesis Support Dossiers

The main unified-synthesis section remains the canonical synthesis unit in the numbered Core 1.3 sequence of the manuscript. The material gathered here is retained as level-wise evidence support: it keeps compact K-anchored theorem-support, empirical, falsifier, and packet dossiers available inside the flagship volume while omitting the synthesis-section introduction and release-gate prose. Section 24 uses the generated synthesis surface for the main K-level exposition. This appendix is the support-dossier wrapper: it gives readers a single inspection location for the level-wise support files without turning them into a second synthesis chapter. The main chapter states the synthesis, while this appendix carries the dossier-oriented inspection copy. Compatibility support files are not input directly by the English platinum master. A build variant must therefore choose either the compatibility synthesis-support master or this appendix wrapper, never both. The full DFS compatibility stream still records those compatibility entries for auditability, but the platinum master does not input the compatibility synthesis-support master. The constants-and-parameters apparatus is part of the Appendix C data apparatus and is addressable there as Appendix C.4; this support appendix does not create a second synthesis or a second top-level constants appendix. Source stamp: `releases/oc_core_1_3/editorial/OC_CORE_1_3_UNIFIED_SYNTHESIS_latest.json`; current repository SHA and UTC generation timestamp are recorded inside that JSON surface so this appendix does not freeze stale build metadata.

### Q.1 Level K0: structural conditions for any universe

Current closure status: `PASS`. Promotion blockers: none.

**Theorem claim.** K0 fixes the non-empty admissible meta-domain required for any lawful continuum and therefore bounds every later K-level before empirical specialization.

**Operator and K-level binding.** The local K0 root process glyphs  $\Psi_0$ ,  $\Phi_0$ , and  $\Lambda_0$  fix distinction generation, relational reconfiguration, and compositional assembly; compact synthesis aliases  $F_0$ ,  $Q_0$ , and  $U_0$  label the corresponding root constraints only where they are defined locally. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K0`.

**Parameter law.**

$$\mu(\Omega(K_0)) > 0, \quad \forall x \in \{1, \dots, 12\} : \text{DoF}(M_x) > 0 \text{ and } \frac{\text{DoF}(K_x)}{\text{DoF}(M_x)} \leq 1, \quad C_{\text{nt}} \geq 1$$

**Observable binding and units.** Meta-admissibility, meta-space compatibility, and the presence of at least one non-trivial lawful cycle. **Observable ids:** `K0::ADMISSIBLE_STATE_MEASURE`, `K0::META_SPACE_COMPATIBILITY_RATIO`, `K0::NONTRIVIAL_CYCLE_PRESENCE`, `MATHEMATICS::EPSILON_STAR`, `MATHEMATICS::MINIMAL_DENOMINATOR`, `MATHEMATICS::MINIMAL_AUGMENTATION_RANK`, `MATHEMATICS::COMPONENT_FIBER_COUNT`, `MATHEMATICS::MOVE_CONNECTIVITY_OR_DEGREE_BOUND`, `MATHEMATICS::STRICT_TERMINALIZED_TOTAL`, `MATHEMATICS::COUNTEREXAMPLE_EXECUTED_TOTAL`, `MATHEMATICS::PYTHON_EVIDENCE_EXECUTED_TOTAL`.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 30.

| Quantity                                               | Symbol                            | Value / range | Units or note                                          |
|--------------------------------------------------------|-----------------------------------|---------------|--------------------------------------------------------|
| Admissible-state measure floor                         | $\mu(\Omega(K_0))$                | $> 0$         | dimensionless<br>lower bound                           |
| Meta-space compatibility ratio for each promoted level | $\text{DoF}(K_x)/\text{DoF}(M_x)$ | $\leq 1$      | dimensionless<br>ratio; applies for each x in 1,...,12 |
| Non-trivial cycle presence                             | $C_{\text{nt}}$                   | 1             | present                                                |

**Falsifier and collapse boundary.** Collapse occurs if the admissible set degenerates, if a later K-level exceeds its meta-space, or if the structural substrate loses its last non-trivial lawful cycle. **Falsifier refs:** `APPENDIX_Q_SYNTHESIS_DOSSIER_K0`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k0.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`, `content/falsifiability/falsifiability_k11.tex`, `content/falsifiability/falsifiability_k12.tex`.

**Closed-domain bindings.** `MATHEMATICS: class THEOREM_NATIVE, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE`. **Packet ref:** `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics`. **Metrics:** `strict_campaign_acceptance_status=PASS_31_OF_31_TERMINALIZED`, `strict_terminalized_total=31`.

## Q.2 Level K1: minimal observable distinctions and energetic axes

Current closure status: `PASS`. Promotion blockers: none.

**Theorem claim.** K1 introduces resolvable energetic and temporal distinctions while preserving the admissible-state floor inherited from K0.

**Operator and K-level binding.** F, G, and U bind contradiction load to resolvable energetic distinctions, monotonic update order, and a non-degenerate observable region. **Kernel refs:** `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTHESIS_DOSSIER_K1`.

**Parameter law.**

$$\Delta E \geq \Theta_{\Delta E}, \quad \frac{dt}{d\tau} > 0, \quad \mu(\Omega(K_1)) > 0$$

**Observable binding and units.** The first observable regime is tracked by energy-gap resolution, monotonic update ordering, and positive configuration-space measure. Observable ids: `K1::ENERGY_GAP_THRESHOLD`, `K1::TEMPORAL_MONOTONICITY`, `K1::CONFIGURATION_MEASURE`, `PHYSICS::FINE_STRUCTURE_CONSTANT_RESIDUAL`, `PHYSICS::HYDROGEN_BALMER_LINE_RESIDUAL`, `PHYSICS::TRANSPORT_SCALING_RESIDUAL`.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 31.

| Quantity                          | Symbol              | Value / range           | Units or note             |
|-----------------------------------|---------------------|-------------------------|---------------------------|
| Minimal resolvable energy scale   | $\Theta_{\Delta E}$ | $10^{-34}$ – $10^{-33}$ | J                         |
| Temporal monotonicity             | $dt/d\tau$          | $> 0$                   | ordered update constraint |
| Configuration-space measure floor | $\mu(\Omega(K_1))$  | $> 0$                   | dimensionless lower bound |

**Falsifier and collapse boundary.** Collapse begins when energetic distinctions are no longer resolvable, temporal order breaks, or the first observable state space degenerates. Falsifier refs: `APPENDIX_Q_SYNTESIS_DOSSIER_K1`, `content/falsifiability/falsifiability_master.tex`, `content/falsifiability/falsifiability_k1.tex`, `content/falsifiability/falsifiability_k2.tex`, `content/falsifiability/falsifiability_k9.tex`, `content/falsifiability/falsifiability_k10.tex`.

**Closed-domain bindings.** `PHYSICS: class THEOREM_NATIVE`, verdict `PASS`, held-out cases 20, theorem packet `COMPLETE`, parameter law `COMPLETE`. Packet ref: `releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics`. Metrics: `covered_case_total=30`, `covered_held_out_case_total=20`, `normalized_error_mean_abs=0.188136`, `normalized_error_p95=0.64778`, `normalized_error_max=1.48`, `tail_breach_count=0`.

### Q.3 Level K2: fields, phases, and large-scale structure

Current closure status: `PASS`. Promotion blockers: none.

**Theorem claim.** K2 binds field configurations, expansion history, and phase thresholds into the first large-scale physical continuum admissible under the kernel operators.

**Operator and K-level binding.** F, H, and R constrain field admissibility, phase transitions, and large-scale stabilization margins inside the physical continuum. Kernel refs: `content/03_model.tex`, `content/10_klevels_full.tex`, `content/11_operators_full.tex`, `APPENDIX_Q_SYNTESIS_DOSSIER_K2`.

**Parameter law.**

$$H_0 > 0, \quad 0 < \Omega_m < 1, \quad |\Omega_k| < 2 \times 10^{-3}, \quad \mathcal{T}_{\text{phase}}(t) \in \{T_c^{\text{EW}}, T_c^{\text{QCD}}\}$$

**Observable binding and units.** Expansion rate, curvature, cosmological density fractions, and critical phase windows tied to physical observables and replay packets. Observable ids: K2::EXPANSION\_RATE, K2::CURVATURE\_BOUND, K2::PHASE\_TRANSITION\_WINDOW, CHEMISTRY::GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY::SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY::KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL, PHYSICS::FINE\_STRUCTURE\_CONSTANT\_RESIDUAL, PHYSICS::HYDROGEN\_BALMER\_LINE\_RESIDUAL, PHYSICS::TRANSPORT\_SCALING\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 32.

| Quantity                | Symbol                              | Value / range                                            | Units or note                      |
|-------------------------|-------------------------------------|----------------------------------------------------------|------------------------------------|
| Hubble parameter today  | $H_0$                               | $67.4 \pm 0.5$                                           | $\text{km s}^{-1} \text{Mpc}^{-1}$ |
| Matter density fraction | $\Omega_m$                          | $0.315 \pm 0.007$                                        | dimensionless                      |
| Critical phase windows  | $T_c^{\text{EW}}, T_c^{\text{QCD}}$ | $\sim 100 \text{ GeV};$<br>$150\text{--}170 \text{ MeV}$ | closed phase set                   |
| Curvature bound         | $\Omega_k$                          | $ \Omega_k  < 2 \times 10^{-3}$                          | dimensionless                      |

**Falsifier and collapse boundary.** K2 collapses when curvature exits the admissible band, phase windows fail to close lawfully, or large-scale physical observables lose replayable stabilization. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K2, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k2.tex, content/falsifiability/falsifiability\_k3.tex, content/falsifiability/falsifiability\_k4.tex, content/falsifiability/falsifiability\_k6.tex, content/falsifiability/falsifiability\_k1.tex, content/falsifiability/falsifiability\_k9.tex, content/falsifiability/falsifiability\_k10.tex.

**Closed-domain bindings.** CHEMISTRY: class THEOREM\_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/chemistry. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=19, normalized\_error\_mean\_abs=0.533285, normalized\_error\_p95=1.0, normalized\_error\_max=1.597772, tail\_breach\_count=0. DRT: class REFUTED, verdict FAIL\_CLOSED, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: none. Metrics: none. PHYSICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/physics. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=20, normalized\_error\_mean\_abs=0.188136, normalized\_error\_p95=0.64778, normalized\_error\_max=1.48, tail\_breach\_count=0.

#### Q.4 Level K3: molecular organization and chemical bonding

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K3 packages bounded molecular organization through lawful bond-length, bond-energy, and activation-energy envelopes derived from lower-level admissibility.

**Operator and K-level binding.** G, H, and R bind composition-sensitive bonding, activation barriers, and molecular-scale stabilization windows. Kernel refs: content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K3.

**Parameter law.**

$$r_{\text{cov}} \in [0.1, 0.2] \text{ nm}, \quad E_{\text{bond}} \in [1, 5] \text{ eV}, \quad E^\ddagger \in [0.1, 1] \text{ eV}$$

**Observable binding and units.** Bond lengths, dissociation energies, dielectric context, and activation barriers for bounded chemistry lanes. Observable ids: K3::BOND\_LENGTH, K3::BOND\_ENERGY, K3::ACTIVATION\_ENERGY, CHEMISTRY::GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY::SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY::KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 33.

| Quantity                   | Symbol            | Value / range | Units or note |
|----------------------------|-------------------|---------------|---------------|
| Bond length (covalent)     | $r_{\text{cov}}$  | 0.1–0.2       | nm            |
| Bond dissociation energy   | $E_{\text{bond}}$ | 1–5           | eV            |
| Reaction activation energy | $E^\ddagger$      | 0.1–1         | eV            |

**Falsifier and collapse boundary.** K3 collapses when bond- and barrier-level regularities cannot be bounded by a non-degenerate parameter law or when chemical observables drift outside admissible molecular windows. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K3, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k2.tex, content/falsifiability/falsifiability\_k3.tex, content/falsifiability/falsifiability\_k4.tex.

**Closed-domain bindings.** CHEMISTRY: class THEOREM\_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/chemistry. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=19, normalized\_error\_mean\_abs=0.533285, normalized\_error\_p95=1.0, normalized\_error\_max=1.597772, tail\_breach\_count=0.

**Q.5 Level K4: compartmental and membrane thresholds**

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K4 closes the first compartmental organization packet by binding membrane thickness, bending stability, and gradient maintenance to lawful biological thresholds.

**Operator and K-level binding.** H, R, and S couple membrane stability, osmotic admissibility, and retained gradients into a bounded protocellular regime. Kernel refs: content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K4.

**Parameter law.**

$$d_{\text{mem}} \in [3, 5] \text{ nm}, \quad \kappa \in [10, 25] k_{\text{B}}T, \quad \Delta pH \in [0.5, 2]$$

$$\gamma_{\text{edge}} \in [5, 20] \text{ pN}, \quad \Pi \in [10^5, 10^6] \text{ Pa}, \quad P_i \in [10^{-12}, 10^{-8}] \text{ m/s}$$

**Observable binding and units.** Membrane thickness, bending modulus, osmotic pressure, proton gradient, and permeability lanes for compartmental organization. Observable ids: K4::MEMBRANE\_THICKNESS, K4::BENDING\_MODULUS, K4::PROTON\_GRADIENT, K4::LINE\_TENSION, K4::OSMOTIC\_PRESSURE\_SCALE, K4::PERMEABILITY\_RANGE, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL, CHEMISTRY::GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY::SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY::KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 34.

| Quantity                 | Symbol                 | Value / range          | Units or note |
|--------------------------|------------------------|------------------------|---------------|
| Membrane thickness       | $d_{\text{mem}}$       | 3–5                    | nm            |
| Membrane bending modulus | $\kappa$               | 10–25                  | $k_B T$       |
| Typical proton gradient  | $\Delta pH$            | 0.5–2                  | pH units      |
| Line tension (edge)      | $\gamma_{\text{edge}}$ | 5–20                   | pN            |
| Osmotic pressure scale   | $\Pi$                  | $10^5$ – $10^6$        | Pa            |
| Permeability range       | $P_i$                  | $10^{-12}$ – $10^{-8}$ | m/s           |

**Falsifier and collapse boundary.** K4 collapses if compartments cannot retain a lawful gradient, if membranes lose structural stability, or if permeability escapes the admissible packet. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K4, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k4.tex, content/falsifiability/falsifiability\_k5.tex, content/falsifiability/falsifiability\_k6.tex, content/falsifiability/falsifiability\_k7.tex, content/falsifiability/falsifiability\_k2.tex, content/falsifiability/falsifiability\_k3.tex.

**Closed-domain bindings.** BIOLOGY: class THEOREM\_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/biology. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=16, normalized\_error\_mean\_abs=0.444222, normalized\_error\_p95=1.030033, normalized\_error\_max=1.438582, tail\_breach\_count=0. CHEMISTRY: class THEOREM\_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/chemistry. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=19, normalized\_error\_mean\_abs=0.533285, normalized\_error\_p95=1.0, normalized\_error\_max=1.597772, tail\_breach\_count=0.

## Q.6 Level K5: excitable and early neural continua

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K5 binds excitability, refractory timing, and membrane potential maintenance into the first lawful neural packet.

**Operator and K-level binding.** F, G, and S stabilize excitable-state transitions, ion-channel gating, and refractory boundaries in bounded neural lanes. Kernel refs: content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K5.

**Parameter law.**

$$C_m = 1 \mu\text{F}/\text{cm}^2, \quad V_{\text{rest}} \in [-75, -60] \text{ mV}, \quad \tau_{\text{ref}} \in [2, 5] \text{ ms}$$

**Observable binding and units.** Membrane capacitance, resting potential, channel conductance, and refractory timing for bounded excitable systems. Observable ids: K5::MEMBRANE\_CAPACITANCE, K5::RESTING\_POTENTIAL, K5::REFRACTORY\_TIME, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 35.

| Quantity                   | Symbol              | Value / range | Units or note             |
|----------------------------|---------------------|---------------|---------------------------|
| Membrane capacitance       | $C_m$               | 1             | $\mu\text{F}/\text{cm}^2$ |
| Resting membrane potential | $V_{\text{rest}}$   | -75 – -60     | mV                        |
| Refractory time            | $\tau_{\text{ref}}$ | 2–5           | ms                        |

**Falsifier and collapse boundary.** K5 collapses when excitability loses bounded thresholds, refractory timing fails, or the neural packet cannot survive counterexample pressure across datasets. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K5, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k4.tex, content/falsifiability/falsifiability\_k5.tex, content/falsifiability/falsifiability\_k6.tex, content/falsifiability/falsifiability\_k7.tex.

**Closed-domain bindings.** BIOLOGY: class THEOREM\_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/biology. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=16, normalized\_error\_mean\_abs=0.444222, normalized\_error\_p95=1.030033, normalized\_error\_max=1.438582, tail\_breach\_count=0.

**Q.7 Level K6: cognitive prediction and representation**

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K6 formalizes bounded cognitive prediction and representation by linking prediction-error thresholds, working-memory span, and adaptation strength.

**Operator and K-level binding.** Q, R, and S bind representation stability, prediction thresholds, and adaptive update constraints in bounded cognitive lanes. Kernel refs: content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K6.

**Parameter law.**

$$\Theta_{\text{pred}} \in [0.05, 0.15], \quad M_{\text{WM}} \in [3, 7], \quad \eta \in [10^{-3}, 10^{-1}]$$

**Observable binding and units.** Prediction error, working-memory span, and adaptation strength for bounded cognitive organization. Observable ids: K6::PREDICTION\_ERROR\_THRESHOLD, K6::WORKING\_MEMORY\_SPAN, K6::HEBBIAN\_STRENGTH, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL.



**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 36.

| Quantity                     | Symbol                 | Value / range         | Units or note    |
|------------------------------|------------------------|-----------------------|------------------|
| Prediction-error threshold   | $\Theta_{\text{pred}}$ | 0.05–0.15             | normalized units |
| Working-memory span          | $M_{\text{WM}}$        | 3–7                   | bound items      |
| Hebbian strength coefficient | $\eta$                 | $10^{-3}$ – $10^{-1}$ | dimensionless    |

**Falsifier and collapse boundary.** K6 collapses when predictive stabilization fails across bounded cognitive observables or when representation thresholds require hidden freedom. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K6, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k4.tex, content/falsifiability/falsifiability\_k5.tex, content/falsifiability/falsifiability\_k6.tex, content/falsifiability/falsifiability\_k7.tex, content/falsifiability/falsifiability\_k2.tex.

**Closed-domain bindings.** BIOLOGY: class THEOREM\_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/biology. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=16, normalized\_error\_mean\_abs=0.444222, normalized\_error\_p95=1.030033, normalized\_error\_max=1.438582, tail\_breach\_count=0. DRT: class REFUTED, verdict FAIL\_CLOSED, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: none. Metrics: none.

## Q.8 Level K7: social coordination and group stability

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K7 closes bounded social coordination through group size, communication bandwidth, and coordination thresholds.

**Operator and K-level binding.** R, S, and U bind coordination thresholds, communication throughput, and conflict costs into admissible social trajectories. Kernel refs: content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K7.

**Parameter law.**

$$N_s \in [120, 180], \quad J_{\text{comm}} \in [1, 5] \text{ bits/s}, \quad \Theta_{\text{coord}} \in [0.2, 0.4] \\ C_{\text{conf}} \in [0.1, 1]$$

**Observable binding and units.** Group stability size, communication throughput, coordination threshold, and conflict cost inside social coordination continua. Observable ids: K7::GROUP\_STABILITY\_SIZE, K7::COMMUNICATION\_BANDWIDTH, K7::COORDINATION\_THRESHOLD, K7::CONFLICT\_COST\_SCALE, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 37.

| Quantity                             | Symbol                  | Value / range | Units or note   |
|--------------------------------------|-------------------------|---------------|-----------------|
| Group stability size                 | $N_s$                   | 120–180       | agents          |
| Information-flow bandwidth per agent | $J_{\text{comm}}$       | 1–5           | bits/s          |
| Coordination threshold               | $\Theta_{\text{coord}}$ | 0.2–0.4       | dimensionless   |
| Conflict cost scale                  | $C_{\text{conf}}$       | 0.1–1         | arbitrary units |

**Falsifier and collapse boundary.** K7 collapses when communication throughput and coordination thresholds cannot stabilize social group trajectories without hidden rescue assumptions. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K7, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k4.tex, content/falsifiability/falsifiability\_k5.tex, content/falsifiability/falsifiability\_k6.tex, content/falsifiability/falsifiability\_k7.tex.

**Closed-domain bindings.** BIOLOGY: class THEOREM\_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/biology. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=16, normalized\_error\_mean\_abs=0.444222, normalized\_error\_p95=1.030033, normalized\_error\_max=1.438582, tail\_breach\_count=0.

## Q.9 Level K8: civilizational flows and infrastructural stability

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K8 binds civilizational energy density, renewal timing, and repair-decay balance into bounded macro-process trajectories and regime-shift packets.

**Operator and K-level binding.** H, R, and U bind throughput, repair-decay balance, and regime-shift boundaries in bounded systems-level observables. Kernel refs: content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K8.

**Parameter law.**

$$E_{\text{dens}} \in [10^2, 10^3] \text{ W/m}^2, \quad \tau_{\text{infra}} \in [20, 50] \text{ years}, \quad R_{\text{rep}} \in [0.8, 1.2] \\ \Theta_I \in [0.1, 0.3]$$

**Observable binding and units.** Energy density, infrastructure renewal, information coherence, and repair-decay balance in bounded civilizational lanes. Observable ids: K8::ENERGY\_CONSUMPTION\_DENSITY, K8::INFRASTRUCTURE\_RENEWAL\_TIME, K8::REPAIR\_TO\_DECAY\_RATIO, K8::INFORMATION\_COHERENCE\_THRESHOLD, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::POPULATION\_TRAJECTORY\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::GDP\_GROWTH\_TURNING\_POINT\_ERROR, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::PROCESS\_SYSTEM\_REGIME\_SCORE.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 38.

| Quantity                        | Symbol                | Value / range       | Units or note    |
|---------------------------------|-----------------------|---------------------|------------------|
| Energy consumption density      | $E_{\text{dens}}$     | $10^2\text{--}10^3$ | W/m <sup>2</sup> |
| Infrastructure renewal time     | $\tau_{\text{infra}}$ | 20–50               | years            |
| Information-coherence threshold | $\Theta_I$            | 0.1–0.3             | dimensionless    |
| Repair-to-decay ratio           | $R_{\text{rep}}$      | 0.8–1.2             | dimensionless    |

**Falsifier and collapse boundary.** K8 collapses when macro-trajectories lose bounded turning-point fidelity, when repair-decay balance drifts outside its band, or when regime-shift packets require discretionary knobs. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K8, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k8.tex.

**Closed-domain bindings.** SYSTEMS\_CIVILIZATIONAL\_PROJECTION: class THEOREM\_NATIVE, verdict PASS, held-out cases 18, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/systems\_civilizational\_projection. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=18, normalized\_error\_mean\_abs=0.56097, normalized\_error\_p95=1.924704, normalized\_error\_max=2.25395, tail\_breach\_count=0.

## Q.10 Level K9: knowledge systems and scientific memory

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K9 formalizes bounded scientific-knowledge continua through anomaly accumulation, paradigm coherence, and model-validation bandwidth.

**Operator and K-level binding.** Q, R, and U bind anomaly handling, paradigm coherence, and memory retention inside knowledge-system packets. Kernel refs: content/03\_model.tex, content/10\_levels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K9.

**Parameter law.**

$$\Theta_{\text{par}} \in [0.15, 0.25], \quad J_{\text{anom}} \in [10^{-4}, 10^{-2}], \quad \tau_{\text{mem}} \in [30, 200] \text{ years} \\ B_{\text{val}} \in [0.1, 1]$$

**Observable binding and units.** Paradigm coherence, anomaly accumulation, validation bandwidth, and scientific-memory half-life. Observable ids: K9::PARADIGM\_COHERENCE\_THRESHOLD, K9::ANOMALY\_ACCUMULATION\_RATE, K9::SCIENTIFIC\_MEMORY\_HALF\_LIFE, K9::MODEL\_VALIDATION\_BANDWIDTH, MATHEMATICS::EPSILON\_STAR, MATHEMATICS::MINIMAL\_DENOMINATOR, MATHEMATICS::MINIMAL\_AUGMENTATION\_RANK, MATHEMATICS::COMPONENT\_FIBER\_COUNT, MATHEMATICS::MOVE\_CONNECTIVITY\_OR\_DEGREE\_BOUND, MATHEMATICS::STRICT\_TERMINALIZED\_TOTAL, MATHEMATICS::COUNTEREXAMPLE\_EXECUTED\_TOTAL, MATHEMATICS::PYTHON\_EVIDENCE\_EXECUTED\_TOTAL, PHYSICS::FINE\_STRUCTURE\_CONSTANT\_RESIDUAL, PHYSICS::HYDROGEN\_BALMER\_LINE\_RESIDUAL, PHYSICS::TRANSPORT\_SCALING\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 39.

| Quantity                     | Symbol                | Value / range         | Units or note |
|------------------------------|-----------------------|-----------------------|---------------|
| Paradigm-coherence threshold | $\Theta_{\text{par}}$ | 0.15–0.25             | dimensionless |
| Anomaly accumulation rate    | $J_{\text{anom}}$     | $10^{-4}$ – $10^{-2}$ | per cycle     |
| Scientific-memory half-life  | $\tau_{\text{mem}}$   | 30–200                | years         |
| Model-validation bandwidth   | $B_{\text{val}}$      | 0.1–1                 | normalized    |

**Falsifier and collapse boundary.** K9 collapses if anomaly load outruns lawful model revision or if scientific memory and validation bandwidth fail to stabilize the knowledge packet. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K9, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k0.tex, content/falsifiability/falsifiability\_k9.tex, content/falsifiability/falsifiability\_k10.tex, content/falsifiability/falsifiability\_k11.tex, content/falsifiability/falsifiability\_k12.tex, content/falsifiability/falsifiability\_k1.tex, content/falsifiability/falsifiability\_k2.tex.

**Closed-domain bindings.** MATHEMATICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/mathematics. Metrics: strict\_campaign\_acceptance\_status=PASS\_31\_OF\_31\_TERMINALIZED, strict\_terminalized\_total=31. PHYSICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/physics. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=20, normalized\_error\_mean\_abs=0.188136, normalized\_error\_p95=0.64778, normalized\_error\_max=1.48, tail\_breach\_count=0.

## Q.11 Level K10: formal systems and recursion

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K10 binds bounded recursion, consistency thresholds, and proof-flow capacity into the formal-system lane of the unified-science closure stack.

**Operator and K-level binding.** F, Q, and U bind recursion depth, proof throughput, and consistency thresholds in formal continua. Kernel refs: content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K10.

**Parameter law.**

$$d_{\text{rec}} \in [10, 10^4], \quad \Theta_{\text{cons}} \in [0.01, 0.05], \quad J_{\text{proof}} \in [1, 10^3]$$

**Observable binding and units.** Recursion depth, consistency threshold, and proof-flow capacity in bounded formal systems. Observable ids: K10::RECURSION\_DEPTH, K10::CONSISTENCY\_THRESHOLD, K10::PROOF\_FLOW\_CAPACITY, MATHEMATICS::EPSILON\_STAR, MATHEMATICS::MINIMAL\_DENOMINATOR, MATHEMATICS::MINIMAL\_AUGMENTATION\_RANK, MATHEMATICS::COMPONENT\_FIBER\_COUNT, MATHEMATICS::MOVE\_CONNECTIVITY\_OR\_DEGREE\_BOUND, MATHEMATICS::STRICT\_TERMINALIZED\_TOTAL, MATHEMATICS::COUNTEREXAMPLE\_EXECUTED\_TOTAL, MATHEMATICS::P

YTHON\_EVIDENCE\_EXECUTED\_TOTAL, PHYSICS::FINE\_STRUCTURE\_CONSTANT\_RESIDUAL, PHYSICS::HYDROGEN\_BALMER\_LINE\_RESIDUAL, PHYSICS::TRANSPORT\_SCALING\_RESIDUAL.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 40.

| Quantity                    | Symbol                 | Value / range      | Units or note |
|-----------------------------|------------------------|--------------------|---------------|
| Recursion depth (effective) | $d_{\text{rec}}$       | 10–10 <sup>4</sup> | steps         |
| Consistency threshold       | $\Theta_{\text{cons}}$ | 0.01–0.05          | dimensionless |
| Proof-flow capacity         | $J_{\text{proof}}$     | 1–10 <sup>3</sup>  | steps/s       |

**Falsifier and collapse boundary.** K10 collapses if formal recursion and proof throughput cannot remain bounded without violating consistency thresholds. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K10, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k0.tex, content/falsifiability/falsifiability\_k9.tex, content/falsifiability/falsifiability\_k10.tex, content/falsifiability/falsifiability\_k11.tex, content/falsifiability/falsifiability\_k12.tex, content/falsifiability/falsifiability\_k1.tex, content/falsifiability/falsifiability\_k2.tex.

**Closed-domain bindings.** MATHEMATICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/mathematics. Metrics: strict\_campaign\_acceptance\_status=PASS\_31\_OF\_31\_TERMINALIZED, strict\_terminalized\_total=31. META ONTOLOGY: class FRAME\_ONLY, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/metaontology. Metrics: none. PHYSICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/physics. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=20, normalized\_error\_mean\_abs=0.188136, normalized\_error\_p95=0.64778, normalized\_error\_max=1.48, tail\_breach\_count=0.

## Q.12 Level K11: meta-theoretical reflexivity

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K11 captures meta-theoretical reflexivity as irreducible scientific work beyond K10 across the closed domain atlas.

**Operator and K-level binding.** Q, R, and S bind meta-coherence, reflexive depth, and cross-landscape coupling under the requirement that no K10-only translation preserves all declared upper-level burdens. Kernel refs: content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K11.

**Parameter law.**

$$\Theta_{\text{meta}} \in [0.02, 0.1], \quad d_{\text{refl}} \in [3, 50], \quad J_X \in [0.1, 1] \\ P_{\text{funct}} \in [10^0, 10^3]$$

**Observable binding and units.** Meta-coherence, reflexive depth, and cross-landscape coupling in the meta-theoretical packet. Observable ids: K11::META\_COHERENCE\_THRESHOLD, K11::REFLEXIVE\_DEPTH, K11::CROSS\_LANDSCAPE\_COUPLING, K11::FUNCTORIAL\_POTENTIAL, MATHEMATICS::EPILON\_STAR, MATHEMATICS::MINIMAL\_DENOMINATOR, MATHEMATICS::MINIMAL\_AUGMENTATION\_RANK, MATHEMATICS::COMPONENT\_FIBER\_COUNT, MATHEMATICS::MOVE\_CONNECTIVITY\_OR\_DEGREE\_BOUND, MATHEMATICS::STRICT\_TERMINALIZED\_TOTAL, MATHEMATICS::COUNTEREXAMPLE\_EXECUTED\_TOTAL, MATHEMATICS::PYTHON\_EVIDENCE\_EXECUTED\_TOTAL, PHYSICS::FINE\_STRUCTURE\_CONSTANT\_RESIDUAL, PHYSICS::HYDROGEN\_BALMER\_LINE\_RESIDUAL, PHYSICS::TRANSPORT\_SCALING\_RESIDUAL, CHEMISTRY::GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY::SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY::KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::POPULATION\_TRAJECTORY\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::GDP\_GROWTH\_TURNING\_POINT\_ERROR, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::PROCESS\_SYSTEM\_REGIME\_SCORE.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 41.

| Quantity                 | Symbol                 | Value / range   | Units or note  |
|--------------------------|------------------------|-----------------|----------------|
| Meta-coherence threshold | $\Theta_{\text{meta}}$ | 0.02–0.1        | dimensionless  |
| Functorial potential     | $P_{\text{funct}}$     | $10^0$ – $10^3$ | abstract units |
| Reflexive depth          | $d_{\text{refl}}$      | 3–50            | levels         |
| Cross-landscape coupling | $J_X$                  | 0.1–1           | dimensionless  |

**Falsifier and collapse boundary.** K11 collapses if an explicit K10-only reduction preserves the declared meta-theoretical burdens or if meta-coherence depends on an excluded lift route. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K11, content/falsifiability/falsifiability\_master.tex, content/falsifiability/falsifiability\_k0.tex, content/falsifiability/falsifiability\_k9.tex, content/falsifiability/falsifiability\_k10.tex, content/falsifiability/falsifiability\_k11.tex, content/falsifiability/falsifiability\_k12.tex, content/falsifiability/falsifiability\_k1.tex, content/falsifiability/falsifiability\_k2.tex, content/falsifiability/falsifiability\_k3.tex, content/falsifiability/falsifiability\_k4.tex, content/falsifiability/falsifiability\_k5.tex, content/falsifiability/falsifiability\_k6.tex, content/falsifiability/falsifiability\_k7.tex, content/falsifiability/falsifiability\_k8.tex.

**Closed-domain bindings.** MATHEMATICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/mathematics. Metrics: strict\_campaign\_acceptance\_status=PASS\_31\_OF\_31\_TERMINALIZED, strict\_terminalized\_total=31. PHYSICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/physics. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=20, normalized\_error\_mean\_abs=0.188136, normalized\_error\_p95=0.64778, normalized\_error\_max=1.48, tail\_breach\_count=0. CHEMISTRY: class THEOREM\_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/chemistry. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=19, normalized\_error\_mean\_abs=0.533285, normalized\_error\_p95=1.0, normalized\_error\_max=1.597772, tail\_breach\_count=0. BIOLOGY: class THEOREM\_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law

COMPLETE. **Packet ref:** releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/biology. **Metrics:** covered\_case\_total=30, covered\_held\_out\_case\_total=16, normalized\_error\_mean\_abs=0.444222, normalized\_error\_p95=1.030033, normalized\_error\_max=1.438582, tail\_breach\_count=0. **SYSTEMS\_CIVILIZATIONAL\_PROJECTION:** class THEOREM\_NATIVE, verdict PASS, held-out cases 18, theorem packet COMPLETE, parameter law COMPLETE. **Packet ref:** releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/systems\_civilizational\_projection. **Metrics:** covered\_case\_total=30, covered\_held\_out\_case\_total=18, normalized\_error\_mean\_abs=0.56097, normalized\_error\_p95=1.924704, normalized\_error\_max=2.25395, tail\_breach\_count=0. **METAONTOLOGY:** class FRAME\_ONLY, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. **Packet ref:** releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/metaontology. **Metrics:** none.

### Q.13 Level K12: global semantic coherence

Current closure status: PASS. Promotion blockers: none.

**Theorem claim.** K12 packages global semantic coherence across the closed domain atlas because cross-domain meaning remains jointly interpretable and no surviving global falsifier breaks the unified synthesis.

**Operator and K-level binding.** Q, R, S, and U bind semantic coherence, global compatibility, and reference stability into the final unified-science synthesis layer. **Kernel refs:** content/03\_model.tex, content/10\_klevels\_full.tex, content/11\_operators\_full.tex, APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K12.

**Parameter law.**

$$C_{\text{uni}} \in [10^3, 10^{12}], \quad \Theta_{\text{uni}} \in [10^{-3}, 10^{-2}], \quad R_{\text{uni}} \in [0.1, 1] \\ T_{\text{uni}} \in [10^2, 10^8]$$

**Observable binding and units.** Universal integration capacity, cross-continuum compatibility, and structural reachability. **Observable ids:** K12::UNIVERSAL\_INTEGRATION\_CAPACITY, K12::CROSS\_CONTINUUM\_COMPATIBILITY, K12::STRUCTURAL\_REACHABILITY, K12::GLOBAL\_TENSION\_BUDGET, MATHEMATICS::EPSILON\_STAR, MATHEMATICS::MINIMAL\_DENOMINATOR, MATHEMATICS::MINIMAL\_AUGMENTATION\_RANK, MATHEMATICS::COMPONENT\_FIBER\_COUNT, MATHEMATICS::MOVE\_CONNECTIVITY\_OR\_DEGREE\_BOUND, MATHEMATICS::STRICT\_TERMINALIZED\_TOTAL, MATHEMATICS::COUNTEREXAMPLE\_EXECUTED\_TOTAL, MATHEMATICS::PYTHON\_EVIDENCE\_EXECUTED\_TOTAL, PHYSICS::FINE\_STRUCTURE\_CONSTANT\_RESIDUAL, PHYSICS::HYDROGEN\_BALMER\_LINE\_RESIDUAL, PHYSICS::TRANSPORT\_SCALING\_RESIDUAL, CHEMISTRY::GAS\_PHASE\_ENTHALPY\_RESIDUAL, CHEMISTRY::SPECTRAL\_TRANSITION\_RESIDUAL, CHEMISTRY::KINETIC\_OR\_EQUILIBRIUM\_RESIDUAL, BIOLOGY::EXPRESSION\_SIGNATURE\_RESIDUAL, BIOLOGY::STATE\_TRANSITION\_ORDER\_ERROR, BIOLOGY::RESPONSE\_ONSET\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::POPULATION\_TRAJECTORY\_RESIDUAL, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::GDP\_GROWTH\_TURNING\_POINT\_ERROR, SYSTEMS\_CIVILIZATIONAL\_PROJECTION::PROCESS\_SYSTEM\_REGIME\_SCORE.

**Numerical instantiation and prediction table.** The table below is a promoted subset of the numerical backing rows in Appendix 42.

| Quantity                       | Symbol                | Value / range             | Units or note |
|--------------------------------|-----------------------|---------------------------|---------------|
| Universal integration capacity | $C_{\text{uni}}$      | $10^3\text{--}10^{12}$    | dimensionless |
| Cross-continuum compatibility  | $\Theta_{\text{uni}}$ | $10^{-3}\text{--}10^{-2}$ | dimensionless |
| Structural reachability        | $R_{\text{uni}}$      | 0.1–1                     | dimensionless |
| Global tension budget          | $T_{\text{uni}}$      | $10^2\text{--}10^8$       | dimensionless |

**Falsifier and collapse boundary.** K12 collapses if cross-domain compatibility fragments, if the unified atlas cannot pass integrability, or if a surviving global falsifier breaks semantic coherence. Falsifier refs: APPENDIX\_Q\_SYNTHESIS\_DOSSIER\_K12, content/falsifiability/falsifiability\_maste r.tex, content/falsifiability/falsifiability\_k0.tex, content/falsifiability/fa lsifiability\_k9.tex, content/falsifiability/falsifiability\_k10.tex, content/fa lsifiability/falsifiability\_k11.tex, content/falsifiability/falsifiability\_k1 2.tex, content/falsifiability/falsifiability\_k1.tex, content/falsifiability/fa lsifiability\_k2.tex, content/falsifiability/falsifiability\_k3.tex, content/fal sifiability/falsifiability\_k4.tex, content/falsifiability/falsifiability\_k5.t ex, content/falsifiability/falsifiability\_k6.tex, content/falsifiability/falsi fiability\_k7.tex, content/falsifiability/falsifiability\_k8.tex.

**Closed-domain bindings.** MATHEMATICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editor ial/dossier\_packages/closure\_bundles/mathematics. Metrics: strict\_campaign\_acce ptance\_status=PASS\_31\_OF\_31\_TERMINALIZED, strict\_terminalized\_total=31. PHYS ICS: class THEOREM\_NATIVE, verdict PASS, held-out cases 20, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure \_bundles/physics. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=20, normalized\_error\_mean\_abs=0.188136, normalized\_error\_p95=0.64778, normalized\_err or\_max=1.48, tail\_breach\_count=0. CHEMISTRY: class THEOREM\_NATIVE, verdict PASS, held-out cases 19, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3 /editorial/dossier\_packages/closure\_bundles/chemistry. Metrics: covered\_case\_to tal=30, covered\_held\_out\_case\_total=19, normalized\_error\_mean\_abs=0.533285, norm alized\_error\_p95=1.0, normalized\_error\_max=1.597772, tail\_breach\_count=0. BIOLOG Y: class THEOREM\_NATIVE, verdict PASS, held-out cases 16, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bu ndles/biology. Metrics: covered\_case\_total=30, covered\_held\_out\_case\_total=16, no rmalized\_error\_mean\_abs=0.444222, normalized\_error\_p95=1.030033, normalized\_erro r\_max=1.438582, tail\_breach\_count=0. SYSTEMS\_CIVILIZATIONAL\_PROJECTION: class THE OREM\_NATIVE, verdict PASS, held-out cases 18, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: releases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/sys tems\_civilizational\_projection. Metrics: covered\_case\_total=30, covered\_held\_out \_case\_total=18, normalized\_error\_mean\_abs=0.56097, normalized\_error\_p95=1.924704, normalized\_error\_max=2.25395, tail\_breach\_count=0. METAONTOLOGY: class FRAME\_ONL Y, verdict PASS, held-out cases 0, theorem packet COMPLETE, parameter law COMPLETE. Packet ref: r eleases/oc\_core\_1\_3/editorial/dossier\_packages/closure\_bundles/metaontology. Metrics: none.



## R Practical Utility and Model Comparison Atlas

This appendix materialises five inspection sections referenced in Section 25: use cases, operational playbooks, same-claim baselines, serious-model comparators, and readiness/frontier status. The governing JSON surface remains the complete machine-readable record.

- **Governing surface:** `logion/k0/governance/status/OC_CORE_1_3_PRACTICAL_UTILITY_ATLAS_latest.json`.
- **Release mirror:** `releases/oc_core_1_3/editorial/OC_CORE_1_3_PRACTICAL_UTILITY_ATLAS_latest.json`.
- **Atlas id:** `OC_CORE_1_3_PRACTICAL_UTILITY_ATLAS`.
- **Schema id:** `OC_CORE_1_3_PRACTICAL_UTILITY_ATLAS_v1`.
- **repository SHA:** recorded at `metadata.repo_sha` in the governing surface.
- **Release timestamp (UTC):** recorded at `metadata.ts_utc` in the governing surface.

The cross-domain comparison rows use two prior comparator families: best-in-class local-model portfolio governance, and complexity-science or multiscale-integration programmes.

### R.1 Full use-case table

| Use-case id                                       | Domain      | Support                                         | Required inputs                                                                                                 | Output / decision                                                                                                                                                  | Scope boundary                                                                                                                                                                                     |
|---------------------------------------------------|-------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MATHEMATICS_USE_001_STRICT_CLOSURE_CERTIFICATION  | Mathematics | formally proved                                 | formal problem statement; admissible exact-question encoding; strict closure and counterexample policy          | A replayable accept or reject result together with minimal denominator, augmentation rank, or surviving counterexample classification.                             | Applies only to the bounded exact question families already represented in the mathematical anchor. It does not claim a general uniqueness theorem beyond the current impossibility boundary.      |
| PHYSICS_USE_001_BOUNDED_RESIDUAL_AUDIT            | Physics     | theorem-native + empirical / held-out validated | official constants or spectra or transport benchmark family; declared held-out split; residual tolerance policy | A pass or fail residual verdict, anomaly prioritisation, and a decision about whether the family remains inside the promoted packet.                               | Applies only to the promoted constants, Balmer spectra, and transport-scaling packet. It does not replace unrestricted microphysical law solving or the full local-derivation stack.               |
| PHYSICS_USE_002_BROADENER_MICROPHYSICAL_EXTENSION | Physics     | frontier program                                | new observable family; new theorem target; new held-out route                                                   | A lawful frontier-workbench campaign definition rather than an already closed prediction engine.                                                                   | No broader microphysical lane is currently promoted by this row. It is a disciplined research program, not a deployable closed packet.                                                             |
| CHEMISTRY_USE_001_COMPOUND_REACTION_PACKET_AUDIT  | Chemistry   | theorem-native + empirical / held-out validated | compound or reaction-class family; pinned reference families; held-out rotation and tolerance policy            | A pass or fail residual verdict, compound or reaction-family screening result, and a decision about whether the lane remains inside the promoted chemistry packet. | Applies only to the promoted thermochemical, spectroscopic, and kinetic packet. It is not a general replacement for unrestricted quantum chemistry or catalyst-specific design.                    |
| CHEMISTRY_USE_002_GENERAL_CATALYST_DISCOVERY      | Chemistry   | frontier program                                | new compound or catalyst family; new observable family; new held-out route                                      | A bounded frontier campaign definition for chemistry extension rather than an already closed design system.                                                        | This row is not a promoted deployment lane. It defines a lawful research program rather than a closed chemistry engine.                                                                            |
| BIOLOGY_USE_001_TRANSITION_AND_RESPONSE_SCREENING | Biology     | theorem-native + empirical / held-out validated | bounded assay lane; multi-dataset held-out split; declared signature or onset observable family                 | A pass or fail signature verdict, a state-transition ordering judgment, and an escalation decision for the biological lane.                                        | Applies only to the promoted expression, state-transition, and response-onset packet. It does not authorize unrestricted whole-organism biology, therapy design, or universal clinical prediction. |

| Use-case id                                                                           | Domain                         | Support                                         | Required inputs                                                                                                         | Output / decision                                                                                                                                           | Scope boundary                                                                                                                                                             |
|---------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIOLOGY_USE_002_MULTI_SCALE_PHENOTYPE_CONTROL_SYSTEMS_USE_001_REGIME_SHIFT_MONITORING | Biology                        | frontier program                                | new biological lane; new observable family; new dataset or assay route                                                  | A bounded frontier campaign definition rather than an already lawful intervention engine.                                                                   | This row is not a promoted deployment lane. It is a frontier biological research program only.                                                                             |
| SYSTEMS_USE_002_UNRESTRICTED_CIVILIZATIONAL_FORECASTING                               | Systems /K8 projection         | theorem-native + empirical / held-out validated | pinned official macro-process series; held-out interval policy; declared trajectory or regime observable family         | A pass or fail trajectory verdict, a turning-point or regime-shift alert, and a decision about whether the lane remains inside the promoted systems packet. | Applies only to the promoted macro-trajectory and regime-shift packet. It does not authorize unrestricted civilizational prediction or unlimited strategic forecasting.    |
| SYSTEMS_USE_002_UNRESTRICTED_CIVILIZATIONAL_FORECASTING                               | Systems /K8 projection         | hypothesis only                                 | new macro-process families; longer horizon interval families; new theorem and comparator program                        | A clearly demarcated hypothesis boundary rather than a promoted prediction product.                                                                         | This row is hypothesis only and may not be used as an operational forecasting or control authority.                                                                        |
| UNIFIED_USE_001_CROSS_DOMAIN_SELECTION                                                | Cross-domain / unified science | operationally supported within bounds           | declared problem class; target observable family; decision horizon or held-out interval; candidate same-claim baselines | A lawful packet-selection decision, a comparator shortlist, and an explicit escalation path when the closed packet is not enough.                           | This row supports bounded cross-domain routing and coordination only. It does not authorize an unrestricted one-shot universal predictor or universal intervention engine. |

## R.2 Use-case trace-anchor matrix

Trace anchors in this appendix use publication-safe alias keys for legacy compatibility filenames. The alias keys are stable release anchors resolved by the review-target manifest and by the generated projection surfaces.

| Use-case id                                      | Trace anchors                                                                                                                                                                                                                                                                | Bundle / comparator hooks                                                                                                                                                                                                                                                              |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MATHEMATICS_USE_001_STRICT_CLOSURE_CERTIFICATION | content/20_oc_core_1_3_theorem_roadmap.tex, content/21_oc_core_1_3_worked_examples.tex, content/24_oc_core_1_3_proof_machinery.tex, appendix/M_oc_core_1_3_proof_machinery_appendix.tex                                                                                      | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/mathematics/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics; Comparators exact enumerative baseline; direct constructive proof baseline.         |
| PHYSICS_USE_001_BOUNDED_RESIDUAL_AUDIT           | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, CHAPTER_25_UNIFIED_SYNTHESIS, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json                                                           | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics; Comparators bounded residual fit without theorem trace; family-wise least-squares baseline. |
| PHYSICS_USE_002_BROADER_MICROPHYSICAL_EXTENSION  | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_FULL_SCIENTIFIC_CLOSURE_PROGRAM_latest.json, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics; Comparators bounded residual fit without theorem trace; family-wise least-squares baseline. |

| Use-case id                                                             | Trace anchors                                                                                                                                                                                                                                                                | Bundle / comparator hooks                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CHEMISTRY_US<br>E_001_COMPOU<br>ND_REACTION_<br>PACKET_AUDIT            | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, CHAPTER_25_UNIFIED_SYNTHESIS_releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json                                                            | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry; Comparators reference-table envelope baseline; reaction-class residual fit baseline.                                         |
| CHEMISTRY_US<br>E_002_GENERA<br>L_CATALYST_D<br>ISCOVERY                | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_FULL_SCIENTIFIC_CLOSURE_PROGRAM_latest.json, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry; Comparators reference-table envelope baseline; reaction-class residual fit baseline.                                         |
| BIOLOGY_USE_<br>001_TRANSITI<br>ON_AND_RESPO<br>NSE_SCREENIN<br>G       | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, CHAPTER_25_UNIFIED_SYNTHESIS_releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json                                                            | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology; Comparators dataset-family signature fit baseline; state-transition score baseline.                                              |
| BIOLOGY_USE_<br>002_MULTI_SC<br>ALE_PHENOTYP<br>E_CONTROL               | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_FULL_SCIENTIFIC_CLOSURE_PROGRAM_latest.json, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology; Comparators dataset-family signature fit baseline; state-transition score baseline.                                              |
| SYSTEMS_USE_<br>001_REGIME_S<br>HIFT_MONITOR<br>ING                     | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, CHAPTER_25_UNIFIED_SYNTHESIS_releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json                                                            | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection; Comparators macro-trend residual baseline; turning-point heuristic baseline. |
| SYSTEMS_USE_<br>002_UNRESTRI<br>CTED_CIVILIZ<br>ATIONAL_FORE<br>CASTING | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_FULL_SCIENTIFIC_CLOSURE_PROGRAM_latest.json, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection; Comparators macro-trend residual baseline; turning-point heuristic baseline. |
| UNIFIED_USE_<br>001_CROSS_DO<br>MAIN_ROUTE_S<br>ELECTION                | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, content/24_oc_core_1_3_proof_machinery.tex, CHAPTER_25_UNIFIED_SYNTHESIS                                                                                    | Comparators best-in-class local-model portfolio governance; multi-model expert routing without shared theorem trace.                                                                                                                                                                                                        |

### R.3 Operational playbook matrix

| Playbook id                                  | Domain                         | When to use                                                                                                                                                                                       | Procedure and failure boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MATHEMATICS_PLAYBOOK_01_STRICT_CLOSURE       | Mathematics                    | Use when the task is to determine whether a bounded exact question already lies inside the promoted mathematical anchor or should be rejected by the strict counterexample route.                 | Procedure: 1. Encode the target object in the admissible exact-question form. 2. Run the declared terminalization and counterexample search. 3. Accept the result only if the replay is deterministic and survivor-free. 4. Escalate to frontier if the question leaves the declared exact scope. Failure boundary: Stop if the object cannot be encoded in the admissible exact-question format or if a surviving counterexample remains after the declared search.                                                                 |
| PHYSICS_PLAYBOOK_001_BOUNDED_RESIDUAL_AUDIT  | Physics                        | Use when the task is to decide whether a constants family, spectral family, or transport family belongs inside the promoted physics packet or should be escalated as an anomaly.                  | Procedure: 1. Map the observable family to the promoted physics packet. 2. Load the pinned official dataset and the locked held-out split. 3. Run the declared residual replay and inspect sign or order constraints. 4. Accept the family only if the replay stays inside the promoted tolerance band. Failure boundary: Stop and demote to frontier if the observable family requires a widened scope, hidden fit freedom, or a post-hoc split change.                                                                             |
| CHEMISTRY_PLAYBOOK_001_PACKET_AUDIT          | Chemistry                      | Use when the task is to decide whether a bounded compound family or reaction class lies inside the promoted chemistry packet or should be escalated as an out-of-envelope case.                   | Procedure: 1. Map the chemistry lane to the promoted packet and observable family. 2. Load the pinned reference tables and the locked held-out split. 3. Run the declared replay on thermochemical, spectral, or kinetic outputs. 4. Accept the lane only if the held-out family remains inside the promoted tolerance band. Failure boundary: Stop and demote to frontier if the lane requires hidden fit freedom, widened scope, or post-hoc split changes.                                                                        |
| BIOLOGY_PLAYBOOK_001_SIGNATURE_SCREENING     | Biology                        | Use when the task is to decide whether a bounded assay lane belongs inside the promoted biological packet or should be escalated as an unresolved biological signature family.                    | Procedure: 1. Map the biological lane to the promoted packet and observable family. 2. Load the locked multi-dataset route and held-out split. 3. Run the declared replay on expression, transition-order, or onset outputs. 4. Accept the lane only if the held-out datasets stay inside the promoted signature envelope. Failure boundary: Stop and demote to frontier if the lane depends on a single dataset family, hidden fit freedom, or a widened biological claim boundary.                                                 |
| SYSTEMS_PLAYBOOK_001_REGIME_SHIFT_MONITORING | Systems /K8 projection         | Use when the task is to decide whether an official macro-process series remains inside the promoted systems packet or should be escalated as a turning-point or regime anomaly.                   | Procedure: 1. Map the macro-process lane to the promoted systems packet. 2. Load the pinned official series and the locked held-out interval. 3. Run the declared residual and turning-point replay. 4. Accept the lane only if held-out intervals remain inside the promoted trajectory and regime envelope. Failure boundary: Stop and demote to frontier if the lane requires out-of-band trajectory behaviour, a widened horizon, or post-hoc interval selection.                                                                |
| UNIFIED_PLAYBOOK_001_ROUTE_SELECTION         | Cross-domain / unified science | Use when a scientific or engineering problem touches more than one domain lane and the team must decide which closed packet, comparator family, and evidence route should govern the next action. | Procedure: 1. Map the target problem to the narrowest lawful observable family. 2. Select the closed packet whose theorem-to-observable map already governs that family. 3. Check same-claim baselines and serious comparators before claiming advantage. 4. Escalate only when the closed packet boundary is reached and the atlas permits the translation. Failure boundary: Stop and demote to frontier if the observable family is outside every closed packet or if two candidate routes produce incompatible bounded verdicts. |

## R.4 Operational playbook trace-anchor matrix

| Playbook id                             | Trace anchors                                                                                                                                                                           | Bundle hooks                                                                                                                                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MATHEMATICS_PLAYBOOK_001_STRICT_CLOSURE | content/20_oc_core_1_3_theorem_roadmap.tex, content/21_oc_core_1_3_worked_examples.tex, content/24_oc_core_1_3_proof_machinery.tex, appendix/M_oc_core_1_3_proof_machinery_appendix.tex | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/mathematics/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics. |

| Playbook id                                  | Trace anchors                                                                                                                                                                                                                                                               | Bundle hooks                                                                                                                                                                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PHYSICS_PLAYBOOK_001_BOUNDED_RESIDUAL_AUDIT  | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_DOMAIN_CLOSURE_BUNDLE_REGISTRY_latest.json, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics.                                                     |
| CHEMISTRY_PLAYBOOK_001_PACKET_AUDIT          | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_DOMAIN_CLOSURE_BUNDLE_REGISTRY_latest.json, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry.                                                 |
| BIOLOGY_PLAYBOOK_001_SIGNATURE_SCREENING     | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_DOMAIN_CLOSURE_BUNDLE_REGISTRY_latest.json, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology.                                                     |
| SYSTEMS_PLAYBOOK_001_REGIME_SHIFT_MONITORING | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_DOMAIN_CLOSURE_BUNDLE_REGISTRY_latest.json, releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json | Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection. |
| UNIFIED_PLAYBOOK_001_ROUTE_SELECTION         | content/22_oc_core_1_3_operationalization_program.tex, content/23_oc_core_1_3_empirical_execution_protocols.tex, content/24_oc_core_1_3_proof_machinery.tex, releases/oc_core_1_3/editorial/OC_CORE_1_3_UNIFIED_SCIENCE_ATLAS_latest.json                                   | No additional bundle hook is declared for this playbook.                                                                                                                                                                                       |

## R.5 Same-claim baseline matrix

| Comparison id                 | Domain      | Comparator families                                            | Verdict / trace anchors                                                                                                                                                                                                                                                                                                             |
|-------------------------------|-------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAME_CLAIM_CLASS::MATHEMATICS | Mathematics | exact enumerative baseline; direct constructive proof baseline | Verdict DECLARED_NO_SUPERIOR_BASLINE; Cost-normalized budget NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION; Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/mathematics/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/mathematics. |

| Comparison id                                       | Domain                 | Comparator families                                                            | Verdict / trace anchors                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------|------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAME_CLAIM_CLASS::PHYSICS                           | Physics                | bounded residual fit without theorem trace; family-wise least-squares baseline | Verdict NO_SIMPLER_SUPERIOR_BASELINE_FOUND; Cost-normalized budget NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION; Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/physics.                                                     |
| SAME_CLAIM_CLASS::CHEMISTRY                         | Chemistry              | reference-table envelope baseline; reaction-class residual fit baseline        | Verdict NO_SIMPLER_SUPERIOR_BASELINE_FOUND; Cost-normalized budget NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION; Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/chemistry.                                                 |
| SAME_CLAIM_CLASS::BIOLOGY                           | Biology                | dataset-family signature fit baseline; state-transition score baseline         | Verdict NO_SIMPLER_SUPERIOR_BASELINE_FOUND; Cost-normalized budget NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION; Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/biology.                                                     |
| SAME_CLAIM_CLASS::SYSTEMS_CIVILIZATIONAL_PROJECTION | Systems /K8 projection | macro-trend residual baseline; turning-point heuristic baseline                | Verdict NO_SIMPLER_SUPERIOR_BASELINE_FOUND; Cost-normalized budget NO_COMPARATOR_MAY_WIN_ON_SAME_CLAIM_CLASS_AFTER_COST_NORMALIZATION; Source bundle releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection/bundle.json; Closure dossier releases/oc_core_1_3/editorial/dossier_packages/closure_bundles/systems_civilizational_projection. |

## R.6 Serious model comparison matrix

| Domain  | Model family                                                  | Problem class                         | Prior strength / fragmentation                                                                                                                                                                                                                                                              | Relation    | OC gain                                                                                                                                                          |
|---------|---------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physics | Gyrokinetic, confinement-scaling, and plasma-transport models | transport or plasma scaling envelopes | Provides strong local transport and confinement heuristics for specific plasma regimes and benchmark devices. These models are powerful in their native lane but do not by themselves close a unified residual grammar spanning constants, spectra, and transport under one promotion rule. | complements | OC adds a shared bounded packet and replay rule that lets transport residual families live beside constants and spectra under one inspected comparison boundary. |

| Domain                        | Model family                                                            | Problem class                                          | Prior strength / fragmentation                                                                                                                                                                                                                                                                                                                       | Relation                         | OC gain                                                                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physics                       | Precision quantum electrodynamics and atomic-structure theory           | fundamental constants and spectral transition families | Delivers high-precision local calculations for relations among constants, atomic transitions, and spectral families inside explicit microscopic equation classes. Selection and justification of the local model family remain lane-specific, and the route does not by itself give a shared kernel-to-observable closure discipline across domains. | integrates                       | OC adds one bounded theorem-to-observable packet that keeps constants, Balmer spectra, and transport residual families under one explicit closure, falsifier, and held-out replay contract. |
| Chemistry                     | Chemical kinetics, transition-state, and equilibrium models             | reaction-class kinetic or equilibrium envelopes        | Explains rate, barrier, and equilibrium behaviour for declared reaction classes with strong domain-specific local formalisms. The family remains reaction-class specific and does not itself unify thermochemistry, spectroscopy, and kinetics under one kernel-bound promotion route.                                                               | integrates                       | OC adds a bounded chemistry packet that keeps those observable families under one theorem, one parameter-law shell, and one held-out replay discipline.                                     |
| Chemistry                     | Quantum chemistry, ab initio, and density-functional families           | thermochemical and spectroscopic envelopes             | Provides molecule-specific energetic and spectroscopic calculations with a mature tool chain for local chemical systems. Method choice, basis dependence, and domain-specific tuning remain lane-specific, and the family does not by itself provide a closure ladder from kernel operators to held-out benchmark governance.                        | remains superior within its lane | OC adds a bounded packet that ties thermochemical, spectroscopic, and kinetic envelopes to one theorem-native route with one falsifier and one replay discipline.                           |
| Biology                       | Excitable-media and response-onset dynamical models                     | response-onset residual lanes                          | Captures threshold and onset dynamics well in specific excitability, signalling, or response families. The family remains local to its chosen mechanism and does not by itself close broader biological signature, transition, and onset lanes under one bounded acceptance bar.                                                                     | complements                      | OC adds a single bounded biological packet that keeps response onset tied to expression and transition-order evidence inside one replay and falsifier framework.                            |
| Biology                       | Gene-regulatory, transcriptomic, and systems-biology state-space models | expression-signature and state-transition lanes        | Provides established state-space, regulatory-network, and transcriptomic modelling inside specific assay and dataset families. Performance is often lane- and dataset-specific, and cross-dataset closure plus theorem-to-observable promotion discipline remain fragmented.                                                                         | integrates                       | OC adds a bounded packet that keeps expression signatures, state-transition order, and response onset under one multi-dataset held-out closure route.                                       |
| Systems /K8 projection        | Systems dynamics, control, and change-point models                      | macro-process trajectories and regime shifts           | Provides strong local tools for trend extraction, control analysis, and regime-change detection in declared systems lanes. These tools are typically lane-specific and do not by themselves close a shared theorem route from kernel operators to official macro-process held-out promotion.                                                         | integrates                       | OC adds a bounded macro-trajectory packet that keeps trajectory residuals, turning points, and regime scores under one theorem-to-observable route and one held-out decision bar.           |
| Cross-domain /unified science | Complexity-science and multiscale integration programs                  | cross-domain integration and multiscale explanation    | Provides rich ideas about emergence, multiscale organisation, and complex adaptive behaviour across many scientific domains. These programs are often strong conceptually but heterogeneous in proof bar, operator discipline, and closed promotion criteria.                                                                                        | complements                      | OC adds a harder closure grammar: explicit theorem packets, parameter laws, held-out replay, falsifiers, and same-claim comparator ledgers for each bounded lane.                           |

| Domain                        | Model family                                                                   | Problem class                                    | Prior strength / fragmentation                                                                                                                                                                                                                 | Relation   | OC gain                                                                                                                                                                         |
|-------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cross-domain /unified science | Best-in-class local-model portfolio governance across separate domain theories | cross-domain scientific planning and explanation | Explains many local scientific lanes well by using separate best-in-class models inside each domain. Cross-domain translation, shared operator meaning, and unified promotion discipline remain fragmented across disconnected model families. | integrates | OC adds one explicit kernel, one K-level ladder, one theorem-to-observable grammar, and one atlas for translating bounded packets across domains under declared scope controls. |

## R.7 Serious model comparison trace matrix

| Comparison id                                          | Domain    | Trace anchors                                                                                                                                                                                                                                                                   | Scope boundary                                                                                                                  |
|--------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::PHYSICS::PLASMA_TRANSPORT_MODELS   | Physics   | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json, content/14_disciplines_extended.tex, content/26_oc_core_1_3_practical_utility.tex   | OC does not replace specialised transport solvers or device-specific plasma modelling outside the promoted packet.              |
| SERIOUS_COMPARATOR::PHYSICS::QUANTUM_ATOMICS_PRECISION | Physics   | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/physics/bundle.json, content/14_disciplines_extended.tex, content/26_oc_core_1_3_practical_utility.tex   | Detailed local derivation outside the promoted residual packet still relies on the standard physics model stack.                |
| SERIOUS_COMPARATOR::CHEMISTRY::KINETICS_AND_TST        | Chemistry | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json, content/14_disciplines_extended.tex, content/26_oc_core_1_3_practical_utility.tex | OC does not erase the need for local kinetic modelling outside the promoted bounded families.                                   |
| SERIOUS_COMPARATOR::CHEMISTRY::QUANTUM_CHEMISTRY       | Chemistry | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/chemistry/bundle.json, content/14_disciplines_extended.tex, content/26_oc_core_1_3_practical_utility.tex | High-precision molecule-specific calculation remains the job of the established chemistry stack outside the promoted OC packet. |
| SERIOUS_COMPARATOR::BIOLOGY::EXCITABILITY_AND_RESPONSE | Biology   | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json, content/14_disciplines_extended.tex, content/26_oc_core_1_3_practical_utility.tex   | Detailed mechanism-specific onset modelling still belongs to the established lane-specific models outside the promoted packet.  |



| Comparison id                                                  | Domain                         | Trace anchors                                                                                                                                                                                                                                                                                           | Scope boundary                                                                                                                       |
|----------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::BIOLOGY::SYSTEMS_BIOLOGY                   | Biology                        | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/biology/bundle.json, content/14_disciplines_extended.tex, content/26_oc_core_1_3_practical_utility.tex                           | OC does not yet replace detailed mechanistic biological modelling outside the promoted assay lanes.                                  |
| SERIOUS_COMPARATOR::SYSTEMS::SYSTEM_DYNAMICS_AND_CHANGE_POINTS | Systems /K8 projection         | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, releases/oc_core_1_3/editorial/science_sources/closure_bundles/systems_civilizational_projection_bundle.json, content/14_disciplines_extended.tex, content/26_oc_core_1_3_practical_utility.tex | OC does not promote unrestricted civilizational prediction or replace every local macro or control model outside the bounded packet. |
| SERIOUS_COMPARATOR::CROSS_DOMAIN::COMPLEXITY_PROGRAMS          | Cross-domain / unified science | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, content/24_oc_core_1_3_proof_machinery.tex, CHAPTER_25_UNIFIED_SYNTHESIS, content/26_oc_core_1_3_practical_utility.tex                                                                          | OC does not exhaust every possible complexity-science question; it narrows attention to the lanes it can lawfully close.             |
| SERIOUS_COMPARATOR::CROSS_DOMAIN::REDUCTIONIST_PATCHWORK       | Cross-domain / unified science | releases/oc_core_1_3/editorial/science_sources/practical_utility/serious_model_comparison_catalog.json, content/24_oc_core_1_3_proof_machinery.tex, CHAPTER_25_UNIFIED_SYNTHESIS, content/26_oc_core_1_3_practical_utility.tex                                                                          | OC does not abolish local domain theories; it governs how bounded closed packets are related and promoted together.                  |

## R.8 Readiness and open-frontier matrix

| Use-case id                                       | Domain      | Readiness                                       | What remains open                                                                                                                                             |
|---------------------------------------------------|-------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MATHEMATICS_USE_001_STRICT_CLOSURE_CERTIFICATION  | Mathematics | formally proved                                 | Global uniqueness beyond the current impossibility boundary remains frontier territory rather than a promoted practical result.                               |
| PHYSICS_USE_001_BOUNDED_RESIDUAL_AUDIT            | Physics     | theorem-native + empirical / held-out validated | Full unrestricted microphysical derivation outside the promoted residual packet remains open and may not be advertised as closed by this use-case row.        |
| PHYSICS_USE_002_BROADENED_MICROPHYSICAL_EXTENSION | Physics     | frontier program                                | OC Core 1.3 does not yet lawfully promote a general OC replacement for the full physics model stack.                                                          |
| CHEMISTRY_USE_001_FOUNDED_REACTION_PACKET_AUDIT   | Chemistry   | theorem-native + empirical / held-out validated | Unrestricted molecule-specific precision chemistry and catalyst-design lanes remain outside the promoted packet and must not be advertised as already closed. |
| CHEMISTRY_USE_002_GENERATED_CATALYST_DISCOVERY    | Chemistry   | frontier program                                | OC Core 1.3 does not yet promote a general OC catalyst-discovery or unrestricted synthetic-chemistry engine.                                                  |

| Use-case id                                             | Domain                         | Readiness                                       | What remains open                                                                                                                         |
|---------------------------------------------------------|--------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| BIOLOGY_USE_001_TRANSITION_AND_RESPONSE_SCREENING       | Biology                        | theorem-native + empirical / held-out validated | Unrestricted phenotype control, broad intervention design, and general clinical deployment remain outside the promoted biological packet. |
| BIOLOGY_USE_002_MULTI_SCALE_PHENOTYPE_CONTROL           | Biology                        | frontier program                                | OC Core 1.3 does not yet promote a general OC biological intervention or whole-phenotype control engine.                                  |
| SYSTEMS_USE_001_REGIME_SHIFT_MONITORING                 | Systems /K8 projection         | theorem-native + empirical / held-out validated | Unrestricted long-horizon civilizational prediction remains outside the promoted systems packet.                                          |
| SYSTEMS_USE_002_UNRESTRICTED_CIVILIZATIONAL_FORECASTING | Systems /K8 projection         | hypothesis only                                 | OC Core 1.3 does not yet promote unrestricted long-horizon civilizational forecasting or control as lawful closed science.                |
| UNIFIED_USE_001_CROSS_DOMAIN_ROUTE_SELECTION            | Cross-domain / unified science | operationally supported within bounds           | Unrestricted cross-domain intervention planning beyond the declared closed packets remains outside lawful promotion.                      |

## R.9 Fragmentation-to-closure matrix

| Comparison id                                        | Domain    | Prior science already did                                                                                                                                         | Fragmentation or limit                                                                                                                                                                               | What OC closes or adds                                                                                                                                                                      | Open boundary                                                                                                                   |
|------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::PHYSICS::PLASMA_TRANSPORT_MODELS | Physics   | Provides strong local transport and confinement heuristics for specific plasma regimes and benchmark devices.                                                     | These models are powerful in their native lane but do not by themselves close a unified residual grammar spanning constants, spectra, and transport under one promotion rule.                        | OC adds a shared bounded packet and replay rule that lets transport residual families live beside constants and spectra under one inspected comparison boundary.                            | OC does not replace specialised transport solvers or device-specific plasma modelling outside the promoted packet.              |
| SERIOUS_COMPARATOR::PHYSICS::QED_ATOM_PRECISION      | Physics   | Delivers high-precision local calculations for relations among constants, atomic transitions, and spectral families inside explicit microscopic equation classes. | Selection and justification of the local model family remain lane-specific, and the route does not by itself give a shared kernel-to-observable closure discipline across domains.                   | OC adds one bounded theorem-to-observable packet that keeps constants, Balmer spectra, and transport residual families under one explicit closure, falsifier, and held-out replay contract. | Detailed local derivation outside the promoted residual packet still relies on the standard physics model stack.                |
| SERIOUS_COMPARATOR::CHEMISTRY::KINETICS_AND_TST      | Chemistry | Explains rate, barrier, and equilibrium behaviour for declared reaction classes with strong domain-specific local formalisms.                                     | The family remains reaction-class specific and does not itself unify thermochemistry, spectroscopy, and kinetics under one kernel-bound promotion route.                                             | OC adds a bounded chemistry packet that keeps those observable families under one theorem, one parameter-law shell, and one held-out replay discipline.                                     | OC does not erase the need for local kinetic modelling outside the promoted bounded families.                                   |
| SERIOUS_COMPARATOR::CHEMISTRY::QUANTUM_CHEMISTRY     | Chemistry | Provides molecule-specific energetic and spectroscopic calculations with a mature tool chain for local chemical systems.                                          | Method choice, basis dependence, and domain-specific tuning remain lane-specific, and the family does not by itself provide a closure ladder from kernel operators to held-out benchmark governance. | OC adds a bounded packet that ties thermochemical, spectroscopic, and kinetic envelopes to one theorem-native route with one falsifier and one replay discipline.                           | High-precision molecule-specific calculation remains the job of the established chemistry stack outside the promoted OC packet. |

| Comparison id                                            | Domain                        | Prior science already did                                                                                                       | Fragmentation or limit                                                                                                                                                    | What OC closes or adds                                                                                                                                                            | Open boundary                                                                                                                        |
|----------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| SERIOUS_COMPARATOR::BIOLOGY::EXCITABILITY_AND_RESPONSE   | Biology                       | Captures threshold and onset dynamics well in specific excitability, signalling, or response families.                          | The family remains local to its chosen mechanism and does not by itself close broader biological signature, transition, and onset lanes under one bounded acceptance bar. | OC adds a single bounded biological packet that keeps response onset tied to expression and transition-order evidence inside one replay and falsifier framework.                  | Detailed mechanism-specific onset modelling still belongs to the established lane-specific models outside the promoted packet.       |
| SERIOUS_COMPARATOR::BIOLOGY::SYSTEMS_BIOLOGY             | Biology                       | Provides established state-space, regulatory-network, and transcriptional modelling inside specific assay and dataset families. | Performance is often lane- and dataset-specific, and cross-dataset closure plus theorem-to-observable promotion discipline remain fragmented.                             | OC adds a bounded packet that keeps expression signatures, state-transition order, and response onset under one multi-dataset held-out closure route.                             | OC does not yet replace detailed mechanistic biological modelling outside the promoted assay lanes.                                  |
| SERIOUS_COMPARATOR::SYSTEMS_DYNAMICS_AND_CHANGE_POINTS   | Systems /K8 projection        | Provides strong local tools for trend extraction, control analysis, and regime-change detection in declared systems lanes.      | These tools are typically lane-specific and do not by themselves close a shared theorem route from kernel operators to official macro-process held-out promotion.         | OC adds a bounded macro-trajectory packet that keeps trajectory residuals, turning points, and regime scores under one theorem-to-observable route and one held-out decision bar. | OC does not promote unrestricted civilizational prediction or replace every local macro or control model outside the bounded packet. |
| SERIOUS_COMPARATOR::CROSS_DOMAIN::COMPLEXITY_PROGRAMS    | Cross-domain /unified science | Provides rich ideas about emergence, multiscale organisation, and complex adaptive behaviour across many scientific domains.    | These programs are often strong conceptually but heterogeneous in proof bar, operator discipline, and closed promotion criteria.                                          | OC adds a harder closure grammar: explicit theorem packets, parameter laws, held-out replay, falsifiers, and same-claim comparator ledgers for each bounded lane.                 | OC does not exhaust every possible complexity-science question; it narrows attention to the lanes it can lawfully close.             |
| SERIOUS_COMPARATOR::CROSS_DOMAIN::REDUCTIONIST_PATCHWORK | Cross-domain /unified science | Explains many local scientific lanes well by using separate best-in-class models inside each domain.                            | Cross-domain translation, shared operator meaning, and unified promotion discipline remain fragmented across disconnected model families.                                 | OC adds one explicit kernel, one K-level ladder, one theorem-to-observable grammar, and one atlas for translating bounded packets across domains under declared scope controls.   | OC does not abolish local domain theories; it governs how bounded closed packets are related and promoted together.                  |

## R.10 Frontier and hypothesis boundary

| Domain                 | Problem class                                                           | Support          | Why this is not yet closed                                                                                                 |
|------------------------|-------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------|
| Physics                | Broader microphysical law extension beyond the promoted residual packet | frontier program | OC Core 1.3 does not yet lawfully promote a general OC replacement for the full physics model stack.                       |
| Chemistry              | General catalyst or out-of-family chemistry design                      | frontier program | OC Core 1.3 does not yet promote a general OC catalyst-discovery or unrestricted synthetic-chemistry engine.               |
| Biology                | Multi-scale phenotype control and unrestricted biological intervention  | frontier program | OC Core 1.3 does not yet promote a general OC biological intervention or whole-phenotype control engine.                   |
| Systems /K8 projection | Unrestricted long-horizon civilizational forecasting and control        | hypothesis only  | OC Core 1.3 does not yet promote unrestricted long-horizon civilizational forecasting or control as lawful closed science. |

## S OC 1.3.1 Science Delta Appendix

- delta\_id=DELTA::WHITE\_SPOT\_CLOSURE; delta\_class=scientific\_closure; status\_1\_3\_1=closure\_companion\_and\_action\_bucket=must\_be\_added
- delta\_id=DELTA::SIMULATION\_CORPUS; delta\_class=simulation\_and\_reproducibility; status\_1\_3\_1=top\_level\_simulation\_corpus; action\_bucket=must\_be\_added
- delta\_id=DELTA::DATA\_MANIFEST\_LAYER; delta\_class=data\_route\_and\_traceability; status\_1\_3\_1=manifest\_and\_action\_bucket=must\_be\_added

- delta\_id=DELTA::MASTER\_MONOGRAPH; delta\_class=release\_authority; status\_1\_3\_1=oc\_core\_1\_3\_1\_master\_p  
action\_bucket=must\_be\_added
- delta\_id=DELTA::ARTICLE\_FAMILY; delta\_class=publication\_surface; status\_1\_3\_1=bounded\_article\_family\_pres  
action\_bucket=must\_be\_added
- delta\_id=DELTA::RELEASE\_AUTOMATION; delta\_class=release\_automation; status\_1\_3\_1=contract\_and\_validat  
action\_bucket=must\_be\_added
- delta\_id=DELTA::NO\_SEND\_OUTBOUND; delta\_class=outbound\_readiness; status\_1\_3\_1=release\_specific\_packag  
action\_bucket=must\_be\_added
- delta\_id=DELTA::PUBLIC\_RELEASE\_STAGE; delta\_class=release\_staging; status\_1\_3\_1=oc\_core\_1\_3\_1\_ready\_n  
action\_bucket=candidate\_for\_1\_3\_1
- delta\_id=DELTA::K11\_K12\_EXTENSION; delta\_class=high\_cost\_theory\_extension; status\_1\_3\_1=mapped\_but\_not  
action\_bucket=must\_remain\_frontier
- delta\_id=DELTA::PAID\_DATA\_ESCALATIONS; delta\_class=data\_escalation; status\_1\_3\_1=still\_optional;  
action\_bucket=too\_expensive\_now

## **T OC 1.3.1 Simulation and Data Appendix**

- lane\_id=LANE::MATHEMATICS; domain=mathematics; strengthening\_status=STRENGTHENED\_NUMERICAL  
primary\_strengthening\_mode=NUMERIC\_PACKET
- lane\_id=LANE::PHYSICS; domain=physics; strengthening\_status=STRENGTHENED\_NUMERICAL  
primary\_strengthening\_mode=NUMERIC\_PACKET
- lane\_id=LANE::CHEMISTRY; domain=chemistry; strengthening\_status=STRENGTHENED\_NUMERICAL  
primary\_strengthening\_mode=NUMERIC\_PACKET
- lane\_id=LANE::K3\_K4\_ORIGINS; domain=k3\_k4\_origins\_of\_life; strengthening\_status=STRENGTHENED\_BY\_S  
primary\_strengthening\_mode=SIMULATION\_CORPUS
- lane\_id=LANE::BIOLOGY\_K4\_K5; domain=biology; strengthening\_status=STRENGTHENED\_BY\_DATA\_ROUTE  
primary\_strengthening\_mode=DATA\_MANIFEST\_BINDING
- lane\_id=LANE::COGNITION\_K6; domain=k6\_cognition; strengthening\_status=STRENGTHENED\_FORMALLY;  
primary\_strengthening\_mode=FORMULA\_AND\_OPERATOR\_PATCH
- lane\_id=LANE::SOCIETY\_K7; domain=k7\_society; strengthening\_status=STRENGTHENED\_FORMALLY;  
primary\_strengthening\_mode=FORMULA\_AND\_OPERATOR\_PATCH
- lane\_id=LANE::SYSTEMS\_K8; domain=k8\_systems; strengthening\_status=STRENGTHENED\_BY\_SIMULATION;  
primary\_strengthening\_mode=SIMULATION\_CORPUS
- lane\_id=LANE::KNOWLEDGE\_K9; domain=k9\_knowledge\_systems; strengthening\_status=STRENGTHENED\_FO  
primary\_strengthening\_mode=FORMULA\_AND\_OPERATOR\_PATCH
- lane\_id=LANE::FORMAL\_K10; domain=k10\_formal\_systems; strengthening\_status=STRENGTHENED\_FORMAL  
primary\_strengthening\_mode=FORMULA\_AND\_OPERATOR\_PATCH
- lane\_id=LANE::META\_K11\_K12; domain=k11\_k12\_meta\_theory; strengthening\_status=PARTIAL\_ONLY;  
primary\_strengthening\_mode=FRONTIER\_AND\_SIMULATION\_MIX

## **U OC 1.3.1 Critique Closure Appendix**

- critique\_id=CRITIQUE::DESKTOP\_CLOSEOUT\_STATE\_DRIFT; processing\_class=REAL\_PACKAGING\_WEAK  
output\_mode=clarification\_unit; publication\_effect=none

- critique\_id=CRITIQUE::PUBLIC\_REPO\_ARCHITECTURE\_DRIFT; processing\_class=REAL\_PACKAGING\_WEA  
output\_mode=clarification\_unit; publication\_effect=refresh public architecture truth and release gate  
explanation
- critique\_id=CRITIQUE::SCOPE\_INFLATION\_CHECK\_\_WEDGE\_001; processing\_class=REQUIRES\_DEMOTIO  
output\_mode=demotion\_decision; publication\_effect=bounded clarification note for reviewer/public  
audience
- critique\_id=CRITIQUE::PUBLIC\_REPO\_SIMULATION\_ABSENCE; processing\_class=REAL\_PACKAGING\_WEA  
output\_mode=clarification\_unit; publication\_effect=bind simulation corpus into OC Core 1.3.1 release  
package

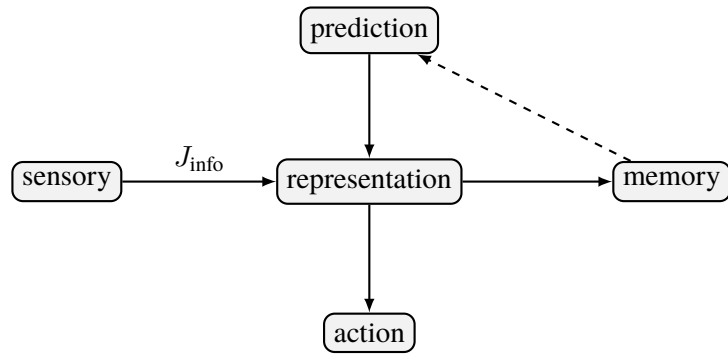


Figure 31: Flows between representations, models and data in  $K_9$ .

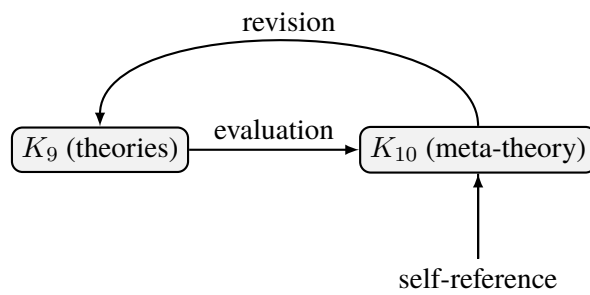


Figure 32: Self-referential structure of meta-theoretical continua at level  $K_{10}$ .